



Operation and Maintenance Manual

320, 323 Excavator

HEX 1-UP (320)
KFE 1-UP (320)
MYK 1-UP (320)
YCP 1-UP (320)
HDT 1-UP (323)
LTN 1-UP (323)
NDL 1-UP (323)
RAZ 1-UP (323)

Language: Original Instructions

Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards, including human factors that can affect safety. This person should also have the necessary training, skills and tools to perform these functions properly.

Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.

Do not operate or perform any lubrication, maintenance or repair on this product, until you verify that you are authorized to perform this work, and have read and understood the operation, lubrication, maintenance and repair information.

Safety precautions and warnings are provided in this manual and on the product. If these hazard warnings are not heeded, bodily injury or death could occur to you or to other persons.

The hazards are identified by the "Safety Alert Symbol" and followed by a "Signal Word" such as "DANGER", "WARNING" or "CAUTION". The Safety Alert "WARNING" label is shown below.



The meaning of this safety alert symbol is as follows:

Attention! Become Alert! Your Safety is Involved.

The message that appears under the warning explains the hazard and can be either written or pictorially presented.

A non-exhaustive list of operations that may cause product damage are identified by "NOTICE" labels on the product and in this publication.

Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. You must not use this product in any manner different from that considered by this manual without first satisfying yourself that you have considered all safety rules and precautions applicable to the operation of the product in the location of use, including site-specific rules and precautions applicable to the worksite. If a tool, procedure, work method or operating technique that is not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that you are authorized to perform this work, and that the product will not be damaged or become unsafe by the operation, lubrication, maintenance or repair procedures that you intend to use.

The information, specifications, and illustrations in this publication are on the basis of information that was available at the time that the publication was written. The specifications, torques, pressures, measurements, adjustments, illustrations, and other items can change at any time. These changes can affect the service that is given to the product. Obtain the complete and most current information before you start any job. Cat dealers have the most current information available.

NOTICE

When replacement parts are required for this product Caterpillar recommends using original Caterpillar® replacement parts.

Other parts may not meet certain original equipment specifications.

When replacement parts are installed, the machine owner/user should ensure that the machine remains in compliance with all applicable requirements.

In the United States, the maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

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Foreword

California Proposition 65 Warning

Diesel engine exhaust and some of its constituents are known to the State of California to cause cancer, birth defects, and other reproductive harm.



WARNING – This product can expose you to chemicals including ethylene glycol, which is known to the State of California to cause birth defects or other reproductive harm. For more information go to:

www.P65Warnings.ca.gov

Do not ingest this chemical. Wash hands after handling to avoid incidental ingestion.



WARNING – This product can expose you to chemicals including lead and lead compounds, which are known to the State of California to cause cancer, birth defects, or other reproductive harm. For more information go to:

www.P65Warnings.ca.gov

Wash hands after handling components that may contain lead.

Literature Information

This manual should be stored in the operator's compartment in the literature holder or seat back literature storage area.

This manual contains safety information, operation instructions, transportation information, lubrication information, and maintenance information.

Some photographs or illustrations in this publication show details or attachments that can be different from your machine. Guards and covers might have been removed for illustrative purposes.

Continuing improvement and advancement of product design might have caused changes to your machine which are not included in this publication. Read, study, and keep this manual with the machine.

Whenever a question arises regarding your machine, or this publication, please consult your Cat dealer for the latest available information.

Safety

The safety section lists basic safety precautions. In addition, this section identifies the text and locations of warning signs and labels used on the machine.

Read and understand the basic precautions listed in the safety section before operating or performing lubrication, maintenance, and repair on this machine.

Operation

The operation section is a reference for the new operator and a refresher for the experienced operator. This section includes a discussion of gauges, switches, machine controls, attachment controls, transportation, and towing information.

Photographs and illustrations guide the operator through correct procedures of checking, starting, operating, and stopping the machine.

Operating techniques outlined in this publication are basic. Skill and techniques develop as the operator gains knowledge of the machine and its capabilities.

Maintenance

The maintenance section is a guide to equipment care. The Maintenance Interval Schedule (MIS) lists the items to be maintained at a specific service interval. Items without specific intervals are listed under the "When Required" service interval. The Maintenance Interval Schedule lists the page number for the step-by-step instructions required to accomplish the scheduled maintenance. Use the Maintenance Interval Schedule as an index or "one safe source" for all maintenance procedures.

Maintenance Intervals

Use the service hour meter to determine servicing intervals. Calendar intervals shown (daily, weekly, monthly, etc.) can be used instead of service hour meter intervals if the calendar intervals provide more convenient servicing schedules and approximate the indicated service hour meter reading. Perform the recommended service at the interval that occurs first.

Under severe, dusty, or wet operating conditions, more frequent lubrication than is specified in the maintenance intervals chart might be necessary.

Perform service on items at multiples of the original requirement. For example, at every 500 service hours or 3 months, also service those items listed under every 250 service hours or monthly and every 10 service hours or daily.

Certified Engine Maintenance

Proper maintenance and repair are essential to keep the engine and machine systems operating correctly. As the heavy-duty off-road diesel engine owner, you are responsible for the performance of the required maintenance listed in the Owner Manual, Operation and Maintenance Manual, and Service Manual.

It is prohibited for any person engaged in the business of repairing, servicing, selling, leasing, or trading engines or machines to remove, alter, or to render inoperative, any emission-related device or element of design installed on or in an engine or machine that is in compliance with all applicable regulations of the intended country to which it has been shipped. Certain elements of the machine and engine such as the exhaust system, fuel system, electrical system, intake air system, and cooling system may be emission-related and should not be altered unless approved by Caterpillar.

Machine Capacity

Additional attachments or modifications may exceed machine design capacity which can adversely affect performance characteristics. Included would be stability and system certifications such as brakes, steering, and rollover protective structures (ROPS). Contact your Cat dealer for further information.

Product Identification Number

Effective First Quarter 2001 the Product Identification Number (PIN) has changed from 8 to 17 characters. To provide uniform equipment identification, construction equipment manufacturers are moving to comply with the latest version of the product identification numbering standard. Non-road machine PINs are defined by ISO 10261. The new PIN format will apply to all machines and generator sets. The PIN plates and frame marking will display the 17 character PIN. The new format will look like the following:

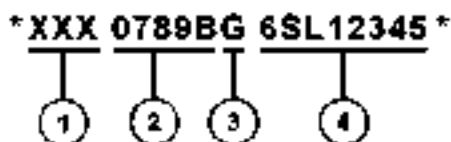


Illustration 1

g03891925

Where:

1. World Manufacturing Code (characters 1-3)

2. Machine Descriptor (characters 4-8)

3. Check Character (character 9)

4. Machine Indicator Section (MIS) or Product Sequence Number (characters 10-17). These were previously referred to as the Serial Number.

Machines and generator sets produced before First Quarter 2001 will maintain their 8 character PIN format.

Components such as engines, transmissions, axles, and work tools will continue to use an 8 character Serial Number (S/N).

Safety Section

i07503031

Safety Messages

SMCS Code: 7000; 7405

There are several specific safety messages on this machine. The exact location of the hazards and the description of the hazards are reviewed in this section. Please become familiarized with all safety messages.

Make sure that all the safety messages are legible. Clean the safety messages or replace the safety messages if you cannot read the words. Replace the illustrations if the illustrations are not visible. When you clean the safety messages, use a cloth, water, and soap. Do not use solvent, gasoline, or other harsh chemicals to clean the safety messages. Solvents, gasoline, or harsh chemicals could loosen the adhesive that secures the safety message. Loose adhesive will allow the safety message to fall.

Replace any safety message that is damaged, or missing. If a safety message is attached to a part that is replaced, install a safety message on the replacement part. Any Cat dealer can provide new safety messages.

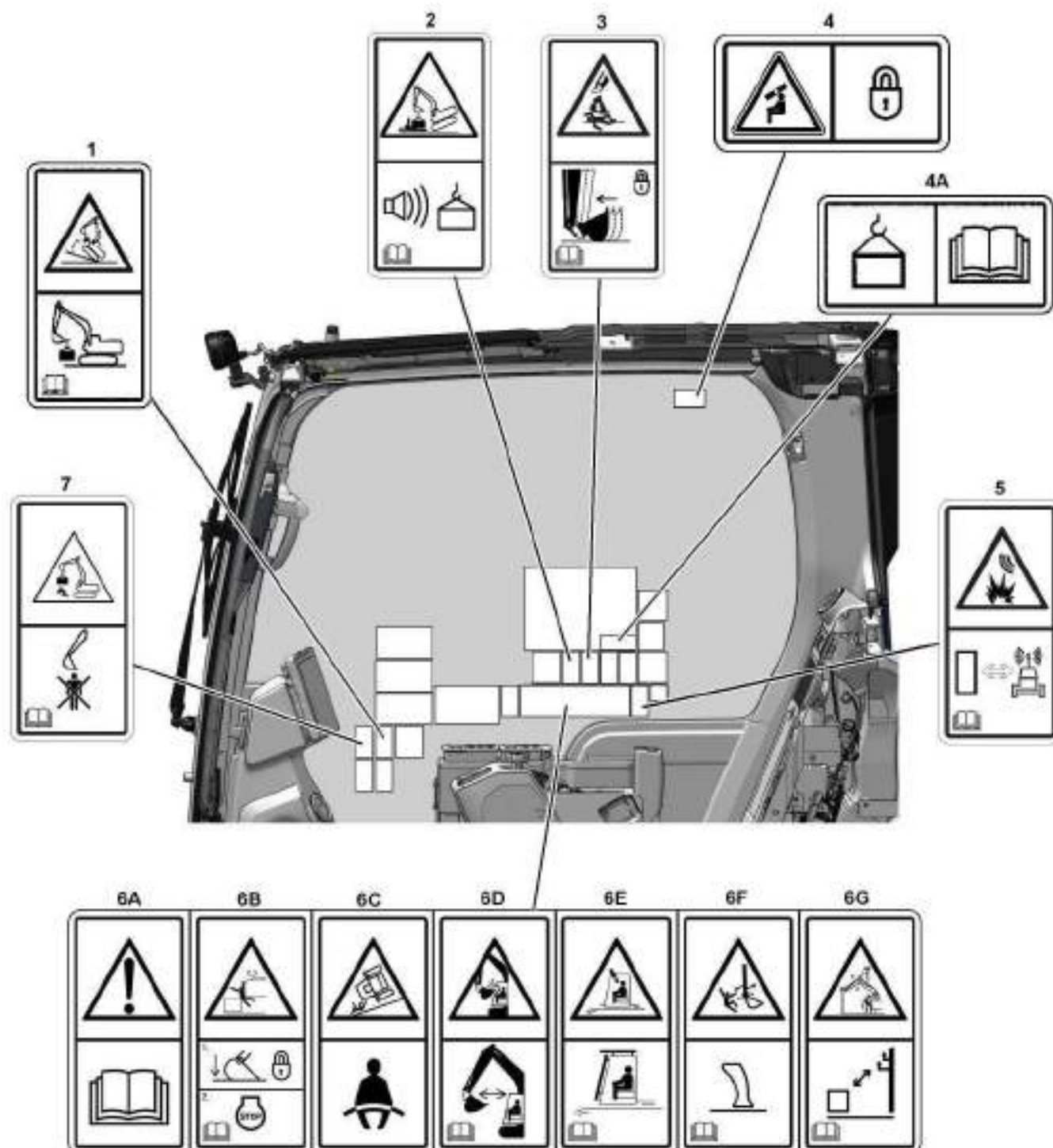


Illustration 2

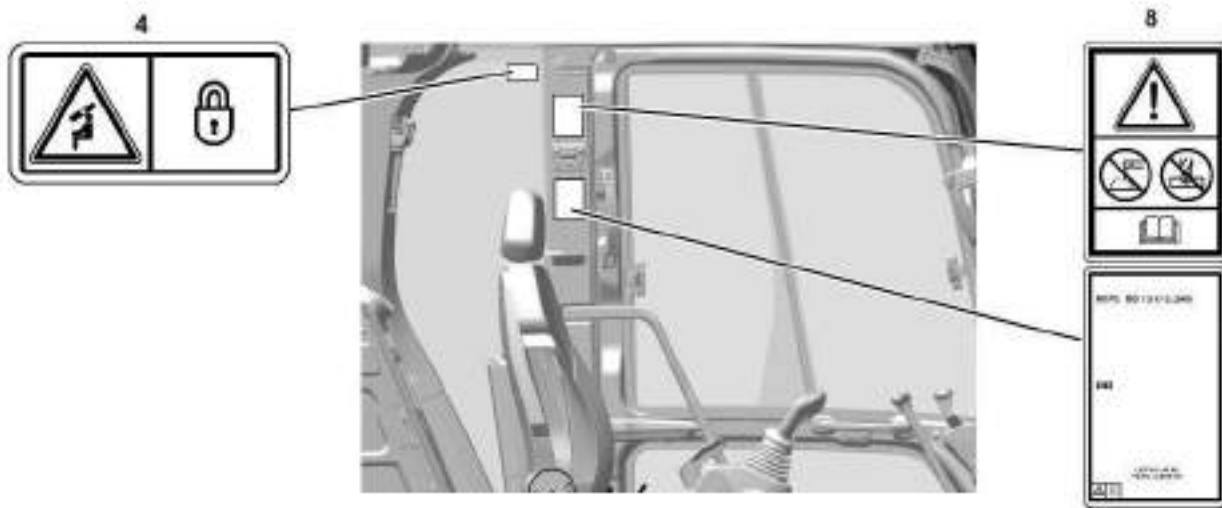


Illustration 3

g06188729

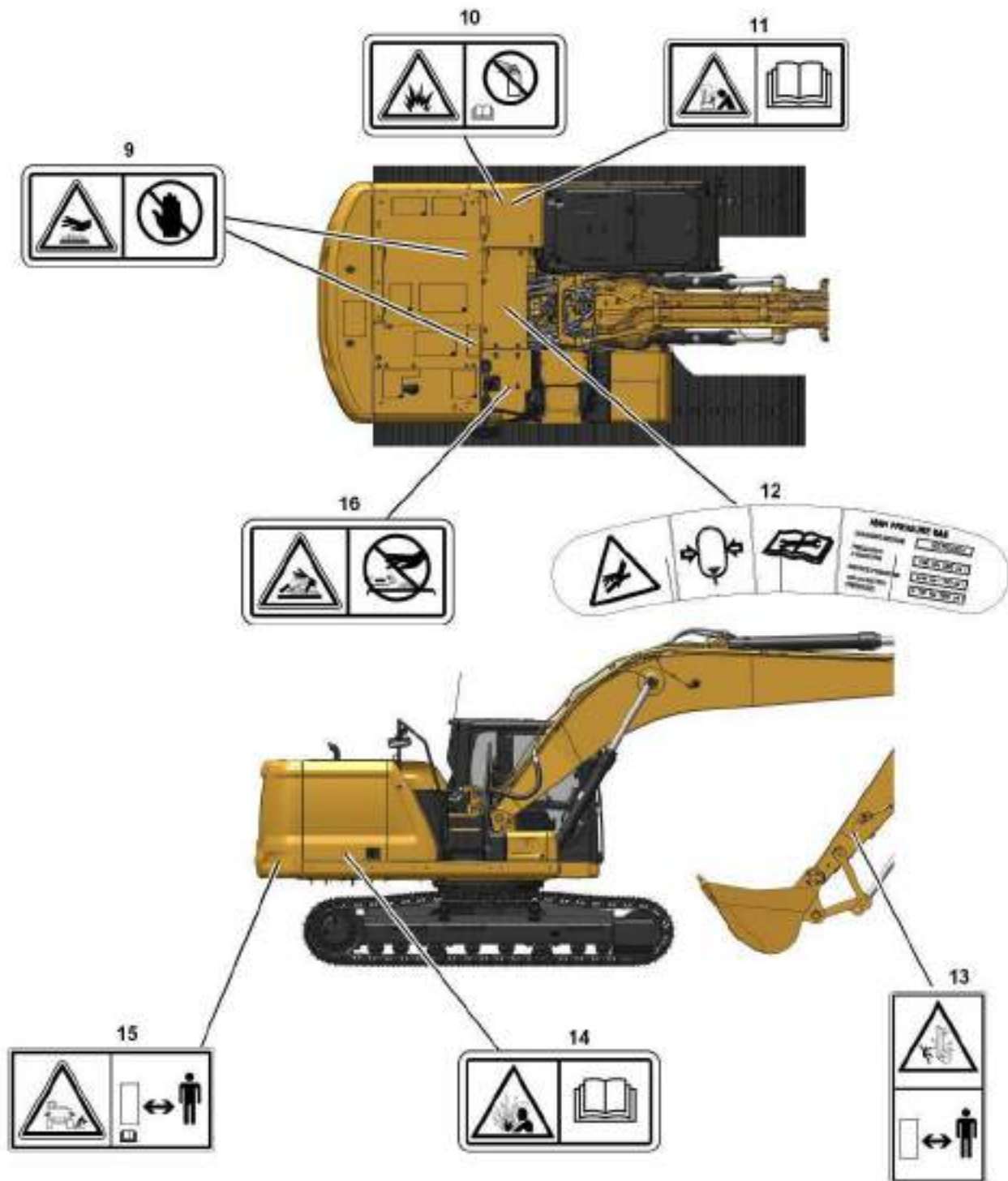


Illustration 4

g06180922



Illustration 5

g06184325

Lifting Level Warning (1)



Illustration 6

g06188532

Overload Warning Device (2)

If equipped, this safety message is located in the cab on the right side window.



Illustration 7

g06224998

WARNING

Overloading the machine could impact the machine's stability which could result in a tipover hazard. A tipover hazard could result in serious injury or death. Always activate the overload warning device before you handle or lift objects.

Refer to Operation and Maintenance Manual, "Operator Controls" for further information.

Crushing Injury (3)

If equipped, this safety message is located in the cab on the right side window.



Illustration 8

g06188540

WARNING

Crush injury. Could cause serious injury or death. Always confirm that the quick coupler is engaged onto the pins. Read the Operator's Manual.

Refer to Operation and Maintenance Manual, "Quick Coupler Operation" for further information.

Crushing Hazard (4)

These safety messages are on the left and right side cab windows.



Illustration 9

g02061339

WARNING

Personal injury can result if the window is not latched in the overhead position; ensure the auto lock is engaged.

Refer to Operation and Maintenance Manual, "Window (Front)" for further information.

Lifting Notice (4A)

If equipped, this safety message is located in the cab on the right side window.

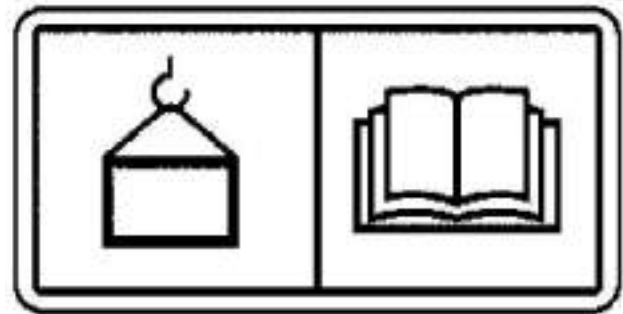


Illustration 10

g06289554

WARNING

When operating the machine's lift tool, read Operation and Maintenance Manual first for lifting capabilities.

Product Link (5)

If equipped, this safety message is located in the cab on the right side window.



Illustration 11

g06188657

WARNING

This machine is equipped with a Caterpillar Product Link communication device. When electric detonators are used, this communication device should be deactivated within 12 m (40 ft) of a blast site for satellite-based systems and within 3 m (10 ft) of a blast site for cellular based systems, or within the distance mandated under applicable legal requirements. Failure to do so could cause interference with blasting operations and result in serious injury or death.

In cases where the type of Product Link module cannot be identified, Caterpillar recommends that the device be disabled no less than 12 m (40 ft) from the blast perimeter.

Refer to Operation and Maintenance Manual, "Product Link" for further information.

Do Not Operate (6A)

This safety message is located in the cab on the right side window.



Illustration 12

g06188661

WARNING

Do not operate or work on this machine unless you have read and understand the instructions and warnings in the Operation and Maintenance Manual. Failure to follow the instructions or heed the warnings could result in injury or death. Contact any Caterpillar dealer for replacement manuals. Proper care is your responsibility.

Crushing Hazard (6B)

If equipped, this safety message is located in the cab on the right side window.



Illustration 13

g02282255

WARNING

Crush Hazard! A machine may move unexpectedly and without warning resulting in personal injury or death.

Before leaving the machine lower the work tool to the ground, lock operator controls, shut off the engine and remove the key.

Seat Belt (6C)

This safety message is located in the cab on the right side window.



Illustration 14

g06188642

WARNING

A seat belt should be worn at all times during machine operation to prevent serious injury or death in the event of an accident or machine overturn. Failure to wear a seat belt during machine operation may result in serious injury or death.

Crushing Hazard (6D)

This safety message is located in the cab on the right side window.



Illustration 15

g06188644

WARNING

Crushing Hazard! Certain machine front linkage combinations (boom, stick, quick coupler, work tool) may require keeping the work tool away from the cab during operation. Personal injury or death may result if the work tool contacts the cab during operation.

Crushing Hazard (6E)

This safety message is located in the cab on the right side window.



Illustration 16

g06188652

WARNING

The impact from objects that strike the front of the cab or the top of the cab could result in a crushing hazard with the potential for personal injury or death.

The front guard and the top guard should be installed on the cab for applications where the hazard of falling objects exist. Read the Operation and Maintenance Manual.

Refer to Operation and Maintenance Manual, "Guards" for further information.

Joystick Controls Alternate Patterns (6F)

If equipped, this safety message is located in the cab on the right side window.



Illustration 17

g06188665

WARNING

Crush Hazard. Improper joystick setting could cause possible unexpected movement of the boom, stick, or worktool which could result in serious injury or death. Confirm that the joystick settings are properly configured before you operate the machine. Read the Operation and Maintenance Manual.

Refer to Operation and Maintenance Manual, "Joystick Controls Alternate Patterns" for further information.

Electrical Power Lines (6G)

This safety message is located in the cab.



Illustration 18

g06188667

DANGER

Electrocution Hazard! Keep the machine and attachments a safe distance from electrical power. Stay clear 3 m (10 ft) plus twice the line insulator length. Read and understand the instructions and warnings in the Operation and Maintenance Manual. Failure to follow the instructions and warnings will cause serious injury or death

Refer to Operation and Maintenance Manual, "Specifications" for further information.

Do Not Lift Over Personnel (7)

This safety message is located in the cab on the right side window.



Illustration 19

g06188697

Do not lift

Do Not Weld or Drill on ROPS (8)



Illustration 20

g06207749

If equipped, this safety message is on the left side pillar in the cab.

WARNING

Structural damage, an overturn, modification, alteration, or improper repair, can impair this structure's protective capability thereby voiding this certification. Do not weld on or drill holes in the structure. Consult a Caterpillar dealer to determine this structure's limitations without voiding its certification.

This machine has been certified to the standards that are listed on the certification plate. The maximum mass of the machine, which includes the operator and the attachments without a payload, should not exceed the mass on the certification plate.

Refer to Operation and Maintenance Manual, "Guards (Operator Protection)" for more information.

Hot Surface (9)

This message is on the outside of the engine hood and on the inside of the engine hood.



Illustration 21

g01372256

WARNING

Engine hood and engine hood parts can be hot while engine is running or immediately after engine shutdown. Hot parts or hot components can cause burns or personal injury. Do not allow these parts to contact your skin, when engine is running or immediately after engine shutdown. Use protective clothing or protective equipment to protect your skin.

Aerosol Starting Aid (10)

This safety message is located near the precleaner. The following information is not applicable to machines that are equipped with an ether starting aid.



Illustration 22

g01372254

WARNING

Explosion hazard! Do not use ether! This machine is equipped with an air inlet heater. Using ether can create explosions or fires that can cause personal injury or death. Read and follow the starting procedure in the Operation and Maintenance Manual.

Refer to Operation and Maintenance Manual, "Engine Starting" for the proper starting procedure.

Jump-Start Cables (11)

This safety message is positioned on the circuit breaker panel.

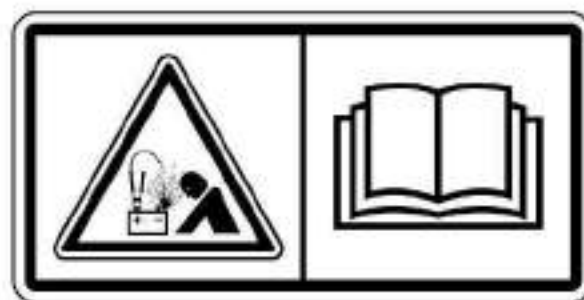


Illustration 23

g01370909

WARNING

Explosion Hazard! Improper jumper cable connections can cause an explosion resulting in serious injury or death. Batteries may be located in separate compartments. Refer to the Operation and Maintenance Manual for the correct jump starting procedure.

Refer to Operation and Maintenance Manual, "Engine Starting with Jump-Start Cables" for further information.

High-Pressure Gas (12)

This safety message is positioned on the accumulator.



Illustration 24

g06188756

WARNING

Pressurized System!

Hydraulic accumulators contain gas and oil under high pressure. **DO NOT** disconnect lines or disassemble any component of a pressurized accumulator. All gas pre-charge must be removed from the accumulator as instructed by the service manual before servicing or disposing of the accumulator or any accumulator component.

Failure to follow the instructions and warnings could result in personal injury or death.

Only use dry nitrogen gas to recharge accumulators. See your Cat dealer for special equipment and detailed information for accumulator service and charging.

Refer to Operation and Maintenance Manual, "Equipment Lowering with Engine Stopped" for further information.

Crushing Hazard (13)

This safety message is on both sides of the stick.

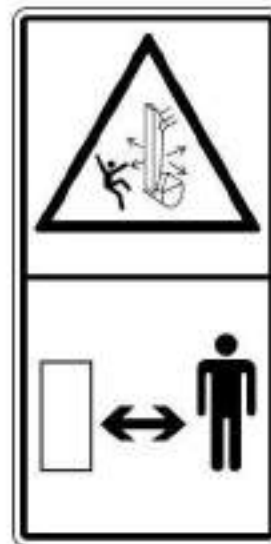


Illustration 25

g01385579

WARNING

A crushing hazard exists when the stick and boom are in motion and when the machine is being used in object handling applications. Failure to stay clear of the stick and boom when the machine is in operation can result in personal injury or death. Stay clear of the stick and boom when the machine is in operation.

Vapor Explosion (14)

If equipped, this safety message is on the storage compartment for the fuel transfer pump.



Illustration 26

g01407639

WARNING

Explosion hazard! Fuel vapors can accumulate in the refueling pump compartment and can be ignited by improper operation of the refueling pump. Failure to follow the operating instructions for the refueling pump could result in personal injury or death. Read and follow the operating instructions for the refueling pump in the Operation and Maintenance Manual.

Refer to Operation and Maintenance Manual, “Fuel Transfer Pump (Refueling)” for further information.

Crushing Hazard (15)

This safety message is on the rear of each side of the machine.

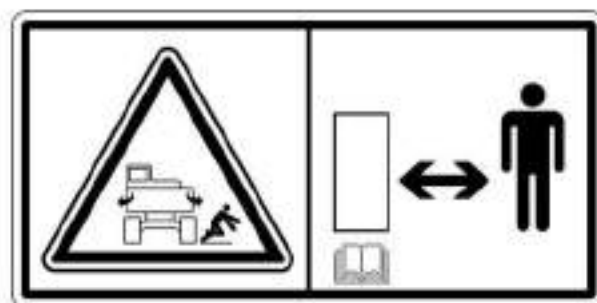


Illustration 27

g06219420

WARNING

Machine swings. Stay back. Crushing hazard could cause serious injury or death.

Relieve Hydraulic Tank Pressure (16)

This safety message is on top of the hydraulic tank.



Illustration 28

g01371640

WARNING

Hot Fluid Under Pressure!

Do NOT remove pressure cap when hot. Hot oil could cause serious injury or death.

Falling Object Guard Structure (17)

If equipped, this safety message is on top of the front falling object guard structure. This safety message is also on the left side of the falling object guard structure on the top of the cab.

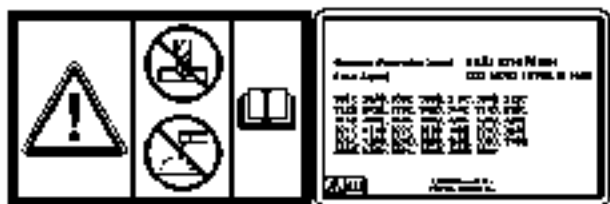


Illustration 29

g02428757



Structural damage, an overturn, modification, alteration, or improper repair can impair this structure's protection capability thereby voiding this certification. Do not weld on or drill holes in the structure. Consult a Caterpillar dealer to determine this structure's limitations without voiding its certification.

i07945148

Additional Messages

SMCS Code: 7000; 7405

There are several specific messages on this machine. The exact location of the messages and the description of the information are reviewed in this section. Become familiar with all messages.

Make sure that all the messages are legible. Clean the messages or replace the messages if you cannot read the words. Replace the illustrations if the illustrations are not legible. When you clean the messages, use a cloth, water, and soap. Do not use solvent, gasoline, or other harsh chemicals to clean the messages. Solvents, gasoline, or harsh chemicals could loosen the adhesive that secures the messages. Loose adhesive will allow the messages to fall.

Replace any message that is damaged, or missing. If a message is attached to a part that is replaced, install a message on the replacement part. Any Cat dealer can provide new messages.

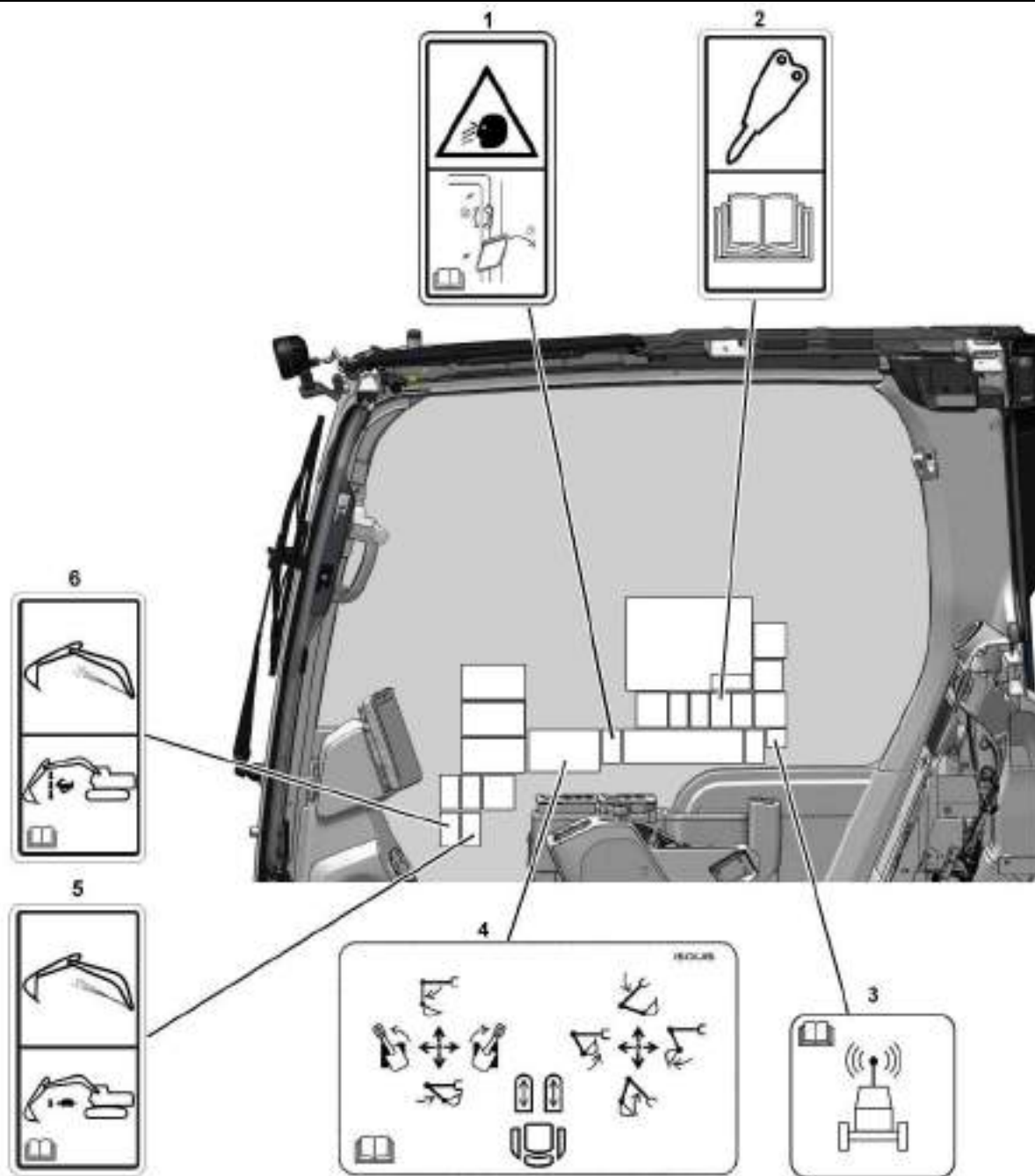
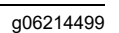


Illustration 30

g06187617

g06189121



For machines equipped with the Cat Grade Control monitor, the monitor must be moved downward before lifting or lowering the front window. The monitor is in the path of the window track in its normal position.



Illustration 33

g06214810

Hammer Operation(2)

This message is on the window on the right side of the cab.

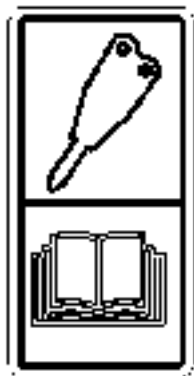


Illustration 34

g06189240

See Operation and Maintenance Manual, "Hammer Operation" within Operation and Maintenance Manual, "Work Tool Control (One-Way Flow)" for instructions on hammer operation.

Also, see Operation and Maintenance Manual, "Hydraulic Hammer Control (Foot Switch)" within Operation and Maintenance Manual, "Work Tool Control (One-Way Flow)" for instructions on hammer operation.

Data Privacy (3)

Illustration 35

g01418953

The Product Link System is a satellite communication device that transmits information regarding the machine back to Caterpillar and Cat dealers and customers. All logged events and diagnostic codes that are available to the Cat Electronic Technician (ET) on the Cat data link can be sent to the satellite. Information can also be sent to the Product Link System. The information is used to improve Cat products and Cat services.

Refer to Operation and Maintenance Manual, "Product Link" for more information.

Joystick Controls Alternate Patterns (4)

If equipped, this message is on the right side window of the cab.

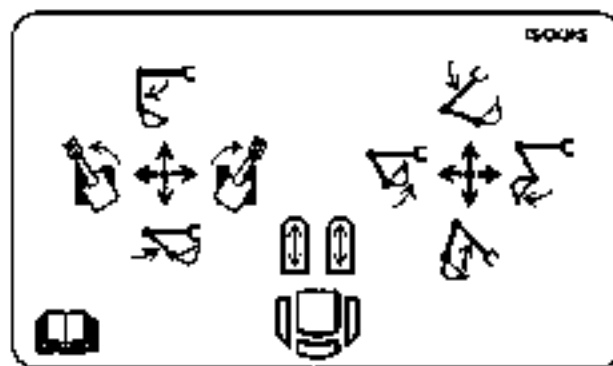


Illustration 36

g06214805

Refer to Operation and Maintenance Manual, "Joystick Controls Alternate Patterns" for further information.

Hose Burst (5)



Illustration 37

g06189238

Hose Burst (6)



Illustration 38

g06189239

Alternate Exit (7)

If equipped, this message is on the rear window of the cab in the lower left-hand corner.



Illustration 39

g06189112

Pull the ring to pull out the seal. Push the window out of the cab and exit through the opening.

Refer to Operation and Maintenance Manual, "Alternate Exit" for further information.

No Step (8)

This message is on various places on the upper structure and covers. It is also on the engine valve cover.

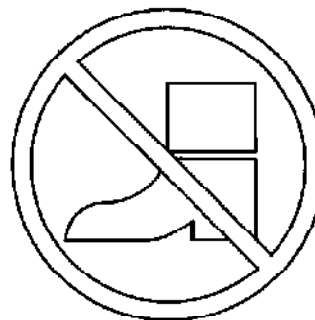


Illustration 40

g00911158

Do not step in this area.

DEF Purge Indicator Lamp (9)

This message is located behind the cab near the disconnect switch and pertains to the Diesel Exhaust Fluid (DEF) system.

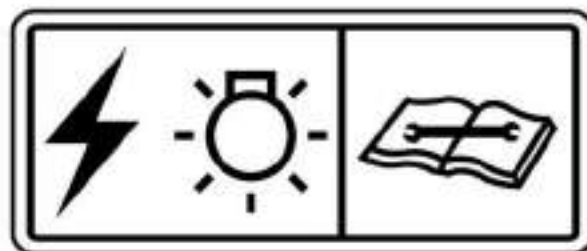


Illustration 41

g03796564

For machines equipped with a circuit that stays activated for DEF purge with the disconnect switch off.

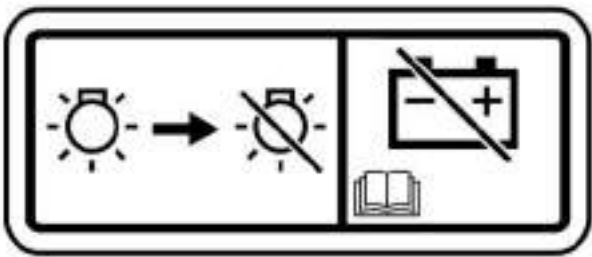


Illustration 42 g03408962

For machines not equipped with a circuit that stays activated for DEF purge with the disconnect switch off.

NOTICE

After the engine is shutdown apply the battery disconnect switch. Applying the battery disconnect too soon will prevent the DEF system from being purged and could cause DEF to freeze in the lines.

Air Conditioner (10)

These messages are positioned on the left door behind the cab.

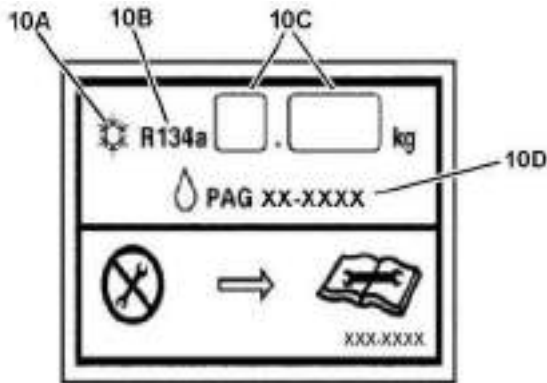


Illustration 43 g06214936

- (10A) Air conditioning symbol
- (10B) R134a (Refrigerant type common name)
- (10C) Refrigerant quantity
- (10D) PAG (polyalkylene glycol) lubricating oil part number

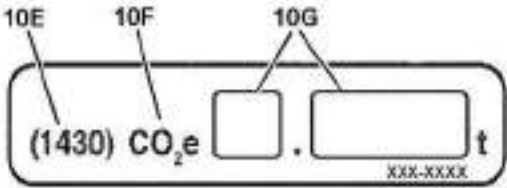


Illustration 44 g06214938

If equipped, this plate provides the below additional European Union required greenhouse gas information.

- (10E) (1430) - This is the Global Warming Potential of R134a
- (10F) CO₂ equivalent
- (10G) The system contains 1.430 metric tonne of CO₂ equivalent



Illustration 45 g06214940

(10H) If equipped, this film provides the required language translations of the text "Contains fluorinated greenhouse gases" for the European Union greenhouse gas regulation.

These messages for the air conditioner system have the appropriate information for the following services: the air conditioner lubricant, the refrigerant charge, and the refrigerant capacity.

Engine Oil Requirements (11)

This message is on top of the engine.



Illustration 46 g06208149

Refer to Operation and Maintenance Manual, “Lubricant Viscosities”.

Diesel Fuel Requirements (12)

This message is located by the fuel tank.

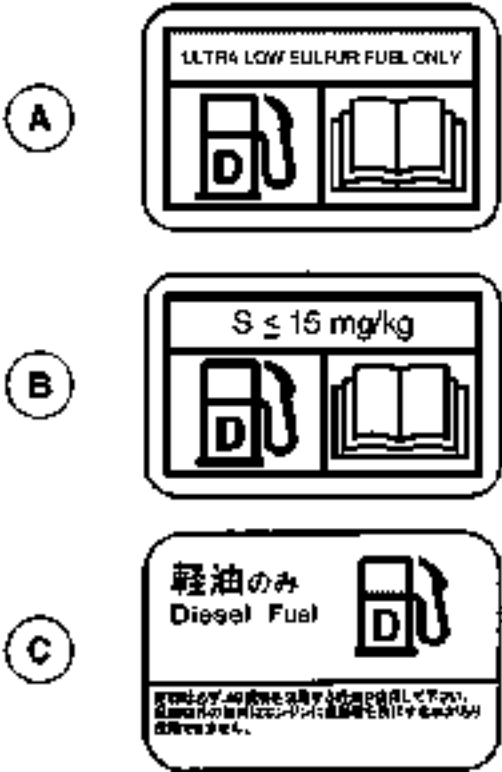


Illustration 47 g03218956

(A) North America film
(B) Europe, Africa, Middle East film
(C) Japan film

Hydraulic Oil Level Check (13)

This message is located in the right access compartment next to the sight gauge for the hydraulic oil .

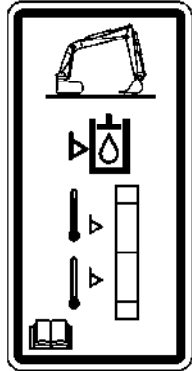


Illustration 48 g01069075

Check hydraulic oil level daily. See Operation and Maintenance Manual, “Hydraulic System Oil Level - Check” for details.

i07746355

General Hazard Information

SMCS Code: 7000

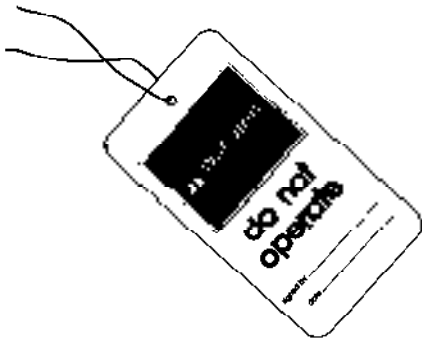


Illustration 49 g00104545

Typical example

Attach a “Do Not Operate” warning tag or a similar warning tag to the start switch or to the controls. Attach the warning tag before you service the equipment or before you repair the equipment. Warning tag SEHS7332 is available from your Cat dealer.

⚠ WARNING

Operating the machine while distracted can result in the loss of machine control. Use extreme caution when using any device while operating the machine. Operating the machine while distracted can result in personal injury or death.

Know the width of your equipment to maintain proper clearance when you operate the equipment near fences or near boundary obstacles.

Be aware of high-voltage power lines and power cables that are buried. If the machine comes in contact with these hazards, serious injury or death may occur from electrocution.

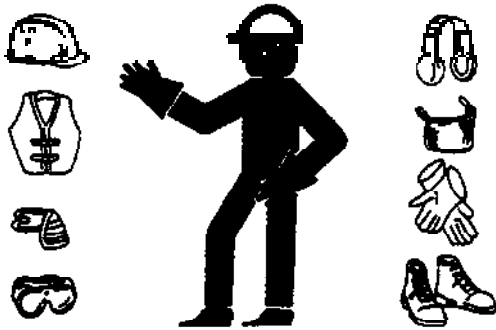


Illustration 50

g00702020

Wear a hard hat, protective glasses, and other protective equipment, as required.

Do not wear loose clothing or jewelry that can snag on controls or on other parts of the equipment.

Make sure that all protective guards and all covers are secured in place on the equipment.

Keep the equipment free from foreign material. Remove debris, oil, tools, and other items from the deck, from walkways, and from steps.

Secure all loose items such as lunch boxes, tools, and other items that are not a part of the equipment.

Know the appropriate work site hand signals and the personnel that are authorized to give the hand signals. Accept hand signals from one person only.

Do not smoke when you service an air conditioner. Also, do not smoke if refrigerant gas may be present. Inhaling the fumes that are released from a flame that contacts air conditioner refrigerant can cause bodily harm or death. Inhaling gas from air conditioner refrigerant through a lighted cigarette can cause bodily harm or death.

Never put maintenance fluids into glass containers. Drain all liquids into a suitable container.

Obey all local regulations for the disposal of liquids.

Use all cleaning solutions with care. Report all necessary repairs.

Do not allow unauthorized personnel on the equipment.

Unless you are instructed otherwise, perform maintenance with the equipment in the servicing position. Refer to Operation and Maintenance Manual for the procedure for placing the equipment in the servicing position.

When you perform maintenance above ground level, use appropriate devices such as ladders or man lift machines. If equipped, use the machine anchorage points and use approved fall arrest harnesses and lanyards.

Pressurized Air and Water

Pressurized air and/or water can cause debris and/or hot water to be blown out. The debris and/or hot water could result in personal injury.

When pressurized air and/or pressurized water is used for cleaning, wear protective clothing, protective shoes, and eye protection. Eye protection includes goggles or a protective face shield.

The maximum air pressure for cleaning purposes must be reduced to 205 kPa (30 psi) when the nozzle is deadheaded and the nozzle is used with an effective chip deflector and personal protective equipment. The maximum water pressure for cleaning purposes must be below 275 kPa (40 psi).

Avoid direct spraying of water on electrical connectors, connections, and components. When using air for cleaning, allow the machine to cool to reduce the possibility of fine debris igniting when re-deposited on hot surfaces.

Trapped Pressure

Pressure can be trapped in a hydraulic system. Releasing trapped pressure can cause sudden machine movement or attachment movement. Use caution if you disconnect hydraulic lines or fittings. High-pressure oil that is released can cause a hose to whip. High-pressure oil that is released can cause oil to spray. Fluid penetration can cause serious injury and possible death.

Fluid Penetration

Pressure can be trapped in the hydraulic circuit long after the machine has been stopped. The pressure can cause hydraulic fluid or items such as pipe plugs to escape rapidly if the pressure is not relieved correctly.

Do not remove any hydraulic components or parts until pressure has been relieved or personal injury may occur. Do not disassemble any hydraulic components or parts until pressure has been relieved or personal injury may occur. Refer to the Service Manual for any procedures that are required to relieve the hydraulic pressure.

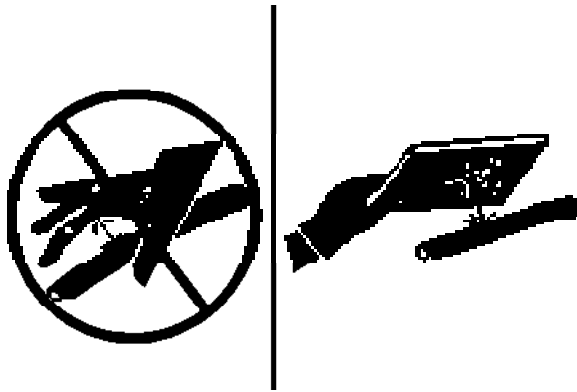


Illustration 51

g00687600

Always use a board or cardboard when you check for a leak. Leaking fluid that is under pressure can penetrate body tissue. Fluid penetration can cause serious injury and possible death. A pin hole leak can cause severe injury. If fluid is injected into your skin, you must get treatment immediately. Seek treatment from a doctor that is familiar with this type of injury.

Containing Fluid Spillage

Care must be taken in order to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the equipment. Prepare to collect the fluid with suitable containers before opening any compartment or disassembling any component that contains fluids.

Refer to Special Publication, NENG2500, "Cat dealer Service Tool Catalog" for the following items:

- Tools that are suitable for collecting fluids and equipment that is suitable for collecting fluids
- Tools that are suitable for containing fluids and equipment that is suitable for containing fluids

Obey all local regulations for the disposal of liquids.

Inhalation



Illustration 52

g02159053

Exhaust

Use caution. Exhaust fumes can be hazardous to your health. If you operate the machine in an enclosed area, adequate ventilation is necessary.

Asbestos Information

Cat equipment and replacement parts that are shipped from Caterpillar are asbestos free. Caterpillar recommends the use of only genuine Cat replacement parts. Use the following guidelines when you handle any replacement parts that contain asbestos or when you handle asbestos debris.

Use caution. Avoid inhaling dust that might be generated when you handle components that contain asbestos fibers. Inhaling this dust can be hazardous to your health. The components that may contain asbestos fibers are brake pads, brake bands, lining material, clutch plates, and some gaskets. The asbestos that is used in these components is bound in a resin or sealed in some way. Normal handling is not hazardous unless airborne dust that contains asbestos is generated.

If dust that may contain asbestos is present, there are several guidelines that should be followed:

- Never use compressed air for cleaning.
- Avoid brushing materials that contain asbestos.
- Avoid grinding materials that contain asbestos.
- Use a wet method in order to clean up asbestos materials.
- A vacuum cleaner that is equipped with a high efficiency particulate air filter (HEPA) can also be used.

- Use exhaust ventilation on permanent machining jobs.
- Wear an approved respirator if there is no other way to control the dust.
- Comply with applicable rules and regulations for the work place. In the United States, use Occupational Safety and Health Administration (OSHA) requirements. These OSHA requirements can be found in “29 CFR 1910.1001”. In Japan, use the requirements found in the “Ordinance on Prevention of Health Impairment due to Asbestos” in addition to the requirements of the Industrial Safety and Health Act.
- Obey environmental regulations for the disposal of asbestos.
- Stay away from areas that might have asbestos particles in the air.

Dispose of Waste Properly

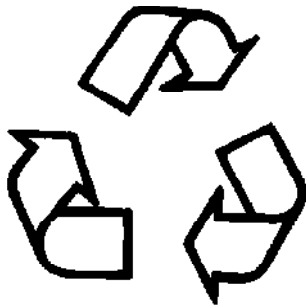


Illustration 53

g00706404

Improperly disposing of waste can threaten the environment. Potentially harmful fluids should be disposed of according to local regulations.

Always use leakproof containers when you drain fluids. Do not pour waste onto the ground, down a drain, or into any source of water.

i01359664

Crushing Prevention and Cutting Prevention

SMCS Code: 7000

Support the equipment properly before you perform any work or maintenance beneath that equipment. Do not depend on the hydraulic cylinders to hold up the equipment. Equipment can fall if a control is moved, or if a hydraulic line breaks.

Do not work beneath the cab of the machine unless the cab is properly supported.

Unless you are instructed otherwise, never attempt adjustments while the machine is moving or while the engine is running.

Never jump across the starter solenoid terminals in order to start the engine. Unexpected machine movement could result.

Whenever there are equipment control linkages the clearance in the linkage area will change with the movement of the equipment or the machine. Stay clear of areas that may have a sudden change in clearance with machine movement or equipment movement.

Stay clear of all rotating and moving parts.

If it is necessary to remove guards in order to perform maintenance, always install the guards after the maintenance is performed.

Keep objects away from moving fan blades. The fan blade will throw objects or cut objects.

Do not use a kinked wire cable or a frayed wire cable. Wear gloves when you handle wire cable.

When you strike a retainer pin with force, the retainer pin can fly out. The loose retainer pin can injure personnel. Make sure that the area is clear of people when you strike a retainer pin. To avoid injury to your eyes, wear protective glasses when you strike a retainer pin.

Chips or other debris can fly off an object when you strike the object. Make sure that no one can be injured by flying debris before striking any object.

i07746334

Burn Prevention

SMCS Code: 7000

Do not touch any part of an operating engine. Allow the engine to cool before any maintenance is performed on the engine. Relieve all pressure in the air system, in the oil system, in the lubrication system, in the fuel system, or in the cooling system before any lines, fittings, or related items are disconnected.

Coolant

When the engine is at operating temperature, the engine coolant is hot. The coolant is also under pressure. The radiator and all lines to the heaters or to the engine contain hot coolant.

Any contact with hot coolant or with steam can cause severe burns. Allow cooling system components to cool before the cooling system is drained.

Check the coolant level only after the engine has been stopped.

Ensure that the filler cap is cool before removing the filler cap. The filler cap must be cool enough to touch with a bare hand. Remove the filler cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Alkali can cause personal injury. Do not allow alkali to contact the skin, the eyes, or the mouth.

Oils

Hot oil and hot components can cause personal injury. Do not allow hot oil to contact the skin. Also, do not allow hot components to contact the skin.

Remove the hydraulic tank filler cap only after the engine has been stopped. The filler cap must be cool enough to touch with a bare hand. Follow the standard procedure in this manual to remove the hydraulic tank filler cap.

Batteries

The liquid in a battery is an electrolyte. Electrolyte is an acid that can cause personal injury. Do not allow electrolyte to contact the skin or the eyes.

Do not smoke while checking the battery electrolyte levels. Batteries give off flammable fumes which can explode.

Always wear protective glasses when you work with batteries. Wash hands after touching batteries. The use of gloves is recommended.

i06179517

Fire Prevention and Explosion Prevention

SMCS Code: 7000



Illustration 54

g00704000

Regeneration

The exhaust gas temperatures during regeneration will be elevated. Follow proper fire prevention instructions and use the disable regeneration function (if equipped) when appropriate.

General

All fuels, most lubricants, and some coolant mixtures are flammable.

To minimize the risk of fire or explosion, Caterpillar recommends the following actions.

Always perform a Walk-Around Inspection, which may help you identify a fire hazard. Do not operate a machine when a fire hazard exists. Contact your Cat dealer for service.

Understand the use of the primary exit and alternative exit on the machine. Refer to Operation and Maintenance Manual, "Alternative Exit".

Do not operate a machine with a fluid leak. Repair leaks and clean up fluids before resuming machine operation. Fluids that are leaking or spilled onto hot surfaces or onto electrical components can cause a fire. A fire may cause personal injury or death.

Remove flammable material such as leaves, twigs, papers, trash, and so on. These items may accumulate in the engine compartment or around other hot areas and hot parts on the machine.

Keep the access doors to major machine compartments closed and access doors in working condition in order to permit the use of fire suppression equipment, in case a fire should occur.

Clean all accumulations of flammable materials such as fuel, oil, and debris from the machine.

Do not operate the machine near any flame.

Keep shields in place. Exhaust shields (if equipped) protect hot exhaust components from oil spray or fuel spray in a break in a line, in a hose, or in a seal. Exhaust shields must be installed correctly.

Do not weld or flame cut on tanks or lines that contain flammable fluids or flammable material. Empty and purge the lines and tanks. Then clean the lines and tanks with a nonflammable solvent prior to welding or flame cutting. Ensure that the components are properly grounded in order to avoid unwanted arcs.

Dust that is generated from repairing nonmetallic hoods or fenders may be flammable and/or explosive. Repair such components in a ventilated area away from open flames or sparks. Use suitable Personal Protection Equipment (PPE).

Inspect all lines and hoses for wear or deterioration. Replace damaged lines and hoses. The lines and the hoses should have adequate support and secure clamps. Tighten all connections to the recommended torque. Damage to the protective cover or insulation may provide fuel for fires.

Store fuels and lubricants in properly marked containers away from unauthorized personnel. Store oily rags and flammable materials in protective containers. Do not smoke in areas that are used for storing flammable materials.



Illustration 55

g03839130

Use caution when you are fueling a machine. Do not smoke while you are fueling a machine. Do not fuel a machine near open flames or sparks. Do not use cell phones or other electronic devices while you are refueling. Always stop the engine before fueling. Fill the fuel tank outdoors. Properly clean areas of spillage.

Avoid static electricity risk when fueling. Ultra low sulfur diesel (ULSD) poses a greater static ignition hazard than earlier diesel formulations with a higher sulfur content. Avoid death or serious injury from fire or explosion. Consult with your fuel or fuel system supplier to ensure that the delivery system is in compliance with fueling standards for proper grounding and bonding practices.

Never store flammable fluids in the operator compartment of the machine.

Battery and Battery Cables



Illustration 56

g03839133

Caterpillar recommends the following in order to minimize the risk of fire or an explosion related to the battery.

Do not operate a machine if battery cables or related parts show signs of wear or damage. Contact your Cat dealer for service.

Follow safe procedures for engine starting with jump-start cables. Improper jumper cable connections can cause an explosion that may result in injury. Refer to Operation and Maintenance Manual, "Engine Starting with Jump Start Cables" for specific instructions.

Do not charge a frozen battery. This may cause an explosion.

Gases from a battery can explode. Keep any open flames or sparks away from the top of a battery. Do not smoke in battery charging areas. Do not use cell phones or other electronic devices in battery charging areas.

Never check the battery charge by placing a metal object across the terminal posts. Use a voltmeter in order to check the battery charge.

Daily inspect battery cables that are in areas that are visible. Inspect cables, clips, straps, and other restraints for damage. Replace any damaged parts. Check for signs of the following, which can occur over time due to use and environmental factors:

- Fraying

- Abrasion
- Cracking
- Discoloration
- Cuts on the insulation of the cable
- Fouling
- Corroded terminals, damaged terminals, and loose terminals

Replace damaged battery cable(s) and replace any related parts. Eliminate any fouling, which may have caused insulation failure or related component damage or wear. Ensure that all components are reinstalled correctly.

An exposed wire on the battery cable may cause a short to ground if the exposed area comes into contact with a grounded surface. A battery cable short produces heat from the battery current, which may be a fire hazard.

An exposed wire on the ground cable between the battery and the disconnect switch may cause the disconnect switch to be bypassed if the exposed area comes into contact with a grounded surface. This may result in an unsafe condition for servicing the machine. Repair components or replace components before servicing the machine.

WARNING

Fire on a machine can result in personal injury or death. Exposed battery cables that come into contact with a grounded connection can result in fires. Replace cables and related parts that show signs of wear or damage. Contact your Cat dealer.

Wiring

Check electrical wires daily. If any of the following conditions exist, replace parts before you operate the machine.

- Fraying
- Signs of abrasion or wear
- Cracking
- Discoloration
- Cuts on insulation
- Other damage

Make sure that all clamps, guards, clips, and straps are reinstalled correctly. This will help to prevent vibration, rubbing against other parts, and excessive heat during machine operation.

Attaching electrical wiring to hoses and tubes that contain flammable fluids or combustible fluids should be avoided.

Consult your Cat dealer for repair or for replacement parts.

Keep wiring and electrical connections free of debris.

Lines, Tubes, and Hoses

Do not bend high-pressure lines. Do not strike high-pressure lines. Do not install any lines that are bent or damaged. Use the appropriate backup wrenches in order to tighten all connections to the recommended torque.

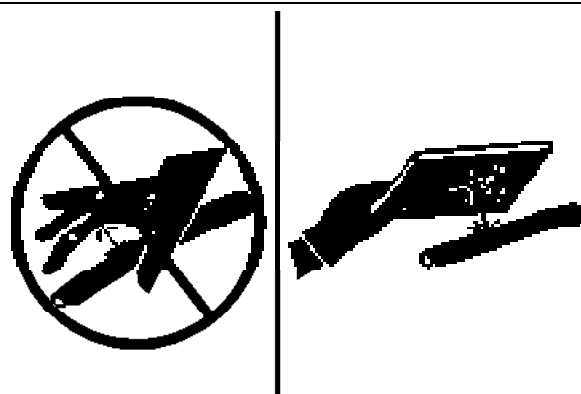


Illustration 57

g00687600

Check lines, tubes, and hoses carefully. Wear Personal Protection Equipment (PPE) in order to check for leaks. Always use a board or cardboard when you check for a leak. Leaking fluid that is under pressure can penetrate body tissue. Fluid penetration can cause serious injury and possible death. A pin hole leak can cause severe injury. If fluid is injected into your skin, you must get treatment immediately. Seek treatment from a doctor that is familiar with this type of injury.

Replace the affected parts if any of the following conditions are present:

- End fittings are damaged or leaking.
- Outer coverings are chafed or cut.
- Wires are exposed.
- Outer coverings are swelling or ballooning.
- Flexible parts of the hoses are kinked.
- Outer covers have exposed embedded armoring.
- End fittings are displaced.

Make sure that all clamps, guards, and heat shields are installed correctly. During machine operation, this will help to prevent vibration, rubbing against other parts, excessive heat, and failure of lines, tubes, and hoses.

Do not operate a machine when a fire hazard exists. Repair any lines that are corroded, loose, or damaged. Leaks may provide fuel for fires. Consult your Cat dealer for repair or for replacement parts. Use genuine Cat parts or the equivalent, for capabilities of both the pressure limit and temperature limit.

Ether

Ether (if equipped) is commonly used in cold-weather applications. Ether is flammable and poisonous.

Only use approved Ether canisters for the Ether dispensing system fitted to your machine, do not spray Ether manually into an engine, follow the correct cold engine starting procedures. Refer to the section in the Operation and Maintenance Manual with the label "Engine Starting".

WARNING

Manually spraying Ether into an engine with a Diesel Particulate Filter (DPF) may result in the accumulation of Ether in the DPF and an explosion. This in conjunction with other factors may result in an injury or death.

Use ether in ventilated areas. Do not smoke while you are replacing an ether cylinder.

Do not store ether cylinders in living areas or in the operator compartment of a machine. Do not store ether cylinders in direct sunlight or in temperatures above 49° C (120.2° F). Keep ether cylinders away from open flames or sparks.

Dispose of used ether cylinders properly. Do not puncture an ether cylinder. Keep ether cylinders away from unauthorized personnel.

Fire Extinguisher

As an additional safety measure, keep a fire extinguisher on the machine.

Be familiar with the operation of the fire extinguisher. Inspect the fire extinguisher and service the fire extinguisher regularly. Follow the recommendations on the instruction plate.

Consider installation of an aftermarket Fire Suppression System, if the application and working conditions warrant the installation.

i07041871

Fire Safety

SMCS Code: 7000

Note: Locate secondary exits and how to use the secondary exits before you operate the machine.

Note: Locate fire extinguishers and how to use a fire extinguisher before you operate the machine.

If you find that you are involved in a machine fire, your safety and that of others on site are the top priority. The following actions should only be performed if the actions do not present a danger or risk to you and any nearby people. Assess the risk of personal injury and move away to a safe distance as soon as you feel unsafe.

Move the machine away from nearby combustible material such as fuel/oil stations, structures, trash, mulch, and timber.

Lower any implements and turn off the engine as soon as possible. If you leave the engine running, the engine will continue to feed a fire. The fire will be fed from any damaged hoses that are attached to the engine or pumps.

If possible, turn the battery disconnect switch to the OFF position. Disconnecting the battery will remove the ignition source in the event of an electrical short. Disconnecting the battery will eliminate a second ignition source if electrical wiring is damaged by the fire, resulting in a short circuit.

Notify emergency personnel of the fire and your location.

If your machine is equipped with a fire suppression system, follow the manufacturers procedure for activating the system.

Note: Fire suppression systems need to be regularly inspected by qualified personnel. You must be trained to operate the fire suppression system.

If you are unable to do anything else, shut off the machine before exiting. By shutting off the machine, fuels will not continue to be pumped into the fire.

If the fire grows out of control, be aware of the following risks:

- Tires on wheeled machines pose a risk of explosion as tires burn. Hot shrapnel and debris can be thrown great distances in an explosion.
- Tanks, accumulators, hoses, and fittings can rupture in a fire, spraying fuels and shrapnel over a large area.

- Remember that nearly all the fluids on the machine are flammable, including coolant and oils. Additionally, plastics, rubbers, fabrics, and resins in fiberglass panels are also flammable.

i06952417

Fire Extinguisher Location

SMCS Code: 7000; 7419



Illustration 58

g06188176

Installation of a fire extinguisher is recommended. Be familiar with the operation of the fire extinguisher. Inspect the fire extinguisher and service the fire extinguisher regularly. Obey the recommendations on the instruction plate.

The recommended location for mounting the fire extinguisher is on the upper handrail on the right side of the machine.

i01329108

Track Information

SMCS Code: 4170; 7000

Track adjusting systems use either grease or oil under high pressure to keep the track under tension.

Grease or oil under high pressure coming out of the relief valve can penetrate the body causing injury or death. Do not watch the relief valve to see if grease or oil is escaping. Watch the track or track adjustment cylinder to see if the track is being loosened.

The pins and bushings in a dry track pin joint can become very hot. It is possible to burn the fingers if there is more than brief contact with these components.

i02546320

High Pressure Fuel Lines

SMCS Code: 1000; 1274; 7000

WARNING

Contact with high pressure fuel may cause fluid penetration and burn hazards. High pressure fuel spray may cause a fire hazard. Failure to follow these inspection, maintenance and service instructions may cause personal injury or death.

The high pressure fuel lines are the fuel lines that are between the high pressure fuel pump and the high pressure fuel manifold and the fuel lines that are between the fuel manifold and cylinder head. These fuel lines are different from fuel lines on other fuel systems.

This is because of the following differences:

- The high pressure fuel lines are constantly charged with high pressure.
- The internal pressures of the high pressure fuel lines are higher than other types of fuel system.
- The high pressure fuel lines are formed to shape and then strengthened by a special process.

Do not step on the high pressure fuel lines. Do not deflect the high pressure fuel lines. Do not bend or strike the high pressure fuel lines. Deformation or damage of the high pressure fuel lines may cause a point of weakness and potential failure.

Do not check the high pressure fuel lines with the engine or the starting motor in operation. After the engine has stopped allow 10 minutes to pass in order to allow the pressure to be purged before any service or repair is performed on the engine fuel lines.

Do not loosen the high pressure fuel lines in order to remove air from the fuel system. This procedure is not required.

Visually inspect the high pressure fuel lines before the engine is started. This inspection should be each day.

If you inspect the engine in operation, always use the proper inspection procedure in order to avoid a fluid penetration hazard. Refer to Operation and Maintenance Manual, "General hazard Information".

- Inspect the high pressure fuel lines for damage, deformation, a nick, a cut, a crease, or a dent.
- Do not operate the engine with a fuel leak. If there is a leak do not tighten the connection in order to stop the leak. The connection must only be tightened to the recommended torque. Refer to Disassembly and Assembly for your engine.

- If the high pressure fuel lines are torqued correctly and the high pressure fuel lines are leaking the high pressure fuel lines must be replaced.
- Ensure that all clips on the high pressure fuel lines are in place. Do not operate the engine with clips that are damaged, missing or loose.
- Do not attach any other item to the high pressure fuel lines.
- Loosened high pressure fuel lines must be replaced. Also removed high pressure fuel lines must be replaced. Refer to Disassembly and Assembly for your engine.

i01122596

Electrical Storm Injury Prevention

SMCS Code: 7000

When lightning is striking in the vicinity of the machine, the operator should never attempt the following procedures:

- Mount the machine.
- Dismount the machine.

If you are in the operator's station during an electrical storm, stay in the operator's station. If you are on the ground during an electrical storm, stay away from the vicinity of the machine.

i00771840

Before Starting Engine

SMCS Code: 1000; 7000

Start the engine only from the operator compartment. Never short across the starter terminals or across the batteries. Shorting could damage the electrical system by bypassing the engine neutral start system.

Inspect the condition of the seat belt and of the mounting hardware. Replace any parts that are worn or damaged. Regardless of appearance, replace the seat belt after three years of use. Do not use a seat belt extension on a retractable seat belt.

Adjust the seat so that full pedal travel can be achieved with the operator's back against the back of the seat.

Make sure that the machine is equipped with a lighting system that is adequate for the job conditions. Make sure that all machine lights are working properly.

Before you start the engine and before you move the machine, make sure that no one is underneath the machine, around the machine, or on the machine. Make sure that the area is free of personnel.

i07746368

Visibility Information

SMCS Code: 7000

Before you start the machine, perform a walk-around inspection in order to ensure that there are no hazards around the machine.

While the machine is in operation, constantly survey the area around the machine in order to identify potential hazards as hazards become visible around the machine.

Your machine may be equipped with visual aids. Some examples of visual aids are Closed Circuit Television (CCTV) and mirrors. Before operating the machine, ensure that the visual aids are in proper working condition and that the visual aids are clean. Adjust the visual aids using the procedures that are located in this Operation and Maintenance Manual. If equipped, the Work Area Vision System shall be adjusted according to Operation and Maintenance Manual, SEBU8157, "Work Area Vision System". If equipped, the Cat Detect Object Detection shall be adjusted according to the Operation and Maintenance Manual, "Cat Detect Object Detection" for your machine.

It may not be possible to provide direct visibility on large machines to all areas around the machine. Appropriate job site organization is required in order to minimize hazards that are caused by restricted visibility. Job site organization is a collection of rules and procedures that coordinates machines and people that work together in the same area. Examples of job site organization include the following:

- Safety instructions
- Controlled patterns of machine movement and vehicle movement
- Workers that direct safe movement of traffic
- Restricted areas
- Operator training
- Warning symbols or warning signs on machines or on vehicles
- A system of communication
- Communication between workers and operators prior to approaching the machine

Modifications of the machine configuration by the user that result in a restriction of visibility shall be evaluated.

i07575957

Restricted Visibility

SMCS Code: 7000

The size and the configuration of this machine may result in areas that cannot be seen when the operator is seated. For restricted visibility areas, an appropriate job site organization must be utilized to minimize hazards of this restricted visibility. For more information regarding job site organization refer to Operation and Maintenance Manual, "Visibility Information".

Illustrations 59 - 62 provide an approximate visual indication of the areas at ground level inside a radius of 12 m (39 ft) from the operator of significant restricted visibility for various machine configurations. Refer to the correct illustration for your machine configuration. All restricted visibility areas less than 300 mm wide may not be shown. These illustrations do not indicate areas of restricted visibility for distances outside of the shown radius. The areas of restricted visibility shown in the illustrations are with the track and work tool of the machine in the Travel position. Illustration 63 shows the position of the work tool in the travel position. The Caterpillar authorized work tool that resulted in the largest visibility restriction was used.

Illustration 59 indicates restricted visibility areas at ground level inside the shown radius from the operator without the use of visual aids that may be optional for this product in some markets.

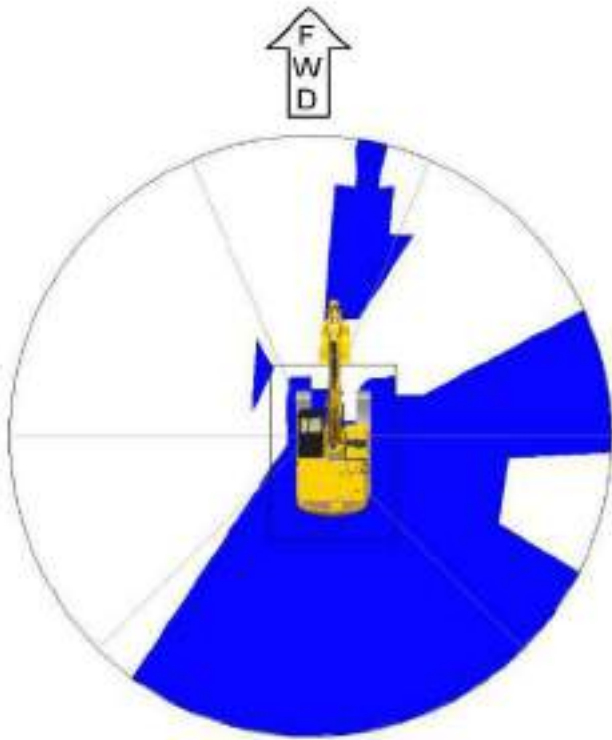


Illustration 59

g06366562

Top view of the machine, ground level visibility without use of optional visual aids

(A) 12 m (39 ft)

Note: The shaded areas indicate the approximate location of areas with significant restricted visibility.

Illustration 60 indicates restricted visibility areas at ground level inside the shown radius from the operator with the use of available rear camera, right side mirror, and left side mirror installed.

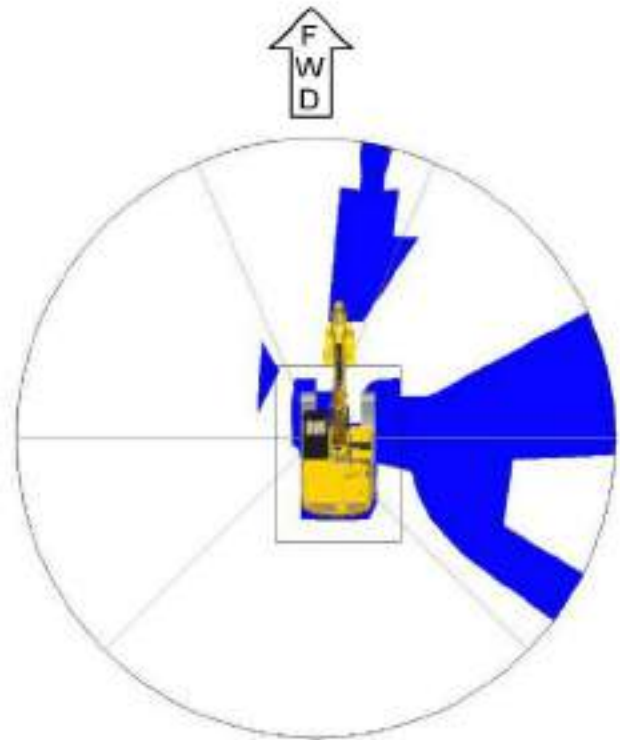


Illustration 60

g06366568

Top view of the machine, ground level visibility with rear camera, left side mirror, and right side mirror

(A) 12 m (39 ft)

Note: The shaded areas indicate the approximate location of areas with significant restricted visibility.

Illustration 61 indicates restricted visibility areas at ground level inside the shown radius from the operator with the use of available rear camera, right side camera, left side mirrors, and left side second mirror.

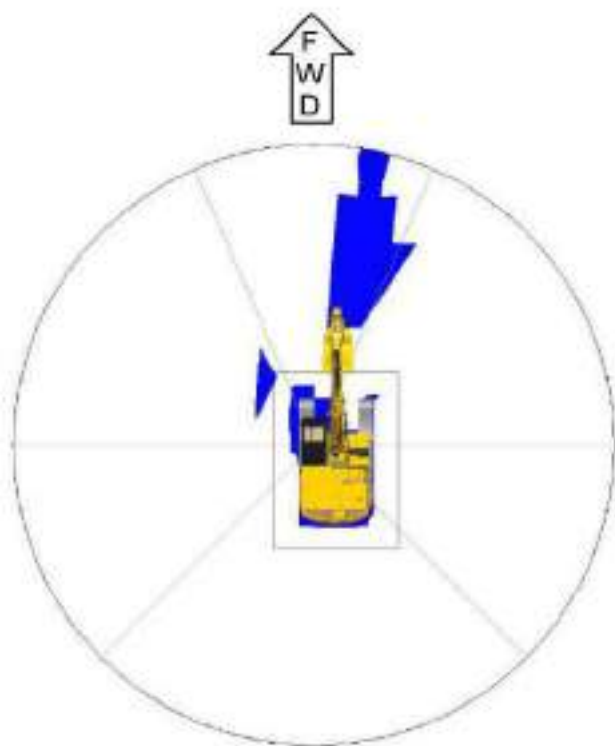


Illustration 61

g06356105

Top view of the machine, ground level visibility with rear camera, right side camera, left side mirror, and left side second mirror.

(A) 12 m (39 ft)

Note: The shaded areas indicate the approximate location of areas with significant restricted visibility.

Illustration 62 indicated restricted visibility areas at ground level inside the shown radius from the operator with the use of front camera, rear camera, right side camera, left side camera installed.

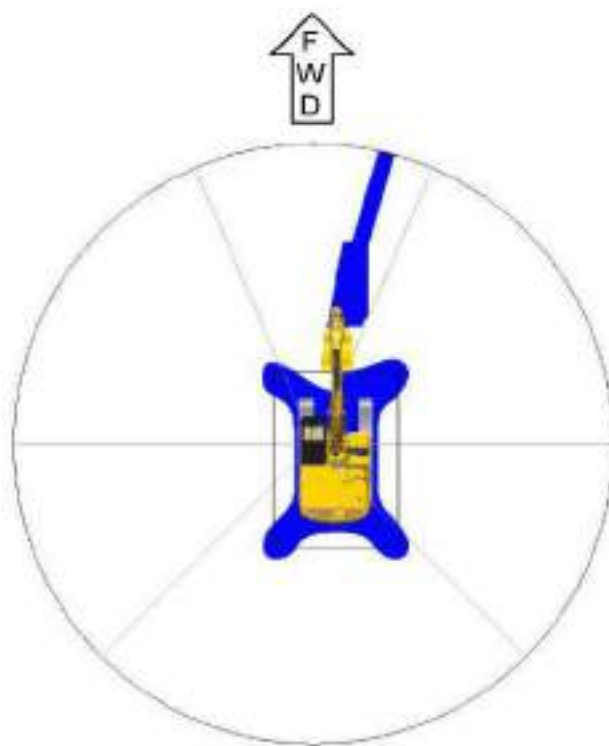


Illustration 62

g06356117

Top view of the machine, 360 visibility with front camera, rear camera, right side camera, and left side camera.

Note: The shaded areas indicate the approximate location of areas with significant restricted visibility.

Restricted visibility is measured when the front linkage of the machine is in the travel position. Illustration 63 shows the machine in the travel position.

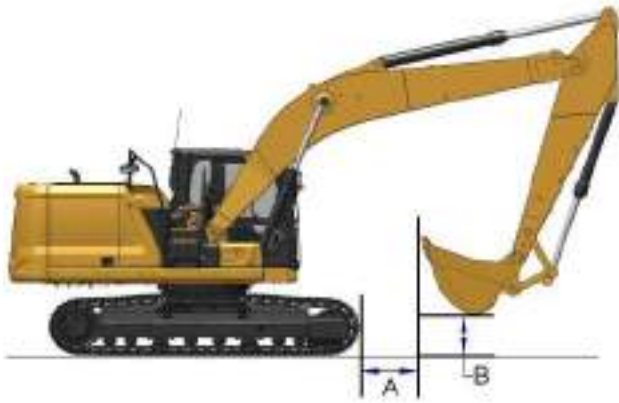


Illustration 63

g06181081

(A) 1 m (3.0 ft) from the front of the machine to the bucket
(B) 0.5 m (1.6 ft) from ground level

i03562260

Engine Starting

SMCS Code: 1000; 7000

If a warning tag is attached to the engine start switch or to the controls, do not start the engine. Also, do not move any controls.

Make sure that you are seated before you start the engine.

Move all hydraulic controls to the HOLD position before you start the engine. Move the hydraulic lockout control to the LOCKED position. For further details on this procedure, refer to Operation and Maintenance Manual, "Operator Controls".

Diesel engine exhaust contains products of combustion which can be harmful to your health. Always run the engine in a well ventilated area. If you are in an enclosed area, vent the exhaust to the outside.

Briefly sound the horn before you start the engine.

i01340061

Before Operation

SMCS Code: 7000

Clear all personnel from the machine and from the area.

Clear all obstacles from the machine's path. Beware of hazards (wires, ditches, etc).

Be sure that all windows are clean. Secure the doors and the windows in the open position or in the shut position.

Adjust the rearview mirrors (if equipped) for the best visibility close to the machine. Make sure that the horn, the travel alarm (if equipped), and all other warning devices are working properly.

Fasten the seat belt securely.

Warm up the engine and the hydraulic oil before operating the machine.

Before moving the machine, check the position of the undercarriage. The normal travel position is with the idler wheels to the front under the cab and the drive sprockets to the rear. When the undercarriage is in the reversed position, the directional controls must be operated in opposite directions.

i04159629

Work Tools

SMCS Code: 6700

Only use work tools that are recommended by Caterpillar for use on Cat machines.

Use of work tools, including buckets, which are outside of Caterpillar's recommendations or specifications for weight, dimensions, flows, pressures, and so on, may result in less-than-optimal vehicle performance, including but not limited to reductions in production, stability, reliability, and component durability. Caterpillar recommends appropriate work tools for our machines to maximize the value our customers receive from our products. Caterpillar understands that special circumstances may lead a customer to use tools outside of our specifications. In these cases, customers must be aware that such choices can reduce vehicle performance and will affect their ability to claim warranty in the event of what a customer may perceive as a premature failure.

Work tools and work tool control systems, that are compatible with your Cat machine, are required for safe machine operation and/or reliable machine operation. If you are in doubt about the compatibility of a particular work tool with your machine, consult your Cat dealer.

Make sure that all necessary guarding is in place on the host machine and on the work tool.

Keep all windows and doors closed on the host machine. A polycarbonate shield must be used when the host machine is not equipped with windows and when a work tool could throw debris.

Do not exceed the maximum operating weight that is listed on the ROPS certification.

If your machine is equipped with an extendable stick, install the transport pin when you are using the following work tools: hydraulic hammers, augers and compactors

Always wear protective glasses. Always wear the protective equipment that is recommended in the operation manual for the work tool. Wear any other protective equipment that is required for the operating environment.

To prevent personnel from being struck by flying objects, ensure that all personnel are out of the work area.

While you are performing any maintenance, any testing, or any adjustments to the work tool stay clear of the following areas: cutting edges, pinching surfaces and crushing surfaces.

Never use the work tool for a work platform.

i07199415

Assist and E-Fence

SMCS Code: 5050

WARNING

Personal Responsibility

When the Assist System is engaged, the Assist System will control various machine functions. The Assist System is not a substitute for the personal judgment of an operator for the present work site situation or work site conditions. Failure to pay attention to work site situations or work site conditions with the Assist System engaged could result in personal injury or death. Always be aware of the work site situation and work site conditions when operating the machine with the Assist System engaged.

WARNING

Be Alert

Due to system limitations with the Assist System, warnings from the Assist System may be issued late, improperly, or not at all. Failure to respond to warnings from the Assist System with appropriate action could result in personal injury or death. Always be alert when operating a machine with the Assist System engaged and be ready to intervene to prevent an accident.

Operation Guidance

- Ground personnel should maintain a safe distance from the machine. Remain out of the swing radius/swing zone, and always check for ground personnel in the swing radius/swing zone. Do not operate with ground personnel in the swing radius/swing zone.
- When setting height or load restrictions, make sure that any work tools that are attached are in the least favorable position.
- Check for the correct operation once the restriction has been set.
- Validate proper settings both initially and periodically during machine operation.
- Do not move the (machine) quickly when close to a restriction setting.
- Do not operate the machine in a reckless manner.
- Do not travel with the restriction set.
- Reset the load restriction if the machine is moved to a different location.
- Always check settings or reestablish desired settings after leaving the machine for any period to ensure desired performance.
- If the proper setting procedure has been followed, but the system does not appear to be functioning properly, consult your Cat dealer.

Hazard recommendations:

- Return the joysticks to the neutral position once a limit has been reached
- Operate the machine at a slow speed
- Do not travel

Lifting operations with the machine

- Ensure that Lift Mode is selected during lifting operations
- Operate the machine at a safe speed
- Take special care when traveling with a load attached

- Do not operate the machine in a reckless manner

i07889511

Operation

SMCS Code: 7000

Machine Operating Temperature Range

The machine must function satisfactorily in the anticipated ambient temperature limits that are encountered during operation. The standard machine configuration is intended for use within an ambient temperature range of -18°C (0°F) to 43°C (109°F). Special configurations for different ambient temperatures may be available. Consult your Cat dealer for additional information on special configurations of your machine.

Limiting Conditions and Criteria

Limiting conditions are immediate issues with this machine that must be addressed prior to continuing operation.

The Operation and Maintenance Manual, Safety Section describes limiting condition criteria for replacing items such as safety messages, seat belt and mounting hardware, lines, tubes, hoses, battery cables and related parts, electrical wires, and repairing any fluid leak.

The Operation and Maintenance Manual, Maintenance Interval Schedule describes limiting condition criteria that require repair or replacement for items (if equipped) such as alarms, horns, braking system, steering system, and rollover protective structures.

The Operation and Maintenance Manual, Monitoring System (if equipped) provides information on limiting condition criteria, including a Warning Category 3 that requires immediate shutdown of the engine.

Critical Failures

The following table provides summary information on several limiting conditions found in this Operation and Maintenance Manual. The table provides criteria and required action for the limiting conditions listed. Each System or Component in this table, together with the respective limiting condition, describes a potential critical failure that must be addressed. Not addressing limiting conditions with required actions may, in conjunction with other factors or circumstances, result in a risk of personal injury or death. If an accident occurs, notify emergency personnel and provide location and description of accident.

Safety Section
Operation

Table 1

System or Component Name	Limiting Condition	Criteria for Action	Required Action
Line, tubes, and hoses	End fittings are damaged or leaking. Outer coverings are chafed or cut. Wires are exposed. Outer coverings are swelling or ballooning. Flexible parts of the hoses are kinked. Outer covers have exposed embedded armoring. End fittings are displaced.	Visible corrosion, loose, or damaged lines, tubes, or hoses. Visible fluid leaks.	Immediately repair any lines, tubes, or hoses that are corroded, loose, or damaged. Immediately repair any leaks as these may provide fuel for fires.
Electrical Wiring	Signs of fraying, abrasion, cracking, discoloration, cuts on the insulation	Visible damage to electrical wiring	Immediately replace damaged wiring
Battery cable(s)	Signs of fraying, abrasion, cracking, discoloration, cuts on the insulation of the cable, fouling, corroded terminals, damaged terminals, and loose terminals	Visible damage to battery cable(s)	Immediately replace damaged battery cables
Operator Protective Structure	Structures that are bent, cracked, or loose. Loose, missing, or damaged bolts.	Visible damage to structure. Loose, missing, or damaged bolts.	Do not operate machine with damaged structure or loose, missing, or damaged bolts. Contact your Cat dealer for inspection and repair or replacement options.
Seat Belt	Worn or damaged seat belt or mounting hardware	Visible wear or damage	Immediately replace parts that are worn or damaged.
Seat Belt	Age of seat belt	Three years after date of installation	Replace seat belt three years after date of installation
Safety Messages	Appearance of safety message	Damage to safety messages making them illegible	Replace the illustrations if illegible.
Audible Warning Device(s) (if equipped)	Sound level of audible warning	Reduced or no audible warning present	Immediately repair or replace audible warning devices not working properly.
Camera(s) (if equipped)	Dirt or debris on camera lens	Dirt or debris obstructing camera view	Clean camera before operating machine.
Cab Windows (if equipped)	Dirt, debris, or damaged windows	Dirt or debris obstructing operator visibility. Any damaged windows.	Clean windows before operating machine. Repair or replace damaged windows before operating machine.
Mirrors (if equipped)	Dirt, debris, or damaged mirror	Dirt or debris obstructing operator visibility. Any damaged mirrors.	Clean mirrors before operating machine. Repair or replace damaged mirrors before operating machine.
Braking System	Inadequate braking performance	System does not pass Braking System - Test(s) included in Maintenance Section or in the Testing and Adjusting Manual	Contact your Cat dealer to inspect and, if necessary, repair the brake system.
Cooling System	The coolant temperature is too high.	Monitoring System displays Warning Category 3	Stop the engine immediately. Check the coolant level and check the radiator for debris. Refer to Operation and Maintenance Manual, Cooling System Coolant Level - Check. Check the fan drive belts for the water pump. Refer to Operation and Maintenance Manual, Belts - Inspect/Adjust/ Replace. Make any necessary repairs.
Engine Oil System	A problem has been detected with the engine oil pressure.	Monitoring System displays Warning Category 3	If the warning stays on during low idle, stop the engine and check the engine oil level. Perform any necessary repairs as soon as possible.
Engine system	An engine fault has been detected by the engine ECM.	Monitoring System displays Warning Category 3	Stop the engine immediately. Contact your Cat dealer for service.
Fuel System	A problem has been detected with the fuel system.	Monitoring System displays Warning Category 3	Stop the engine. Determine the cause of the fault and perform any necessary repairs.
Hydraulic Oil System	The hydraulic oil temperature is too high.	Monitoring System displays Warning Category 3	Stop the engine immediately. Check the hydraulic oil level and check the hydraulic oil cooler for debris. Perform any necessary repairs as soon as possible.

(continued)

(Table 1, contd)

System or Component Name	Limiting Condition	Criteria for Action	Required Action
Steering System	A problem has been detected with the steering system. (If equipped with steering system monitoring.)	Monitoring System displays Warning Category 3	Move machine to a safe location and stop the engine immediately. Contact your Cat dealer to inspect and, if necessary, repair the steering system.
Overall Machine	Machine service is required.	Monitoring System displays Warning Category 3	Stop the engine immediately. Contact your Cat dealer for service.

Machine Operation

Only operate the machine while you are in a seat. The seat belt must be fastened while you operate the machine. Only operate the controls while the engine is running.

Check for proper operation of all controls and of all protective devices while you operate the machine slowly in an open area.

When the machine is moving watch the clearance of the boom. Uneven ground can cause the boom to move in all directions.

Make sure that no personnel will be endangered before you move the machine. Do not allow riders on the machine unless the machine has an additional seat with a seat belt.

Report any machine damage that was noted during machine operation. Make any necessary repairs.

Never use the work tool for a work platform.

Hold attachments approximately 40 cm (15 inches) above ground level while you drive the machine. Do not drive the machine close to an overhang, to the edge of a cliff, or to the edge of an excavation.

If the machine begins to sideslip on a grade, immediately dump the load and turn the machine downhill.

Be careful to avoid any ground condition which could cause the machine to tip. Tipping can occur when you work on hills, on banks, or on slopes. Tipping can also occur when you cross ditches, ridges, or other unexpected obstructions.

When possible, operate the machine up slopes and down slopes with the final drive sprockets facing down the slope. Avoid operating the machine across the slope. Place the heaviest end of the machine uphill when you are working on an incline.

Keep the machine under control. Do not overload the machine beyond capacity.

Avoid changing the direction of travel on a slope. Changing the direction of travel on a slope could result in tipping or side slipping of the machine.

Bring the load close to the machine before traveling any distances.

Bring the load close to the machine before swinging the load.

Lifting capacity decreases as the load is moved further from the machine.

Make sure that the towing eyes and the towing devices are adequate for your needs.

Only connect trailing equipment to a drawbar or to a hitch.

Never straddle a wire cable. Never allow other personnel to straddle a wire cable.

When you maneuver in order to connect the equipment, make sure that no personnel are between the machine and trailing equipment. Block up the hitch of the trailing equipment in order to align the equipment with the drawbar.

Check the local regulations, state codes, and/or directives of the job site for a specific minimum distance from obstacles.

Before you operate the machine, check with local utilities for the locations of underground pipes and for the locations of buried cables.

Know the maximum dimensions of your machine.

Watch the load at all times.

Do not operate the machine without the counterweight. The machine can tip when the boom is over the side.

The clamshell, the grapple, or the magnet can swing in all directions. Move the joysticks in a continuous motion. Failure to move the joysticks in a continuous motion can cause the clamshell, the grapple, or the magnet to swing into the cab or into a person in the work area. This will result in personal injury.

Certain machine front linkage combinations (boom, stick, quick coupler, work tool) can allow the work tool to contact the machine undercarriage, swing frame, boom, boom hydraulic cylinder and or the cab. Be aware of the position of the work tool while you operate the machine.

Shut down the machine until damaged or non-functioning visibility aid(s) is repaired (if applicable) or until appropriate job site organization is used to minimize hazards that are caused by any resulting restricted visibility.

Machine Operation when the Machine is not Completely Assembled

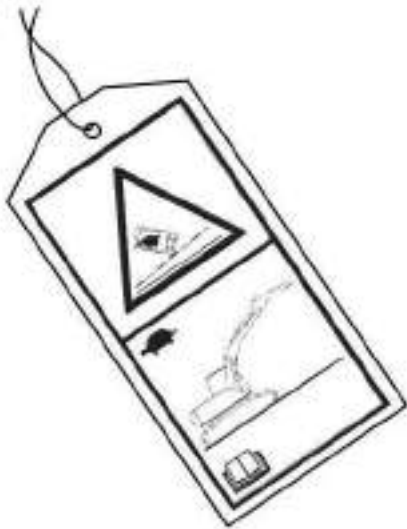


Illustration 64

g02202544

Attach the tag to the controls of the machine. When the tag is attached to the controls, operate the machine as described below.

If the machine needs to be operated without the boom, stick, and/or counterweight being installed, the machine should be operated slowly on flat, stable ground or pavement by qualified operators. Avoid any machine operations which could affect machine stability, including the swing function. The ROPS structural certification depends on the support of the boom, stick, and counterweight in the event of a machine tip over or a machine rollover incident.

i06299648

Engine Stopping

SMCS Code: 1000; 7000

Do not stop the engine immediately after the machine has been operated under load. Stopping the engine immediately can cause overheating and accelerated wear of engine components.

After the machine is parked and the parking brake is engaged, allow the engine to run at low idle for 5 minutes before shutdown. Running the engine allows hot areas of the engine to cool gradually.

i06304391

Lifting Objects

SMCS Code: 7000

There may be local regulations and/or government regulations that govern the use of machines which lift heavy objects. Obey all local and government regulations.

If this machine is used to lift objects within an area that is controlled by the European Directive "2006/42/EC", the machine must be equipped with a boom lowering control valve, a stick lowering control valve, and an overload warning device.

If this machine is used to lift objects within Japan, Japanese regulations require the machine to be equipped with a shovel crane configuration.

Contact your Cat dealer for additional information

i07749631

Demolition

SMCS Code: 6700

There maybe local regulations and/or government regulations that govern the use of machines which are designed and used as demolition machinery.

Note: Obey all local and government regulations.

Demolition machinery is designed for demolishing by pushing or pulling, or fragmenting. Demolition is done by crushing or shearing, buildings and/or other civil engineering structures and component parts and/or separating the resultant debris.

If this machine is used for demolition within an area that is controlled by the European Directive 2006/42/EC, the machine must be equipped with:

- Rollover Protective Structure (ROPS, not required for demolition excavators)
- Boom Lowering Control Valve (BLCV) / Stick Lowering Control Valve (SLCV)
- Top Guard / Front Guard
- Bottom / Motor / Swivel Guard
- EN 356 class P5A front window glass
- If a roof window is used to provide visibility to the working area, then roof window shall be equipped with motorized windscreen wipers and washers.

Demolition applications may generate flying debris. Ensure that there are no personnel in the area around the machine where flying debris may travel.

Demolition applications may generate airborne dust that can be hazardous to your health. If you operate the machine in a dust generating applications, use appropriate safeguarding or adequate ventilation to minimize risk.

i07946718

Parking

SMCS Code: 7000

The hydraulic system controls remain pressurized if the accumulator is charged. This condition is true even when the engine is not running. The hydraulic control system pressure should decrease in a short time (approximately 1 minute). While the hydraulic controls maintain a charge, the hydraulic work tools and machine controls remain functional.

There can be residual pressure within the hydraulic system even when the accumulator is empty. Refer to this Operation and Maintenance Manual, "System Pressure Release" before any service is performed to the hydraulic system.

Machine movement that is sudden and unexpected will occur if any of the controls are moved. Machine movement that is sudden and unexpected, can cause personal injury or death.

Always move the hydraulic lockout control to the LOCKED position before you shut off the engine or immediately after the engine stops running.

Park the machine on a hard, level surface. If you must park the machine on a grade, chock the tracks of the machine.



Illustration 65

g06181120

Place the machine in the servicing position.

Note: Make sure that all work tools are in the recommended servicing position before servicing the machine.

Stop the engine.

Turn the engine start switch to the OFF position.

Turn the battery disconnect switch to the OFF position, if you do not intend on operating the machine for an extended period. This will prevent drainage of the battery. A battery short circuit, any current draw from certain components, and vandalism can cause drainage of the battery.

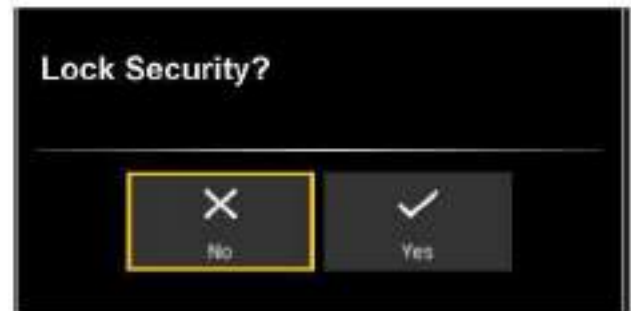


Illustration 66

g06217247

Note: Do not turn off the battery disconnect switch until 5 seconds have elapsed after turning the engine start switch to the OFF position. Do not turn off the battery disconnect switch while the "Lock Security?" screen is displayed on the monitor. Both conditions would prevent the Diesel Exhaust Fluid (DEF) system from being purged and could cause the fluid to freeze in the lines.

Install barriers or lighting as required to prevent interference in road traffic.

Select places free of danger by flooding and other water damage.

i07746366

Slope Operation

SMCS Code: 7000

Machines that are operating safely in various applications depend on these criteria: the machine model, configuration, machine maintenance, operating speed of the machine, conditions of the terrain, fluid levels, and tire inflation pressures. The most important criteria are the skill and judgment of the operator.

A well trained operator that follows the instructions in the Operation and Maintenance Manual has the greatest impact on stability. Operator training provides a person with the following abilities: observation of working and environmental conditions, feel for the machine, identification of potential hazards and operating the machine safely by making appropriate decisions.

When you work on side hills and when you work on slopes, consider the following important points:

Speed of travel – At higher speeds, forces of inertia tend to make the machine less stable.

Roughness of terrain or surface – The machine may be less stable with uneven terrain.

Direction of travel – Avoid operating the machine across the slope. When possible, operate the machine up the slopes and operate the machine down the slopes. Place the heaviest end of the machine uphill when you are working on an incline.

Mounted equipment – Balance of the machine may be impeded by the following components: equipment that is mounted on the machine, machine configuration, weights, and counterweights.

Nature of surface – Ground that has been newly filled with earth may collapse from the weight of the machine.

Surface material – Rocks and moisture of the surface material may drastically affect the machine's traction and machine's stability. Rocky surfaces may promote side slipping of the machine.

Slippage due to excessive loads – This may cause downhill tracks or downhill tires to dig into the ground, which will increase the angle of the machine.

Width of tracks or tires – Narrower tracks or narrower tires further increase the digging into the ground which causes the machine to be less stable.

Implements attached to the drawbar – This may decrease the weight on the uphill tracks. This may also decrease the weight on the uphill tires. The decreased weight will cause the machine to be less stable.

Height of the working load of the machine – When the working loads are in higher positions, the stability of the machine is reduced.

Operated equipment – Be aware of performance features of the equipment in operation and the effects on machine stability.

Operating techniques – Keep all attachments or pulled loads low to the ground for optimum stability.

Machine systems have limitations on slopes – Slopes can affect the proper function and operation of the various machine systems. These machine systems are needed for machine control.

Note: Operators with lots of experience and proper equipment for specific applications are also required. Safe operation on steep slopes may also require special machine maintenance. Refer to Lubricant Viscosities and Refill Capacities in this manual for the proper fluid level requirements and intended machine use. Fluids must be at the correct levels to ensure that systems will operate properly on a slope.

i08229294

Equipment Lowering with Engine Stopped

SMCS Code: 7000-II

Before lowering any equipment with the engine stopped, clear the area around the equipment of all personnel. The procedure to use will vary with the type of equipment to be lowered. Keep in mind most systems use a high pressure fluid or air to raise or lower equipment. The procedure will cause high pressure air, hydraulic, or some other media to be released in order to lower the equipment. Wear appropriate personal protective equipment and follow the established procedure in the Operation and Maintenance Manual, "Equipment Lowering with Engine Stopped" in the Operation Section of the manual.

i07865151

Sound Information and Vibration Information

SMCS Code: 7000

Sound Level Information

The declared dynamic operator sound pressure level is 71 dB(A) when "ISO 6396: 2008" is used to measure the value for an enclosed cab. The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds. The measurement was conducted with the cab doors and the cab windows closed. The cab was properly installed and maintained.

The declared exterior sound power level L_{WA} is 99 dB (A) for 320 and 100 dB(A) for 323 when the value is measured according to the dynamic test procedures and the conditions that are specified in "ISO 6395:2008". The measurement was conducted at 70 % of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds.

The declared sound levels listed above include both measurement uncertainty and uncertainty due to production variation.

Hearing protection may be needed when the machine is operated with an open operator station for extended periods or in a noisy environment. Hearing protection may be needed when the machine is operated with a cab that is not properly maintained, or when the doors and windows are open for extended periods or in a noisy environment.

Sound Level Information for Machines in European Union Countries and in Countries that Adopt the “EU Directives”

The declared dynamic operator sound pressure level is 71 dB(A) when “ISO 6396: 2008” is used to measure the value for an enclosed cab. The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds. The measurement was conducted with the cab doors and the cab windows closed. The cab was properly installed and maintained.

The declared exterior sound power level (LWA) when the value is measured according to the dynamic test procedures and conditions that are specified in “ISO 6395:2008” are as follows:

- 320 - 99 dB(A)
- 323 - 100 dB(A)

The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds.

The declared sound levels listed above include both measurement uncertainty and uncertainty due to production variation.

Sound Level Information for Machines in Eurasian Economic Union Countries

The declared dynamic operator sound pressure level is 71 dB(A) when “ISO 6396:2008” is used to measure the value for an enclosed cab. The measurement was conducted at 70 % of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds. The measurement was conducted with the cab doors and the cab windows closed.

The declared exterior sound power level L_{WA} is 99 dB (A) for 320 and 100 dB(A) for 323 when the value is measured according to the dynamic test procedures and the conditions that are specified in “ISO 6395:2008”. The measurement was conducted at 70 % of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds.

The declared sound levels listed above include both measurement uncertainty and uncertainty due to production variation.

“The European Union Physical Agents (Vibration) Directive 2002/44/EC”

Vibration Data for Track Type Excavator

Information Concerning Hand/Arm Vibration Level

When the machine is operated according to the intended use, the hand/arm vibration of this machine is below 2.5 meter per second squared.

Information Concerning Whole Body Vibration Level

This section provides vibration data and a method for estimating the vibration level for track type excavators.

Note: Vibration levels are influenced by many different parameters. Many items are listed below.

- Operator training, behavior, mode, and stress
- Job site organization, preparation, environment, weather, and material
- Machine type, quality of the seat, quality of the suspension system, attachments, and condition of the equipment

It is not possible to get precise vibration levels for this machine. The expected vibration levels can be estimated with the information in Table 2 to calculate the daily vibration exposure. A simple evaluation of the machine application can be used.

Estimate the vibration levels for the three vibration directions. For typical operating conditions, use the average vibration levels as the estimated level. With an experienced operator and smooth terrain, subtract the Scenario Factors from the average vibration level to obtain the estimated vibration level. For aggressive operations and severe terrain, add the Scenario Factors to the average vibration level to obtain the estimated vibration level.

Note: All vibration levels are in meter per second squared.

Table 2

"ISO Reference Table A - Equivalent vibration levels of whole body vibration emission for earthmoving equipment."							
Machine Type	Typical Operating Activity	Vibration Levels			Scenario Factors		
		X axis	Y axis	Z axis	X axis	Y axis	Z axis
Track Type Excavators	excavating	0,44	0,27	0,30	0,24	0,16	0,17
	hydraulic breaker application	0,53	0,31	0,55	0,30	0,18	0,28
	mining application	0,65	0,42	0,61	0,21	0,15	0,32
	transfer	0,48	0,32	0,79	0,19	0,20	0,23

Note: Refer to "ISO/TR 25398 Mechanical Vibration - Guideline for the assessment of exposure to whole body vibration of ride on operated earthmoving machines" for more information about vibration. This publication uses data that is measured by international institutes, organizations, and manufacturers. This document provides information about the whole body exposure of operators of earthmoving equipment. Refer to Operation and Maintenance Manual, SEBU8257, "The European Union Physical Agents (Vibration) Directive 2002/44/EC" for more information about machine vibration levels.

The Cat suspension seat meets the criteria of "ISO 7096". This standard represents vertical vibration level under severe operating conditions.

Guidelines for Reducing Vibration Levels on Earthmoving Equipment

Properly adjust machines. Properly maintain machines. Operate machines smoothly. Maintain the conditions of the terrain. The following guidelines can help reduce the whole body vibration level:

1. Use the right type and size of machine, equipment, and attachments.
2. Maintain machines according to the manufacturers recommendations.
 - a. Tire pressures
 - b. Brake and steering systems
 - c. Controls, hydraulic system, and linkages
3. Keep the terrain in good condition.
 - a. Remove any large rocks or obstacles.
 - b. Fill any ditches and holes.
 - c. Provide machines and schedule time to maintain the conditions of the terrain.
4. Use a seat that meets "ISO 7096". Keep the seat maintained and adjusted.
 - a. Adjust the seat and suspension for the weight and the size of the operator.

- b. Inspect and maintain the seat suspension and adjustment mechanisms.

5. Perform the following operations smoothly.

- a. Steer
- b. Brake
- c. Accelerate.
- d. Shift the gears.

6. Move the attachments smoothly.

7. Adjust the machine speed and the route to minimize the vibration level.

- a. Drive around obstacles and rough terrain.
- b. Slow down when necessary to go over rough terrain.

8. Minimize vibrations for a long work cycle or a long travel distance.

- a. Use machines that are equipped with suspension systems.
- b. Use the ride control system on track type excavators.
- c. If no ride control system is available, reduce speed to prevent bounce.
- d. Haul the machines between workplaces.

9. Less operator comfort may be caused by other risk factors. The following guidelines can be effective to provide better operator comfort:

- a. Adjust the seat and adjust the controls to achieve good posture.
- b. Adjust the mirrors to minimize twisted posture.
- c. Provide breaks to reduce long periods of sitting.
- d. Avoid jumping from the cab.
- e. Minimize repeated handling of loads and lifting of loads.

- f. Minimize any shocks and impacts during sports and leisure activities.

i07746359

Sources

The vibration information and the calculation procedure are based on "ISO/TR 25398 Mechanical Vibration - Guideline for the assessment of exposure to whole body vibration of ride on operated earthmoving machines". Harmonized data is measured by international institutes, organizations, and manufacturers.

This literature provides information about assessing the whole body vibration exposure of operators of earthmoving equipment. The method is based on measured vibration emission under real working conditions for all machines.

Check the original directive. This document summarizes part of the content of the applicable law. This document is not meant to substitute the original sources. Other parts of these documents are based on information from the United Kingdom Health and Safety Executive.

Refer to Operation and Maintenance Manual, SEBU8257, "The European Union Physical Agents (Vibration) Directive 2002/44/EC" for more information about vibration.

Consult your local Cat dealer for more information about machine features that minimize vibration levels. Consult your local Cat dealer about safe machine operation.

Use the following web site to find your local dealer:

Caterpillar, Inc.
www.cat.com

i07746362

Operator Station

SMCS Code: 7300; 7301; 7325

Any modifications to the inside of the operator station should not project into the operator space or into the space for the companion seat (if equipped). The addition of a radio, fire extinguisher, and other equipment must be installed so that the defined operator space and the space for the companion seat (if equipped) is maintained. Any item that is brought into the cab should not project into the defined operator space or the space for the companion seat (if equipped). A lunch box or other loose items must be secured. Objects must not pose an impact hazard in rough terrain or in the event of a rollover.

Guards (Operator Protection)

SMCS Code: 7000; 7150

There are different types of guards that are used to protect the operator. The machine and the machine application determine the type of guard that should be used.

A daily inspection of the guards is required in order to check for structures that are bent, cracked, or loose. Never operate a machine with a damaged structure.

The operator becomes exposed to a hazardous situation if the machine is used improperly or if poor operating techniques are used. This situation can occur even though a machine is equipped with an appropriate protective guard. Follow the established operating procedures that are recommended for your machine.

Rollover Protective Structure (ROPS), Falling Object Protective Structure (FOPS) or Tip Over Protection Structure (TOPS)

The ROPS/FOPS Structure (if equipped) on your machine is specifically designed, tested and certified for that machine. Any alteration or any modification to the ROPS/FOPS Structure could weaken the structure. This places the operator into an unprotected environment. Modifications or attachments that cause the machine to exceed the weight that is stamped on the certification plate also place the operator into an unprotected environment. Excessive weight may inhibit the brake performance, the steering performance and the ROPS. The protection that is offered by the ROPS/FOPS Structure will be impaired if the ROPS/FOPS Structure has structural damage. Damage to the structure can be caused by an overturn, a falling object, a collision, etc.

Do not mount items (fire extinguishers, first aid kits, work lights, etc) by welding brackets to the ROPS/FOPS Structure or by drilling holes in the ROPS/FOPS Structure. Welding brackets or drilling holes in the ROPS/FOPS Structures can weaken the structures. Consult your Cat dealer for mounting guidelines.

The Tip Over Protection Structure (TOPS) is another type of guard that is used on mini hydraulic excavators. This structure protects the operator in the event of a tipover. The same guidelines for the inspection, the maintenance and the modification of the ROPS/FOPS Structure are required for the Tip Over Protection Structure.

Other Guards (If Equipped)

Protection from flying objects and/or falling objects is required for special applications. Logging applications and demolition applications are two examples that require special protection.

A front guard needs to be installed when a work tool that creates flying objects is used. Mesh front guards that are approved by Caterpillar or polycarbonate front guards that are approved by Caterpillar are available for machines with a cab or an open canopy. On machines that are equipped with cabs, the windows should also be closed. Safety glasses are recommended when flying hazards exist for machines with cabs and machines with open canopies.

If the work material extends above the cab, top guards and front guards should be used. Typical examples of this type of application are listed below:

- Demolition applications
- Rock quarries
- Forestry products

Additional guards may be required for specific applications or work tools. The Operation and Maintenance Manual for your machine or your work tool will provide specific requirements for the guards. Refer to Operation Maintenance manual, "Demolition" for additional information. Consult your Cat dealer for additional information.

Product Information Section

General Information

i07105852

Regulatory Information (Japan)

SMCS Code: 7000

Qualifications for Machine Operation

The following qualifications are required for the operation of this machine:

Excavation and Loading

Completion of the construction machines (for land leveling, hauling, loading, and excavation) operation skill training course. (Qualification by the Industrial Safety and Health Act)

Demolition

Completion of the construction machines (for demolition) operation skill training course. (Qualification by the Industrial Safety and Health Act)

Mining Jobs

Certification by the Director General or Deputy Director General of Bureau of Mine Safety after completion of the safety training course. (Qualification by the Mine Safety Act)

Crane Slings for the Bucket with a Hook

Completion of the special slinging training for the crane for loads weighing less than 1 ton. (Qualification by the Industrial Safety and Health Act)

Trailer Transportation

In principle, this machine should be transported by a trailer. Select the appropriate trailer regarding the machine weight and measurements shown in the major specifications in the specification part of this manual. Be aware machine weight and transportation measurements differ depending on the various types of attachments.

- In the event heavy items are to be transported, observe the related laws. These laws include Road Traffic Law, Road Laws, Road Transportation Vehicle Laws, and Vehicle Restriction Laws.

- Conduct prior investigation of the road width, ground clearance of road/railway bridges, weight restrictions etc. of the planned transportation route, to confirm the viability of the transportation execution.

Load

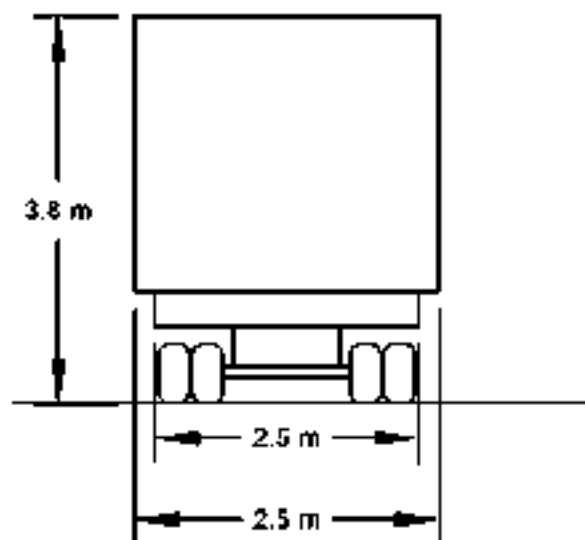


Illustration 67

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- Not more than 3.8 m (12 ft 6 inch)
- Not more than 2.5 m (8 ft 2 inch)(Safety Standard)
- Not more than 2.5 m (8 ft 2 inch) (Vehicle Restriction Laws)
- Items that protrude out are not allowed. (Government ordinance for Road Traffic Laws)

Transportation weight and measurements are restricted by the Vehicle Restriction Laws. If the actual weight/measurements exceed the limitation figures, you must submit the restriction relaxation request to the pertinent governmental agencies. For details, consult your Cat dealer.

Table 3

Total Length	Not more than 12 m (39 ft 4 inch)
Total Width (A)	Not more than 2.5 m (8 ft 2 inch)

(continued)

(Table 3, contd)

Total Height (B)	Not more than 3.8 m (12 ft 6 inch) when loaded on the trailer.
Total Weight	20 to 25 ton (depending on road, axle, and vehicle length)

Operation of Construction Equipment and the Governing Laws and Regulations

NOTICE

Various laws and regulations, including Industrial Safety and Health Act, are enforced to ensure prevention of injuries on and around construction equipment and safe and comfortable operation of equipment. Be sure to obey them.

NOTICE

The notices regarding machine operation, inspection, maintenance, and safety contained in this manual are applicable only to cases in which the machine is used for the specified jobs. It is impossible for this kind of manual to cover every kind of operation. Therefore, the content of this manual does not necessarily explain all possible cases. Be sure to pay careful attention also to the items not covered by this manual and confirm the safety before starting jobs to prevent human injury and machine damage accidents.

Qualification of Operators

Operation of construction equipment is limited to persons who have any of the following licenses by law.

Note: Employers will face imprisonment up to a maximum of 6 months or a fine of up to a maximum of five hundred thousand yen if they let unqualified personnel operate equipment. Unqualified operators will also be fined up to a maximum of five hundred thousand yen.

- One who completed an operating skill course for vehicle-type construction equipment at a registered training institution.
- One who passed the construction equipment and technologies license examination (Type 1-3) defined by the Construction Industry Law.
- One who completed an operating training course for construction equipment defined by the Vocational Training Law.

- One who took a special training (rules and skills) at a registered training institution to operate equipment weighing less than 3 tons.
- With an auto-drivers license, an operator does not need to complete an operating skill course for construction equipment to operate equipment on the roads that apply to the rules of the Road Traffic Act. However, the operator needs to complete the course to engage in snow clearing or excavating on the roads.
- The operator must be qualified under the Mine Safety Act to operate construction equipment in a mine.

Acquisition of the Qualifications

The company offers training courses for construction machine operation, in addition to other skills. For details, contact the company's dealer in your area.

Regarding machine operation qualifications, also refer to the laws related to the construction machines shown at the end of this manual.

Subsidy System

Small-to-medium-sized construction business companies are eligible to receive a subsidy for a part of training fees and wages when they have their employees attend a training course to improve skills.

Operation of Construction Equipment and the Governing Laws and Regulations

NOTICE

Information about operating skill course for vehicle-type construction equipment (for ground leveling, transporting, loading, excavating).

Industrial Safety and Health Act requires operators of construction equipment weight 3 tons and over to acquire a certificate of completion of an operating skill course. Registered with and authorized by the respective directors general of the regional labor bureaus, we offer operating skill courses for vehicle-type construction equipment and special trainings.

Request for Periodical Self-Inspection

Rules of Periodical Self-Inspection

The employer shall, as provided for by the Ordinance of the Ministry of Health, Labor and Welfare, conduct self-inspection periodically. The employer shall keep the records of the results in respect to construction equipment such as tractor shovels and power shovels, etc., specified by Cabinet Order. (from Article 45, Industrial Safe and Health Act)

Ordinance on Industrial Safety and Hygiene

Periodical self-inspections Article 167

(1) The employer shall, as regards a vehicle type construction machine, carry out self-inspections for the following matters periodically once every period within a year. However, this shall not apply to the non-use period of a vehicle type construction machine, which is not used for a period exceeding 1 year.

(2) The employer shall, as regards a vehicle type construction machine set forth in the proviso of the proceeding paragraph, carry out self-inspection for abnormalities in each part of a construction machine before resuming the operation.

Periodical self-inspections Article 168

(1) The employer shall, as regards a vehicle type construction machine, carry out self-inspections for the following matters periodically once every period within a month. However, this shall not apply to the non-use period of a vehicle type construction machine, which is not used for a period exceeding one month:

- (i) Abnormalities in a brake, a clutch, a controlling device, and working devices.
- (ii) Damage in a wire, rope, and a chain
- (iii) Damage in a bucket, a dipper, etc.

(2) The employer shall, as regards to the vehicle type construction machine set forth in the proviso of the preceding paragraph, carry out self-inspection for the matters listed in each item of the same paragraph before resuming the operation.

Record of Periodical Self-Inspections Article 169

The employer shall, when having carried out the self-inspections set forth in the preceding two Articles, record the results and retain the records for 3 years.

Specified Self-Inspection Article 169-2

The specified self-inspection pertaining to the vehicle type construction machine shall be the self-inspection (prescribed by Article 167) and carried out by qualified personnel. The employer shall, when having carried out the specified self-inspection pertaining to a vehicle type construction machine, affix an inspection sticker stating the month and year when the said specified self-inspection was carried out at a readily visible location of the said machine.

- Caterpillar Japan has a supporting program for self-inspection as a registered inspection agency. Qualified personnel and inspection equipment are available to help customers who do not conduct internal inspections or do not have time to conduct the specified self-inspections. Contact a Cat dealer near you for details.
- Maintenance and inspection record book for a record-saving purpose can be purchased at Caterpillar Japan.
- Penalty: Employer who fails to carry out self-inspections and to record the results will face a fine of up to five hundred thousand yen.

Checkup before Commencing the Work Article 170

The employer shall, when carrying out the work using a vehicle type construction machine, check functions of a brake and a clutch before commencing the work for the day.

Other Rules

Besides qualification for operating equipment and self inspections, the following obligations are set forth in the Industrial Safety and Health Act:

- To conduct health and safety training for new recruits and shop foremen.
- To appoint the operation leader or supervisor, and establish health and safety management system.
- To inform employees of a chain of command at the worksite, communication and signal rules, traveling route of equipment, speed limits, signs of restricted areas, etc. for securing safety in the workplace.

The Industrial Safety and Health Act further also set obligations related to mechanical structures and rental activities of equipment.

Safety comes before anything else. Establish a workplace where no injuries occur by observing the governing laws and by referring to this manual, specifically the descriptions on safety.

Construction Equipment and Environmental Laws

Prohibition of Emissions and Obligations to Recover Fluorocarbons

Law Concerning the Recovery and Destruction of Fluorocarbons (Enforcement date: April 1, 2015)

Being emitted into the atmosphere, Fluorocarbons, used as refrigerants of air conditioning, destroy the ozone layer and accelerate the global warming as a cause of environmental destruction. Follow the instructions below required by law when handling air conditioners to protect the global environment.

1. 1. Do not arbitrarily emit the encapsulated refrigerant installed on the product into the atmosphere.
2. 2. Recover the encapsulated refrigerant when disposing of the product.

Note: Violators of the law will face a maximum one-year imprisonment or a fine up to a maximum of five hundred thousand yen.

When you need to fill, recover a refrigerant or dispose of a product with an encapsulated refrigerant installed, please ask a filling-recovery operator registered with the government of the local prefecture as "class-1 filling-recovery operator." And carry out the simple inspection of air conditioner and keep the record.

Class-1 Specified products sold after October 1, 2015 shall have the label inside of the cab showing the type and quantity of refrigerant, GWP (Global Warming Potential), and precautions for use. (Refer to the fluorocarbon label in the OMM safety section)

Standard Certificate of Transfer

Dear Customers

Japan Construction Equipment Manufacturers Association

Standard Certificate of Transfer

Issued by the Japan Construction Equipment Manufacturers Association

Standard Certificate of Transfer issued by the Japan Construction Equipment Manufacturers Association proves the ownership of your equipment. Request us to issue the certificate as a proof of transfer of ownership.

Commercial transactions of construction equipment are generally made on a long-term installment plan basis with a special provision of reservation of ownership that the seller retains the ownership of the sold equipment until the buyer completely pays off the installments.

Ownership of some construction equipment can be proved with a vehicle inspection certificate, but the certificate is not issued for most of the equipment. Therefore, the buyer will need to present a third party with a proof of ownership of the sold equipment.

Japan Construction Equipment Manufacturers Association launched a system of standard certificate of transfer in 1971 to normalize trading in construction equipment and establishes a business practice relating to transfer of ownership. Customers are kindly requested to understand the intent of the system and request your seller to issue a certificate of transfer.

1. About the standard certificate of transfer

- a. Japan Construction Equipment Manufacturers Association (hereinafter referred to as CEMA) sets the rules and form of standard certificate of transfer (hereinafter referred to as certificate of transfer), and members of the CEMA issue the certificate of transfer. A certificate of transfer proves the ownership of equipment.

2. Purpose of issuance

- a. A certificate of transfer will be issued for the purpose of clarifying the ownership of equipment and preventing misconduct such as trades of stolen equipment or fraud.

3. Issuer

- a. A certificate of transfer will be issued by a distributor (Primary transferer) who sells new construction equipment and is authorized by the CEMA.

4. Eligibility

- a. A certificate of transfer will be issued for the equipment, which is sold by CEMA-member distributors and defined as construction equipment by the CEMA

5. Issuance

- a. A certificate of transfer will be issued and directly given to a buyer upon the buyer's request when he/she buys eligible equipment from an issuer.
- b. A certificate of transfer may not be issued for the equipment, which was sold as new merchandise more than 10 years ago.
- c. A certificate of transfer is not permitted to substitute a vehicle inspection certificate.

6. Prohibition of reissuance

- a. Certificate of transfer should be safely stored as it will not be reissued under any circumstances.

7. In case a certificate description runs out of space
- Discretionary page/s to the certificate will be valid with a tally seal of the issuer at the joint of two pages.

Contact CEMA-member companies or distributors for more details of the system.

Industrial Safety and Health Act

Article 164 (Extracted) of Industrial Safety and Health Act (Restriction on use Other Than Main Application)

Article 164

Business Operator must not use construction machineries of vehicle type for applications other than main application of the applicable construction machineries of vehicle type such as: lifting cargos by hydraulic excavator or lifting/lowering workers using the clamshell.

[2] The previous clause will not be applied for any of the following cases:

- In performing cargo lifting, any one of the following may be applicable.
 - Cannot be avoided due to the nature of the work or necessary in view of performing work in safe.
 - When working with attachments installed for metals of hook or shackle etc or other devices for lifting application applicable to any one of the following as implements for boom or bucket etc
 - Enough strength is retained bearable for loads to be applied.
 - Load lifted up is not feared to be dropped from the applicable instrument used, due to provided locking device is in use or etc.
 - Load not feared of disengaging from the implement.
- In performing work other than cargo lifting, nothing is feared to do harm to the workers.

[3] The business operator must take the following measures, in performing cargo lifting work applicable to Items 1a and 1b of Step 1 above. To prevent any danger of workers from contact with lifted cargo, drop of lifted cargo or turnover or falling down of construction machineries of vehicle type.

- Designate one person who issues a sign as well as setting up fixed signs related to cargo lifting work, and follow his signs.
- Perform work on a flat ground.
- Keep any worker away from any place where is feared to cause any danger to worker due to contact with a cargo or drop of lifted cargo.

[4] Do not perform any work applying load exceeding the allowed rated max load specified according to structure or materials of the applicable construction machineries of the vehicle type.

[5] In using wire rope in slinging device, use wire rope applicable to every item of the followings.

- Safety coefficient is 6 or more. (The safety coefficient here must be the same as specified in Article 213 item 2 in Safety Rules on Crane Works (Article 34 in Ordinance of Ministry of Labor, 1972) etc. Hereinafter called as "Crane Rules")
- Among wire rope 1 strands, numbers of cut strands (other than filler) are less than 10%.
- Reduction of diameter is 7% or less than nominal diameter.
- Free from kinking.
- Free from badly collapse and corrosion.

[6] In using lifting chain as slinging device, the chain is applicable to every item of the followings.

- Safety coefficient is 5 or more.
- Elongation is 5% or less than the length when the applicable lifting chain was fabricated.
- Reduction of diameter of the cross section of link is 10% or less than diameter of cross section of the applicable link when the applicable lifting chain was manufactured.
- Free from cracks.

[7] In using those other than wire rope and lifting chain as slinging device, they must be free from bad damage and corrosion.

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Specifications

SMCS Code: 7000

Intended Use

The intended use of this machine is for excavating with a bucket or working with approved work tools. The machine should be operated with the undercarriage in a stationary position since the upper structure is normally capable of 360 degree swing with mounted equipment. This machine can be used in object handling applications that are within the lift capacity of the machine. When this machine is used in object handling applications, ensure that the machine is properly configured and operated properly. Obey any local governmental regulations and regional governmental regulations. Only lift objects from approved lifting points and with approved lifting devices.

Expected Life

The expected life, defined as total machine hours, of this machine is dependent upon many factors including the machine owner's desire to rebuild the machine back to factory specifications. The expected life interval of this machine is 8,000 service hours. The expected life interval corresponds to the service hours to engine overhaul. Service hours to engine overhaul may vary based on overall machine duty cycle. At the expected life interval, remove the machine from operation and consult your Cat dealer for inspect, repair, rebuild, install remanufactured, install new components, or disposal options and to establish a new expected life interval. If a decision is made to remove this machine from service, refer to Operation and Maintenance Manual, "Decommissioning and Disposal".

The following items are required to obtain an economical expected life of this machine:

- Perform regular preventive maintenance procedures as described in the Operation and Maintenance Manual.
- Perform machine inspections as described in the Operation and Maintenance Manual and correct any problems discovered.
- Perform system testing as described in the Operation and Maintenance Manual and correct any problems discovered.
- Ensure that machine application conditions comply with Caterpillar's recommendations.

- Ensure that the operating weight does not exceed limits set by manufacturer.
- Ensure that all frame cracks are identified, inspected, and repaired to prevent further development.

Application/Configuration Restrictions

The maximum travel operating slope for machine lubrication is 35 degrees.

The operator station is ROPS certified up to a mass of 27800 kg (61300 lb) per ISO 12117-2:2008.

Specification Data

Specifications

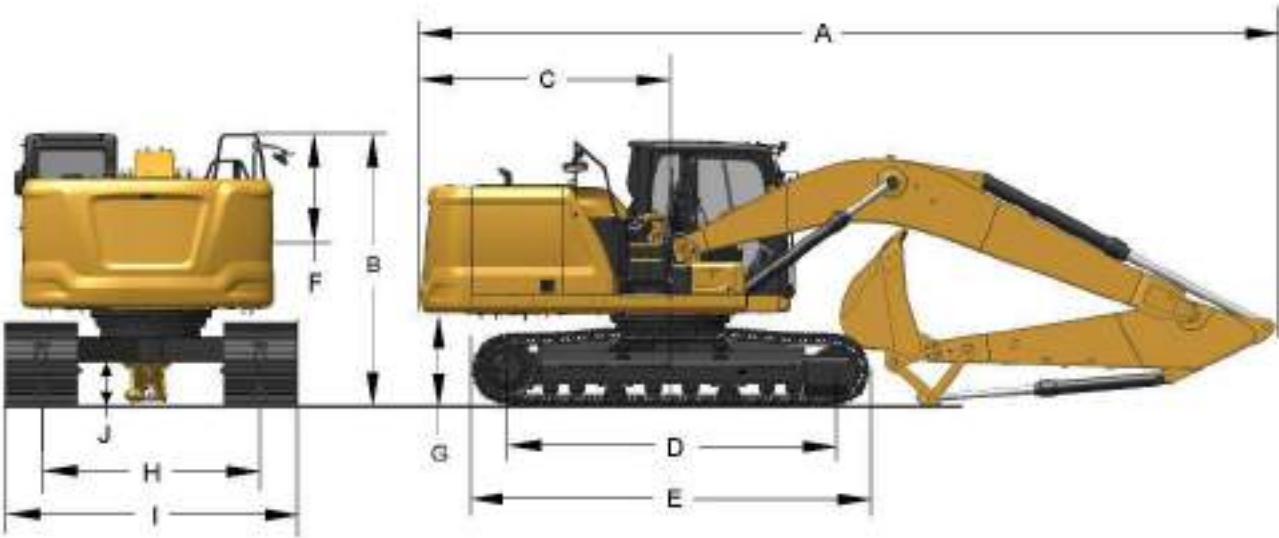


Illustration 68

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Table 4

Model	320 ⁽¹⁾ 323 ⁽²⁾						
Boom	Reach Boom				Variable Angle (VA) Boom		Super Long Reach Boom
	5.7 m (18 ft 8 inch)				2.8 m (9 ft 2 inch)		8.8 m (29 ft)
Stick	2.9 m (9 ft 6 inch)	2.5 m (8 ft 2 inch)	3.9 m (12 ft 10 inch)	3.7 m (12 ft 1 inch)	2.9 m (9 ft 6 inch)	2.5 m (8 ft 2 inch)	6.3 m (20 ft 7 inch)
320 Approximate Weight ⁽³⁾	18008 kg (39700 lb)	17962 kg (39600 lb)	18688 kg (41200 lb)	18688 kg (41200 lb)	18461 kg (40700 lb)	18416 kg (40600 lb)	19278 kg (42500 lb)
323 Approximate Weight ⁽³⁾	19550 kg (43100 lb)	19731 kg (43500 lb)	19731 kg (43500 lb)	19731 kg (43500 lb)	20185 kg (44500 lb)	20003 kg (44100 lb)	21092 kg (46500 lb)
Overall Length (A)	3162 mm (10 ft 4 inch)		3467 mm (11 ft 4 inch)	3467 mm (11 ft 4 inch)	3002 mm (9 ft 8 inch)	2907 mm (9 ft 5 inch)	3185 mm (10 ft 4 inch)
Overall Height (B)	2960 mm (9 ft 7 inch)						
Swing Radius (C)	2833 mm (9 ft 3 inch)						
Length to Center of Rollers (D)	3650 mm (12 ft)						
Length of Track (E)	4450 mm (14 ft 6 inch)						
Handrail Height (F)	2954 mm (9 ft 7 inch)						

(continued)

Product Information Section
Specifications

(Table 4, contd)

Counterweight Clearance (G)	1048 mm (3 ft 4 inch)
Track Gauge (H)	2380 mm (7 ft 8 inch)
Overall Width (I) with 600 mm (24 inch) Shoe	2980 mm (9 ft 9 inch)
Overall Width (I) with 700 mm (28 inch) Shoe	3080 mm (10 ft 1 inch)
Overall Width (I) with 790 mm (31 inch) Shoe	3170 mm (10 ft 5 inch)
Overall Width (I) with 900 mm (36 inch) Shoe	3280 mm (10 ft 10 inch)
Ground Clearance (J)	450 mm (1 ft 6 inch)

- (1) Approximate weight for machines equipped with 790 mm (31 inch) triple grouser track shoes, a 4700 kg (10400 lb) counterweight, and without a quick coupler.
(2) Approximate weight for machines equipped with 790 mm (31 inch) triple grouser track shoes, a 5400 kg (11900 lb) counterweight, and without a quick coupler.
(3) 10 percent full fuel tank.

Working Ranges and Forces

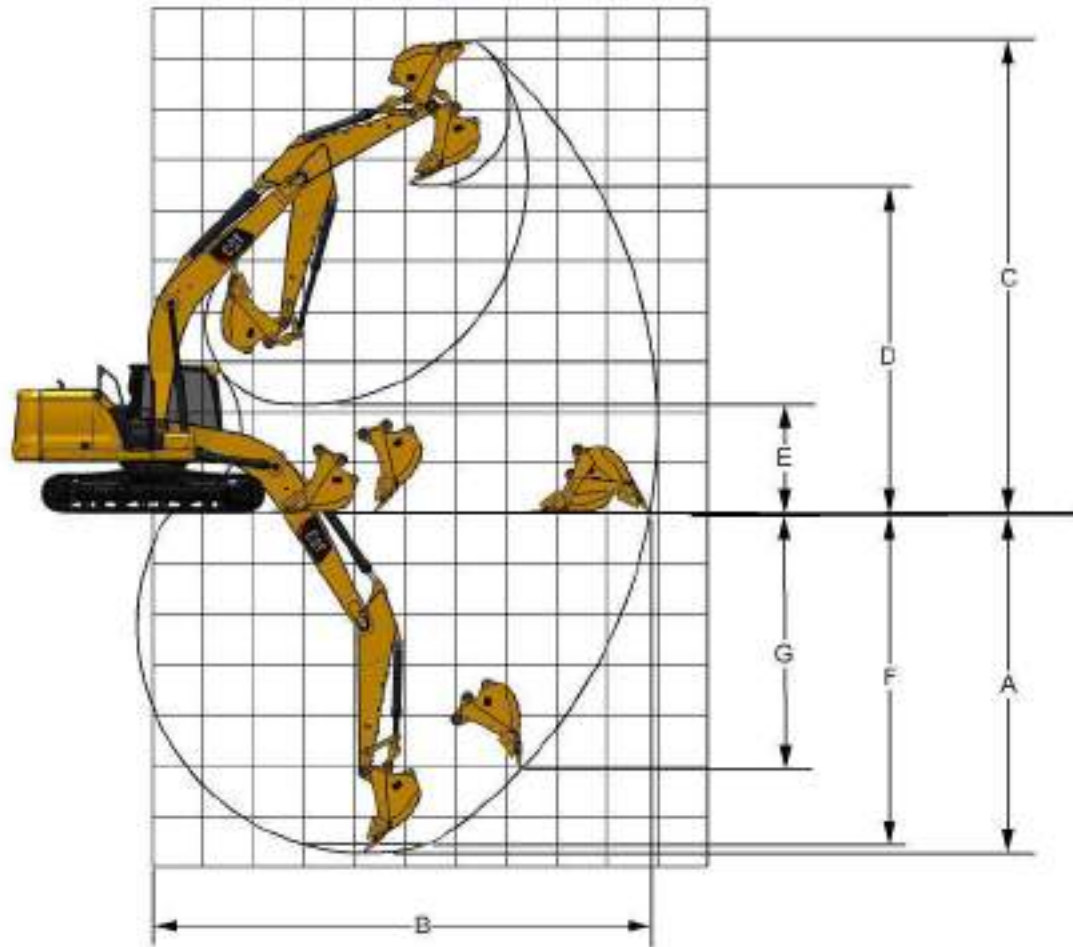


Illustration 69
Reach Boom

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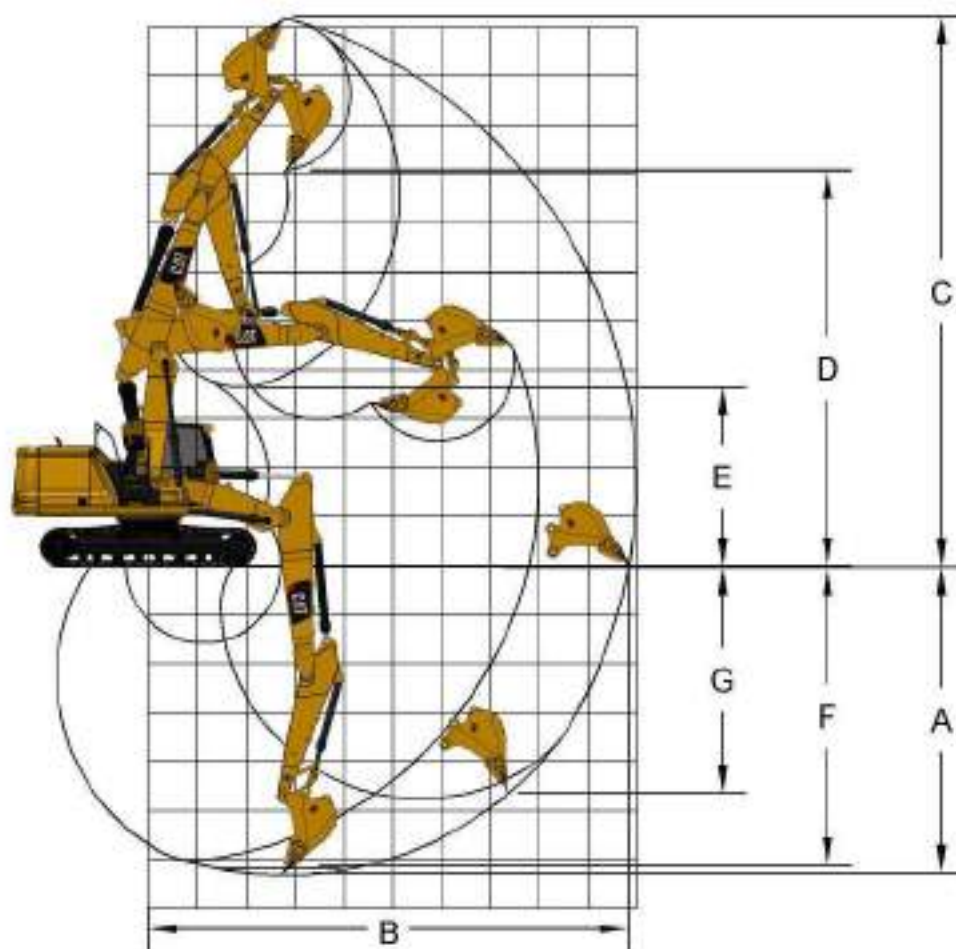


Illustration 70
VA Boom

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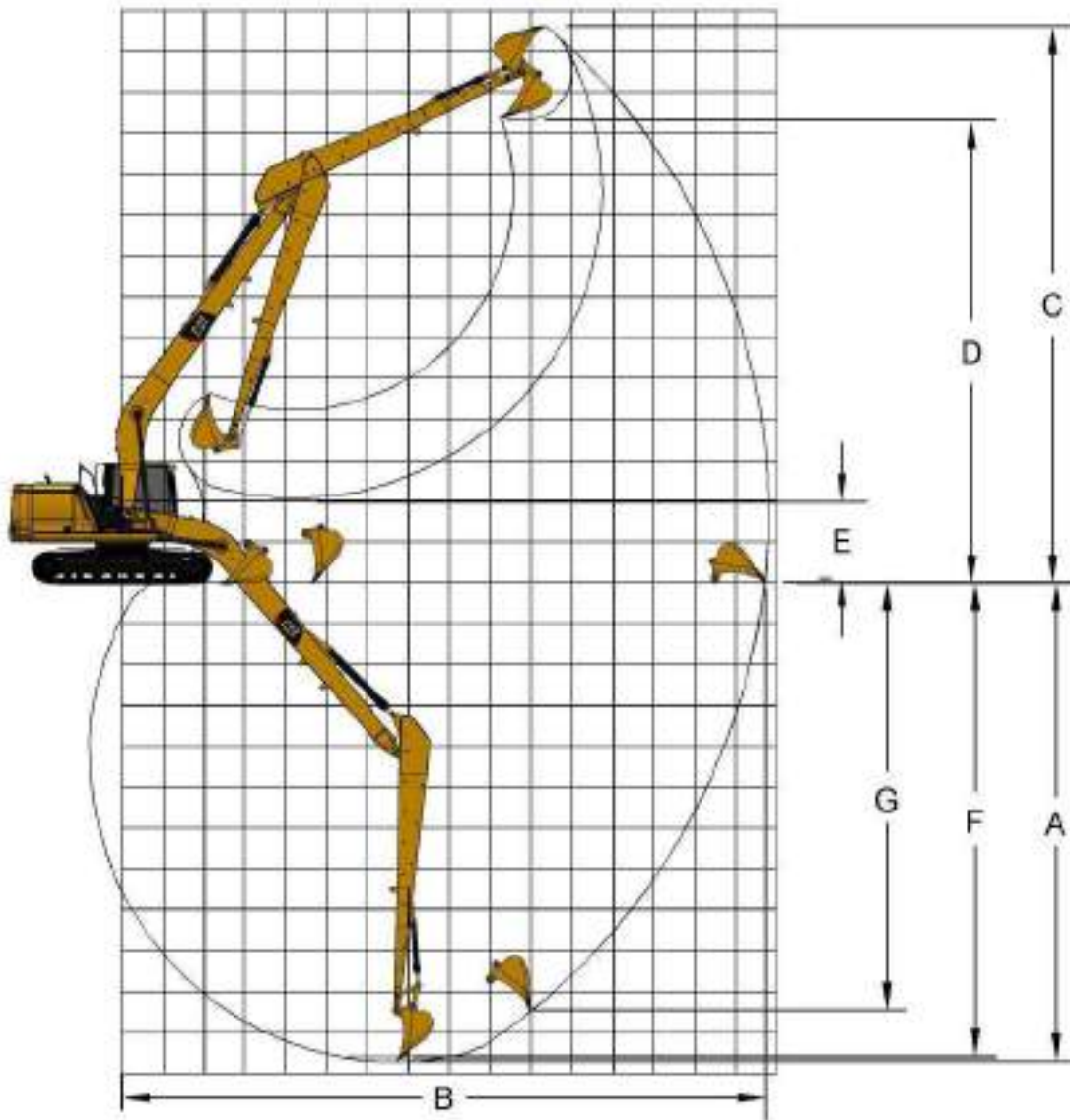


Illustration 71

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Super Long Reach (SLR)

320

Note: All dimensions are approximate and may vary depending on bucket selection.

Product Information Section
Specifications

Table 5

320								
Boom Options		Reach Boom			VA Boom		Super Long Reach (SLR) Boom	
		5.7m (19 ft)			2.8m (9 ft) Stub 3.3 m (10 ft) Fore		8.85 m (29 ft)	
Stick Options		Reach Stick			Reach Stick		SLR Stick	
		R2.9B1 (9 ft 6 in)	R2.9B1 (9 ft 6 in)	R2.5B1 (8 ft 2 in)	R2.9B1 (9 ft 6 in)	R2.5B1 (8 ft 2 in)	6.28A (20 ft 7 in)	6.28A (20 ft 7 in)
Maximum digging depth (A)	mm	6720	6620	6300	6700	6280	11540	11690
Maximum digging depth (A)	ft	22.0	21.7	20.7	22.0	20.6	37.9	38.4
Maximum Reach at Ground Line (B)	mm	9860	9760	9470	10210	9810	15570	15730
Maximum Reach at Ground Line (B)	ft	32.3	32.0	31.1	33.5	32.2	51.1	51.6
Maximum cutting height (C)	mm	9370	9430	9170	11520	11180	13540	13610
Maximum cutting height (C)	ft	30.7	30.9	30.1	37.8	36.7	44.4	44.7
Maximum Loading Height (D)	mm	6490	6590	6290	8410	8070	11440	11290
Maximum Loading Height (D)	ft	21.3	21.6	20.6	27.6	26.5	37.5	37.0
Minimum Loading height (E)	mm	2170	2270	2590	3260	3660	2240	2080
Minimum Loading height (E)	ft	7.1	7.4	8.5	10.7	12.0	7.3	6.8
Maximum Depth Cut for 2440 mm (8ft) Level Bottom (F)	mm	6550	6450	6110	6600	6180	11440	11590
Maximum Depth Cut for 2440 mm (8ft) Level Bottom (F)	ft	21.5	21.2	20.0	21.7	20.3	37.5	38.0
Maximum Vertical Wall Digging Depth (G)	mm	5190	5960	4800	5060	4680	11020	10560
Maximum Vertical Wall Digging Depth (G)	ft	17.0	19.6	15.7	16.6	15.4	36.2	34.6
Bucket Digging Force (SAE)	kN	134	133	134	134	134	62	54
Bucket Digging Force (SAE)	lbf	295	294	295	295	295	136	119
Bucket Digging Force (ISO)	kN	150	148	150	150	150	62	60
Bucket Digging Force (ISO)	lbf	332	326	332	332	332	136	133
Stick Digging Force (SAE)	kN	103	105	114	103	114	49	48
Stick Digging Force (SAE)	lbf	228	232	251	228	251	108	106
Stick Digging Force (ISO)	kN	106	108	118	106	118	49	49
Stick Digging Force (ISO)	lbf	235	238	260	235	260	108	107
Bucket type								
Bucket Capacity	m ³	1.19	0.80	1.19	1.19	1.19	0.57	0.53
Bucket Capacity	yd ³	1.56	1.05	1.56	1.56	1.56	0.75	0.69
Bucket Tip Radius	mm	1570	1470	1570	1570	1570	1070	1230
Bucket Tip Radius	ft	5.2	4.8	5.2	5.2	5.2	3.5	4.0

323

Note: All dimensions are approximate and may vary depending on bucket selection.

Table 6

323								
Boom Options		Reach Boom			VA Boom		Super Long Reach (SLR) Boom	
		5.7 m (19 ft)			2.8m (9 ft) Stub 3.3 m (10 ft) Fore		8.85 m (29 ft)	
		Reach Stick			Reach Stick		SLR Stick	
Maximum digging depth (A)	mm	6730	6310	7670	6700	6290	11540	11690
Maximum digging depth (A)	ft	22.1	20.7	25.2	22.0	20.6	37.9	38.4
Maximum Reach at Ground Line (B)	mm	9870	9470	10770	10220	9820	15570	15730
Maximum Reach at Ground Line (B)	ft	32.4	31.1	35.3	33.5	32.2	51.1	51.6
Maximum cutting height (C)	mm	9450	9250	9910	11540	11200	13540	13610
Maximum cutting height (C)	ft	31.0	30.3	32.5	37.9	36.7	44.4	44.7
Maximum Loading Height (D)	mm	6480	6280	6930	8380	8040	11440	11290
Maximum Loading Height (D)	ft	21.3	20.6	22.7	27.5	26.4	37.5	37.0
Minimum Loading height (E)	mm	2160	2580	1220	3250	3650	2240	2080
Minimum Loading height (E)	ft	7.1	8.5	4.0	10.7	12.0	7.3	6.8
Maximum Depth Cut for 2440 mm (8ft) Level Bottom (F)	mm	6560	6120	7530	6610	6190	11440	11590
Maximum Depth Cut for 2440 mm (8ft) Level Bottom (F)	ft	21.5	20.1	24.7	21.7	20.3	37.5	38.0
Maximum Vertical Wall Digging Depth (G)	mm	5740	5340	6640	5380	4980	11020	10560
Maximum Vertical Wall Digging Depth (G)	ft	18.8	17.5	21.8	17.7	16.3	36.2	34.6
Bucket Digging Force (SAE)	kN	124	124	124	124	124	62	54
Bucket Digging Force (SAE)	lbf	274	274	274	274	274	136	119
Bucket Digging Force (ISO)	kN	140	140	140	140	140	62	60

(continued)

Product Information Section
Specifications

(Table 6, contd)

323								
Boom Options		Reach Boom			VA Boom		Super Long Reach (SLR) Boom	
		5.7 m (19 ft)			2.8m (9 ft) Stub 3.3 m (10 ft) Fore		8.85 m (29 ft)	
		Reach Stick			Reach Stick		SLR Stick	
Bucket Digging Force (ISO)	lbf	309	309	309	309	309	136	133
Stick Digging Force (SAE)	kN	104	114	85	104	114	49	48
Stick Digging Force (SAE)	lbf	228	252	188	228	252	108	106
Stick Digging Force (ISO)	kN	107	118	87	107	118	49	49
Stick Digging Force (ISO)	lbf	235	260	193	235	260	108	107
Bucket type								
Bucket Capacity	m ³	1.30	1.30	1.30	1.30	1.30	0.57	0.53
Bucket Capacity	yd ³	1.70	1.70	1.70	1.70	1.70	0.75	0.69
Bucket Tip Radius	mm	1580	1580	1580	1580	1580	1070	1230
Bucket Tip Radius	ft	5.2	5.2	5.2	5.2	5.2	3.5	4.0

i08271900

Specifications

(Shovel Crane Specifications)

SMCS Code: 7000

S/N: HEX1–Up



WARNING

Failure to comply to the rated load can cause possible personal injury or property damage. This includes the risk of unintended boom lowering. Review the rated load of a particular work tool before performing any operation. Make adjustments to the rated load as necessary for non-standard configurations.

Japan regulations require a shovel crane configuration to lift certain objects. A shovel crane configuration uses a rated load capacity. Refer to the Rated Load specifications below.

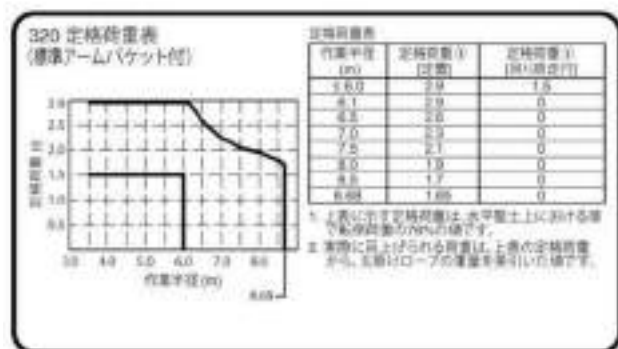
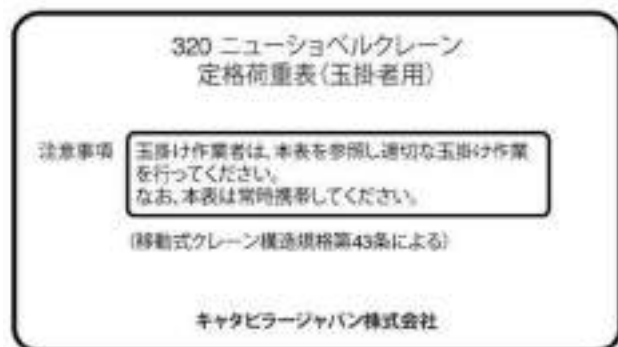
Refer to this Operation and Maintenance Manual, "Shovel Crane Control" for additional information.

Refer to this Operation and Maintenance Manual, "Shovel Crane Operation" for additional information.

Consult your Cat® dealer for additional information.

Rated Load Specification

320



A stick is designed to match only one specific family of buckets.

Note: The selection of a compatible boom-stick-bucket combination is a guide. Work tools, uneven ground conditions, soft ground conditions, or poor ground conditions have effects on machine performance. The operator is responsible for being aware of these effects.

Consult your Cat dealer for information on selecting the correct boom-stick-bucket combination.

The following tables show various compatible boom-stick-bucket combinations. Select an optimum combination according to the working conditions and according to the type of work that is being done.

Illustration 72

g06310968

i07174905

Boom/Stick/Bucket Combinations (323)

SMCS Code: 6000; 6700

S/N: NDL1-Up

S/N: LTN1-Up

S/N: HDT1-Up

S/N: RAZ1-Up

This machine can be equipped with a large variety of boom-stick-bucket combinations to meet the needs of various applications.

Buckets are grouped into different families by capacity. As a rule, use a bucket with a smaller capacity when you are using a longer stick and/or a longer boom. Conversely, use a bucket with a larger capacity when you are using a shorter stick and/or a shorter boom. This rule ensures better machine stability and protection against structural machine damage.

Table 7

Excavator without a Quick Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Gener- al Duty - Ca- pacity (GDC)	B	600 mm (24 inch)	0.55 m³ (0.72 yd³)	619 kg (1365 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		750 mm (30 inch)	0.75 m³ (0.98 yd³)	710 kg (1565 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)		
		900 mm (36 inch)	0.95 m³ (1.24 yd³)	787 kg (1735 lb)		(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)	(1)		
		1050 mm (42 inch)	1.16 m³ (1.52 yd³)	848 kg (1870 lb)		(1)	(1)	(3)	(1)	(2)	(1)	(1)	(1)	(1)		
		1200 mm (48 inch)	1.38 m³ (1.80 yd³)	926 kg (2041 lb)		(2)	(3)	(4)	(3)	(3)	(1)	(1)	(3)	(2)	(1)	(2)
		1350 mm (54 inch)	1.59 m³ (2.08 yd³)	1004 kg (2213 lb)		(3)	(5)	(5)	(4)	(4)	(2)	(5)	(5)	(3)	(2)	(3)
Gener- al Duty - Ca- pacity (GDC) - Wide Tip	B	600 mm (24 inch)	0.55 m³ (0.72 yd³)	633 kg (1395 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		750 mm (30 inch)	0.75 m³ (0.98 yd³)	731 kg (1612 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		900 mm (36 inch)	0.95 m³ (1.24 yd³)	813 kg (1793 lb)		(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1050 mm (42 inch)	1.16 m³ (1.52 yd³)	895 kg (1973 lb)		(1)	(2)	(4)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)

(continued)

(Table 7, contd)

Excavator without a Quick Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1200 mm (48 inch)	1.38 m³ (1.80 yd³)	979 kg (2158 lb)		(2)	(3)	(6)	(3)	(3)	(1)	(1)	(3)	(2)	(1)	(2)
		1350 mm (54 inch)	1.59 m³ (2.08 yd³)	1063 kg (2343 lb)		(3)	(5)	(5)	(4)	(4)	(2)	(5)	(5)	(3)	(2)	(3)
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	826 kg (1821 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	880 kg (1940 lb)		(1)	(1)	(3)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	972 kg (2143 lb)		(1)	(2)	(5)	(2)	(2)	(1)	(1)	(5)	(1)	(1)	(1)
		1350 mm (53 inch)	1.38 m³ (1.81 yd³)	1054 kg (2324 lb)		(5)	(5)	(5)	(3)	(5)	(5)	(5)	(5)	(5)	(1)	(5)
Heavy Duty - Power (HDP)	B	900 mm (36 inch)	0.79 m³ (1.03 yd³)	842 kg (1856 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1050 mm (41 inch)	0.96 m³ (1.26 yd³)	907 kg (2000 lb)		(1)	(1)	(3)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(continued)

(Table 7, contd)

Excavator without a Quick Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1200 mm (48 inch)	1.14 m³ (1.49 yd³)	993 kg (2189 lb)		(5)	(5)	(5)	(1)	(5)	(5)	(5)	(5)	(1)	(5)	
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)		
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	889 kg (1960 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)		
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	964 kg (2125 lb)		(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)	(1)		
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	1053 kg (2321 lb)		(1)	(1)	(5)	(1)	(2)	(1)	(1)	(5)	(1)	(1)	(1)
Severe Duty - Power (SDP)	B	900 mm (36 inch)	0.79 m³ (1.03 yd³)	908 kg (2002 lb)	90	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
Mud / Clean- up (M/ CU)	B	1700 mm (67 inch)	1.60 m³ (2.09 yd³)	979 kg (2158 lb)	100	(3)	(4)	(1)	(3)	(4)	(2)	(2)	(1)	(3)	(2)	(3)
Ditch Clean- ing (DC)	B	1500 mm (59 inch)	1.01 m³ (1.32 yd³)	652 kg (1437 lb)	100	(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1800 mm (71 inch)	1.24 m³ (1.62 yd³)	740 kg (1631 lb)		(1)	(2)	(3)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)

(continued)

(Table 7, contd)

Excavator without a Quick Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Ditch Clean- ing - Tilt (DCT)	B	1500 mm (59 inch)	0.90 m³ (1.18 yd³)	948 kg (2090 lb)	100	(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1800 mm (71 inch)	1.11 m³ (1.45 yd³)	1063 kg (2343 lb)		(1)	(2)	(4)	(2)	(2)	(1)	(1)	(2)	(1)	(1)	
		1800 mm (71 inch)	1.40 m³ (1.83 yd³)	1148 kg (2531 lb)		(3)	(3)	(1)	(3)	(4)	(1)	(2)	(4)	(2)	(2)	(2)
		2000 mm (79 inch)	1.23 m³ (1.61 yd³)	1132 kg (2496 lb)		(2)	(3)	(1)	(2)	(3)	(1)	(1)	(3)	(2)	(1)	(1)
Gener- al Duty (GD)	312	900 mm (35 inch)	0.53 m³ (0.69 yd³)	403 kg (888 lb)	100	-	-	-	-	-	-	-	-	-	-	
Ditch Clean- ing (DC) Long Reach	312	1200 mm (48 inch)	0.57 m³ (0.74 yd³)	386 kg (851 lb)	100	-	-	-	-	-	-	-	-	-	-	
Maximum load pin-on (payload + bucket)						3330 kg (7341 lb)	3100 kg (6834 lb)	2445 kg (5390 lb)	3165 kg (6978 lb)	2935 kg (6471 lb)	3980 kg (8774 lb)	3710 kg (8179 lb)	2985 kg (6581 lb)	2840 kg (6261 lb)	3810 kg (8400 lb)	3545 kg (7815 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

(5) Not Recommended

(6) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

Table 8

Excavator with Pin Grabber Coupler (North America)													
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323							
						4200 kg (9260 lb)					5400 kg (11905 lb)		
						Reach Boom			HD Reach Boom		Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick
General Duty - Capacity (GDC)	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	619 kg (1365 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	710 kg (1565 lb)		(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	787 kg (1735 lb)		(1)	(1)	(3)	(1)	(2)	(1)	(1)	(1)
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	848 kg (1869 lb)		(2)	(4)	(5)	(2)	(4)	(1)	(1)	(4)
		1200 mm (48 inch)	1.38 m ³ (1.80 yd ³)	926 kg (2041 lb)		(4)	(3)	(5)	(3)	(3)	(1)	(2)	(3)
		1350 mm (53 inch)	1.59 m ³ (2.08 yd ³)	1004 kg (2213 lb)		(3)	(5)	(6)	(3)	(5)	(4)	(4)	(5)
General Duty - Capacity (GDC) - Wide Tip	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	633 kg (1395 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	731 kg (1611 lb)		(1)	(1)	(2)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	813 kg (1792 lb)		(1)	(1)	(3)	(1)	(2)	(1)	(1)	(1)
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	895 kg (1973 lb)		(2)	(4)	(5)	(4)	(4)	(1)	(1)	(4)

(continued)

(Table 8, contd)

Excavator with Pin Grabber Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1200 mm (48 inch)	1.38 m³ (1.80 yd³)	979 kg (2158 lb)		(4)	(3)	(6)	(3)	(3)	(2)	(2)	(3)	(4)	(2)	(4)
		1350 mm (53 inch)	1.59 m³ (2.08 yd³)	1063 kg (2343 lb)		(3)	(5)	(6)	(5)	(5)	(4)	(4)	(5)	(3)	(4)	(3)
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	826 kg (1821 lb)		(1)	(1)	(4)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1050 mm (41 inch)	1.00 m³ (1.31 yd³)	880 kg (1940 lb)		(1)	(2)	(3)	(2)	(2)	(1)	(1)	(2)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	972 kg (2143 lb)		(2)	(4)	(5)	(4)	(3)	(1)	(1)	(3)	(2)	(1)	(2)
		1350 mm (53 inch)	1.38 m³ (1.81 yd³)	1054 kg (2324 lb)		(3)	(3)	(6)	(3)	(5)	(2)	(4)	(3)	(4)	(2)	(4)
Heavy Duty - Power (HDP)	B	900 mm (36 inch)	0.79 m³ (1.03 yd³)	842 kg (1856 lb)	100	(1)	(1)	(4)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1050 mm (41 inch)	0.96 m³ (1.26 yd³)	907 kg (2000 lb)		(1)	(2)	(3)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)

(continued)

(Table 8, contd)

Excavator with Pin Grabber Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1200 mm (48 inch)	1.14 m³ (1.49 yd³)	993 kg (2189 lb)		(2)	(4)	(5)	(4)	(4)	(1)	(1)	(4)	(2)	(1)	(2)
Heavy Duty - Pin Grab- ber Per- for- mance (HD - PGP)	B	600 mm (24 inch)	0.44 m³ (0.57 yd³)	676 kg (1490 lb)	90	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.60 m³ (0.79 yd³)	778 kg (1715 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.76 m³ (1.00 yd³)	864 kg (1905 lb)		(1)	(1)	(4)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1050 mm (42 inch)	0.93 m³ (1.22 yd³)	928 kg (2046 lb)		(1)	(2)	(3)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)
		1200 mm (48 inch)	1.11 m³ (1.45 yd³)	1016 kg (2234 lb)		(2)	(4)	(5)	(4)	(4)	(1)	(1)	(4)	(2)	(1)	(1)
		1350 mm (53 inch)	1.28 m³ (1.67 yd³)	1104 kg (2434 lb)		(4)	(3)	(6)	(3)	(3)	(1)	(2)	(3)	(4)	(2)	(4)
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	889 kg (1960 lb)		(1)	(1)	(4)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(continued)

(Table 8, contd)

Excavator with Pin Grabber Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	964 kg (2125 lb)		(1)	(1)	(3)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	1053 kg (2321 lb)		(2)	(4)	(5)	(4)	(4)	(1)	(1)	(4)	(2)	(1)	(1)
Severe Duty - Power (SDP)	B	900 mm (36 inch)	0.79 m³ (1.03 yd³)	908 kg (2002 lb)	90	(1)	(1)	(4)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
Mud / Clean- up (M/ CU)	B	1700 mm (67 inch)	1.60 m³ (2.09 yd³)	979 kg (2158 lb)	100	(3)	(5)	(6)	(3)	(5)	(4)	(4)	(5)	(3)	(4)	(3)
Ditch Clean- ing (DC)	B	1500 mm (59 inch)	1.01 m³ (1.32 yd³)	652 kg (1437 lb)	100	(1)	(1)	(4)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)
		1800 mm (71 inch)	1.24 m³ (1.62 yd³)	740 kg (1631 lb)		(2)	(4)	(5)	(4)	(4)	(1)	(1)	(4)	(2)	(1)	(1)

(continued)

(Table 8, contd)

Excavator with Pin Grabber Coupler (North America)																
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323										
						4200 kg (9260 lb)					5400 kg (11905 lb)					
						Reach Boom			HD Reach Boom		Reach Boom			HD Reach Boom		
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	3.9 m (12 ft 8 inch) Stick	2.9 m (9 ft 6 inch) TRS Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Ditch Clean- ing - Tilt (DCT)	B	1500 mm (59 inch)	0.90 m³ (1.18 yd³)	948 kg (2090 lb)	100	(1)	(1)	(3)	(1)	(2)	(1)	(1)	(2)	(1)	(1)	(1)
		1800 mm (71 inch)	1.11 m³ (1.45 yd³)	1063 kg (2343 lb)		(2)	(4)	(5)	(4)	(3)	(1)	(1)	(4)	(2)	(1)	(2)
		1800 mm (71 inch)	1.40 m³ (1.83 yd³)	1148 kg (2531 lb)		(3)	(3)	(6)	(3)	(5)	(2)	(4)	(5)	(3)	(4)	(4)
		2000 mm (79 inch)	1.23 m³ (1.61 yd³)	1132 kg (2496 lb)		(4)	(3)	(6)	(3)	(3)	(1)	(2)	(3)	(4)	(2)	(4)
Maximum load pin-on (payload + bucket)						6920 kg (1525- 5 lb)	6413 kg (1413- 8 lb)	4969 kg (1095- 4 lb)	6556 kg (1445- 3 lb)	6049 kg (1333- 6 lb)	8353 kg (18415 lb)	7758 kg (17102 lb)	6159 kg (13579 lb)	5839 kg (12874 lb)	7978 kg (17588 lb)	7394 kg (16300 lb)

- (1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.
 (2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.
 (3) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.
 (4) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.
 (5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.
 (6) Not Recommended

Table 9

Excavator without a Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Gener- al Duty (GD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	551 kg (1215 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(continued)

(Table 9, contd)

Excavator without a Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	622 kg (1371 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	668 kg (1473 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	803 kg (1770 lb)		(1)	(1)	(1)	(2)	(2)	(3)	(1)	(1)	(1)	(1)
		1300 mm (51 inch)	1.30 m³ (1.71 yd³)	835 kg (1841 lb)		(1)	(2)	(2)	(3)	(3)	(3)	(1)	(1)	(1)	(1)
		1400 mm (55 inch)	1.43 m³ (1.87 yd³)	870 kg (1918 lb)		(4)	(4)	(3)	(4)	(4)	(4)	(4)	(4)	(1)	(4)
Heavy Duty (HD)	B	1050 mm (42 inch)	1.00 m³ (1.31 yd³)	Yet to Verify	100	(1)	(1)	(1)	(1)	(1)	(2)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	907 kg (1999 lb)		(1)	(2)	(2)	(2)	(3)	(3)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	918 kg (2024 lb)		(1)	(2)	(2)	(2)	(3)	(3)	(1)	(1)	(1)	(1)
		1300 mm (52 inch)	1.30 m³ (1.71 yd³)	962 kg (2120 lb)		(2)	(2)	(2)	(3)	(3)	(5)	(1)	(1)	(1)	(1)
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(continued)

(Table 9, contd)

Excavator without a Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	889 kg (1960 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	964 kg (2125 lb)		(1)	(1)	(1)	(1)	(2)	(1)	(1)	(1)	(1)	
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	1053 kg (2321 lb)		(1)	(1)	(1)	(2)	(2)	(3)	(1)	(1)	(1)	(1)
Severe Duty (SDV)	B	1200 mm (48 inch)	1.20 m³ (1.57 yd³)	1001 kg (2207 lb)	90	(1)	(1)	(1)	(2)	(2)	(3)	(1)	(1)	(1)	(1)
Ditch Clean- ing (DC)	B	2000 mm (78 inch)	1.22 m³ (1.60 yd³)	869 kg (1916 lb)	100	(1)	(2)	(2)	(2)	(3)	(3)	(1)	(1)	(1)	(1)
Ditch Clean- ing - Tilt (DCT)	B	2000 mm (78 inch)	1.23 m³ (1.61 yd³)	1096 kg (2416 lb)	100	(2)	(2)	(2)	(3)	(3)	(5)	(1)	(1)	(1)	(1)
General Duty (GD)	312	900 mm (35 inch)	0.53 m³ (0.69 yd³)	403 kg (888 lb)	100	-	-	-	-	-	-	-	-	-	-
Ditch Clean- ing (DC) Long Reach	312	1200 mm (48 inch)	0.57 m³ (0.74 yd³)	386 kg (851 lb)	100	-	-	-	-	-	-	-	-	-	-
Maximum load pin-on (payload + bucket)						3330 kg (7341 lb)	3100 kg (6834 lb)	3165 kg (6978 lb)	2935 kg (6471 lb)	2790 kg (6151 lb)	2600 kg (5732 lb)	3980 kg (8774 lb)	3710 kg (8179 lb)	3810 kg (8400 lb)	3545 kg (7815 lb)

(continued)

(Table 9, contd)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.

(4) Not Recommended

(5) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

Table 10

Excavator with Pin Grabber Coupler (Europe)															
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Utility Duty (UD) (No Adjuster)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	564 kg (1243 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1000 mm (39 inch)	0.93 m ³ (1.22 yd ³)	735 kg (1620 lb)	100	(1)	(1)	(1)	(1)	(2)	(3)	(1)	(1)	(1)	(1)
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	914 kg (2015 lb)	100	(3)	(4)	(4)	(4)	(5)	(5)	(2)	(2)	(2)	(3)
General Duty (GD) - EU	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	551 kg (1215 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	622 kg (1371 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	668 kg (1473 lb)		(1)	(1)	(1)	(1)	(1)	(2)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	803 kg (1770 lb)		(2)	(3)	(2)	(3)	(4)	(4)	(1)	(1)	(1)	(1)
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	835 kg (1841 lb)		(3)	(3)	(3)	(4)	(4)	(5)	(1)	(2)	(1)	(2)

(continued)

(Table 10, contd)

Excavator with Pin Grabber Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1400 mm (55 inch)	1.43 m³ (1.87 yd³)	870 kg (1918 lb)		(3)	(4)	(4)	(4)	(5)	(5)	(2)	(2)	(2)	(3)
Gener- al Duty (GD) (No Adjust- er)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	546 kg (1204 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	617 kg (1361 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1000 mm (39 inch)	0.93 m³ (1.22 yd³)	710 kg (1565 lb)		(1)	(1)	(1)	(1)	(2)	(3)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	799 kg (1761 lb)		(2)	(3)	(2)	(3)	(4)	(4)	(1)	(1)	(1)	(1)
		1400 mm (55 inch)	1.43 m³ (1.87 yd³)	866 kg (1909 lb)		(3)	(4)	(4)	(4)	(5)	(5)	(2)	(2)	(2)	(3)
		1500 mm (60 inch)	1.58 m³ (2.06 yd³)	906 kg (1997 lb)		(4)	(4)	(4)	(5)	(5)	(5)	(2)	(3)	(3)	(3)
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	826 kg (1821 lb)		(1)	(1)	(1)	(1)	(1)	(2)	(1)	(1)	(1)	(1)

(continued)

(Table 10, contd)

Excavator with Pin Grabber Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	880 kg (1940 lb)		(1)	(2)	(2)	(2)	(3)	(4)	(1)	(1)	(1)	(1)
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	880 kg (1940 lb)		(1)	(2)	(2)	(2)	(3)	(4)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	907 kg (1999 lb)		(2)	(3)	(3)	(3)	(4)	(4)	(1)	(1)	(1)	(2)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	918 kg (2024 lb)		(2)	(3)	(3)	(4)	(4)	(5)	(1)	(1)	(1)	(2)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	972 kg (2141 lb)		(2)	(3)	(3)	(4)	(4)	(5)	(1)	(1)	(1)	(2)
		1300 mm (52 inch)	1.30 m³ (1.71 yd³)	962 kg (2120 lb)		(3)	(4)	(3)	(4)	(4)	(5)	(1)	(2)	(2)	(2)
		1350 mm (54 inch)	1.38 m³ (1.81 yd³)	1054 kg (2322 lb)		(4)	(4)	(4)	(5)	(5)	(5)	(2)	(3)	(2)	(3)
Severe Duty (SD)	B	1050 mm (42 inch)	1.00 m³ (1.31 yd³)	964 kg (2125 lb)	90	(1)	(1)	(1)	(2)	(3)	(3)	(1)	(1)	(1)	(1)
Severe Duty (SDV)	B	1200 mm (48 inch)	1.20 m³ (1.57 yd³)	1001 kg (2207 lb)	90	(2)	(3)	(3)	(3)	(4)	(4)	(1)	(1)	(1)	(1)
Ditch Clean- ing (DC)	B	2000 mm (78 inch)	1.22 m³ (1.60 yd³)	869 kg (1916 lb)	100	(2)	(3)	(3)	(3)	(4)	(4)	(1)	(1)	(1)	(2)

(continued)

(Table 10, contd)

Excavator with Pin Grabber Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Ditch Clean- ing - Tilt (DCT)	B	2000 mm (78 inch)	1.23 m³ (1.61 yd³)	1096 kg (2416 lb)	100	(3)	(4)	(4)	(4)	(5)	(5)	(1)	(2)	(2)	(2)
Maximum load pin-on (payload + bucket)						6920 kg (15255 lb)	6413 kg (14138 lb)	6556 kg (14453 lb)	6049 kg (13336 lb)	5729 kg (12631 lb)	5310 kg (11707 lb)	8353 kg (18415 lb)	7758 kg (17102 lb)	7978 kg (17588 lb)	7394 kg (16300 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.(5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

Table 11

Excavator with CW-40 Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Gener- al Duty (GD) - CW40	B	900 mm (36 inch)	0.81 m³ (1.06 yd³)	658 kg (1450 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	704 kg (1551 lb)		(1)	(1)	(1)	(1)	(2)	(2)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	764 kg (1683 lb)		(1)	(2)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)

(continued)

(Table 11, contd)

Excavator with CW-40 Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1300 mm (51 inch)	1.30 m³ (1.71 yd³)	804 kg (1772 lb)		(2)	(3)	(3)	(3)	(4)	(4)	(1)	(1)	(1)	(2)
Heavy Duty (HD) - CW40	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	611 kg (1348 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	877 kg (1933 lb)		(2)	(2)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)
		1300 mm (52 inch)	1.30 m³ (1.71 yd³)	931 kg (2052 lb)		(5)	(5)	(3)	(5)	(5)	(5)	(5)	(5)	(1)	(5)
Sever Duty (SD)- CW40	B	1200 mm (48 inch)	1.20 m³ (1.57 yd³)	985 kg (2171 lb)	90	(1)	(2)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)
Ditch Clean- ing (DC) CW40	B	2100 mm (83 inch)	1.29 m³ (1.69 yd³)	793 kg (1748 lb)	100	(2)	(3)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)
		2100 mm (83 inch)	1.46 m³ (1.91 yd³)	812 kg (1790 lb)		(3)	(3)	(3)	(4)	(4)	(6)	(1)	(2)	(2)	(2)
		1800 mm (72 inch)	1.50 m³ (1.96 yd³)	777 kg (1713 lb)		(3)	(3)	(3)	(4)	(4)	(6)	(1)	(2)	(2)	(2)
		1800 mm (72 inch)	1.50 m³ (1.96 yd³)	826 kg (1821 lb)		(3)	(4)	(3)	(4)	(4)	(6)	(1)	(2)	(2)	(2)
		2100 mm (83 inch)	1.76 m³ (2.31 yd³)	867 kg (1911 lb)		(4)	(4)	(4)	(6)	(6)	(6)	(3)	(3)	(3)	(3)

(continued)

(Table 11, contd)

Excavator with CW-40 Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Ditch Clean- ing-Tilt (DCT)- CW40	B	2000 mm (79 inch)	1.23 m³ (1.61 yd³)	1161 kg (2560 lb)	100	(3)	(3)	(3)	(4)	(4)	(6)	(1)	(2)	(1)	(2)
Maximum load pin-on (payload + bucket)						3188 kg (7028 lb)	2951 kg (6506 lb)	3019 kg (6656 lb)	2784 kg (6138 lb)	2642 kg (5825 lb)	2445 kg (5390 lb)	3849 kg (8486 lb)	3573 kg (7877 lb)	3680 kg (8113 lb)	3406 kg (7509 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

(5) Not Recommended

(6) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

Table 12

Excavator with CW-40S Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Gener- al Duty (GD) - CW40S	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	503 kg (1109 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	588 kg (1297 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	655 kg (1444 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)		

(continued)

(Table 12, contd)

Excavator with CW-40S Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	770 kg (1697 lb)		(2)	(2)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)
		1300 mm (51 inch)	1.30 m³ (1.71 yd³)	801 kg (1765 lb)		(2)	(3)	(3)	(3)	(4)	(4)	(1)	(1)	(1)	(2)
		1400 mm (55 inch)	1.43 m³ (1.87 yd³)	837 kg (1845 lb)		(3)	(4)	(3)	(4)	(4)	(5)	(1)	(2)	(2)	(2)
Heavy Duty (HD) - CW40S	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	584 kg (1288 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	874 kg (1927 lb)		(2)	(3)	(2)	(3)	(4)	(4)	(1)	(1)	(1)	(1)
		1300 mm (52 inch)	1.30 m³ (1.71 yd³)	929 kg (2048 lb)		(6)	(6)	(3)	(6)	(6)	(6)	(6)	(6)	(1)	(6)
Ditch Clean- ing (DC) CW40S	B	2000 mm (78 inch)	1.22 m³ (1.60 yd³)	715 kg (1576 lb)	100	(2)	(2)	(2)	(3)	(3)	(4)	(1)	(1)	(1)	(1)
		2200 mm (87 inch)	1.36 m³ (1.78 yd³)	769 kg (1695 lb)		(2)	(3)	(3)	(4)	(4)	(4)	(1)	(2)	(1)	(2)

(continued)

(Table 12, contd)

Excavator with CW-40S Quick Coupler (Europe)															
Buck- et Type	Link- age	Width of Buck- et	Capaci- ty of Bucket	Weight of Bucket	Fill (%)	323									
						4200 kg (9260 lb)						5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		VA Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
Ditch Clean- ing-Tilt (DCT)- CW40S	B	2000 mm (79 inch)	1.23 m ³ (1.61 yd ³)	1142 kg (2518 lb)	100	(3)	(4)	(3)	(4)	(4)	(5)	(1)	(2)	(2)	(2)
Maximum load pin-on (payload + bucket)						3099 kg (6832 lb)	2869 kg (6325 lb)	2934 kg (6468 lb)	2704 kg (5961 lb)	2559 kg (5642 lb)	2369 kg (5223 lb)	3749 kg (8265 lb)	3749 kg (7670 lb)	3579 kg (7890 lb)	3314 kg (7306 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.(5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

(6) Not Recommended

Table 13

Excavator with Pin-ON (No Quick Coupler) (Japan)													
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323							
						4200 kg (9260 lb)						5400 kg (11905 lb)	
						Reach Boom		HD Reach Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
General Duty-Excavation (GDX)	B	950 mm (37 inch)	0.80 m ³ (1.04 yd ³)	688 kg (1517 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1050 mm (41 inch)	0.90 m ³ (1.17 yd ³)	721 kg (1590 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1150 mm (45 inch)	1.00 m ³ (1.31 yd ³)	753 kg (1660 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)

(continued)

(Table 13, contd)

Excavator with Pin-ON (No Quick Coupler) (Japan)													
Bucket Type	Link-age	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323							
						4200 kg (9260 lb)				5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
General Duty-Excavation (GDX)-ESCO	B	1000 mm (40 inch)	0.80 m³ (1.04 yd³)	715 kg (1575 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1100 mm (44 inch)	0.90 m³ (1.17 yd³)	749 kg (1650 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
Heavy Duty-Excavation (HDX)	B	950 mm (37 inch)	0.80 m³ (1.04 yd³)	798 kg (1760 lb)	100	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
		1050 mm (41 inch)	0.90 m³ (1.17 yd³)	837 kg (1846 lb)		(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
Maximum load pin-on (payload + bucket)						3330 kg (7341 lb)	3100 kg (6834 lb)	3165 kg (6978 lb)	2935 kg (6471 lb)	3980 kg (8774 lb)	3710 kg (8179 lb)	3810 kg (8400 lb)	3545 kg (7815 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.

Table 14

Excavator with Pin Grabber Coupler (Japan)													
Bucket Type	Link-age	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323							
						4200 kg (9260 lb)				5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
General Duty-Excavation (GDX)	B	950 mm (37 inch)	0.80 m ³ (1.04 yd ³)	688 kg (1517 lb)	100	(1)	(2)	(2)	(2)	(2)	(2)	(2)	(2)
		1050 mm (41 inch)	0.90 m ³ (1.17 yd ³)	721 kg (1590 lb)		(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)

(continued)

(Table 14, contd)

Excavator with Pin Grabber Coupler (Japan)													
Bucket Type	Link-age	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	323							
						4200 kg (9260 lb)				5400 kg (11905 lb)			
						Reach Boom		HD Reach Boom		Reach Boom		HD Reach Boom	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick
		1150 mm (45 inch)	1.00 m³ (1.31 yd³)	753 kg (1660 lb)		(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)
General Duty-Excavation (GDX)-ESCO	B	1000 mm (40 inch)	0.80 m³ (1.04 yd³)	715 kg (1575 lb)	100	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)
		1100 mm (44 inch)	0.90 m³ (1.17 yd³)	749 kg (1650 lb)		(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)
Heavy Duty-Excavation (HDX)	B	950 mm (37 inch)	0.80 m³ (1.04 yd³)	798 kg (1760 lb)	100	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(2)
		1050 mm (41 inch)	0.90 m³ (1.17 yd³)	837 kg (1846 lb)		(2)	(2)	(2)	(3)	(2)	(2)	(2)	(2)
Maximum load pin-on (payload + bucket)						6920 kg (15255 lb)	6413 kg (14138 lb)	6556 kg (14453 lb)	6049 kg (13336 lb)	8353 kg (18415 lb)	7758 kg (17102 lb)	7978 kg (17588 lb)	7394 kg (16300 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.

i07865211

Boom/Stick/Bucket Combinations

SMCS Code: 6000; 6700

S/N: KFE1-Up

S/N: MYK1-Up

S/N: YCP1-Up

S/N: HEX1-Up

This machine can be equipped with a large variety of boom-stick-bucket combinations to meet the needs of various applications.

Buckets are grouped into different families by capacity. As a rule, use a bucket with a smaller capacity when you are using a longer stick and/or a longer boom. Conversely, use a bucket with a larger capacity when you are using a shorter stick and/or a shorter boom. This rule ensures better machine stability and protection against structural machine damage.

A stick is designed to match only one specific family of buckets.

Note: The selection of a compatible boom-stick-bucket combination is a guide. Work tools, uneven ground conditions, soft ground conditions, or poor ground conditions have effects on machine performance. The operator is responsible for being aware of these effects.

Consult your Cat dealer for information on selecting the correct boom-stick-bucket combination.

The following tables show various compatible boom-stick-bucket combinations. Select an optimum combination according to the working conditions and according to the type of work that is being done.

320

Table 15

Excavator without Quick Coupler (North America)									
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320			
						4200 kg (9260 lb)			
						Reach Boom		VA Boom	SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	
General Duty - Capacity (GDC)	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	619 kg (1365 lb)	100	(1)	(1)	(1)	-
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	710 kg (1565 lb)		(1)	(1)	(1)	-
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	787 kg (1735 lb)		(1)	(1)	(1)	(2)
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	848 kg (1869 lb)		(1)	(2)	(3)	(3)
		1200 mm (48 inch)	1.38 m ³ (1.80 yd ³)	926 kg (2041 lb)		(2)	(3)	(4)	(4)
		1350 mm (53 inch)	1.59 m ³ (2.08 yd ³)	1004 kg (2213 lb)		(3)	(5)	(6)	(5)
General Duty - Capacity (GDC) - Wide Tip	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	633 kg (1395 lb)	100	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	731 kg (1611 lb)		(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	813 kg (1792 lb)		(1)	(1)	(1)	(2)
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	895 kg (1973 lb)		(1)	(2)	(3)	(3)
		1200 mm (48 inch)	1.38 m ³ (1.80 yd ³)	979 kg (2158 lb)		(3)	(3)	(4)	(4)
		1350 mm (53 inch)	1.59 m ³ (2.08 yd ³)	1063 kg (2343 lb)		(4)	(5)	(6)	(5)
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	826 kg (1821 lb)		(1)	(1)	(1)	(1)
		1050 mm (41 inch)	1.00 m ³ (1.31 yd ³)	880 kg (1940 lb)		(1)	(1)	(2)	(4)

(continued)

Product Information Section
Boom/Stick/Bucket Combinations

(Table 15, contd)

Excavator without Quick Coupler (North America)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	972 kg (2143 lb)		(1)	(1)	(2)	(4)	-
		1350 mm (53 inch)	1.38 m ³ (1.81 yd ³)	1054 kg (2324 lb)		(5)	(5)	(5)	(5)	-
Heavy Duty - Power (HDP)	B	900 mm (36 inch)	0.79 m ³ (1.03 yd ³)	842 kg (1856 lb)	100	(1)	(1)	(1)	(1)	-
		1050 mm (41 inch)	0.96 m ³ (1.26 yd ³)	907 kg (2000 lb)		(1)	(1)	(2)	(2)	-
		1200 mm (48 inch)	1.14 m ³ (1.49 yd ³)	993 kg (2189 lb)		(5)	(5)	(5)	(5)	-
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	889 kg (1960 lb)		(1)	(1)	(1)	(1)	-
		1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	964 kg (2125 lb)		(1)	(1)	(2)	(2)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	1053 kg (2321 lb)		(1)	(2)	(3)	(4)	-
Severe Duty - Power (SDP)	B	900 mm (36 inch)	0.79 m ³ (1.03 yd ³)	908 kg (2002 lb)	90	(1)	(1)	(1)	(1)	-
Mud / Cleanup (M/CU)	B	1700 mm (67 inch)	1.60 m ³ (2.09 yd ³)	979 kg (2158 lb)	100	(3)	(4)	(6)	(6)	-
Ditch Cleaning (DC)	B	1500 mm (59 inch)	1.01 m ³ (1.32 yd ³)	652 kg (1437 lb)	100	(1)	(1)	(1)	(2)	-
		1800 mm (71 inch)	1.24 m ³ (1.62 yd ³)	740 kg (1631 lb)		(1)	(2)	(3)	(3)	-
Ditch Cleaning - Tilt (DCT)	B	1500 mm (59 inch)	0.90 m ³ (1.18 yd ³)	948 kg (2090 lb)	100	(1)	(1)	(1)	(2)	-
		1800 mm (71 inch)	1.11 m ³ (1.45 yd ³)	1063 kg (2343 lb)		(1)	(2)	(3)	(4)	-
		1800 mm (71 inch)	1.40 m ³ (1.83 yd ³)	1148 kg (2531 lb)		(3)	(4)	(4)	(6)	-
		2000 mm (79 inch)	1.23 m ³ (1.61 yd ³)	1132 kg (2496 lb)		(2)	(3)	(4)	(4)	-

(continued)

(Table 15, contd)

Excavator without Quick Coupler (North America)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty (GD)	312	900 mm (35 inch)	0.53 m ³ (0.69 yd ³)	403 kg (888 lb)	100	-	-	-	-	(6)
Ditch Cleaning (DC) Long Reach	312	1200 mm (48 inch)	0.57 m ³ (0.74 yd ³)	386 kg (851 lb)	100	-	-	-	-	(6)
Maximum load pin-on (payload + bucket)						3235 kg (7132 lb)	3010 kg (6636 lb)	2705 kg (5963 lb)	2520 kg (5556 lb)	800 kg (1764 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

(5) Not Recommended

(6) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

Table 16

Excavator with Pin Grabber Coupler (North America)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty - Capacity (GDC)	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	619 kg (1365 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	710 kg (1565 lb)		(1)	(1)	(1)	(2)	-
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	787 kg (1735 lb)		(1)	(2)	(3)	(4)	-
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	848 kg (1869 lb)		(2)	(3)	(4)	(5)	-
		1200 mm (48 inch)	1.38 m ³ (1.80 yd ³)	926 kg (2041 lb)		(4)	(4)	(5)	(5)	-
		1350 mm (53 inch)	1.59 m ³ (2.08 yd ³)	1004 kg (2213 lb)		(4)	(5)	(6)	(6)	-

(continued)

Product Information Section
Boom/Stick/Bucket Combinations

(Table 16, contd)

Excavator with Pin Grabber Coupler (North America)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty - Capacity (GDC) - Wide Tip	B	600 mm (24 inch)	0.55 m ³ (0.72 yd ³)	633 kg (1395 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.75 m ³ (0.98 yd ³)	731 kg (1611 lb)		(1)	(1)	(1)	(2)	-
		900 mm (36 inch)	0.95 m ³ (1.24 yd ³)	813 kg (1792 lb)		(1)	(2)	(3)	(4)	-
		1050 mm (41 inch)	1.16 m ³ (1.52 yd ³)	895 kg (1973 lb)		(3)	(3)	(4)	(5)	-
		1200 mm (48 inch)	1.38 m ³ (1.80 yd ³)	979 kg (2158 lb)		(4)	(4)	(5)	(6)	-
		1350 mm (53 inch)	1.59 m ³ (2.08 yd ³)	1063 kg (2343 lb)		(5)	(5)	(6)	(6)	-
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	826 kg (1821 lb)		(1)	(1)	(2)	(3)	-
		1050 mm (41 inch)	1.00 m ³ (1.31 yd ³)	880 kg (1940 lb)		(1)	(1)	(3)	(4)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	972 kg (2143 lb)		(3)	(4)	(5)	(5)	-
		1350 mm (53 inch)	1.38 m ³ (1.81 yd ³)	1054 kg (2324 lb)		(4)	(4)	(5)	(6)	-
Heavy Duty - Power (HDP)	B	900 mm (36 inch)	0.79 m ³ (1.03 yd ³)	842 kg (1856 lb)	100	(1)	(1)	(2)	(3)	-
		1050 mm (41 inch)	0.96 m ³ (1.26 yd ³)	907 kg (2000 lb)		(1)	(2)	(3)	(4)	-
		1200 mm (48 inch)	1.14 m ³ (1.49 yd ³)	993 kg (2189 lb)		(3)	(3)	(4)	(5)	-
Heavy Duty - Pin Grabber Performance (HD - PGP)	B	600 mm (24 inch)	0.44 m ³ (0.57 yd ³)	676 kg (1490 lb)	90	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.60 m ³ (0.79 yd ³)	778 kg (1715 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.76 m ³ (1.00 yd ³)	864 kg (1905 lb)		(1)	(1)	(2)	(3)	-

(continued)

(Table 16, contd)

Excavator with Pin Grabber Coupler (North America)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
		1050 mm (42 inch)	0.93 m³ (1.22 yd³)	928 kg (2046 lb)		(1)	(2)	(3)	(4)	-
		1200 mm (48 inch)	1.11 m³ (1.45 yd³)	1016 kg (2234 lb)		(3)	(3)	(4)	(5)	-
		1350 mm (53 inch)	1.28 m³ (1.67 yd³)	1104 kg (2434 lb)		(4)	(4)	(5)	(6)	-
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m³ (0.61 yd³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m³ (0.84 yd³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m³ (1.06 yd³)	889 kg (1960 lb)		(1)	(1)	(2)	(3)	-
		1050 mm (42 inch)	1.00 m³ (1.31 yd³)	964 kg (2125 lb)		(1)	(2)	(3)	(4)	-
		1200 mm (48 inch)	1.19 m³ (1.56 yd³)	1053 kg (2321 lb)		(3)	(3)	(4)	(5)	-
Severe Duty - Power (SDP)	B	900 mm (36 inch)	0.79 m³ (1.03 yd³)	908 kg (2002 lb)	90	(1)	(1)	(1)	(2)	-
Mud / Cleanup (M/CU)	B	1700 mm (67 inch)	1.60 m³ (2.09 yd³)	979 kg (2158 lb)	100	(4)	(5)	(6)	(6)	-
Ditch Cleaning (DC)	B	1500 mm (59 inch)	1.01 m³ (1.32 yd³)	652 kg (1437 lb)	100	(1)	(2)	(3)	(3)	-
		1800 mm (71 inch)	1.24 m³ (1.62 yd³)	740 kg (1631 lb)		(2)	(3)	(4)	(5)	-
Ditch Cleaning - Tilt (DCT)	B	1500 mm (59 inch)	0.90 m³ (1.18 yd³)	948 kg (2090 lb)	100	(1)	(2)	(3)	(4)	-
		1800 mm (71 inch)	1.11 m³ (1.45 yd³)	1063 kg (2343 lb)		(3)	(4)	(5)	(5)	-
		1800 mm (71 inch)	1.40 m³ (1.83 yd³)	1148 kg (2531 lb)		(4)	(5)	(6)	(6)	-
		2000 mm (79 inch)	1.23 m³ (1.61 yd³)	1132 kg (2496 lb)		(4)	(4)	(5)	(6)	-
Maximum load pin-on (payload + bucket)						6710 kg (14793 lb)	6214 kg (13700 lb)	5542 kg (12218 lb)	5134 kg (11319 lb)	1342 kg (2959 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.

(continued)

Product Information Section
Boom/Stick/Bucket Combinations

(Table 16, contd)

(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.

(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.

(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

(5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

(6) Not Recommended

Table 17

Excavator without Quick Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty (GD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	551 kg (1215 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	622 kg (1371 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	668 kg (1473 lb)		(1)	(1)	(1)	(1)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	803 kg (1770 lb)		(1)	(2)	(3)	(3)	-
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	835 kg (1841 lb)		(2)	(3)	(3)	(4)	-
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	870 kg (1918 lb)		(5)	(5)	(5)	(5)	-
Heavy Duty (HD)	B	1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	27 kg (59 lb)	100	(1)	(1)	(2)	(3)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	907 kg (1999 lb)		(1)	(2)	(3)	(4)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	918 kg (2024 lb)		(2)	(2)	(3)	(4)	-
		1300 mm (52 inch)	1.30 m ³ (1.71 yd ³)	962 kg (2120 lb)		(2)	(3)	(4)	(4)	-
Severe Duty (SD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	694 kg (1530 lb)	90	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	802 kg (1768 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	889 kg (1960 lb)		(1)	(1)	(1)	(1)	-
		1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	964 kg (2125 lb)		(1)	(1)	(2)	(2)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	1053 kg (2321 lb)		(1)	(2)	(3)	(4)	-

(continued)

(Table 17, contd)

Excavator without Quick Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
Severe Duty (SDV)	B	1200 mm (48 inch)	1.20 m ³ (1.57 yd ³)	1001 kg (2207 lb)	90	(1)	(2)	(3)	(3)	-
Ditch Cleaning (DC)	B	2000 mm (78 inch)	1.22 m ³ (1.60 yd ³)	869 kg (1916 lb)	100	(2)	(2)	(3)	(4)	-
Ditch Cleaning - Tilt (DCT)	B	2000 mm (78 inch)	1.23 m ³ (1.61 yd ³)	1096 kg (2416 lb)	100	(2)	(3)	(4)	(4)	-
General Duty (GD)	312	900 mm (35 inch)	0.53 m ³ (0.69 yd ³)	403 kg (888 lb)	100	-	-	-	-	757
Ditch Cleaning (DC) Long Reach	312	1200 mm (48 inch)	0.57 m ³ (0.74 yd ³)	386 kg (851 lb)	100	-	-	-	-	733
Maximum load pin-on (payload + bucket)						3235 kg (7132 lb)	3010 kg (6636 lb)	2705 kg (5963 lb)	2520 kg (5556 lb)	800 kg (1764 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

(5) Not Recommended

Table 18

Excavator with Pin Grabber Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
Utility Duty (UD) (No Adjuster)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	564 kg (1243 lb)	100	(1)	(1)	(1)	(1)	-
		1000 mm (39 inch)	0.93 m ³ (1.22 yd ³)	735 kg (1620 lb)		(1)	(1)	(2)	(3)	-
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	914 kg (2015 lb)		(4)	(4)	(5)	(5)	-

(continued)

Product Information Section
Boom/Stick/Bucket Combinations

(Table 18, contd)

Excavator with Pin Grabber Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty (GD) - EU	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	551 kg (1215 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	622 kg (1371 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	668 kg (1473 lb)		(1)	(1)	(1)	(2)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	803 kg (1770 lb)		(2)	(3)	(4)	(5)	-
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	835 kg (1841 lb)		(3)	(4)	(4)	(5)	-
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	870 kg (1918 lb)		(4)	(4)	(5)	(5)	-
General Duty (GD) (No Adjuster)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	546 kg (1204 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	617 kg (1360 lb)		(1)	(1)	(1)	(1)	-
		1000 mm (39 inch)	0.93 m ³ (1.22 yd ³)	710 kg (1565 lb)		(1)	(1)	(2)	(3)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	799 kg (1761 lb)		(2)	(3)	(4)	(5)	-
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	866 kg (1909 lb)		(4)	(4)	(5)	(5)	-
		1500 mm (59 inch)	1.58 m ³ (2.06 yd ³)	906 kg (1997 lb)		(4)	(5)	(5)	(6)	-
Heavy Duty (HD)	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	649 kg (1431 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (30 inch)	0.64 m ³ (0.84 yd ³)	748 kg (1649 lb)		(1)	(1)	(1)	(1)	-
		900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	826 kg (1821 lb)		(1)	(1)	(2)	(3)	-
		1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	880 kg (1940 lb)		(2)	(2)	(3)	(4)	-
		1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	880 kg (1940 lb)		(2)	(2)	(3)	(4)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	907 kg (1999 lb)		(3)	(3)	(4)	(5)	-

(continued)

(Table 18, contd)

Excavator with Pin Grabber Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	918 kg (2024 lb)		(3)	(3)	(4)	(5)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	972 kg (2143 lb)		(3)	(4)	(5)	(5)	-
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	962 kg (2121 lb)		(3)	(4)	(5)	(5)	-
		1350 mm (53 inch)	1.38 m ³ (1.81 yd ³)	1054 kg (2324 lb)		(4)	(4)	(5)	(6)	-
Severe Duty (SD)	B	1050 mm (41 inch)	1.00 m ³ (1.31 yd ³)	964 kg (2125 lb)	90	(1)	(2)	(3)	(4)	-
Severe Duty (SDV)	B	1200 mm (48 inch)	1.20 m ³ (1.57 yd ³)	1001 kg (2207 lb)	90	(2)	(3)	(4)	(5)	-
Ditch Cleaning (DC)	B	2000 mm (78 inch)	1.22 m ³ (1.60 yd ³)	869 kg (1916 lb)	100	(3)	(3)	(4)	(5)	-
Ditch Cleaning - Tilt (DCT)	B	2000 mm (78 inch)	1.23 m ³ (1.61 yd ³)	1096 kg (2416 lb)	100	(3)	(4)	(5)	(6)	-
Maximum load pin-on (payload + bucket)						6710 kg (14793 lb)	6214 kg (13700 lb)	5542 kg (12218 lb)	5134 kg (11319 lb)	1342 kg (2959 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.(5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

(6) Not Recommended

Table 19

Excavator with CW-40 Quick Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty (GD) - CW40	B	900 mm (36 inch)	0.81 m ³ (1.06 yd ³)	658 kg (1451 lb)	100	(1)	(1)	(1)	(1)	-

(continued)

Product Information Section
Boom/Stick/Bucket Combinations

(Table 19, contd)

Excavator with CW-40 Quick Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
		1050 mm (42 inch)	1.00 m ³ (1.31 yd ³)	704 kg (1552 lb)		(1)	(1)	(2)	(3)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	764 kg (1684 lb)		(2)	(2)	(3)	(4)	-
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	804 kg (1772 lb)		(3)	(3)	(4)	(4)	-
Heavy Duty (HD) - CW40	B	600 mm (24 inch)	0.46 m ³ (0.31 yd ³)	611 kg (1347 lb)	100	(1)	(1)	(1)	(1)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	877 kg (1933 lb)		(2)	(3)	(4)	(4)	-
		1300 mm (52 inch)	1.30 m ³ (1.71 yd ³)	931 kg (2052 lb)		(5)	(5)	(5)	(5)	-
Severe Duty (SD) CW40	B	1200 mm (48 inch)	1.20 m ³ (1.57 yd ³)	985 kg (2171 lb)	90	(2)	(3)	(4)	(4)	-
Ditch Cleaning (DC) CW40	B	2100 mm (83 inch)	1.29 m ³ (1.69 yd ³)	793 kg (1748 lb)	100	(2)	(3)	(4)	(4)	-
		2100 mm (83 inch)	1.46 m ³ (1.91 yd ³)	812 kg (1790 lb)		(2)	(3)	(4)	(6)	-
		2100 mm (83 inch)	1.76 m ³ (2.31 yd ³)	867 kg (1911 lb)		(4)	(6)	(6)	(5)	-
		1800 mm (71 inch)	1.50 m ³ (1.96 yd ³)	777 kg (1713 lb)		(2)	(4)	(4)	(6)	-
		1800 mm (71 inch)	1.50 m ³ (1.96 yd ³)	826 kg (1821 lb)		(2)	(4)	(6)	(6)	-
Ditch Cleaning - Tilt (DCT) CW40	B	2000 mm (78 inch)	1.23 m ³ (1.61 yd ³)	1161 kg (2560 lb)	100	(2)	(4)	(6)	(6)	-
Maximum load pin-on (payload + bucket)						3034 kg (6689 lb)	2808 kg (6191 lb)	2499 kg (5509 lb)	2311 kg (5095 lb)	554 kg (1221 lb)

- (1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.
 (2) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.
 (3) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.
 (4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.
 (5) Not Recommended
 (6) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

Table 20

Excavator with CW-40S Quick Coupler (Europe)										
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320 Long				
						4200 kg (9260 lb)				4700 kg (10362 lb)
						Reach Boom		VA Boom		SLR
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	
General Duty (GD) - CW40S	B	600 mm (24 inch)	0.46 m ³ (0.61 yd ³)	503 kg (1109 lb)	100	(1)	(1)	(1)	(1)	-
		750 mm (29 inch)	0.64 m ³ (0.84 yd ³)	588 kg (1296 lb)		(1)	(1)	(1)	(1)	-
		900 mm (35 inch)	0.81 m ³ (1.06 yd ³)	655 kg (1444 lb)		(1)	(1)	(1)	(1)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	770 kg (1697 lb)		(2)	(3)	(3)	(4)	-
		1300 mm (51 inch)	1.30 m ³ (1.71 yd ³)	801 kg (1766 lb)		(3)	(3)	(4)	(4)	-
		1400 mm (55 inch)	1.43 m ³ (1.87 yd ³)	837 kg (1845 lb)		(3)	(4)	(4)	(5)	-
Heavy Duty (HD) - CW40S	B	600 mm (24 inch)	0.46 m ³ (0.31 yd ³)	611 kg (1347 lb)	100	(1)	(1)	(1)	(1)	-
		1200 mm (48 inch)	1.19 m ³ (1.56 yd ³)	877 kg (1933 lb)		(2)	(3)	(4)	(4)	-
		1300 mm (52 inch)	1.30 m ³ (1.71 yd ³)	931 kg (2052 lb)		(6)	(6)	(6)	(6)	-
Ditch Cleaning (DC) CW40S	B	2000 mm (79 inch)	1.22 m ³ (1.60 yd ³)	715 kg (1576 lb)	100	(2)	(3)	(3)	(4)	-
		2200 mm (87 inch)	1.36 m ³ (1.78 yd ³)	769 kg (1695 lb)		(3)	(3)	(4)	(5)	-
Ditch Cleaning - Tilt (DCT) CW40S	B	2000 mm (78 inch)	1.23 m ³ (1.61 yd ³)	1142 kg (2518 lb)	100	(3)	(4)	(5)	(5)	-
Maximum load pin-on (payload + bucket)						3004 kg (6623 lb)	2779 kg (6127 lb)	2474 kg (5454 lb)	2289 kg (5046 lb)	-

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.(5) 900 kg/m³ (1500 lb/yd³) is the maximum density of material.

(6) Not Recommended

Product Information Section
Boom/Stick/Bucket Combinations

Table 21

Excavator without Quick Coupler (Japan)									
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320			
						3700 kg (8157 lb)			
						Reach Boom		Reach Boom HD	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) HD Stick	2.9 m (9 ft 6 inch) HD Stick
General Duty - Excavation (GDX)	B	950 mm (37 inch)	0.80 m³ (1.04 yd³)	688 kg (1517 lb)	100	(1)	(1)	(1)	(1)
		1050 mm (41 inch)	0.90 m³ (1.17 yd³)	721 kg (1589 lb)		(1)	(1)	(1)	(2)
		1150 mm (45 inch)	1.00 m³ (1.31 yd³)	753 kg (1660 lb)		(2)	(2)	(2)	(3)
General Duty - Excavation (GDX) - ESCO	B	1000 mm (40 inch)	0.80 m³ (1.04 yd³)	715 kg (1576 lb)	100	(1)	(1)	(1)	(1)
		1100 mm (44 inch)	0.90 m³ (1.17 yd³)	749 kg (1651 lb)		(1)	(2)	(2)	(2)
Heavy Duty - Excavation (HDX)	B	950 mm (37 inch)	0.80 m³ (1.04 yd³)	798 kg (1759 lb)	100	(1)	(1)	(1)	(2)
		1050 mm (41 inch)	0.90 m³ (1.17 yd³)	837 kg (1845 lb)		(1)	(2)	(2)	(3)
Maximum load pin-on (payload + bucket)						2695 kg (5941 lb)	2505 kg (5523 lb)	2525 kg (5567 lb)	2335 kg (5148 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.

(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.

(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.

Table 22

Excavator with Pin Grabber Coupler (Japan)								
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320		
						3700 kg (8157 lb)		
						Reach Boom		Reach Boom HD
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) HD Stick
General Duty - Excavation (GDX)	B	950 mm (37 inch)	0.80 m ³ (1.04 yd ³)	688 kg (1517 lb)	100	(1)	(2)	(3)
		1050 mm (41 inch)	0.90 m ³ (1.17 yd ³)	721 kg (1589 lb)		(2)	(3)	(4)
		1150 mm (45 inch)	1.00 m ³ (1.31 yd ³)	753 kg (1660 lb)		(3)	(4)	(4)
General Duty - Excavation (GDX) - ESCO	B	1000 mm (40 inch)	0.80 m ³ (1.04 yd ³)	715 kg (1576 lb)	100	(2)	(2)	(3)

(continued)

(Table 22, contd)

Excavator with Pin Grabber Coupler (Japan)									
Bucket Type	Linkage	Width of Bucket	Capacity of Bucket	Weight of Bucket	Fill (%)	320			
						3700 kg (8157 lb)			
						Reach Boom		Reach Boom HD	
						2.5 m (8 ft 2 inch) Stick	2.9 m (9 ft 6 inch) Stick	2.5 m (8 ft 2 inch) HD Stick	2.9 m (9 ft 6 inch) HD Stick
		1100 mm (44 inch)	0.90 m³ (1.17 yd³)	749 kg (1651 lb)		(2)	(3)	(3)	(4)
Heavy Duty - Excavation (HDX)	B	950 mm (37 inch)	0.80 m³ (1.04 yd³)	798 kg (1759 lb)	100	(2)	(3)	(3)	(4)
		1050 mm (41 inch)	0.90 m³ (1.17 yd³)	837 kg (1845 lb)		(3)	(4)	(4)	(4)
Maximum load pin-on (payload + bucket)						5520 kg (12169 lb)	5101 kg (11246 lb)	5145 kg (11343 lb)	4726 kg (10419 lb)

(1) 2100 kg/m³ (3500 lb/yd³) is the maximum density of material.(2) 1800 kg/m³ (3000 lb/yd³) is the maximum density of material.(3) 1500 kg/m³ (2500 lb/yd³) is the maximum density of material.(4) 1200 kg/m³ (2000 lb/yd³) is the maximum density of material.

i08287322

Lifting Capacities (323 Reach Boom/Variable Angle Boom/Super Long Reach Boom)

SMCS Code: 7000

S/N: NDL1-Up

S/N: LTN1-Up

S/N: HDT1-Up

S/N: RAZ1-Up

WARNING

Failure to comply to the rated load can cause possible personal injury or property damage. This includes the risk of unintended boom lowering. Review the rated load of a particular work tool before performing any operation. Make adjustments to the rated load as necessary for non-standard configurations.

There may be local regulations and/or government regulations that govern the use of excavators which lift heavy objects. Obey all local and government regulations.

Lifting capacities should be used as a guide. Work tools, uneven ground conditions, soft ground conditions, or poor ground conditions have effects on lifting capacities. The operator is responsible for being aware of these effects.

The lifting capacities are defined by "ISO 10567 2007". The lifting capacities are defined as the lower value of 75% of the static tipping capacity or 87% of the hydraulic lift capacity.

Note: Lifting capacities are based on a standard machine with the following conditions:

- Lift point: Stick nose without bucket
- Lubricants full
- Fuel tank full
- Steel track
- Complete cab with a 75 kg (165 lb) operator

Lifting capacities will vary with different work tools and attachments. The weight of a work tool attachment must be subtracted from the lift capacity. Consult your Cat dealer regarding the lifting capacities for specific work tools and attachments.

This machine may be equipped with various sticks. Lifting capacities may vary between the different sticks. Measure the distance on the stick between the boom hinge pin and the work tool hinge pin. This distance will inform you of the size of the stick that is equipped on the machine.

Use the lifting eye that is provided on the linkage to lift objects. When the lifting eye is used, the connection must be made with a sling or shackle.

Note: Japan regulations require a shovel crane configuration to lift certain objects. A shovel crane has a rated load capacity, therefore, the lift capacities discussed below do not apply to a shovel crane configuration. Contact your Cat dealer for additional information.

Note: In European countries, regulations require an overload warning device and a boom and stick lowering control valve if more than 1000 kg (2200 lb) is lifted during object handling applications. Regulations also require an overload warning device and a boom lowering control valve if a force that is greater than 40000 N·m (29500 lb ft) is created during object handling applications. If the machine is not equipped with these devices, even if the hydraulic lift capacity is capable, do not exceed a load of 1000 kg (2200 lb). Do not exceed a force of 40000 N·m (29500 lb ft) in European object handling applications.

Configuration Identification

Note: Each component has a stamp to identify the configuration affecting lifting capacity.

The owner will need to check the machine configuration to identify the correct lifting capacity.

The configuration identifier will be located with the part number stamped on the component. Refer to the following table for the abbreviation of the configuration.

Table 23

Configuration Identification		
Component	Configuration	Abbreviation
Front	Reach Boom	R
	Mass Boom	M
	Variable Angle Boom	VA
	Super Long Reach Boom	SLR
	Standard	STD
	Short Reach Stick	SR
	Severe Duty Bucket	SD
	Heavy Duty	HD
	Semi-Heavy Duty	S-HD
	Extreme Special	ES

(Table 23, contd)

Configuration Identification		
Component	Configuration	Abbreviation
	Thumb Ready Stick	TR
Undercarriage	Short Undercarriage (Crawler)	STD
	Long Undercarriage (Crawler)	LC
	Long Narrow Undercarriage (Crawler)	LN
Cylinder	Standard	-
	Heavy Lift	HL
Counterweight	Metric Ton (tonne)	t ⁽¹⁾

(1) Counterweight stamp indicates metric ton. (example 1.0t = 1000 kg)

Symbols Found in the Lifting Capacity Charts

Below are symbols that are commonly found on lifting capacity charts for track excavators.

Note: Depending on the machine configuration, some symbols may not be used.

(mm)
(inch) **Measurements are provided in millimeters and inches**



Lift Capacities are provided in kilograms and pounds



Load is limited by hydraulic lifting capacity rather than by a tipping load



Lift point radius



Lift point height



Lifting capacity over the front of the machine



Lifting capacity over the side of the machine

(continued)



Heavy Lift ON

**Reach Boom with a 2.5 m
(8 ft 2 inch) Stick and 4200 kg
(9260 lb) Counterweight**

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

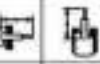
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							5650 223
6000 240				+ 5350 + 2100	+ 5350 + 1700	+ 4350 + 5600	6030 239
4500 180			+ 8700 + 14500	+ 8700 + 12500	+ 8000 + 9500	+ 4200 + 5400	5770 228
3000 120			+ 8450 + 12750	+ 8900 + 11050	+ 5650 + 12300	+ 4350 + 5600	5650 223
1500 60			+ 9500 + 21450	+ 13000 + 15750	+ 4900 + 10600	+ 4800 + 10250	5800 229
0			+ 10050 + 23000	+ 13000 + 16400	+ 4800 + 10000	+ 5200 + 11400	5200 205
-1500 -60		+ 10250 + 25800	+ 10250 + 25800	+ 10250 + 22700	+ 10250 + 16350	+ 5700 + 12800	5350 211
-3000 -120		+ 13750 + 28800	+ 13750 + 28800	+ 13750 + 20650	+ 13750 + 16100	+ 6300 + 13800	4430 175
-4500 -180			+ 7750 + 14550	+ 7750 + 14550		+ 6200 + 13550	6200 245

Illustration 73

g06224929

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

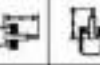
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							5650 223
6000 240				+ 5900 + 12900	+ 5600 + 11800	+ 4750 + 10500	6030 239
4500 180			+ 7400 + 15950	+ 8350 + 13850	+ 5200 + 11500	+ 4850 + 10250	5770 228
3000 120			+ 8300 + 20000	+ 7200 + 16700	+ 6700 + 11000	+ 4750 + 10450	5650 223
1500 60			+ 11000 + 23100	+ 7900 + 15750	+ 4900 + 10600	+ 5600 + 11150	5800 229
0			+ 11700 + 25150	+ 7900 + 16400	+ 4800 + 10300	+ 5600 + 11000	5200 205
-1500 -60		+ 13000 + 27300	+ 13000 + 27300	+ 13000 + 25100	+ 13000 + 16350	+ 5700 + 12800	5350 211
-3000 -120		+ 14500 + 31500	+ 14500 + 31500	+ 14500 + 22900	+ 14500 + 16550	+ 6300 + 13800	4430 175
-4500 -180			+ 7900 + 16150	+ 7900 + 16150		+ 6300 + 13800	6200 245

Illustration 74

g06224933

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

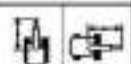
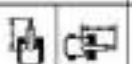
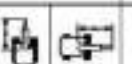
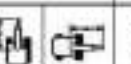
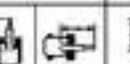
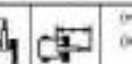
(mm) (inch)	580 68	3906 120	4580 180	6800 240	7500 300		(mm) (inch)
7500 300							4750 187
6800 240				5350 211	6350 250		4350 171
4580 180			8780 345	6700 264	5300 209	4300 169	3750 148
3906 120			8450 332	7750 304	6500 256	5900 232	4350 171
1800 60			9050 356	7300 287	4800 189	5900 232	4650 183
8 0			21450 844	18700 736	16750 658	12100 476	10250 403
8 0			23000 905	16250 638	14450 570	11900 468	10250 403
-1800 -60		11300 445	11350 447	18180 716	16350 646	10200 401	8050 316
-3906 -120		25800 1012	25800 1012	32180 1261	16350 646	10200 401	8050 316
-4580 -180		13150 518	13150 518	16150 634	14450 570	11900 468	10250 403
-7500 -300		26500 1043	26500 1043	32450 1272	16350 646	10200 401	8050 316

Illustration 75

g06224922

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

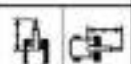
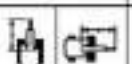
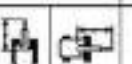
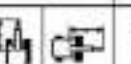
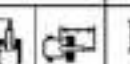
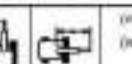
(mm) (inch)	580 68	3906 120	4580 180	6800 240	7500 300		(mm) (inch)
7500 300							5150 203
6800 240				5900 232	5500 217		4750 187
4580 180			7490 295	6350 250	5050 199	4850 191	3750 148
3906 120			8380 329	7750 304	6500 256	5790 227	4750 187
1800 60			11000 433	7800 307	4800 189	5600 220	4650 183
8 0			23180 912	18700 736	16750 658	12100 476	10250 403
8 0			25150 988	16250 638	14450 570	11900 468	10250 403
-1800 -60		12900 508	12900 508	19480 766	16350 646	10200 401	8050 316
-3906 -120		27300 1070	27300 1070	35180 1381	16350 646	10200 401	8050 316
-4580 -180		14550 574	14550 574	18150 716	16350 646	10200 401	8050 316
-7500 -300		31500 1240	31500 1240	32450 1272	16350 646	10200 401	8050 316

Illustration 76

g06224926

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							4750 187
6000 240				5350 211	5350 211		4350 171
4500 180			5150 203	5150 203	4800 189	4800 189	4250 167
3000 120			4450 175	4450 175	4000 157	4000 157	3750 148
1500 60			3080 121	3080 121	2700 106	2700 106	2450 97
0			23450 923	23450 923	21100 831	21100 831	19250 758
-1500 -60			19650 776	19650 776	17800 697	17800 697	16050 630
-3000 -120			16650 657	16650 657	15000 590	15000 590	13800 543
-4500 -180			13650 538	13650 538	12200 478	12200 478	11050 434

Illustration 77

g06224885

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							5150 203
6000 240				5900 232	5900 232		4750 187
4500 180			7400 291	7400 291	5200 205	5200 205	4650 183
3000 120			6300 248	6300 248	4750 187	4750 187	4250 167
1500 60			5180 204	5180 204	4000 157	4000 157	3750 148
0			41700 1638	41700 1638	37800 1483	37800 1483	34500 1358
-1500 -60			37900 1483	37900 1483	34100 1339	34100 1339	31050 1223
-3000 -120			34100 1339	34100 1339	30400 1197	30400 1197	27550 1084
-4500 -180			30300 1197	30300 1197	26700 1053	26700 1053	24050 947

Illustration 78

g06224886

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							* 4700 * 4700 9900 * 16550 * 16550 220
6000 240				* 6300 * 6300 * 11700 19650			* 4300 * 4300 9830 * 9800 * 9800 270
4500 180			* 8700 * 8700 * 14100 * 14100	* 5750 5300 * 12550 11350	* 4800 3750		* 4250 3700 7570 * 9400 8150 300
3000 120			* 8400 7050 * 18150 16450	* 6900 5050 * 14150 12000	6400 3600		* 4350 3300 7900 * 9000 7350 320
1500 60			* 9000 7200 * 21450 15450	* 7300 4850 * 15750 10450	5500 3000		* 4650 3050 8950 * 16250 11000 320
0			* 10650 8950 * 23000 16000	7500 4700 * 16150 12100	6450 3500		* 5100 3300 7950 * 12550 1250 310
-1500 -60		* 11300 * 11300 * 25000 * 25000	* 10100 8050 * 22100 14900	7800 4600 * 16850 13050			8600 3600 7300 * 12350 7900 290
-3000 -120		* 13150 * 13150 * 28500 * 28500	* 9150 7050 * 20400 16100	* 7850 4750 * 16100 13200			* 8300 4300 6470 * 15900 9600 260
-4500 -180			* 7150 * 7150 * 14550 * 14550				* 6200 * 6200 4900 * 13000 * 13000 200

Illustration 79

g06224934

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							* 5150 * 5150 9800 * 11500 * 11500 320
6000 240				* 5900 5450 * 12500 11500			* 4750 4400 6830 * 10500 9800 270
4500 180			* 7400 * 7400 * 16300 * 16300	* 6300 5300 * 13050 11350	* 5250 3750		* 4650 3700 7670 * 10250 8150 300
3000 120			* 8300 1850 * 20000 14450	* 7200 5850 * 15600 10800	5600 3850		* 4750 3350 7900 * 10450 7350 320
1500 60			* 11300 1200 * 23700 15450	7700 4850 * 16500 16450	6000 3500		8800 3200 8800 * 10950 7500 320
0			* 11700 8950 * 25350 16000	7500 4700 * 16150 12100	6450 3500		* 5100 3300 7950 * 12550 1250 310
-1500 -60		* 12000 * 12000 * 27300 * 27300	* 11100 8950 * 25100 14900	7400 4650 * 16050 10850			8800 3600 7350 * 12350 7900 290
-3000 -120		* 14550 13600 * 31500 29100	* 10050 7850 * 22300 16150	7550 4750 * 16250 13200			8800 4300 6470 * 15100 9650 260
-4500 -180			* 7300 7300 * 16050 16000				* 6300 6300 4900 * 15800 14450 200

Illustration 80

g06224936

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7000 3000							* 4750 * 4750 6600 * 10500 * 10500 226
6000 240				* 5350 * 5350 * 11700 * 11700			* 4350 * 4350 6830 * 9600 * 9600 276
4500 180			* 8100 * 8100 * 14400 * 14400	* 6700 9400 * 12550 11700	* 4800 3800		* 4250 3900 7570 * 9400 8400 300
3000 120			* 8400 1800 * 18150 16350	* 6800 5200 * 14150 11250	* 5800 3800 * 12350 8100		* 4350 3450 7900 * 9600 7000 320
1500 60			* 8900 1400 * 21450 16000	* 7300 5900 * 16750 13600	* 5700 3700 * 12300 7900		* 4600 3300 8800 * 10250 7350 320
0 0			* 18050 1200 * 23000 16500	* 7800 4000 * 16750 13450	* 5800 3800 * 12350 7800		* 4600 3400 7800 * 11450 7500 310
-1500 60		* 11350 * 11350 * 25800 * 25800	* 18000 1200 * 22100 16450	* 7700 4900 * 16600 13350			* 10800 3700 7350 * 12800 8200 290
-3000 120		* 13150 * 13150 * 28800 * 28800	* 18000 1200 * 28400 16900	* 7800 4900 * 16100 13550			* 10800 3400 6470 * 13800 8300 290
-4500 180			* 7150 * 7150 * 14950 * 14950				* 8200 * 8200 4900 * 17550 * 17550 230

Illustration 81

g06224887

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7000 3000							* 5150 * 5150 9500 * 11800 * 11800 228
6000 240				* 5900 5600 * 12900 12000			* 4750 4550 6830 * 10600 10100 276
4500 180			* 7400 * 7400 * 15950 * 15950	* 6350 5450 * 13850 11700	* 5200 3800		* 4650 3600 7570 * 10250 8400 300
3000 120			* 8500 2800 * 20300 16900	* 7200 5200 * 16000 11250	* 5800 3800 * 12300 8100		* 4750 3450 7900 * 10450 7600 320
1500 60			* 11500 1400 * 23100 18000	* 7900 5900 * 17100 13600	* 5700 3700 * 12300 7900		* 5190 3300 8800 * 11150 7350 320
0 0			* 11100 1200 * 25300 16500	* 7800 4000 * 16750 13450	* 5800 3800 * 12350 7800		* 5300 3400 7800 * 11650 7500 310
-1500 60		* 13000 * 13000 * 27300 * 27300	* 11000 1200 * 25100 16450	* 7750 4900 * 16600 13350			* 5800 3700 7350 * 12800 8200 290
-3000 120		* 14500 14050 * 31800 30050	* 18500 1200 * 32800 16900	* 7800 4900 * 16700 13600			* 7800 4450 6470 * 16400 9800 290
-4500 180			* 7100 7050 * 16500 16300				* 6800 6550 4900 * 15800 14900 230

Illustration 82

g06224889

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							* 4750 * 4750 5600 * 18850 * 18550 220
6000 240				* 5350 * 5350 * 11700 * 11700			* 4350 * 4350 8800 * 8600 * 8600 270
4500 180			* 6700 * 6700 * 14800 * 14800	* 6700 10400 * 13500 11600	* 4800 3900		* 4250 3850 7570 * 8400 8500 260
3000 120			* 8400 7600 * 18150 17150	* 8300 5300 * 14150 11400	* 5800 3850		* 4250 3500 7960 * 8600 7700 320
1500 60			* 8800 7600 * 21450 16200	* 7300 5000 * 15750 10900	8900 2700		* 4000 3400 8000 * 18250 14500 320
0 0			* 10400 7300 * 23800 15700	* 7900 4500 * 18050 10600	5700 3650		* 3200 3450 7860 * 11450 7800 310
-1500 -60		* 11300 * 11300 * 25800 * 25800	* 10800 7250 * 22700 15850	* 7900 4900 * 18050 10500			10400 3750 7350 12950 8300 250
-3000 -120		* 13150 * 13150 * 28600 * 28600	* 11100 7350 * 20600 16850	* 7900 4950 * 18100 10700			* 8300 4500 8470 * 13900 10000 260
-4500 -180			* 7100 * 7100 * 14550 * 14550				* 4200 * 6300 4900 * 13050 * 13550 200

Illustration 83

g06224891

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							* 5150 * 5150 5600 * 11500 * 11500 220
6000 240				* 5900 5850 * 12900 12150			* 4750 4800 6800 * 10500 10250 270
4500 180			* 7400 * 7400 * 16950 * 16950	* 6300 5300 * 13050 11850	* 5250 3100		* 4500 3850 7570 * 10250 10000 260
3000 120			* 9000 7600 * 20800 17150	* 7200 5300 * 15600 11400	5800 3850		* 4750 3500 7960 * 10450 7700 320
1500 60			* 11800 7600 * 23700 16200	* 8050 5800 * 17300 11900	6800 2700		* 4700 3400 8060 * 11150 7450 320
0 0			* 11700 7300 * 20350 15700	* 7900 4900 * 16600 10600	5700 3650		* 3200 3450 7860 * 11300 7800 310
-1500 -60		* 12000 * 12000 * 27300 * 27300	* 11800 7250 * 19850 15850	* 7800 4900 * 16050 10500			8900 3750 7350 12950 8300 250
-3000 -120		* 14500 14200 * 31000 30400	* 10650 7350 * 22600 16850	* 7900 4950 * 16700 10700			* 7900 4500 8470 * 15400 10000 260
-4500 -180			* 7900 * 7900 * 16650 * 16650				* 6000 * 6450 4580 * 15000 * 15000 200

Illustration 84

g06224893

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

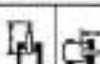
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							4750 228
6000 240				5350 210	5350 210		4350 210
4500 180			6700 264	6700 264	5600 220	4800 190	4250 168
3000 120			8450 333	8450 333	6500 256	5600 220	4350 171
1500 60			9650 380	9650 380	7300 287	6000 236	4650 183
0			10650 418	10650 418	7800 307	6500 256	5000 197
-1500 -60		11350 448	11350 448	10400 408	7900 311	6600 260	5100 201
-3000 -120		12150 480	12150 480	11100 435	8000 315	6700 264	5200 205
-4500 -180		12950 510	12950 510	11800 465	8100 319	6800 268	5300 209

Illustration 85

g06224895

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

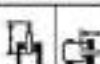
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							5150 228
6000 240				5900 236	5900 236		4750 210
4500 180			7400 291	7400 291	6350 250	5200 205	4850 190
3000 120			9300 366	9300 366	7200 283	6000 236	4750 187
1500 60			11600 457	11600 457	8000 315	6800 268	5000 197
0			13100 515	13100 515	8800 347	7600 299	5450 215
-1500 -60		12000 472	12000 472	11000 435	8900 351	7700 303	5500 219
-3000 -120		12700 500	12700 500	11700 461	9000 354	7800 307	5600 221
-4500 -180		13500 530	13500 530	12500 492	9100 358	7900 311	5700 225

Illustration 86

g06224896

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Reach Boom with a 2.9 m (9 ft 6 inch) Stick and 4200 kg (9260 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Load (metric)	1500 80	3000 120	4500 180	6000 240	7500 300			
								(mm) (inch)
7500 300								4550 179
6000 240								4350 171
4500 180								4150 163
3000 120								3950 155
1500 60								3750 147
0								3550 139
-1500 -60								3350 131
-3000 -120								3150 123
-4500 -180								2950 115
-6000 -240								2750 107
-7500 -300								2550 100
-9000 -360								2350 92
-10500 -420								2150 84
-12000 -480								1950 76
-13500 -540								1750 68
-15000 -600								1550 60
-16500 -660								1350 52
-18000 -720								1150 44
-19500 -780								950 36
-21000 -840								750 28
-22500 -900								550 20
-24000 -960								350 12
-25500 -1020								150 4
-27000 -1080								-50 -2
-28500 -1140								-250 -10
-30000 -1200								-450 -18
-31500 -1260								-650 -26
-33000 -1320								-850 -34
-34500 -1380								-1050 -42
-36000 -1440								-1250 -50
-37500 -1500								-1450 -58
-39000 -1560								-1650 -66
-40500 -1620								-1850 -74
-42000 -1680								-2050 -82
-43500 -1740								-2250 -90
-45000 -1800								-2450 -98
-46500 -1860								-2650 -106
-48000 -1920								-2850 -114
-49500 -1980								-3050 -122
-51000 -2040								-3250 -130
-52500 -2100								-3450 -138
-54000 -2160								-3650 -146
-55500 -2220								-3850 -154
-57000 -2280								-4050 -162
-58500 -2340								-4250 -170
-60000 -2400								-4450 -178
-61500 -2460								-4650 -186
-63000 -2520								-4850 -194
-64500 -2580								-5050 -202
-66000 -2640								-5250 -210
-67500 -2700								-5450 -218
-69000 -2760								-5650 -226
-70500 -2820								-5850 -234
-72000 -2880								-6050 -242
-73500 -2940								-6250 -250
-75000 -3000								-6450 -258
-76500 -3060								-6650 -266
-78000 -3120								-6850 -274
-79500 -3180								-7050 -282
-81000 -3240								-7250 -290
-82500 -3300								-7450 -298
-84000 -3360								-7650 -306
-85500 -3420								-7850 -314
-87000 -3480								-8050 -322
-88500 -3540								-8250 -330
-90000 -3600								-8450 -338
-91500 -3660								-8650 -346
-93000 -3720								-8850 -354
-94500 -3780								-9050 -362
-96000 -3840								-9250 -370
-97500 -3900								-9450 -378
-99000 -3960								-9650 -386
-100500 -4020								-9850 -394
-102000 -4080								-10050 -402
-103500 -4140								-10250 -410
-105000 -4200								-10450 -418
-106500 -4260								-10650 -426
-108000 -4320								-10850 -434
-109500 -4380								-11050 -442
-111000 -4440								-11250 -450
-112500 -4500								-11450 -458
-114000 -4560								-11650 -466
-115500 -4620								-11850 -474
-117000 -4680								-12050 -482
-118500 -4740								-12250 -490
-120000 -4800								-12450 -498
-121500 -4860								-12650 -506
-123000 -4920								-12850 -514
-124500 -4980								-13050 -522
-126000 -5040								-13250 -530
-127500 -5100								-13450 -538
-129000 -5160								-13650 -546
-130500 -5220								-13850 -554
-132000 -5280								-14050 -562
-133500 -5340								-14250 -570
-135000 -5400								-14450 -578
-136500 -5460								-14650 -586
-138000 -5520								-14850 -594
-139500 -5580								-15050 -602
-141000 -5640								-15250 -610
-142500 -5700								-15450 -618
-144000 -5760								-15650 -626
-145500 -5820								-15850 -634
-147000 -5880								-16050 -642
-148500 -5940								-16250 -650
-150000 -6000								-16450 -658
-151500 -6060								-16650 -666
-153000 -6120								-16850 -674
-154500 -6180								-17050 -682
-156000 -6240								-17250 -690
-157500 -6300								-17450 -698
-159000 -6360								-17650 -706
-160500 -6420								-17850 -714
-162000 -6480								-18050 -722
-163500 -6540								-18250 -730
-165000 -6600								-18450 -738
-166500 -6660								-18650 -746
-168000 -6720								-18850 -754
-169500 -6780								-19050 -762
-171000 -6840								-19250 -770
-172500 -6900								-19450 -778
-174000 -6960								-19650 -786
-175500 -7020								-19850 -794
-177000 -7080								-20050 -802
-178500 -7140								-20250 -810
-180000 -7200								-20450 -818
-181500 -7260								-20650 -826
-183000 -7320								-20850 -834
-184500 -7380								-21050 -842
-186000 -7440								-21250 -850
-187500 -7500								-21450 -858
-189000 -7560								-21650 -866
-190500 -7620								-21850 -874
-192000 -7680								-22050 -882
-193500 -7740								-22250 -890
-195000 -7800								-22450 -898
-196500 -7860								-22650 -906
-198000 -7920								-22850 -914
-199500 -7980								-23050 -922
-201000 -8040								-23250 -930
-202500 -8100								-23450 -938
-204000 -8160								-23650 -946
-205500 -8220								-23850 -954
-207000 -8280								-24050 -962
-208500 -8340								-24250 -970
-210000 -8400								-24450 -978
-211500 -8460								-24650 -986
-213000 -8520								-24850 -994
-214500 -8580								-25050 -1002
-216000 -8640								-25250 -1010
-217500 -8700								-25450 -1018
-219000 -8760								-25650 -1026
-220500 -8820								-25850 -1034
-222000 -8880								-26050 -1042
-223500 -8940								-26250 -1050
-225000 -9000								-26450 -1058
-226500 -9060								-26650 -1066
-228000 -9120								-26850 -1074
-229500 -9180								-27050 -1082
-231000 -9240								-27250 -1090
-232500 -9300								-27450 -1098
-234000 -9360								-27650 -1106
-235500 -9420								-27850 -1114
-237000 -9480								-28050 -1122
-238500 -9540								-28250 -1130
-240000 -9600								-28450 -1138
-241500 -9660								-28650 -1146
-243000 -9720								-28850 -1154
-244500 -9780								-29050 -1162
-246000 -9840								-29250 -1170
-247500 -9900								-29450 -1178
-249000 -9960								-29650 -1186
-250500 -10020								-29850 -1194
-252000 -10080								-30050 -1202
-253500 -10140								-30250 -1210
-255000 -10200								-30450 -1218
-256500 -10260								

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

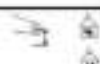
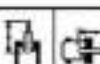
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
1500 300				4550 4550		3950 6750	6750 340
6000 240				4950 10850	4950 10850	3850 8050	7290 290
4500 180				5450 11800	5450 11700	3850 11200	7330 320
3000 120			7900 17050	7900 17000	6250 13500	5450 11850	8390 330
1500 60			9600 20850	7400 16950	7050 15300	5650 12750	9450 340
0 0		6250 14350	6250 14350	7100 22700	7850 18500	5550 11950	8290 330
-1500 -300	6700 14500	6700 14300	10800 24450	10800 24450	10600 22950	10900 16150	11900 7780
-6000 -240	11450 25650	11450 25650	14050 30450	13790 29390	9950 21450	7100 15300	6950 3050
-4500 -180		11200 24050	11200 24050	9850 17650	7300 16750		5900 220

Illustration 89

g06224811

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

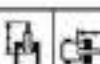
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
1500 300				4550 4550		4300 8550	6750 240
6000 240				5450 11850	5450 11850	4000 8800	7290 290
4500 180				6000 13000	5450 11700	5650 12350	7330 320
3000 120			8700 18750	7900 17000	6850 14900	5750 12400	8390 330
1500 60			10890 22900	7400 16950	7800 16900	4950 10700	9450 340
0 0		8800 18200	8800 18200	7800 25850	7850 18500	5550 11950	8290 330
-1500 -300	7100 15900	7100 15900	11400 25300	11400 25300	11100 25350	11800 16150	11900 7780
-6000 -240	12100 27790	12100 27790	15900 33800	15700 29300	10950 23700	7100 15300	6950 3050
-4500 -180		12400 28250	12400 28250	8950 19300	7300 16750		5900 220

Illustration 90

g06224818

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

Reach (mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 91

g06224766

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 92

g06224771

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Level (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Level (inch)
1500 300							1500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0 0							0 0
-600 -60							-600 -60
-1200 -120							-1200 -120
-1800 -180							-1800 -180
-2400 -240							-2400 -240
-3000 -300							-3000 -300
-3600 -360							-3600 -360
-4200 -420							-4200 -420
-4800 -480							-4800 -480
-5400 -540							-5400 -540
-6000 -600							-6000 -600
-6600 -660							-6600 -660
-7200 -720							-7200 -720
-7800 -780							-7800 -780
-8400 -840							-8400 -840
-9000 -900							-9000 -900
-9600 -960							-9600 -960
-10200 -1020							-10200 -1020
-10800 -1080							-10800 -1080
-11400 -1140							-11400 -1140
-12000 -1200							-12000 -1200
-12600 -1260							-12600 -1260
-13200 -1320							-13200 -1320
-13800 -1380							-13800 -1380
-14400 -1440							-14400 -1440
-15000 -1500							-15000 -1500
-15600 -1560							-15600 -1560
-16200 -1620							-16200 -1620
-16800 -1680							-16800 -1680
-17400 -1740							-17400 -1740
-18000 -1800							-18000 -1800
-18600 -1860							-18600 -1860
-19200 -1920							-19200 -1920
-19800 -1980							-19800 -1980
-20400 -2040							-20400 -2040
-21000 -2100							-21000 -2100
-21600 -2160							-21600 -2160
-22200 -2220							-22200 -2220
-22800 -2280							-22800 -2280
-23400 -2340							-23400 -2340
-24000 -2400							-24000 -2400
-24600 -2460							-24600 -2460
-25200 -2520							-25200 -2520
-25800 -2580							-25800 -2580
-26400 -2640							-26400 -2640
-27000 -2700							-27000 -2700
-27600 -2760							-27600 -2760
-28200 -2820							-28200 -2820
-28800 -2880							-28800 -2880
-29400 -2940							-29400 -2940
-30000 -3000							-30000 -3000
-30600 -3060							-30600 -3060
-31200 -3120							-31200 -3120
-31800 -3180							-31800 -3180
-32400 -3240							-32400 -3240
-33000 -3300							-33000 -3300
-33600 -3360							-33600 -3360
-34200 -3420							-34200 -3420
-34800 -3480							-34800 -3480
-35400 -3540							-35400 -3540
-36000 -3600							-36000 -3600
-36600 -3660							-36600 -3660
-37200 -3720							-37200 -3720
-37800 -3780							-37800 -3780
-38400 -3840							-38400 -3840
-39000 -3900							-39000 -3900
-39600 -3960							-39600 -3960
-40200 -4020							-40200 -4020
-40800 -4080							-40800 -4080
-41400 -4140							-41400 -4140
-42000 -4200							-42000 -4200
-42600 -4260							-42600 -4260
-43200 -4320							-43200 -4320
-43800 -4380							-43800 -4380
-44400 -4440							-44400 -4440
-45000 -4500							-45000 -4500
-45600 -4560							-45600 -4560
-46200 -4620							-46200 -4620
-46800 -4680							-46800 -4680
-47400 -4740							-47400 -4740
-48000 -4800							-48000 -4800
-48600 -4860							-48600 -4860
-49200 -4920							-49200 -4920
-49800 -4980							-49800 -4980
-50400 -5040							-50400 -5040
-51000 -5100							-51000 -5100
-51600 -5160							-51600 -5160
-52200 -5220							-52200 -5220
-52800 -5280							-52800 -5280
-53400 -5340							-53400 -5340
-54000 -5400							-54000 -5400
-54600 -5460							-54600 -5460
-55200 -5520							-55200 -5520
-55800 -5580							-55800 -5580
-56400 -5640							-56400 -5640
-57000 -5700							-57000 -5700
-57600 -5760							-57600 -5760
-58200 -5820							-58200 -5820
-58800 -5880							-58800 -5880
-59400 -5940							-59400 -5940
-60000 -6000							-60000 -6000
-60600 -6060							-60600 -6060
-61200 -6120							-61200 -6120
-61800 -6180							-61800 -6180
-62400 -6240							-62400 -6240
-63000 -6300							-63000 -6300
-63600 -6360							-63600 -6360
-64200 -6420							-64200 -6420
-64800 -6480							-64800 -6480
-65400 -6540							-65400 -6540
-66000 -6600							-66000 -6600
-66600 -6660							-66600 -6660
-67200 -6720							-67200 -6720
-67800 -6780							-67800 -6780
-68400 -6840							-68400 -6840
-69000 -6900							-69000 -6900
-69600 -6960							-69600 -6960
-70200 -7020							-70200 -7020
-70800 -7080							-70800 -7080
-71400 -7140							-71400 -7140
-72000 -7200							-72000 -7200
-72600 -7260							-72600 -7260
-73200 -7320							-73200 -7320
-73800 -7380							-73800 -7380
-74400 -7440							-74400 -7440
-75000 -7500							-75000 -7500
-75600 -7560							-75600 -7560
-76200 -7620							-76200 -7620
-76800 -7680							-76800 -7680
-77400 -7740							-77400 -7740
-78000 -7800							-78000 -7800
-78600 -7860							-78600 -7860
-79200 -7920							-79200 -7920
-79800 -7980							-79800 -7980
-80400 -8040							-80400 -8040
-81000 -8100							-81000 -8100
-81600 -8160							-81600 -8160
-82200 -8220							-82200 -8220
-82800 -8280							-82800 -8280
-83400 -8340							-83400 -8340
-84000 -8400							-84000 -8400
-84600 -8460							-84600 -8460
-85200 -8520							-85200 -8520
-85800 -8580							-85800 -8580
-86400 -8640							-86400 -8640
-87000 -8700							-87000 -8700
-87600 -8760							-87600 -8760
-88200 -8820							-88200 -8820
-88800 -8880							-88800 -8880
-89400 -8940							-89400 -8940
-90000 -9000							-90000 -9000
-90600 -9060							-90600 -9060
-91200 -9120							-91200 -9120
-91800 -9180							-91800 -9180
-92400 -9240							-92400 -9240
-93000 -9300							-93000 -9300
-93600 -9360							-93600 -9360
-94200 -9420							-94200 -9420
-94800 -9480							-94800 -9480
-95400 -9540							-95400 -9540
-96000 -9600							-96000 -9600
-96600 -9660							-96600 -9660
-97200 -9720							-97200 -9720
-97800 -9780							-97800 -9780
-98400 -9840							-98400 -9840
-99000 -9900							-99000 -9900
-99600 -9960							-99600 -9960
-100200 -10020							-100200 -10020
-100800 -10080							-100800 -10080
-101400 -10140							-101400 -10140
-102000 -10200							-102000 -10200
-102600 -10260							-102600 -10260
-103200 -10320							-103200 -10320
-103800 -10380							-103800 -10380
-104400 -10440							-104400 -10440
-105000 -10500							-105000 -10500
-105600 -10560							-105600 -10560
-106200 -10620							-106200 -10620
-106800 -10680							-106800 -10680
-107400 -10740							-107400 -10740
-108000 -10800							-108000 -10800
-108600 -10860</							

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

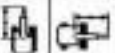
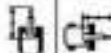
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							(mm) (inch)
7500 300							3950 155
6000 240							3650 143
4500 180							3350 131
3000 120							3050 119
1500 60							2750 107
0							2450 96
-1500 -60							2150 84
-3000 -120							1850 73
-4500 -180							1550 61
-6000 -240							1250 49
-7500 -300							950 37
-9000 -360							650 25
-10500 -420							350 14
-12000 -480							50 2

Illustration 95

g06224774

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

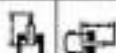
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							(mm) (inch)
7500 300							4300 169
6000 240							4000 157
4500 180							3700 145
3000 120							3400 133
1500 60							3100 121
0							2800 109
-1500 -60							2500 97
-3000 -120							2200 85
-4500 -180							1900 73
-6000 -240							1600 61
-7500 -300							1300 49
-9000 -360							1000 37
-10500 -420							700 25
-12000 -480							400 14
-13500 -540							100 4

Illustration 96

g06224789

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							3550 140
6000 240							3550 140
4500 180							3550 140
3000 120							3550 140
1500 60							3550 140
0							3550 140
-1500 -60							3550 140
-3000 -120							3550 140
-4500 -180							3550 140
-6000 -240							3550 140
-7500 -300							3550 140

Illustration 97

g06224797

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

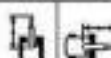
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							4000 160
6000 240							4000 160
4500 180							4000 160
3000 120							4000 160
1500 60							4000 160
0							4000 160
-1500 -60							4000 160
-3000 -120							4000 160
-4500 -180							4000 160
-6000 -240							4000 160
-7500 -300							4000 160

Illustration 98

g06226043

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
1500 300							3990 * 3950 6750
6000 240							3850 * 3850 7290
4500 180							3690 * 3690 7050
3000 120							3550 * 3550 6360
1500 60							3390 * 3390 6450
0							3250 * 3250 340
-600 0							3110 * 3110 6260
-1500 -60							2970 * 2970 1780
-3000 -120							2830 * 2830 310
-4500 -180							2690 * 2690 260

Illustration 99

g06224800

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
1500 300							4300 * 4300 6750
6000 240							4000 * 4000 7290
4500 180							3800 * 3800 7350
3000 120							3600 * 3600 8360
1500 60							3400 * 3400 8450
0							3200 * 3200 340
-600 0							3000 * 3000 6260
-1500 -60							2800 * 2800 1780
-3000 -120							2600 * 2600 310
-4500 -180							2400 * 2400 260

Illustration 100

g06224801

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Reach Boom with a 3.9 m (12 ft 8 inch) Stick and 4200 kg (9260 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Stick (inch)
7500 300								2700 250
6000 240					4050 1800			2550 2300
4500 180					4300 1950			2500 2250
3000 120					5350 2450			2500 2250
1500 60					6700 3000			2500 2250
0					8250 3600			2500 2250
-1500 -60					10000 4200			2500 2250
-3000 -120					11900 4800			2500 2250
-4500 -180					13900 5400			2500 2250
-6000 -240					16000 6000			2500 2250
-7500 -300					18200 6600			2500 2250

Illustration 101

g06224643

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Stick (inch)
7500 300								3000 280
6000 240					4500 1800			2800 2600
4500 180					4750 1950			2750 2550
3000 120					5900 2450			2750 2550
1500 60					7300 3000			2750 2550
0					8900 3600			2750 2550
-1500 -60					10700 4200			2750 2550
-3000 -120					12700 4800			2750 2550
-4500 -180					14800 5400			2750 2550
-6000 -240					17000 6000			2750 2550
-7500 -300					19300 6600			2750 2550

Illustration 102

g06224645

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Load (inch)
7500 300									2700 100
6000 240					4000 160	3500 140			2500 100
4500 180					4300 170	3800 150			2500 100
3000 120					4300 170	3800 150			2500 100
1500 60					4300 170	3800 150			2500 100
0					4300 170	3800 150			2500 100
-1500 -60					4300 170	3800 150			2500 100
-3000 -120					4300 170	3800 150			2500 100
-4500 -180					4300 170	3800 150			2500 100
-6000 -240					4300 170	3800 150			2500 100
-7500 -300					4300 170	3800 150			2500 100

Illustration 103

g06224640

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Load (inch)
7500 300									2700 100
6000 240					4500 180	3500 140			2500 100
4500 180					4750 190	3750 150			2750 110
3000 120					4750 190	3750 150			2750 110
1500 60					4750 190	3750 150			2750 110
0					4750 190	3750 150			2750 110
-1500 -60					4750 190	3750 150			2750 110
-3000 -120					4750 190	3750 150			2750 110
-4500 -180					4750 190	3750 150			2750 110
-6000 -240					4750 190	3750 150			2750 110
-7500 -300					4750 190	3750 150			2750 110

Illustration 104

g06224641

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

Reach inch	1500 60		3000 120		4500 180		6000 240		7500 300		9000 360				
														Reach inch	
1500 300														* 2700 * 6000	1320 290
3000 240									* 4190 * 6750	3900 6400				* 2550 * 6600	8300 330
4500 180									* 4300 * 6250	3850 6250				* 2500 * 5500	8820 350
6000 120					* 4400 * 6650	* 4400 * 6650	* 5300 * 8250	5200 7850	* 4700 * 8250	3750 7850	* 3600 * 6500	3750 5900	* 2550 * 5900	9250 370	
7500 60			* 5250 * 22000	* 5250 * 21000	* 6200 * 19000	* 6250 * 18500	* 4900 * 17850	* 5250 * 7150	* 3550 * 6750	* 3550 * 6750	* 2700 * 5850	2550 6950	2550 6950	9250 370	
9000 0			* 7600 * 17400	* 7600 * 17400	* 8550 * 22900	* 8550 * 22900	* 5050 * 18000	4950 7300	5400 7300	* 3600 * 6600	* 2900 * 6600	2550 8600	2550 8600	9800 360	
10500 -60	* 5700 * 12700	* 5700 * 11700	* 3800 * 22250	* 3800 * 22250	* 4700 * 16550	* 4700 * 16550	4900 16850	5300 7150	5300 7150	* 3300 * 6750	* 3300 * 6750	2700 5950	2700 5950	8730 350	
12000 -120	* 8800 * 19800	* 8800 * 18800	* 5850 * 30850	* 5850 * 27800	* 6750 * 21800	* 6750 * 20250	4500 7450	5300 7150	5300 7150	* 4050 * 6000	* 4050 * 6000	3750 8800	3750 8800	9800 320	
13500 -180	* 12300 * 23550	* 12300 * 23550	* 11750 * 26300	* 11750 * 26300	* 7550 * 26000	* 7550 * 24950	* 6700 * 8950	4950 5850	4950 5850	* 5450 * 5300	* 5450 * 5300	3950 8600	3950 8600	8870 270	
15000 -240					* 8450 * 8450	* 8450 * 8450						* 5700 * 5700	* 5700 * 5700	4910	

Illustration 105

g06224603

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	1500 60		3000 120		4500 180		6000 240		7500 300		9000 360				Reach (inch)
														Reach (inch)	
1500 300															3000 6600 3000 6600 1320 290
3000 240									4500 6600	3900 6400					2600 6750 2600 6750 8300 330
4500 180									4750 6400	3850 6250					2750 6050 2750 6050 8520 350
6000 120					7600 8250	7600 8250	8250 12100	8200 10200	5250 14400	3750 7850	3650 7850	2750 5500	2600 5700	2650 5800	9250 370
7500 60			5600 22500	5600 22500	6200 19000	6250 18500	4900 17850	5250 7150	5550 8950	2550 7600	4200 8500	2700 5750	2550 6450	2550 6550	9300 370
9000 0			8100 18450	8100 18450	8700 23750	8700 23750	4950 18000	5400 7300	4950 10600	3400 3900	4750 5850	2600 5750	3200 6450	2550 5600	9800 360
10500 -60			10500 12500	10500 12500	11800 22550	11800 22550	6950 16550	7350 16850	4500 21800	5300 20250	3350 7150		3650 6000	2750 5300	8730 350
12000 -120			14600 21800	14600 21800	16450 30750	16450 27800	8750 21800	9250 20250	4500 6750	5300 9450	3350 7150		4450 6050	3750 6600	8800 320
13500 -180			18650 26150	18650 26150	20500 33550	20500 30550	9750 26000	10250 24950	4500 5850	5300 5850	3350 7150		5050 5300	4050 5300	8870 270
15000 -240					12450 6450	12450 6450									4910

Illustration 106

g06224611

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (lb)	500 80	1000 120	1500 180	2000 240	2500 300	3000 360	3500 420	4000 480	4500 540	5000 600	5500 660	6000 720	6500 780	7000 840	7500 900	8000 960	8500 1020	9000 1080	9500 1140	10000 1200	10500 1260	11000 1320	11500 1380	12000 1440	12500 1500	13000 1560	13500 1620	14000 1680	14500 1740	15000 1800	15500 1860	16000 1920	16500 1980	17000 2040	17500 2100	18000 2160	18500 2220	19000 2280	19500 2340	20000 2400	20500 2460	21000 2520	21500 2580	22000 2640	22500 2700	23000 2760	23500 2820	24000 2880	24500 2940	25000 3000	25500 3060	26000 3120	26500 3180	27000 3240	27500 3300	28000 3360	28500 3420	29000 3480	29500 3540	30000 3600	30500 3660	31000 3720	31500 3780	32000 3840	32500 3900	33000 3960	33500 4020	34000 4080	34500 4140	35000 4200	35500 4260	36000 4320	36500 4380	37000 4440	37500 4500	38000 4560	38500 4620	39000 4680	39500 4740	40000 4800	40500 4860	41000 4920	41500 4980	42000 5040	42500 5100	43000 5160	43500 5220	44000 5280	44500 5340	45000 5400	45500 5460	46000 5520	46500 5580	47000 5640	47500 5700	48000 5760	48500 5820	49000 5880	49500 5940	50000 6000	50500 6060	51000 6120	51500 6180	52000 6240	52500 6300	53000 6360	53500 6420	54000 6480	54500 6540	55000 6600	55500 6660	56000 6720	56500 6780	57000 6840	57500 6900	58000 6960	58500 7020	59000 7080	59500 7140	60000 7200	60500 7260	61000 7320	61500 7380	62000 7440	62500 7500	63000 7560	63500 7620	64000 7680	64500 7740	65000 7800	65500 7860	66000 7920	66500 7980	67000 8040	67500 8100	68000 8160	68500 8220	69000 8280	69500 8340	70000 8400	70500 8460	71000 8520	71500 8580	72000 8640	72500 8700	73000 8760	73500 8820	74000 8880	74500 8940	75000 9000	75500 9060	76000 9120	76500 9180	77000 9240	77500 9300	78000 9360	78500 9420	79000 9480	79500 9540	80000 9600	80500 9660	81000 9720	81500 9780	82000 9840	82500 9900	83000 9960	83500 10020	84000 10080	84500 10140	85000 10200	85500 10260	86000 10320	86500 10380	87000 10440	87500 10500	88000 10560	88500 10620	89000 10680	89500 10740	90000 10800	90500 10860	91000 10920	91500 10980	92000 11040	92500 11100	93000 11160	93500 11220	94000 11280	94500 11340	95000 11400	95500 11460	96000 11520	96500 11580	97000 11640	97500 11700	98000 11760	98500 11820	99000 11880	99500 11940	100000 12000	100500 12060	101000 12120	101500 12180	102000 12240	102500 12300	103000 12360	103500 12420	104000 12480	104500 12540	105000 12600	105500 12660	106000 12720	106500 12780	107000 12840	107500 12900	108000 12960	108500 13020	109000 13080	109500 13140	110000 13200	110500 13260	111000 13320	111500 13380	112000 13440	112500 13500	113000 13560	113500 13620	114000 13680	114500 13740	115000 13800	115500 13860	116000 13920	116500 13980	117000 14040	117500 14100	118000 14160	118500 14220	119000 14280	119500 14340	120000 14400	120500 14460	121000 14520	121500 14580	122000 14640	122500 14700	123000 14760	123500 14820	124000 14880	124500 14940	125000 15000	125500 15060	126000 15120	126500 15180	127000 15240	127500 15300	128000 15360	128500 15420	129000 15480	129500 15540	130000 15600	130500 15660	131000 15720	131500 15780	132000 15840	132500 15900	133000 15960	133500 16020	134000 16080	134500 16140	135000 16200	135500 16260	136000 16320	136500 16380	137000 16440	137500 16500	138000 16560	138500 16620	139000 16680	139500 16740	140000 16800	140500 16860	141000 16920	141500 16980	142000 17040	142500 17100	143000 17160	143500 17220	144000 17280	144500 17340	145000 17400	145500 17460	146000 17520	146500 17580	147000 17640	147500 17700	148000 17760	148500 17820	149000 17880	149500 17940	150000 18000	150500 18060	151000 18120	151500 18180	152000 18240	152500 18300	153000 18360	153500 18420	154000 18480	154500 18540	155000 18600	155500 18660	156000 18720	156500 18780	157000 18840	157500 18900	158000 18960	158500 19020	159000 19080	159500 19140	160000 19200	160500 19260	161000 19320	161500 19380	162000 19440	162500 19500	163000 19560	163500 19620	164000 19680	164500 19740	165000 19800	165500 19860	166000 19920	166500 19980	167000 20040	167500 20100	168000 20160	168500 20220	169000 20280	169500 20340	170000 20400	170500 20460	171000 20520	171500 20580	172000 20640	172500 20700	173000 20760	173500 20820	174000 20880	174500 20940	175000 21000	175500 21060	176000 21120	176500 21180	177000 21240	177500 21300	178000 21360	178500 21420	179000 21480	179500 21540	180000 21600	180500 21660	181000 21720	181500 21780	182000 21840	182500 21900	183000 21960	183500 22020	184000 22080	184500 22140	185000 22200	185500 22260	186000 22320	186500 22380	187000 22440	187500 22500	188000 22560	188500 22620	189000 22680	189500 22740	190000 22800	190500 22860	191000 22920	191500 22980	192000 23040	192500 23100	193000 23160	193500 23220	194000 23280	194500 23340	195000 23400	195500 23460	196000 23520	196500 23580	197000 23640	197500 23700	198000 23760	198500 23820	199000 23880	199500 23940	200000 24000	200500 24060	201000 24120	201500 24180	202000 24240	202500 24300	203000 24360	203500 24420	204000 24480	204500 24540	205000 24600	205500 24660	206000 24720	206500 24780	207000 24840	207500 24900	208000 24960	208500 25020	209000 25080	209500 25140	210000 25200	210500 25260	211000 25320	211500 25380	212000 25440	212500 25500	213000 25560	213500 25620	214000 25680	214500 25740	215000 25800	215500 25860	216000 25920	216500 25980	217000 26040	217500 26100	218000 26160	218500 26220	219000 26280	219500 26340	220000 26400	220500 26460	221000 26520	221500 26580	222000 26640	222500 26700	223000 26760	223500 26820	224000 26880	224500 26940	225000 27000	225500 27060	226000 27120	226500 27180	227000 27240	227500 27300	228000 27360	228500 27420	229000 27480	229500 27540	230000 27600	230500 27660	231000 27720	231500 27780	232000 27840	232500 27900	233000 27960	233500 28020	234000 28080	234500 28140	235000 28200	235500 28260	236000 28320	236500 28380	237000 28440	237500 28500	238000 28560	238500 28620	239000 28680	239500 28740	240000 28800	240500 28860	241000 28920	241500 28980	242000 29040	242500 29100	243000 29160	243500 29220	244000 29280	244500 29340	245000 29400	245500 29460	246000 29520	246500 29580	247000 29640	247500 29700	248000 29760	248500 29820	249000 29880	249500 29940	250000 30000	250500 30060	251000 30120	251500 30180	252000 30240	252500 30300	253000 30360	253500 30420	254000 30480	254500 30540	255000 30600	255500 30660	256000 30720	256500 30780	257000 30840	257500 30900	258000 30960	258500 31020	259000 31080	259500 31140	260000 31200	260500 31260	261000 31320	261500 31380	262000 31440	262500 31500	263000 31560	263500 31620	264000 31680	264500 31740	265000 31800	265500 31860	266000 31920	266500 31980	267000 32040	267500 32100	268000 32160	268500 32220	269000 32280	269500 32340	270000 32400	270500 32460	271000 32520	271500 32580	272000 32640	272500 32700	273000 32760	273500 32820	274000 32880	274500 32940	275000 33000	275500 33060	276000 33120	276500 33180	277000 33240	277500 33300	278000 33360	278500 33420	279000 33480	279500 33540	280000 33600	280500 33660	281000 33720	281500 33780	282000 33840	282500 33900	283000 33960	283500 34020	284000 34080	284500 34140	285000 34200	285500 34260	286000 34320	286500 34380	287000 34440	287500 34500	288000 34560	288500 34620	289000 34680	289500 34740	290000 34800	290500 34860	291000 34920	291500 34980	292000 35040	292500 35100	293000 35160	293500 35220	294000 35280	294500 35340	295000 35400	295500 35460	296000 35520	296500 35580	297000 35640	297500 35700	298000 35760	298500 35820	299000 35880	299500 35940	300000 36000	300500 36060	301000 36120	301500 36180	302000 36240	302500 36300	303000 36360	303500 36420	304000 36480	304500 36540	305000 36600	305500 36660	306000 36720	306500 36780	307000 36840	307500 36900	308000 36960	308500 37020	309000 37080	309500 37140	310000 37200	310500 37260	311000 37320	311500 37380	312000 37440	312500 37500	313000 37560	313500 37620	314000 37680	314500 37740	315000 37800	315500 37860	316000 37920	316500 37980	317000 38040	317500 38100	318000 38160	318500 38220	319000 38280	319500 38340	320000 38400	320500 38460	321000 38520	321500 38580	322000 38640	322500 38700	323000 38760	323500 38820	324000 38880	324500 38940	325000 39000	325500 39060	326000 39120	326500 39180	327000 39240	327500 39300	328000 39360	328500 39420	329000 39480	329500 39540	330000 39600	330500 39660	331000 39720	331500 39780	332000 39840	332500 39900	333000 39960	333500 40020	334000 40080	334500 40140	335000 40200	335500 40260	336000 40320	336500 40380	337000 40440	337500 40500	338000 40560	338500 40620	339000 40680	339500 40740	340000 40800	340500 40860	341000 40920	341500 40980	342000 41040	342500 41100	343000 41160	343500 41220	344000 41280	344500 41340	345000 41400	345500 41460	346000 41520	346500 41580	347000 41640	347500 41700	348000 41760	348500 41820	349000 41880
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Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Load (inch)
7500 300									7500 300
6000 240					4050 160	4050 160			6000 240
4500 180					4350 175	3500 140			4500 180
3000 120				6400 255	6400 255	5300 210	5300 210	4300 170	3000 120
1500 60		8250 330	8250 330	8000 315	7550 300	5250 210	4650 185	4050 160	1500 60
0		7500 300	7500 300	7000 280	6550 260	5500 220	4850 195	4250 170	0
-1500 -60	5700 225	5700 225	5000 200	4000 160	3500 140	3400 135			-1500 -60
-3000 -120	8900 350	8900 350	8250 325	7650 305	7450 295	5400 210	4750 185	4150 165	-3000 -120
-4500 -180	11900 470	11900 470	10900 430	10150 405	9850 390	7800 310	7150 285	6550 260	-4500 -180
-6000 -240	12900 510	12900 510	11900 470	11150 445	10850 430	8800 340	8150 315	7550 295	-6000 -240
-7500 -300	13900 545	13900 545	12900 505	12150 475	11850 465	9800 380	9150 355	8550 335	-7500 -300

Illustration 109

g06224613

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Load (inch)
7500 300									7500 300
6000 240					4500 180	4500 180			6000 240
4500 180					4750 190	3800 150			4500 180
3000 120				7000 280	7000 280	5250 210	4650 185	4050 160	3000 120
1500 60		8800 340	8800 340	8000 315	7550 300	5250 210	4650 185	4050 160	1500 60
0		8000 310	8000 310	7000 280	6550 260	5500 220	4850 195	4250 170	0
-1500 -60	6050 240	6050 240	5400 210	4400 175	3900 155	3800 150			-1500 -60
-3000 -120	9400 370	9400 370	8750 340	8150 320	7950 310	5900 230	5250 205	4650 185	-3000 -120
-4500 -180	12400 490	12400 490	11400 450	10650 425	10350 415	8300 320	7650 295	7050 275	-4500 -180
-6000 -240	13400 525	13400 525	12400 485	11650 460	11350 450	9300 350	8650 330	8050 310	-6000 -240
-7500 -300	14400 560	14400 560	13400 520	12650 500	12350 490	10300 390	9650 370	9050 350	-7500 -300

Illustration 110

g06224615

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Reach (inch)	2500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Reach (inch)
7500 300								7500 300
6800 240					4150 165	4150 165		6800 240
4500 180					4300 170	3950 155		4500 180
3900 120				8400 330	8400 330	8300 320		3900 120
3600 80		3250 125	3250 125	8300 320	8300 320	8250 315	5250 205	3600 80
0 0		7900 310	7600 295	7500 290	7500 290	7500 290	7150 280	0 0
-600 -24	5700 225	5700 225	5800 228	6000 232	6250 245	6500 255		-600 -24
-3800 -150	8800 345	8800 345	10400 408	10500 410	10500 410	10500 410	8450 325	-3800 -150
-4500 -180	12900 505	12900 505	13500 525	13500 525	13500 525	13500 525	10500 410	-4500 -180
-6000 -236	20500 805	20500 805	20500 805	20500 805	20500 805	20500 805	17000 665	-6000 -236

Illustration 111

g06224620

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	2500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Reach (inch)
7500 300								7500 300
6800 240					4500 175	4500 175		6800 240
4500 180					4700 185	3950 155		4500 180
3900 120				7100 275	7100 275	7000 270		3900 120
3600 80		3800 145	3800 145	8300 320	8300 320	8250 315	5250 205	3600 80
0 0		7900 310	7600 295	7500 290	7500 290	7500 290	7150 280	0 0
-600 -24	6950 275	6950 275	7000 278	7250 290	7500 300	7750 310		-600 -24
-3800 -150	9400 340	9400 340	10400 408	10500 410	10500 410	10500 410	8450 325	-3800 -150
-4500 -180	12650 500	12650 500	13500 525	13500 525	13500 525	13500 525	10500 410	-4500 -180
-6000 -236	20750 810	20750 810	20750 810	20750 810	20750 810	20750 810	17000 665	-6000 -236

Illustration 112

g06224624

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Load (inch)
								
1500 300								* 2300 * 2700 7300
6000 240					* 4050 * 4050			* 6000 * 6000 290
4500 180					* 6750 * 6750			* 2550 * 2550 8300
3000 120					* 4300 * 4000			* 5600 * 5600 300
1500 60					* 3350 * 3600			* 2500 * 2500 8900
1500 60								* 5800 * 5800 390
1500 60								* 2350 * 2350 9250
1500 60								* 5900 * 5900 300
1500 60								* 2350 * 2350 9500
1500 60								* 5950 * 5950 370
1500 60								* 2350 * 2350 9800
1500 60								* 5950 * 5950 360
1500 60								* 2350 * 2350 10100
1500 60								* 5950 * 5950 380
1500 60								* 2350 * 2350 10400
1500 60								* 5950 * 5950 390
1500 60								* 2350 * 2350 10700
1500 60								* 5950 * 5950 400
1500 60								* 2350 * 2350 11000
1500 60								* 5950 * 5950 410
1500 60								* 2350 * 2350 11300
1500 60								* 5950 * 5950 420
1500 60								* 2350 * 2350 11600
1500 60								* 5950 * 5950 430
1500 60								* 2350 * 2350 11900
1500 60								* 5950 * 5950 440
1500 60								* 2350 * 2350 12200
1500 60								* 5950 * 5950 450
1500 60								* 2350 * 2350 12500
1500 60								* 5950 * 5950 460
1500 60								* 2350 * 2350 12800
1500 60								* 5950 * 5950 470
1500 60								* 2350 * 2350 13100
1500 60								* 5950 * 5950 480
1500 60								* 2350 * 2350 13400
1500 60								* 5950 * 5950 490

Illustration 113

g06224626

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360		Load (inch)
								
1500 300								* 3000 * 3000 7300
6000 240					* 4500 * 4500			* 6000 * 6000 290
4500 180					* 6750 * 6750			* 2550 * 2550 8300
3000 120					* 4750 * 4000			* 5600 * 5600 300
1500 60					* 3400 * 3600			* 2500 * 2500 8900
1500 60								* 5850 * 5850 390
1500 60								* 2350 * 2350 9250
1500 60								* 5900 * 5900 300
1500 60								* 2350 * 2350 9500
1500 60								* 5950 * 5950 370
1500 60								* 2350 * 2350 9800
1500 60								* 5950 * 5950 360
1500 60								* 2350 * 2350 10100
1500 60								* 5950 * 5950 380
1500 60								* 2350 * 2350 10400
1500 60								* 5950 * 5950 390
1500 60								* 2350 * 2350 10700
1500 60								* 5950 * 5950 400
1500 60								* 2350 * 2350 11000
1500 60								* 5950 * 5950 410
1500 60								* 2350 * 2350 11300
1500 60								* 5950 * 5950 420
1500 60								* 2350 * 2350 11600
1500 60								* 5950 * 5950 430
1500 60								* 2350 * 2350 11900
1500 60								* 5950 * 5950 440
1500 60								* 2350 * 2350 12200
1500 60								* 5950 * 5950 450
1500 60								* 2350 * 2350 12500
1500 60								* 5950 * 5950 460
1500 60								* 2350 * 2350 12800
1500 60								* 5950 * 5950 470
1500 60								* 2350 * 2350 13100
1500 60								* 5950 * 5950 480
1500 60								* 2350 * 2350 13400
1500 60								* 5950 * 5950 490

Illustration 114

g06224637

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Reach Boom with a 2.5 m (8 ft 2 inch) Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (inch)
7500 300								7500 300
6000 240				5350 210	5350 210			6000 240
4500 180			6700 265	6700 265	5750 225	5750 225	4600 180	4500 180
3000 120			8450 330	8450 330	6950 275	6950 275	4350 165	3000 120
1500 60			9950 390	9950 390	7350 290	7350 290	4050 150	1500 60
0			10650 415	10650 415	7800 305	7800 305	3850 145	0
-1500 -60		10950 425	11350 445	10900 425	7800 305	6850 265		-1500 -60
-3000 -120		13150 515	13150 515	10950 430	7050 275	6850 265		-3000 -120
-4500 -180		20500 805	20500 805	10000 390	6700 260	6250 240		-4500 -180
-6000 -240			7850 305	7850 305				-6000 -240

Illustration 115

g06225424

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (inch)
7500 300								7500 300
6000 240				5350 210	5350 210			6000 240
4500 180			7400 285	7400 285	6250 240	6250 240	4600 180	4500 180
3000 120			8900 345	8900 345	7200 280	7200 280	4350 165	3000 120
1500 60			10000 390	10000 390	7550 295	7550 295	4050 150	1500 60
0			10700 415	10700 415	7800 305	7800 305	3850 145	0
-1500 -60		12000 465	12000 465	10800 420	7000 270	6850 265		-1500 -60
-3000 -120		14550 565	14550 565	10950 430	7050 275	6850 265		-3000 -120
-4500 -180		21900 835	21900 835	10000 390	6700 260	6250 240		-4500 -180
-6000 -240			7900 305	7900 305				-6000 -240

Illustration 116

g06225425

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Stick (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							4750 * 4750 5800 10500 * 10500 220
6000 240				5350 * 5350 11700 * 11700			4350 * 4350 6830 8800 * 8800 270
4500 180			8700 * 8700 14500 * 14500	5750 * 5750 12550 * 12550	4800 4400		4250 * 4250 7570 8400 * 8400 300
3000 120			8450 * 8450 18150 * 18150	5950 * 5950 12850 * 12850	5850 4350		4350 * 4350 4000 7960 8600 * 8600 320
1500 60			9950 * 9950 23450 * 23450	7300 * 7300 15750 * 15750	8000 4250		4850 * 4850 8080 8450 * 8450 320
0 0			10450 * 10450 23800 * 23800	8300 * 8300 18250 * 18250	9600 4200		5200 * 5200 7860 8850 * 8850 310
-1500 -60		10350 * 10350 25800 * 25800	10500 * 10500 22700 * 22700	8250 * 8250 17750 * 17750			8750 * 8750 4300 7350 13900 * 13900 290
-3000 -120		13150 * 13150 28500 * 28500	8350 * 8350 20950 * 20950	7050 * 7050 15150 * 15150			8300 * 8300 5150 6470 13900 * 13900 260
-4500 -180			7750 * 7750 14350 * 14350				8200 * 8200 4980 13550 * 13550 300

Illustration 117

g06226077

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

Stick (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							5750 * 5750 5800 11500 * 11500 220
6000 240				5900 * 5900 12900 * 12900			4750 * 4750 6830 10500 * 10500 270
4500 180			7400 * 7400 15350 * 15350	6350 * 6350 13300 * 13300	5250 4400		4650 * 4650 7570 10250 * 10250 300
3000 120			9900 * 9900 20800 * 20800	7200 * 7200 15600 * 15600	6250 4350		4750 * 4750 4000 7960 10450 * 10450 320
1500 60			11900 * 11900 23700 * 23700	8500 * 8500 17400 * 17400	8400 4250		5100 * 5100 8080 11150 * 11150 320
0 0			11700 * 11700 25350 * 25350	8600 * 8600 18600 * 18600	8350 4200		5700 * 5700 7860 12500 * 12500 310
-1500 -60		10000 * 10000 27300 * 27300	7800 * 7800 25100 * 25100	6800 * 6800 16850 * 16850			8500 * 8500 4300 7350 14400 * 14400 290
-3000 -120		14550 * 14550 31500 * 31500	8950 * 8950 22900 * 22900	7800 * 7800 16700 * 16700			7000 * 7000 5150 6470 15400 * 15400 260
-4500 -180			7300 * 7300 16550 * 16550				8050 * 8050 4980 15000 * 15000 300

Illustration 118

g06225423

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
1900 300							4750 10050
5800 240				5350 11700	5350 11700		4350 8800
4500 180			6700 14500	5750 12550	4800 10650	4450 9400	4250 8650
3800 120			8450 18550	6900 15150	5850 12850	4350 9350	4050 8000
1500 80			9950 21950	7300 16150	6000 13250	4250 9350	3850 8500
0 0			10650 23000	8300 18350	7000 15350	5200 12450	3950 10050
-500 -80		1050 25800	11350 25800	10500 22700	8250 18300		4150 13000
-3800 -120		13150 28500	13150 28500	10350 22950	7050 15550		6300 13900
-4500 -180			7850 17350	7850 17350			6300 13900

Illustration 119

g06225400

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

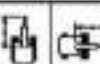
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							5750 11500
6000 240				5900 12900	5900 12900		4750 10500
4500 180			7400 16350	6350 13950	5250 11650	4450 9400	4250 8650
3800 120			8900 19600	7200 15850	6250 13850	4350 9350	4050 8000
1500 80			10600 23300	8650 19050	7350 16150	6450 14250	5100 11550
0 0			10700 23300	8800 19300	7500 16300	6600 14400	5200 11650
-500 -80		12000 26700	12000 26700	10600 23300	8650 19050		6550 14400
-3800 -120		14550 31900	14550 31900	10650 23350	8700 19100		7000 15400
-4500 -180			7900 17350	7900 17350			6850 15000

Illustration 120

g06225404

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

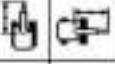
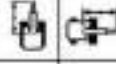
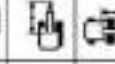
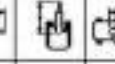
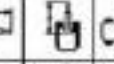
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Counterweight (inch)
7500 300							4750 10550
6000 240				5350 11700	5350 11700		4350 9600
4500 180			6700 14500	6700 14500	5750 12550	4850 10800	4250 9400
3000 120			8450 18850	8450 18850	6050 13350	5850 12900	4050 9000
1500 60			9950 21950	9950 21950	6800 15050	6500 14500	3900 8600
0			10650 23900	10650 23900	7800 17250	6200 13950	3200 7150
-1500 -60		10950 25900	11350 25900	10500 22700	8850 19550		4350 9600
-3000 -120		13850 30900	13850 30900	10500 23500	7050 15500		5200 11500
-4500 -180			7850 17350	7850 17350			6200 13550

Illustration 121

g06225407

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

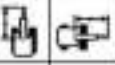
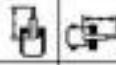
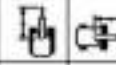
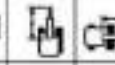
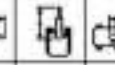
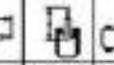
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Counterweight (inch)
7500 300							5750 12550
6000 240				5300 11500	5300 11500		4750 10500
4500 180			7400 16350	7400 16350	6250 13850	5250 11600	4400 9800
3000 120			9300 20800	9300 20800	7200 15800	6250 13850	4050 9000
1500 60			10600 23700	10600 23700	8000 17600	6500 14500	3900 8600
0			11700 25750	11700 25750	8800 19400	6450 14250	3200 7150
-1500 -60		12000 27300	12000 27300	10500 23100	8850 19550		4350 9600
-3000 -120		14550 31500	14550 31500	10500 23500	7050 15500		5200 11500
-4500 -180			7900 17350	7900 17350			6200 13550

Illustration 122

g06225409

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

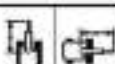
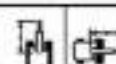
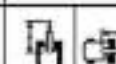
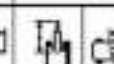
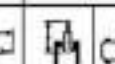
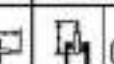
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							4750 1850
6000 240				5350 11700	5350 11700		4350 1600
4500 180			6700 14500	6700 14500	5750 12550	4000 14500	4250 1400
3000 120			8450 18150	8450 18150	6900 13750	5550 12350	4350 1300
1500 60			9950 21450	9950 21450	7300 15750	6000 13050	4050 10250
0			10650 23000	10650 23000	7900 16350	6200 13450	4300 11450
-1500		11350 25800	11350 25800	10500 22700	7800 16850	5700 12300	4150 12900
-3000		11850 26500	11850 26500	10500 22700	7800 16850	5700 12300	4150 12900
-4500			7850 14950	7850 14950			6200 13550
-6000							

Illustration 123

g06225413

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

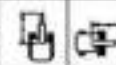
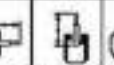
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							5150 1650
6000 240				5300 12500	5300 12500		4750 1650
4500 180			7900 15350	7900 15350	6350 13600	5250 12350	4650 1300
3000 120			9000 20800	9000 20800	7200 13750	6250 13650	4750 10450
1500 60			10900 23700	10900 23700	8000 15700	6600 14200	5100 11750
0			11700 25350	11700 25350	8800 16350	6800 14050	5700 12500
-1500		12000 27300	12000 27300	11600 25300	8600 16850	6700 12300	4700 14500
-3000		14500 31500	14500 31500	10900 22800	7800 16700	5750 12450	7000 15400
-4500			7300 16550	7300 16550			8550 15000
-6000							

Illustration 124

g06225415

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

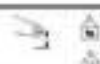
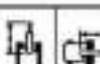
(mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300							4750 187
6000 240				5350 211	5350 211		4350 171
4500 180			6700 264	6700 264	4800 190	4800 190	4250 167
3000 120			8450 333	8450 333	5850 230	5850 230	4050 160
1500 60			9850 388	9850 388	6850 270	6850 270	3850 152
0 0			10850 426	10850 426	7850 309	7850 309	3650 144
-1500 -60		11850 466	11850 466	12850 505	12850 505		3450 136
-3000 -120		13850 543	13850 543	14850 583	14850 583		3250 128
-4500 -180		15850 620	15850 620	16850 660	16850 660		3050 120

Illustration 125

g06225416

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300							5150 203
6000 240				5500 217	5500 217		4750 187
4500 180			7400 291	7400 291	5250 207	5250 207	4550 179
3000 120			9000 354	9000 354	6250 246	6250 246	4350 171
1500 60			10800 424	10800 424	7250 285	7250 285	4150 163
0 0			11700 460	11700 460	8250 324	8250 324	3950 155
-1500 -60		12600 496	12600 496	13600 533	13600 533		3750 147
-3000 -120		14500 573	14500 573	15500 612	15500 612		3550 139
-4500 -180		16400 650	16400 650	17400 690	17400 690		3350 131

Illustration 126

g06225417

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Reach Boom with a 2.9 m (9 ft 5 inch) Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

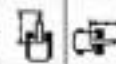
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							6150 240
6000 240				4350 168	4550 178		7230 283
4500 180				5430 212	5450 213	5100 200	7300 287
3000 120			7300 287	8250 323	8050 315	5450 213	8360 329
1500 60			9600 376	10500 410	10500 410	5950 233	9450 370
0 0		8250 323	10500 410	10500 410	12700 500	8950 350	13400 526
-1500 -60	6700 263	6700 263	10600 414	10500 410	11800 460	8950 350	14000 549
-3000 -120	11450 450	11450 450	14050 552	13300 520	13300 520	11800 460	18750 738
-4500 -180	25650 1009	25650 1009	30450 1198	27400 1076	27400 1076		40300 1579
-6000 -240		11200 438	11200 438	8050 315			5900 230

Illustration 127

g06225386

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

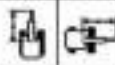
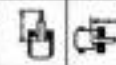
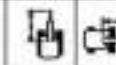
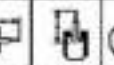
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							6150 240
6000 240				5450 212	5450 213		7230 283
4500 180				6000 235	6000 235	5650 221	7300 287
3000 120			8700 340	8250 323	8050 315	5450 213	8360 329
1500 60			10500 410	10500 410	10500 410	5950 233	9450 370
0 0		8600 335	10600 414	10500 410	12700 500	8950 350	13400 526
-1500 -60	7100 278	7100 278	11400 448	10700 417	10700 417	8950 350	14000 549
-3000 -120	12100 472	12100 472	15900 618	14950 580	14950 580	11800 460	18750 738
-4500 -180	27650 1089	27650 1089	33600 1308	29700 1120	29700 1120		40300 1579
-6000 -240		12400 485	12400 485	8950 350			5900 230

Illustration 128

g06225389

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

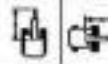
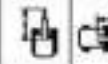
Reach (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							6150 240
6000 240				4350 10850	4350		3650 8050
4500 180				5450 11800	5450	3300 8850	3600 7900
3000 120			7300 17050	6250 13500	5450 11850	4400 9450	3650 8050
1500 60			8600 20650	10500 18550	5850 12300	4300 9000	3650 7950
0 0		6250 14350	6250 14350	10500 22700	10500 17550	8550 12100	3700 8700
+7500 300	6700 16900	6700 14300	10800 24450	10800 22350	10800 17750	8550 11950	4000 8750
+6000 240	11850 25650	11850 23950	14050 30450	14050 27450	14050 17300	8550 12050	4000 10300
+4500 180		7030 24080	7030 24080	8050 17150	8050		6700 10400

Illustration 129

g06225383

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

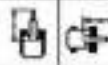
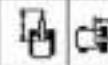
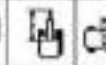
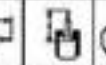
Reach (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							6150 240
6000 240				5450 11850	5450		4000 8800
4500 180				5850 13000	5850	4500 9650	3600 8600
3000 120			8700 18750	8850 14000	8080 13180	4400 9450	4000 8000
1500 60			10250 22300	10500 18550	10500 12500	4300 9000	4250 7550
0 0		6600 15200	6600 15200	10800 25950	10800 17550	8350 12100	3700 8700
+7500 300	7100 15800	7100 15800	11400 25900	11700 25350	10850 17650	8350 13680	4000 12150
+6000 240	12100 27150	12100 27150	15500 33600	15500 23700	15500 17900	8350 12050	4000 10300
+4500 180		12400 26950	12400 26950	13950 19900	13950		6750 10250

Illustration 130

g06225385

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

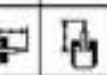
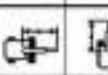
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300					4550 180	4550 180	3350 132
6000 240					4950 198	4950 198	3650 144
4500 180					5450 216	5450 216	3950 156
3000 120					5950 238	5950 238	4250 168
1500 60					6450 256	6450 256	4550 180
0 0					6950 276	6950 276	4850 192
-1500 -60					7450 296	7450 296	5150 204
-3000 -120					7950 316	7950 316	5450 216
-4500 -180					8450 336	8450 336	5750 228
-6000 -240					8950 356	8950 356	6050 240
-7500 -300					9450 376	9450 376	6350 252

Illustration 131

g06225253

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

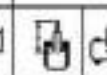
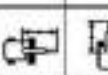
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300					4350 171	4350 171	3150 124
6000 240					4750 188	4750 188	3450 136
4500 180					5150 205	5150 205	3750 148
3000 120					5550 222	5550 222	4050 160
1500 60					5950 239	5950 239	4350 172
0 0					6350 256	6350 256	4650 184
-1500 -60					6750 273	6750 273	4950 196
-3000 -120					7150 290	7150 290	5250 208
-4500 -180					7550 307	7550 307	5550 220
-6000 -240					7950 324	7950 324	5850 232
-7500 -300					8350 341	8350 341	6150 244

Illustration 132

g06225255

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

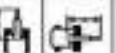
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0							0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180
-6000 -240							-6000 -240
-7500 -300							-7500 -300

Illustration 133

g06225388

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

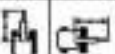
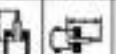
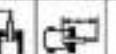
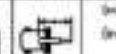
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0							0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180
-6000 -240							-6000 -240
-7500 -300							-7500 -300

Illustration 134

g06225389

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

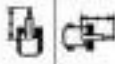
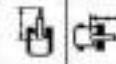
Reach (mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Stick (mm) (inch)
7500 300							6150 240
6000 240				4350 170	4350 170		7230 285
4500 180				3450 135	3450 135	3800 150	7330 288
3000 120			7300 287	7300 287	5250 207	5250 207	8380 330
1500 60			3600 142	3600 142	2950 116	2950 116	9450 370
0 0		6250 247	6250 247	10500 413	10500 413	12700 500	10450 411
-1500 -60	6700 264	6700 264	10800 421	10800 421	8100 315	8100 315	7780 306
-3000 -120	11450 450	11450 450	14050 552	14050 552	13200 518	13200 518	6350 250
-4500 -180			11200 437	8950 350			5800 228
-6000 -240			24000 940	17150 667			220

Illustration 135

g06225256

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

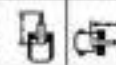
Reach (mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Stick (mm) (inch)
7500 300							6150 240
6000 240				5450 215	5450 215		7230 285
4500 180				3900 153	3900 153	3800 150	7330 288
3000 120			5700 224	5700 224	6080 239	6080 239	8380 330
1500 60			10250 400	10250 400	8450 331	8450 331	9450 370
0 0		6600 259	6600 259	11800 461	11800 461	13250 521	10450 411
-1500 -60	7100 278	7100 278	11400 445	11400 445	8650 337	8650 337	7780 306
-3000 -120	12100 474	12100 474	15500 606	15500 606	13650 536	13650 536	6350 250
-4500 -180			12400 488	8950 350			5800 228
-6000 -240			26550 1043	19900 776			220

Illustration 136

g06225258

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

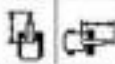
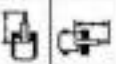
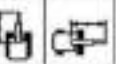
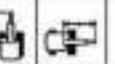
Reach (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0							0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180

Illustration 137

g06225259

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

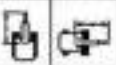
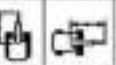
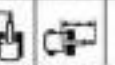
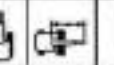
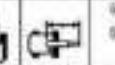
Reach (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0							0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180

Illustration 138

g06225260

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
								
7500 300						4550	4550	3050 120
6000 240						4350	4350	3050 120
4500 180						10850	10850	3050 120
3000 120						5450	5450	3050 120
1500 60						11800	11800	3050 120
0						10000	10000	3050 120
-1500 -60						7900	7900	3050 120
-3000 -120						10250	10250	3050 120
-4500 -180						3600	3600	3050 120
-6000 -240						20850	20850	3050 120
-7500 -300						8250	8250	3050 120
-9000 -360						10900	10900	3050 120
-10500 -420						13250	13250	3050 120
-12000 -480						15600	15600	3050 120
-13500 -540						17950	17950	3050 120
-15000 -600						20300	20300	3050 120
-16500 -660						22650	22650	3050 120
-18000 -720						25000	25000	3050 120
-19500 -780						27350	27350	3050 120
-21000 -840						29700	29700	3050 120
-22500 -900						32050	32050	3050 120
-24000 -960						34400	34400	3050 120
-25500 -1020						36750	36750	3050 120
-27000 -1080						39100	39100	3050 120
-28500 -1140						41450	41450	3050 120
-30000 -1200						43800	43800	3050 120
-31500 -1260						46150	46150	3050 120
-33000 -1320						48500	48500	3050 120
-34500 -1380						50850	50850	3050 120
-36000 -1440						53200	53200	3050 120
-37500 -1500						55550	55550	3050 120
-39000 -1560						57900	57900	3050 120
-40500 -1620						60250	60250	3050 120
-42000 -1680						62600	62600	3050 120
-43500 -1740						64950	64950	3050 120
-45000 -1800						67300	67300	3050 120
-46500 -1860						69650	69650	3050 120
-48000 -1920						72000	72000	3050 120
-49500 -1980						74350	74350	3050 120
-51000 -2040						76700	76700	3050 120
-52500 -2100						79050	79050	3050 120
-54000 -2160						81400	81400	3050 120
-55500 -2220						83750	83750	3050 120
-57000 -2280						86100	86100	3050 120
-58500 -2340						88450	88450	3050 120
-60000 -2400						90800	90800	3050 120
-61500 -2460						93150	93150	3050 120
-63000 -2520						95500	95500	3050 120
-64500 -2580						97850	97850	3050 120
-66000 -2640						100200	100200	3050 120
-67500 -2700						102550	102550	3050 120
-69000 -2760						104900	104900	3050 120
-70500 -2820						107250	107250	3050 120
-72000 -2880						109600	109600	3050 120
-73500 -2940						111950	111950	3050 120
-75000 -3000						114300	114300	3050 120
-76500 -3060						116650	116650	3050 120
-78000 -3120						119000	119000	3050 120
-79500 -3180						121350	121350	3050 120
-81000 -3240						123700	123700	3050 120
-82500 -3300						126050	126050	3050 120
-84000 -3360						128400	128400	3050 120
-85500 -3420						130750	130750	3050 120
-87000 -3480						133100	133100	3050 120
-88500 -3540						135450	135450	3050 120
-90000 -3600						137800	137800	3050 120
-91500 -3660						140150	140150	3050 120
-93000 -3720						142500	142500	3050 120
-94500 -3780						144850	144850	3050 120
-96000 -3840						147200	147200	3050 120
-97500 -3900						149550	149550	3050 120
-99000 -3960						151900	151900	3050 120
-100500 -4020						154250	154250	3050 120
-102000 -4080						156600	156600	3050 120
-103500 -4140						158950	158950	3050 120
-105000 -4200						161300	161300	3050 120
-106500 -4260						163650	163650	3050 120
-108000 -4320						166000	166000	3050 120
-109500 -4380						168350	168350	3050 120
-111000 -4440						170700	170700	3050 120
-112500 -4500						173050	173050	3050 120
-114000 -4560						175400	175400	3050 120
-115500 -4620						177750	177750	3050 120
-117000 -4680						180100	180100	3050 120
-118500 -4740						182450	182450	3050 120
-120000 -4800						184800	184800	3050 120
-121500 -4860						187150	187150	3050 120
-123000 -4920						189500	189500	3050 120
-124500 -4980						191850	191850	3050 120
-126000 -5040						194200	194200	3050 120
-127500 -5100						196550	196550	3050 120
-129000 -5160						198900	198900	3050 120
-130500 -5220						201250	201250	3050 120
-132000 -5280						203600	203600	3050 120
-133500 -5340						205950	205950	3050 120
-135000 -5400						208300	208300	3050 120
-136500 -5460						210650	210650	3050 120
-138000 -5520						213000	213000	3050 120
-139500 -5580						215350	215350	3050 120
-141000 -5640						217700	217700	3050 120
-142500 -5700						220050	220050	3050 120
-144000 -5760						222400	222400	3050 120
-145500 -5820						224750	224750	3050 120
-147000 -5880						227100	227100	3050 120
-148500 -5940						229450	229450	3050 120
-150000 -6000						231800	231800	3050 120
-151500 -6060						234150	234150	3050 120
-153000 -6120						236500	236500	3050 120
-154500 -6180						238850	238850	3050 120
-156000 -6240						241200	241200	3050 120
-157500 -6300						243550	243550	3050 120
-159000 -6360						245900	245900	3050 120
-160500 -6420						248250	248250	3050 120
-162000 -6480						250600	250600	3050 120
-163500 -6540						252950	252950	3050 120
-165000 -6600						255300	255300	3050 120
-166500 -6660						257650	257650	3050 120
-168000 -6720						260000	260000	3050 120
-169500 -6780						262350	262350	3050 120
-171000 -6840						264700	264700	3050 120
-172500 -6900						267050	267050	3050 120
-174000 -6960						269400	269400	3050 120
-175500 -7020						271750	271750	3050 120
-177000 -7080						274100	274100	3050 120
-178500 -7140						276450	276450	3050 120
-180000 -7200						278800	278800	3050 120
-181500 -7260						281150	281150	3050 120
-183000 -7320						283500	283500	3050 120
-184500 -7380						285850	285850	3050 120
-186000 -7440						288200	288200	3050 120
-187500 -7500						290550	290550	3050 120
-189000 -7560						292900	292900	3050 120
-190500 -7620						295250	295250	3050 120
-192000 -7680						297600	297600	3050 120
-193500 -7740						300000	300000	3050 120
-195000 -7800						302350	302350	3050 120
-196500 -7860						304700	304700	3050 120
-198000 -7920						307050	307050	3050 120
-199500 -7980						309400	309400	3050 120
-201000 -8040						311750	311750	3050 120
-202500 -8100						314100	314100	3050 120
-204000 -8160						316450	316450	3050 120
-205500 -8220						318800	318800	3050 120
-207000 -8280						321150	321150	3050 120
-208500 -8340						323500	323500	3050 120
-210000 -8400						325850	325850	3050 120
-211500 -8460						328200	328200	3050 120
-213000 -8520						330550	330550	3050 120
-214500 -8580						332900	332900	3050 120
-216000 -8640								

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

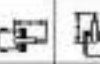
Boom (inch)	1508 60	3000 120	4180 165	5000 248	7500 300		
							(mm) (inch)
7500 300							* 2790 * 3700 7320 * 6080 * 6300 396
6000 240							* 4080 * 4280 8360 * 8750 * 8750 5600 330
4100 165							* 4300 * 4300 2580 2820 * 5380 * 5380 3180 350
3000 120			* 6480 * 6480 1308 1308 * 12850 * 12850 11400 11400	* 5380 * 5380 4700 4300 * 9280 9300 5680 5680			2050 2500 3050 370
1500 60		* 3290 * 3290 8300 8300 * 20080 * 20080 17900 17900	* 4250 * 4250 4250 4250 * 11600 12380 11360 8850	* 5280 5790 5280 4150 * 11360 12380 11360 8850			2180 2700 3230 370
8 8		* 3580 * 3580 1650 1650 * 13180 * 13180 20900 17600	* 8280 8280 1850 1850 * 11250 11900 11250 11900	* 5780 5780 4350 4350 * 12380 12380 11360 8850			2980 2980 5480 5480 9160 360
-1100 -40	* 5700 * 5700 8880 8880 * 12700 * 12700 22250 22250	* 8880 * 8880 10300 10300 * 22250 * 22250 22250 22250	* 10300 10300 11500 11500 * 14150 14150 11500 11500	* 11500 11500 11500 11500 * 14150 14150 11500 11500	* 11500 11500 11500 11500 * 14150 14150 11500 11500		3390 3250 3250 3250 3250 3250 3250 3250
-3000 -120	* 8880 * 8880 10300 10300 * 14150 * 14150 20900 20900	* 10300 * 10300 12800 12800 * 20900 * 20900 20900 20900	* 12800 12800 14150 14150 * 20900 20900 14150 14150	* 14150 14150 14150 14150 * 20900 20900 14150 14150	* 14150 14150 14150 14150 * 20900 20900 14150 14150		4050 4050 4050 4050 4050 4050 4050 4050
-4100 -165	* 12800 * 12800 17600 17600 * 20900 * 20900 20900 20900	* 17600 * 17600 20900 20900 * 20900 * 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900		5480 4500 4500 4500 4500 4500 4500 4500
-6000 -240			* 6480 * 6480 1308 1308				* 6780 * 6780 4810

Illustration 141

g06225248

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

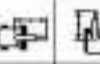
Boom (inch)	1508 60	3000 120	4180 165	5000 248	7500 300		
							(mm) (inch)
7500 300							* 3800 * 3800 7300 290 * 6800 * 6800 5680 396
6000 240							* 4400 * 4400 8360 8360 * 8800 * 8800 5680 330
4100 165							* 4700 4450 2750 2750 8920 * 5400 5680 3180 3180 350
3000 120			* 7180 * 7180 1080 1080 * 15250 * 15250 11200 11200	* 5850 * 5850 5050 4250 * 12780 12780 11400 8380	* 6250 4380 2800 2800 * 11400 8380 6100 6100		4550 3880 2880 2880 5650 5650 370
1500 60		* 3290 * 3290 8300 8300 * 21980 * 21980 18800 18800	* 4250 * 4250 4250 4250 * 11800 12380 11360 8850	* 5780 5780 4350 4350 * 12380 12380 11360 8850	* 5800 5800 4350 4350 * 12380 12380 11360 8850		2840 2840 5440 5440 9320 360
8 8		* 3580 * 3580 1650 1650 * 13180 * 13180 20900 17600	* 8280 8280 1850 1850 * 11250 11900 11250 11900	* 5780 5780 4350 4350 * 12380 12380 11360 8850	* 5780 5780 4350 4350 * 12380 12380 11360 8850		2980 2980 5480 5480 9160 360
-1100 -40	* 5700 * 5700 8880 8880 * 12700 * 12700 22250 22250	* 8880 * 8880 10300 10300 * 22250 * 22250 22250 22250	* 10300 10300 11500 11500 * 14150 14150 11500 11500	* 11500 11500 11500 11500 * 14150 14150 11500 11500	* 11500 11500 11500 11500 * 14150 14150 11500 11500		3390 3250 3250 3250 3250 3250 3250 3250
-3000 -120	* 8880 * 8880 10300 10300 * 14150 * 14150 20900 20900	* 10300 * 10300 12800 12800 * 20900 * 20900 20900 20900	* 12800 12800 14150 14150 * 20900 20900 14150 14150	* 14150 14150 14150 14150 * 20900 20900 14150 14150	* 14150 14150 14150 14150 * 20900 20900 14150 14150		4050 4050 4050 4050 4050 4050 4050 4050
-4100 -165	* 12800 * 12800 17600 17600 * 20900 * 20900 20900 20900	* 17600 * 17600 20900 20900 * 20900 * 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900	* 20900 20900 20900 20900 * 20900 20900 20900 20900		5480 4500 4500 4500 4500 4500 4500 4500
-6000 -240			* 6480 * 6480 1308 1308				* 6780 * 6780 4810

Illustration 142

g06225249

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			(mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180
-6000 -240								-6000 -240
-7500 -300								-7500 -300

Illustration 143

g06225102

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			(mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180
-6000 -240								-6000 -240
-7500 -300								-7500 -300

Illustration 144

g06225226

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

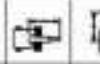
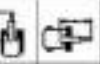
Reach (mm)	323 80	323 130	4190 180	6006 348	7508 369		
							(mm) (inch)
7508 300							* 2730 * 3708 7326 * 6080 * 6308 396
6006 248							* 4098 * 4358 * 2940 * 2568 8306 * 8758 * 8758 * 5880 * 5808 370
4190 180							* 4308 * 4308 * 2580 * 2508 8826 * 5398 * 5398 * 3180 * 3048 358
3230 128							* 6480 * 6480 * 1308 * 1308 * 4700 4358 * 2640 * 2508 9358 * 13858 * 13858 * 11458 * 11458 * 9288 370
1908 86							* 5298 * 5298 * 8380 * 8380 * 6258 6750 * 5298 4158 * 2780 * 2708 9238 * 20880 * 20880 * 17880 * 17880 * 11608 12358 * 11368 8858 * 6660 * 6458 370
0 0							* 3580 * 3580 * 1658 8288 * 1858 9580 * 5788 4858 * 2980 * 2968 9168 * 13180 * 13180 * 20980 17680 * 11258 17880 * 12388 9658 * 6480 * 6408 368
-1908 -86	* 5708 * 5708 * 8880 * 8880 * 10388 8080 * 11808 6380 * 8888 3858 * 3380 3258 8738 * 12708 * 12708 * 22550 * 22550 * 22550 11150 * 16258 17880 * 13888 8808 * 7380 7188 358						
-3006 -128	* 8808 * 8808 * 13080 * 13080 * 16158 7580 * 14858 5280 * 1718 3958 * 4080 3958 8808 * 19888 * 19888 * 30880 * 30880 * 21880 17180 * 11158 11458 * 12388 8808 * 3080 8808 328						
-4190 -188	* 12988 * 12988 * 17180 * 17180 * 2158 8180 * 6708 5480 * 12080 18188 270 * 20888 * 20888 * 28588 * 28588 * 18880 17480 * 11258 11830						* 6780 * 6780 4818
-6006 -248							

Illustration 145

g06225089

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (mm)	323 80	323 130	4190 180	6006 348	7508 369		
							(mm) (inch)
7508 300							* 2600 * 3080 7328 * 6680 * 6680 398
6006 248							* 4188 * 4308 * 2880 * 2188 8308 * 8808 * 8808 * 6158 * 6158 338
4190 180							* 4718 4458 * 2750 * 2750 8528 * 16488 1808 * 8858 * 6080 358
3006 128							* 7188 * 7188 * 1858 * 1858 * 5288 4358 * 2808 * 2680 9258 * 15258 * 15258 * 12708 * 12708 * 11688 9308 * 6188 * 6188 378
1908 86							* 9888 * 9888 * 8380 8080 * 6808 6750 * 5888 4158 * 2958 * 2680 9238 * 25880 * 25880 * 18880 16688 * 11808 12358 * 12688 8858 * 6458 * 6458 378
0 0							* 8188 * 8188 * 10788 8288 * 11808 9580 6288 4858 * 3080 3068 9168 * 16458 * 16458 * 22158 17888 * 11808 17880 11268 9658 * 7658 8780 368
-1908 -86	* 8858 * 8858 * 10480 * 10480 * 13558 8080 * 1308 6380 6188 3858 * 3380 3258 8738 * 13808 * 13808 * 23580 * 23580 * 24880 17180 * 11908 17880 11158 8808 * 5858 7188 358						
-3006 -128	* 9408 * 9408 * 14480 * 14480 * 1708 7580 * 1308 5380 6188 3858 * 4458 3880 8808 * 21908 * 21908 * 32780 32788 * 26358 17180 * 17808 11458 12288 8408 * 9888 8080 328						
-4190 -188	* 13858 * 13858 * 18058 * 18058 * 18188 8180 * 7808 6480 * 13080 18188 278 * 28758 * 28758 * 31858 * 31858 * 21758 17480 * 11808 11830						* 6388 * 6388 4818
-6006 -248							

Illustration 146

g06225092

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							+ 2700 + 2700 7500 + 6000 + 6000 300
6000 240					+ 4300 + 4300 + 2600 + 2600 8300 + 8700 + 8700 + 9600 + 9600 330		
4500 180					+ 4300 + 4300 + 2500 + 2500 8500 + 9350 + 9350 + 9500 + 9500 350		
3000 120			+ 6400 + 6400 + 5300 + 5300 + 4700 + 4700 + 3500 + 3500 8250 + 13850 + 13850 + 14450 + 14450 + 15250 + 15250 + 16050 + 16050 370				
1500 60		+ 9050 + 9050 + 8300 + 8300 + 6200 + 6200 + 5250 + 5250 + 4100 + 4100 + 2700 + 2700 8330 + 22000 + 22000 + 17000 + 17000 + 13600 + 13600 + 11100 + 11100 + 8600 + 8600 + 5950 + 5950 + 4850 + 4850 370					
0 0		+ 7900 + 7900 + 6650 + 6650 + 4950 + 4950 + 4400 + 4400 + 3950 + 3950 + 2900 + 2900 + 2500 + 2500 8160 + 11000 + 11000 + 10000 + 10000 + 9250 + 9250 + 8400 + 8400 + 7650 + 7650 + 6400 + 6400 + 5400 + 5400 360					
-1100 -60	+ 5700 + 5700 + 2800 + 2800 + 10300 + 10300 + 7800 + 7800 + 6200 + 6200 + 3800 + 3800 + 3300 + 3300 8730 + 12700 + 12700 + 22250 + 22250 + 16900 + 16900 + 14200 + 14200 + 11300 + 11300 + 8350 + 8350 + 7300 + 7300 390						
-3000 -120	+ 8900 + 8900 + 13000 + 13000 + 16150 + 16150 + 1800 + 1800 + 7450 + 7450 + 5200 + 5200 + 3750 + 3750 + 4000 + 4000 8080 + 18000 + 18000 + 30000 + 30000 + 36050 + 36050 + 41650 + 41650 + 47200 + 47200 + 52800 + 52800 + 58400 + 58400 330						
-4500 -180	+ 12000 + 12000 + 17050 + 17050 + 21150 + 21150 + 1850 + 1850 + 6700 + 6700 + 5300 + 5300 + 3900 + 3900 + 5600 + 5600 8670 + 23050 + 23050 + 28200 + 28200 + 33350 + 33350 + 38400 + 38400 + 43450 + 43450 + 48500 + 48500 + 53550 + 53550 270						
-6000 -240			+ 6450 + 6450				+ 6700 + 6700 8870

Illustration 147

g06225250

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

boom (inch)	1500 60		3000 120		4500 180		6000 240		7500 300				
												(mm) (inch)	
7500 300												+ 3600 + 3600 7320 + 6600 + 6600 298	
6000 240									+ 4900 4900 + 2800 + 2800 8300 + 9600 9600 + 11500 + 11500 338				
4500 180									+ 4700 4400 + 2750 + 2750 8520 + 10400 9400 + 10500 + 10500 358				
3000 120					+ 7100 + 7100 + 5850 + 5850 + 4850 + 4850 + 4200 + 4200 + 3600 + 3600 + 2800 + 2800 + 2000 + 2000 8250 + 13250 + 13250 + 15350 + 15350 + 17450 + 17450 + 19550 + 19550 + 21650 + 21650 + 23750 + 23750 + 25850 + 25850 378								
1500 60		+ 9000 + 9000 + 8200 + 8200 + 6100 + 6100 + 5150 + 5150 + 4100 + 4100 + 2950 + 2950 + 2000 + 2000 8238 + 21900 + 21900 + 19000 + 19000 + 16150 + 16150 + 13300 + 13300 + 10450 + 10450 + 7600 + 7600 + 4750 + 4750 + 1900 + 1900 378											
0 0		+ 8100 + 8100 + 6700 + 6700 + 5400 + 5400 + 4100 + 4100 + 2800 + 2800 + 2000 + 2000 + 1500 + 1500 8160 + 18400 + 18400 + 15550 + 15550 + 12700 + 12700 + 9850 + 9850 + 6950 + 6950 + 4050 + 4050 + 1150 + 1150 + 1500 + 1500 368											
-1100 -60	+ 6500 + 6500 + 5050 + 5050 + 3600 + 3600 + 2150 + 2150 + 1700 + 1700 + 1250 + 1250 + 800 + 800 + 350 + 350 8738 + 13300 + 13300 + 10450 + 10450 + 7600 + 7600 + 4750 + 4750 + 1900 + 1900 + 1500 + 1500 + 1050 + 1050 + 650 + 650 318												
-3000 -120	+ 9400 + 9400 + 7450 + 7450 + 5500 + 5500 + 3550 + 3550 + 2600 + 2600 + 1650 + 1650 + 1100 + 1100 + 600 + 600 8808 + 21100 + 21100 + 18250 + 18250 + 15400 + 15400 + 12550 + 12550 + 9700 + 9700 + 6850 + 6850 + 3950 + 3950 + 1000 + 1000 328												
-4500 -180	+ 12600 + 12600 + 10650 + 10650 + 8700 + 8700 + 6750 + 6750 + 4800 + 4800 + 2850 + 2850 + 1900 + 1900 + 1400 + 1400 8878 + 23750 + 23750 + 20900 + 20900 + 18050 + 18050 + 15200 + 15200 + 12350 + 12350 + 9500 + 9500 + 6650 + 6650 + 3750 + 3750 278												
-6000 -240				+ 7100 + 7100								+ 6700 + 6700 8878	

Illustration 148

g06225251

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

[illegible]

Illustration 149

g06225093

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

jezd (cm)	1500 60	2000 120	4000 160	6000 240	7500 300		jezd (cm)
							
7000 300							3000 240
6000 240							2400 190
4000 160							1600 130
3000 120							1200 100
2000 80							800 70
1500 60							600 50
1000 40							400 30
800 30							300 20
600 20							200 10
400 10							100 5
300 5							50 2
200 2							20 1
100 1							10 0.5
50 0.5							5 0.2
20 0.2							2 0.1
10 0.1							1 0.05
5 0.05							0.5 0.02
2 0.02							0.2 0.01
1 0.01							0.1 0.005
0.5 0.005							0.05 0.002
0.2 0.002							0.02 0.001
0.1 0.001							0.01 0.0005
0.05 0.0005							0.005 0.0002
0.02 0.0002							0.002 0.0001
0.01 0.0001							0.001 0.00005
0.005 0.00005							0.0005 0.00002
0.002 0.00002							0.0002 0.00001
0.001 0.00001							0.0001 0.000005
0.0005 0.000005							0.00005 0.000002
0.0002 0.000002							0.00002 0.000001
0.0001 0.000001							0.00001 0.0000005
0.00005 0.0000005							0.000005 0.0000002
0.00002 0.0000002							0.000002 0.0000001
0.00001 0.0000001							0.000001 0.00000005
0.000005 0.00000005							0.0000005 0.00000002
0.000002 0.00000002							0.0000002 0.00000001
0.000001 0.00000001							0.0000001 0.000000005
0.0000005 0.000000005							0.00000005 0.000000002
0.0000002 0.000000002							0.00000002 0.000000001
0.0000001 0.000000001							0.00000001 0.0000000005
0.00000005 0.0000000005							0.000000005 0.0000000002
0.00000002 0.0000000002							0.000000002 0.0000000001
0.00000001 0.0000000001							0.000000001 0.00000000005
0.000000005 0.00000000005							0.0000000005 0.00000000002
0.000000002 0.00000000002							0.0000000002 0.00000000001
0.000000001 0.00000000001							0.0000000001 0.000000000005
0.0000000005 0.000000000005							0.00000000005 0.000000000002
0.0000000002 0.000000000002							0.00000000002 0.000000000001
0.0000000001 0.000000000001							0.00000000001 0.0000000000005
0.00000000005 0.0000000000005							0.000000000005 0.0000000000002
0.00000000002 0.0000000000002							0.000000000002 0.0000000000001
0.00000000001 0.0000000000001							0.0

Illustration 150

g06225094

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

(mm) (inch)	1508 60	3058 120	4180 165	5800 228	7508 300				(mm) (inch)
									
7508 300							* 2780 * 2708	7300	
8008 318							* 8080 * 6808	2900	
4508 180						* 4098 * 4098	* 2180 * 2180	8388	
3008 120						* 8758 * 8758	* 5680 * 5680	3300	
1508 60						* 4308 * 4308	* 2580 * 2580	8820	
60						* 9358 * 9358	* 5580 * 5580	2800	
8						* 4708 * 4708	* 2580 * 2580	9250	
8						* 8808 * 8808	* 5680 * 5680	3700	
-1808 -68						* 5258 * 5258	* 2180 * 2180	9330	
-3008 -120						* 11358 * 11358	* 9808 * 9808	3700	
-4508 -180						* 12680 * 12680	* 11358 * 11358	3600	
-6008 -236						* 15208 * 15208	* 12180 * 12180	3600	
-7508 -295						* 17808 * 17808	* 14350 * 14350	3500	
-9008 -354						* 20408 * 20408	* 16880 * 16880	3500	
-10508 -413						* 23008 * 23008	* 19408 * 19408	3500	
-12008 -472						* 25608 * 25608	* 21908 * 21908	3500	
-13508 -531						* 28208 * 28208	* 24408 * 24408	3500	
-15008 -590						* 30808 * 30808	* 27008 * 27008	3500	
-16508 -649						* 33408 * 33408	* 29608 * 29608	3500	
-18008 -708						* 36008 * 36008	* 32208 * 32208	3500	
-19508 -767						* 38608 * 38608	* 34808 * 34808	3500	
-21008 -826						* 41208 * 41208	* 37408 * 37408	3500	
-22508 -885						* 43808 * 43808	* 40008 * 40008	3500	
-24008 -944						* 46408 * 46408	* 42608 * 42608	3500	
-25508 -1003						* 49008 * 49008	* 45208 * 45208	3500	
-27008 -1062						* 51608 * 51608	* 47808 * 47808	3500	
-28508 -1121						* 54208 * 54208	* 50408 * 50408	3500	
-30008 -1180						* 56808 * 56808	* 53008 * 53008	3500	
-31508 -1239						* 59408 * 59408	* 55608 * 55608	3500	
-33008 -1298						* 62008 * 62008	* 58208 * 58208	3500	
-34508 -1357						* 64608 * 64608	* 60808 * 60808	3500	
-36008 -1416						* 67208 * 67208	* 63408 * 63408	3500	
-37508 -1475						* 69808 * 69808	* 66008 * 66008	3500	
-39008 -1534						* 72408 * 72408	* 68608 * 68608	3500	
-40508 -1593						* 75008 * 75008	* 71208 * 71208	3500	
-42008 -1652						* 77608 * 77608	* 73808 * 73808	3500	
-43508 -1711						* 80208 * 80208	* 76408 * 76408	3500	
-45008 -1770						* 82808 * 82808	* 79008 * 79008	3500	
-46508 -1829						* 85408 * 85408	* 81608 * 81608	3500	
-48008 -1888						* 88008 * 88008	* 84208 * 84208	3500	
-49508 -1947						* 90608 * 90608	* 86808 * 86808	3500	
-51008 -2006						* 93208 * 93208	* 89408 * 89408	3500	
-52508 -2065						* 95808 * 95808	* 92008 * 92008	3500	
-54008 -2124						* 98408 * 98408	* 94608 * 94608	3500	
-55508 -2183						* 101008 * 101008	* 97208 * 97208	3500	
-57008 -2242						* 103608 * 103608	* 99808 * 99808	3500	
-58508 -2301						* 106208 * 106208	* 102408 * 102408	3500	
-60008 -2360						* 108808 * 108808	* 105008 * 105008	3500	
-61508 -2419						* 111408 * 111408	* 107608 * 107608	3500	
-63008 -2478						* 114008 * 114008	* 110208 * 110208	3500	
-64508 -2537						* 116608 * 116608	* 112808 * 112808	3500	
-66008 -2596						* 119208 * 119208	* 115408 * 115408	3500	
-67508 -2655						* 121808 * 121808	* 118008 * 118008	3500	
-69008 -2714						* 124408 * 124408	* 120608 * 120608	3500	
-70508 -2773						* 127008 * 127008	* 123208 * 123208	3500	
-72008 -2832						* 129608 * 129608	* 125808 * 125808	3500	
-73508 -2891						* 132208 * 132208	* 128408 * 128408	3500	
-75008 -2950						* 134808 * 134808	* 131008 * 131008	3500	
-76508 -3009						* 137408 * 137408	* 133608 * 133608	3500	
-78008 -3068						* 140008 * 140008	* 136208 * 136208	3500	
-79508 -3127						* 142608 * 142608	* 138808 * 138808	3500	
-81008 -3186						* 145208 * 145208	* 141408 * 141408	3500	
-82508 -3245						* 147808 * 147808	* 144008 * 144008	3500	
-84008 -3304						* 150408 * 150408	* 146608 * 146608	3500	
-85508 -3363						* 153008 * 153008	* 149208 * 149208	3500	
-87008 -3422						* 155608 * 155608	* 151808 * 151808	3500	
-88508 -3481						* 158208 * 158208	* 154408 * 154408	3500	
-90008 -3540						* 160808 * 160808	* 157008 * 157008	3500	
-91508 -3599						* 163408 * 163408	* 159608 * 159608	3500	
-93008 -3658						* 166008 * 166008	* 162208 * 162208	3500	
-94508 -3717						* 168608 * 168608	* 164808 * 164808	3500	
-96008 -3776						* 171208 * 171208	* 167408 * 167408	3500	
-97508 -3835						* 173808 * 173808	* 170008 * 170008	3500	
-99008 -3894						* 176408 * 176408	* 172608 * 172608	3500	
-100508 -3953						* 179008 * 179008	* 175208 * 175208	3500	
-102008 -4012						* 181608 * 181608	* 177808 * 177808	3500	
-103508 -4071						* 184208 * 184208	* 180408 * 180408	3500	
-105008 -4130						* 186808 * 186808	* 183008 * 183008	3500	
-106508 -4189						* 189408 * 189408	* 185608 * 185608	3500	
-108008 -4248						* 192008 * 192008	* 188208 * 188208	3500	
-109508 -4307						* 194608 * 194608	* 190808 * 190808	3500	
-111008 -4366						* 197208 * 197208	* 193408 * 193408	3500	
-112508 -4425						* 199808 * 199808	* 196008 * 196008	3500	
-114008 -4484						* 202408 * 202408	* 198608 * 198608	3500	
-115508 -4543						* 205008 * 205008	* 201208 * 201208	3500	
-117008 -4602						* 207608 * 207608	* 203808 * 203808	3500	
-118508 -4661						* 210208 * 210208	* 206408 * 206408	3500	
-120008 -4720						* 212808 * 212808	* 209008 * 209008	3500	
-121508 -4779						* 215408 * 215408	* 211608 * 211608	3500	
-123008 -4838						* 218008 * 218008	* 214208 * 214208	3500	
-124508 -4897						* 220608 * 220608	* 216808 * 216808	3500	
-126008 -4956						* 223208 * 223208	* 219408 * 219408	3500	
-127508 -5015						* 225808 * 225808	* 222008 * 222008	3500	
-129008 -5074						* 228408 * 228408	* 224608 * 224608	3500	
-130508 -5133						* 231008 * 231008	* 227208 * 227208	3500	
-132008 -5192						* 233608 * 233608	* 229808 * 229808	3500	
-133508 -5251						* 236208 * 236208	* 232408 * 232408	3500	
-135008 -5310						* 238808 * 238808	* 235008 * 235008	3500	
-136508 -5369						* 241408 * 241408	* 237608 * 237608	3500	
-138008 -5428						* 244008 * 244008	* 240208 * 240208	3500	
-139508 -5487						* 246608 * 246608	* 242808 * 242808	3500	
-141008 -5546						* 249208 * 249208	* 245408 * 245408	3500	
-142508 -5605						* 251808 * 251808	* 248008 * 248008	3500	
-144008 -5664						* 254408 * 254408	* 250608 * 250608	3500	
-145508 -5723						* 257008 * 257008	* 253208 * 253208	3500	
-147008 -5782						* 259608 * 259608	* 255808 * 255808	3500	
-148508 -5841						* 262208 * 262208	* 258408 * 258408	3500	
-150008 -5900						* 264808 * 264808	* 261008 * 261008	3500	
-151508 -5959						* 267408 * 267408	* 263608 * 263608	3500	
-153008 -6018						* 270008 * 270008	* 266208 * 266208	3500	
-154508 -6077						* 272608 * 272608	* 268808 * 268808	3500	
-156008 -6136						* 275208 * 275208	* 271408 * 271408	3500	
-157508 -6195						* 277808 * 277808	* 274008 * 274008	3500	
-159008 -6254						* 280408 * 280408	* 276608 * 276608	3500	
-160508 -6313						* 283008 * 283008	* 279208 * 279208	3500	
-162008 -6372						* 285608 * 285608	* 281808 * 281808	3500	
-163508 -6431						* 288208 * 288208	* 284408 * 284408	3500	
-165008 -6490						* 290808 * 290808	* 287008 * 287008	3500	
-166508 -6549						* 293408 * 293408	* 289608 * 289608	3500	
-168008 -6608						* 296008 * 296008	* 292208 * 292208	3500	
-169508 -6667						* 298608 * 298608	* 294808 * 294808	3500	
-171008 -6726						* 301208 * 301208	* 297408 * 297408	3500	
-172508 -6785						* 303808 * 303808	* 300008 * 300008	3500	
-174008 -6844						* 306408 * 306408	* 302608 * 302608	3500	
-175508 -6903						* 309008 * 309008	* 305208 * 305208	3500	
-177008 -6962						* 311608 * 311608	* 307808 * 307808	3500	
-178508 -7021						* 314208 * 314208	* 310408 * 310408	3500	
-180008 -7080						* 316808 * 316808	* 313008 * 313008	3500	
-181508 -7139						* 319408 * 319408	* 315608 * 315608	3500	
-183008 -7198						* 322008 * 322008	* 318208 * 318208</		

Illustration 151

g06225095

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

1508 (60")	1508 60		3058 120		4180 165		5800 228		7508 300				(mm) (inch)
												(mm) (inch)	
7508 300												* 3080 * 3008 7320 * 5680 * 6008 290	
8008 318									* 4580 * 4508 * 2080 * 2008 8388 * 8680 * 8680 * 8150 * 8158 330				
4008 100									* 4780 4558 * 2150 * 2158 8820 * 16480 1608 * 9050 * 9058 350				
3008 120					* 7180 * 7180 * 1858 * 1880 * 5258 8808 * 5258 8808 * 2080 * 2008 9250 * 15250 * 15250 * 12708 * 12780 * 11680 9808 * 8150 * 8158 370								
1508 60		* 5080 * 5080 * 8380 8288 * 8908 1880 * 5080 4258 * 2580 * 2308 9330 * 25380 * 25380 * 15880 1608 * 11808 13650 * 10680 9008 * 4480 * 6888 370											
8 8		* 8180 * 8180 * 10780 8888 * 1808 9680 * 5380 4158 * 2580 3158 9180 * 16480 * 16480 * 27158 1608 * 11808 12180 * 11658 9808 * 7080 9888 360											
-1808 -68	* 6858 * 6858 * 10480 * 10480 * 11358 8288 * 8308 9880 6388 4858 * 3650 3388 8730 * 13688 * 13688 * 23580 * 23580 * 24880 17080 * 11808 11880 13588 8758 * 8050 7308 350												
-3008 -120	* 8480 * 8480 * 14450 * 14458 * 11208 8180 * 8308 14350 6388 4858 * 4450 3758 3680 * 31188 * 31188 * 33750 * 33758 * 26258 13550 * 17808 11780 13588 8758 * 8050 8308 330												
-4508 -180	* 13888 * 13888 * 14580 * 14588 * 12180 8388 * 7880 1880 * 30758 * 30758 * 31380 * 31388 * 21758 17880 * 11808 11880 * 6480 * 6480											* 9080 4880 6870 * 13380 10408 270	
-608 -24					* 7180 * 7180							* 6180 * 6180 1545	

Illustration 152

g06225096

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

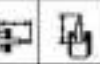
(mm) (inch)	1500 59	3000 118	4500 169	6000 236	7500 308		
							(mm) (inch)
1100 390						* 2790 * 9000	7320 290
9000 240					* 4300 * 8700	* 4000 * 8100	8300 330
4100 160					* 4300 * 9300	* 2500 * 5500	8520 330
3000 120			* 4400 * 11000	* 6000 * 11400	* 6700 * 11400	* 4700 * 9600	8250 320
1100 60		* 3000 * 22000	* 5200 * 20000	* 6300 * 17000	* 6300 * 16000	* 6200 * 11300	8300 320
0 0		* 1600 * 17400	* 3600 * 17400	* 5600 * 20900	* 7600 * 15200	* 5700 * 12300	8100 360
-1100 60	* 6700 * 12700	* 6700 * 12700	* 8800 * 22200	* 10300 * 22200	* 13000 * 17000	* 13000 * 11000	8700 360
-3000 -120	* 8900 * 18900	* 8900 * 18900	* 12000 * 31000	* 15100 * 17700	* 17400 * 16100	* 17000 * 12300	8000 330
-4100 -160	* 12000 * 28000	* 12000 * 28000	* 17000 * 20300	* 21000 * 19500	* 23000 * 11200	* 23000 * 10100	8670 270
-6000 -240				* 4400 * 4400		* 4700 * 4700	8670 270

Illustration 153

g06225097

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	1500 59	3000 118	4500 169	6000 236	7500 308		
							(mm) (inch)
7500 300						* 3000 * 9600	7320 290
9000 240					* 4300 * 9000	* 4000 * 8100	8300 330
4100 160					* 4700 * 10400	* 2700 * 6000	8520 330
3000 120			* 7100 * 15200	* 7100 * 15200	* 6000 * 12700	* 5200 * 11600	8250 320
1100 60		* 3000 * 25000	* 5200 * 21000	* 6300 * 16000	* 6300 * 16000	* 6200 * 11300	8300 320
0 0		* 1600 * 16400	* 3600 * 21100	* 5600 * 16000	* 7600 * 12000	* 5700 * 11000	8100 360
-1100 60	* 6700 * 12000	* 6700 * 12000	* 8800 * 22500	* 10300 * 22500	* 13000 * 17000	* 13000 * 11000	8700 360
-3000 -120	* 8900 * 21100	* 8900 * 21100	* 12000 * 30700	* 15100 * 17700	* 17400 * 16100	* 17000 * 12300	8000 330
-4100 -160	* 12000 * 30700	* 12000 * 30700	* 17000 * 31000	* 21000 * 21000	* 23000 * 11200	* 23000 * 10100	8670 270
-6000 -240				* 4400 * 4400		* 4700 * 4700	8670 270

Illustration 154

g06225098

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.5 m (8 ft 2 inch) Stick and 4200 kg (9260 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

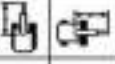
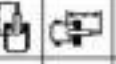
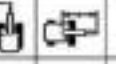
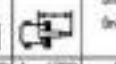
Reach (mm) (inch)	500 60	3000 120	4500 90	6000 240	7500 300		Load (mm) (inch)
7500 300							4700 185
6000 240				5200 205	5200 205		4300 169
4500 90			8900 350	8900 350	4750 187	3700 146	4000 157
3000 120			8250 325	8250 325	5550 219	3600 142	4300 169
1500 60			8700 340	8700 340	5500 217	3500 138	4600 181
0			10950 430	10950 430	5450 215	3450 136	5000 197
-1500 -60		11300 445	10300 405	8850 345	4600 181		5800 228
-3000 -120		12800 500	9500 375	8950 350	4650 183		6800 268
-4500 -180			8950 350				10000 394

Illustration 155

g06225072

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

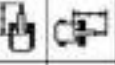
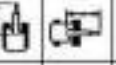
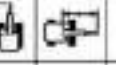
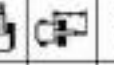
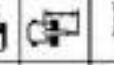
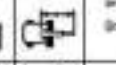
Reach (mm) (inch)	500 60	3000 120	4500 90	6000 240	7500 300		Load (mm) (inch)
7500 300							5100 201
6000 240				5750 226	5450 215		4700 185
4500 90			7250 285	7250 285	5150 203	3700 146	4900 193
3000 120			8100 315	8100 315	5950 234	3600 142	4700 185
1500 60			8750 340	8750 340	5500 217	3500 138	5000 197
0			11850 465	11850 465	5450 215	3450 136	5800 228
-1500 -60		11950 471	11300 445	9850 388	4600 181		6800 268
-3000 -120		13450 530	10500 410	9000 354	4650 183		7800 307
-4500 -180			9000 354				10000 394

Illustration 156

g06225075

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

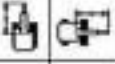
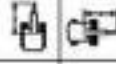
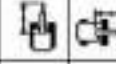
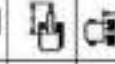
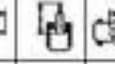
Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							4700 18400
6000 240				5200 19450	5200 19450		4300 16900
4500 180			6550 14150	5500 12100	4750 10400	3700 8050	4000 15900
3000 120			6250 11750	5350 10550	5050 12000	3600 7750	4300 16450
1500 60			5700 20950	7150 15350	4600 10300	2500 11850	4600 18050
0			10950 22400	6900 14800	1500 6150	3450 11700	5100 18000
-1500 -60		10300 25650	10000 22300	6850 14700	1450 6050		3600 12300
-3000 -120		12800 27700	5250 20800	6350 14350	4650 10050		6100 13450
-4500 -180			6850 14400				5350 13000

Illustration 157

g06225066

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

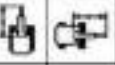
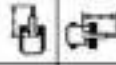
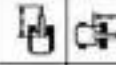
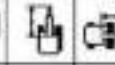
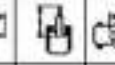
Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							5100 11950
6000 240				5750 12600	5450 11700		4700 10350
4500 180			7250 15600	6200 13500	5100 11400	3700 8050	4600 10100
3000 120			3100 19600	7050 16500	5050 12000	3600 7750	4700 10350
1500 60			10750 23150	7150 15350	4600 10300	2500 11850	4600 10350
0			11450 24750	6900 14800	1500 6150	3450 11700	5100 18000
-1500 -60		11850 27150	11000 24450	6850 14700	1450 6050		3600 12300
-3000 -120		14150 30950	10250 22000	6350 14350	4650 10050		6100 13450
-4500 -180			7650 16900				5350 14500

Illustration 158

g06225070

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

(mm) (inch)	7500 60	3000 120	4500 90	6000 240	7500 300			(mm) (inch)
7500 300								4700 30400
6000 240				5200 16450	5200 16450			4300 3450
4500 90			6850 14750	5900 11350	4750 3700			4100 3250
3000 120			6050 11750	6350 10850	5500 12000			4300 3450
1500 60			8000 20950	7150 5350	5900 10300			4800 10100
0			10250 22400	6300 16300	7500 18900			5100 11000
-1500 -60		11300 25650	10300 22300	6950 16700	4600 10250			5600 12300
-3000 -120		12000 27700	5200 20800	6950 16950	4600 10050			6100 13450
-4500 -180			6850 14400					5550 13000

Illustration 159

g06224991

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

(mm) (inch)	7500 60	3000 120	4500 90	6000 240	7500 300			(mm) (inch)
7500 300								5100 11950
6000 240				5750 12600	5450 11700			4700 10350
4500 90			7250 15800	6200 12550	5100 10250			4600 10100
3000 120			8100 17800	7000 15050	5650 12100			4700 10250
1500 60			10750 23850	7150 15350	5900 10300			4850 10950
0			11450 24750	6300 16300	7500 18900			5100 11000
-1500 -60		11350 27150	10300 24450	6950 16700	4600 10250			5600 12300
-3000 -120		14150 30950	5200 20800	6950 16950	4650 10050			6000 14500
-4500 -180			7650 16800					6600 14300

Illustration 160

g06224993

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

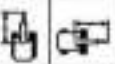
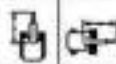
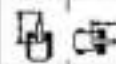
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							4700 190
6000 240				5200 210	5200 210		4300 170
4500 180			8350 330	8350 330	5450 215	4750 190	4000 160
3000 120			8250 325	8350 330	5200 210	5500 220	4300 170
1500 60			20950 800	37000 1450	71000 2800	57000 2250	4600 180
0			16950 660	71000 2800	47500 1875	56000 2200	5100 200
-1500 -60		10300 400	11300 440	10300 400	7550 300	4700 185	5750 225
-3000 -120		12800 500	12800 500	5250 205	8850 350	4800 190	6000 235
-4500 -180		27700 1080	27700 1080	20900 820	18400 725	10350 410	13450 530
-6000 -240			8850 345	8850 345			5550 218
-7500 -300			14600 575	14600 575			10000 395

Illustration 161

g06224996

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

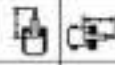
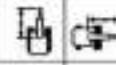
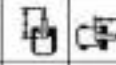
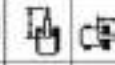
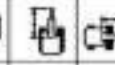
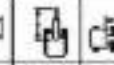
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							5700 225
6000 240				5750 225	5600 220		4700 185
4500 180			7250 285	8250 325	5200 210	5450 215	4600 180
3000 120			15600 615	16600 655	10900 425	11700 460	10600 415
1500 60			9700 380	18500 725	37000 1450	59000 2300	4700 185
0			10750 420	19600 765	49000 1925	67000 2625	6000 235
-1500 -60		11850 465	11850 465	10300 400	7700 300	4700 185	5750 225
-3000 -120		14750 580	13900 535	5250 205	7800 305	4800 190	6000 235
-4500 -180		30850 1210	29750 1165	22200 870	18200 715	10350 410	13450 530
-6000 -240			7650 300	7650 300			5550 218
-7500 -300			13000 510	13000 510			10000 395

Illustration 162

g06224999

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Reach inch	500 80	3000 120	4500 180	6000 240	7500 300			Reach inch
7500 300								5900 230
6000 240				5200 205	5200 205			5900 230
4500 180			8350 330	8550 335	8750 345	8950 355	9150 365	5900 230
3000 120			8250 325	8450 330	8650 335	8850 340	9050 345	5900 230
1500 60			8150 320	8350 325	8550 330	8750 335	8950 340	5900 230
0			8050 315	8250 320	8450 325	8650 330	8850 335	5900 230
-1500 -60		11000 430	11300 440	11600 450	11900 460	12200 470	12500 480	5900 230
-3000 -120		12800 500	13100 510	13400 520	13700 530	14000 540	14300 550	5900 230
-4500 -180		14600 570	14900 580	15200 590	15500 600	15800 610	16100 620	5900 230

Illustration 163

g06225018

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Reach inch	500 80	3000 120	4500 180	6000 240	7500 300			Reach inch
7500 300								5900 230
6000 240				5150 200	5150 200			5900 230
4500 180			8250 320	8450 325	8650 330	8850 335	9050 340	5900 230
3000 120			8150 315	8350 320	8550 325	8750 330	8950 335	5900 230
1500 60			8050 310	8250 315	8450 320	8650 325	8850 330	5900 230
0			7950 305	8150 310	8350 315	8550 320	8750 325	5900 230
-1500 -60		11350 440	11650 450	11950 460	12250 470	12550 480	12850 490	5900 230
-3000 -120		13150 510	13450 520	13750 530	14050 540	14350 550	14650 560	5900 230
-4500 -180		14950 580	15250 590	15550 600	15850 610	16150 620	16450 630	5900 230

Illustration 164

g06225048

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.9 m (9 ft 6 inch) Stick and 4200 kg (9260 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							3850 150
6000 240							3600 140
4500 180							3400 130
3000 120							3200 120
1500 60							3000 110
0 0							2800 100
-1500 -60							2600 90
-3000 -120							2400 80
-4500 -180							2200 70

Illustration 165

g06224959

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							4250 165
6000 240							4000 155
4500 180							3850 150
3000 120							3600 140
1500 60							3400 130
0 0							3200 120
-1500 -60							3000 110
-3000 -120							2800 100
-4500 -180							2600 90

Illustration 166

g06224973

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

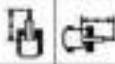
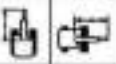
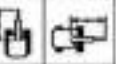
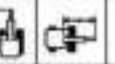
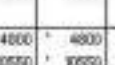
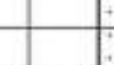
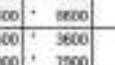
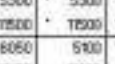
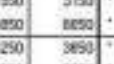
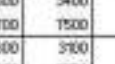
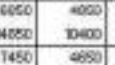
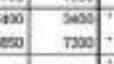
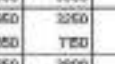
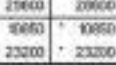
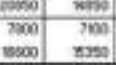
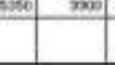
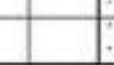
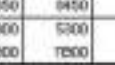
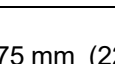
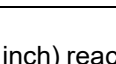
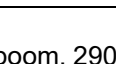
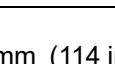
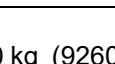
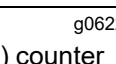
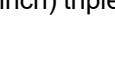
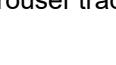
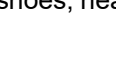
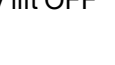
Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 167

g06224952

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

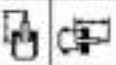
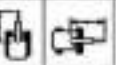
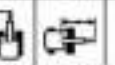
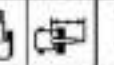
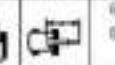
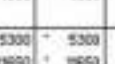
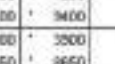
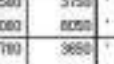
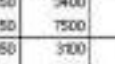
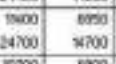
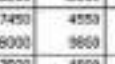
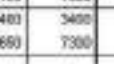
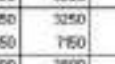
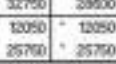
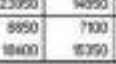
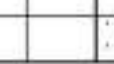
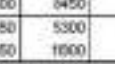
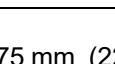
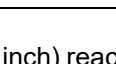
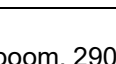
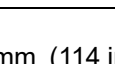
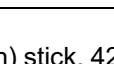
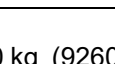
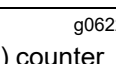
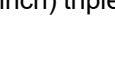
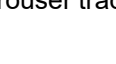
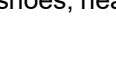
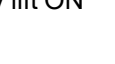
Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0								0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 168

g06224953

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

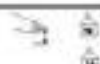
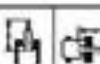
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				4500 180	4500 180		3850 152
6000 240				4800 192	4800 192		3600 142
4500 180				5300 210	5300 210	4950 198	3350 133
3000 120			7150 281	7700 302	8050 318	8650 339	8350 329
1500 60			8350 329	8900 348	9450 369	10050 396	9750 385
0 0		6200 244	6200 244	10250 400	10750 420	11250 439	11750 460
-1500 -60	6800 268	6800 268	10350 408	10900 426	11450 446	12050 471	11750 460
-3000 -120	10350 408	10350 408	13650 536	14200 556	14750 576	15350 601	15050 590
-4500 -180		10850 426	10850 426	15350 601	15900 621	16450 642	16150 631

Illustration 169

g06226115

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

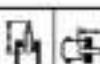
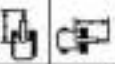
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				4500 180	4500 180		4250 167
6000 240				5300 210	5300 210		3900 153
4500 180				6050 239	6050 239	5650 222	3550 139
3000 120			8850 348	9400 366	9950 387	10550 411	10250 401
1500 60			10050 396	10600 416	11150 437	11750 460	11450 449
0 0		8550 336	8550 336	11800 460	12350 486	12900 509	13450 530
-1500 -60	7000 274	7000 274	10350 408	10900 426	11450 446	12050 471	11750 460
-3000 -120	10550 411	10550 411	13850 536	14400 556	14950 576	15550 601	15250 590
-4500 -180		11050 426	11050 426	15550 601	16100 621	16650 642	16350 631

Illustration 170

g06224977

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Reach (inch)	1500 60	3000 121	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0							0
-600							-600
-1200							-1200
-1800							-1800
-2400							-2400
-3000							-3000
-3600							-3600
-4200							-4200
-4800							-4800
-5400							-5400
-6000							-6000
-6600							-6600
-7200							-7200
-7800							-7800
-8400							-8400
-9000							-9000
-9600							-9600
-10200							-10200
-10800							-10800
-11400							-11400
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-12600							-12600
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-25800							-25800
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-29400							-29400
-30000							-30000
-30600							-30600
-31200							-31200
-31800							-31800
-32400							-32400
-33000							-33000
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-36600							-36600
-37200							-37200
-37800							-37800
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-68400							-68400
-69000							-69000
-69600							-69600
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-70800							-70800
-71400							-71400
-72000							-72000
-72600							-72600
-73200							-73200
-73800							-73800
-74400							-74400
-75000							-75000
-75600							-75600
-76200							-76200
-76800							-76800
-77400							-77400
-78000							-78000
-78600							-78600
-79200							-79200
-79800							-79800
-80400							-80400
-81000							-81000
-81600							-81600
-82200							-82200
-82800							-82800
-83400							-83400
-84000							-84000
-84600							-84600
-85200							-85200
-85800							-85800
-86400							-86400
-87000							-87000
-87600							-87600
-88200							-88200
-88800							-88800
-89400							-89400
-90000							-90000
-90600							-90600
-91200							-91200
-91800							-91800
-92400							-92400
-93000							-93000
-93600							-93600
-94200							-94200
-94800							-94800
-95400							-95400
-96000							-96000
-96600							-96600
-97200							-97200
-97800							-97800
-98400							-98400
-99000							-99000
-99600							-99600
-100200							-100200
-100800							-100800
-101400							-101400
-102000							-102000
-102600							-102600
-103200							-103200
-103800							-103800
-104400							-104400
-105000							-105000
-105600							-105600
-106200							-106200
-106800							-106800
-107400							-107400
-108000							-108000
-108600							-108600
-109200							-109200
-109800							-109800
-110400							-110400
-111000							-111000
-111600							-111600
-112200							-112200
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-113400							-113400
-114000							-114000
-114600							-114600
-115200							-115200
-115800							-115800
-116400							-116400
-117000							-117000
-117600							-117600
-118200							-118200
-118800							-118800
-119400							-119400
-120000							-120000
-120600							-120600
-121200							-121200
-121800							-121800
-122400							-122400
-123000							-123000
-123600							-123600
-124200							-124200
-124800							-124800
-125400							-125400
-126000							-126000
-126600							-126600
-127200							-127200
-127800							-127800
-128400							-128400
-129000							-129000
-129600							-129600
-130200							-130200
-130800							-130800
-131400							-131400
-132000							-132000
-132600							-132600
-133200							-133200
-133800							-133800
-134400							-134400
-135000							-135000
-135600							-135600
-136200							-136200
-136800							-136800
-137400							-137

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								4750 187.5
6000 240				5300 209	5300 209			4350 171
4500 180			8700 343	8700 343	8700 343	4450 174	4450 174	4250 167
3000 120			8400 327	8400 327	8400 327	4350 171	4350 171	4000 157
1500 60			8900 348	8900 348	8900 348	4250 167	4250 167	3950 155
0			10550 413	10550 413	10550 413	4150 163	4150 163	3850 151
-1500 -60		11350 445	10400 403	10400 403	10400 403	4050 159	4050 159	3750 147
-3000 -120		13000 508	10300 398	10300 398	10300 398	3950 155	3950 155	3650 143
-4500 -180			10650 414	10650 414	10650 414	3850 151	3850 151	3550 139

Illustration 173

g06225629

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								5750 226.6
6000 240				5850 230	5850 230			4750 187.5
4500 180			7350 288	7350 288	7350 288	4450 174	4450 174	4350 171
3000 120			8250 322	8250 322	8250 322	4350 171	4350 171	4000 157
1500 60			8900 348	8900 348	8900 348	4250 167	4250 167	3950 155
0			10550 413	10550 413	10550 413	4150 163	4150 163	3850 151
-1500 -60		12000 473	10450 405	10450 405	10450 405	4050 159	4050 159	3750 147
-3000 -120		14350 548	10450 398	10450 398	10450 398	3950 155	3950 155	3650 143
-4500 -180			10650 414	10650 414	10650 414	3850 151	3850 151	3550 139

Illustration 174

g06225631

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

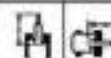
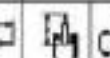
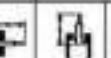
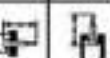
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							4750 * 4750 5600 10550 * 10550 220
6000 240				5300 * 5300 11700 * 11700			4350 * 4350 6830 9600 * 9600 270
4500 180			6700 * 6700 14450 * 14450	5750 * 5750 12500 * 12500	4600 * 4600	4450	4250 * 4250 7570 9400 * 9400 300
3000 120			8400 * 8400 18950 * 18950	6900 * 6900 15550 * 15550	5600 * 5600 12250 * 12250	4350 * 4350	3950 * 3950 7960 8750 * 8750 320
1500 60			9900 * 9900 21900 * 21900	8200 * 8200 18550 * 18550	6750 * 6750 15050 * 15050	5350 * 5350	4650 * 4650 9050 10250 * 10250 320
0			10950 * 10950 22800 * 22800	8250 * 8250 18700 * 18700	6600 * 6600 14600 * 14600	4150 * 4150	3900 * 3900 7860 8600 * 8600 310
-1500 -60		10950 * 10950 25900 * 25900	11350 * 11350 22900 * 22900	8200 * 8200 18650 * 18650	7700 * 7700 17100 * 17100		6050 * 6050 7350 13350 * 13350 290
-3000 -120		13000 * 13000 28950 * 28950	14500 * 14500 20400 * 20400	8300 * 8300 18550 * 18550	7900 * 7900 17550 * 17550		6250 * 6250 8470 13750 * 13750 280
-4500 -180			7950 * 7950 17700 * 17700				6700 * 6700 4980 15300 * 15300 200

Illustration 175

g06225620

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

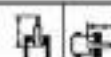
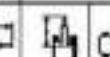
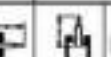
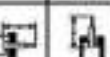
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							5150 * 5150 5600 11600 * 11600 220
6000 240				5050 * 5050 11200 * 11200			4750 * 4750 6830 10600 * 10600 270
4500 180			7350 * 7350 16300 * 16300	6250 * 6250 13750 * 13750	5250 * 5250	4450	4350 * 4350 7570 10250 * 10250 300
3000 120			9250 * 9250 19900 * 19900	7750 * 7750 17100 * 17100	6200 * 6200 13850 * 13850	4350 * 4350	3950 * 3950 7960 8750 * 8750 320
1500 60			10900 * 10900 23900 * 23900	9000 * 9000 19700 * 19700	6750 * 6750 15050 * 15050	5350 * 5350	4650 * 4650 9050 10250 * 10250 320
0			11650 * 11650 25850 * 25850	8250 * 8250 18700 * 18700	6600 * 6600 14600 * 14600	4150 * 4150	3900 * 3900 7860 8600 * 8600 310
-1500 -60		12000 * 12000 27300 * 27300	11450 * 11450 24850 * 24850	8200 * 8200 18650 * 18650	8950 * 8950 19900 * 19900		6050 * 6050 7350 14400 * 14400 290
-3000 -120		14350 * 14350 31850 * 31850	13450 * 13450 22550 * 22550	8300 * 8300 18550 * 18550	7700 * 7700 17550 * 17550		6300 * 6300 8470 15200 * 15200 280
-4500 -180			7800 * 7800 17300 * 17300				6750 * 6750 4980 14800 * 14800 200

Illustration 176

g06225627

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							4750 10550
6000 240				5300 11700	5300 11700		4350 9600
4500 180			6700 14450	6700 14450	5750 12500	4650 10300	4250 9400
3000 120			8400 18650	8400 18650	6000 13200	5600 12250	4350 9600
1500 60			8900 19300	8900 19300	7250 16000	5950 13000	4650 10300
0			10550 22800	10550 22800	7700 17000	6400 14000	5200 11450
-1500 -60		10550 22800	11350 25000	8200 18000	7700 17000	5550 12250	4250 9400
-3000 -120		13000 28950	14500 31950	8300 18300	7900 17300	5600 12250	4350 9600
-4500 -180			13950 30400	10900 23600	8350 18350		4750 10550
			14700 32200	11700 25700			5100 11200

Illustration 177

g06225607

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							5750 12500
6000 240				5850 12900	5850 12900		4750 10550
4500 180			7350 16300	7350 16300	6200 13400	5250 11450	4650 10300
3000 120			8250 18200	8000 17500	6000 13200	6300 13950	4750 10550
1500 60			10900 23900	8900 19300	7250 16000	6400 14000	5950 13000
0			11850 25950	10550 22800	8500 18650	6750 14750	6400 14000
-1500 -60		10000 21900	10000 21900	8950 19550	8550 18650	5550 12250	4250 9400
-3000 -120		14350 31650	14350 31650	8450 18550	7700 17000	5600 12250	4350 9600
-4500 -180			13950 30400	10900 23600	8350 18350		4750 10550
			14700 32200	11700 25700			5100 11200

Illustration 178

g06225611

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

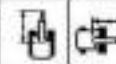
Load (inch)	500 60		3000 120		4500 180		6000 240		7500 300				Load (inch)		
													Load (inch)		
7500 300													4750 187.5	4750 187.5	5900 232
6000 240							5300 209	5300 209					4350 171	4350 171	5600 220
4500 180					6700 264	6700 264	5750 226	5750 226	4000 157	4250 167	4250 167	4250 167	3600 141	3600 141	4500 177
3000 120					8400 331	8400 331	6900 272	6900 272	5000 196	4300 168	4350 171	4350 171	3500 138	3500 138	4400 173
1500 60					8900 350	8900 350	7250 285	7250 285	5350 210	4200 163	4650 183	4650 183	3000 120	3000 120	4300 167
0 0					10950 427	10950 427	7700 301	7700 301	6150 242	4100 161	5200 205	5200 205	2050 81	2050 81	4200 165
-750 -30					12800 504	12800 504	8700 338	8700 338	6300 246	4050 158	5150 202	5150 202	1450 57	1450 57	4100 161
-1500 -60					15800 616	15800 616	10400 407	10400 407	7700 301	5450 213			8050 317	8050 317	4000 157
-3000 -120					25800 1030	25800 1030	22500 885	22500 885	17400 680	17750 697			13350 523	13350 523	3250 128
-4500 -180					33000 1320	33000 1320	29400 1168	29400 1168	22400 885	22750 897			16250 642	16250 642	4050 158
-6000 -240					40400 1616	40400 1616	36400 1432	36400 1432	28400 1136	28750 1150			20250 798	20250 798	3000 120
-7500 -300					47800 1912	47800 1912	43400 1712	43400 1712	34400 1376	34750 1390			24250 958	24250 958	2050 81

Illustration 179

g06225632

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 60		3000 120		4500 180		6000 240		7500 300				Load (inch)		
													Load (inch)		
7500 300													5150 203	5150 203	5900 232
6000 240							5850 230	5850 230					4750 187.5	4750 187.5	5600 220
4500 180					7350 288	7350 288	6350 250	6350 250	5250 206	4350 171			4850 190	4300 168	5400 212
3000 120					8250 324	8250 324	7150 282	5500 213	6200 240	4300 168			5250 206	3900 153	5350 209
1500 60					10900 427	10900 427	9500 375	8200 320	13550 536	9200 360			10450 410	8650 338	10000 394
0					11850 464	11850 464	10000 394	8200 320	13000 510	9000 354			11000 433	9000 354	10000 394
-750 -30					2550 100	1500 59	8450 331	7850 308	13450 526	8950 352			12500 492	8500 331	9150 360
-1500 -60					12000 454	12000 454	11650 457	8950 347	14450 568	10450 410			14150 558	10450 410	10450 410
-3000 -120					14350 558	14350 558	10450 410	8200 320	17350 682	12000 472			16500 644	12000 472	12000 472
-4500 -180					2180 85	3180 125	22350 885	16500 644	28850 1136	17500 690			26000 1024	17500 690	17500 690
-6000 -240					3800 149	3800 149							6750 263	6750 263	6750 263
-7500 -300					16350 644	16350 644							14800 578	14800 578	14800 578

Illustration 180

g06225633

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							4750 187
6000 240				5300 210	5300 210		4350 171
4500 180			6700 268	6700 268	6700 268	4900 196	4250 168
3000 120			8400 333	8400 333	8400 333	5800 230	4050 160
1500 60			9800 392	9800 392	9800 392	6600 262	3850 152
0			10900 436	10900 436	10900 436	7200 287	3650 144
-1500 -60		10300 413	10300 413	10300 413	10300 413	7800 310	3450 136
-3000 -120		13000 520	13000 520	13000 520	13000 520	8600 339	3250 128
-4500 -180		16150 646	16150 646	16150 646	16150 646	9600 384	3050 120

Illustration 181

g06225613

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							5150 203
6000 240				5350 214	5350 214		4750 187
4500 180			7350 293	7350 293	7350 293	5200 208	4400 173
3000 120			9350 370	9350 370	9350 370	6200 248	4050 160
1500 60			10900 436	10900 436	10900 436	7200 287	3850 152
0			11850 470	11850 470	11850 470	7800 310	3650 144
-1500 -60		12650 501	12650 501	12650 501	12650 501	8400 333	3450 136
-3000 -120		14350 568	14350 568	14350 568	14350 568	9200 364	3250 128
-4500 -180		16350 646	16350 646	16350 646	16350 646	10200 404	3050 120

Illustration 182

g06225616

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

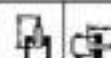
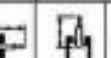
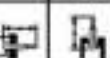
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							4150 1650
6000 240				5300 2100	5300 2100		4350 1650
4500 180			6700 2650	6700 2650	5750 2250	4800 1900	4250 1650
3000 120			8400 3300	8400 3300	6200 2450	5600 2200	4350 1650
1500 60			10900 4300	10900 4300	7250 2850	5950 2350	4650 1800
0			10950 4300	10950 4300	7700 3000	6400 2500	5200 2050
-1500 -60		10950 4300	11350 4450	11350 4450	7700 3000	5750 2250	6050 2350
-3000 -120		25900 10000	25900 10000	22900 8950	16350 6350	12400 4800	13350 5200
-4500 -180		20950 8200	20950 8200	20400 7900	14350 5550	12550 4850	13750 5300
-6000 -240			7950 3100	7950 3100			6700 2600
-7500 -300			14700 5700	14700 5700			13300 5100

Illustration 183

g06225618

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

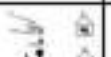
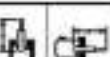
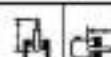
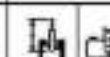
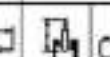
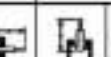
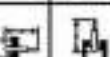
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300							4150 1650
6000 240				5850 2300	5850 2300		4750 1850
4500 180			7350 2900	7350 2900	6350 2450	5250 2050	4650 1800
3000 120			9250 3600	9250 3600	7150 2800	6200 2400	4750 1850
1500 60			10900 4300	10900 4300	8000 3100	6800 2600	5150 2000
0			11850 4550	11850 4550	8500 3300	7300 2800	5700 2200
-1500 -60		10000 3950	10000 3950	11450 4400	11450 4400	8550 3250	6700 2600
-3000 -120		27300 10500	27300 10500	24850 9550	18450 7000	12400 4800	14800 5700
-4500 -180		14350 5500	14350 5500	14350 5500	12550 4850	10200 4000	15200 5900
-6000 -240			7950 3100	7950 3100			6700 2600
-7500 -300			14700 5700	14700 5700			13300 5100

Illustration 184

g06225619

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.5 m (8 ft 2 inch) Extended Stick and 5400 kg (11905 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (mm) (inch)
7500 300							4700 185
6000 240				5200 205	5200 205		4300 169
4500 180			6550 258	6550 258	5600 220	4750 187	4200 165
3000 120			8250 324	8250 324	6350 250	5950 234	4300 169
1500 60			9750 384	9750 384	7300 287	5900 232	4900 193
0			10950 429	10950 429	8250 324	6850 270	5100 201
0			12400 488	12400 488	9300 366	7900 311	6350 250
-1500 -60		10380 408	10380 408	8050 316	7550 297		5950 234
-3000 -120		25650 1006	25650 1006	17300 678	16350 642		10350 405
-4500 -180		32880 1287	32880 1287	20550 783	18550 725		13450 529
-6000 -240			6850 266	6850 266			5950 234
-7500 -300			14400 562	14400 562			10300 404

Illustration 185

g06225677

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (mm) (inch)
7500 300							5100 201
6000 240				5750 226	5750 226		4700 185
4500 180			7250 285	7250 285	6200 244	5150 202	4900 193
3000 120			9100 358	9100 358	7000 276	6100 239	4700 185
1500 60			10750 421	10750 421	7950 311	6300 248	5000 197
0			12350 485	12350 485	8950 353	7000 276	6350 250
0			14150 550	14150 550	10350 405	8250 324	7900 311
-1500 -60		10380 408	10380 408	8050 316	7550 297		5950 234
-3000 -120		27150 1066	27150 1066	17300 678	16350 642		14150 550
-4500 -180		30650 1206	30650 1206	22250 873	19550 725		14950 590
-6000 -240			7650 285	7650 285			8800 345
-7500 -300			16000 625	16000 625			14900 586

Illustration 186

g06225679

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Grade (Grade)	500 60	3000 120	4500 180	6000 240	7500 300		Grade (Grade)
7500 300							Grade (Grade)
7500 300							4700 18400
6000 240				5200 11450	5200 11450		4300 9450
4500 180			6550 14150	5600 12200	5600 12200	4750 10300	4200 9250
3000 120			6550 11750	6350 13600	5500 12000	4250 9100	4300 9450
1500 60			9700 20950	6350 11350	5600 12100	5800 12850	4500 10100
0 0			10050 22400	8100 11400	1550 16350	5450 11750	6000 13000
-1500 -60		10300 25850	10000 22300	8050 11300	1550 16350	5400 11600	5950 13050
-3000 -120		12800 27700	10500 20800	8150 11550	1650 14600	5450 11600	6100 13450
-4500 -180			8850 14400	6850 14400			5950 13000

Illustration 187

g06225674

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

Grade (Grade)	500 60	3000 120	4500 180	6000 240	7500 300		Grade (Grade)
7500 300							Grade (Grade)
7500 300							5100 11350
6000 240				5750 12600	5750 12600		4700 10350
4500 180			7250 15800	6200 13500	6100 13150	5100 10100	4500 9400
3000 120			9100 19600	7000 15200	5850 12800	6100 13250	4700 10350
1500 60			10750 23550	8350 11550	7050 12100	5900 12600	5000 11000
0 0			11450 24750	8100 11400	8350 10100	6250 11750	5800 12400
-1500 -60		11900 27150	11900 24450	8050 11300	8400 10150	5400 11600	6000 13050
-3000 -120		14150 30650	10250 22000	8150 11550	7600 16000	5450 11600	6100 13450
-4500 -180			7850 16500	7850 16500			6000 13000

Illustration 188

g06225675

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300	 	 	 	 	 	 	7500 300
6000 240				5200 205	5200 205		6000 240
4500 180			6500 255	6500 255	4700 185	4250 165	4500 180
3000 120			8200 325	8200 325	5500 215	4250 165	3000 120
1500 60			9700 385	9700 385	5900 235	4500 185	1500 60
0			10900 430	10900 430	6000 240	5750 225	0
-1500 -60		10300 410	10300 410	8050 315	5400 210		-1500 -60
-3000 -120		25850 1015	22300 880	18350 725			-3000 -120
-4500 -180		27700 1085	20800 815	16800 660			-4500 -180
-6000 -240			8850 345				-6000 -240

Illustration 189

g06225654

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300	 	 	 	 	 	 	7500 300
6000 240				5750 225	5750 225		6000 240
4500 180			7750 305	7750 305	5100 200	4250 165	4500 180
3000 120			8900 345	8900 345	5700 220	4250 165	3000 120
1500 60			10750 420	10750 420	5900 230	4500 185	1500 60
0			11850 465	11850 465	6250 245	5800 225	0
-1500 -60		11950 468	10300 400	8400 320	5400 210		-1500 -60
-3000 -120		27150 1065	24450 955	18150 725			-3000 -120
-4500 -180		30950 1215	22500 885	16200 660			-4500 -180
-6000 -240			8900 345				-6000 -240

Illustration 190

g06225656

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

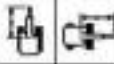
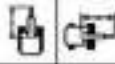
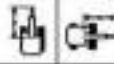
Boom (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Boom (inch)
7500 300							7500 300
6000 240				5200 19450	5200 19450		6000 240
4500 180			8350 14150	8350 14150	5600 12200	4750 12200	4500 180
3000 120			8250 11750	8350 11750	5800 12400	5580 12000	3000 120
1500 60			8700 20950	8300 11700	5550 12350	5680 12850	1500 60
0			10950 22400	7950 17100	5250 12250	6080 13000	0
-1500 -60		11300 25850	10300 22100	7950 17050	5300 12450		-1500 -60
-3000 -120		12800 27700	10200 20900	8050 17300	5400 14800		-3000 -120
-4500 -180			8850 14600	8850 14600			-4500 -180

Illustration 191

g06225682

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

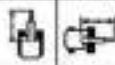
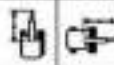
Boom (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Boom (inch)
7500 300							7500 300
6000 240				5750 12800	5750 12800		6000 240
4500 180			8750 15600	8350 13000	5600 13000	4750 12200	4500 180
3000 120			8900 19600	8750 15200	5800 13400	5580 13250	3000 120
1500 60			10750 23150	8300 17000	5550 12350	5680 13400	1500 60
0			10950 24750	7950 17100	5250 12250	6080 13200	0
-1500 -60		11350 27150	10300 24450	7950 18100	5300 14450		-1500 -60
-3000 -120		14150 30950	10250 22500	8050 17300	5400 15200		-3000 -120
-4500 -180			8850 18000	8850 18000			-4500 -180

Illustration 192

g06225683

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

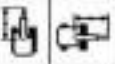
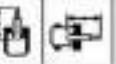
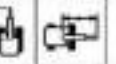
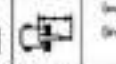
Boom (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Boom (inch)
7500 300							7500 300
6000 240				5200 205	5200 205		6000 240
4500 180			8350 330	8350 330	5600 220	4750 185	4500 180
3000 120			8250 325	8350 330	5580 219	4300 168	3000 120
1500 60			8700 340	8400 335	5680 223	4200 164	1500 60
0			10950 425	8200 315	7590 300	5550 217	0
-1500 -60			22400 880	16250 625	15500 600	13000 510	-1500 -60
-3000 -120		11300 445	25850 995	22100 860	17550 680	16350 640	-3000 -120
-4500 -180		27700 1080	28900 1120	17800 695	14800 575	12000 470	-4500 -180

Illustration 193

g06225657

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

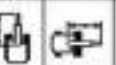
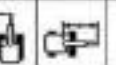
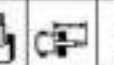
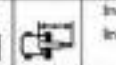
Boom (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Boom (inch)
7500 300							7500 300
6000 240				5750 225	5750 225		6000 240
4500 180			12300 475	8250 315	6200 235	5780 227	4500 180
3000 120			8900 340	8350 320	5580 217	4300 168	3000 120
1500 60			8700 335	8400 325	5680 220	4200 164	1500 60
0			10950 415	8200 310	7590 295	5550 213	0
-1500 -60			24750 960	17800 685	16000 630	13650 530	-1500 -60
-3000 -120		11800 465	27850 1075	24450 935	18050 720	16550 660	-3000 -120
-4500 -180		30650 1200	32200 1240	17800 685	14800 570	12000 465	-4500 -180

Illustration 194

g06225663

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			
								Load (inch)
7500 300								4700 * 4700 5600 10400 * 10400 220
6000 240				5200 * 5200 11450 * 11450				4300 * 4300 6830 9450 * 9450 270
4500 180			8350 * 8350 14750 * 14750	5900 * 5900 12200 * 12200	4750 * 4750 4450			4000 * 4200 7570 3250 * 3250 300
3000 120			8250 * 8250 17750 * 17750	6350 * 6350 12550 * 12550	5580 * 5580 9350	4350		4300 * 3950 7980 3450 * 3750 320
1500 60			9700 * 9700 20950 * 20950	8250 * 8250 15350 * 15350	5750 * 5750 12850	4350 * 4350 9100		4600 * 3850 8050 3050 * 3400 320
0			10950 * 10950 22400 * 22400	8300 * 8300 16250 * 16250	5600 * 5600 12050	4150 * 4150 8250		5150 * 3900 7860 3600 * 3600 310
-1500 -60		10300 * 10300 25850 * 25850	8250 * 8250 22100 * 22100	7550 * 7550 16350 * 16350	5550 * 5550 11550			5550 * 4250 7350 3350 * 3400 290
-3000 -120		12800 * 12800 27700 * 27700	8350 * 8350 20900 * 20900	8850 * 8850 14800 * 14800	5600 * 5600 12100			6100 * 5100 6470 3450 * 3350 280
-4500 -180			8850 * 8850 14800 * 14800					5550 * 5550 4980 13000 * 13000 200

Illustration 195

g06225664

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			
								Load (inch)
7500 300								5700 * 5700 5800 11350 * 11350 220
6000 240				5750 * 5750 12600 * 12600				4700 * 4700 6830 10350 * 10350 270
4500 180			1250 * 1250 15800 * 15800	8250 * 8250 13450 * 13450	5750 * 5750 4450			4600 * 4350 7570 3050 * 3050 300
3000 120			9700 * 9700 19600 * 19600	1000 * 1000 12550 * 12550	6100 * 6100 13250	4350		4700 * 3950 7980 3450 * 3750 320
1500 60			10750 * 10750 22150 * 22150	8650 * 8650 15350 * 15350	5750 * 5750 13350	4350 * 4350 9100		5000 * 3950 8050 3400 * 3400 320
0			11450 * 11450 24750 * 24750	8350 * 8350 16350 * 16350	5600 * 5600 12050	4150 * 4150 8250		5600 * 3900 7860 3600 * 3600 310
-1500 -60		11950 * 11950 27150 * 27150	10300 * 10300 24450 * 24450	8250 * 8250 17150 * 17150	5550 * 5550 11550			6000 * 4250 7350 3450 * 3400 290
-3000 -120		14150 * 14150 30850 * 30850	10350 * 10350 22200 * 22200	8350 * 8350 16200 * 16200	5600 * 5600 12100			6000 * 5100 6470 3450 * 3350 280
-4500 -180			7950 * 7950 16300 * 16300					6000 * 5600 4980 14500 * 14500 200

Illustration 196

g06225667

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

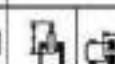
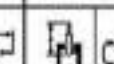
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240				5200 205	5200 205		6000 240
4500 180			6550 260	6550 260	5600 220	4750 185	4500 180
3000 120			8250 325	8250 325	6900 275	5900 235	3000 120
1500 60			9750 385	9750 385	8400 330	7400 290	1500 60
0			10950 435	10950 435	9600 380	8600 340	0
-1500 -60		10300 405	10300 405	8350 330	7550 300		-1500 -60
-3000 -120		12800 505	12800 505	10450 415	8650 340		-3000 -120
-4500 -180			16850 665	16850 665			-4500 -180

Illustration 197

g06225669

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

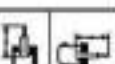
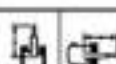
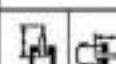
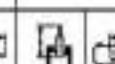
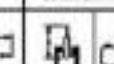
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							7500 300
6000 240				5750 225	5750 225		6000 240
4500 180			1250 50	1250 50	6200 245	5450 215	4500 180
3000 120			1900 75	1900 75	8100 320	7100 280	3000 120
1500 60			2750 105	2750 105	10550 415	9400 370	1500 60
0			3450 135	3450 135	12550 495	11400 450	0
-1500 -60		11950 470	11950 470	8400 330	7600 300		-1500 -60
-3000 -120		14450 565	14450 565	10550 415	9700 380		-3000 -120
-4500 -180			18500 725	18500 725			-4500 -180

Illustration 198

g06225670

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.9 m (9 ft 6 inch) Normal Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

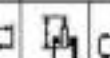
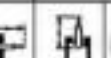
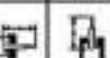
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300				4950 195	4950 195		3050 120
6000 240				4950 195	4950 195		3850 152
4500 180				5400 213	5400 213	4500 177	3850 152
3000 120			7900 311	8050 317	8050 317	4400 173	3850 152
1500 60			9500 374	9600 378	9600 378	4250 167	3850 152
0 0		6250 246	6250 246	10400 409	7600 299	4100 161	4300 169
-1500 -60	6700 264	6700 264	10600 417	10500 409	7750 303	4050 159	4300 169
-3000 -120	14300 560	14300 560	24450 963	22750 896	16900 664	13050 515	4300 169
-4500 -180	25650 1010	25650 1010	30100 1180	21100 829	17300 677	12550 493	4300 169

Illustration 199

g06225578

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

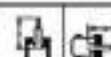
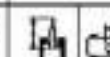
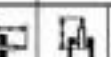
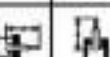
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300				4950 195	4950 195		4300 169
6000 240				5450 214	5450 214		4000 157
4500 180				5950 234	5950 234	4800 189	4000 157
3000 120			8700 343	8700 343	8700 343	4400 173	4000 157
1500 60			10700 417	10700 417	10700 417	4250 167	4000 157
0 0		8800 346	8800 346	12900 508	8300 327	4150 163	4700 185
-1500 -60	7100 279	7100 279	11800 464	11800 464	8600 338	4050 159	4700 185
-3000 -120	15100 594	15100 594	25300 993	25300 993	16900 664	13050 515	4700 185
-4500 -180	27780 1094	27780 1094	33250 1306	23450 924	17350 677	12550 493	4700 185

Illustration 200

g06225580

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

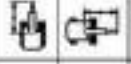
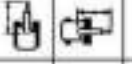
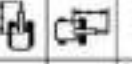
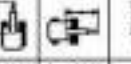
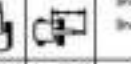
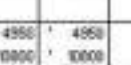
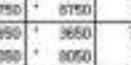
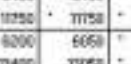
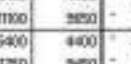
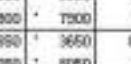
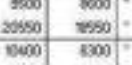
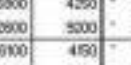
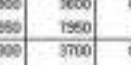
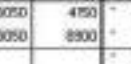
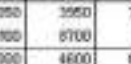
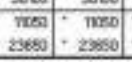
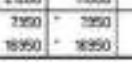
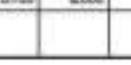
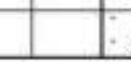
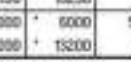
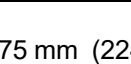
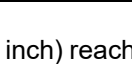
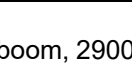
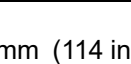
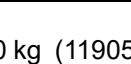
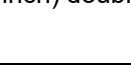
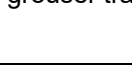
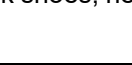
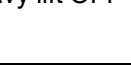
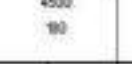
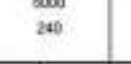
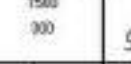
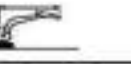
Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							6750 240
6000 240							7230 230
4500 180							7350 320
3000 120							8360 330
1500 60							8460 340
0 0							8260 330
+500 +60							7760 310
+3000 +120							6960 280
+4500 +180							5900 220

Illustration 201

g06225572

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

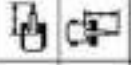
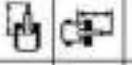
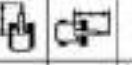
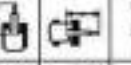
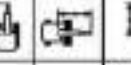
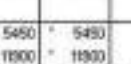
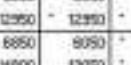
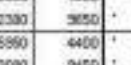
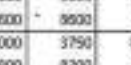
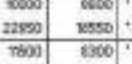
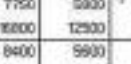
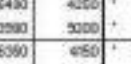
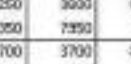
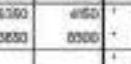
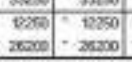
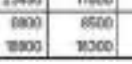
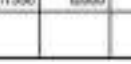
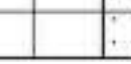
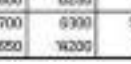
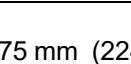
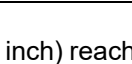
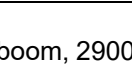
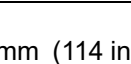
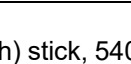
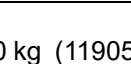
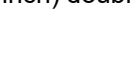
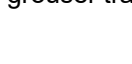
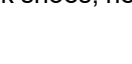
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7500 300							6750 240
6000 240							7230 230
4500 180							7350 320
3000 120							8360 330
1500 60							8460 340
0 0							8260 330
+500 +60							7760 310
+3000 +120							6960 280
+4500 +180							5900 220

Illustration 202

g06225575

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

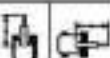
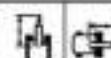
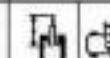
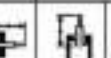
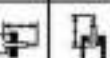
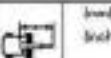
Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300				4550 4550		3550 8750	9550 340
6000 240				4550 10800	4550 10800	3550 8050	7230 230
4500 180				5400 11750	5400 11750	4500 9850	7330 320
3000 120			7900 10950	8200 13050	6050 11750	4400 9450	8360 330
1500 60			9500 20950	9600 18550	7000 12500	5800 12800	9450 340
0 0		6250 14350	6250 14350	10400 22300	1800 11300	5600 12300	4500 9050
-1500 -60	6700 16900	6700 14300	10800 24450	10900 22750	8200 11850	5550 16600	6050 13050
-3000 -120	11450 25850	11450 25850	13900 30400	13900 30400	9800 21300	8250 11800	6250 13200
-4500 -180		10950 23850	10950 23850	7950 16950	7950 16950	6000 13200	5800 10200

Illustration 203

g06225536

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

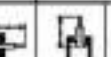
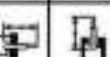
Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300				4550 4550		4300 8550	6750 240
6000 240				5450 11800	5450 11800	4000 8000	7230 230
4500 180				5550 12550	5550 12550	4500 9850	7330 320
3000 120			6700 16700	6850 14800	6050 13050	4400 9450	8360 330
1500 60			10600 22850	10600 18550	7150 16800	5800 12500	9450 340
0 0		6800 15200	6800 15200	11900 24850	1800 11300	5600 12300	4500 9050
-1500 -60	7100 15900	7100 15900	11400 25300	11400 25300	7800 16550	6350 11900	6050 13050
-3000 -120	12100 27100	12100 27100	15350 33290	15350 33290	10850 23490	8250 11800	6250 13200
-4500 -180		12250 26200	12250 26200	8800 18800	8800 18800	6700 14650	5800 10200

Illustration 204

g06225537

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
1500 300							
5000 240							
4500 180							
3000 120							
1500 60							
0							
-6000 -90							
-3000 -120							
-1500 -60							
-4500 -180							

Illustration 205

g06225583

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							Load (inch)
1500 300							
5000 240							
4500 180							
3000 120							
1500 60							
0							
-6000 -90							
-3000 -120							
-1500 -60							
-4500 -180							

Illustration 206

g06225586

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

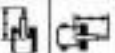
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							3050 120
6000 240							3050 120
4500 180							3050 120
3000 120							3050 120
1500 60							3050 120
0							3050 120
-1500 -60							3050 120
-3000 -120							3050 120
-4500 -180							3050 120
-6000 -240							3050 120
-7500 -300							3050 120

Illustration 207

g06225538

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

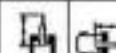
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
7500 300							3050 120
6000 240							3050 120
4500 180							3050 120
3000 120							3050 120
1500 60							3050 120
0							3050 120
-1500 -60							3050 120
-3000 -120							3050 120
-4500 -180							3050 120
-6000 -240							3050 120
-7500 -300							3050 120

Illustration 208

g06225539

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

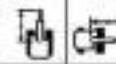
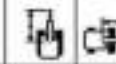
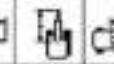
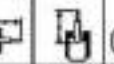
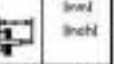
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6750 240
6000 240				4350 10000	4350 10000			6750 240
4500 180				5400 11750	5400 11750	5100 11000	4800 10000	7350 320
3000 120			7900 18350	7900 18350	6200 13350	5400 11750	4500 10000	8390 330
1500 60			3500 20550	3500 20550	2900 12600	2400 10000	1900 8550	8450 340
0		6250 14350	6250 14350	10400 22500	7800 16350	5750 12400	4300 10000	8290 330
-1500 -60	5700 14300	5700 14300	10600 24450	10600 24450	8450 22750	5700 12200	4250 10000	7780 310
-3000 -120	11450 25050	11450 25050	13900 30100	13900 30100	9800 21000	7300 16300	5700 12300	6950 280
-4500 -180		11050 23650	11050 23650	7350 16350	7350 16350		6000 13200	5900 220

Illustration 209

g06225542

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

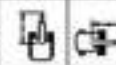
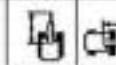
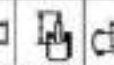
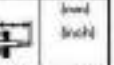
Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6750 240
6000 240				4350 11800	4350 11800			6750 240
4500 180				5450 12550	5450 12550	5100 11000	4800 10000	7350 320
3000 120			8700 19700	8700 19700	6850 13350	5950 13000	4500 10000	8390 330
1500 60			10800 22150	10800 22150	8900 12600	7750 10000	6400 8550	8450 340
0		6800 15200	6800 15200	11900 24850	8500 18350	5750 12400	4300 10000	8290 330
-1500 -60	7100 15800	7100 15800	11400 25900	11400 25900	9800 21000	5700 12200	4250 10000	7780 310
-3000 -120	12100 27150	12100 27150	15350 33250	15350 33250	10850 23450	8050 17300	5700 12300	6950 280
-4500 -180		12250 26200	12250 26200	8800 19800	8800 19800		6100 14650	5900 220

Illustration 210

g06225543

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6750 240
6000 240								5250 210
4500 180								3750 150
3000 120								2250 90
1500 60								750 30
0 0								0 0
-1500 -60								-750 -30
-3000 -120								-2250 -90
-4500 -180								-3750 -150

Illustration 211

g06225545

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6750 240
6000 240								5250 210
4500 180								3750 150
3000 120								2250 90
1500 60								750 30
0 0								0 0
-1500 -60								-750 -30
-3000 -120								-2250 -90
-4500 -180								-3750 -150

Illustration 212

g06225549

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.9 m (9 ft 6 inch) Extended Stick and 5400 kg (11905 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

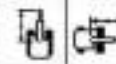
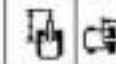
Load (mm)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							3050 120
6000 240							3000 118
4500 180							2950 116
3000 120							2900 114
1500 60							2850 112
0 0							2800 110
-1500 -60							2750 108
-3000 -120							2700 106
-4500 -180							2650 104

Illustration 213

g06225643

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

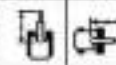
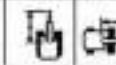
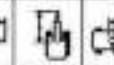
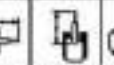
Load (mm)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							3050 120
6000 240							3000 118
4500 180							2950 116
3000 120							2900 114
1500 60							2850 112
0 0							2800 110
-1500 -60							2750 108
-3000 -120							2700 106
-4500 -180							2650 104

Illustration 214

g06225644

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

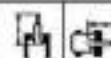
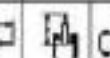
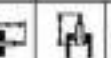
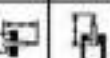
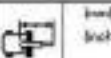
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300				4500 180	4500 180		3050 120
5500 240				4800 192	4800 192		3600 144
4500 180				5300 212	5300 212	4350 174	3500 140
3000 120			7750 310	8050 322	8050 322	5250 210	3800 152
1500 60			8350 334	8650 346	8650 346	5850 234	3450 138
0		6200 248	8200 328	10050 402	10400 416	6000 240	3250 130
-1500 -60	6400 256	6400 256	10700 428	10700 428	10700 428	5200 208	3000 120
-3000 -120	14750 582	14750 582	24300 972	24300 972	24300 972	11750 462	3350 132
-4500 -180	25500 1008	25500 1008	29600 1184	29600 1184	29600 1184	15500 612	3900 156

Illustration 215

g06225649

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

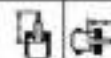
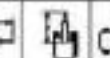
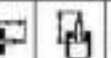
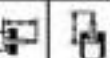
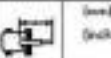
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
7500 300				4300 172	4300 172		4250 168
5500 240				5300 212	5300 212		3500 140
4500 180				5850 234	5850 234	4350 174	3650 144
3000 120			8950 354	9350 370	9350 370	5850 234	3550 140
1500 60			10050 402	10400 416	10400 416	6250 250	3450 138
0		6950 274	8950 354	10050 402	10400 416	6400 256	3250 130
-1500 -60	7000 276	7000 276	11350 450	11350 450	11350 450	5300 212	3000 120
-3000 -120	15650 618	15650 618	25750 1022	25750 1022	25750 1022	11750 462	3350 132
-4500 -180	27000 1068	27000 1068	32750 1290	32750 1290	32750 1290	15500 612	3900 156

Illustration 216

g06225650

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

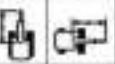
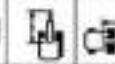
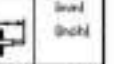
Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Reach (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0 0								0 0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 217

g06225646

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

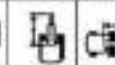
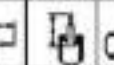
Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Reach (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0 0								0 0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 218

g06225648

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 2.9 m (9 ft 6 inch) TRS Extended Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6150 240
6000 240								7230 230
4500 180								7350 320
3000 120								8380 330
1500 60								8450 340
0								8260 330
-1500 -60								7780 310
-3000 -120								6950 280
-4500 -180								5600 220

Illustration 219

g06225699

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
7500 300								6150 240
6000 240								7230 230
4500 180								7350 320
3000 120								8380 330
1500 60								8450 340
0								8260 330
-1500 -60								7780 310
-3000 -120								6950 280
-4500 -180								5600 220

Illustration 220

g06225700

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

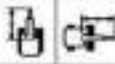
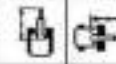
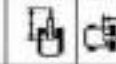
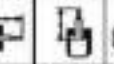
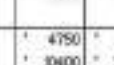
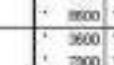
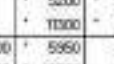
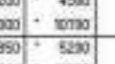
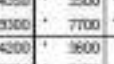
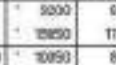
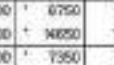
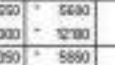
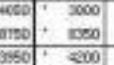
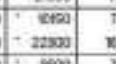
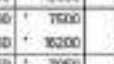
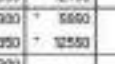
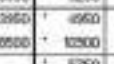
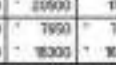
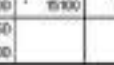
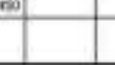
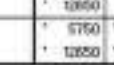
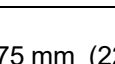
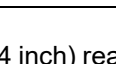
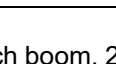
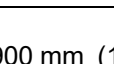
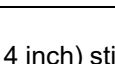
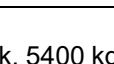
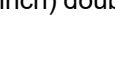
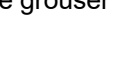
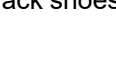
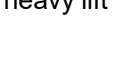
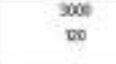
(mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							8150 323
6000 240							7250 286
4500 180							7550 300
3000 120							8360 330
1500 60							8450 333
0 0							8280 326
-1500 -60							7780 306
-3000 -120							8880 349
-4500 -180							8800 346

Illustration 221

g06225696

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

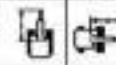
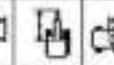
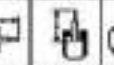
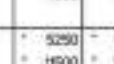
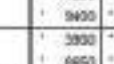
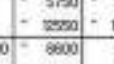
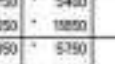
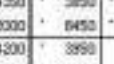
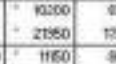
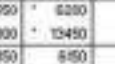
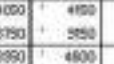
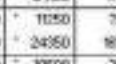
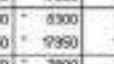
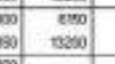
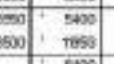
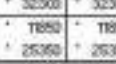
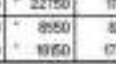
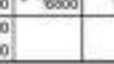
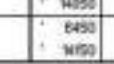
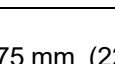
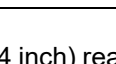
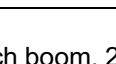
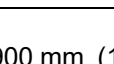
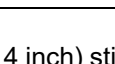
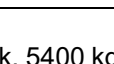
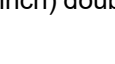
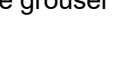
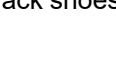
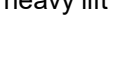
(mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							8150 323
6000 240							7250 286
4500 180							7550 300
3000 120							8360 330
1500 60							8450 333
0 0							8280 326
-1500 -60							7780 306
-3000 -120							8880 349
-4500 -180							8800 346

Illustration 222

g06225697

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

Reach (mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300			
								Reach (mm) (inch)
7500 300				4500 180	4500 180			3050 120
6000 240				4750 188	4750 188			3000 118
4500 180				5200 205	5200 205	4300 170	4350 171	2900 114
3000 120			7800 307	8000 315	5550 219	5200 205	4200 165	3500 138
1500 60			8900 350	9100 358	6150 242	5600 220	4500 177	3450 136
0 0		8200 323	8200 323	10950 430	10950 430	5400 213	4000 157	3250 128
-1500 -60	6600 260	6600 260	10700 417	10700 417	8150 320	5850 230	4850 191	3750 148
-3000 -120	11350 447	11350 447	13450 529	13450 529	9900 390	7850 309	6550 258	4450 175
-4500 -180	22500 885	22500 885	25150 986	25150 986	13650 538	11500 453	9550 375	5900 232

Illustration 223

g06225687

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300			
								Reach (mm) (inch)
7500 300				4300 169	4300 169			4250 167
6000 240				5250 207	5250 207			3900 153
4500 180				6500 256	6500 256	5400 213	4350 171	3850 151
3000 120			8400 327	8400 327	5850 230	5750 226	4300 169	3550 139
1500 60			10200 398	10200 398	7500 295	5600 220	4100 161	3450 136
0 0		6550 258	6550 258	11850 463	11850 463	5150 203	4000 157	3500 138
-1500 -60	7000 276	7000 276	11950 469	11950 469	8300 326	6150 242	3950 155	3750 148
-3000 -120	12050 473	12050 473	14900 584	14900 584	9950 398	7800 307	6450 254	4450 175
-4500 -180	27000 1060	27000 1060	32300 1268	32300 1268	17500 688	14500 573	11550 453	5900 232

Illustration 224

g06225688

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							3550 140
6000 240				4500 180	4500 180		3000 120
4500 180				4750 190	4750 190		2500 100
3000 120				5200 208	5200 208		2000 80
1500 60				5650 224	5650 224		1500 60
0 0				6100 240	6100 240		1000 40
-1500 -60				6550 256	6550 256		500 20
-3000 -120				7000 272	7000 272		0 0
-4500 -180				7450 288	7450 288		

Illustration 225

g06225701

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
7500 300							4250 168
6000 240				4500 180	4500 180		3800 150
4500 180				5250 208	5250 208		3350 132
3000 120				6000 240	6000 240		2900 114
1500 60				6750 268	6750 268		2450 96
0 0				7500 296	7500 296		2000 80
-1500 -60				8250 324	8250 324		1550 62
-3000 -120				9000 352	9000 352		1100 44
-4500 -180				9750 380	9750 380		650 26

Illustration 226

g06225703

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

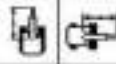
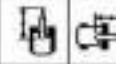
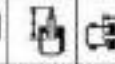
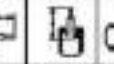
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							3850 152
6000 240				4750 187	4750 187		3600 142
4500 180				5200 205	5200 205	4500 177	3500 138
3000 120			7800 307	7800 307	5950 234	5200 205	3800 150
1500 60			8400 331	8400 331	6750 266	5950 234	3600 142
0 0		6200 244	6200 244	7800 307	7800 307	5950 234	3500 138
-1500 -60	8900 350	8900 350	10700 421	10700 421	12500 492	12500 492	4600 181
-3000 -120	13500 531	13500 531	13450 528	13450 528	15200 598	15200 598	5750 227
-4500 -180	18000 709	18000 709	17850 701	17850 701	19500 768	19500 768	6850 270

Illustration 227

g06225689

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

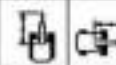
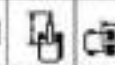
(mm) (inch)	500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300							4250 167
6000 240				5250 207	5250 207		3900 153
4500 180				5750 227	5750 227	4500 177	3850 152
3000 120			8400 331	8400 331	5950 234	5200 205	3800 150
1500 60			8900 350	8900 350	6750 266	5950 234	3600 142
0 0		6950 273	6950 273	8400 331	8400 331	5950 234	3500 138
-1500 -60	7000 276	7000 276	10950 428	10950 428	12500 492	12500 492	4600 181
-3000 -120	12050 475	12050 475	14300 562	14300 562	15200 598	15200 598	5750 227
-4500 -180	17000 669	17000 669	16850 661	16850 661	19500 768	19500 768	6850 270

Illustration 228

g06225690

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Reach (mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Reach (mm) (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0 0							0 0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180

Illustration 229

g06225691

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

Reach (mm) (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Reach (mm) (inch)
7500 300							7500 300
6000 240							6000 240
4500 180							4500 180
3000 120							3000 120
1500 60							1500 60
0 0							0 0
-1500 -60							-1500 -60
-3000 -120							-3000 -120
-4500 -180							-4500 -180

Illustration 230

g06225692

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

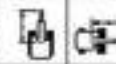
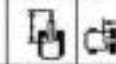
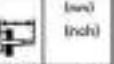
Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0 0								0 0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 231

g06225694

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

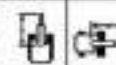
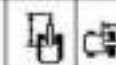
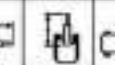
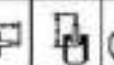
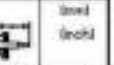
Reach (mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Reach (mm) (inch)
7500 300								7500 300
6000 240								6000 240
4500 180								4500 180
3000 120								3000 120
1500 60								1500 60
0 0								0 0
-1500 -60								-1500 -60
-3000 -120								-3000 -120
-4500 -180								-4500 -180

Illustration 232

g06225695

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

HD Reach Boom with a 3.9 m (12 ft 8 inch) Stick and 5400 kg (11905 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Load (lb/ft)	800 80	3000 100	4500 150	6000 240	7500 300	9000 360		Load (lb/ft)
7500 300								2700 1200
6000 240					4050 1800	4050 1800		2550 1150
4500 180					4250 1900	4250 1900		2550 1150
3000 120				6400 2900	6400 2900	4750 2150		2550 1150
1500 60		3250 1450	3250 1450	6250 2800	6250 2800	4750 2150	4050 1800	2550 1150
0 0		22000 10000	22000 10000	17000 7700	17000 7700	10300 4650	7750 3500	5850 2650
-1500 -60	5700 2550	5700 2550	8800 4000	8800 4000	10200 4600	10200 4600	3850 1750	3300 1500
-3000 -120	8800 4000	8800 4000	13600 6200	13600 6200	13600 6200	13600 6200	4050 1800	3850 1750
-4500 -180	12300 5550	12300 5550	13000 6000	13000 6000	13000 6000	13000 6000	5750 2600	4500 2050
-6000 -240	23050 10500	23050 10500	27350 12400	27350 12400	27350 12400	27350 12400	7800 3500	10700 4850

Illustration 233

g06225501

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Load (lb/ft)	800 80	3000 100	4500 150	6000 240	7500 300	9000 360		Load (lb/ft)
7500 300								3000 1320
6000 240					4500 2025	4500 2025		2800 1280
4500 180					4750 2150	4750 2150		2750 1250
3000 120				7400 3350	7400 3350	5200 2350		3850 1750
1500 60		3800 1725	3800 1725	6250 2800	6250 2800	4750 2150	4050 1800	2550 1150
0 0		22000 10000	22000 10000	17000 7700	17000 7700	10300 4650	7750 3500	5850 2650
-1500 -60	8050 3650	8050 3650	10400 4725	10400 4725	12500 5675	12500 5675	4450 2025	3850 1750
-3000 -120	13500 6150	13500 6150	14400 6525	14400 6525	17000 7700	17000 7700	5850 2650	4450 2025
-4500 -180	21000 9500	21000 9500	22250 10125	22250 10125	24000 10900	24000 10900	7800 3500	10700 4850
-6000 -240	30750 13900	30750 13900	33000 15000	33000 15000	35250 15900	35250 15900	10750 4850	13100 5950

Illustration 234

g06225504

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Level (inch)	500 80	3000 120	4500 180	6000 240	7500 300	9000 360		Level (inch)
7500 300								7500 300
6000 240					4150 165	4150 165		6000 240
4500 180					4250 165	4250 165		4500 180
3000 120				8400 330	8400 330	8300 300	8300 300	3000 120
1500 60		3250 125	3250 125	6150 230	6150 230	6000 210	6000 210	1500 60
0 0		7900 310	7600 290	5850 220	5700 210	5550 200	5550 200	0 0
-1500 -60	5700 225	5700 225	3800 145	3200 120	3050 115	2850 105	2850 105	-1500 -60
-3000 -120	8800 335	8800 335	5850 220	5050 185	4900 180	4700 170	4700 170	-3000 -120
-4500 -180	12900 500	12900 500	8050 300	6900 255	6750 250	6550 240	6550 240	-4500 -180
-6000 -240	20500 800	20500 800	12550 460	10750 400	10600 400	10400 390	10400 390	-6000 -240

Illustration 235

g06225493

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

Level (inch)	500 80	3000 120	4500 180	6000 240	7500 300	9000 360		Level (inch)
7500 300								7500 300
6000 240					4150 165	4150 165		6000 240
4500 180					4250 165	4250 165		4500 180
3000 120				8400 330	8400 330	8300 300	8300 300	3000 120
1500 60		3250 125	3250 125	6150 230	6150 230	6000 210	6000 210	1500 60
0 0		7900 310	7600 290	5850 220	5700 210	5550 200	5550 200	0 0
-1500 -60	5700 225	5700 225	3800 145	3200 120	3050 115	2850 105	2850 105	-1500 -60
-3000 -120	8800 335	8800 335	5850 220	5050 185	4900 180	4700 170	4700 170	-3000 -120
-4500 -180	12900 500	12900 500	8050 300	6900 255	6750 250	6550 240	6550 240	-4500 -180
-6000 -240	20500 800	20500 800	12550 460	10750 400	10600 400	10400 390	10400 390	-6000 -240

Illustration 236

g06225499

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

Level Inch/ft	2500 60	3000 70	4500 100	6000 140	7500 180	9000 220	10500 260	12000 300	13500 340	15000 380	16500 420	18000 460	19500 500	21000 540	22500 580	24000 620	25500 660	27000 700	28500 740	30000 780	31500 820	33000 860	34500 900	36000 940	37500 980	39000 1020	40500 1060	42000 1100	43500 1140	45000 1180	46500 1220	48000 1260	49500 1300	51000 1340	52500 1380	54000 1420	55500 1460	57000 1500	58500 1540	60000 1580	61500 1620	63000 1660	64500 1700	66000 1740	67500 1780	69000 1820	70500 1860	72000 1900	73500 1940	75000 1980	76500 2020	78000 2060	79500 2100	81000 2140	82500 2180	84000 2220	85500 2260	87000 2300	88500 2340	90000 2380	91500 2420	93000 2460	94500 2500	96000 2540	97500 2580	99000 2620	100500 2660	102000 2700	103500 2740	105000 2780	106500 2820	108000 2860	109500 2900	111000 2940	112500 2980	114000 3020	115500 3060	117000 3100	118500 3140	120000 3180	121500 3220	123000 3260	124500 3300	126000 3340	127500 3380	129000 3420	130500 3460	132000 3500	133500 3540	135000 3580	136500 3620	138000 3660	139500 3700	141000 3740	142500 3780	144000 3820	145500 3860	147000 3900	148500 3940	150000 3980	151500 4020	153000 4060	154500 4100	156000 4140	157500 4180	159000 4220	160500 4260	162000 4300	163500 4340	165000 4380	166500 4420	168000 4460	169500 4500	171000 4540	172500 4580	174000 4620	175500 4660	177000 4700	178500 4740	180000 4780	181500 4820	183000 4860	184500 4900	186000 4940	187500 4980	189000 5020	190500 5060	192000 5100	193500 5140	195000 5180	196500 5220	198000 5260	199500 5300	201000 5340	202500 5380	204000 5420	205500 5460	207000 5500	208500 5540	210000 5580	211500 5620	213000 5660	214500 5700	216000 5740	217500 5780	219000 5820	220500 5860	222000 5900	223500 5940	225000 5980	226500 6020	228000 6060	229500 6100	231000 6140	232500 6180	234000 6220	235500 6260	237000 6300	238500 6340	240000 6380	241500 6420	243000 6460	244500 6500	246000 6540	247500 6580	249000 6620	250500 6660	252000 6700	253500 6740	255000 6780	256500 6820	258000 6860	259500 6900	261000 6940	262500 6980	264000 7020	265500 7060	267000 7100	268500 7140	270000 7180	271500 7220	273000 7260	274500 7300	276000 7340	277500 7380	279000 7420	280500 7460	282000 7500	283500 7540	285000 7580	286500 7620	288000 7660	289500 7700	291000 7740	292500 7780	294000 7820	295500 7860	297000 7900	298500 7940	300000 7980	301500 8020	303000 8060	304500 8100	306000 8140	307500 8180	309000 8220	310500 8260	312000 8300	313500 8340	315000 8380	316500 8420	318000 8460	319500 8500	321000 8540	322500 8580	324000 8620	325500 8660	327000 8700	328500 8740	330000 8780	331500 8820	333000 8860	334500 8900	336000 8940	337500 8980	339000 9020	340500 9060	342000 9100	343500 9140	345000 9180	346500 9220	348000 9260	349500 9300	351000 9340	352500 9380	354000 9420	355500 9460	357000 9500	358500 9540	360000 9580	361500 9620	363000 9660	364500 9700	366000 9740	367500 9780	369000 9820	370500 9860	372000 9900	373500 9940	375000 9980	376500 10020	378000 10060	379500 10100	381000 10140	382500 10180	384000 10220	385500 10260	387000 10300	388500 10340	390000 10380	391500 10420	393000 10460	394500 10500	396000 10540	397500 10580	399000 10620	400500 10660	402000 10700	403500 10740	405000 10780	406500 10820	408000 10860	409500 10900	411000 10940	412500 10980	414000 11020	415500 11060	417000 11100	418500 11140	420000 11180	421500 11220	423000 11260	424500 11300	426000 11340	427500 11380	429000 11420	430500 11460	432000 11500	433500 11540	435000 11580	436500 11620	438000 11660	439500 11700	441000 11740	442500 11780	444000 11820	445500 11860	447000 11900	448500 11940	450000 11980	451500 12020	453000 12060	454500 12100	456000 12140	457500 12180	459000 12220	460500 12260	462000 12300	463500 12340	465000 12380	466500 12420	468000 12460	469500 12500	471000 12540	472500 12580	474000 12620	475500 12660	477000 12700	478500 12740	480000 12780	481500 12820	483000 12860	484500 12900	486000 12940	487500 12980	489000 13020	490500 13060	492000 13100	493500 13140	495000 13180	496500 13220	498000 13260	499500 13300	501000 13340	502500 13380	504000 13420	505500 13460	507000 13500	508500 13540	510000 13580	511500 13620	513000 13660	514500 13700	516000 13740	517500 13780	519000 13820	520500 13860	522000 13900	523500 13940	525000 13980	526500 14020	528000 14060	529500 14100	531000 14140	532500 14180	534000 14220	535500 14260	537000 14300	538500 14340	540000 14380	541500 14420	543000 14460	544500 14500	546000 14540	547500 14580	549000 14620	550500 14660	552000 14700	553500 14740	555000 14780	556500 14820	558000 14860	559500 14900	561000 14940	562500 14980	564000 15020	565500 15060	567000 15100	568500 15140	570000 15180	571500 15220	573000 15260	574500 15300	576000 15340	577500 15380	579000 15420	580500 15460	582000 15500	583500 15540	585000 15580	586500 15620	588000 15660	589500 15700	591000 15740	592500 15780	594000 15820	595500 15860	597000 15900	598500 15940	600000 15980	601500 16020	603000 16060	604500 16100	606000 16140	607500 16180	609000 16220	610500 16260	612000 16300	613500 16340	615000 16380	616500 16420	618000 16460	619500 16500	621000 16540	622500 16580	624000 16620	625500 16660	627000 16700	628500 16740	630000 16780	631500 16820	633000 16860	634500 16900	636000 16940	637500 16980	639000 17020	640500 17060	642000 17100	643500 17140	645000 17180	646500 17220	648000 17260	649500 17300	651000 17340	652500 17380	654000 17420	655500 17460	657000 17500	658500 17540	660000 17580	661500 17620	663000 17660	664500 17700	666000 17740	667500 17780	669000 17820	670500 17860	672000 17900	673500 17940	675000 17980	676500 18020	678000 18060	679500 18100	681000 18140	682500 18180	684000 18220	685500 18260	687000 18300	688500 18340	690000 18380	691500 18420	693000 18460	694500 18500	696000 18540	697500 18580	699000 18620	700500 18660	702000 18700	703500 18740	705000 18780	706500 18820	708000 18860	709500 18900	711000 18940	712500 18980	714000 19020	715500 19060	717000 19100	718500 19140	720000 19180	721500 19220	723000 19260	724500 19300	726000 19340	727500 19380	729000 19420	730500 19460	732000 19500	733500 19540	735000 19580	736500 19620	738000 19660	739500 19700	741000 19740	742500 19780	744000 19820	745500 19860	747000 19900	748500 19940	750000 19980	751500 20020	753000 20060	754500 20100	756000 20140	757500 20180	759000 20220	760500 20260	762000 20300	763500 20340	765000 20380	766500 20420	768000 20460	769500 20500	771000 20540	772500 20580	774000 20620	775500 20660	777000 20700	778500 20740	780000 20780	781500 20820	783000 20860	784500 20900	786000 20940	787500 20980	789000 21020	790500 21060	792000 21100	793500 21140	795000 21180	796500 21220	798000 21260	799500 21300	801000 21340	802500 21380	804000 21420	805500 21460	807000 21500	808500 21540	810000 21580	811500 21620	813000 21660	814500 21700	816000 21740	817500 21780	819000 21820	820500 21860	822000 21900	823500 21940	825000 21980	826500 22020	828000 22060	829500 22100	831000 22140	832500 22180	834000 22220	835500 22260	837000 22300	838500 22340	840000 22380	841500 22420	843000 22460	844500 22500	846000 22540	847500 22580	849000 22620	850500 22660	852000 22700	853500 22740	855000 22780	856500 22820	858000 22860	859500 22900	861000 22940	862500 22980	864000 23020	865500 23060	867000 23100	868500 23140	870000 23180	871500 23220	873000 23260	874500 23300	876000 23340	877500 23380	879000 23420	880500 23460	882000 23500	883500 23540	885000 23580	886500 23620	888000 23660	889500 23700	891000 23740	892500 23780	894000 23820	895500 23860	897000 23900	898500 23940	900000 23980	901500 24020	903000 24060	904500 24100	906000 24140	907500 24180	909000 24220	910500 24260	912000 24300	913500 24340	915000 24380	916500 24420	918000 24460	919500 24500	921000 24540	922500 24580	924000 24620	925500 24660	927000 24700	928500 24740	930000 24780	931500 24820	933000 24860	934500 24900	936000 24940	937500 24980	939000 25020	940500 25060	942000 25100	943500 25140	945000 25180	946500 25220	948000 25260	949500 25300	951000 25340	952500 25380	954000 25420	955500 25460	957000 25500	958500 25540	960000 25580	961500 25620	963000 25660	964500 25700	966000 25740	967500 25780	969000 25820	970500 25860	972000 25900	973500 25940	975000 25980	976500 26020	978000 26060	979500 26100	981000 26140	982500 26180	984000 26220	985500 26260	987000 26300	988500 26340	990000 26380	991500 26420	993000 26460	994500 26500	996000 26540	997500 26580	999000 26620	1000500 26660	1002000 26700	1003500 26740	1005000 26780	1006500 26820	1008000 26860	1009500 26900	1011000 26940	1012500 26980	1014000 27020	1015500 27060	1017000 27100	1018500 27140	1020000 27180	1021500 27220	1023000 27260	1024500 27300	1026000 27340	1027500 27380	1029000 27420	1030500 27460	1032000 27500	1033500 27540	1035000 27580	1036500 27620	1038000 27660	1039500 27
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600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Stick (inch)
7500 300									7320
9000 240					4050 159	4190 165			8300
4500 180					4250 167	4290 169			8320
3000 120				6400 252	6400 252	6400 252	4700 185	4290 169	8250
1500 60		3250 127	3290 129	4290 169	4290 169	4290 169	4100 162	4100 162	8230
0 0		7900 311	7900 311	7900 311	7900 311	7900 311	7900 311	7900 311	8160
-7500 -300	5700 224	5700 224	5700 224	5700 224	5700 224	5700 224	5700 224	5700 224	8730
-9000 -240	8900 350	8900 350	8900 350	8900 350	8900 350	8900 350	8900 350	8900 350	9300
-4500 -180	12500 492	12500 492	12500 492	12500 492	12500 492	12500 492	12500 492	12500 492	9870
-3000 -120	16000 630	16000 630	16000 630	16000 630	16000 630	16000 630	16000 630	16000 630	10440
-1500 -60	19500 764	19500 764	19500 764	19500 764	19500 764	19500 764	19500 764	19500 764	11010
-0 0	23000 900	23000 900	23000 900	23000 900	23000 900	23000 900	23000 900	23000 900	11580

Illustration 239

g06225507

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360			Stick (inch)
7500 300									7320
9000 240					4500 177	4500 177			8300
4500 180					4750 187	4800 190			8320
3000 120				7000 276	7000 276	7000 276	5200 205	4290 169	8250
1500 60		3800 149	3800 149	4800 189	4800 189	4800 189	4600 181	4600 181	8230
0 0		8000 311	8000 311	8000 311	8000 311	8000 311	8000 311	8000 311	8160
-7500 -300	6050 238	6050 238	6050 238	6050 238	6050 238	6050 238	6050 238	6050 238	8730
-9000 -240	9200 360	9200 360	9200 360	9200 360	9200 360	9200 360	9200 360	9200 360	9300
-4500 -180	12750 501	12750 501	12750 501	12750 501	12750 501	12750 501	12750 501	12750 501	9870
-3000 -120	16200 635	16200 635	16200 635	16200 635	16200 635	16200 635	16200 635	16200 635	10440
-1500 -60	19700 771	19700 771	19700 771	19700 771	19700 771	19700 771	19700 771	19700 771	11010
-0 0	23200 910	23200 910	23200 910	23200 910	23200 910	23200 910	23200 910	23200 910	11580

Illustration 240

g06225512

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

level inch	1000 80		3000 125		4500 160		6800 240		7580 300		5000 380				level inch
															
1500 300													* 2700	* 2700	1120
													* 6000	* 6000	280
6800 240									* 4090	* 4090			* 2550	* 2550	1000
									* 6750	* 6750			* 5600	* 5600	330
4500 160									* 4250	* 4250			* 2500	* 2500	880
									* 3350	* 3350			* 5500	* 5500	350
3000 125					* 8400	* 8400	* 6300	* 6300	* 4700	* 4400	* 3600	* 3300	* 2550	* 2550	820
					* 10500	* 10500	* 8450	* 8450	* 6200	* 5450	* 4500	* 4500	* 3600	* 3600	370
5000 60			* 3250	* 3250	* 8250	* 8250	* 6200	* 5800	* 5200	* 4250	* 4050	* 3250	* 2700	* 2700	1130
			* 23500	* 23500	* 17000	* 17000	* 15450	* 15450	* 10800	* 9800	* 7700	* 6900	* 5850	* 5850	370
0			* 7600	* 7600	* 9600	* 9300	* 7000	* 6550	* 5850	* 4100	* 3800	* 3150	* 2300	* 2300	880
0			* 12400	* 12400	* 20750	* 19050	* 6150	* 6050	* 12200	* 9800	* 8800	* 8600	* 8400	* 8400	360
-5000	* 5700	* 5700	* 9800	* 9800	* 9200	* 8950	* 1450	* 5400	* 5850	* 4000			* 3300	* 3250	870
-60	* 12700	* 12700	* 22250	* 22250	* 22050	* 17300	* 8100	* 7600	* 12700	* 9800			* 7300	* 7200	350
-3800	* 8900	* 8900	* 15850	* 15850	* 10350	* 8300	* 1400	* 5350	* 5700	* 4000			* 4050	* 3700	880
-05	* 19900	* 19900	* 30950	* 30950	* 27700	* 71250	* 6350	* 1500	* 12200	* 9800			* 9000	* 8150	120
-4500	* 12300	* 12300	* 13000	* 13000	* 3000	* 8750	* 8600	* 5450					* 5250	* 4600	880
-950	* 23950	* 23950	* 27850	* 27850	* 9350	* 15550	* 14050	* 1750					* 18000	* 16250	270
-6000					* 4350	* 4350							* 5600	* 5600	490

Illustration 241

g06225458

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

level inch)	800 80		3000 125		4500 180		6800 240		7580 300		5000 380				level inch)
7500 300															7320
6800 240									4500	4500			2800	2800	6300
4500 180									3600	3600			9150	9150	3300
3000 120									4750	4580			2750	2750	8820
800 30									3030	3150			8050	8050	3500
500 20					7400	7400	8850	8850	5200	4480	3650	3300	3800	3800	8250
300 10					6250	6250	8050	8050	4050	3450	2760	2060	3150	3150	6750
150 5									5750	4250	4400	3250	2950	2950	6330
75 3					3800	3800	3750	4750	6300	5800	5750	4250	3650	3650	5700
37 1.5					23300	23300	9700	8900	34500	12500	10500	9000	8600	8600	4450
18 0.75					9100	9100	8800	8300	7750	6950	6250	4100	4150	3150	3100
9 0.375					8450	8450	22500	10050	9800	10500	9800	7250	6600	7050	6600
-7500 -300		6050	6950	93400	93400	3500	6050	6250	5400	6300	4000		3050	3350	8730
-6800 -240		13500	13500	23500	23500	24400	17300	17950	16000	13350	3800		8050	7200	3500
-6000 -200		5800	5800	9490	9490	1100	8000	8200	8350	8200	4000		4450	3700	8800
-4500 -180		2100	2100	32750	32750	24050	17250	17750	15000	11350	3600		3950	3150	3200
-3000 -120		13850	13850	94400	94400	8000	3750	3250	5450				8000	4900	8870
-1500 -60		30750	30750	31800	31800	21500	15550	16850	17750				11950	10250	2700
-800 -30						7050	7050						6250	6250	4980

Illustration 242

g06225460

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Reach (inch)	800 #0		3000 120		4500 180		6000 240		7500 300		9000 360				Reach (inch)	
																
7500 300														* 2700 * 3000	2700 3000	7320 290
6000 240									* 4050 * 4190	* 4050 * 4190				* 2550 * 3000	2550 3000	6300 250
4500 180									* 4250 * 4290	* 4250 * 4290				* 2500 * 2500	2500 2500	6820 260
3000 120					* 8400 * 10600	* 8400 * 10600	* 9300 * 11450	* 9300 * 11450	* 4700 * 6000	* 4450 * 5550	* 3500 * 4500	* 3550 * 4500	* 2550 * 3600	* 2550 * 3600	2550 3600	8250 320
1500 60		* 3250 * 2280	* 3250 * 2280	* 3250 * 2280	* 3250 * 2280	* 3250 * 2280	* 3250 * 2280	* 3250 * 2280	* 5200 * 7000	* 4300 * 5200	* 4050 * 5200	* 3250 * 4050	* 2700 * 3250	* 2700 * 3250	2700 3250	5250 370
0 0		* 7600 * 10400	* 7600 * 10400	* 7600 * 10400	* 7600 * 10400	* 7600 * 10400	* 7600 * 10400	* 7600 * 10400	* 9850 * 12100	* 9850 * 12100	* 9850 * 12100	* 9850 * 12100	* 9850 * 12100	* 9850 * 12100	* 9850 * 12100	9850 12100
-1500 -60	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 5700 * 12700	* 8850 * 11300	* 8850 * 11300	* 8850 * 11300	* 8850 * 11300	* 8850 * 11300	* 8850 * 11300	* 8850 * 11300	8850 11300
-3000 -120	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 8800 * 15900	* 11300 * 16950	* 11300 * 16950	* 11300 * 16950	* 11300 * 16950	* 11300 * 16950	* 11300 * 16950	* 11300 * 16950	11300 16950
-4500 -180	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 12900 * 23950	* 16950 * 27000	* 16950 * 27000	* 16950 * 27000	* 16950 * 27000	* 16950 * 27000	* 16950 * 27000	* 16950 * 27000	16950 27000
-6000 -240					* 1350 * 4350	* 1350 * 4350								* 9500 * 9500	9500 9500	4910

Illustration 243

g06225463

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	800 #0		3000 120		4500 180		6000 240		7500 300		9000 360				Reach (inch)	
															Reach (inch)	
7500 300																7320 290
6000 240																6300 250
4500 180																5550 215
3000 120																4800 180
1500 60																4050 158
0 0																3300 128
-1500 -60																2550 100
-3000 -120																1800 70
-4500 -180																1050 40
-6000 -240																300 11
-7500 -300																-300 -11
-9000 -360																-1050 -40
-10500 -420																-2300 -90
-12000 -480																-3550 -140
-13500 -540																-4800 -188
-15000 -600																-6050 -238
-16500 -660																-7300 -285
-18000 -720																-8550 -335
-19500 -780																-9800 -382
-21000 -840																-11050 -430
-22500 -900																-12300 -478
-24000 -960																-13550 -525
-25500 -1020																-14800 -573
-27000 -1080																-16050 -620
-28500 -1140																-17300 -668
-30000 -1200																-18550 -715
-31500 -1260																-19800 -763
-33000 -1320																-21050 -810
-34500 -1380																-22300 -858
-36000 -1440																-23550 -905
-37500 -1500																-24800 -953
-39000 -1560																-26050 -1000
-40500 -1620																-27300 -1048
-42000 -1680																-28550 -1095
-43500 -1740																-29800 -1143
-45000 -1800																-31050 -1190
-46500 -1860																-32300 -1238
-48000 -1920																-33550 -1285
-49500 -1980																-34800 -1333
-51000 -2040																-36050 -1380
-52500 -2100																-37300 -1428
-54000 -2160																-38550 -1475
-55500 -2220																-39800 -1523
-57000 -2280																-41050 -1570
-58500 -2340																-42300 -1618
-60000 -2400																-43550 -1665
-61500 -2460																-44800 -1713
-63000 -2520																-46050 -1760
-64500 -2580																-47300 -1808
-66000 -2640																-48550 -1855
-67500 -2700																-49800 -1903
-69000 -2760																-51050 -1950
-70500 -2820																-52300 -1998
-72000 -2880																-53550 -2045
-73500 -2940																-54800 -2093
-75000 -3000																-56050 -2140
-76500 -3060																-57300 -2188
-78000 -3120																-58550 -2235
-79500 -3180																-59800 -2283
-81000 -3240																-61050 -2330
-82500 -3300																-62300 -2378
-84000 -3360																-63550 -2425
-85500 -3420																-64800 -2473
-87000 -3480																-66050 -2520
-88500 -3540																-67300 -2568
-90000 -3600																-68550 -2615
-91500 -3660																-69800 -2663
-93000 -3720																-71050 -2710
-94500 -3780																-72300 -2758
-96000 -3840																-73550 -2805
-97500 -3900																-74800 -2853
-99000 -3960																-76050 -2900
-100500 -4020																-77300 -2948
-102000 -4080																-78550 -2995
-103500 -4140																-79800 -3043
-105000 -4200																-81050 -3090
-106500 -4260																-82300 -3138
-108000 -4320																-83550 -3185
-109500 -4380																-84800 -3233
-111000 -4440																-86050 -3280
-112500 -4500																-87300 -3328
-114000 -4560																-88550 -3375
-115500 -4620																-89800 -3423
-117000 -4680																-91050 -3470
-118500 -4740																-92300 -3518
-120000 -4800																-93550 -3565
-121500 -4860																-94800 -3613
-123000 -4920																-96050 -3660
-124500 -4980																-97300 -3708
-126000 -5040																-98550 -3755
-127500 -5100																-99800 -3803
-129000 -5160																-101050 -3850
-130500 -5220																-102300 -3898
-132000 -5280																-103550 -3945
-133500 -5340																-104800 -3993
-135000 -5400																-106050 -4040
-136500 -5460																-107300 -4088
-138000 -5520																-108550 -4135
-139500 -5580																-109800 -4183
-141000 -5640																-111050 -4230
-142500 -5700																-112300 -4278
-144000 -5760																-113550 -4325
-145500 -5820																-114800 -4373
-147000 -5880																-116050 -4420
-148500 -5940																-117300 -4468
-150000 -6000																-118550 -4515
-151500 -6060																-119800 -4563
-153000 -6120																-121050 -4610
-154500 -6180																-122300 -4658
-156000 -6240																-123550 -4705
-157500 -6300																-124800 -4753
-159000 -6360																-126050 -4800
-160500 -6420																-127300 -4848
-162000 -6480																-128550 -4895
-163500 -6540																-129800 -4943
-165000 -6600																-131050 -4990
-166500 -6660																-132300 -5038
-168000 -6720																-133550 -5085
-169500 -6780																-134800 -5133
-171000 -6840																

Illustration 244

g06225465

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Reach (inch)	1500 40		3000 125		4500 160		6000 240		7500 300		9000 360				Reach (inch)		
															Reach (inch)		
7500 300															* 2700 * 6000	* 2700 * 6000	1320 290
6000 240									* 4190 * 6750	* 4090 * 6750					* 2550 * 6600	* 2550 * 6600	8300 330
4500 180									* 4290 * 5350	* 4290 * 5350					* 2500 * 5500	* 2500 * 5500	8820 350
3000 120					* 8400 * 10600	* 8400 * 10600	* 5300 * 8450	* 5300 * 8450	* 4700 * 8200	* 4500 * 8050	* 3600 * 8500	* 3400 * 8500	* 2550 * 6600	* 2550 * 6600	* 2550 * 6600	* 2550 * 6600	8250 370
1500 60			* 3250 * 21800	* 3250 * 21800	* 3250 * 19800	* 3250 * 18450	* 3250 * 17000	* 5200 * 8900	* 5200 * 8900	* 4350 * 7150	* 4050 * 7150	* 3350 * 7850	* 2700 * 5850	* 2700 * 5850	* 2700 * 5850	* 2700 * 5850	7700 370
0			* 7600 * 17400	* 7600 * 17400	* 9600 * 20750	* 9600 * 19250	* 9600 * 17800	* 5700 * 10200	* 5850 * 10200	* 4200 * 8600	* 3850 * 8600	* 3250 * 8600	* 2900 * 8400	* 2900 * 8400	* 2900 * 8400	* 2900 * 8400	8160 360
-7500 -60	* 5700 * 12700	* 5700 * 12700	* 3800 * 22250	* 3800 * 22250	* 3000 * 22050	* 3000 * 21750	* 1450 * 8100	* 9350 * 17600	* 9350 * 17600	* 8850 * 12300	* 8850 * 12300	* 4100 * 8800	* 3300 * 7300	* 3300 * 7300	* 3300 * 7300	* 3300 * 7300	8730 350
-3000 -120	* 8800 * 19800	* 8800 * 19800	* 5850 * 30950	* 5850 * 30950	* 5050 * 21700	* 5050 * 21700	* 5050 * 21700	* 5900 * 10200	* 5900 * 10200	* 4100 * 8800	* 4100 * 8800	* 4050 * 8000	* 4050 * 8000	* 4050 * 8000	* 4050 * 8000	* 4050 * 8000	8900 320
-4500 -180	* 12900 * 29350	* 12900 * 29350	* 13900 * 27850	* 13900 * 27850	* 3000 * 19350	* 3000 * 19350	* 3000 * 19350	* 8600 * 10200	* 8600 * 10200				* 5350 * 7800	* 5350 * 7800	* 5350 * 7800	* 5350 * 7800	8570 270
-6000 -240					* 4350 * 4350								* 5500 * 5500	* 5500 * 5500			4910

Illustration 245

g06225466

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 40		3000 120		4500 160		6000 240		7500 300		9000 360				Load (inch)	
															Load (inch)	
7500 300															3000 6800 3000 6800 7320 290	
6000 240									4500 8600	4500 8600					2850 6450 2850 6450 5300 320	
4500 180									4750 8050	4750 8050					2750 6050 2750 6050 5520 350	
3000 120					7100 8250	7100 8250	5850 8250	5850 8250	5200 8350	4500 7850	3800 7850	3000 6450	3000 6450	2600 6450	8250 370	
1500 60			3000 21300	3000 21300	3150 19700	3150 18300	3150 16900	3150 15500	5750 9300	4250 8800	3300 7100	2550 6450	2550 6450	2250 6450	8320 370	
0			8100 18450	8100 18450	10800 22950	10800 20250	10800 18850	10800 17450	6250 10550	4200 8000	3650 7250	3000 6950	3000 6950	2650 6950	8960 360	
-7500 -300		8050 25500	8050 25500	10400 27550	10400 27550	7000 24400	7000 24400	7000 24400	6250 10750	5550 9750	4400 8800			3800 7400	8730 350	
-6000 -240		1400 27000	1400 27000	14450 32750	14450 32750	7100 24250	7100 24250	7100 24250	5200 10800	4350 8900				4450 8300	8600 320	
-4500 -180		12650 30750	12650 30750	14400 35000	14400 35000	10800 25000	10800 25000	10800 25000						5000 10500	4750 10500 8570 270	
-3000 -120														5250 5250		4910

Illustration 246

g06225468

Lift Chart Above: 323, 5675 mm (224 inch) reach boom, 3900 mm (154 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Variable Angle Boom with a 2.5 m (8 ft 2 inch) Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

Level (mm)	500 50		1000 100		1500 150		2000 200		2500 250				Level (mm)
													Level (mm)
5000 350													5250 3600
7500 300					6250 1600	6250 1600	4800 1200	4800 1200					4800 1200
6000 240					6750 1600	6750 1600	4350 1100	4350 1100					4400 1100
4500 180					10400 2200	10400 2200	8550 1400	8550 1400	4300 1100	4300 1100	4400 1100	4350 1100	4350 1100
3000 120					9450 2200	9450 2200	5900 1300	5900 1300	4050 1050	4050 1050	4400 1100	4350 1100	4350 1100
1500 60					8450 1800	8450 1800	6050 1300	6050 1300	4800 1000	4800 1000	5100 1100	4900 1050	4900 1050
0 0					3250 1050	3250 1050	5550 1000	5550 1000	5250 1000	5450 1000	3900 900	4800 950	4800 950
-1500 -60					10400 2200	10400 2200	8550 1400	8550 1400	8150 1350	8350 1350	4400 1100	4300 1050	4300 1050
-3000 -120					16150 3650	16150 3650	16850 3650	16850 3650	4250 1050	4250 1050		3600 900	3600 900
-4500 -180					20900 4600	20900 4600	21200 4600	21200 4600				8800 1950	8800 1950

Illustration 247

g06226166

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Size (mm)	500 20		1000 40		1500 60		2000 80		2500 100		3000 120		3500 140		4000 160		4500 180		5000 200		5500 220		6000 240		6500 260		7000 280		7500 300				Size (mm)																																																																																																																																																																																																																																																																																												
																																																																																																																																																																																																																																																																																																																													

Illustration 248

g06226168

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Reach (inch)	800 80	3000 120	4500 180	6000 240	7500 300		Reach (inch)
5000 360							4110 320
7500 300			5550 4000	6550 4900			6060 240
6000 240			4750 3500	5750 4350			5230 250
4500 180		3000 2250	4000 3000	5000 3850	6000 4750	7000 5650	7930 610
3000 120		1450 1100	2450 1900	3450 2700	4450 3400	5450 4200	6390 500
1500 60		700 550	1200 950	2200 1750	3200 2500	4200 3300	5140 400
0 0	3250 2650	3250 2650	6850 5450	8850 7050	10850 8650	12850 10450	14850 11700
-1500 -60	1000 2200	1000 2200	8850 6950	7150 5650	5200 4200	3800 3000	2400 1900
-3000 -120	6250 3650	6250 3650	8850 7400	5550 4700	4250 3500		1700 1300
-4500 -180	2050 2050	5400 2050					4850 350

Illustration 249

g06225901

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

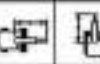
Reach (inch)	800 80	3000 120	4500 180	6000 240	7500 300		Reach (inch)
5000 360							4110 320
7500 300			5550 4200	6550 4900			6060 240
6000 240			4750 3500	5750 4350			5230 250
4500 180		3000 24750	4000 33000	5000 40500	6000 48500	7000 56500	7930 640
3000 120		1450 23400	2450 16050	3450 22000	4450 27500	5450 33000	6390 510
1500 60		700 11600	1200 7000	2200 13500	3200 16000	4200 19000	5140 410
0 0	3800 21000	3800 21000	8400 42500	8800 44500	10800 54500	12800 64500	14800 7400
-1500 -60	1100 24250	1100 24250	3800 21300	3800 21300	5800 30300	5800 30300	8800 7100
-3000 -120	7200 38450	7200 38450	3050 27800	3050 27800	4800 26000		5700 2700
-4500 -180	21250 21250	8500 21250					4850 350

Illustration 250

g06225902

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe

Reach (inch)	100 40	300 120	450 180	600 240	750 300		Reach (inch)
3000 360							3350 10500
7500 300			6550 16800	6750 17200	4350 11000	4550 11500	4850 12500
6000 240			6750 17200	4350 11000	3550 8000	3750 8500	4000 9000
4500 180		1600 22550	1600 22550	6550 16800	4200 9500	4200 9500	4400 10000
3000 120		1650 23000	1650 23000	5300 11700	4050 9000	4050 9000	4250 9500
1500 60		1650 23000	1650 23000	6550 16800	4050 9000	4050 9000	4250 9500
0 0	3250 20900	3250 20900	6050 14450	6050 14450	5250 11800	5450 12300	5650 12800
-1500 -60	10400 22900	10400 22900	8850 18900	8850 18900	5250 11800	5450 12300	5650 12800
-3000 -120	16250 36350	16250 36350	8950 19400	8950 19400	5250 11800	5450 12300	5650 12800
-4500 -180	20950 36350	20950 36350	14000 31000	14000 31000	5250 11800	5450 12300	5650 12800

Illustration 251

g06225837

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Reach (inch)	100 40	300 120	450 180	600 240	750 300		Reach (inch)
3000 360							3350 10500
7500 300			1700 16200	1700 16200	5250 11800	5250 11800	5450 12300
6000 240			1700 16200	1700 16200	4750 10500	4750 10500	4950 11000
4500 180		1600 22550	1600 22550	7200 16050	4650 10250	4650 10250	4850 10750
3000 120		1600 22550	1600 22550	6550 16800	4650 10250	4650 10250	4850 10750
1500 60		1650 23000	1650 23000	6550 16800	4650 10250	4650 10250	4850 10750
0 0	5900 21000	5900 21000	8400 18250	8400 18250	5250 11800	5450 12300	5650 12800
-1500 -60	10000 24250	10000 24250	8400 18250	8400 18250	5250 11800	5450 12300	5650 12800
-3000 -120	17200 36450	17200 36450	8400 18250	8400 18250	5250 11800	5450 12300	5650 12800
-4500 -180	21250 36450	21250 36450	16000 35000	16000 35000	5250 11800	5450 12300	5650 12800

Illustration 252

g06225840

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

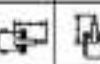
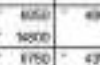
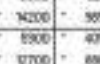
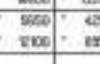
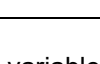
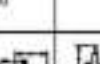
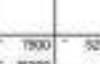
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
8000 360							4700 190
7000 300							6000 240
6000 240							7200 290
4500 180							7800 310
3000 120							8300 330
1500 60							8300 330
0							8200 320
-500							7700 310
-1000							6700 270
-1500							4700 190

Illustration 253

g06225911

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

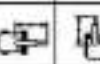
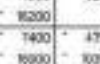
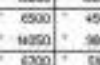
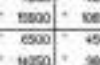
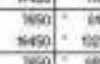
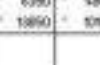
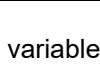
Reach (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
8000 360							4700 190
7000 300							6000 240
6000 240							7200 290
4500 180							7800 310
3000 120							8300 330
1500 60							8300 330
0							8200 320
-500							7700 310
-1000							6700 270
-1500							4700 190

Illustration 254

g06225913

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 160							5950 19000 4700 160
1700 300			6750 14800	6750 14800			4900 10950 6000 240
8000 340			6750 14800	6750 14800	4350 3500		4400 3700 4400 230
4500 180		10400 22550	8400 22550	6550 14200	6550 14200	4200 3650	4200 3650 4200 310
3000 120		3450 2100	3450 2100	5300 11700	5300 11700	4050 3600	4050 3600 4050 300
1500 60		3450 13800	3450 13800	6050 13000	6050 13000	4900 11800	4900 11800 4900 320
0 0	3250 21850	3250 21850	6050 10450	6050 10450	6050 10450	5450 10300	5450 10300 5450 330
-1500 -60	10400 22300	10400 22300	8850 18900	8850 18900	6750 10250	5300 3550	4000 3800 4000 270
-3000 -120	3450 30350	3450 30350	6550 15400	6550 15400	4250 3350		3600 3000 3600 270
-4500 -180	20950 20950	20950 20950	3450 20300	3450 20300			4500 15550 4500 160

Illustration 255

g06225880

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

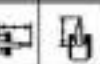
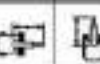
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 160							5950 19000 4700 160
1700 300			7600 16200	7600 16200	5250 3250		5250 12000 6000 240
8000 340			7600 16000	7600 16000	4750 3350		4800 10350 4800 230
4500 180		11400 24750	11400 24750	7200 15500	7200 15500	4850 3850	4850 3850 4850 310
3000 120		10900 25400	10900 25400	6500 14050	6500 14050	5000 1150	4700 10350 4700 300
1500 60		6850 14650	6850 14650	6700 14400	6700 14400	5050 12100	4950 10000 4950 320
0 0	3800 21300	3800 21300	6400 14250	6400 14250	6000 13250	5350 11950	5700 11500 5700 330
-1500 -60	10400 24250	10400 24250	8850 21000	8850 21000	6850 14800	5300 11350	4500 3900 4500 270
-3000 -120	3450 36450	3450 36450	6550 21800	6550 21800	4850 10700		4300 3050 4300 270
-4500 -180	22250 22250	22250 22250	6500 20850	6500 20850			7300 15400 7300 160

Illustration 256

g06225881

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Load (kg)	500 50		800 80		1000 100		1500 150		2000 200		2500 250				Load (kg)	
																
5000 500																
3000 300																
7500 750																
2000 200																
6000 600																
2400 240																
4500 450																
800 80																
3000 300																
100 10																
5000 500																
800 80																
5 0.5																
5 0.5																
-1500 -150																
-400 -40																
-3000 -300																
-600 -60																
-4000 -400																
-800 -80																

Illustration 257

g06225884

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

Load (kg)	1500 50		3000 100		4500 150		6000 200		7500 300				Load (kg)	
 9000 360													470	
1800 300					* 7500 * 18200	* 7500 * 18200	* 5280 * 5280					* 5500 * 14800	* 5500 * 14800	150
6000 240					* 7400 * 18000	* 7400 * 18000	* 4750 * 10250	* 4750 * 10250				* 5200 * 18800	* 5200 * 18800	8000 240
4500 180					* 18000 * 24750	* 18000 * 24750	* 7200 * 9900	* 7200 * 9900	* 4850 * 10650	* 4850 * 10650	* 4400 * 9400	* 4850 * 10250	* 4850 * 9900	9000 280
3800 120					* 10000 * 23400	* 10000 * 23400	* 8500 * 14050	* 8500 * 14050	* 4500 * 9800	* 4500 * 9800	* 4250 * 11850	* 4700 * 10350	* 4500 * 9000	8000 330
1600 60					* 8850 * 14850	* 8850 * 14850	* 4700 * 14400	* 4700 * 14400	* 5100 * 7800	* 5100 * 7800	* 5550 * 12100	* 4800 * 10850	* 4550 * 7700	8300 330
0 0	* 3900 * 21000	* 3900 * 21000	* 6400 * 14250	* 6400 * 14250	* 8500 * 17850	* 8500 * 17850	* 5800 * 12250	* 5400 * 7650	* 5100 * 10050	* 4300 * 8850	* 5750 * 11000	* 5750 * 10000	8200 330	
-1500 -60	* 10900 * 24050	* 10900 * 24050	* 5950 * 20000	* 5950 * 20000	* 8550 * 16900	* 8550 * 16900	* 6850 * 14600	* 5350 * 11300	* 4950 * 9300	* 4500 * 8650	* 4500 * 9600	* 3900 * 8550	7700 340	
-3800 -120	* 17000 * 38450	* 17000 * 38450	* 8050 * 28900	* 8050 * 28900	* 4350 * 15850	* 4350 * 15850	* 4800 * 10900	* 4600 * 10100			* 4400 * 3650	* 4100 * 9050	6700 270	
-4500 -180	* 23900 * 23900	* 23900 * 23900	* 15900 * 22950	* 15900 * 22950							* 7350 * 17400	* 7350 * 24000	4900 270	

Illustration 258

g06225888

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

900 mm (35 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Stick (inch)	2500 80	3000 120	4500 90	6000 240	7500 300		
							Stick (inch)
3000 90							3350 106
2500 80							2850 90
2000 65							2350 74
1500 50							1850 58
1000 35							1350 42
500 20							850 26
0 0							350 14
-500 -20							-150 -6
-1000 -35							-650 -26
-1500 -50							-1150 -42
-2000 -65							-1650 -58
-2500 -80							-2150 -74
-3000 -90							-2650 -90

Illustration 259

g06225889

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Stick (inch)	2500 80	3000 120	4500 90	6000 240	7500 300		
							Stick (inch)
3000 90							3350 106
2500 80							2850 90
2000 65							2350 74
1500 50							1850 58
1000 35							1350 42
500 20							850 26
0 0							350 14
-500 -20							-150 -6
-1000 -35							-650 -26
-1500 -50							-1150 -42
-2000 -65							-1650 -58
-2500 -80							-2150 -74
-3000 -90							-2650 -90

Illustration 260

g06225894

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2500 mm (98 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Variable Angle Boom with a 2.9 m (9 ft 6 inch) Stick and 5400 kg (11905 lb) Counterweight

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Single Grouser Shoe (14 mm (0.6 inch) thickness)

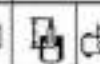
Load inch	600 40		3800 120		4900 180		8000 240		7500 300				
													inch feet
3300 360					* 4700 * 12750	* 4700 * 12750					* 4750 * 12750	* 4750 * 12750	4880 190
7900 300					* 6500 * 14200	* 6500 * 14200	* 3950 * 8650	* 3950 * 8650			* 3950 * 8650	* 3950 * 8650	8620 340
6800 240					* 6700 * 14950	* 6700 * 14950	* 4750 * 10550	* 4750 * 10550	* 3750 * 8200	* 3750 * 8200	* 3700 * 8100	* 3700 * 8100	7630 300
4900 180			* 8500 * 23850	* 8500 * 23850	* 6850 * 15250	* 6850 * 15250	* 3650 * 8050	* 3650 * 8050	* 3650 * 8050	* 3650 * 8050	* 3600 * 7900	* 3600 * 7900	8090 320
3800 120	* 17800 * 17800	* 17800 * 17800	* 8000 * 25850	* 8000 * 25850	* 6400 * 13800	* 6400 * 13800	* 3550 * 8350	* 3550 * 8350	* 2950 * 6500	* 2950 * 6500	* 2950 * 6500	* 2950 * 6500	6710 260
1900 60			* 17500 * 13050	* 17500 * 13050	* 5800 * 12100	* 5800 * 12100	* 4350 * 9450	* 4350 * 9450	* 4450 * 9700	* 4450 * 9700	* 4450 * 9700	* 4450 * 9700	8730 340
0 0	* 8500 * 14700	* 8500 * 14700	* 8300 * 16700	* 8300 * 16700	* 14000 * 15800	* 14000 * 15800	* 5750 * 12000	* 5750 * 12000	* 5050 * 10850	* 5050 * 10850	* 4950 * 10750	* 4950 * 10750	3250 1200
-1900 -60	* 8250 * 14350	* 8250 * 14350	* 8250 * 16350	* 8250 * 16350	* 13900 * 15700	* 13900 * 15700	* 5650 * 11900	* 5650 * 11900	* 4950 * 10750	* 4950 * 10750	* 4850 * 10650	* 4850 * 10650	3160 1200
-3800 -120	* 13250 * 25800	* 13250 * 25800	* 13250 * 25800	* 13250 * 25800	* 6250 * 31400	* 6250 * 31400	* 4500 * 30450	* 4500 * 30450			* 3250 * 1200	* 3250 * 1200	1340 290
-4900 -180	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 1950 * 31750	* 1950 * 31750	* 1900 * 31500	* 1900 * 31500			* 4850 * 12300	* 4850 * 12300	5390 210

Illustration 261

g06225723

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

Load inch	600 40		1800 120		4900 180		8000 240		7500 300				
												inch feet	
3300 360				* 5100 * 13700	* 5100 * 13700					* 5200 * 13700	* 5200 * 13700	4880 190	
7900 300				* 7100 * 15900	* 7100 * 15900	* 4300 * 9450	* 4300 * 9450			* 4350 * 9450	* 4350 * 9450	8620 340	
6800 240				* 7300 * 16350	* 7300 * 16350	* 5250 * 11500	* 5250 * 11500	* 4050 * 8900	* 4050 * 8900	* 4000 * 8850	* 4000 * 8850	7630 300	
4900 180			* 8500 * 25000	* 8500 * 25000	* 7000 * 15750	* 5450 * 12000	* 5450 * 12000	* 4000 * 8800	* 4000 * 8800	* 3900 * 8450	* 3900 * 8450	8090 320	
3800 120	* 18000 * 18000	* 18000 * 27700	* 18000 * 27700	* 18000 * 27700	* 6300 * 13900	* 4250 * 9250	* 4250 * 9250	* 4350 * 9450	* 4350 * 9450	* 3250 * 6500	* 3250 * 6500	6710 260	
1900 60			* 8100 * 13650	* 8100 * 13650	* 6200 * 13400	* 4800 * 10450	* 4800 * 10450	* 4850 * 10700	* 4850 * 10700	* 4450 * 9700	* 4450 * 9700	8730 340	
0 0	* 8900 * 14850	* 8900 * 16850	* 8900 * 16850	* 8900 * 16850	* 14000 * 15800	* 5700 * 12000	* 5700 * 12000	* 5050 * 10850	* 5050 * 10850	* 3900 * 8450	* 3900 * 8450	3250 1200	
-1900 -60	* 8750 * 14750	* 8750 * 16750	* 8750 * 16750	* 8750 * 16750	* 13900 * 15700	* 5600 * 11900	* 5600 * 11900	* 4950 * 10750	* 4950 * 10750	* 3850 * 8350	* 3850 * 8350	3160 1200	
-3800 -120	* 14050 * 31850	* 14050 * 31850	* 14050 * 31850	* 14050 * 31850	* 9950 * 33500	* 7300 * 31900	* 7300 * 31900	* 5500 * 30450	* 5500 * 30450			1340 290	
-4900 -180	* 22050 * 43300	* 22050 * 43300	* 22050 * 43300	* 22050 * 43300	* 2150 * 31750	* 2150 * 31750	* 2150 * 31750			* 5450 * 12300	* 5450 * 12300	5390 210	

Illustration 262

g06225724

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift ON

600 mm (24 inch) Double Grouser Shoe (13 mm (0.5 inch) thickness)

Load (inch)	500 40	3800 120	4900 180	8000 240	7500 300		
							inch
3000 360			4700 180	4700 180			4750 180
7000 300			6500 160	6500 160	3350 130	3350 130	3350 130
6000 240			6700 160	6700 160	3750 150	3750 150	3750 150
4900 180		6500 160	6500 160	6500 160	6500 240	6500 240	6500 240
3800 120	2980 118	2980 118	2980 118	2980 118	2980 118	2980 118	2980 118
1500 60		1750 70	1750 70	1750 70	1750 70	1750 70	1750 70
0 0	8500 330	8500 330	8500 330	8500 330	8500 330	8500 330	8500 330
-1000 -80	8250 320	8250 320	8250 320	8250 320	8250 320	8250 320	8250 320
-3000 -120	5250 200	5250 200	5250 200	5250 200	5250 200	5250 200	5250 200
-4000 -160	20450 730	20450 730	20450 730	20450 730	20450 730	20450 730	20450 730

Illustration 263

g06225720

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

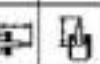
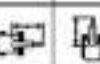
Load (inch)	500 40	3800 120	4900 180	8000 240	7500 300		
							inch
3000 360			5100 200	5100 200			5200 200
7000 300			7100 280	7100 280	4350 170	4350 170	4350 170
6000 240			7300 280	7300 280	4750 190	4750 190	4750 190
4900 180		6500 160	6500 160	6500 160	6500 240	6500 240	6500 240
3800 120	2980 118	2980 118	2980 118	2980 118	2980 118	2980 118	2980 118
1500 60		1750 70	1750 70	1750 70	1750 70	1750 70	1750 70
0 0	8500 330	8500 330	8500 330	8500 330	8500 330	8500 330	8500 330
-1000 -80	8250 320	8250 320	8250 320	8250 320	8250 320	8250 320	8250 320
-3000 -120	5250 200	5250 200	5250 200	5250 200	5250 200	5250 200	5250 200
-4000 -160	20450 730	20450 730	20450 730	20450 730	20450 730	20450 730	20450 730

Illustration 264

g06225722

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift ON

Product Information Section

323 Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) Triple Grouser Shoe

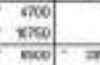
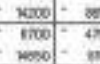
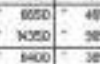
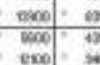
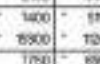
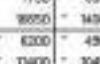
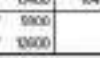
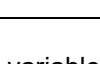
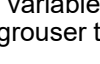
Load (lb/kN)	100 45	300 135	450 203	600 270	750 338		Load (lb/kN)
3000 1360							4800 2180
1100 500							8800 3950
6000 2700							7800 3500
4500 2030							8350 3750
3800 1700							8710 3900
3000 1360							9500 4270
0 0							8810 3950
+1500 +680							8750 3920
-3000 -1360							7340 3300
-4500 -2030							5350 2400
-5500 -2480							4300 1930

Illustration 265

g06225705

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

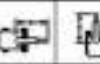
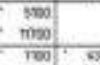
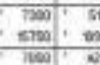
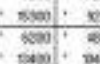
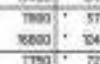
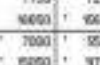
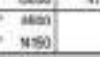
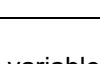
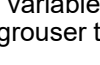
Load (lb/kN)	100 45	300 135	450 203	600 270	750 338		Load (lb/kN)
3000 1360							4880 2190
1100 500							8620 3880
6000 2700							7690 3450
4500 2030							8360 3750
3800 1700							8710 3900
3000 1360							9500 4270
0 0							8810 3950
+1500 +680							8750 3920
-3000 -1360							7340 3300
-4500 -2030							5350 2400
-5500 -2480							4300 1930

Illustration 266

g06225707

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

600 mm (24 inch) Triple Grouser Shoe (10 mm (0.4 inch) thickness)

Load (metric)	2500 55	3000 66	4500 99	6000 132	7500 165	Stick Diagram	Load (metric)
3600 360							4700 4700
1700 300							4700 4700
6800 240							4700 4700
4800 180							4700 4700
3600 120							4700 4700
1900 60							4700 4700
0 0							4700 4700
-1600 -60							4700 4700
-3600 -120							4700 4700
-4800 -180							4700 4700

Illustration 267

g06225726

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

Load (metric)	2500 55	3000 66	4500 99	6000 132	7500 165	Stick Diagram	Load (metric)
3600 360							5200 5200
1700 300							5200 5200
6800 240							5200 5200
4800 180							5200 5200
3600 120							5200 5200
1900 60							5200 5200
0 0							5200 5200
-1600 -60							5200 5200
-3600 -120							5200 5200
-4800 -180							5200 5200

Illustration 268

g06225727

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift ON

700 mm (28 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							Load (inch)
3000 120			* 4700 * 18750	* 4700 * 18750		* 4750 * 18750	4880 190
7000 300			* 6500 * 14200	* 6500 * 14200		* 6550 * 14200	6620 260
6000 240			* 6700 * 14950	* 6700 * 14950	* 4750 * 11750	* 4750 * 11750	4880 190
4500 180		* 6500 * 22850	* 6500 * 22850	* 6500 * 22850	* 6550 * 22850	* 6550 * 22850	6620 260
3000 120	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	17800 710
1500 60		* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	* 17800 * 17800	17800 710
0 0	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	8500 330
-1500 -60	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	8500 330
-3000 -120	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	12250 490
-4500 -180	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	20450 800

Illustration 269

g06225709

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							Load (inch)
3000 120			* 5100 * 17000	* 5100 * 17000		* 5200 * 17000	4880 190
7000 300			* 7100 * 25000	* 7100 * 25000		* 7150 * 25000	6620 260
6000 240			* 7300 * 25800	* 7300 * 25800	* 4750 * 11750	* 4750 * 11750	4880 190
4500 180		* 6500 * 25000	* 6500 * 25000	* 6500 * 25000	* 6550 * 25000	* 6550 * 25000	6620 260
3000 120	* 6000 * 6000	* 6000 * 6000	* 6000 * 6000	* 6000 * 6000	* 6000 * 6000	* 6000 * 6000	6000 230
1500 60		* 6000 * 13850	* 6000 * 13850	* 6000 * 13850	* 6000 * 13850	* 6000 * 13850	6000 230
0 0	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	8500 330
-1500 -60	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	* 8500 * 14700	8500 330
-3000 -120	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	* 12250 * 25800	12250 490
-4500 -180	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	* 20450 * 44550	20450 800

Illustration 270

g06225711

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) triple grouser track shoes, heavy lift ON

790 mm (31 inch) Triple Grouser Shoe (12.5 mm (0.50 inch) thickness)

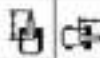
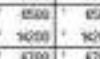
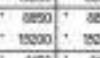
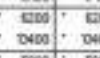
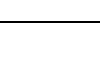
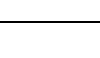
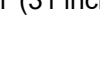
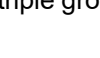
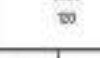
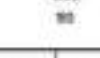
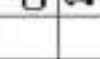
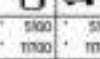
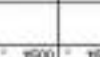
Reach (inch)	323 80	323 120	4500 180	6000 240	7500 300		Reach (inch)
5000 360							4800 190
7500 300							6020 260
6000 240							7000 300
4500 180							8350 330
3000 120							8750 350
1500 60							9570 360
0							9670 340
-2000 -60							8100 320
-3000 -120							7340 250
-4500 -180							5350 210

Illustration 271

g06225712

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift OFF

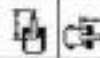
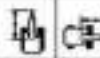
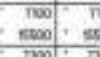
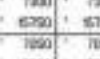
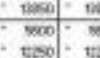
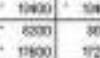
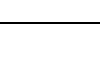
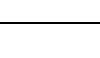
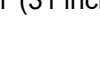
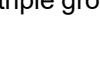
Reach (inch)	323 80	323 120	4500 180	6000 240	7500 300		Reach (inch)
5000 360							4880 190
7500 300							6020 260
6000 240							7000 300
4500 180							8360 330
3000 120							8750 350
1500 60							9570 360
0							9670 340
-2000 -60							8100 320
-3000 -120							7340 250
-4500 -180							5350 210

Illustration 272

g06225715

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 790 mm (31 inch) triple grouser track shoes, heavy lift ON

[illegible]

q06225716

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift OFF

Load lb/kl	1500 60	3000 120	4500 180	6000 240	7500 300		Reach ft/kl
							
3000			* 5800	* 5100			* 5200
260			* 11100	* 9700			* 11100
7500			* 7800	* 7100	* 4300	* 4300	* 4300
300			* 8500	* 8500	* 4450	* 4450	* 4450
6000			* 7300	* 7300	* 5250	* 5250	* 4900
240			* 8250	* 8250	* 4300	* 4300	* 4650
4500		* 1600	* 1600	* 7300	* 7300	* 5150	* 5150
180		* 25000	* 25000	* 6750	* 6750	* 4900	* 4900
3000		* 1000	* 1000	* 7050	* 7050	* 4250	* 4250
120	* 18000	* 18000	* 23000	* 1600	* 1600	* 4350	* 4350
1600			* 600	* 600	* 8200	* 8200	* 4900
60		* 13950	* 13950	* 10400	* 10400	* 8450	* 8450
0	* 6900	* 6900	* 5800	* 5800	* 4200	* 4200	* 4500
0	* 14850	* 14850	* 12250	* 12250	* 10600	* 10600	* 9800
-5000	* 6750	* 6750	* 5000	* 5000	* 3300	* 3300	* 4450
-60	* 15450	* 15450	* 20450	* 20450	* 17500	* 17500	* 17500
-3000	* 14000	* 14000	* 3950	* 3950	* 7000	* 7000	* 5500
-120	* 37050	* 37050	* 27600	* 27600	* 26050	* 26050	* 26050
-4500	* 22050	* 22050	* 10250	* 10250	* 4500	* 4500	* 5450
-80	* 43500	* 43500	* 22780	* 22780	* 14780	* 14780	* 12200

q06225719

Lift Chart Above: 323, 2686 / 3300 mm (106 / 130 inch) variable angle boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) triple grouser track shoes, heavy lift ON

Super Long Reach Boom with a 6.28 m (20 ft 7 inch) Stick and 5400 kg (11905 lb) Counterweight

600 mm (24 inch) Heavy-Duty Triple Grouser Shoe

Index Index	2010 10	2011 11	2012 12	2013 13	2014 14	2015 15	2016 16	2017 17	2018 18	2019 19	2020 20	2021 21	2022 22	2023 23	2024 24	2025 25	2026 26	2027 27	2028 28	2029 29	2030 30	2031 31	2032 32	2033 33	2034 34	2035 35	2036 36	2037 37	2038 38	2039 39	2040 40	2041 41	2042 42	2043 43	2044 44	2045 45	2046 46	2047 47	2048 48	2049 49	2050 50	2051 51	2052 52	2053 53	2054 54	2055 55	2056 56	2057 57	2058 58	2059 59	2060 60	2061 61	2062 62	2063 63	2064 64	2065 65	2066 66	2067 67	2068 68	2069 69	2070 70	2071 71	2072 72	2073 73	2074 74	2075 75	2076 76	2077 77	2078 78	2079 79	2080 80	2081 81	2082 82	2083 83	2084 84	2085 85	2086 86	2087 87	2088 88	2089 89	2090 90	2091 91	2092 92	2093 93	2094 94	2095 95	2096 96	2097 97	2098 98	2099 99	2100 100	2101 101	2102 102	2103 103	2104 104	2105 105	2106 106	2107 107	2108 108	2109 109	2110 110	2111 111	2112 112	2113 113	2114 114	2115 115	2116 116	2117 117	2118 118	2119 119	2120 120	2121 121	2122 122	2123 123	2124 124	2125 125	2126 126	2127 127	2128 128	2129 129	2130 130	2131 131	2132 132	2133 133	2134 134	2135 135	2136 136	2137 137	2138 138	2139 139	2140 140	2141 141	2142 142	2143 143	2144 144	2145 145	2146 146	2147 147	2148 148	2149 149	2150 150	2151 151	2152 152	2153 153	2154 154	2155 155	2156 156	2157 157	2158 158	2159 159	2160 160	2161 161	2162 162	2163 163	2164 164	2165 165	2166 166	2167 167	2168 168	2169 169	2170 170	2171 171	2172 172	2173 173	2174 174	2175 175	2176 176	2177 177	2178 178	2179 179	2180 180	2181 181	2182 182	2183 183	2184 184	2185 185	2186 186	2187 187	2188 188	2189 189	2190 190	2191 191	2192 192	2193 193	2194 194	2195 195	2196 196	2197 197	2198 198	2199 199	2200 200	2201 201	2202 202	2203 203	2204 204	2205 205	2206 206	2207 207	2208 208	2209 209	2210 210	2211 211	2212 212	2213 213	2214 214	2215 215	2216 216	2217 217	2218 218	2219 219	2220 220	2221 221	2222 222	2223 223	2224 224	2225 225	2226 226	2227 227	2228 228	2229 229	2230 230	2231 231	2232 232	2233 233	2234 234	2235 235	2236 236	2237 237	2238 238	2239 239	2240 240	2241 241	2242 242	2243 243	2244 244	2245 245	2246 246	2247 247	2248 248	2249 249	2250 250	2251 251	2252 252	2253 253	2254 254	2255 255	2256 256	2257 257	2258 258	2259 259	2260 260	2261 261	2262 262	2263 263	2264 264	2265 265	2266 266	2267 267	2268 268	2269 269	2270 270	2271 271	2272 272	2273 273	2274 274	2275 275	2276 276	2277 277	2278 278	2279 279	2280 280	2281 281	2282 282	2283 283	2284 284	2285 285	2286 286	2287 287	2288 288	2289 289	2290 290	2291 291	2292 292	2293 293	2294 294	2295 295	2296 296	2297 297	2298 298	2299 299	2300 300	2301 301	2302 302	2303 303	2304 304	2305 305	2306 306	2307
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Illustration 275

g06357087

Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) heavy duty triple grouser track shoes, heavy lift OFF

(mm) Quot.)	1000 mm		1200 mm		1600 mm		1900 mm		2300 mm		2600 mm		3000 mm		3300 mm		3600 mm		3900 mm		4200 mm		4500 mm		4800 mm		5100 mm		5400 mm		5700 mm		6000 mm		6300 mm		6600 mm		6900 mm		7200 mm		7500 mm		7800 mm		8100 mm		8400 mm		8700 mm		9000 mm		9300 mm		9600 mm		9900 mm		10200 mm		10500 mm		10800 mm		11100 mm		11400 mm		11700 mm		12000 mm		12300 mm		12600 mm		12900 mm		13200 mm		13500 mm		13800 mm		14100 mm		14400 mm		14700 mm		15000 mm		15300 mm		15600 mm		15900 mm		16200 mm		16500 mm		16800 mm		17100 mm		17400 mm		17700 mm		18000 mm		18300 mm		18600 mm		18900 mm		19200 mm		19500 mm		19800 mm		20100 mm		20400 mm		20700 mm		21000 mm		21300 mm		21600 mm		21900 mm		22200 mm		22500 mm		22800 mm		23100 mm		23400 mm		23700 mm		24000 mm		24300 mm		24600 mm		24900 mm		25200 mm		25500 mm		25800 mm		26100 mm		26400 mm		26700 mm		27000 mm		27300 mm		27600 mm		27900 mm		28200 mm		28500 mm		28800 mm		29100 mm		29400 mm		29700 mm		30000 mm		30300 mm		30600 mm		30900 mm		31200 mm		31500 mm		31800 mm		32100 mm		32400 mm		32700 mm		33000 mm		33300 mm		33600 mm		33900 mm		34200 mm		34500 mm		34800 mm		35100 mm		35400 mm		35700 mm		36000 mm		36300 mm		36600 mm		36900 mm		37200 mm		37500 mm		37800 mm		38100 mm		38400 mm		38700 mm		39000 mm		39300 mm		39600 mm		39900 mm		40200 mm		40500 mm	
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g06357093

Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 700 mm (28 inch) heavy duty triple grouser track shoes, heavy lift OFF

790 mm (31 inch) Heavy-Duty Triple Grouser Shoe

Boom (inch)	8000 32	10000 39	12000 47	14000 55	16000 63	18000 71	20000 79	22000 87	24000 95	26000 103	28000 111	30000 119	32000 127	34000 135	36000 143	38000 151	40000 159	42000 167	44000 175	46000 183	48000 191	50000 199	52000 207	54000 215	56000 223	58000 231	60000 239	62000 247	64000 255	66000 263	68000 271	70000 279	72000 287	74000 295	76000 303	78000 311	80000 319	82000 327	84000 335	86000 343	88000 351	90000 359	92000 367	94000 375	96000 383	98000 391	100000 399	102000 407	104000 415	106000 423	108000 431	110000 439	112000 447	114000 455	116000 463	118000 471	120000 479	122000 487	124000 495	126000 503	128000 511	130000 519	132000 527	134000 535	136000 543	138000 551	140000 559	142000 567	144000 575	146000 583	148000 591	150000 599	152000 607	154000 615	156000 623	158000 631	160000 639	162000 647	164000 655	166000 663	168000 671	170000 679	172000 687	174000 695	176000 703	178000 711	180000 719	182000 727	184000 735	186000 743	188000 751	190000 759	192000 767	194000 775	196000 783	198000 791	200000 799	202000 807	204000 815	206000 823	208000 831	210000 839	212000 847	214000 855	216000 863	218000 871	220000 879	222000 887	224000 895	226000 903	228000 911	230000 919	232000 927	234000 935	236000 943	238000 951	240000 959	242000 967	244000 975	246000 983	248000 991	250000 999	252000 1007	254000 1015	256000 1023	258000 1031	260000 1039	262000 1047	264000 1055	266000 1063	268000 1071	270000 1079	272000 1087	274000 1095	276000 1103	278000 1111	280000 1119	282000 1127	284000 1135	286000 1143	288000 1151	290000 1159	292000 1167	294000 1175	296000 1183	298000 1191	300000 1199	302000 1207	304000 1215	306000 1223	308000 1231	310000 1239	312000 1247	314000 1255	316000 1263	318000 1271	320000 1279	322000 1287	324000 1295	326000 1303	328000 1311	330000 1319	332000 1327	334000 1335	336000 1343	338000 1351	340000 1359	342000 1367	344000 1375	346000 1383	348000 1391	350000 1399	352000 1407	354000 1415	356000 1423	358000 1431	360000 1439	362000 1447	364000 1455	366000 1463	368000 1471	370000 1479	372000 1487	374000 1495	376000 1503	378000 1511	380000 1519	382000 1527	384000 1535	386000 1543	388000 1551	390000 1559	392000 1567	394000 1575	396000 1583	398000 1591	400000 1599	402000 1607	404000 1615	406000 1623	408000 1631	410000 1639	412000 1647	414000 1655	416000 1663	418000 1671	420000 1679	422000 1687	424000 1695	426000 1703	428000 1711	430000 1719	432000 1727	434000 1735	436000 1743	438000 1751	440000 1759	442000 1767	444000 1775	446000 1783	448000 1791	450000 1799	452000 1807	454000 1815	456000 1823	458000 1831	460000 1839	462000 1847	464000 1855	466000 1863	468000 1871	470000 1879	472000 1887	474000 1895	476000 1903	478000 1911	480000 1919	482000 1927	484000 1935	486000 1943	488000 1951	490000 1959	492000 1967	494000 1975	496000 1983	498000 1991	500000 1999	502000 2007	504000 2015	506000 2023	508000 2031	510000 2039	512000 2047	514000 2055	516000 2063	518000 2071	520000 2079	522000 2087	524000 2095	526000 2103	528000 2111	530000 2119	532000 2127	534000 2135	536000 2143	538000 2151	540000 2159	542000 2167	544000 2175	546000 2183	548000 2191	550000 2199	552000 2207	554000 2215	556000 2223	558000 2231	560000 2239	562000 2247	564000 2255	566000 2263	568000 2271	570000 2279	572000 2287	574000 2295	576000 2303	578000 2311	580000 2319	582000 2327	584000 2335	586000 2343	588000 2351	590000 2359	592000 2367	594000 2375	596000 2383	598000 2391	600000 2399	602000 2407	604000 2415	606000 2423	608000 2431	610000 2439	612000 2447	614000 2455	616000 2463	618000 2471	620000 2479	622000 2487	624000 2495	626000 2503	628000 2511	630000 2519	632000 2527	634000 2535	636000 2543	638000 2551	640000 2559	642000 2567	644000 2575	646000 2583	648000 2591	650000 2599	652000 2607	654000 2615	656000 2623	658000 2631	660000 2639	662000 2647	664000 2655	666000 2663	668000 2671	670000 2679	672000 2687	674000 2695	676000 2703	678000 2711	680000 2719	682000 2727	684000 2735	686000 2743	688000 2751	690000 2759	692000 2767	694000 2775	696000 2783	698000 2791	700000 2799	702000 2807	704000 2815	706000 2823	708000 2831	710000 2839	712000 2847	714000 2855	716000 2863	718000 2871	720000 2879	722000 2887	724000 2895	726000 2903	728000 2911	730000 2919	732000 2927	734000 2935	736000 2943	738000 2951	740000 2959	742000 2967	744000 2975	746000 2983	748000 2991	750000 2999	752000 3007	754000 3015	756000 3023	758000 3031	760000 3039	762000 3047	764000 3055	766000 3063	768000 3071	770000 3079	772000 3087	774000 3095	776000 3103	778000 3111	780000 3119	782000 3127	784000 3135	786000 3143	788000 3151	790000 3159	792000 3167	794000 3175	796000 3183	798000 3191	800000 3199	802000 3207	804000 3215	806000 3223	808000 3231	810000 3239	812000 3247	814000 3255	816000 3263	818000 3271	820000 3279	822000 3287	824000 3295	826000 3303	828000 3311	830000 3319	832000 3327	834000 3335	836000 3343	838000 3351	840000 3359	842000 3367	844000 3375	846000 3383	848000 3391	850000 3399	852000 3407	854000 3415	856000 3423	858000 3431	860000 3439	862000 3447	864000 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4111	1030000 4119	1032000 4127	1034000 4135	1036000 4143	1038000 4151	1040000 4159	1042000 4167	1044000 4175	1046000 4183	1048000 4191	1050000 4199	1052000 4207	1054000 4215	1056000 4223	1058000 4231	1060000 4239	1062000 4247	1064000 4255	1066000 4263	1068000 4271	1070000 4279	1072000 4287	1074000 4295	1076000 4303	1078000 4311	1080000 4319	1082000 4327	1084000 4335	1086000 4343	1088000 4351	1090000 4359	1092000 4367	1094000 4375	1096000 4383	1098000 4391	1100000 4399	1102000 4407	1104000 4415	1106000 4423	1108000 4431	1110000 4439	1112000 4447	1114000 4455	1116000 4463	1118000 4471	1120000 4479	1122000 4487	1124000 4495	1126000 4503	1128000 4511	1130000 4519	1132000 4527	1134000 4535	1136000 4543	1138000 4551	1140000 4559	1142000 4567	1144000 4575	1146000 4583	1148000 4591	1150000 4599	1152000 4607	1154000 4615	1156000 4623	1158000 4631	1160000 4639	1162000 4647	1164000 4655	1166000 4663	1168000 4671	1170000 4679	1172000 4687	1174000 4695	1176000 4703	1178000 4711	1180000 4719	1182000 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(mm) Inch	1000 in		1200 in		1400 in		1600 in		1800 in		2000 in		2200 in		2400 in		2600 in		2800 in		3000 in		3200 in		3400 in		3600 in		3800 in		4000 in		4200 in		4400 in		4600 in		4800 in		5000 in		5200 in		5400 in		5600 in		5800 in		6000 in		6200 in		6400 in		6600 in		6800 in		7000 in		7200 in		7400 in		7600 in		7800 in		8000 in		8200 in		8400 in		8600 in		8800 in		9000 in		9200 in		9400 in		9600 in		9800 in		10000 in		10200 in		10400 in		10600 in		10800 in		11000 in		11200 in		11400 in		11600 in		11800 in		12000 in		12200 in		12400 in		12600 in		12800 in		13000 in		13200 in		13400 in		13600 in		13800 in		14000 in		14200 in		14400 in		14600 in		14800 in		15000 in		15200 in		15400 in		15600 in		15800 in		16000 in		16200 in		16400 in		16600 in		16800 in		17000 in		17200 in		17400 in		17600 in		17800 in		18000 in		18200 in		18400 in		18600 in		18800 in		19000 in		19200 in		19400 in		19600 in		19800 in		20000 in		20200 in		20400 in		20600 in		20800 in		21000 in		21200 in		21400 in		21600 in		21800 in		22000 in		22200 in		22400 in		22600 in		22800 in		23000 in		23200 in		23400 in		23600 in		23800 in		24000 in		24200 in		24400 in		24600 in		24800 in		25000 in		25200 in		25400 in		25600 in		25800 in		26000 in		26200 in		26400 in		26600 in		26800 in		27000 in		27200 in		27400 in
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g06357268

Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 900 mm (35 inch) heavy duty triple grouser track shoes, heavy lift OFF

600 mm (24 inch) Single Grouser Shoe

(mm) (inch)	800 31	1000 39	1200 47	1400 55	1600 63	1800 71	2000 79	2200 87	2400 94	2600 102	2800 110	3000 118	3200 126	3400 134	3600 142	3800 150	4000 157	4200 165	4400 173	4600 181	4800 189	5000 197	5200 205	5400 213	5600 221	5800 229	6000 237	6200 245	6400 253	6600 261	6800 269	7000 277	7200 285	7400 293	7600 301	7800 309	8000 317	8200 325	8400 333	8600 341	8800 349	9000 357	9200 365	9400 373	9600 381	9800 389	10000 397	10200 405	10400 413	10600 421	10800 429	11000 437	11200 445	11400 453	11600 461	11800 469	12000 477	12200 485	12400 493	12600 501	12800 509	13000 517	13200 525	13400 533	13600 541	13800 549	14000 557	14200 565	14400 573	14600 581	14800 589	15000 597	15200 605	15400 613	15600 621	15800 629	16000 637	16200 645	16400 653	16600 661	16800 669	17000 677	17200 685	17400 693	17600 701	17800 709	18000 717	18200 725	18400 733	18600 741	18800 749	19000 757	19200 765	19400 773	19600 781	19800 789	20000 797	20200 805	20400 813	20600 821	20800 829	21000 837	21200 845	21400 853	21600 861	21800 869	22000 877	22200 885	22400 893	22600 901	22800 909	23000 917	23200 925	23400 933	23600 941	23800 949	24000 957	24200 965	24400 973	24600 981	24800 989	25000 997	25200 1005	25400 1013	25600 1021	25800 1029	26000 1037	26200 1045	26400 1053	26600 1061	26800 1069	27000 1077	27200 1085	27400 1093	27600 1101	27800 1109	28000 1117	28200 1125	28400 1133	28600 1141	28800 1149	29000 1157	29200 1165	29400 1173	29600 1181	29800 1189	30000 1197	30200 1205	30400 1213	30600 1221	30800 1229	31000 1237	31200 1245	31400 1253	31600 1261	31800 1269	32000 1277	32200 1285	32400 1293	32600 1301	32800 1309	33000 1317	33200 1325	33400 1333	33600 1341	33800 1349	34000 1357	34200 1365	34400 1373	34600 1381	34800 1389	35000 1397	35200 1405	35400 1413	35600 1421	35800 1429	36000 1437	36200 1445	36400 1453	36600 1461	36800 1469	37000 1477	37200 1485	37400 1493	37600 1501	37800 1509	38000 1517	38200 1525	38400 1533	38600 1541	38800 1549	39000 1557	39200 1565	39400 1573	39600 1581	39800 1589	40000 1597	40200 1605	40400 1613	40600 1621	40800 1629	41000 1637	41200 1645	41400 1653	41600 1661	41800 1669	42000 1677	42200 1685	42400 1693	42600 1701	42800 1709	43000 1717	43200 1725	43400 1733	43600 1741	43800 1749	44000 1757	44200 1765	44400 1773	44600 1781	44800 1789	45000 1797	45200 1805	45400 1813	45600 1821	45800 1829	46000 1837	46200 1845	46400 1853	46600 1861	46800 1869	47000 1877	47200 1885	47400 1893	47600 1901	47800 1909	48000 1917	48200 1925	48400 1933	48600 1941	48800 1949	49000 1957	49200 1965	49400 1973	49600 1981	49800 1989	50000 1997	50200 2005	50400 2013	50600 2021	50800 2029	51000 2037	51200 2045	51400 2053	51600 2061	51800 2069	52000 2077	52200 2085	52400 2093	52600 2101	52800 2109	53000 2117	53200 2125	53400 2133	53600 2141	53800 21
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Illustration 279

g06357297

Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) single grouser track shoes, heavy lift OFF

600 mm (24 inch) Double Grouser Shoe

Unit (mm)	800		1000		1200		1400		1600		1800		2000		2200		2400		2600		2800		3000		3200		3400		3600		3800		4000		4200		4400		4600		4800		5000		5200		5400		5600		5800		6000		6200		6400		6600		6800		7000		7200		7400		7600		7800		8000		8200		8400		8600		8800		9000		9200		9400		9600		9800		10000		10200		10400		10600		10800		11000		11200		11400		11600		11800		12000		12200		12400		12600		12800		13000		13200		13400		13600		13800		14000		14200		14400		14600		14800		15000		15200		15400		15600		15800		16000		16200		16400		16600		16800		17000		17200		17400		17600		17800		18000		18200		18400		18600		18800		19000		19200		19400		19600		19800		20000		20200		20400		20600		20800		21000		21200		21400		21600		21800		22000		22200		22400		22600		22800		23000		23200		23400		23600		23800		24000		24200		24400		24600		24800		25000		25200		25400		25600		25800		26000		26200		26400		26600		26800		27000		27200		27400		27600		27800		28000		28200		28400		28600		28800		29000		29200		29400		29600		29800		30000		30200		30400		30600		30800		31000		31200		31400		31600		31800		32000		32200		32400		32600		32800		33000		33200		33400		33600		33800		34000		34200		34400		34600		34800		35000		35200		35400		35600		35800		36000		36200		36400		36600		36800		37000		37200		37400		37600		37800		38000		38200		38400		38600		38800		39000		39200		39400		39600		39800		40000		40200		40400		40600		40800		41000		41200		41400		41600		41800		42000		42200		42400		42600		42800		43000		43200		43400		43600		43800		44000		44200		44400		44600		44800		45000		45200		45400		45600		45800		46000		46200		46400		46600		46800		47000		47200		47400		47600		47800		48000		48200		48400		48600		48800		49000		49200		49400		49600		49800		50000		50200		50400		50600		50800		51000		51200		51400		51600		51800		52000		52200		52400		52600		52800		53000		53200		53400		53600		53800		54000		54200		54400		54600		54800		55000		55200		55400		55600		55800		56000		56200		56400		56600		56800		57000		57200		57400		57600		57800		58000		58200		58400		58600		58800		59000		59200		59400		59600		59800		60000		60200		60400		60600		60800		61000		61200		61400		61600		61800		62000		62200		62400		62600		62800		63000		63200		63400		63600		63800		64000		64200		64400		64600		64800		65000		65200		65400		65600		65800		66000		66200		66400		66600		66800		67000		67200		67400		67600		67800		68000		68200		68400		68600		68800		69000		69200		69400		69600		69800		70000		70200		70400		70600		70800		71000		71200		71400		71600		71800		72000		72200		72400		72600		72800		73000		73200		73400		73600		73800		74000		74200		74400		74600		74800		75000		75200		75400		75600		75800		76000		76200		76400		76600		76800		77000		77200		77400		77600		77800		78000		78200		78400		78600		78800		79000		79200		79400		79600		79800		80000		80200		80400		80600		80800		81000		81200		81400		81600		81800		82000		82200		82400		82600		82800		83000		83200		83400		83600		83800		84000		84200		84400		84600		84800		85000		85200		85400		85600		85800		86000		86200		86400		86600		86800		87000		87200		87400		87600		87800		88000		88200		88400		88600		88800		89000		89200		89400		89600		89800		90000		90200		90400		90600		90800		91000		91200		91400		91600		91800		92000		92200		92400		92600		92800		93000		93200		93400		93600		93800		94000		94200		94400		94600		94800		95000		95200		95400		95600		95800		96000		96200		96400		96600		96800		97000		97200		97400		97600		97800		98000		98200		98400		98600		98800		99000		99200		99400		99600		99800		100000		100200		100400		100600		100800		101000		101200		101400		101600		101800		102000		102200		102400		102600		102800		103000		103200		103400		103600		103800		104000		104200		104400		104600		104800		105000		105200		105400		105600		105800		106000		106200		106400		106600		106800		107000		107200		107400		107600		107800		108000		108200		108400		108600		108800		109000		109200		109400		109600		109800		110000		110200		110400		110600		110800		111000		111200		111400		111600		111800		112000		112200		112400		112600		112800		113000		113200		113400		113600		113800		114000		114200		114400		114600		114800		115000		115200		115400		115600		115800		116000		116200		116400		116600		116800		117000		117200		117400		117600		117800		118000		118200		118400		118600		118800		119000		119200		119400		119600		119800		120000		120200		120400		120600		120800		121000		121200		121400		121600		121800		122000		122200		122400		122600		122800		123000		123200		123400		123600		123800		124000		124200		124400		124600		124800		125000		125200		125400		125600		125800		126000		126200		126400		126600		126800		127000		127200		127400		127600		127800		128000		128200		128400		128600		128800		129000		129200		129400		129600		129800		130000		130200		130400		130600		130800		131000		131200		131400		131600		131800		132000		132200		132400		132600		132800		133000		133200		133400		133600		133800		134000		134200		134400		134600		134800		135000		135200		135400		135600		135800		136000		136200		136400		136600		136800		137000		137200		137400		137600		137800		138000		138200		138400		138600		138800		139000		13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Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) double grouser track shoes, heavy lift OFF

600 mm (24 inch) Triple Grouser Shoe

Grout (C.A.)	800 mm	1000 mm	1200 mm	1400 mm	1600 mm	1800 mm	2000 mm	2200 mm	2400 mm	2600 mm	2800 mm	3000 mm	3200 mm	3400 mm	3600 mm	3800 mm	4000 mm	4200 mm	4400 mm	4600 mm	4800 mm	5000 mm	5200 mm	5400 mm	5600 mm	5800 mm	6000 mm	6200 mm	6400 mm	6600 mm	6800 mm	7000 mm	7200 mm	7400 mm	7600 mm	7800 mm	8000 mm	8200 mm	8400 mm	8600 mm	8800 mm	9000 mm	9200 mm	9400 mm	9600 mm	9800 mm	10000 mm	10200 mm	10400 mm	10600 mm	10800 mm	11000 mm	11200 mm	11400 mm	11600 mm	11800 mm	12000 mm	12200 mm	12400 mm	12600 mm	12800 mm	13000 mm	13200 mm	13400 mm	13600 mm	13800 mm	14000 mm	14200 mm	14400 mm	14600 mm	14800 mm	15000 mm	15200 mm	15400 mm	15600 mm	15800 mm	16000 mm	16200 mm	16400 mm	16600 mm	16800 mm	17000 mm	17200 mm	17400 mm	17600 mm	17800 mm	18000 mm	18200 mm	18400 mm	18600 mm	18800 mm	19000 mm	19200 mm	19400 mm	19600 mm	19800 mm	20000 mm	20200 mm	20400 mm	20600 mm	20800 mm	21000 mm	21200 mm	21400 mm	21600 mm	21800 mm	22000 mm	22200 mm	22400 mm	22600 mm	22800 mm	23000 mm	23200 mm	23400 mm	23600 mm	23800 mm	24000 mm	24200 mm	24400 mm	24600 mm	24800 mm	25000 mm	25200 mm	25400 mm	25600 mm	25800 mm	26000 mm	26200 mm	26400 mm	26600 mm	26800 mm	27000 mm	27200 mm	27400 mm	27600 mm	27800 mm	28000 mm	28200 mm	28400 mm	28600 mm	28800 mm	29000 mm	29200 mm	29400 mm	29600 mm	29800 mm	30000 mm	30200 mm	30400 mm	30600 mm	30800 mm	31000 mm	31200 mm	31400 mm	31600 mm	31800 mm	32000 mm	32200 mm	32400 mm	32600 mm	32800 mm	33000 mm	33200 mm	33400 mm	33600 mm	33800 mm	34000 mm	34200 mm	34400 mm	34600 mm	34800 mm	35000 mm	35200 mm	35400 mm	35600 mm	35800 mm	36000 mm	36200 mm	36400 mm	36600 mm	36800 mm	37000 mm	37200 mm	37400 mm	37600 mm	37800 mm	38000 mm	38200 mm	38400 mm	38600 mm	38800 mm	39000 mm	39200 mm	39400 mm	39600 mm	39800 mm	40000 mm	40200 mm	40400 mm	40600 mm	40800 mm	41000 mm	41200 mm	41400 mm	41600 mm	41800 mm	42000 mm	42200 mm	42400 mm	42600 mm	42800 mm	43000 mm	43200 mm	43400 mm	43600 mm	43800 mm	44000 mm	44200 mm	44400 mm	44600 mm	44800 mm	45000 mm	45200 mm	45400 mm	45600 mm	45800 mm	46000 mm	46200 mm	46400 mm	46600 mm	46800 mm	47000 mm	47200 mm	47400 mm	47600 mm	47800 mm	48000 mm	48200 mm	48400 mm	48600 mm	48800 mm	49000 mm	49200 mm	49400 mm	49600 mm	49800 mm	50000 mm	50200 mm	50400 mm	50600 mm	50800 mm	51000 mm	51200 mm	51400 mm	51600 mm	51800 mm	52000 mm	52200 mm	52400 mm	52600 mm	52800 mm	53000 mm	53200 mm	53400 mm	53600 mm	53800 mm	54000 mm	54200 mm	54400 mm	54600 mm	54800 mm	55000 mm	55200 mm	55400 mm	55600 mm	55800 mm	56000 mm	56200 mm	56400 mm	56600 mm	56800 mm	57000 mm	57200 mm	57400 mm	57600 mm	57800 mm	58000 mm	58200 mm	58400 mm	58600 mm	58800 mm	59000 mm	59200 mm	59400 mm	59600 mm	59800 mm	60000 mm	60200 mm	60400 mm	60600 mm	60800 mm	61000 mm	61200 mm	61400 mm	61600 mm	61800 mm	62000 mm	62200 mm	62400 mm	62600 mm	62800 mm	63000 mm	63200 mm	63400 mm	63600 mm	63800 mm	64000 mm	64200 mm	64400 mm	64600 mm	64800 mm	65000 mm	65200 mm	65400 mm	65600 mm	65800 mm	66000 mm	66200 mm	66400 mm	66600 mm	66800 mm	67000 mm	67200 mm	67400 mm	67600 mm	67800 mm	68000 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mm	111200 mm	111400 mm	111600 mm	111800 mm	112000 mm	112200 mm	112400 mm	112600 mm	112800 mm	113000 mm	113200 mm	113400 mm	113600 mm	113800 mm	114000 mm	114200 mm	114400 mm	114600 mm	114800 mm	115000 mm	115200 mm	115400 mm	115600 mm	115800 mm	116000 mm	116200 mm	116400 mm	116600 mm	116800 mm	117000 mm	117200 mm	117400 mm	117600 mm	117800 mm	118000 mm	118200 mm	118400 mm	118600 mm	118800 mm	119000 mm	119200 mm	119400 mm	119600 mm	119800 mm	120000 mm	120200 mm	120400 mm	120600 mm	120800 mm	121000 mm	121200 mm	121400 mm	121600 mm	121800 mm	122000 mm	122200 mm	122400 mm	122600 mm	122800 mm	123000 mm	123200 mm	123400 mm	123600 mm	123800 mm	124000 mm	124200 mm	124400 mm	124600 mm	124800 mm	125000 mm	125200 mm	125400 mm	125600 mm	125800 mm	126000 mm	126200 mm	126400 mm	126600 mm	126800 mm	127000 mm	127200 mm	127400 mm	127600 mm	127800 mm	128000 mm	128200 mm	128400 mm	128600 mm	128800 mm	129000 mm	129200 mm	129400 mm	129600 mm	129800 mm	130000 mm	130200 mm	130400 mm	130600 mm	130800 mm	131000 mm	131200 mm	131400 mm	131600 mm	131800 mm	132000 mm	132200 mm	132400 mm	132600 mm	132800 mm	133000 mm	133200 mm	133400 mm	133600 mm	133800 mm	134000 mm	134200 mm	134400 mm	134600 mm	134800 mm	135000 mm	135200 mm	135400 mm	135600 mm	135800 mm	136000 mm	136200 mm	136400 mm	136600 mm	136800 mm	137000 mm	137200 mm	137400 mm	137600 mm	137800 mm	138000 mm	138200 mm	138400 mm	138600 mm	138800 mm	139000 mm	139200 mm	139400 mm	139600 mm	139800 mm	140000 mm	140200 mm	140400 mm	140600 mm	140800 mm	141000 mm	141200 mm	141400 mm	141600 mm	141800 mm	142000 mm	142200 mm	142400 mm	142600 mm	142800 mm	143000 mm	143200 mm	143400 mm	143600 mm	143800 mm	144000 mm	144200 mm	144400 mm	144600 mm	144800 mm	145000 mm	145200 mm	145400 mm	145600 mm	145800 mm	146000 mm	146200 mm	146400 mm	146600 mm	146800 mm	147000 mm	147200 mm	147400 mm	147600 mm	147800 mm	148000 mm	148200 mm	148400 mm	148600 mm	148800 mm	149000 mm	149200 mm	149400 mm	149600 mm	149800 mm	150000 mm	150200 mm	150400 mm	150600 mm	150800 mm	151000 mm	151200 mm	151400 mm	151600 mm	151800 mm	152000 mm	152200 mm	152400 mm	152600 mm	152800 mm	153000 mm	153200 mm	153400 mm	153600 mm	153800 mm	154000 mm	154200 mm	154400 mm	154600 mm	154800 mm	155000 mm	155200 mm	155400 mm	155600 mm	155800 mm	156000 mm	156200 mm	156400 mm	156600 mm	156800 mm	157000 mm	157200 mm	157400 mm	157600 mm	157800 mm	158000 mm	158200 mm	158400 mm	158600 mm	158800 mm	159000 mm	159200 mm	159400 mm	159600 mm	159800 mm	160000 mm	160200 mm	160400 mm	160600 mm	160800 mm	161000 mm	161200 mm	161400 mm	161600 mm	161800 mm	162000 mm	162200 mm	162400 mm	162600 mm	162800 mm	163000 mm	163200 mm	163400 mm	163600 mm	163800 mm	164000 mm	164200 mm	164400 mm	164600 mm	164800 mm	165000 mm	165200 mm	165400 mm	165600 mm	165800 mm	166000 mm	166200 mm	166400 mm	166600 mm	166800 mm	167000 mm	167200 mm	167400 mm	167600 mm	167800 mm	168000 mm	168200 mm	168400 mm	168600 mm	168800 mm	169000 mm	169200 mm	169400 mm	169600 mm	169800 mm	170000 mm	170200 mm	170400 mm	170600 mm	170800 mm	171000 mm	171200 mm	171400 mm	171600 mm	171800 mm	172000 mm	172200 mm	172400 mm	172600 mm	172800 mm	173000 mm	173200 mm	173400 mm	173600 mm	173800 mm	174000 mm	174200 mm	174400 mm	174600 mm	174800 mm	175000 mm	175200 mm	175400 mm	175600 mm	175800 mm	176000 mm	176200 mm	176400 mm	176600 mm	176800 mm	177000 mm	177200 mm	177400 mm	177600 mm	177800 mm	178000 mm	178200 mm	178400 mm	178600 mm	178800 mm	179000 mm	179200 mm	179400 mm	179600 mm	179800 mm	180000 mm	180200 mm	180400 mm	180600 mm	180800 mm	181000 mm	181200 mm	181400 mm	181600 mm	181800 mm	182000 mm	182200 mm	182400 mm	182600 mm	182800 mm	183000 mm	183200 mm	183400 mm	183600 mm	183800 mm	184000 mm	184200 mm	184400 mm	184600 mm	184800 mm	185000 mm	
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Illustration 281

g06357360

Lift Chart Above: 323, 8850 mm (348 inch) Super Long Reach boom, 2900 mm (114 inch) stick, 5400 kg (11905 lb) counter weight, and 600 mm (24 inch) triple grouser track shoes, heavy lift OFF

i08287326

Lifting Capacities

(320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom)

SMCS Code: 7000

S/N: KFE1-Up

S/N: MYK1-Up

S/N: YCP1-Up

S/N: HEX1-Up



Failure to comply to the rated load can cause possible personal injury or property damage. This includes the risk of unintended boom lowering. Review the rated load of a particular work tool before performing any operation. Make adjustments to the rated load as necessary for non-standard configurations.

There may be local regulations and/or government regulations that govern the use of excavators which lift heavy objects. Obey all local and government regulations.

Lifting capacities should be used as a guide. Work tools, uneven ground conditions, soft ground conditions, or poor ground conditions have effects on lifting capacities. The operator is responsible for being aware of these effects.

The lifting capacities are defined by “ISO 10567 2007”. The lifting capacities are defined as the lower value of 75% of the static tipping capacity or 87% of the hydraulic lift capacity.

Note: Lifting capacities are based on a standard machine with the following conditions:

- Lift point: Stick nose without bucket
- Lubricants full
- Fuel tank full
- Steel track
- Complete cab with a 75 kg (165 lb) operator

Lifting capacities will vary with different work tools and attachments. The weight of a work tool attachment must be subtracted from the lift capacity. Consult your Cat dealer regarding the lifting capacities for specific work tools and attachments.

This machine may be equipped with various sticks. Lifting capacities may vary between the different sticks. Measure the distance on the stick between the boom hinge pin and the work tool hinge pin. This distance will inform you of the size of the stick that is equipped on the machine.

Use the lifting eye that is provided on the linkage to lift objects. When the lifting eye is used, the connection must be made with a sling or shackle.

Note: Japan regulations require a shovel crane configuration to lift certain objects. A shovel crane has a rated load capacity, therefore, the lift capacities discussed below do not apply to a shovel crane configuration. Contact your Cat dealer for additional information.

Note: In European countries, regulations require an overload warning device and a boom and stick lowering control valve if more than 1000 kg (2200 lb) is lifted during object handling applications. Regulations also require an overload warning device and a boom lowering control valve if a force that is greater than 40000 N·m (29500 lb ft) is created during object handling applications. If the machine is not equipped with these devices, even if the hydraulic lift capacity is capable, do not exceed a load of 1000 kg (2200 lb). Do not exceed a force of 40000 N·m (29500 lb ft) in European object handling applications.

Configuration Identification

Note: Each component has a stamp to identify the configuration affecting lifting capacity.

The owner will need to check the machine configuration to identify the correct lifting capacity.

The configuration identifier will be located with the part number stamped on the component. Refer to the following table for the abbreviation of the configuration.

Table 24

Configuration Identification		
Component	Configuration	Abbreviation
Front	Reach Boom	R
	Mass Boom	M
	Variable Angle Boom	VA
	Super Long Reach Boom	SLR
	Standard	STD
	Short Reach Stick	SR
	Severe Duty Bucket	SD
	Heavy Duty	HD
	Semi-Heavy Duty	S-HD
	Extreme Special	ES
	Thumb Ready Stick	TR
Undercarriage	Short Undercarriage (Crawler)	STD
	Long Undercarriage (Crawler)	LC
	Long Narrow Undercarriage (Crawler)	LN
Cylinder	Standard	-
	Heavy Lift	HL
Counterweight	Metric Ton (tonne)	t ⁽¹⁾

⁽¹⁾ Counterweight stamp indicates metric ton. (example 1.0t = 1000 kg)

Symbols Found in the Lifting Capacity Charts

Below are symbols that are commonly found on lifting capacity charts for track excavators.

Note: Depending on the machine configuration, some symbols may not be used.

(mm)
(inch) **Measurements are provided in millimeters and inches**



Lift Capacities are provided in kilograms and pounds



Load is limited by hydraulic lifting capacity rather than by a tipping load



Lift point radius



Lift point height



Lifting capacity over the front of the machine



Lifting capacity over the side of the machine



Heavy Lift ON

**Reach Boom with a 2.5 m
(8 ft 2 inch) Stick and 4200 kg
(9260 lb) Counterweight**

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) SG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300							* 4750 * 4750 5600 * 10660 * 10660 220
6000 240				* 5360 * 5360 * 11700 11800			* 4350 * 4350 6800 * 9600 * 9600 210
4500 180			* 6700 * 6700 * 14000 * 14000	* 5200 * 5200 * 12500 11300	* 4800 3700		* 4250 3650 7570 * 9400 8100 300
3000 120			* 8400 1900 * 18150 14400	* 6000 5200 * 14100 10850	5800 3650 12100 7800		* 4250 3350 7960 * 9600 7300 320
1500 60			* 9900 1100 * 21450 16400	* 7300 4800 * 16750 10750	5500 3550 11850 7600		* 4850 3200 8050 * 10250 7600 320
0			* 10000 6900 * 23000 14900	7000 4700 16100 10350	5450 3500 11700 7500		0100 3200 7840 11250 7200 310
-1500 -60		* 11350 * 11350 * 26800 * 26800	* 16500 6900 * 22700 14850	7450 4850 16500 10950			5800 3650 7350 12350 7850 290
-3000 -120		* 13150 * 13150 * 28500 * 28500	* 20500 7000 * 26000 15050	* 7950 4700 * 13100 10150			* 6300 4300 6470 * 13900 9600 260
-4500 -180			* 7100 * 7100 * 14550 * 14550				* 6800 * 6800 4950 * 13650 * 13650 290

Illustration 282

g06464516

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300							* 5190 * 5190 6900 * 11580 * 11580 220
6000 240				* 5980 5400 * 12980 11800			* 4750 4350 6850 * 10580 9750 210
4500 180			* 7000 * 7000 * 15950 * 15950	* 5360 5360 * 13650 11300	* 4250 3700		* 4640 3650 7570 * 10250 8100 300
3000 120			* 9300 1900 * 20000 16400	* 7200 5850 * 16000 10850	6900 3650 12100 7800		* 4750 3350 7960 * 10450 7300 320
1500 60			* 11900 1100 * 23700 16400	7000 4800 16200 10350	5500 3550 11850 7600		0100 3200 8050 10950 7350 320
0			* 11700 6900 * 25950 14900	7400 4700 16150 10550	5450 3500 11700 7500		0100 3200 7840 11250 7200 310
-1500 -60		* 12000 * 12000 * 27780 * 27780	* 15500 6900 * 25100 14850	7450 4850 16050 10950			5800 3650 7350 12350 7850 290
-3000 -120		* 14150 13550 * 31180 28800	* 19500 7000 * 32800 15050	7600 4700 16250 10150			6080 4300 6470 15180 9600 260
-4500 -180			* 7900 * 7900 * 16550 * 16550				* 6900 * 6900 4950 * 15080 * 15080 290

Illustration 283

g06464524

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift ON

600 mm (24 inch) DG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300						* 4700 * 4700	9000
6000 240				* 5300 * 5300		* 18550 * 18550	220
4500 180				* 11700 11800		* 4350 * 4350	5030
3000 120						* 3900 * 3900	270
1500 60						* 4250 3600	7570
0						* 8600 8100	380
-1500 -60						* 4350 3200	7960
-3000 -120						* 8000 7300	320
-4500 -180						* 4950 3200	8050
-6000 -240						* 16250 7050	320
-7500 -300						* 6100 3200	7660
-9000 -360						* 11200 7200	370
-10500 -410						* 5800 3500	7350
-12000 -470						* 12300 7000	280
-13500 -530						* 4300 4300	8470
-15000 -590						* 13900 8500	280
-16500 -650						* 6300 4200	8980
-18000 -710						* 13550 13550	260

Illustration 284

g06464526

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
7500 300						* 9150 * 9150	8880
6000 240				* 5900 5400		* 11500 * 11500	220
4500 180				* 12900 11800		* 4750 4350	5030
3000 120						* 3050 3150	270
1500 60						* 4650 3600	7570
0						* 10250 9100	380
-1500 -60						* 4750 3300	7960
-3000 -120						* 10400 7300	320
-4500 -180						* 4950 3200	8050
-6000 -240						* 10550 7050	320
-7500 -300						* 6100 3200	7660
-9000 -360						* 11200 7200	370
-10500 -410						* 5800 3500	7350
-12000 -470						* 12300 7000	280
-13500 -530						* 4300 4300	8470
-15000 -590						* 13900 8500	280
-16500 -650						* 6300 4200	8980
-18000 -710						* 13550 13550	260

Illustration 285

g06464527

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300						* 4750 * 4750	5600
6000 240				* 5300 212.5	5300 212.5	* 5050 * 5050	220
4500 180			* 6700 263.8	* 6700 263.8	* 6700 263.8	* 6700 263.8	8830
3000 120			* 8050 316.9	* 8050 316.9	* 8050 316.9	* 8050 316.9	270
1500 60			* 9400 368.1	* 9400 368.1	* 9400 368.1	* 9400 368.1	7570
0			* 10750 423.0	* 10750 423.0	* 10750 423.0	* 10750 423.0	300
-1500 -60			* 12100 476.4	* 12100 476.4	* 12100 476.4	* 12100 476.4	7960
-3000 -120			* 13450 529.8	* 13450 529.8	* 13450 529.8	* 13450 529.8	300
-4500 -180			* 14800 583.2	* 14800 583.2	* 14800 583.2	* 14800 583.2	7960
			* 16150 636.6	* 16150 636.6	* 16150 636.6	* 16150 636.6	310
			* 17500 690.0	* 17500 690.0	* 17500 690.0	* 17500 690.0	7350
			* 18850 743.4	* 18850 743.4	* 18850 743.4	* 18850 743.4	290
			* 20200 796.8	* 20200 796.8	* 20200 796.8	* 20200 796.8	6470
			* 21550 850.2	* 21550 850.2	* 21550 850.2	* 21550 850.2	260
			* 22900 903.6	* 22900 903.6	* 22900 903.6	* 22900 903.6	4950
			* 24250 957.0	* 24250 957.0	* 24250 957.0	* 24250 957.0	200

Illustration 286

g06464530

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300						* 5150 * 5150	5600
6000 240				* 5800 228.3	5800 228.3	* 5550 * 5550	220
4500 180			* 7150 281.3	* 7150 281.3	* 7150 281.3	* 7150 281.3	8830
3000 120			* 8500 334.6	* 8500 334.6	* 8500 334.6	* 8500 334.6	270
1500 60			* 9850 387.8	* 9850 387.8	* 9850 387.8	* 9850 387.8	7570
0			* 11200 439.9	* 11200 439.9	* 11200 439.9	* 11200 439.9	300
-1500 -60			* 12550 493.3	* 12550 493.3	* 12550 493.3	* 12550 493.3	7960
-3000 -120			* 13900 546.7	* 13900 546.7	* 13900 546.7	* 13900 546.7	300
-4500 -180			* 15250 600.0	* 15250 600.0	* 15250 600.0	* 15250 600.0	7960
			* 16600 653.4	* 16600 653.4	* 16600 653.4	* 16600 653.4	310
			* 17950 706.8	* 17950 706.8	* 17950 706.8	* 17950 706.8	7350
			* 19300 760.2	* 19300 760.2	* 19300 760.2	* 19300 760.2	290
			* 20650 813.6	* 20650 813.6	* 20650 813.6	* 20650 813.6	6470
			* 22000 867.0	* 22000 867.0	* 22000 867.0	* 22000 867.0	260
			* 23350 920.4	* 23350 920.4	* 23350 920.4	* 23350 920.4	4950
			* 24700 973.8	* 24700 973.8	* 24700 973.8	* 24700 973.8	200

Illustration 287

g06464532

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) TG track shoes, heavy lift ON

700 mm (28 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4180 165	6000 240	7500 300		
							(mm) (inch)
7500 300						+ 4750 + 4750	5600
6000 240				+ 5360 + 5360		+ 13650 + 13650	220
4180 165				+ 11700 11800		+ 4350 + 4350	6830
3000 120			+ 4700 + 4700	+ 6760 6860	+ 4800 4800	+ 3600 + 3600	270
1500 60			+ 14500 + 14500	+ 10550 11300		+ 4350 + 4350	7570
0			+ 8450 7600	+ 6600 6860	6860 3660	+ 4350 + 4350	7960
-1500 -60			+ 18150 16350	+ 14150 10850	10850 7880	+ 3800 + 3800	320
-3000 -120			+ 28500 27650	+ 23500 20250	20250 11850	+ 4850 + 4850	8910
-4180 -165			+ 41800 40000	+ 34800 30800	30800 18400	+ 10250 + 10250	320
-6000 -240			+ 53600 51800	+ 44600 38600	38600 22600	+ 11200 + 11200	7960
-7500 -300			+ 67600 65800	+ 56600 48600	48600 29600	+ 13800 + 13800	7360
-8450 -335			+ 76000 74200	+ 64000 56000	56000 33600	+ 15800 + 15800	290
-10850 -425			+ 90500 88700	+ 77500 67500	67500 40500	+ 18800 + 18800	6470
-14150 -555			+ 110500 108700	+ 95500 83500	83500 50500	+ 23800 + 23800	260
-18150 -715			+ 141500 139700	+ 121500 107500	107500 65500	+ 30800 + 30800	4960
-22650 -895			+ 176500 174700	+ 151500 133500	133500 81500	+ 39800 + 39800	200

Illustration 288

g06464534

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4180 165	6000 240	7500 300		
							(mm) (inch)
7500 300						+ 5150 + 5150	5600
6000 240				+ 5800 5400		+ 11500 + 11500	220
4180 165				+ 12900 11800		+ 4750 + 4750	6830
3000 120			+ 7400 + 7400	+ 6350 6350	+ 4350 4350	+ 18500 + 18500	270
1500 60			+ 18100 + 18100	+ 10950 11300		+ 4650 + 4650	7570
0			+ 8800 7600	+ 7300 6860	6860 3660	+ 4750 + 4750	7960
-1500 -60			+ 20800 18350	+ 13600 10850	10850 7880	+ 18450 + 18450	320
-3000 -120			+ 31000 27650	+ 20450 16350	16350 11850	+ 28900 + 28900	8910
-4180 -165			+ 41700 38000	+ 30400 24400	24400 14400	+ 38400 + 38400	320
-6000 -240			+ 53600 50000	+ 40600 33600	33600 20600	+ 48900 + 48900	7960
-7500 -300			+ 67600 64000	+ 51600 42600	42600 25600	+ 60900 + 60900	7360
-8450 -335			+ 76000 72400	+ 58600 48600	48600 29600	+ 68900 + 68900	290
-10850 -425			+ 90500 86900	+ 70600 58600	58600 35600	+ 83900 + 83900	6470
-14150 -555			+ 110500 106900	+ 86600 72600	72600 44600	+ 103900 + 103900	260
-18150 -715			+ 130500 126900	+ 106600 89600	89600 54600	+ 128900 + 128900	4960
-22650 -895			+ 150500 146900	+ 126600 106600	106600 64600	+ 158900 + 158900	200

Illustration 289

g06464535

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7000 300						* 4700 * 4700	5600 220
6000 240				* 5350 * 5350		* 4300 * 4300	6030 270
4500 180			* 6700 * 6700	* 5700 * 5700	* 4800 * 4800	* 4200 * 4200	7570 300
3000 120			* 8450 * 8450	* 7700 * 7700	* 6900 * 6900	* 6100 * 6100	7960 320
1500 60			* 10550 * 10550	* 9550 * 9550	* 8550 * 8550	* 7550 * 7550	8060 320
0 0			* 11650 * 11650	* 10650 * 10650	* 9650 * 9650	* 8650 * 8650	7960 310
-1500 -60		* 11500 * 11500	* 10500 * 10500	* 9500 * 9500	* 8500 * 8500	* 7500 * 7500	7560 300
-3000 -120		* 12150 * 12150	* 11150 * 11150	* 10150 * 10150	* 9150 * 9150	* 8150 * 8150	6470 260
-4500 -180			* 12650 * 12650	* 11650 * 11650	* 10650 * 10650	* 9650 * 9650	5580 220

Illustration 290

g06464537

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300						* 5150 * 5150	5080 200
6000 240				* 5900 * 5900		* 4750 * 4750	6030 270
4500 180			* 7400 * 7400	* 6300 * 6300	* 5200 * 5200	* 4050 * 4050	7570 300
3000 120			* 9000 * 9000	* 7900 * 7900	* 6800 * 6800	* 5700 * 5700	7960 320
1500 60			* 10700 * 10700	* 9700 * 9700	* 8700 * 8700	* 7700 * 7700	8060 320
0 0			* 11700 * 11700	* 10700 * 10700	* 9700 * 9700	* 8700 * 8700	7960 310
-1500 -60		* 12000 * 12000	* 11000 * 11000	* 10000 * 10000	* 9000 * 9000	* 8000 * 8000	7560 300
-3000 -120		* 14050 * 14050	* 13050 * 13050	* 12050 * 12050	* 11050 * 11050	* 10050 * 10050	6470 260
-4500 -180			* 15050 * 15050	* 14050 * 14050	* 13050 * 13050	* 12050 * 12050	5580 220

Illustration 291

g06464540

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift ON

900 mm (35 inch) TG shoe

(mm) (inch)	1500 60		3000 120		4500 180		6000 240		7500 300				(mm) (inch)
													
7500 300													5000 200
6000 240							* 6360 * 2500	* 6360 * 2500				* 4750 * 1870	3000 120
4500 180					* 6700 * 2630	* 6700 * 2630	* 6700 * 2630	* 6700 * 2630	* 4800 1900	* 3900 1550	* 4200 1650	* 3800 1500	7500 300
3000 120					* 9450 * 3650	* 9450 * 3650	* 9450 * 3650	* 9450 * 3650	* 6600 2600	* 5200 2050	* 4300 1690	* 3500 1380	7500 300
1500 60					* 18150 * 7050	* 18150 * 7050	* 18150 * 7050	* 18150 * 7050	* 12550 5000	* 8200 3240	* 9600 3760	* 7600 2990	8000 3150
0					* 23450 * 9050	* 23450 * 9050	* 23450 * 9050	* 23450 * 9050	* 12400 4840	* 8000 3130	* 16200 6300	* 7400 2910	3000 120
0					* 13650 * 5200	* 13650 * 5200	* 13650 * 5200	* 13650 * 5200	* 5700 2250	* 3600 1410	* 6300 2480	* 3400 1330	7500 300
-1500 -60			* 11350 * 4400	* 11350 * 4400	* 19500 * 7450	* 19500 * 7450	* 19500 * 7450	* 19500 * 7450	* 12250 4800	* 7800 3030	* 6800 2680	* 3700 1450	7500 300
-3000 -120			* 28800 * 11150	* 28800 * 11150	* 32700 * 12600	* 32700 * 12600	* 32700 * 12600	* 32700 * 12600	* 19500 7600	* 12900 5000	* 12900 5000	* 8200 3200	2900 1140
-4500 -180			* 28500 * 10950	* 28500 * 10950	* 32650 * 12550	* 32650 * 12550	* 32650 * 12550	* 32650 * 12550	* 19100 7400	* 12600 4950	* 12600 4950	* 8200 3190	2900 1140

Illustration 292

q06464541

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
+2							
7500						* 5100	* 5100
300						* 11500	* 11500
6000				* 5900	5900		* 4750
240				* 12900	12900		* 10500
4500			* 7600	* 1400	* 6300	6300	* 6300
180			* 16600	* 16600	* 11800	11800	* 10500
3000			* 9500	9500	* 13000	13000	* 4700
120			* 20000	11100	* 11800	11300	* 19400
1500			* 11000	1450	6000	5800	* 5180
60			* 22700	9100	17250	10850	* 11100
9			* 11700	1340	9840	4800	6700
9			* 20300	16800	16900	16550	12250
-1500		* 12000	* 12000	* 11500	1200	7800	4850
-60		* 27300	* 27300	* 25100	11550	10900	10450
-3000		* 14100	14100	* 10500	1300	7800	4850
-120		* 31100	30250	* 22900	11750	* 6700	10600
-4500				* 7900	1800		
-900				* 10500	14000		

Illustration 293

g06464543

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift ON

Reach Boom with a 2.9 m (9 ft 6 inch) Stick and 4200 kg (9260 lb) Counterweight

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) SG shoe

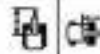
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				+ 4500 + 4500		+ 3900 + 8750	8100 240
6000 240				+ 4950 + 10950		+ 3950 + 8050	7290 230
4500 180				+ 5400 + 10950	+ 6100 + 12000	+ 3600 + 7900	7990 300
3000 120			+ 7950 + 17050	+ 7750 + 16750	+ 6050 + 13000	+ 5450 + 8100	3100 8300
1500 60			+ 9600 + 20050	+ 7250 + 15500	+ 4850 + 10450	+ 3550 + 7050	3300 8400
0 0		+ 8350 + 14350	+ 6350 + 14350	+ 10400 + 22750	+ 6950 + 15050	+ 4700 + 11700	3400 8700
-1500 -60	+ 8700 + 14350	+ 6700 + 14350	+ 18900 + 24450	+ 18900 + 25500	+ 10400 + 14500	+ 5450 + 11050	3300 7250
-3000 -120	+ 11650 + 26800	+ 11650 + 26800	+ 14350 + 38450	+ 13400 + 28700	+ 9550 + 21450	+ 6050 + 13000	3900 8500
-4500 -180		+ 11200 + 24000	+ 11200 + 24000	+ 8950 + 17150	+ 7150 + 15450	+ 6100 + 13400	5300 11900

Illustration 294

g06464544

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift OFF

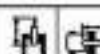
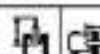
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				+ 4500 + 4500		+ 4300 + 9650	8130 243
6000 240				+ 5450 + 11950		+ 4800 + 8800	7290 239
4500 180				+ 6000 + 13000	+ 6550 + 12350	+ 3900 + 8100	7990 305
3000 120			+ 7750 + 18750	+ 6950 + 16750	+ 5150 + 12150	+ 4800 + 8900	3130 8250
1500 60			+ 18550 + 22850	+ 7250 + 15500	+ 4850 + 10450	+ 3550 + 7050	3350 8450
0 0		+ 8900 + 15200	+ 6900 + 15200	+ 11650 + 25050	+ 6950 + 15050	+ 4700 + 11700	3450 8750
-1500 -60	+ 7100 + 15800	+ 7100 + 15800	+ 11400 + 35900	+ 11400 + 25900	+ 6950 + 14500	+ 5150 + 11350	3300 7250
-3000 -120	+ 12100 + 27100	+ 12100 + 27100	+ 15000 + 33000	+ 13400 + 27700	+ 10500 + 18050	+ 6100 + 13500	3850 8650
-4500 -180		+ 12400 + 26500	+ 12400 + 26500	+ 8950 + 18050	+ 7150 + 15450	+ 6150 + 14850	5350 11950

Illustration 295

g06464544

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift ON

600 mm (24 inch) DG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			(mm) (inch)
								
7500 300								
6000 240								
4500 180								
3000 120								
1500 60								
0								
-1500 -60								
-3000 -120								
-4500 -180								

Illustration 296

g06464547

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			(mm) (inch)
								
7500 300								
6000 240								
4500 180								
3000 120								
1500 60								
0								
-1500 -60								
-3000 -120								
-4500 -180								

Illustration 297

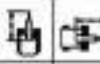
g06464550

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300							
9000 360							
4500 180							
3000 120							
1500 60							
0 0							
-1500 -60							
-3000 -120							
-4500 -180							
-1500 -60							
0 0							
-1500 -60							
-3000 -120							
-4500 -180							
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-3000 -120							
-4500 -180							
-1500 -60							
0 0							
-1500 -60							
-3000 -120							
-4500 -180							

700 mm (28 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							
6000 240							
4500 180							
3000 120							
1500 60							
0							
-1500 -60							
-3000 -120							
-4500 -180							

Illustration 300

g06464556

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		
							(mm) (inch)
7500 300							
6000 240							
4500 180							
3000 120							
1500 60							
0							
-1500 -60							
-3000 -120							
-4500 -180							

Illustration 301

g06464557

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) TG shoe

(mm) (inch)	1500 60		3000 120		4500 180		6000 240		7500 300				(mm) (inch)
													(mm) (inch)
7500 300							* 4500 180	* 4500 180			* 3850 152	* 3850 152	6150 242
6000 240							* 4000 160	* 4000 160			* 3450 136	* 3450 136	7250 285
4500 180							* 3400 134	* 3400 134	* 5100 201	3800 150	* 3400 134	* 3400 134	7950 312
3000 120					* 2900 114	* 2900 114	* 3300 130	* 3300 130	* 11800 465	8800 347	* 8800 347	* 8800 347	8350 328
1500 60					* 2400 94	* 2400 94	* 3000 118	* 3000 118	* 11800 465	8800 347	* 8800 347	* 8800 347	8750 342
0					* 2000 79	* 2000 79	* 15300 598	* 15300 598	12600 496	7750 305	* 8550 337	* 8550 337	9150 358
8			* 6000 236	* 6000 236	* 10600 417	* 10600 417	7600 299	4700 185	8400 327	3600 142	* 4300 169	* 4300 169	9550 372
3			* 14350 565	* 14350 565	* 22700 894	* 22700 894	18400 724	18250 718	11850 463	7900 307	* 5450 215	* 5450 215	9950 388
-1500 -60	* 8700 343	* 8700 343	* 10800 421	* 10800 421	* 10180 398	* 10180 398	7550 297	4700 185	8400 327	3500 138	* 5850 230	* 5850 230	7750 305
-3000 -120	* 11600 457	* 11600 457	* 24450 960	* 24450 960	* 22560 887	* 22560 887	10060 394	6200 244	11800 463	7550 297	* 11400 449	* 11400 449	7350 289
-4500 -180	* 26600 1048	* 26600 1048	* 30450 1196	* 30450 1196	* 21450 843	* 21450 843	18180 711	* 10850 423	* 18150 710		* 12300 484	* 12300 484	6800 268

Illustration 302

g06464558

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1000 60	3000 120	4500 180	6000 240	7500 300			(mm) (inch)		
										
+8										
7500 300				* 4500 180	* 4050 160			* 4300 169	* 4300 169	8150 320
6000 240				* 5450 219	* 5450 219			* 5650 223	* 5650 223	7250 285
4500 180				* 6300 252	* 6300 252	* 6650 262	3800 150	* 3900 154	* 3900 154	7950 312
3000 120			* 8700 343	* 8700 343	* 8700 343	* 9050 354	3800 150	* 3900 154	* 3900 154	8350 328
1500 60			* 10550 415	* 10550 415	* 10550 415	* 10900 425	3800 150	* 3900 154	* 3900 154	8750 342
0			* 11900 469	* 11900 469	* 11900 469	* 12250 482	3800 150	* 3900 154	* 3900 154	9150 358
-1500 -60	* 7100 278	* 7100 278	* 11400 445	* 11400 445	* 11400 445	* 11750 460	3800 150	* 3900 154	* 3900 154	9550 372
-3000 -120	* 12100 472	* 12100 472	* 15500 609	* 15500 609	* 15500 609	* 15850 620	3800 150	* 3900 154	* 3900 154	9950 388
-4500 -180			* 13400 512	* 13400 512	* 13400 512	* 13750 531	3800 150	* 3900 154	* 3900 154	10350 402

Illustration 303

g06464559

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift ON

900 mm (35 inch) TG shoe

(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				* 4300 * 4300		* 3900 * 3900	8100 320
6000 240				* 4300 * 4300		* 3900 * 3900	7200 280
4500 180				* 5400 * 5400	* 6100 3900	* 5600 3500	7900 310
3000 120			* 7000 * 7000	* 6200 5300	* 5400 3850	* 5000 3500	6300 250
1500 60			* 13000 * 13000	* 11800 11800	* 11800 8200	* 8000 7200	3300 130
0 0		* 8200 * 8200	* 10400 7800	* 7800 4800	* 6700 3800	* 4700 3200	8200 320
-1500 -60	* 6700 * 6700	* 18000 * 18000	* 10000 7200	* 7800 4800	* 5700 3600	* 3400 2400	7700 300
-3000 -120	* 11400 * 11400	* 14000 14000	* 9600 7200	* 7000 4300		* 5000 4000	6800 270
-4500 -180		* 11200 * 11200	* 8000 7000			* 6100 5000	6800 270

Illustration 304

g06464561

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift OFF

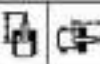
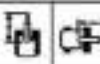
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
7500 300				* 4300 * 4300		* 3900 * 3900	8100 320
6000 240				* 5400 * 5400		* 4900 * 4900	7200 280
4500 180				* 11800 * 11800	* 6600 3900	* 5600 3500	7900 310
3000 120			* 8700 8000	* 6900 5300	* 5600 3850	* 4900 3500	6300 250
1500 60			* 18700 * 18700	* 14800 11400	* 12700 8200	* 8800 7200	3300 130
0 0		* 8800 * 8800	* 11600 7800	* 7800 4800	* 6700 3800	* 4700 3200	8200 320
-1500 -60	* 7100 * 7100	* 19000 * 19000	* 10000 7200	* 7800 4800	* 5700 3600	* 3400 2400	7700 300
-3000 -120	* 12100 * 12100	* 15000 14000	* 9600 7200	* 7000 4300		* 5000 4000	6800 270
-4500 -180		* 12400 * 12400	* 8000 7000			* 6100 5000	6800 270

Illustration 305

g06464565

Lift Chart Above: 320, 5675 mm (224 inch) reach boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift ON

Variable Angle Boom with a 2.5 m (8 ft 2 inch) Stick and 4200 kg (9260 lb) Counterweight

600 mm (24 inch) SG shoe

Load Shed	500 60		3000 120		4500 180		6000 240		7500 300				Int'l Trk	
														
3000 360														
7500 360					6950	6050	4300	4300				3950	5250	47
					8800	14500						13600	15600	5
3000 240												4050	4900	680
												10950	10950	24
3000 240					6750	6750	4350	4350				4450	3650	723
					8800	14500	5500	5500				1700	5650	25
4500 180			10400	10400	8550	8550	4200	4200	4400	3500		4250	3250	750
			23500	23500	14200	14200	3650	3650	8900	3900		3400	1780	3
3000 120			9400	9400	5900	5900	4350	4350	4600	3400		4350	2580	530
			21250	21250	12700	12700	5500	5500	8200	7400		3250	1640	30
6000 60			6400	6400	6950	6050	4600	4600	5700	3300		4900	2600	830
			15800	15800	10000	12000	10000	9700	11600	7100		3350	1650	30
0 0	3050	3050	6050	6050	6950	8400	8550	4300	5250	3200		4650	2350	62
	20150	20150	13450	13450	11300	11750	12300	5000	10300	6000		9700	6250	30
-1500 -40	10400	10400	6050	6050	1750	6400	8150	4300	4400	3200		4050	3100	715
	22900	22900	8900	8900	8950	12750	12750	3000	3200	6000		3950	6650	3
-3000 -120	6250	6250	1050	1050	3850	5650	4250	4250				3650	3600	670
	36350	36350	15400	15400	12300	13100	8950	8950				8000	3000	21
-4500 -360	20250	20250	20400	20400								3250	5500	45
			20300	20300								9900	5000	

Illustration 306

g06464566

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift OFF

Level [inch]	500 G	1000 G	1500 G	2000 G	2500 G	3000 G	Level [inch]
5000							5000
4500							4500
4000							4000
3500							3500
3000							3000
2500							2500
2000							2000
1500							1500
1000							1000
500							500
0							0
-500							-500
-1000							-1000
-1500							-1500
-2000							-2000
-2500							-2500
-3000							-3000
-3500							-3500
-4000							-4000
-4500							-4500
-5000							-5000

Illustration 307

g06464570

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift ON

600 mm (24 inch) DG shoe

Load (lb/ft)	1500 66	3000 132	4500 198	6000 264	7500 330		
							(mm) (inch)
9000							
360							
7600							
300							
6000							
240							
4500							
180							
3000							
120							
1500							
60							
0							
0							
-2500							
-60							
-3000							
-120							
-4500							
-180							

Illustration 308

g06464572

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift OFF

Load (lb/ft)	1500 66	3000 132	4500 198	6000 264	7500 330		
							(mm) (inch)
9000							
360							
7600							
300							
6000							
240							
4500							
180							
3000							
120							
1500							
60							
0							
0							
-2500							
-60							
-3000							
-120							
-4500							
-180							

Illustration 309

g06464575

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) TG shoe

Reach (inch)	500 60	3000 120	4500 180	6000 240	7500 300		Reach (inch)
5000 300							5000 300
4500 240							4500 240
4000 180							4000 180
3500 120							3500 120
3000 60							3000 60
2500 0							2500 0
2000 -60							2000 -60
1500 -120							1500 -120
1000 -180							1000 -180
500 -240							500 -240
0 -300							0 -300
-500 -360							-500 -360
-1000 -420							-1000 -420
-1500 -480							-1500 -480
-2000 -540							-2000 -540
-2500 -600							-2500 -600
-3000 -660							-3000 -660
-3500 -720							-3500 -720
-4000 -780							-4000 -780
-4500 -840							-4500 -840
-5000 -900							-5000 -900
-5500 -960							-5500 -960
-6000 -1020							-6000 -1020
-6500 -1080							-6500 -1080
-7000 -1140							-7000 -1140
-7500 -1200							-7500 -1200
-8000 -1260							-8000 -1260
-8500 -1320							-8500 -1320
-9000 -1380							-9000 -1380
-9500 -1440							-9500 -1440
-10000 -1500							-10000 -1500
-10500 -1560							-10500 -1560
-11000 -1620							-11000 -1620
-11500 -1680							-11500 -1680
-12000 -1740							-12000 -1740
-12500 -1800							-12500 -1800
-13000 -1860							-13000 -1860
-13500 -1920							-13500 -1920
-14000 -1980							-14000 -1980
-14500 -2040							-14500 -2040
-15000 -2100							-15000 -2100
-15500 -2160							-15500 -2160
-16000 -2220							-16000 -2220
-16500 -2280							-16500 -2280
-17000 -2340							-17000 -2340
-17500 -2400							-17500 -2400
-18000 -2460							-18000 -2460
-18500 -2520							-18500 -2520
-19000 -2580							-19000 -2580
-19500 -2640							-19500 -2640
-20000 -2700							-20000 -2700
-20500 -2760							-20500 -2760
-21000 -2820							-21000 -2820
-21500 -2880							-21500 -2880
-22000 -2940							-22000 -2940
-22500 -3000							-22500 -3000
-23000 -3060							-23000 -3060
-23500 -3120							-23500 -3120
-24000 -3180							-24000 -3180
-24500 -3240							-24500 -3240
-25000 -3300							-25000 -3300
-25500 -3360							-25500 -3360
-26000 -3420							-26000 -3420
-26500 -3480							-26500 -3480
-27000 -3540							-27000 -3540
-27500 -3600							-27500 -3600
-28000 -3660							-28000 -3660
-28500 -3720							-28500 -3720
-29000 -3780							-29000 -3780
-29500 -3840							-29500 -3840
-30000 -3900							-30000 -3900
-30500 -3960							-30500 -3960
-31000 -4020							-31000 -4020
-31500 -4080							-31500 -4080
-32000 -4140							-32000 -4140
-32500 -4200							-32500 -4200
-33000 -4260							-33000 -4260
-33500 -4320							-33500 -4320
-34000 -4380							-34000 -4380
-34500 -4440							-34500 -4440
-35000 -4500							-35000 -4500
-35500 -4560							-35500 -4560
-36000 -4620							-36000 -4620
-36500 -4680							-36500 -4680
-37000 -4740							-37000 -4740
-37500 -4800							-37500 -4800
-38000 -4860							-38000 -4860
-38500 -4920							-38500 -4920
-39000 -4980							-39000 -4980
-39500 -5040							-39500 -5040
-40000 -5100							-40000 -5100
-40500 -5160							-40500 -5160
-41000 -5220							-41000 -5220
-41500 -5280							-41500 -5280
-42000 -5340							-42000 -5340
-42500 -5400							-42500 -5400
-43000 -5460							-43000 -5460
-43500 -5520							-43500 -5520
-44000 -5580							-44000 -5580
-44500 -5640							-44500 -5640
-45000 -5700							-45000 -5700
-45500 -5760							-45500 -5760
-46000 -5820							-46000 -5820
-46500 -5880							-46500 -5880
-47000 -5940							-47000 -5940
-47500 -6000							-47500 -6000
-48000 -6060							-48000 -6060
-48500 -6120							-48500 -6120
-49000 -6180							-49000 -6180
-49500 -6240							-49500 -6240
-50000 -6300							-50000 -6300
-50500 -6360							-50500 -6360
-51000 -6420							-51000 -6420
-51500 -6480							-51500 -6480
-52000 -6540							-52000 -6540
-52500 -6600							-52500 -6600
-53000 -6660							-53000 -6660
-53500 -6720							-53500 -6720
-54000 -6780							-54000 -6780
-54500 -6840							-54500 -6840
-55000 -6900							-55000 -6900
-55500 -6960							-55500 -6960
-56000 -7020							-56000 -7020
-56500 -7080							-56500 -7080
-57000 -7140							-57000 -7140
-57500 -7200							-57500 -7200
-58000 -7260							-58000 -7260
-58500 -7320							-58500 -7320
-59000 -7380							-59000 -7380
-59500 -7440							-59500 -7440
-60000 -7500							-60000 -7500
-60500 -7560							-60500 -7560
-61000 -7620							-61000 -7620
-61500 -7680							-61500 -7680
-62000 -7740							-62000 -7740
-62500 -7800							-62500 -7800
-63000 -7860							-63000 -7860
-63500 -7920							-63500 -7920
-64000 -7980							-64000 -7980
-64500 -8040							-64500 -8040
-65000 -8100							-65000 -8100
-65500 -8160							-65500 -8160
-66000 -8220							-66000 -8220
-66500 -8280							-66500 -8280
-67000 -8340							-67000 -8340
-67500 -8400							-67500 -8400
-68000 -8460							-68000 -8460
-68500 -8520							-68500 -8520
-69000 -8580							-69000 -8580
-69500 -8640							-69500 -8640
-70000 -8700							-70000 -8700
-70500 -8760							-70500 -8760
-71000 -8820							-71000 -8820
-71500 -8880							-71500 -8880
-72000 -8940							-72000 -8940
-72500 -9000							-72500 -9000
-73000 -9060							-73000 -9060
-73500 -9120							-73500 -9120
-74000 -9180							-74000 -9180
-74500 -9240							-74500 -9240
-75000 -9300							-75000 -9300
-75500 -9360							-75500 -9360
-76000 -9420							-76000 -9420
-76500 -9480							-76500 -9480
-77000 -9540							-77000 -9540
-77500 -9600							-77500 -9600
-78000 -9660							-78000 -9660
-78500 -9720							-78500 -9720
-79000 -9780							-79000 -9780
-79500 -9840							-79500 -9840
-80000 -9900							-80000 -9900
-80500 -9960							-80500 -9960
-81000 -10020							-81000 -10020
-81500 -10080							-81500 -10080
-82000 -10140							-82000 -10140
-82500 -10200							-82500 -10200
-83000 -10260							-83000 -10260
-83500 -10320							-83500 -10320
-84000 -10380							-84000 -10380
-84500 -10440							-84500 -10440
-85000 -10500							-85000 -10500
-85500 -10560							-85500 -10560
-86000 -10620							-86000 -10620
-86500 -10680							-86500 -10680
-87000 -10740							-87000 -10740
-87500 -10800							-87500 -10800
-88000 -10860							-88000 -10860
-88500 -10920							-88500 -10920
-89000 -10980							-89000 -10980
-89500 -11040							

700 mm (28 inch) TG shoe

Level (feet)	500 60	3000 120	4500 180	6000 240	7500 300		Level (feet)
3000 360							3000 360
2900 300							2900 300
2800 340							2800 340
4500 180							4500 180
3000 120							3000 120
2500 60							2500 60
0							0
-500							-500
-60							-60
-3000							-3000
-120							-120
-4500							-4500
-180							-180

Illustration 312

g06464580

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift OFF

Level (feet)	500 60	3000 120	4500 180	6000 240	7500 300		Level (feet)
3000 360							3000 360
2900 300							2900 300
2800 340							2800 340
4500 180							4500 180
3000 120							3000 120
2500 60							2500 60
0							0
-500							-500
-60							-60
-3000							-3000
-120							-120
-4500							-4500
-180							-180

Illustration 313

g06464582

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) TG shoe

Load (mmF)	500 60	3000 120	4500 180	6000 240	7500 300		Load (mmF)
							Load (mmF)
3000 360							5800 700
7500 360			8800 10600	9800 11800	10800 12800		13800 16800
6000 240			8750 10600	9750 11600	10750 12600		13750 16750
4500 180		10400 12300	10400 12300	9550 11200	4200 3500	4400 3600	4200 3500
3000 120		3450 21200	3450 21200	5300 11700	4050 8800	4050 8800	4350 9800
1500 60		9450 13600	9450 13600	9050 13000	4600 10000	4600 10000	4600 10000
0	5250 20800	5250 20800	6050 10450	8050 11700	5550 10300	4400 9400	5550 10300
-1500 -60	10400 22500	10400 22500	9850 10900	7750 10250	4350 3300	4400 3200	4350 3300
-3000 -120	8250 36350	8250 36350	8850 10400	9850 12100	4250 8250		3600 8000
-4500 -180	10850 30850	10850 30850	9400 20300	9400 20300			6000 15550

Illustration 314

g06464583

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift OFF

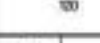
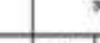
Load (mmF)	500 60	3000 120	4500 180	6000 240	7500 300		Load (mmF)
							Load (mmF)
3000 360							5800 700
7500 360			7500 10200	5250 10200			5200 1000
6000 240			7400 10000	4750 10050			4600 9750
4500 180		10400 24750	10400 24750	9550 10900	4050 3050	4400 3600	4050 3500
3000 120		3400 23400	3400 23400	5300 14050	4050 5800	4050 5800	4350 10250
1500 60		9400 16550	9400 16550	9050 14400	4600 11000	4600 11000	4600 11000
0	5800 21000	5800 21000	6400 14250	8500 13950	9800 10200	5800 9400	9800 10200
-1500 -60	10800 24250	10800 24250	9850 21000	7750 10900	4350 3300	4400 3200	4350 3300
-3000 -120	8200 38450	8200 38450	8850 23800	9850 13650	4250 8050		3600 8450
-4500 -180	10850 32250	10850 32250	9400 23850	9400 23850			6000 17400

Illustration 315

g06464586

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift ON

900 mm (35 inch) TG shoe

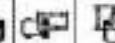
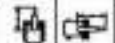
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
								
9000 360								5000 196
7500 300								4800 189
6000 240								4600 182
4500 180								4400 175
3000 120								4200 168
1500 60								4000 161
0								3800 154
-1500 -60								3600 147
-3000 -120								3400 140
-4500 -180								3200 133
-6000 -240								3000 126
-7500 -300								2800 119
-9000 -360								2600 112

Illustration 316

g06464587

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift OFF

Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
								
9000 360								5000 196
7500 300								4800 189
6000 240								4600 182
4500 180								4400 175
3000 120								4200 168
1500 60								4000 161
0								3800 154
-1500 -60								3600 147
-3000 -120								3400 140
-4500 -180								3200 133
-6000 -240								3000 126
-7500 -300								2800 119
-9000 -360								2600 112

Illustration 317

g06464590

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2500 mm (98 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift ON

Variable Angle Boom with a 2.9 m (9 ft 6 inch) Stick and 4200 kg (9260 lb) Counterweight

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) SG shoe

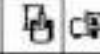
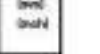
Load (metric)	1500 60		3000 120		4500 180		6000 240		7500 300				Load (metric)
													
3000 360					4700 11750	4700 11750					4700 11750	4700 11750	4880 120
1500 300					9500 14200	9500 14200	3950 8650	3950 8650			9900 8900	9900 8900	6600 260
6000 240					1700 14650	1700 14650	4750 8750	4750 8750	3750 3950	3750 3950	8800 8800	8800 8800	7600 300
4500 90			10500 22950	10500 22950	9550 14350	9550 14350	6550 3850	6550 3850	3650 3650	3650 3650	9600 7900	9600 7900	6350 330
3000 120	11350	11350	10000 21500	10000 21500	8400 11900	8400 11900	3950 6350	3950 6350	3950 3950	3950 3950	8800 7900	8800 7900	6700 350
1500 60			6750 13650	6750 13650	5600 12100	5600 12100	4350 3400	4350 3400	4450 3350	4450 3350	8800 8300	8800 8300	5700 300
0 0	8500 14100	8500 14100	5300 11600	5300 11600	7400 16400	7400 16400	5750 12000	5750 12000	5950 13300	5950 13300	8800 8800	8800 8800	6600 340
-1500 -60	8250 8350	8250 8350	6500 14350	6500 14350	8350 17700	8350 17700	6500 14000	6500 14000	4250 13300	4250 13300	8900 8900	8900 8900	6100 320
-3000 -120	11250 12600	11250 12600	8800 19000	8800 19000	12100 17400	12100 17400	4300 10450	4300 10450			10900 12200	10900 12200	7300 290
-4500 -180	10400 14350	10400 14350	10400 19750	10400 19750	16000 15600	16000 15600					8800 10900	8800 10900	5300 270

Illustration 318

g06464593

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift OFF

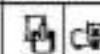
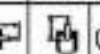
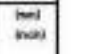
Load (metric)	1500 60		3000 120		4500 180		6000 240		7500 300				Load (metric)
													
3000 360					9100 17700	9100 17700					9200 17700	9200 17700	4880 120
1500 300					17800 15500	17800 15500	5450 3450	5450 3450			9300 8900	9300 8900	6600 260
6000 240					1300 18950	1300 18950	5250 8900	5250 8900	4280 3880	4280 3880	9300 8900	9300 8900	7600 300
4500 90			19500 25050	19500 25050	1300 17700	1300 17700	5750 12000	5750 12000	4080 3680	4080 3680	9600 7900	9600 7900	6350 330
3000 120	19000	19000	10000 23700	10000 23700	10500 15300	10500 15300	5250 5250	5250 5250	4360 3960	4360 3960	9600 8700	9600 8700	6700 350
1500 60			13850 13850	13850 13850	13400 13400	13400 13400	10450 5850	10450 5850	10200 7150	10200 7150	9800 9800	9800 9800	5600 300
0 0	8800 14100	8800 14100	5600 12250	5600 12250	8450 15600	8450 15600	6700 15000	6700 15000	6250 11500	6250 11500	8800 8800	8800 8800	6600 340
-1500 -60	8750 15400	8750 15400	9000 20450	9000 20450	13000 26200	13000 26200	1200 15400	1200 15400	5250 11500	5250 11500	8800 8800	8800 8800	6100 320
-3000 -120	14000 17050	14000 17050	13950 27800	13950 27800	15000 18800	15000 18800	6500 17700	6500 17700	4300		10900 8200	10900 8200	7300 290
-4500 -180	12250 14900	12250 14900	16250 22750	16250 22750	25000 14150	25000 14150					5400 12300	5400 12300	5350 270

Illustration 319

g06464595

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift ON

600 mm (24 inch) DG shoe

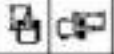
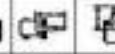
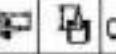
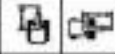
Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 360			4700 11750	4700 10750		4750 10750	4800 190
2500 300			5500 14200	5900 13800	3950 8850	3950 8850	6400 290
2000 240			5700 14850	6700 15500	4750 10750	4750 10750	7800 300
1500 180		10500 22950	10900 23950	14800 32950	14800 32950	14800 32950	3000 6350
1000 120	17350 17350	10000 21550	10900 23550	14800 32500	14800 32500	14800 32500	2750 5950
500 60		5700 12550	5700 12550	5900 12900	5900 12900	5900 12900	2050 4500
0	8900 19100	8900 19100	8900 19100	8900 19100	8900 19100	8900 19100	2850 6250
-500 -60	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-1000 -120	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-1500 -180	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-2000 -240	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-2500 -300	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-3000 -360	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-3500 -420	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-4000 -480	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-4500 -540	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-5000 -600	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500

Illustration 320

g06464596

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift OFF

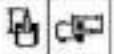
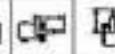
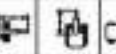
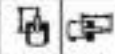
Load (inch)	500 80	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 360			5700 12550	5700 12550		5700 12550	4800 190
2500 300			5700 12550	5700 12550	4000 8850	4000 8850	6400 290
2000 240			5700 12550	5700 12550	4000 8850	4000 8850	7800 300
1500 180		10500 22950	10900 23950	14800 32950	14800 32950	14800 32950	3000 6350
1000 120	17350 17350	10000 21550	10900 23550	14800 32500	14800 32500	14800 32500	2750 5950
500 60		5700 12550	5700 12550	5900 12900	5900 12900	5900 12900	2050 4500
0	8900 19100	8900 19100	8900 19100	8900 19100	8900 19100	8900 19100	2850 6250
-500 -60	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-1000 -120	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-1500 -180	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-2000 -240	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-2500 -300	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-3000 -360	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-3500 -420	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-4000 -480	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-4500 -540	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500
-5000 -600	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	5250 11750	2050 4500

Illustration 321

g06464598

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) TG shoe

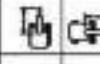
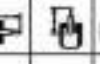
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 360			4700 10700	4700 5750		4750 10750	4900 10900
2600 300			5500 13200	5500 6650	3950 9650	3950 9650	5600 13600
6000 240			5700 13800	6700 16080	4750 8850	4750 8850	3750 7750
4500 180		10500 22050	5550 12300	6550 14200	4850 3850	4850 3850	3600 7300
3000 120	11950 17350	10000 21500	6400 13900	3850 8350	3850 8350	3550 7250	3700 7900
1500 60		10550 22050	1200 12100	3400 3400	3350 3350	3350 7050	3550 7650
0	6500 14100	8500 18100	1300 18500	6300 13000	5150 10200	5050 10050	4750 10300
-1500 -60	5250 10250	5250 10250	5800 12500	5250 11450	4800 10000	4800 9200	3500 8200
-3000 -120	13250 25800	13250 25800	8850 19200	8200 18400	4250 9450		3250 7200
-4500 -180	20450 44550	20450 44550	9150 20100	8000 17600			4050 9000

Illustration 322

g06464601

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) TG track shoes, heavy lift OFF

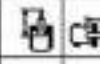
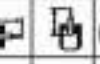
Load (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		Load (inch)
							
3000 360			5700 11700	5700 11700		5200 11000	4900 10900
2600 300			7100 15500	7100 15500	4000 9400	4000 9400	4750 10500
6000 240			7300 15900	7300 15900	5250 8800	5250 8800	4000 7700
4500 180		11500 25000	7300 15700	7300 15700	5750 10300	5750 10300	3800 8600
3000 120	18000 23700	11000 23700	7050 15300	7050 15300	4250 5250	4250 5250	3450 7750
1500 60		11850 15850	10400 13400	10400 13400	5650 10300	5650 10300	3100 6850
0	6500 14100	8500 18100	1300 17600	6300 13600	5150 10400	5050 10200	4750 10050
-1500 -60	51750 10450	51750 10450	5300 20450	5250 20450	4700 18000	4700 18000	3700 8200
-3000 -120	14000 31350	14000 31350	8850 21800	8200 18500	4250 10750		3700 8200
-4500 -180	22050 49300	22050 49300	9250 21750	8000 18150			5450 12000

Illustration 323

g06464603

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 600 mm (24 inch) TG track shoes, heavy lift ON

700 mm (28 inch) TG shoe

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
								
9000								
360								
7500								
300								
6000								
240								
4500								
180								
3000								
120								
1500								
60								
0								
0								
-1500								
-60								
-3000								
-120								
-4500								
-180								

Illustration 324

g06464604

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift OFF

Load (inch)	500 60	3000 120	4500 180	6000 240	7500 300			Load (inch)
								
9000								
360								
7500								
300								
6000								
240								
4500								
180								
3000								
120								
1500								
60								
0								
0								
-1500								
-60								
-3000								
-120								
-4500								
-180								

Illustration 325

g06464605

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

790 mm (31 inch) TG shoe

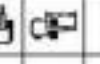
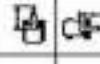
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
5000 360			+ 4700 + 1750	- 4700 - 1750		+ 4750 + 10750	- 4750 - 10050
7500 300			+ 5500 + 14200	- 5500 - 14200	- 3900 - 8800	+ 5850 + 6800	- 5850 - 6800
6000 240			+ 6700 + 14950	- 6700 - 14950	- 4750 - 8200	+ 3750 + 7950	- 3750 - 7950
4500 180		+ 9500 + 22850	+ 9500 + 22850	+ 8650 + 10780	- 4650 - 8600	+ 3650 + 7600	- 3650 - 7600
3000 120	+ 17350 + 17350	+ 18000 + 18000	+ 14400 + 14400	+ 13800 + 13800	- 3900 - 3900	+ 3550 + 7500	- 3550 - 7500
1500 60		+ 6700 + 10050	+ 6700 + 10050	+ 5800 + 12000	- 4300 - 3400	+ 3450 + 7250	- 3450 - 7250
0	+ 6900 + 14100	+ 6900 + 14100	+ 5300 + 10500	+ 5300 + 10500	- 5700 - 10000	+ 3250 + 7000	- 4700 - 5850
-1500 -60	+ 6250 + 10350	+ 6250 + 10350	+ 5500 + 10350	+ 5500 + 10350	- 4300 - 10000	+ 3200 + 6900	- 3200 - 6900
-3000 -120	+ 13250 + 29800	+ 13250 + 29800	+ 8950 + 18200	+ 8950 + 18200	- 4900 - 10400	+ 3250 + 7200	- 3250 - 7200
-4500 -180	+ 20450 + 44550	+ 20450 + 44550	+ 13500 + 31750	+ 13500 + 31750	- 5900 - 13000	+ 4650 + 10000	- 4650 - 10000

Illustration 326

g06464619

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift OFF

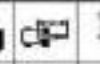
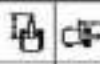
(mm) (inch)	1500 60	3000 120	4500 180	6000 240	7500 300		(mm) (inch)
							
8000 380			+ 5100 + 11700	- 5100 - 11700		+ 5200 + 11700	- 5200 - 11700
7500 300			+ 5700 + 11500	- 5700 + 10500	- 4300 + 5450	+ 4300 + 9400	- 4300 - 9400
6000 240			+ 7300 + 11800	- 7300 + 10800	- 4200 + 8800	+ 4200 + 8800	- 4200 - 8800
4500 180		+ 11500 + 26000	+ 11500 + 26000	+ 7300 + 11700	- 5150 + 10900	+ 4000 + 8800	- 4000 - 8800
3000 120	+ 19800 + 19800	+ 11000 + 23700	+ 7300 + 11300	+ 7050 + 11300	- 4200 + 5250	+ 3650 + 7900	- 3650 - 7900
1500 60		+ 6100 + 13550	+ 6100 + 13550	+ 4200 + 13400	- 4000 + 10300	+ 3600 + 7200	- 3600 - 7200
0	+ 6900 + 11800	+ 6900 + 11800	+ 5300 + 10200	+ 5300 + 10200	- 5700 + 12400	+ 3250 + 7000	- 4700 + 6900
-1500 -60	+ 5750 + 10400	+ 5750 + 10400	+ 5000 + 10400	+ 5000 + 10400	- 4300 + 11500	+ 3200 + 6900	- 3200 - 6900
-3000 -120	+ 11800 + 31350	+ 11800 + 31350	+ 8900 + 21600	+ 8900 + 21600	- 4900 + 11750	+ 3700 + 7150	- 3700 - 7150
-4500 -180	+ 20900 + 49300	+ 20900 + 49300	+ 13500 + 32150	+ 13500 + 32150	- 5900 + 11150	+ 4650 + 10300	- 4650 - 10300

Illustration 327

g06464622

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift ON

900 mm (35 inch) TG shoe

[mm] [inch]	1500 60	3000 120	4500 180	6000 240	7500 300		[mm] [inch]
							
9000 360			+ 4700 + 1850	+ 4700 + 1850		+ 4700 + 1850	4800 190
7500 300			+ 6600 + 2600	+ 6600 + 2600		+ 6600 + 2600	4400 170
6000 240			+ 8700 + 3400	+ 8700 + 3400		+ 8700 + 3400	3900 150
4500 180			+ 10800 + 4200	+ 10800 + 4200		+ 10800 + 4200	3400 130
3000 120			+ 12900 + 4800	+ 12900 + 4800		+ 12900 + 4800	2900 110
1500 60			+ 15000 + 5400	+ 15000 + 5400		+ 15000 + 5400	2400 90
0 0			+ 17100 + 6300	+ 17100 + 6300		+ 17100 + 6300	1900 70
-1500 -60			+ 19200 + 7200	+ 19200 + 7200		+ 19200 + 7200	1400 50
-3000 -120			+ 21300 + 8100	+ 21300 + 8100		+ 21300 + 8100	900 30
-4500 -180			+ 23400 + 9000	+ 23400 + 9000		+ 23400 + 9000	400 10
-6000 -240			+ 25500 + 9900	+ 25500 + 9900		+ 25500 + 9900	-100 -10
-7500 -300			+ 27600 + 10800	+ 27600 + 10800		+ 27600 + 10800	-200 -20
-9000 -360			+ 29700 + 11700	+ 29700 + 11700		+ 29700 + 11700	-300 -30

Illustration 328

g06464624

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy liftOFF

[mm] [inch]	1500 60	3000 120	4500 180	6000 240	7500 300		[mm] [inch]
							
9000 360			+ 4700 + 1850	+ 4700 + 1850		+ 4700 + 1850	4800 190
7500 300			+ 6600 + 2600	+ 6600 + 2600		+ 6600 + 2600	4400 170
6000 240			+ 8700 + 3400	+ 8700 + 3400		+ 8700 + 3400	3900 150
4500 180			+ 10800 + 4200	+ 10800 + 4200		+ 10800 + 4200	3400 130
3000 120			+ 12900 + 4800	+ 12900 + 4800		+ 12900 + 4800	2900 110
1500 60			+ 15000 + 5400	+ 15000 + 5400		+ 15000 + 5400	2400 90
0 0			+ 17100 + 6300	+ 17100 + 6300		+ 17100 + 6300	1900 70
-1500 -60			+ 19200 + 7200	+ 19200 + 7200		+ 19200 + 7200	1400 50
-3000 -120			+ 21300 + 8100	+ 21300 + 8100		+ 21300 + 8100	900 30
-4500 -180			+ 23400 + 9000	+ 23400 + 9000		+ 23400 + 9000	400 10
-6000 -240			+ 25500 + 9900	+ 25500 + 9900		+ 25500 + 9900	-100 -10
-7500 -300			+ 27600 + 10800	+ 27600 + 10800		+ 27600 + 10800	-200 -20
-9000 -360			+ 29700 + 11700	+ 29700 + 11700		+ 29700 + 11700	-300 -30

Illustration 329

g06464625

Lift Chart Above: 320, 2686 mm / 3300 mm (106 inch / 130 inch) Variable angle boom, 2920 mm (115 inch) stick, 4200 kg (9260 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift ON

Super Long Reach Boom with a 6.28 m (20 ft 7 inch) Stick and 4700 kg (10362 lb) Counterweight

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) SG shoe

mm (inch)	1100 43	1300 51	1500 59	1700 67	1900 75	2100 83	2300 91	2500 99	2700 107	2900 115	3100 123	3300 131	3500 139	3700 147	3900 155	4100 163	4300 171	4500 179	4700 187	4900 195	5100 203	5300 211	5500 219	5700 227	5900 235	6100 243	6300 251	6500 259	6700 267	6900 275	7100 283	7300 291	7500 299	7700 307	7900 315	8100 323	8300 331	8500 339	8700 347	8900 355	9100 363	9300 371	9500 379	9700 387	9900 395	10100 403	10300 411	10500 419	10700 427	10900 435	11100 443	11300 451	11500 459	11700 467	11900 475	12100 483	12300 491	12500 499	12700 507	12900 515	13100 523	13300 531	13500 539	13700 547	13900 555	14100 563	14300 571	14500 579	14700 587	14900 595	15100 603	15300 611	15500 619	15700 627	15900 635	16100 643	16300 651	16500 659	16700 667	16900 675	17100 683	17300 691	17500 699	17700 707	17900 715	18100 723	18300 731	18500 739	18700 747	18900 755	19100 763	19300 771	19500 779	19700 787	19900 795	20100 803	20300 811	20500 819	20700 827	20900 835	21100 843	21300 851	21500 859	21700 867	21900 875	22100 883	22300 891	22500 899	22700 907	22900 915	23100 923	23300 931	23500 939	23700 947	23900 955	24100 963	24300 971	24500 979	24700 987	24900 995	25100 1003	25300 1011	25500 1019	25700 1027	25900 1035	26100 1043	26300 1051	26500 1059	26700 1067	26900 1075	27100 1083	27300 1091	27500 1099	27700 1107	27900 1115	28100 1123	28300 1131	28500 1139	28700 1147	28900 1155	29100 1163	29300 1171	29500 1179	29700 1187	29900 1195	30100 1203	30300 1211	30500 1219	30700 1227	30900 1235	31100 1243	31300 1251	31500 1259	31700 1267	31900 1275	32100 1283	32300 1291	32500 1299	32700 1307	32900 1315	33100 1323	33300 1331	33500 1339	33700 1347	33900 1355	34100 1363	34300 1371	34500 1379	34700 1387	34900 1395	35100 1403	35300 1411	35500 1419	35700 1427	35900 1435	36100 1443	36300 1451	36500 1459	36700 1467	36900 1475	37100 1483	37300 1491	37500 1499	37700 1507	37900 1515	38100 1523	38300 1531	38500 1539	38700 1547	38900 1555	39100 1563	39300 1571	39500 1579	39700 1587	39900 1595	40100 1603	40300 1611	40500 1619	40700 1627	40900 1635	41100 1643	41300 1651	41500 1659	41700 1667	41900 1675	42100 1683	42300 1691	42500 1699	42700 1707	42900 1715	43100 1723	43300 1731	43500 1739	43700 1747	43900 1755	44100 1763	44300 1771	44500 1779	44700 1787	44900 1795	45100 1803	45300 1811	45500 1819	45700 1827	45900 1835	46100 1843	46300 1851	46500 1859	46700 1867	46900 1875	47100 1883	47300 1891	47500 1899	47700 1907	47900 1915	48100 1923	48300 1931	48500 1939	48700 1947	48900 1955	49100 1963	49300 1971	49500 1979	49700 1987	49900 1995	50100 2003	50300 2011	50500 2019	50700 2027	50900 2035	51100 2043	51300 2051	51500 2059	51700 2067	51900 2075	52100 2083	52300 2091	52500 2099	52700 2107	52900 2115	53100 2123	53300 2131	53500 2139	53700 2147	53900 2155	54100 2163	54300 2171	54500 2179	54700 2187	54900 2195	55100 2203	55300 2211	55500 2219	55700 2227	55900 2235	56100 2243	56300 2251	56500 2259	56700 2267	56900 2275	57100 2283	57300 2291	57500 2299	57700 2307	57900 2315	58100 2323	58300 2331	58500 2339	58700 2347	58900 2355	59100 2363	59300 2371	59500 2379	59700 2387	59900 2395	60100 2403	60300 2411	60500 2419	60700 2427	60900 2435	61100 2443	61300 2451	61500 2459	61700 2467	61900 2475	62100 2483	62300 2491	62500 2499	62700 2507	62900 2515	63100 2523	63300 2531	63500 2539	63700 2547	63900 2555	64100 2563	64300 2571	64500 2579	64700 2587	64900 2595	65100 2603	65300 2611	65500 2619	65700 2627	65900 2635	66100 2643	66300 2651	66500 2659	66700 2667	66900 2675	67100 2683	67300 2691	67500 2699	67700 2707	67900 2715	68100 2723	68300 2731	68500 2739	68700 2747	68900 2755	69100 2763	69300 2771	69500 2779	69700 2787	69900 2795	70100 2803	70300 2811	70500 2819	70700 2827	70900 2835	71100 2843	71300 2851	71500 2859	71700 2867	71900 2875	72100 2883	72300 2891	72500 2899	72700 2907	72900 2915	73100 2923	73300 2931	73500 2939	73700 2947	73900 2955	74100 2963	74300 2971	74500 2979	74700 2987	74900 2995	75100 3003	75300 3011	75500 3019	75700 3027	75900 3035	76100 3043	76300 3051	76500 3059	76700 3067	76900 3075	77100 3083	77300 3091	77500 3099	77700 3107	77900 3115	78100 3123	78300 3131	78500 3139	78700 3147	78900 3155	79100 3163	79300 3171	79500 3179	79700 3187	79900 3195	80100 3203	80300 3211	80500 3219	80700 3227	80900 3235	81100 3243	81300 3251	81500 3259	81700 3267	81900 3275	82100 3283	82300 3291	82500 3299	82700 3307	82900 3315	83100 3323	83300 3331	83500 3339	83700 3347	83900 3355	84100 3363	84300 3371	84500 3379	84700 3387	84900 3395	85100 3403	85300 3411	85500 3419	85700 3427	85900 3435	86100 3443	86300 3451	86500 3459	86700 3467	86900 3475	87100 3483	87300 3491	87500 3499	87700 3507	87900 3515	88100 3523	88300 3531	88500 3539	88700 3547	88900 3555	89100 3563	89300 3571	89500 3579	89700 3587	89900 3595	90100 3603	90300 3611	90500 3619	90700 3627	90900 3635	91100 3643	91300 3651	91500 3659	91700 3667	91900 3675	92100 3683	92300 3691	92500 3699	92700 3707	92900 3715	93100 3723	93300 3731	93500 3739	93700 3747	93900 3755	94100 3763	94300 3771	94500 3779	94700 3787	94900 3795	95100 3803	95300 3811	95500 3819	95700 3827	95900 3835	96100 3843	96300 3851	96500 3859	96700 3867	96900 3875	97100 3883	97300 3891	97500 3899	97700 3907	97900 3915	98100 3923	98300 3931	98500 3939	98700 3947	98900 3955	99100 3963	99300 3971	99500 3979	99700 3987	99900 3995	100100 4003	100300 4011	100500 4019	100700 4027	100900 4035	101100 4043	101300 4051	101500 4059	101700 4067	101900 4075	102100 4083	102300 4091	102500 4099	102700 4107	102900 4115	103100 4123	103300 4131	103500 4139	103700 4147	103900 4155	104100 4163	104300 4171	104500 4179	104700 4187	104900 4195	105100 4203	105300 4211	105500 4219	105700 4227	105900 4235	106100 4243	106300 4251	106500 4259	106700 4267	106900 4275	107100 4283	107300 4291	107500 4299	107700 4307	107900 4315	108100 4323	108300 4331	108500 4339	108700 4347	108900 4355	109100 4363	109300 4371	109500 4379	109700 4387	109900 4395	110100 4403	110300 4411	110500 4419	110700 4427	110900 4435	111100 4443	111300 4451	111500 4459	111700 4467	111900 4475	112100 4483	112300 4491	112500 4499	112700 4507	112900 4515	113100 4523	113300 4531	113500 4539	113700 4547	113900 4555	114100 4563	114300 4571	114500 4579	114700 4587	114900 4595	115100 4603	115300 4611	115500 4619	115700 4627	115900 4635	116100 4643	116300 4651	116500 4659	116700 4667	116900 4675	117100 4683	117300 4691	117500 4699	117700 4707	117900 4715	118100 4723	118300 4731	118500 4739	118700 4747	118900 4755	119100 4763	119300 4771	119500 4779	119700 4787	119900 4795	120100 4803	120300 4811	120500 4819	120700 4827	120900 4835	121100 4843	121300 4851	121500 4859	121700 4867	121900 4875	122100 4883	122300 4891	122500 4899	122700 4907	122900 4915	123100 4923	123300 4931	123500 4939	123700 4947	123900 4955	124100 4963	124300 4971	124500 4979	124700 4987	124900 4995	125100 5003	125300 5011	125500 5019	125700 5027	125900 5035	126100 5043	126300 5051	126500 5059	126700 5067	126900 5075	127100 5083	127300 5091	127500 5099	127700 5107	127900 5115	128100 5123	128300 5131	128500 5139	128700 5147	128900 5155	129100 5163	129300 5171	129500 5179	129700 5187	129900 5195	130100 5203	130300 5211	130500 5219	130700 5227	130900 5235	131100 5243	131300 5251	131500 5259	131700 5267	131900 5275	132100 5283	132300 5291	132500 5299	132700 5307	132900 5315	133100 5323	133300 5331	133500 5339	133700 5347	133900 5355	134100 5363	134300 5371	134500 5379	134700 5387	134900 5395	135100 5403	135300 5411	135500 5419	135700 5427	135900 5435	136100 5443	136300 5451	136500 5459	136700 5467	136900 5475	137100 5483	137300 5491	137500 5499	137700 5507	137900 5515	138100 5523	138300 5531	138500 5539	138700 5547	138900 5555	139100 5563	139300 5571	139500 5579	139700 5587	139900 5595	140100 5603	140300 5611	140500 5619	140700 5627	140900 5635	141100 5643	141300 5651	141500 5659	141700 5667	141900 5675	142100 5683	142300 5691	142500 5699	142700 5707	142900 5715	143100 5723	143300 5731	143500 5739	143700 5747	143900 5755	144100 5763	144300 5771	144500 5779	144700 5787	144900 5795	145100 5803	145300 5811	145500 5819	145700 5827	145900 5835	146100 5843	146300 5851	146500 5859	146700 5867	146900 5875	147100 5883	147300 5891	147500 5899	147700 5907	147900 5915	148100 5923	148300 5931	148500 5939	148700 5947	148900 5955	149100 5963	149300 5971	149500 5979	149700 5987	149900 5995	150100 6003	150300 6011	150500 6019	150700 6027	150900 6035	151100 6043	151300 6051	151500 6059	151700 6067
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320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

[illegible]

Illustration 331

q06225569

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 600 mm (24 inch) SG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) DG shoe

(mm) (inch)	1500 60	2000 120	2500 100	3000 240	3500 100	4000 360	4500 420	5000 480	5500 640		(mm) (inch)
12000 480											1200 480
10000 400											1000 400
8000 320											8000 320
7000 280											7000 280
6000 240											6000 240
5000 200											5000 200
4000 160											4000 160
3000 120											3000 120
2000 80											2000 80
1000 40											1000 40
0											0
-1000 -40											-1000 -40
-2000 -80											-2000 -80
-3000 -120											-3000 -120
-4000 -160											-4000 -160
-5000 -200											-5000 -200
-6000 -240											-6000 -240
-7000 -280											-7000 -280
-8000 -320											-8000 -320

Illustration 332

g06225573

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift OFF

Product Information Section
320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

Stick (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360	10500 420	12000 480	13500 540		(mm) (inch)
12000 480											1200 1200 10360 3000 3000 400
10000 420							2300 2300 4700 4700				1200 1200 10000 2800 2800 400
9000 360							2400 2400 5400 5400	3000 3000			1100 1100 10000 2400 2400 500
7500 300							2500 2500 6000 6000	2000 2000			1100 1100 10410 2400 2400 500
6000 240							2700 2700 6200 6200	3000 3000	1800 1800	1400 1400	1100 1100 10370 2400 2400 500
4500 180						2100 2100 8000 8000	2000 2000 6300 6300	2700 2700 6800 6800	1800 1800 3000 3000	1400 1400 3000 3000	1100 1100 10410 2400 2400 500
3000 120		4700 4700 11800 11800	6700 6700 16800 16800	8700 8700 18700 18700	10700 10700 20700 20700	12700 12700 22700 22700	14700 14700 24700 24700	16700 16700 26700 26700	18700 18700 28700 28700	20700 20700 30700 30700	1100 1100 10400 2400 2400 500
1500 60			6700 6700 16800 16800	8700 8700 18700 18700	10700 10700 20700 20700	12700 12700 22700 22700	14700 14700 24700 24700	16700 16700 26700 26700	18700 18700 28700 28700	20700 20700 30700 30700	1100 1100 10400 2400 2400 500
0		2000 2000 4000 4000	4000 4000 8000 8000	6000 6000 12000 12000	8000 8000 16000 16000	10000 10000 20000 20000	12000 12000 24000 24000	14000 14000 28000 28000	16000 16000 32000 32000	18000 18000 36000 36000	1100 1100 10400 2400 2400 500
-1500 -60	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500
-3000 -120	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500
-4500 -180	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500
-6000 -240	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500
-7500 -300	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500
-9000 -360	2100 2100 4000 4000	2100 2100 4000 4000	2700 2700 5100 5100	2700 2700 5100 5100	4800 4800 9600 9600	4800 4800 9600 9600	6900 6900 13800 13800	6900 6900 13800 13800	9000 9000 18000 18000	9000 9000 18000 18000	1100 1100 10400 2400 2400 500

Illustration 333

g06225577

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 600 mm (24 inch) DG track shoes, heavy lift ON

Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

600 mm (24 inch) TG shoe

mm (inch)	1100 40	2000 120	4000 160	6000 240	7500 300	9000 360	11000 430	12000 480	12500 540		mm (inch)
12000 480											1200 480
11500 450							2200 870	2300 910			1300 510
10000 390							2200 870	2300 910	2400 950	1400 550	1100 430
7500 300							2200 870	2300 910	2400 950	1400 550	1100 430
5000 190							2200 870	2300 910	2400 950	1400 550	1100 430
2400 90							2200 870	2300 910	2400 950	1400 550	1100 430
1800 70							2200 870	2300 910	2400 950	1400 550	1100 430
1200 40							2200 870	2300 910	2400 950	1400 550	1100 430
600 20							2200 870	2300 910	2400 950	1400 550	1100 430
0 0							2200 870	2300 910	2400 950	1400 550	1100 430
-1500 -60							2200 870	2300 910	2400 950	1400 550	1100 430
-3000 -120							2200 870	2300 910	2400 950	1400 550	1100 430
-4500 -180							2200 870	2300 910	2400 950	1400 550	1100 430
-6000 -240							2200 870	2300 910	2400 950	1400 550	1100 430
-7500 -300							2200 870	2300 910	2400 950	1400 550	1100 430
-9000 -360							2200 870	2300 910	2400 950	1400 550	1100 430
-10500 -420							2200 870	2300 910	2400 950	1400 550	1100 430
-12000 -480							2200 870	2300 910	2400 950	1400 550	1100 430
-13500 -540							2200 870	2300 910	2400 950	1400 550	1100 430
-15000 -600							2200 870	2300 910	2400 950	1400 550	1100 430
-16500 -660							2200 870	2300 910	2400 950	1400 550	1100 430
-18000 -720							2200 870	2300 910	2400 950	1400 550	1100 430
-19500 -780							2200 870	2300 910	2400 950	1400 550	1100 430
-21000 -840							2200 870	2300 910	2400 950	1400 550	1100 430
-22500 -900							2200 870	2300 910	2400 950	1400 550	1100 430
-24000 -960							2200 870	2300 910	2400 950	1400 550	1100 430
-25500 -1020							2200 870	2300 910	2400 950	1400 550	1100 430
-27000 -1080							2200 870	2300 910	2400 950	1400 550	1100 430
-28500 -1140							2200 870	2300 910	2400 950	1400 550	1100 430
-30000 -1200							2200 870	2300 910	2400 950	1400 550	1100 430
-31500 -1260							2200 870	2300 910	2400 950	1400 550	1100 430
-33000 -1320							2200 870	2300 910	2400 950	1400 550	1100 430
-34500 -1380							2200 870	2300 910	2400 950	1400 550	1100 430
-36000 -1440							2200 870	2300 910	2400 950	1400 550	1100 430
-37500 -1500							2200 870	2300 910	2400 950	1400 550	1100 430
-39000 -1560							2200 870	2300 910	2400 950	1400 550	1100 430
-40500 -1620							2200 870	2300 910	2400 950	1400 550	1100 430
-42000 -1680							2200 870	2300 910	2400 950	1400 550	1100 430
-43500 -1740							2200 870	2300 910	2400 950	1400 550	1100 430
-45000 -1800							2200 870	2300 910	2400 950	1400 550	1100 430
-46500 -1860							2200 870	2300 910	2400 950	1400 550	1100 430
-48000 -1920							2200 870	2300 910	2400 950	1400 550	1100 430
-49500 -1980							2200 870	2300 910	2400 950	1400 550	1100 430
-51000 -2040							2200 870	2300 910	2400 950	1400 550	1100 430
-52500 -2100							2200 870	2300 910	2400 950	1400 550	1100 430
-54000 -2160							2200 870	2300 910	2400 950	1400 550	1100 430
-55500 -2220							2200 870	2300 910	2400 950	1400 550	1100 430
-57000 -2280							2200 870	2300 910	2400 950	1400 550	1100 430
-58500 -2340							2200 870	2300 910	2400 950	1400 550	1100 430
-60000 -2400							2200 870	2300 910	2400 950	1400 550	1100 430
-61500 -2460							2200 870	2300 910	2400 950	1400 550	1100 430
-63000 -2520							2200 870	2300 910	2400 950	1400 550	1100 430
-64500 -2580							2200 870	2300 910	2400 950	1400 550	1100 430
-66000 -2640							2200 870	2300 910	2400 950	1400 550	1100 430
-67500 -2700							2200 870	2300 910	2400 950	1400 550	1100 430
-69000 -2760							2200 870	2300 910	2400 950	1400 550	1100 430
-70500 -2820							2200 870	2300 910	2400 950	1400 550	1100 430
-72000 -2880							2200 870	2300 910	2400 950	1400 550	1100 430
-73500 -2940							2200 870	2300 910	2400 950	1400 550	1100 430
-75000 -3000							2200 870	2300 910	2400 950	1400 550	1100 430
-76500 -3060							2200 870	2300 910	2400 950	1400 550	1100 430
-78000 -3120							2200 870	2300 910	2400 950	1400 550	1100 430
-79500 -3180							2200 870	2300 910	2400 950	1400 550	1100 430
-81000 -3240							2200 870	2300 910	2400 950	1400 550	1100 430
-82500 -3300							2200 870	2300 910	2400 950	1400 550	1100 430
-84000 -3360							2200 870	2300 910	2400 950	1400 550	1100 430
-85500 -3420							2200 870	2300 910	2400 950	1400 550	1100 430
-87000 -3480							2200 870	2300 910	2400 950	1400 550	1100 430
-88500 -3540							2200 870	2300 910	2400 950	1400 550	1100 430
-90000 -3600							2200 870	2300 910	2400 950	1400 550	1100 430
-91500 -3660							2200 870	2300 910	2400 950	1400 550	1100 430
-93000 -3720							2200 870	2300 910	2400 950	1400 550	1100 430
-94500 -3780							2200 870	2300 910	2400 950	1400 550	1100 430
-96000 -3840							2200 870	2300 910	2400 950	1400 550	1100 430
-97500 -3900							2200 870	2300 910	2400 950	1400 550	1100 430
-99000 -3960							2200 870	2300 910	2400 950	1400 550	1100 430
-100500 -4020							2200 870	2300 910	2400 950	1400 550	1100 430
-102000 -4080							2200 870	2300 910	2400 950	1400 550	1100 430
-103500 -4140							2200 870	2300 910	2400 950	1400 550	1100 430
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-108000 -4320							2200 870	2300 910	2400 950	1400 550	1100 430
-109500 -4380							2200 870	2300 910	2400 950	1400 550	1100 430
-111000 -4440							2200 870	2300 910	2400 950	1400 550	1100 430
-112500 -4500							2200 870	2300 910	2400 950	1400 550	1100 430
-114000 -4560							2200 870	2300 910	2400 950	1400 550	1100 430
-115500 -4620							2200 870	2300 910	2400 950	1400 550	1100 430
-117000 -4680							2200 870	2300 910	2400 950	1400 550	1100 430
-118500 -4740							2200 870	2300 910	2400 950	1400 550	1100 430
-120000 -4800							2200 870	2300 910	2400 950	1400 550	1100 430
-121500 -4860							2200 870	2300 910	2400 950	1400 550	1100 430
-123000 -4920							2200 870	2300 910	2400 950	1400 550	1100 430
-124500 -4980							2200 870	2300 910	2400 950	1400 550	1100 430
-126000 -5040							2200 870	2300 910	2400 950	1400 550	1100 430
-127500 -5100							2200 870	2300 910	2400 950	1400 550	1100 430
-129000 -5160							2200 870	2300 910	2400 950	1400 550	1100 430
-130500 -5220							2200 870	2300 910	2400 950	1400 550	1100 430
-132000 -5280							2200 870	2300 910	2400 950	1400 550	1100 430
-133500 -5340							2200 870	2300 910	2400 950	1400 550	1100 430
-135000 -5400							2200 870	2300 910	2400 950	1400 550	1100 430
-136500 -5460							2200 870	2300 910	2400 950	1400 550	1100 430
-138000 -5520							2200 870	2300 910	2400 950	1400 550	1100 430
-139500 -5580							2200 870	2300 910	2400 950	1400 550	1100 430
-141000											

Product Information Section
320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

(mm) (inch)	1100 43	1200 47	1300 51	1400 55	1500 59	1600 63	1700 67	1800 71	1900 75	2000 79	2100 83	2200 87	2300 91	2400 95	2500 99	2600 103	2700 107	2800 111	2900 115	3000 119	3100 123	3200 127	3300 131	3400 135	3500 139	3600 143	3700 147	3800 151	3900 155	4000 159	4100 163	4200 167	4300 171	4400 175	4500 179	4600 183	4700 187	4800 191	4900 195	5000 199	5100 203	5200 207	5300 211	5400 215	5500 219	5600 223	5700 227	5800 231	5900 235	6000 239	6100 243	6200 247	6300 251	6400 255	6500 259	6600 263	6700 267	6800 271	6900 275	7000 279	7100 283	7200 287	7300 291	7400 295	7500 299	7600 303	7700 307	7800 311	7900 315	8000 319	8100 323	8200 327	8300 331	8400 335	8500 339	8600 343	8700 347	8800 351	8900 355	9000 359	9100 363	9200 367	9300 371	9400 375	9500 379	9600 383	9700 387	9800 391	9900 395	10000 399	10100 403	10200 407	10300 411	10400 415	10500 419	10600 423	10700 427	10800 431	10900 435	11000 439	11100 443	11200 447	11300 451	11400 455	11500 459	11600 463	11700 467	11800 471	11900 475	12000 479	12100 483	12200 487	12300 491	12400 495	12500 499	12600 503	12700 507	12800 511	12900 515	13000 519	13100 523	13200 527	13300 531	13400 535	13500 539	13600 543	13700 547	13800 551	13900 555	14000 559	14100 563	14200 567	14300 571	14400 575	14500 579	14600 583	14700 587	14800 591	14900 595	15000 599	15100 603	15200 607	15300 611	15400 615	15500 619	15600 623	15700 627	15800 631	15900 635	16000 639	16100 643	16200 647	16300 651	16400 655	16500 659	16600 663	16700 667	16800 671	16900 675	17000 679	17100 683	17200 687	17300 691	17400 695	17500 699	17600 703	17700 707	17800 711	17900 715	18000 719	18100 723	18200 727	18300 731	18400 735	18500 739	18600 743	18700 747	18800 751	18900 755	19000 759	19100 763	19200 767	19300 771	19400 775	19500 779	19600 783	19700 787	19800 791	19900 795	20000 799	20100 803	20200 807	20300 811	20400 815	20500 819	20600 823	20700 827	20800 831	20900 835	21000 839	21100 843	21200 847	21300 851	21400 855	21500 859	21600 863	21700 867	21800 871	21900 875	22000 879	22100 883	22200 887	22300 891	22400 895	22500 899	22600 903	22700 907	22800 911	22900 915	23000 919	23100 923	23200 927	23300 931	23400 935	23500 939	23600 943	23700 947	23800 951	23900 955	24000 959	24100 963	24200 967	24300 971	24400 975	24500 979	24600 983	24700 987	24800 991	24900 995	25000 999	25100 1003	25200 1007	25300 1011	25400 1015	25500 1019	25600 1023	25700 1027	25800 1031	25900 1035	26000 1039	26100 1043	26200 1047	26300 1051	26400 1055	26500 1059	26600 1063	26700 1067	26800 1071	26900 1075	27000 1079	27100 1083	27200 1087	27300 1091	27400 1095	27500 1099	27600 1103	27700 1107	27800 1111	27900 1115	28000 1119	28100 1123	28200 1127	28300 1131	28400 1135	28500 1139	28600 1143	28700 1147	28800 1151	28900 1155	29000 1159	29100 1163	29200 1167	29300 1171	29400 1175	29500 1179	29600 1183	29700 1187	29800 1191	29900 1195	30000 1199	30100 1203	30200 1207	30300 1211	30400 1215	30500 1219	30600 1223	30700 1227	30800 1231	30900 1235	31000 1239	31100 1243	31200 1247	31300 1251	31400 1255	31500 1259	31600 1263	31700 1267	31800 1271	31900 1275	32000 1279	32100 1283	32200 1287	32300 1291	32400 1295	32500 1299	32600 1303	32700 1307	32800 1311	32900 1315	33000 1319	33100 1323	33200 1327	33300 1331	33400 1335	33500 1339	33600 1343	33700 1347	33800 1351	33900 1355	34000 1359	34100 1363	34200 1367	34300 1371	34400 1375	34500 1379	34600 1383	34700 1387	34800 1391	34900 1395	35000 1399	35100 1403	35200 1407	35300 1411	35400 1415	35500 1419	35600 1423	35700 1427	35800 1431	35900 1435	36000 1439	36100 1443	36200 1447	36300 1451	36400 1455	36500 1459	36600 1463	36700 1467	36800 1471	36900 1475	37000 1479	37100 1483	37200 1487	37300 1491	37400 1495	37500 1499	37600 1503	37700 1507	37800 1511	37900 1515	38000 1519	38100 1523	38200 1527	38300 1531	38400 1535	38500 1539	38600 1543	38700 1547	38800 1551	38900 1555	39000 1559	39100 1563	39200 1567	39300 1571	39400 1575	39500 1579	39600 1583	39700 1587	39800 1591	39900 1595	40000 1599	40100 1603	40200 1607	40300 1611	40400 1615	40500 1619	40600 1623	40700 1627	40800 1631	40900 1635	41000 1639	41100 1643	41200 1647	41300 1651	41400 1655	41500 1659	41600 1663	41700 1667	41800 1671	41900 1675	42000 1679	42100 1683	42200 1687	42300 1691	42400 1695	42500 1699	42600 1703	42700 1707	42800 1711	42900 1715	43000 1719	43100 1723	43200 1727	43300 1731	43400 1735	43500 1739	43600 1743	43700 1747	43800 1751	43900 1755	44000 1759	44100 1763	44200 1767	44300 1771	44400 1775	44500 1779	44600 1783	44700 1787	44800 1791	44900 1795	45000 1799	45100 1803	45200 1807	45300 1811	45400 1815	45500 1819	45600 1823	45700 1827	45800 1831	45900 1835	46000 1839	46100 1843	46200 1847	46300 1851	46400 1855	46500 1859	46600 1863	46700 1867	46800 1871	46900 1875	47000 1879	47100 1883	47200 1887	47300 1891	47400 1895	47500 1899	47600 1903	47700 1907	47800 1911	47900 1915	48000 1919	48100 1923	48200 1927	48300 1931	48400 1935	48500 1939	48600 1943	48700 1947	48800 1951	48900 1955	49000 1959	49100 1963	49200 1967	49300 1971	49400 1975	49500 1979	49600 1983	49700 1987	49800 1991	49900 1995	50000 1999	50100 2003	50200 2007	50300 2011	50400 2015	50500 2019	50600 2023	50700 2027	50800 2031	50900 2035	51000 2039	51100 2043	51200 2047	51300 2051	51400 2055	51500 2059	51600 2063	51700 2067	51800 2071	51900 2075	52000 2079	52100 2083	52200 2087	52300 2091	52400 2095	52500 2099	52600 2103	52700 2107	52800 2111	52900 2115	53000 2119	53100 2123	53200 2127	53300 2131	53400 2135	53500 2139	53600 2143	53700 2147	53800 2151	53900 2155	54000 2159	54100 2163	54200 2167	54300 2171	54400 2175	54500 2179	54600 2183	54700 2187	54800 2191	54900 2195	55000 2199	55100 2203	55200 2207	55300 2211	55400 2215	55500 2219	55600 2223	55700 2227	55800 2231	55900 2235	56000 2239	56100 2243	56200 2247	56300 2251	56400 2255	56500 2259	56600 2263	56700 2267	56800 2271	56900 2275	57000 2279	57100 2283	57200 2287	57300 2291	57400 2295	57500 2299	57600 2303	57700 2307	57800 2311	57900 2315	58000 2319	58100 2323	58200 2327	58300 2331	58400 2335	58500 2339	58600 2343	58700 2347	58800 2351	58900 2355	59000 2359	59100 2363	59200 2367	59300 2371	59400 2375	59500 2379	59600 2383	59700 2387	59800 2391	59900 2395	60000 2399	60100 2403	60200 2407	60300 2411	60400 2415	60500 2419	60600 2423	60700 2427	60800 2431	60900 2435	61000 2439	61100 2443	61200 2447	61300 2451	61400 2455	61500 2459	61600 2463	61700 2467	61800 2471	61900 2475	62000 2479	62100 2483	62200 2487	62300 2491	62400 2495	62500 2499	62600 2503	62700 2507	62800 2511	62900 2515	63000 2519	63100 2523	63200 2527	63300 2531	63400 2535	63500 2539	63600 2543	63700 2547	63800 2551	63900 2555	64000 2559	64100 2563	64200 2567	64300 2571	64400 2575	64500 2579	64600 2583	64700 2587	64800 2591	64900 2595	65000 2599	65100 2603	65200 2607	65300 2611	65400 2615	65500 2619	65600 2623	65700 2627	65800 2631	65900 2635	66000 2639	66100 2643	66200 2647	66300 2651	66400 2655	66500 2659	66600 2663	66700 2667	66800 2671	66900 2675	67000 2679	67100 2683	67200 2687	67300 2691	67400 2695	67500 2699	67600 2703	67700 2707	67800 2711	67900 2715	68000 2719	68100 2723	68200 2727	68300 2731	68400 2735	68500 2739	68600 2743	68700 2747	68800 2751	68900 2755	69000 2759	69100 2763	69200 2767	69300 2771	69400 2775	69500 2779	69600 2783	69700 2787	69800 2791	69900 2795	70000 2799	70100 2803	70200 2807	70300 2811	70400 2815	70500 2819	70600 2823	70700 2827	70800 2831	70900 2835	71000 2839	71100 2843	71200 2847	71300 2851	71400 2855	71500 2859	71600 2863	71700 2867	71800 2871	71900 2875	72000 2879	72100 2883	72200 2887	72300 2891	72400 2895	72500 2899	72600 2903	72700 2907	72800 2911	72900 2915	73000 2919	73100 2923	73200 2927	73300 2931	73400 2935	73500 2939	73600 2943	73700 2947	73800 2951	73900 2955	74000 2959	74100 2963	74200 2967	74300 2971	74400 2975	74500 2979	74600 2983	74700 2987	74800 2991	74900 2995	75000 2999	75100 3003	75200 3007	75300 3011	75400 3015	75500 3019	75600 3023	75700 3027	75800 3031	75900 3035	76000 3039	76100 3043	76200 3047	76300 3051	76400 3055	76500 3059	76600 3063	76700 3067	76800 3071	76900 3075	77000 3079	77100 3083	77200 3087	77300 3091	77400 3095	77500 3099	77600 3103	77700 3107	77800 3111	77900 3115	78000 3119	78100 3123	78200 3127	78300 3131	78400 3135	78500 3139	78600 3143	78700 3147	78800 3151	78900 3155	79000 3159	79100 3163	79200 3167	79300 3171	79400 3175	79500 31
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Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

700 mm (28 inch) TG shoe

3000 (100")	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360	10500 420	12000 480	13500 540		
											(mm) (inch)
11000 430											" 1280 " 1280 13080 " 2800 " 2800 420
11800 420							" 3200 " 3200 " 4700 " 4700				" 1300 " 1300 11880 " 2850 " 2850 480
10100 380							" 3200 " 3200 " 3300 " 3300 " 4800 " 4800 " 5050 " 5050	" 1900 " 1900 " 3800 " 3800			" 1150 " 1150 12000 " 2450 " 2450 530
11500 380							" 3200 " 3200 " 3250 " 3250 " 4800 " 4800 " 4900 " 4900	" 1900 " 1900 " 3800 " 3800			" 1150 " 1150 13410 " 2450 " 2450 630
10000 340							" 2400 " 2400 " 2500 " 2500 " 3200 " 3200 " 3300 " 3300	" 1900 " 1900 " 4000 " 4000	" 1800 " 1800 " 3050 " 3050	" 1450 " 1450 " 3000 " 3000	" 1150 " 1150 13070 " 2400 " 2400 580
10500 380						" 2800 " 2800 " 2900 " 2900 " 4000 " 4000 " 4100 " 4100	" 2600 " 2600 " 2700 " 2700 " 3600 " 3600 " 3700 " 3700	" 1900 " 1900 " 4000 " 4000	" 2300 " 2300 " 3000 " 3000	" 1400 " 1400 " 3000 " 3000	" 1150 " 1150 14040 " 2400 " 2400 570
10000 320		" 4750 " 4750 " 5050 " 5050 " 11800 " 11800	" 5050 " 5050 " 5350 " 5350 " 12800 " 12800	" 5350 " 5350 " 5650 " 5650 " 13800 " 13800	" 5650 " 5650 " 5950 " 5950 " 14800 " 14800	" 5950 " 5950 " 6250 " 6250 " 15800 " 15800	" 6250 " 6250 " 6550 " 6550 " 16800 " 16800	" 6550 " 6550 " 6850 " 6850 " 17800 " 17800	" 6850 " 6850 " 7150 " 7150 " 18800 " 18800	" 7150 " 7150 " 7450 " 7450 " 19800 " 19800	" 1150 " 1150 14050 " 2900 " 2900 580
11000 400		" 4750 " 4750 " 5050 " 5050 " 11800 " 11800	" 5050 " 5050 " 5350 " 5350 " 12800 " 12800	" 5350 " 5350 " 5650 " 5650 " 13800 " 13800	" 5650 " 5650 " 5950 " 5950 " 14800 " 14800	" 5950 " 5950 " 6250 " 6250 " 15800 " 15800	" 6250 " 6250 " 6550 " 6550 " 16800 " 16800	" 6550 " 6550 " 6850 " 6850 " 17800 " 17800	" 6850 " 6850 " 7150 " 7150 " 18800 " 18800	" 7150 " 7150 " 7450 " 7450 " 19800 " 19800	" 1150 " 1150 14880 " 2900 " 2900 680
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-1500 -60	" 2100 " 2100 " 4100 " 4100 " 4300 " 4300	" 2100 " 2100 " 4100 " 4100 " 4300 " 4300	" 4100 " 4100 " 6100 " 6100 " 6300 " 6300	" 4100 " 4100 " 6100 " 6100 " 6300 " 6300	" 6100 " 6100 " 8100 " 8100 " 8300 " 8300	" 6100 " 6100 " 8100 " 8100 " 8300 " 8300	" 8100 " 8100 " 10100 " 10100 " 10300 " 10300	" 8100 " 8100 " 10100 " 10100 " 10300 " 10300	" 10100 " 10100 " 12100 " 12100 " 12300 " 12300	" 10100 " 10100 " 12100 " 12100 " 12300 " 12300	" 1350 " 1350 14030 " 2000 " 2000 580
-3000 -120	" 3050 " 3050 " 5050 " 5050 " 5250 " 5250	" 3050 " 3050 " 5050 " 5050 " 5250 " 5250	" 5050 " 5050 " 7050 " 7050 " 7250 " 7250	" 5050 " 5050 " 7050 " 7050 " 7250 " 7250	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 1500 " 1500 12790 " 2500 " 2500 550
-4500 -180	" 4050 " 4050 " 6050 " 6050 " 6250 " 6250	" 4050 " 4050 " 6050 " 6050 " 6250 " 6250	" 6050 " 6050 " 8050 " 8050 " 8250 " 8250	" 6050 " 6050 " 8050 " 8050 " 8250 " 8250	" 8050 " 8050 " 10050 " 10050 " 10250 " 10250	" 8050 " 8050 " 10050 " 10050 " 10250 " 10250	" 10050 " 10050 " 12050 " 12050 " 12250 " 12250	" 10050 " 10050 " 12050 " 12050 " 12250 " 12250	" 12050 " 12050 " 14050 " 14050 " 14250 " 14250	" 12050 " 12050 " 14050 " 14050 " 14250 " 14250	" 1700 " 1700 13170 " 2700 " 2700 530
-6000 -240	" 5050 " 5050 " 7050 " 7050 " 7250 " 7250	" 5050 " 5050 " 7050 " 7050 " 7250 " 7250	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 13050 " 13050 " 15050 " 15050 " 15250 " 15250	" 13050 " 13050 " 15050 " 15050 " 15250 " 15250	" 2000 " 2000 14030 " 3000 " 3000 480
-7500 -280	" 6050 " 6050 " 8050 " 8050 " 8250 " 8250	" 6050 " 6050 " 8050 " 8050 " 8250 " 8250	" 8050 " 8050 " 10050 " 10050 " 10250 " 10250	" 8050 " 8050 " 10050 " 10050 " 10250 " 10250	" 10050 " 10050 " 12050 " 12050 " 12250 " 12250	" 10050 " 10050 " 12050 " 12050 " 12250 " 12250	" 12050 " 12050 " 14050 " 14050 " 14250 " 14250	" 12050 " 12050 " 14050 " 14050 " 14250 " 14250	" 14050 " 14050 " 16050 " 16050 " 16250 " 16250	" 14050 " 14050 " 16050 " 16050 " 16250 " 16250	" 2500 " 2500 11040 " 3500 " 3500 440
-9000 -360	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 7050 " 7050 " 9050 " 9050 " 9250 " 9250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 9050 " 9050 " 11050 " 11050 " 11250 " 11250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 11050 " 11050 " 13050 " 13050 " 13250 " 13250	" 13050 " 13050 " 15050 " 15050 " 15250 " 15250	" 13050 " 13050 " 15050 " 15050 " 15250 " 15250	" 15050 " 15050 " 17050 " 17050 " 17250 " 17250	" 15050 " 15050 " 17050 " 17050 " 17250 " 17250	" 2700 " 2700 8800 " 3700 " 3700 580

Illustration 336

g06225587

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift OFF

Product Information Section
320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

Stick (inch)	1500 60	3000 120	4500 180	6000 240	7500 300	9000 360	10500 420	12000 480	13500 540		(mm)
10000											1200
400											1200
10000											1200
420											1200
9000											1200
360											1200
7000											1200
360											1200
6000											1200
240											1200
4500											1200
180											1200
9000											1200
120											1200
1000											1200
80											1200
0											1200
0											1200
-1000											1200
-40											1200
-9000											1200
-120											1200
-1000											1200
-180											1200
-4000											1200
-240											1200
-7000											1200
-200											1200
-6000											1200
-360											1200

Illustration 337

g06225589

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 700 mm (28 inch) TG track shoes, heavy lift ON

Model (inch)	1500 65	2000 125	4000 185	6000 245	7500 300	9000 360	13500 420	15000 480	17000 540	19000 600	21000 660	23000 720	25000 780	27000 840	29000 900	31000 960	33000 1020	35000 1080	37000 1140	39000 1200	41000 1260	43000 1320	45000 1380	47000 1440	49000 1500	51000 1560	53000 1620	55000 1680	57000 1740	59000 1800	61000 1860	63000 1920	65000 1980	67000 2040	69000 2100	71000 2160	73000 2220	75000 2280	77000 2340	79000 2400	81000 2460	83000 2520	85000 2580	87000 2640	89000 2700	91000 2760	93000 2820	95000 2880	97000 2940	99000 3000	101000 3060	103000 3120	105000 3180	107000 3240	109000 3300	111000 3360	113000 3420	115000 3480	117000 3540	119000 3600	121000 3660	123000 3720	125000 3780	127000 3840	129000 3900	131000 3960	133000 4020	135000 4080	137000 4140	139000 4200	141000 4260	143000 4320	145000 4380	147000 4440	149000 4500	151000 4560	153000 4620	155000 4680	157000 4740	159000 4800	161000 4860	163000 4920	165000 4980	167000 5040	169000 5100	171000 5160	173000 5220	175000 5280	177000 5340	179000 5400	181000 5460	183000 5520	185000 5580	187000 5640	189000 5700	191000 5760	193000 5820	195000 5880	197000 5940	199000 6000	201000 6060	203000 6120	205000 6180	207000 6240	209000 6300	211000 6360	213000 6420	215000 6480	217000 6540	219000 6600	221000 6660	223000 6720	225000 6780	227000 6840	229000 6900	231000 6960	233000 7020	235000 7080	237000 7140	239000 7200	241000 7260	243000 7320	245000 7380	247000 7440	249000 7500	251000 7560	253000 7620	255000 7680	257000 7740	259000 7800	261000 7860	263000 7920	265000 7980	267000 8040	269000 8100	271000 8160	273000 8220	275000 8280	277000 8340	279000 8400	281000 8460	283000 8520	285000 8580	287000 8640	289000 8700	291000 8760	293000 8820	295000 8880	297000 8940	299000 9000	301000 9060	303000 9120	305000 9180	307000 9240	309000 9300	311000 9360	313000 9420	315000 9480	317000 9540	319000 9600	321000 9660	323000 9720	325000 9780	327000 9840	329000 9900	331000 9960	333000 10020	335000 10080	337000 10140	339000 10200	341000 10260	343000 10320	345000 10380	347000 10440	349000 10500	351000 10560	353000 10620	355000 10680	357000 10740	359000 10800	361000 10860	363000 10920	365000 10980	367000 11040	369000 11100	371000 11160	373000 11220	375000 11280	377000 11340	379000 11400	381000 11460	383000 11520	385000 11580	387000 11640	389000 11700	391000 11760	393000 11820	395000 11880	397000 11940	399000 12000	401000 12060	403000 12120	405000 12180	407000 12240	409000 12300	411000 12360	413000 12420	415000 12480	417000 12540	419000 12600	421000 12660	423000 12720	425000 12780	427000 12840	429000 12900	431000 12960	433000 13020	435000 13080	437000 13140	439000 13200	441000 13260	443000 13320	445000 13380	447000 13440	449000 13500	451000 13560	453000 13620	455000 13680	457000 13740	459000 13800	461000 13860	463000 13920	465000 13980	467000 14040	469000 14100	471000 14160	473000 14220	475000 14280	477000 14340	479000 144
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g06225593

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 790 mm (31 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1100 43	1300 51	1500 59	1700 67	1900 75	2100 83	2300 91	2500 98	2700 106	2900 114	3100 122	3300 130	3500 138	3700 146	3900 154	4100 162	4300 170	4500 178	4700 186	4900 194	5100 202	5300 210	5500 218	5700 226	5900 234	6100 242	6300 250	6500 258	6700 266	6900 274	7100 282	7300 290	7500 298	7700 306	7900 314	8100 322	8300 330	8500 338	8700 346	8900 354	9100 362	9300 370	9500 378	9700 386	9900 394	10100 402	10300 410	10500 418	10700 426	10900 434	11100 442	11300 450	11500 458	11700 466	11900 474	12100 482	12300 490	12500 498	12700 506	12900 514	13100 522	13300 530	13500 538	13700 546	13900 554	14100 562	14300 570	14500 578	14700 586	14900 594	15100 602	15300 610	15500 618	15700 626	15900 634	16100 642	16300 650	16500 658	16700 666	16900 674	17100 682	17300 690	17500 698	17700 706	17900 714	18100 722	18300 730	18500 738	18700 746	18900 754	19100 762	19300 770	19500 778	19700 786	19900 794	20100 802	20300 810	20500 818	20700 826	20900 834	21100 842	21300 850	21500 858	21700 866	21900 874	22100 882	22300 890	22500 898	22700 906	22900 914	23100 922	23300 930	23500 938	23700 946	23900 954	24100 962	24300 970	24500 978	24700 986	24900 994	25100 1002	25300 1010	25500 1018	25700 1026	25900 1034	26100 1042	26300 1050	26500 1058	26700 1066	26900 1074	27100 1082	27300 1090	27500 1098	27700 1106	27900 1114	28100 1122	28300 1130	28500 1138	28700 1146	28900 1154	29100 1162	29300 1170	29500 1178	29700 1186	29900 1194	30100 1202	30300 1210	30500 1218	30700 1226	30900 1234	31100 1242	31300 1250	31500 1258	31700 1266	31900 1274	32100 1282	32300 1290	32500 1298	32700 1306	32900 1314	33100 1322	33300 1330	33500 1338	33700 1346	33900 1354	34100 1362	34300 1370	34500 1378	34700 1386	34900 1394	35100 1402	35300 1410	35500 1418	35700 1426	35900 1434	36100 1442	36300 1450	36500 1458	36700 1466	36900 1474	37100 1482	37300 1490	37500 1498	37700 1506	37900 1514	38100 1522	38300 1530	38500 1538	38700 1546	38900 1554	39100 1562	39300 1570	39500 1578	39700 1586	39900 1594	40100 1602	40300 1610	40500 1618	40700 1626	40900 1634	41100 1642	41300 1650	41500 1658	41700 1666	41900 1674	42100 1682	42300 1690	42500 1698	42700 1706	42900 1714	43100 1722	43300 1730	43500 1738	43700 1746	43900 1754	44100 1762	44300 1770	44500 1778	44700 1786	44900 1794	45100 1802	45300 1810	45500 1818	45700 1826	45900 1834	46100 1842	46300 1850	46500 1858	46700 1866	46900 1874	47100 1882	47300 1890	47500 1898	47700 1906	47900 1914	48100 1922	48300 1930	48500 1938	48700 1946	48900 1954	49100 1962	49300 1970	49500 1978	49700 1986	49900 1994	50100 2002	50300 2010	50500 2018	50700 2026	50900 2034	51100 2042	51300 2050	51500 2058	51700 2066	51900 2074	52100 2082	52300 2090	52500 2098	52700 2106	52900 2114	53100 2122	53300 2130	53500 2138	53700 2146	53900 2154	54100 2162	54300 2170	54500 2178	54700 2186	54900 2194	55100 2202	55300 2210	55500 2218	55700 2226	55900 2234	56100 2242	56300 2250	56500 2258	56700 2266	56900 2274	57100 2282	57300 2290	57500 2298	57700 2306	57900 2314	58100 2322	58300 2330	58500 2338	58700 2346	58900 2354	59100 2362	59300 2370	59500 2378	59700 2386	59900 2394	60100 2402	60300 2410	60500 2418	60700 2426	60900 2434	61100 2442	61300 2450	61500 2458	61700 2466	61900 2474	62100 2482	62300 2490	62500 2498	62700 2506	62900 2514	63100 2522	63300 2530	63500 2538	63700 2546	63900 2554	64100 2562	64300 2570	64500 2578	64700 2586	64900 2594	65100 2602	65300 2610	65500 2618	65700 2626	65900 2634	66100 2642	66300 2650	66500 2658	66700 2666	66900 2674	67100 2682	67300 2690	67500 2698	67700 2706	67900 2714	68100 2722	68300 2730	68500 2738	68700 2746	68900 2754	69100 2762	69300 2770	69500 2778	69700 2786	69900 2794	70100 2802	70300 2810	70500 2818	70700 2826	70900 2834	71100 2842	71300 2850	71500 2858	71700 2866	71900 2874	72100 2882	72300 2890	72500 2898	72700 2906	72900 2914	73100 2922	73300 2930	73500 2938	73700 2946	73900 2954	74100 2962	74300 2970	74500 2978	74700 2986	74900 2994	75100 3002	75300 3010	75500 3018	75700 3026	75900 3034	76100 3042	76300 3050	76500 3058	76700 3066	76900 3074	77100 3082	77300 3090	77500 3098	77700 3106	77900 3114	78100 3122	78300 3130	78500 3138	78700 3146	78900 3154	79100 3162	79300 3170	79500 3178	79700 3186	79900 3194	80100 3202	80300 3210	80500 3218	80700 3226	80900 3234	81100 3242	81300 3250	81500 3258	81700 3266	81900 3274	82100 3282	82300 3290	82500 3298	82700 3306	82900 3314	83100 3322	83300 3330	83500 3338	83700 3346	83900 3354	84100 3362	84300 3370	84500 3378	84700 3386	84900 3394	85100 3402	85300 3410	85500 3418	85700 3426	85900 3434	86100 3442	86300 3450	86500 3458	86700 3466	86900 3474	87100 3482	87300 3490	87500 3498	87700 3506	87900 3514	88100 3522	88300 3530	88500 3538	88700 3546	88900 3554	89100 3562	89300 3570	89500 3578	89700 3586	89900 3594	90100 3602	90300 3610	90500 3618	90700 3626	90900 3634	91100 3642	91300 3650	91500 3658	91700 3666	91900 3674	92100 3682	92300 3690	92500 3698	92700 3706	92900 3714	93100 3722	93300 3730	93500 3738	93700 3746	93900 3754	94100 3762	94300 3770	94500 3778	94700 3786	94900 3794	95100 3802	95300 3810	95500 3818	95700 3826	95900 3834	96100 3842	96300 3850	96500 3858	96700 3866	96900 3874	97100 3882	97300 3890	97500 3898	97700 3906	97900 3914	98100 3922	98300 3930	98500 3938	98700 3946	98900 3954	99100 3962	99300 3970	99500 3978	99700 3986	99900 3994	100100 4002	100300 4010	100500 4018	100700 4026	100900 4034	101100 4042	101300 4050	101500 4058	101700 4066	101900 4074	102100 4082	102300 4090	102500 4098	102700 4106	102900 4114	103100 4122	103300 4130	103500 4138	103700 4146	103900 4154	104100 4162	104300 4170	104500 4178	104700 4186	104900 4194	105100 4202	105300 4210	105500 4218	105700 4226	105900 4234	106100 4242	106300 4250	106500 4258	106700 4266	106900 4274	107100 4282	107300 4290	107500 4298	107700 4306	107900 4314	108100 4322	108300 4330	108500 4338	108700 4346	108900 4354	109100 4362	109300 4370	109500 4378	109700 4386	109900 4394	110100 4402	110300 4410	110500 4418	110700 4426	110900 4434	111100 4442	111300 4450	111500 4458	111700 4466	111900 4474	112100 4482	112300 4490	112500 4498	112700 4506	112900 4514	113100 4522	113300 4530	113500 4538	113700 4546	113900 4554	114100 4562	114300 4570	114500 4578	114700 4586	114900 4594	115100 4602	115300 4610	115500 4618	115700 4626	115900 4634	116100 4642	116300 4650	116500 4658	116700 4666	116900 4674	117100 4682	117300 4690	117500 4698	117700 4706	117900 4714	118100 4722	118300 4730	118500 4738	118700 4746	118900 4754	119100 4762	119300 4770	119500 4778	119700 4786	119900 4794	120100 4802	120300 4810	120500 4818	120700 4826	120900 4834	121100 4842	121300 4850	121500 4858	121700 4866	121900 4874	122100 4882	122300 4890	122500 4898	122700 4906	122900 4914	123100 4922	123300 4930	123500 4938	123700 4946	123900 4954	124100 4962	124300 4970	124500 4978	124700 4986	124900 4994	125100 5002	125300 5010	125500 5018	125700 5026	125900 5034	126100 5042	126300 5050	126500 5058	126700 5066	126900 5074	127100 5082	127300 5090	127500 5098	127700 5106	127900 5114	128100 5122	128300 5130	128500 5138	128700 5146	128900 5154	129100 5162	129300 5170	129500 5178	129700 5186	129900 5194	130100 5202	130300 5210	130500 5218	130700 5226	130900 5234	131100 5242	131300 5250	131500 5258	131700 5266	131900 5274	132100 5282	132300 5290	132500 5298	132700 5306	132900 5314	133100 5322	133300 5330	133500 5338	133700 5346	133900 5354	134100 5362	134300 5370	134500 5378	134700 5386	134900 5394	135100 5402	135300 5410	135500 5418	135700 5426	135900 5434	136100 5442	136300 5450	136500 5458	136700 5466	136900 5474	137100 5482	137300 5490	137500 5498	137700 5506	137900 5514	138100 5522	138300 5530	138500 5538	138700 5546	138900 5554	139100 5562	139300 5570	139500 5578	139700 5586	139900 5594	140100 5602	140300 5610	140500 5618	140700 5626	140900 5634	141100 5642	141300 5650	141500 5658	141700 5666	141900 5674	142100 5682	142300 5690	142500 5698	142700 5706	142900 5714	143100 5722	143300 5730	143500 5738	143700 5746	143900 5754	144100 5762	144300 5770	144500 5778	144700 5786	144900 5794	145100 5802	145300 5810	145500 5818	145700 5826	145900 5834	146100 5842	146300 5850	146500 5858	146700 5866	146900 5874	147100 5882	147300 5890	147500 5898	147700 5906	147900 5914	148100 5922	148300 5930	148500 5938	148700 5946	148900 5954	149100 5962	149300 5970	149500 5978	149700 5986	149900 5994	150100 6002	150300 6010	150500 6018	150700 6026	150900 6034	151100 6042	151300 6050	151500 6058	151700 6066	151900 6074	152100 6082	1523
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Product Information Section

320 - Reach Boom/Variable Angle Boom/Super Long Reach Boom

900 mm (35 inch) TG shoe

9000 (2425)	11500 85	13000 125	14500 165	16000 245	17500 305	19000 365	19500 420	12000 400	13500 545		12000 (3048)
											
12000 480											1200 480
11800 420							2700 4700	2300 4700			1300 2800 11800 480
10800 360							3200 4600	2200 4050	2000 3950		1100 2450 10800 480
10600 360							3200 4500	2200 4050	2000 3950		1100 2450 10600 480
10400 360							3200 4400	2200 4050	2000 3950		1100 2450 10400 480
10200 360							3200 4300	2200 4050	2000 3950		1100 2450 10200 480
10000 360							3200 4200	2200 4050	2000 3950		1100 2450 10000 480
9800 360							3200 4100	2200 4050	2000 3950		1100 2450 9800 480
9600 360							3200 4000	2200 4050	2000 3950		1100 2450 9600 480
9400 360							3200 3900	2200 4050	2000 3950		1100 2450 9400 480
9200 360							3200 3800	2200 4050	2000 3950		1100 2450 9200 480
9000 360							3200 3700	2200 4050	2000 3950		1100 2450 9000 480
8800 360							3200 3600	2200 4050	2000 3950		1100 2450 8800 480
8600 360							3200 3500	2200 4050	2000 3950		1100 2450 8600 480
8400 360							3200 3400	2200 4050	2000 3950		1100 2450 8400 480
8200 360							3200 3300	2200 4050	2000 3950		1100 2450 8200 480
8000 360							3200 3200	2200 4050	2000 3950		1100 2450 8000 480
7800 360							3200 3100	2200 4050	2000 3950		1100 2450 7800 480
7600 360							3200 3000	2200 4050	2000 3950		1100 2450 7600 480
7400 360							3200 2900	2200 4050	2000 3950		1100 2450 7400 480
7200 360							3200 2800	2200 4050	2000 3950		1100 2450 7200 480
7000 360							3200 2700	2200 4050	2000 3950		1100 2450 7000 480
6800 360							3200 2600	2200 4050	2000 3950		1100 2450 6800 480
6600 360							3200 2500	2200 4050	2000 3950		1100 2450 6600 480
6400 360							3200 2400	2200 4050	2000 3950		1100 2450 6400 480
6200 360							3200 2300	2200 4050	2000 3950		1100 2450 6200 480
6000 360							3200 2200	2200 4050	2000 3950		1100 2450 6000 480
5800 360							3200 2100	2200 4050	2000 3950		1100 2450 5800 480
5600 360							3200 2000	2200 4050	2000 3950		1100 2450 5600 480
5400 360							3200 1900	2200 4050	2000 3950		1100 2450 5400 480
5200 360							3200 1800	2200 4050	2000 3950		1100 2450 5200 480
5000 360							3200 1700	2200 4050	2000 3950		1100 2450 5000 480
4800 360							3200 1600	2200 4050	2000 3950		1100 2450 4800 480
4600 360							3200 1500	2200 4050	2000 3950		1100 2450 4600 480
4400 360							3200 1400	2200 4050	2000 3950		1100 2450 4400 480
4200 360							3200 1300	2200 4050	2000 3950		1100 2450 4200 480
4000 360							3200 1200	2200 4050	2000 3950		1100 2450 4000 480
3800 360							3200 1100	2200 4050	2000 3950		1100 2450 3800 480
3600 360							3200 1000	2200 4050	2000 3950		1100 2450 3600 480
3400 360							3200 900	2200 4050	2000 3950		1100 2450 3400 480
3200 360							3200 800	2200 4050	2000 3950		1100 2450 3200 480

Illustration 340

g06225598

Lift Chart Above: 320, 8850 mm (348 inch) Super long reach boom, 6283 mm (247 inch) stick, 4700 kg (10362 lb) counter weight, and 900 mm (35 inch) TG track shoes, heavy lift OFF

(mm) (inch)	1100 43	1300 51	1500 59	1700 67	1900 75	2100 83	2300 91	2500 98	2700 106	2900 114	3100 122	3300 130	3500 138	3700 146	3900 154	4100 162	4300 170	4500 178	4700 186	4900 194	5100 202	5300 210	5500 218	5700 226	5900 234	6100 242	6300 250	6500 258	6700 266	6900 274	7100 282	7300 290	7500 298	7700 306	7900 314	8100 322	8300 330	8500 338	8700 346	8900 354	9100 362	9300 370	9500 378	9700 386	9900 394	10100 402	10300 410	10500 418	10700 426	10900 434	11100 442	11300 450	11500 458	11700 466	11900 474	12100 482	12300 490	12500 498	12700 506	12900 514	13100 522	13300 530	13500 538	13700 546	13900 554	14100 562	14300 570	14500 578	14700 586	14900 594	15100 602	15300 610	15500 618	15700 626	15900 634	16100 642	16300 650	16500 658	16700 666	16900 674	17100 682	17300 690	17500 698	17700 706	17900 714	18100 722	18300 730	18500 738	18700 746	18900 754	19100 762	19300 770	19500 778	19700 786	19900 794	20100 802	20300 810	20500 818	20700 826	20900 834	21100 842	21300 850	21500 858	21700 866	21900 874	22100 882	22300 890	22500 898	22700 906	22900 914	23100 922	23300 930	23500 938	23700 946	23900 954	24100 962	24300 970	24500 978	24700 986	24900 994	25100 1002	25300 1010	25500 1018	25700 1026	25900 1034	26100 1042	26300 1050	26500 1058	26700 1066	26900 1074	27100 1082	27300 1090	27500 1098	27700 1106	27900 1114	28100 1122	28300 1130	28500 1138	28700 1146	28900 1154	29100 1162	29300 1170	29500 1178	29700 1186	29900 1194	30100 1202	30300 1210	30500 1218	30700 1226	30900 1234	31100 1242	31300 1250	31500 1258	31700 1266	31900 1274	32100 1282	32300 1290	32500 1298	32700 1306	32900 1314	33100 1322	33300 1330	33500 1338	33700 1346	33900 1354	34100 1362	34300 1370	34500 1378	34700 1386	34900 1394	35100 1402	35300 1410	35500 1418	35700 1426	35900 1434	36100 1442	36300 1450	36500 1458	36700 1466	36900 1474	37100 1482	37300 1490	37500 1498	37700 1506	37900 1514	38100 1522	38300 1530	38500 1538	38700 1546	38900 1554	39100 1562	39300 1570	39500 1578	39700 1586	39900 1594	40100 1602	40300 1610	40500 1618	40700 1626	40900 1634	41100 1642	41300 1650	41500 1658	41700 1666	41900 1674	42100 1682	42300 1690	42500 1698	42700 1706	42900 1714	43100 1722	43300 1730	43500 1738	43700 1746	43900 1754	44100 1762	44300 1770	44500 1778	44700 1786	44900 1794	45100 1802	45300 1810	45500 1818	45700 1826	45900 1834	46100 1842	46300 1850	46500 1858	46700 1866	46900 1874	47100 1882	47300 1890	47500 1898	47700 1906	47900 1914	48100 1922	48300 1930	48500 1938	48700 1946	48900 1954	49100 1962	49300 1970	49500 1978	49700 1986	49900 1994	50100 2002	50300 2010	50500 2018	50700 2026	50900 2034	51100 2042	51300 2050	51500 2058	51700 2066	51900 2074	52100 2082	52300 2090	52500 2098	52700 2106	52900 2114	53100 2122	53300 2130	53500 2138	53700 2146	53900 2154	54100 2162	54300 2170	54500 2178	54700 2186	54900 2194	55100 2202	55300 2210	55500 2218	55700 2226	55900 2234	56100 2242	56300 2250	56500 2258	56700 2266	56900 2274	57100 2282	57300 2290	57500 2298	57700 2306	57900 2314	58100 2322	58300 2330	58500 2338	58700 2346	58900 2354	59100 2362	59300 2370	59500 2378	59700 2386	59900 2394	60100 2402	60300 2410	60500 2418	60700 2426	60900 2434	61100 2442	61300 2450	61500 2458	61700 2466	61900 2474	62100 2482	62300 2490	62500 2498	62700 2506	62900 2514	63100 2522	63300 2530	63500 2538	63700 2546	63900 2554	64100 2562	64300 2570	64500 2578	64700 2586	64900 2594	65100 2602	65300 2610	65500 2618	65700 2626	65900 2634	66100 2642	66300 2650	66500 2658	66700 2666	66900 2674	67100 2682	67300 2690	67500 2698	67700 2706	67900 2714	68100 2722	68300 2730	68500 2738	68700 2746	68900 2754	69100 2762	69300 2770	69500 2778	69700 2786	69900 2794	70100 2802	70300 2810	70500 2818	70700 2826	70900 2834	71100 2842	71300 2850	71500 2858	71700 2866	71900 2874	72100 2882	72300 2890	72500 2898	72700 2906	72900 2914	73100 2922	73300 2930	73500 2938	73700 2946	73900 2954	74100 2962	74300 2970	74500 2978	74700 2986	74900 2994	75100 3002	75300 3010	75500 3018	75700 3026	75900 3034	76100 3042	76300 3050	76500 3058	76700 3066	76900 3074	77100 3082	77300 3090	77500 3098	77700 3106	77900 3114	78100 3122	78300 3130	78500 3138	78700 3146	78900 3154	79100 3162	79300 3170	79500 3178	79700 3186	79900 3194	80100 3202	80300 3210	80500 3218	80700 3226	80900 3234	81100 3242	81300 3250	81500 3258	81700 3266	81900 3274	82100 3282	82300 3290	82500 3298	82700 3306	82900 3314	83100 3322	83300 3330	83500 3338	83700 3346	83900 3354	84100 3362	84300 3370	84500 3378	84700 3386	84900 3394	85100 3402	85300 3410	85500 3418	85700 3426	85900 3434	86100 3442	86300 3450	86500 3458	86700 3466	86900 3474	87100 3482	87300 3490	87500 3498	87700 3506	87900 3514	88100 3522	88300 3530	88500 3538	88700 3546	88900 3554	89100 3562	89300 3570	89500 3578	89700 3586	89900 3594	90100 3602	90300 3610	90500 3618	90700 3626	90900 3634	91100 3642	91300 3650	91500 3658	91700 3666	91900 3674	92100 3682	92300 3690	92500 3698	92700 3706	92900 3714	93100 3722	93300 3730	93500 3738	93700 3746	93900 3754	94100 3762	94300 3770	94500 3778	94700 3786	94900 3794	95100 3802	95300 3810	95500 3818	95700 3826	95900 3834	96100 3842	96300 3850	96500 3858	96700 3866	96900 3874	97100 3882	97300 3890	97500 3898	97700 3906	97900 3914	98100 3922	98300 3930	98500 3938	98700 3946	98900 3954	99100 3962	99300 3970	99500 3978	99700 3986	99900 3994	100100 4002	100300 4010	100500 4018	100700 4026	100900 4034	101100 4042	101300 4050	101500 4058	101700 4066	101900 4074	102100 4082	102300 4090	102500 4098	102700 4106	102900 4114	103100 4122	103300 4130	103500 4138	103700 4146	103900 4154	104100 4162	104300 4170	104500 4178	104700 4186	104900 4194	105100 4202	105300 4210	105500 4218	105700 4226	105900 4234	106100 4242	106300 4250	106500 4258	106700 4266	106900 4274	107100 4282	107300 4290	107500 4298	107700 4306	107900 4314	108100 4322	108300 4330	108500 4338	108700 4346	108900 4354	109100 4362	109300 4370	109500 4378	109700 4386	109900 4394	110100 4402	110300 4410	110500 4418	110700 4426	110900 4434	111100 4442	111300 4450	111500 4458	111700 4466	111900 4474	112100 4482	112300 4490	112500 4498	112700 4506	112900 4514	113100 4522	113300 4530	113500 4538	113700 4546	113900 4554	114100 4562	114300 4570	114500 4578	114700 4586	114900 4594	115100 4602	115300 4610	115500 4618	115700 4626	115900 4634	116100 4642	116300 4650	116500 4658	116700 4666	116900 4674	117100 4682	117300 4690	117500 4698	117700 4706	117900 4714	118100 4722	118300 4730	118500 4738	118700 4746	118900 4754	119100 4762	119300 4770	119500 4778	119700 4786	119900 4794	120100 4802	120300 4810	120500 4818	120700 4826	120900 4834	121100 4842	121300 4850	121500 4858	121700 4866	121900 4874	122100 4882	122300 4890	122500 4898	122700 4906	122900 4914	123100 4922	123300 4930	123500 4938	123700 4946	123900 4954	124100 4962	124300 4970	124500 4978	124700 4986	124900 4994	125100 5002	125300 5010	125500 5018	125700 5026	125900 5034	126100 5042	126300 5050	126500 5058	126700 5066	126900 5074	127100 5082	127300 5090	127500 5098	127700 5106	127900 5114	128100 5122	128300 5130	128500 5138	128700 5146	128900 5154	129100 5162	129300 5170	129500 5178	129700 5186	129900 5194	130100 5202	130300 5210	130500 5218	130700 5226	130900 5234	131100 5242	131300 5250	131500 5258	131700 5266	131900 5274	132100 5282	132300 5290	132500 5298	132700 5306	132900 5314	133100 5322	133300 5330	133500 5338	133700 5346	133900 5354	134100 5362	134300 5370	134500 5378	134700 5386	134900 5394	135100 5402	135300 5410	135500 5418	135700 5426	135900 5434	136100 5442	136300 5450	136500 5458	136700 5466	136900 5474	137100 5482	137300 5490	137500 5498	137700 5506	137900 5514	138100 5522	138300 5530	138500 5538	138700 5546	138900 5554	139100 5562	139300 5570	139500 5578	139700 5586	139900 5594	140100 5602	140300 5610	140500 5618	140700 5626	140900 5634	141100 5642	141300 5650	141500 5658	141700 5666	141900 5674	142100 5682	142300 5690	142500 5698	142700 5706	142900 5714	143100 5722	143300 5730	143500 5738	143700 5746	143900 5754	144100 5762	144300 5770	144500 5778	144700 5786	144900 5794	145100 5802	145300 5810	145500 5818	145700 5826	145900 5834	146100 5842	146300 5850	146500 5858	146700 5866	146900 5874	147100 5882	147300 5890	147500 5898	147700 5906	147900 5914	148100 5922	148300 5930	148500 5938	148700 5946	148900 5954	149100 5962	149300 5970	149500 5978	149700 5986	149900 5994	150100 6002	150300 6010	150500 6018	150700 6026	150900 6034	151100 6042	151300 6050	151500 6058	151700 6066	151900 6074	152100 6082	1523
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Identification Information

i08271901

Plate Locations and Film Locations

SMCS Code: 1000; 7000

The Product Identification Number (PIN) will be used to identify a powered machine that is designed for an operator to ride.

Caterpillar products such as engines, transmissions, and major attachments that are not designed for an operator to ride are identified by Serial Numbers.

For quick reference, record the identification numbers in the spaces that are provided below the illustration.

Product Identification Number (PIN) Plate



Illustration 342

g06184412

The PIN plate is positioned on the front of the machine, close to the operator compartment.

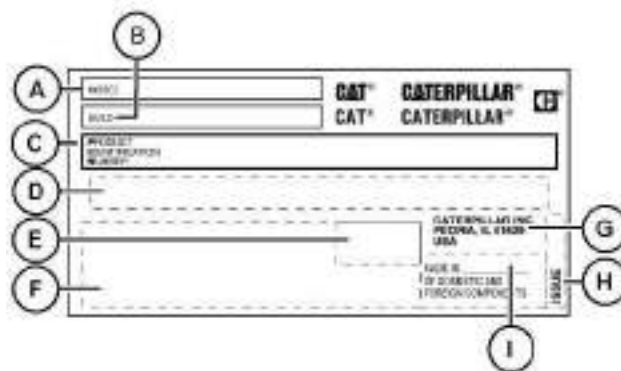


Illustration 343

q06201159

Manufacturer Name and Address _____

Model (A)_____

Build (B)_____

Product Identification Number (C)_____

Bar Code (D)_____

Month and Year of Manufacture Plate (If Required)
(E) _____

Regional Certification Plate (If Required) (F)_____

Address of Manufacturer (G)_____

Issue (H)_____

Country of Origin Info Plate (If Required) (I)_____

Local regulation may require documentation of the month and/or year of manufacture in the Operation and Maintenance Manual. Comply with these regulations

European Union



Illustration 344

g06201193

This plate is positioned on the bottom-left side of the plate for the PIN.

Note: The CE plate is on machines that are certified to the European Union requirements that were effective then.

For machines compliant to 2006/42/EC, the following information is stamped onto the CE plate. For quick reference, record this information in the spaces that are provided below.

- Engine Power Primary Engine (kW) _____
- Engine Power for Additional Engine (kW) (If Equipped) _____
- Typical Machine Operating Weight for European Market (kg) _____
- Year of Construction _____
- Machine Type _____



Illustration 345

g06201198

This plate is positioned on the bottom-left side of the plate for the PIN.

Note: The CE plate is on machines that are certified to the European Union requirements that were effective then.

For machines compliant to 98/37/EC and 89/392/EEC, the following information is stamped onto the CE plate. For quick reference, record this information in the spaces that are provided below.

- Engine Power Primary Engine (kW) _____
- Typical Machine Operating Weight for European Market (kg) _____
- Year _____

For manufacturer name and address and the country of origin, see the PIN plate.

Eurasian Economic Union

For machines compliant to the Eurasian Economic Union requirements, the EAC plate is positioned on or near the Product Identification Number (PIN) plate (see “Product Information Section” of the machine Operation and Maintenance Manual). The EAC plate is placed on machines certified to the Eurasian Economic Union requirements effective at the time of market entry.

Note: One of the below plates may be installed on the machine.



Illustration 346 g06094564

If equipped, the Month and Year of Manufacture are on the PIN plate.



Illustration 347 g06532250

If equipped, the following information is stamped onto the EAC plate. For quick reference, record this information in the spaces provided below.

- Primary Engine Power (kW)_____
- Additional Engine Power (kW) _____
- Typical Machine Operating Weight for Eurasian Economic Union (kg)_____
- Month and Year of Manufacture_____
- Machine Type _____

Manufacturer Information

Manufacturer:

Caterpillar Inc.,
100 N.E. Adams Street
Peoria, Illinois 61629, USA

Entity authorized by the manufacturer at the territory of Eurasian Economic Union:

Caterpillar Eurasia LLC
75, Sadovnicheskaya Emb.
Moscow 115035, Russia

Machine Specification Film

The machine specification film is on machines that are going into Japan.

The Japanese Industrial Safety and Health Act requires machine specifications to be displayed on a film that can easily be seen by the operator.

If equipped, this film will be on the cab door.



Illustration 348 g06178867

Typical example

Electromagnetic Emissions

Note: This label is on machines that are going into Canada.



Illustration 349

g06063443

If equipped, this label is located next to the PIN plate. This label verifies that the product meets the requirements of ICES-002 Issue 6. Compliance to ICES-002 Issue 6 is accomplished by meeting electromagnetic emissions industry standard CISPR-12.

Engine Serial Number

This label is on the engine.

Engine Serial Number _____

Sound Certification Film

If equipped, this label is on the cab door.



Illustration 350

g03533258

For 320



Illustration 351

g01447432

For 323

If equipped, this label will be on the cab door. The certification label is used to verify the environmental sound certification of the machine to the requirements of the European Union. The value that is listed on the label indicates the guaranteed exterior sound power level L_{WA} at the time of manufacture for the conditions that are specified in "2000/14/EC".



Illustration 352

g03105800

- (A) Low Noise Film
- (B) Super Low Noise Film

If equipped, these certification labels are used to verify the Japan Ministry of Land, Infrastructure, Transportation, and Tourism (MLIT) noise designation according to the Japan "Designation Rule of Low Noise Type Construction Machine".

Low Noise (A) – Verifies that the Japan "MLIT" designates the machine as a "Low Noise" type construction machine.

Super Low Noise (B) – Verifies that the Japan "MLIT" designates the machine as a "Super Low Noise" type construction machine.

i05757951

Emissions Certification Film

SMCS Code: 1000; 7000; 7405

Note: This information is pertinent in Japan.



Illustration 353

g03654940

2014 certification label example

A certification label is located on the cab door.

The certification label verifies that the machine conforms to the 2014 Japan Nonroad Special Motor Vehicle Exhaust Regulation.

i08085827

Emissions Certification Film

SMCS Code: 1000; 7000; 7405

Consult your Cat dealer for an Emission Control Warranty Statement.

The emission certification film is on the engine.

Declaration of Conformity

SMCS Code: 1000; 7000

Table 25

An EC or EU Declaration of Conformity document was provided with the machine if it was manufactured to comply with specific requirements for the European Union. In order to determine the details of the applicable Directives, review the complete EC or EU Declaration of Conformity provided with the machine. The extract shown below from an EC or EU Declaration of Conformity for machines that are declared compliant to "2006/42/EC" applies only to those machines originally "CE" marked by the manufacturer listed and which have not since been modified.

ORIGINAL EC or EU DECLARATION OF CONFORMITY

Manufacturer: Caterpillar Inc., 100 N.E. Adams Street, Peoria, Illinois 61629, USA

Person authorized to compile the Technical File and to communicate relevant part (s) of the Technical File to the Authorities of European Union Member States on request:

Standards & Regulations Manager, Caterpillar France SAS
40 Avenue Leon-Blum 38000 Grenoble, France

I, the undersigned, _____, hereby certify that the construction equipment specified hereunder

Description:	Generic Denomination:	Earth - moving Equipment
	Function:	Hydraulic Excavator
	Model/Type:	320,323
	Serial Number:	
	Commercial Name:	Caterpillar

Fulfills all the relevant provisions of the following Directives

Directives	Notified Body	Document No.	
2000/14/EC amended by 2005/88/EC, Note (1)			
2006/42/EC	N/A		
2014/30/EU	N/A		

Note (1) Guaranteed Sound Power Level - ____ dB (A) Annex VI
Representative Equipment Type Sound Power Level - ____ dB (A)
[Engine Power per ISO 14396 - ____ kW, Rated engine speed - ____ rpm
Technical Documentation accessible through person listed above authorized to compile the Technical File

Done at:

Signature

Date:

Name/Position

Note: The above information was correct as of June 2017, but may be subject to change, please refer to the individual declaration of conformity issued with the machine for exact details.

Operation Section

Before Operation

i07103304

Mounting and Dismounting

SMCS Code: 6700; 7000



Illustration 354

g06224270

Typical example

Mount the machine and dismount the machine only at locations that have steps and/or handholds. Before you mount the machine, clean the steps and the handholds. Inspect the steps and handholds. Make all necessary repairs.

Face the machine whenever you get on the machine and whenever you get off the machine.

Maintain a three-point contact with the steps and with the handholds.

Note: Three-point contact can be two feet and one hand. Three-point contact can also be one foot and two hands.

Do not mount a moving machine. Do not dismount a moving machine. Never jump off the machine. Do not carry tools or supplies when you try to mount the machine or when you try to dismount the machine. Use a hand line to pull equipment onto the platform. Do not use any controls as handholds when you enter the operator compartment or when you exit the operator compartment.

Machine Access System Specifications

The machine access system has been designed to meet the intent of the technical requirements in "ISO 2867 Earth-moving Machinery – Access Systems". The access system provides for operator access to the operator station and to conduct the maintenance procedures described in Maintenance section.

Alternate Exit

Machines that are equipped with cabs have alternate exits. For additional information, see Operation and Maintenance Manual, "Alternate Exit".

i08019624

Daily Inspection

SMCS Code: 1000; 6319; 6700; 7000

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

At operating temperature, the engine coolant is hot and under pressure.

Steam can cause personal injury.

Check the coolant level only after the engine has been stopped and the cooling system pressure cap is cool enough to touch with your bare hand.

Remove the cooling system pressure cap slowly to relieve pressure.

Cooling system conditioner contains alkali. Avoid contact with the skin and eyes to prevent personal injury.

NOTICE

Accumulated grease and oil on a machine is a fire hazard. Remove this debris with steam cleaning or high pressure water, at least every 1000 hours or each time any significant quantity of oil is spilled on a machine.

For maximum service life of the machine, perform a thorough walk-around inspection before you mount the machine and before you start the engine.

First 100 Hours

Daily, perform the procedures that are applicable to your machine:

- Operation and Maintenance Manual, "Boom and Stick Linkage - Lubricate"

- Operation and Maintenance Manual, “Bucket Linkage - Lubricate”

Daily Basis

Daily, perform the procedures that are applicable to your machine:

- Operation and Maintenance Manual, “Cooling System Coolant Level - Check”
- Operation and Maintenance Manual, “Engine Oil Level - Check”
- Operation and Maintenance Manual, “Fuel System Water Separator - Drain”
- Operation and Maintenance Manual, “Fuel Tank Water and Sediment - Drain”
- Operation and Maintenance Manual, “Hydraulic System Oil Level - Check”
- Operation and Maintenance Manual, “Indicators and Gauges - Test”
- Operation and Maintenance Manual, “Seat Belt - Inspect”
- Operation and Maintenance Manual, “Track Adjustment - Inspect”
- Operation and Maintenance Manual, “Travel Alarm - Test”

Refer to Operation and Maintenance Manual, “Maintenance Interval Schedule” for all maintenance recommendations.

Note: Watch closely for leaks. If you observe a leak, find the source of the leak and correct the leak. If you suspect a leak or you observe a leak, check the fluid levels more frequently.



Illustration 355

g06181597

Inspect the attachment control linkage, attachment cylinders, and attachment for damage or excessive wear. Make any necessary repairs.

Inspect the lights for broken bulbs and for broken lenses. Replace any broken bulbs and any broken lenses.

Inspect the engine compartment for any trash buildup. Remove any trash buildup from the engine compartment.

Inspect the cooling system for any leaks, for faulty hoses and for any trash buildup. Correct any leaks. Remove any trash from the radiator.

Inspect all the belts for the engine attachments. Replace any belts that are worn, frayed, or broken.



Illustration 356

g06181620

Inspect the hydraulic system for leaks. Inspect the tank, cylinder rod seals, hoses, tubes, plugs, connections, and fittings. Correct any leaks in the hydraulic system.

Inspect the tubes and hoses along the boom and stick for wear and leaks. Replace any hoses or tubes that are worn or leak.

Inspect the differential and the final drives for leaks. Make any necessary repairs.

Inspect the swing drive for leaks.

Make sure that all covers and guards are securely attached. Inspect the covers and the guards for damage.

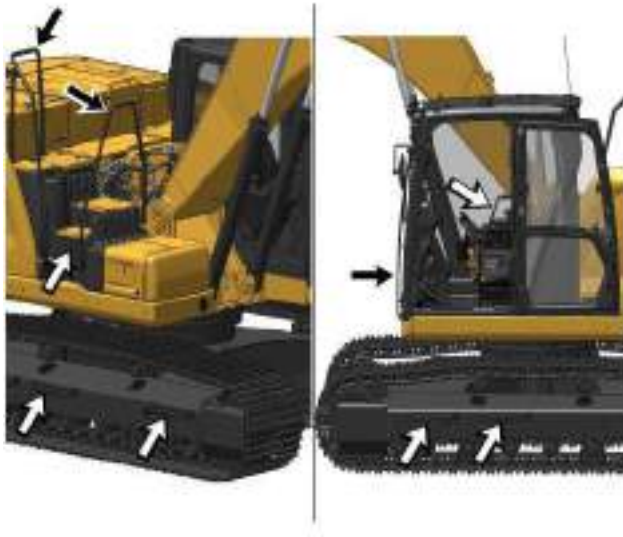


Illustration 357

g06181696

Inspect the steps, the walkways, and the handholds.
Clean the steps, the walkways, and the handholds.
Make any necessary repairs.

Inspect the operator compartment for trash buildup.
Check for trash buildup under the floorplate and on
the crankcase guard. Keep these areas clean.

Adjust the mirrors to achieve the best visibility.

Machine Operation

i06952448

Alternate Exit

SMCS Code: 7310

Rear Window with Ring Seal (If Equipped)



Illustration 358

g06187008



Alternate Exit – The rear window serves as an alternate exit.

To remove the rear window, pull ring (3) and completely remove the window seal, then push out the glass. Climb through the rear window opening to exit the cab.

Rear Window with Lever (If Equipped)



Illustration 359

g06213470

Inside lever in latched position



Alternate Exit – The rear window serves as an alternate exit.

To remove the rear window, rotate handle from its latched position, then push out the glass. Climb through the rear window opening to exit the cab.

i07383859



Illustration 360

g06213471

The window is also equipped with an outside handle. If the operator is unable, outside personnel can rotate the outside handle and pull the window out.

Seat

SMCS Code: 5258-025; 7312-025; 7324; 7327

Comfort Level Seat (If Equipped)



Illustration 361

g06225151

- (1) Headrest
- (2) Backrest adjuster
- (3) Seat and console fore and aft adjuster
- (4) Indicator
- (5) Seat height adjustment lever
- (6) Seat fore and aft lever

The operator can adjust the height of headrest (1). To adjust the headrest, hold the headrest with both hands. Move the headrest up and down. Release the headrest when the desired position is attained. The headrest will remain in the desired position.

Pull up on backrest adjuster (2) to release the lock. Move the backrest to the desired position and then release the adjuster.

Lift up on fore and aft adjuster (3) to release the seat from the locked position. Adjust the seat and console forward or rearward to the desired position and then release the lever to lock the seat.

Use seat height adjustment lever (5) to adjust the seat for the operator's height and weight. Move the adjuster switch to the "+" symbol to raise the height. Ratchet the seat upward without sitting in the seat. Then sit in the seat to check the color of indicator (4). When the indicator is showing green, the seat is in the right range for the operator. Further adjustment can be made as long as the indicator stays green.

To lower the seat, move the adjust switch to the "-" symbol. Ratchet the adjuster downward without sitting in the seat. Then sit in the seat to check the color of indicator (4). When the indicator is showing green, the seat is in the right range for the operator. Further adjustment can be made as long as the indicator stays green.

Pull upward on seat fore and aft lever (6) to release the cushion lock. Adjust the seat cushion forward or backward to the desired position and then release the lock to lock the cushion in place.

Deluxe Level Seat (If Equipped)



Illustration 362

g06303504

- (1) Adjustment lever for the lumbar support
- (2) Seat heater switch
- (3) Adjustment lever for reclining the seat
- (5) Indicator
- (6) Seat and console fore and aft lever
- (7) Seat height adjustment
- (8) Seat cushion angle adjuster
- (9) Console height adjustment
- (10) Seat fore and aft adjuster
- (11) Seat cushion fore and aft adjuster
- (12) Headrest

The lumbar support is located in the back of the seat. Turn knob (1) (if equipped) counterclockwise to increase the force of the lumbar support. To decrease the lumbar support, continue to turn the knob counterclockwise.

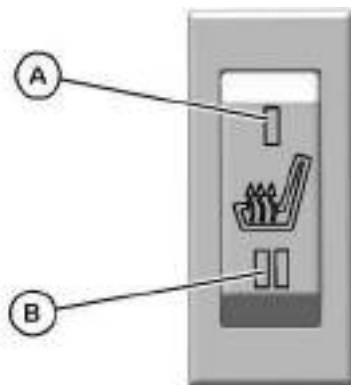


Illustration 363

g06251600

(A) Low heat
(B) High heat

Seat heater switch (2) is a three position switch. The middle position of the switch is off. Press the top position of the switch for low heat. Press the bottom of the switch for high heat.

 WARNING

Heat-induced burns can occur when some people use a seat heater. Do not use the seat heater if you have a reduced ability to sense temperature changes, a reduced ability to feel pain, or have sensitive skin.

Pull up lever (3) to change the angle of the backrest. With the backrest in the desired position. Release the lever.

Push in adjuster (11) to adjust the fore/aft position of the seat cushion.

Pull lever (6) to adjust the seat and the console forward and backward.

Operate seat height adjustment (7) to adjust the seat and the console to the desired height. Pull up the lever to raise the height of the seat. Push down on the lever to lower the seat. Indicator (5) will turn green when the seat is at the optimum height for the operator's height and weight. If the indicator is yellow, further adjustment is required.

Note: Do not operate the seat compressor for over a minute or damage can result.

Use handle (9) to adjust the height of the console. When the lever is pulled forward, a gear is released. The operator can rotate the lever freely. Release the lever to return to the original position.

To adjust the seat forward or backward, pull up lever (10) and hold the lever. Move the seat to the desired position. To lock the seat in the selected position, release the lever.

Push the adjuster (8) to adjust the tilt angle of the seat cushion.

The operator can adjust the height of headrest (12). To adjust the headrest, hold the headrest with both hands. Move the headrest up and down. Release the headrest when the desired position is attained. The headrest will remain in the desired position.

Premium Level Seat (If Equipped)



Illustration 364

g06251562

- (1) Seat heater switch
- (2) Cooling switch
- (3) Adjustment lever for reclining the seat
- (4) Fore/Aft isolator activate/deactivate lever
- (5) Seat and console fore and aft lever
- (6) Seat cushion angle adjuster
- (7) Seat height adjustment
- (8) Adjustable damper
- (9) Seat cushion fore and aft adjuster
- (10) Console height adjustment
- (11) Seat fore and aft adjuster
- (12) Headrest

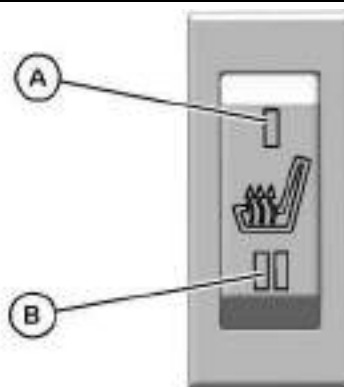


Illustration 365

g06251600

- (A) Low heat
- (B) High heat

Seat heater switch (1) is a three position switch. The middle position of the switch is off. Press the top position of the switch for low heat. Press the bottom of the switch for high heat.

WARNING

Heat-induced burns can occur when some people use a seat heater. Do not use the seat heater if you have a reduced ability to sense temperature changes, a reduced ability to feel pain, or have sensitive skin.

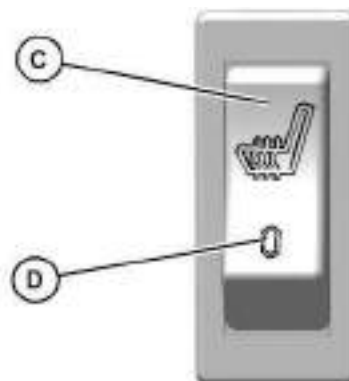


Illustration 366

g06251623

- (C) Cooling on
- (D) Cooling off

To cool the seat, press the top of cooling switch (2). Press the bottom of the switch to turn the cooling off.

Pull up lever (3) to change the angle of the backrest. With the backrest in the desired position. Release the lever.

Push in adjuster (9) to adjust the fore/aft position of the seat cushion.

Pull lever (5) to adjust the seat and the console forward and backward.

Fore/Aft isolator activate/deactivate lever (4) allows the operator to lock the seat and console or allow the seat and console to float with the movement of the machine. With the adjuster pointing to the front of the machine, the console is locked. Rotate the lever to the rear to unlock the console and allow it to float.

Operate seat height adjustment (7) to adjust the seat and the console to the desired height. Pull up the lever to raise the height of the seat. Push down on the lever to lower the seat.

Use adjustable damper (8) to stiffen or loosen the seat suspension.

Note: Do not operate the seat compressor for over a minute or damage can result.

Use handle (10) to adjust the height of the console. When the lever is pulled forward, a gear is released. The operator can rotate the lever freely. Release the lever to return to the original position.

To adjust the seat forward or backward, pull up lever (11) and hold the lever. Move the seat to the desired position. To lock the seat in the selected position, release the lever.

Push adjuster (6) to adjust the tilt angle of the seat cushion.

The operator can adjust the height of headrest (12). To adjust the headrest, hold the headrest with both hands. Move the headrest up and down. Release the headrest when the desired position is attained. The headrest will remain in the desired position.

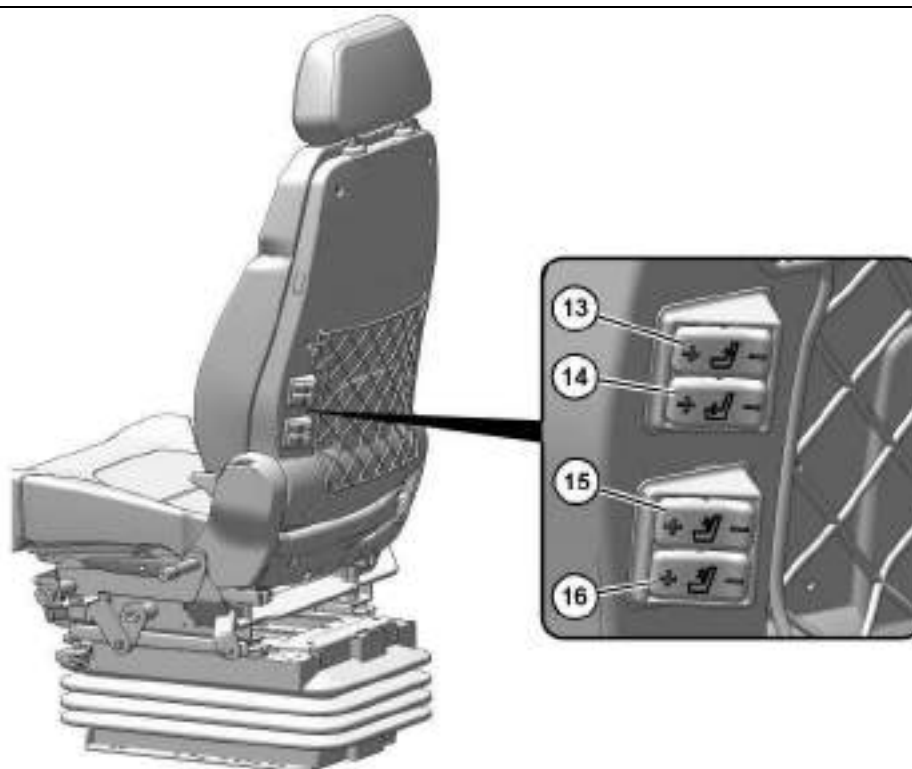


Illustration 367

g06251575

(13) Seat back side bolster control
(14) Seat cushion side bolster control

(15) Upper lumbar control
(16) Lower lumbar control

The lumbar controls on the back of the seat back all use two position switches. Press the + symbol to increase lumbar support. Press the - symbol to decrease lumbar support.

i07092308

Seat Belt

SMCS Code: 7327

Note: This machine was equipped with a seat belt when the machine was shipped from Caterpillar. At the time of installation, the seat belt and the instructions for installation of the seat belt meet the SAE J386 and ISO 6683 standards. Consult your Cat dealer for all replacement parts.

Always check the condition of the seat belt and the condition of the mounting hardware before you operate the machine.

Seat Belt Adjustment for Retractable Seat Belts

Fastening The Seat Belt



Illustration 368

g06223891

Pull seat belt (2) out of retractor (1) in a continuous motion.

Fasten seat belt catch (3) into buckle (4). Make sure that the seat belt is placed low across the lap of the operator.

The retractor will adjust the belt length and the retractor will lock in place. The comfort ride sleeve will allow the operator to have limited movement.

Releasing The Seat Belt



Illustration 369

g06223894

Push the release button on the buckle to release the seat belt. The seat belt will automatically retract into the retractor.

Extension of the Seat Belt

WARNING

When using retractable seat belts, do not use seat belt extensions, or personal injury or death can result.

The retractor system may or may not lock up depending on the length of the extension and the size of the person. If the retractor does not lock up, the seat belt will not retain the person.

Longer, non-retractable seat belts and extensions for the non-retractable seat belts are available.

Caterpillar requires only non-retractable seat belts to be used with a seat belt extension.

Consult your Cat dealer for longer seat belts and for information on extending the seat belts.

i07098988

Selective Catalytic Reduction Warning System

SMCS Code: 1091-WXX; 7400

The selective catalytic reduction (SCR) system is a system used to reduce NOx emissions from the engine. Diesel exhaust fluid (DEF) is pumped from the DEF tank and is sprayed into the exhaust stream. The DEF reacts with the SCR catalyst to reduce NOx and leaves a nitrogen and water vapor. The Exhaust Gas Recirculation (EGR) system cools, measures, and introduces recirculated exhaust gas into the intake manifold to aid in NOx reduction.

NOTICE

Stopping the engine immediately after the engine has been working under load can result in overheating of SCR components.

Refer to the Operation and Maintenance Manual, "Engine Stopping" procedure to allow the engine to cool and to prevent excessive temperatures in the turbocharger housing and the DEF injector.

Note: With the operator key in the OFF position, the purge system will activate.

Definitions

Observe the following definitions.

Self-correct – Fault condition no longer exists. An active fault code will no longer be active.

Notification – Action taken by the system to alert the operator of pending Inducement.

Inducement – Engine derates, vehicle speed limits, or other actions intended to prompt the operator to repair or maintain the emission control system.

Inducement Trigger – Fault conditions that result in activation of the inducement strategy. DEF level inducement faults have a diagnostic fault code. DEF quality fault, SCR tampering fault, SCR system fault, and EGR system faults will all have a related diagnostic fault code along with an inducement diagnostic fault code.

First occurrence – A DEF quality fault, SCR tampering fault, SCR system fault, or an EGR system fault becomes active for the first time.

Repeat occurrence – If any DEF quality fault, SCR tampering fault, SCR system fault, or an EGR system fault becomes active again within 40 hours of the first occurrence.

Safe Harbor Mode – Safe harbor mode is a 20 minute engine run time period. The engine can be operated with full power after reaching a level 3 inducement. Once in level 3 inducement, the operator can perform a key cycle and the engine will enter safe harbor mode. Safe harbor mode can only be implemented once. Safe harbor mode is not allowed for DEF level inducements.



Illustration 370

g06223861

DEF Level Normal

Inducement Strategy for DEF Level

If multiple warnings are present in the system, the most important problem is shown first. Press the right key or press the left key to view all the warnings that are present in the machine. If no keys are pressed within 5 seconds, the display will return to the most important problem.

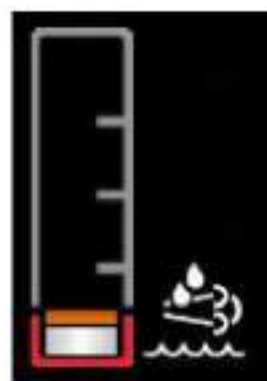


Illustration 371

g06223864

DEF Level Low

If the DEF level falls below 19%, "DEF Level Low" message will be displayed on the monitor. To avoid further inducements, turn the key to the OFF position and add DEF to the DEF tank.

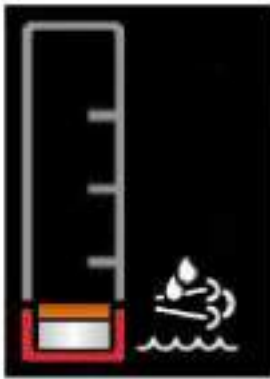


Illustration 372

g06223865

DEF Level Low and Emission Fault

If the DEF level falls below 12.5%, a level 1 inducement event will occur. a “DEF Level Low” message and an “Emission Fault” message will appear on the monitor.

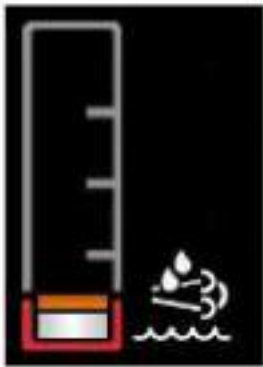


Illustration 373

g06223866

DEF Level Low, Emission Fault, and Flashing Action Lamp

If the DEF level is below 6%, a level 2 inducement event will occur. a “DEF Level Low” message and an “Emission Fault” message will appear on the monitor. The action lamp on the monitor will flash. If the ECM is configured to “Reduced Performance” and the DEF level has reached 0%, the machine will be limited to 75% torque.



Illustration 374

g06223883

DEF Level Low, Emission Fault, Flashing Action Lamp, and Audible Alarm

If the ECM is configured to “Reduced Performance” and the DEF tank has been emptied of all DEF, the engine will be in a level 3 final inducement. If the ECM is configured to “Reduced Time” and the DEF level is 0%, the engine will be in a level 3 final inducement. Prior to final inducement a “DEF Level Low” message, and an “Emission Fault” message will appear on the monitor. The action lamp on the monitor will flash and an audible alarm will sound 20 seconds prior to the final inducement. The engine will be taken to low idle or will be shut down. Once shut down, the engine can be restarted for 5 minute periods at reduced speed and torque. If set to idle, the engine will idle indefinitely at reduced torque. The amber indicator next to the DEF level gauge on the dash will remain lit.

Note: Turn the key to the OFF position and add DEF to the DEF tank to reset the DEF level inducement.

Inducement Strategy for DEF Quality, Tampering, SCR System Fault, and Impeded EGR

If multiple warnings are present in the system, the most important problem is shown first. Press the right key or press the left key to view all the warnings that are present in the machine. If no keys are pressed within 5 seconds, the display will return to the most important problem.



Illustration 375

g03623190

A “Engine Fault Check Engine” message and an “Emission Fault” message will appear on the monitor for a fault resulting from poor DEF quality, SCR system tampering, SCR system fault, or an EGR system fault. If the fault is the result of poor DEF quality, SCR system tampering, or SCR system fault, a first occurrence will result in a level 1 inducement for a duration of 2.5 hours. The level 1 inducement will also illuminate the check engine lamp. Repeat occurrences will result in a level 1 inducement duration of 5 minutes. If the fault is the result of an EGR system fault, a first occurrence will result in a level 1 inducement for a duration of 35 hours. Repeat occurrences will result in a level 1 inducement duration of 48 minutes.

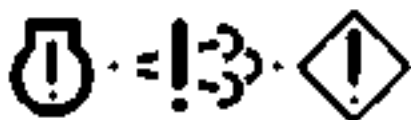


Illustration 376

g03623191

If a fault condition exists for the entire duration of inducement level 1, the strategy advances to inducement level 2. A “Engine Fault Check Engine” message, and an “Emission Fault” message will appear on the monitor and the action lamp on the monitor will flash for a fault resulting from poor DEF quality, SCR system tampering, SCR system fault, or an EGR system fault. For poor DEF quality, SCR system tampering, and SCR system faults, the duration for a level 2 inducement is 70 minutes for the first occurrence. For EGR system faults, the duration for a level 2 inducement is 60 minutes for the first occurrence. Repeat occurrences for poor DEF quality, SCR system tampering, and SCR system faults will result in a level 2 inducement of 5 minutes. Repeat occurrences for EGR system faults will result in a level 2 inducement of 60 minutes.

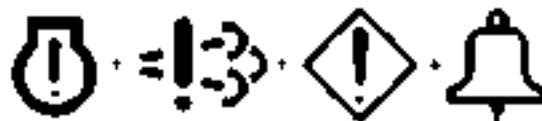


Illustration 377

g03623193

If a fault condition exists for the entire duration of inducement level 2, the strategy advances to inducement level 3. A “Engine Fault Check Engine” message, and an “Emission Fault” message will appear on the monitor, the action lamp will flash, and an audible alarm will sound 20 seconds prior to the level 3 inducement. The engine will be taken to low idle or will be shut down. After the level 3 inducement you may cycle the key, which will allow 20 minutes of engine run time with full torque. After 20 minutes, the engine will be in level 3 final inducement and will allow idle only or be shut down until the issue has been resolved. Once shut down, the engine can be restarted for 5 minute periods at reduced speed and torque. If set to idle, the engine will idle indefinitely at reduced torque.

Note: Contact your Cat dealer for repairs if a fault occurs.

i08036134

Operator Controls

SMCS Code: 7300; 7301; 7451

Note: Your machine may not be equipped with all the controls that are described in this topic.

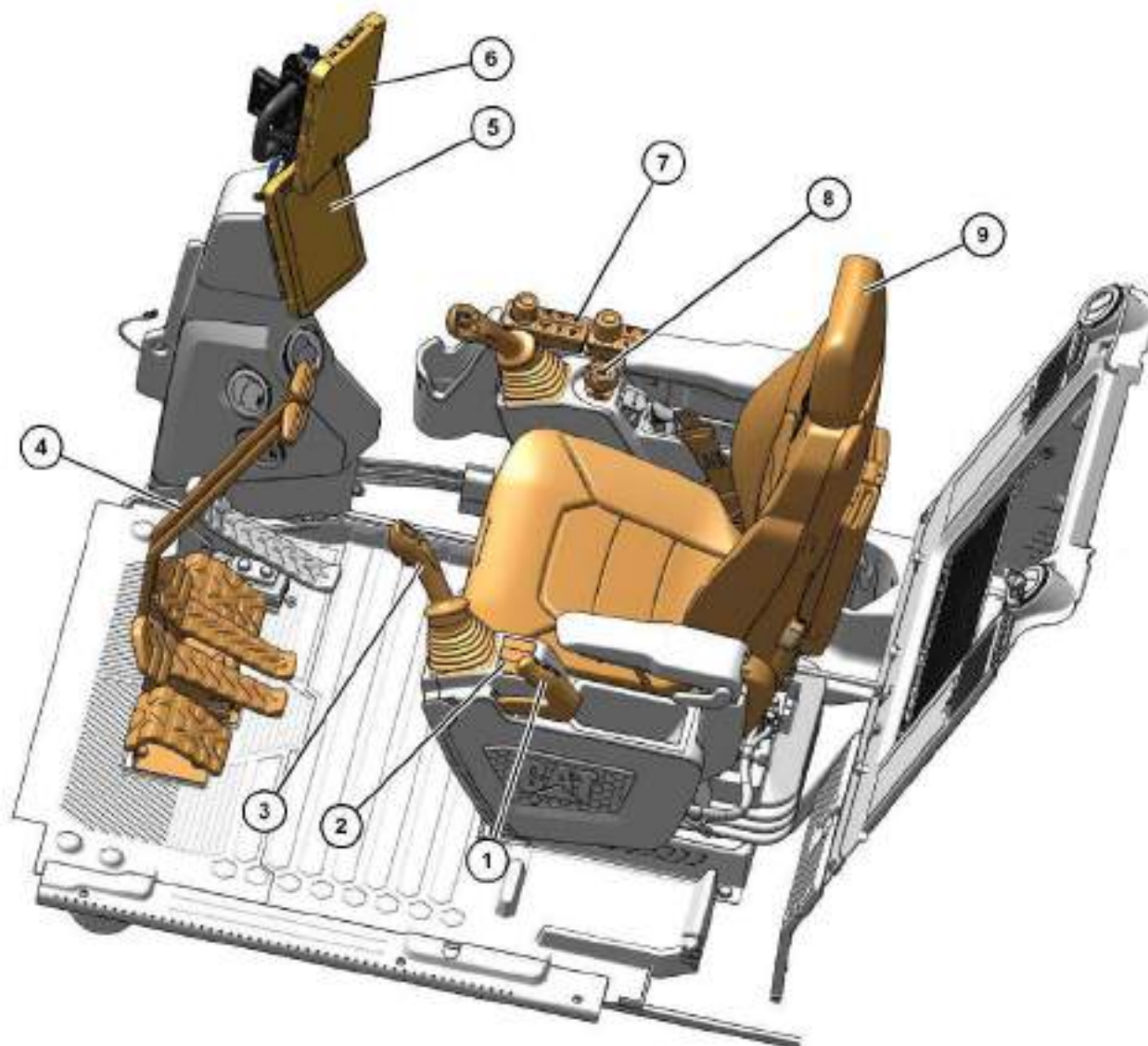


Illustration 378

g06178344

- (1) Hydraulic lockout control
 (2) Left side switch panel
 (3) Joystick controls

- (4) Travel controls
 (5) Monitor
 (6) Cat Grade Control Monitor (If Equipped)

- (7) Right side switch panel
 (8) Engine start switch
 (9) Operator seat

Hydraulic Lockout Control (1)

The lever for the hydraulic lockout control is at the left side of the left console.



Locked – Move the travel levers/pedals and move the joysticks to the HOLD (center) position. Move the lever for the hydraulic lockout control backward to the LOCKED position. All the factory installed hydraulic controls will become inoperable.



Unlocked – Move the lever for the hydraulic lockout control forward to the UNLOCKED position. All the factory installed hydraulic controls will become operable.

Note: Make sure that the lever for the hydraulic lockout control is in the LOCKED position before attempting to start the engine. If the lever is in the UNLOCKED position, the engine start switch will not function.



Tilt – For machines equipped with the tilt-up console, pull the lever to the rearmost position to release the console lock and tilt the console upward for easier exit and entry.

Left Side Switch Panel (2)



Illustration 379

g06219690

Beacon Light Switch (2A) (If equipped)



Beacon Light Switch – Push the top of the switch to turn on the beacon light. Push the bottom of the switch to turn off the beacon light.

Quick Coupler Control (2B) (If equipped)



If equipped, the switch for the quick coupler control is on the left console. The switch is equipped with a spring loaded lock button. To operate the switch, the lock button must be pushed forward to release the switch. With the lock held forward, press the rear of the switch downward to uncouple the bucket or work tool. Press the button again to attach the bucket or work tool.

Note: An alarm will sound whenever the switch has been activated to lock or unlock a work tool.

For further details, see Operation and Maintenance Manual, “Quick Coupler Operation”.

Joystick Controls (3)

The joystick control is used to control the functions of the machine implements. For more information on the individual functions of the joysticks, refer to Operation and Maintenance Manual, “Joystick Controls”.

Travel Control (4)



Illustration 380

g06178249

Position for normal travel

- (A) Rear of machine
- (B) Final drive
- (C) Idler

When you travel, make sure that final drive sprockets (B) are under the rear of the machine.

Stop – Release the travel levers/pedals to stop the machine. When you release the travel levers/pedals from any position, the travel levers/pedals will return to the CENTER position. The travel brakes will be applied.

Move both of the travel levers or both of the travel pedals equally in the same direction to travel straight.

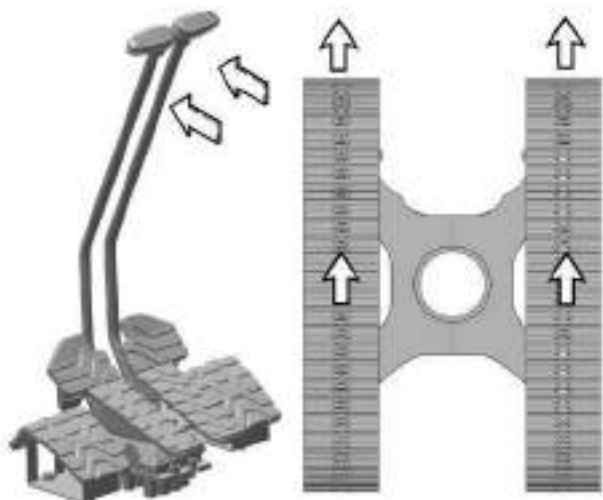


Illustration 381
FORWARD travel

g06178269

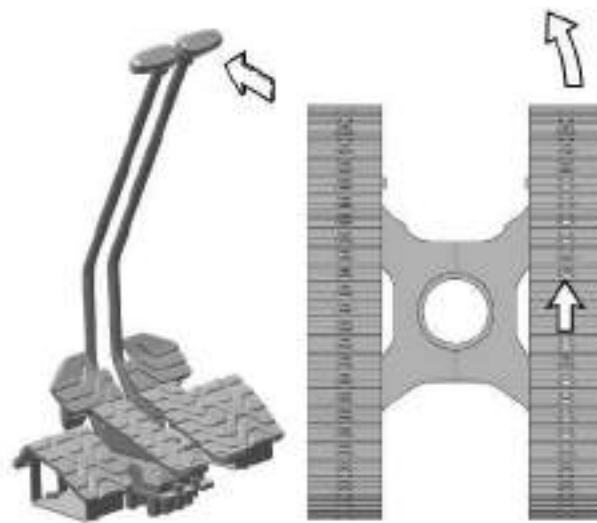


Illustration 383
Pivot left turn (FORWARD)

g06178288

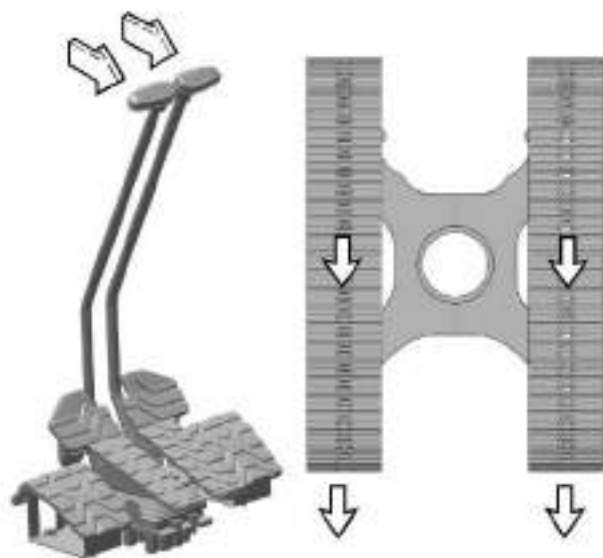


Illustration 382
REVERSE travel

g06178283

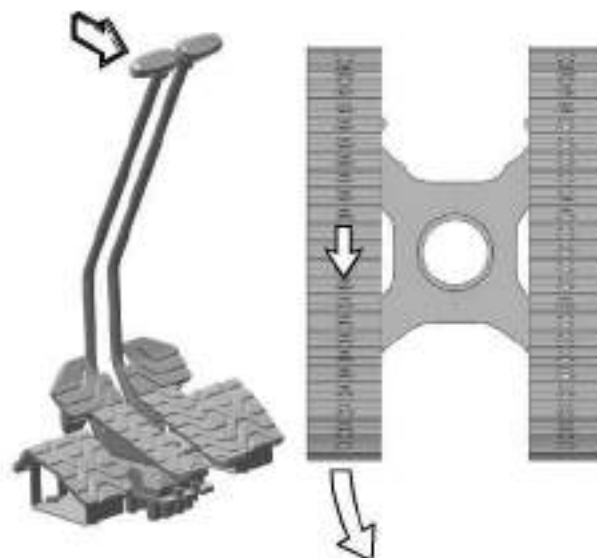


Illustration 384
Pivot Left Turn (REVERSE)

g06178294

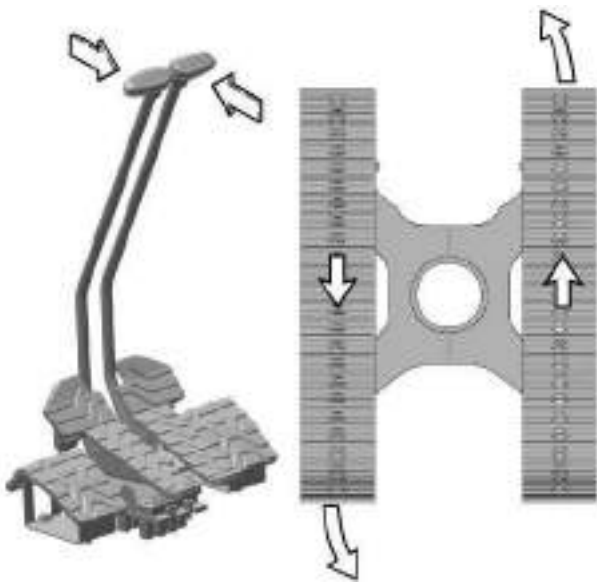


Illustration 385

g06178300

Counterrotate turn (LEFT)

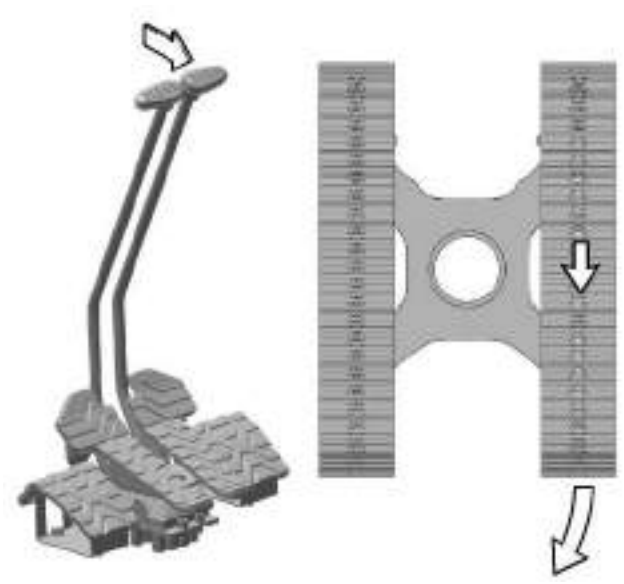


Illustration 387

g06178308

Pivot right turn (REVERSE)

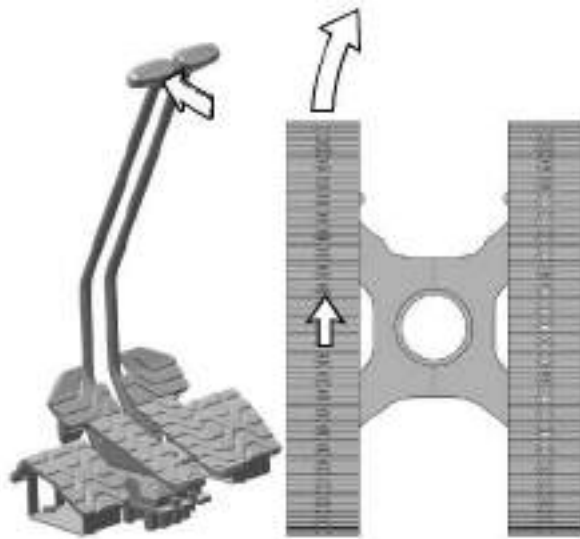


Illustration 386

g06178305

Pivot right turn (FORWARD)

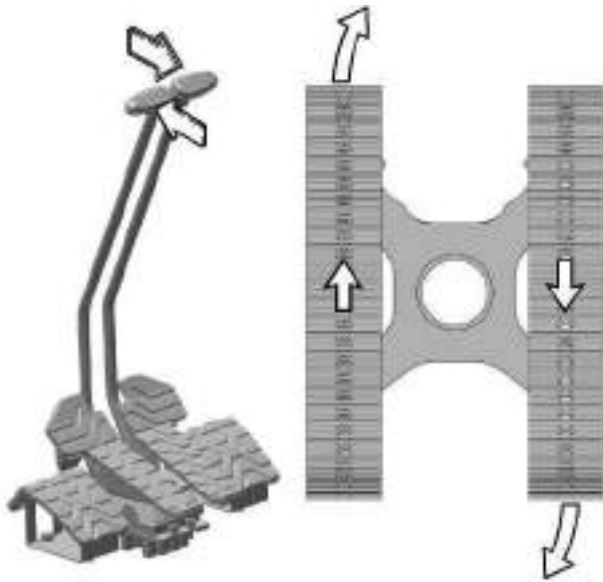


Illustration 388

g06178313

Counterrotate turn (RIGHT)

Monitor (5)

The monitor is used to display various operating information of the machine. For more information on the operation of the monitor, refer to Operation and Maintenance Manual, "Monitoring System".

Cat Grade Control Monitor (6) (if equipped)

Monitor (6) is used to display Cat Grade Control operating information. Cat Grade Control is designed to calculate bucket tip position and provide visual and audible feedback to the operator. This feedback allows the operator to achieve the desired grade safer, quicker and more accurately than traditional grading techniques. For more information on the operation of the monitor, refer to Operation and Maintenance Manual, M0082987, "Cat 2D and 3D GRADE System for Next Gen Hydraulic Excavators".

Engine Start Switch (8)

NOTICE

The engine start switch must be in the ON position and the engine must be running in order to maintain electrical functions and hydraulic functions. This procedure must be followed in order to prevent serious machine damage.

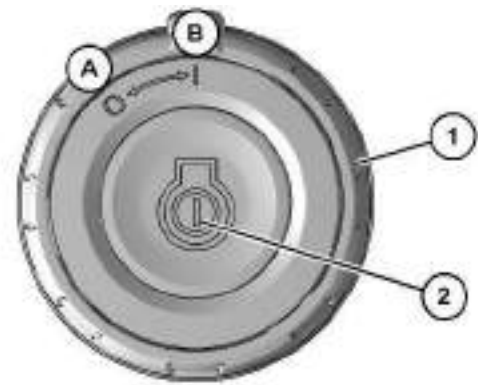


Illustration 389

g06180554

- (A) Off
(B) On
(1) Engine start ring
(2) Start button



OFF – Turn the engine start ring (1) to the OFF position (A) to stop the engine.



ON – To activate the electrical circuits in the cab and enable engine starting, turn the engine start ring (1) clockwise to the ON position (B).



START – To start the engine, enter the code on the monitor. Press start button (2). After the engine starts, release the button.

Note: Pressing the start button with the engine on will also turn off the engine.

Engine Idle Shutdown

If the operator has not operated the machine for a period of time, this function shuts down the engine. Engine Idle Shutdown does not shut down other systems, such as the AC, which can run down the battery after idle shutdown. This function comes disabled from the factory but can be enabled and adjusted in the monitoring system. Refer to Operation and Maintenance Manual, "Monitoring System".

Note: Engine Idle Shutdown may be required for local regulations.

The Engine Idle Shutdown (EIS) shuts down the engine if the following conditions are met:

- The control levers are in neutral.
- The engine coolant temperature is greater than 70° C (158° F).
- The battery voltage is greater than 24.5V
- The engine speed is less than 2000 rpm.

- The ambient temperature range is between 0° C (32° F) and 30° C (86° F).
- The Back Pressure Valve (BPV) is not working.

Note: If any service tests or calibrations are running, the machine will not enter into engine idle shutdown.



Engine Idle Shut down – The action lamp will illuminate and the monitor will display a message 20 seconds before the engine shuts down. An operator can cancel the shutdown by pressing a button on the monitor or move one of the controls.

Operator's Seat (9)

There are different options for operators seats. Each operator seat and console have various adjustments to meet a wide range of operators. For more information, refer to Operation and Maintenance Manual, "Seat".

Tilt-Up Console (9A)

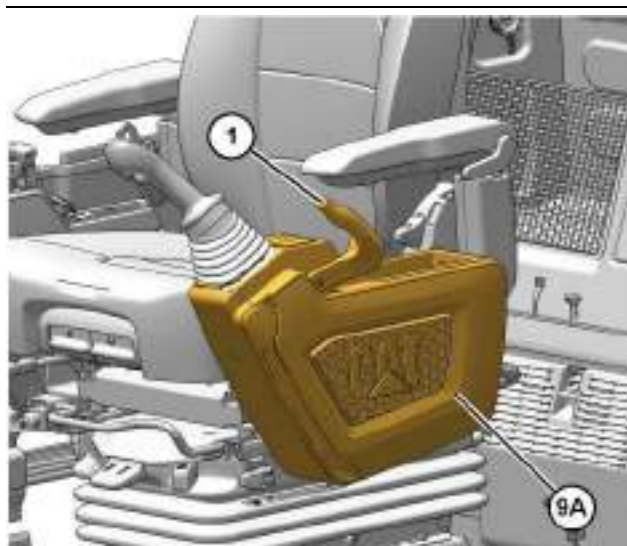


Illustration 390

g06226523

- (1) Hydraulic lockout control
(2) Tilt-up console

Some optional seats are equipped with a tilt-up console. The console can be tilted upward for easier exit and entry. The console is unlocked by pulling the hydraulic lockout control (1) to the rearmost position. The console will then tilt upward. Simply push the console downward until the console locks into place when ready for use.

Right Side Switch Panel (7)

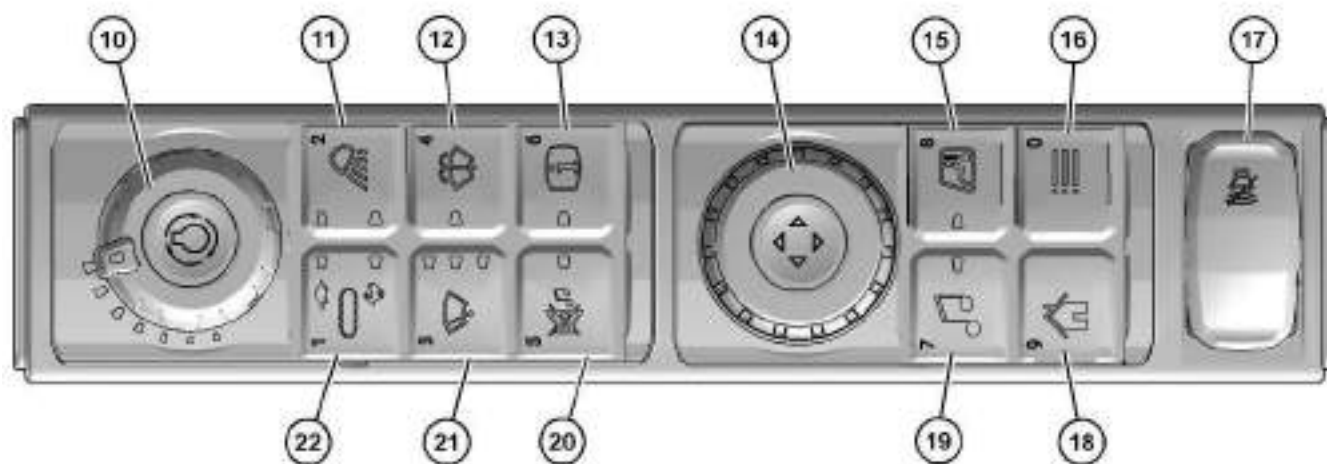


Illustration 391

g06178333

Right side switch panel

- | | | |
|--|--------------------------------------|---------------------------|
| (10) Engine speed / power mode control | (15) Heating and air conditioning | (20) Radio mute switch |
| (11) Light switch | (16) Next menu | (21) Window wiper |
| (12) Window washer | (17) Travel alarm mute (If Equipped) | (22) Travel speed control |
| (13) Operator Information | (18) Home | |
| (14) Jog dial | (19) Radio control | |

Note: In addition to the intended functions, the buttons on the switch panel are numbered from 0-9. The numbered buttons can be used to enter numbers into the monitor for screens such as the passcode screen.

Engine Speed / Power Mode Control (10)

Engine Speed Control – Turn the dial to control the engine speed (engine rpm). Select the desired position from the seven available positions. Turn the dial counterclockwise to decrease the engine speed (engine rpm). Turn the dial clockwise to increase the engine speed (engine rpm).



Power Mode Control – Push in the dial to change the power mode settings. The Power Mode Control allows the operator to choose what power mode to operate the engine. The modes that can be selected are: “ECONOMY”, “SMART”, and “POWER”.

Note: The default power mode setting can be set within the monitor. For more information, refer to Operation and Maintenance Manual, “Monitoring System”.

Note: “ECONOMY” mode is **not** available on GC models.

Light Switch (11)

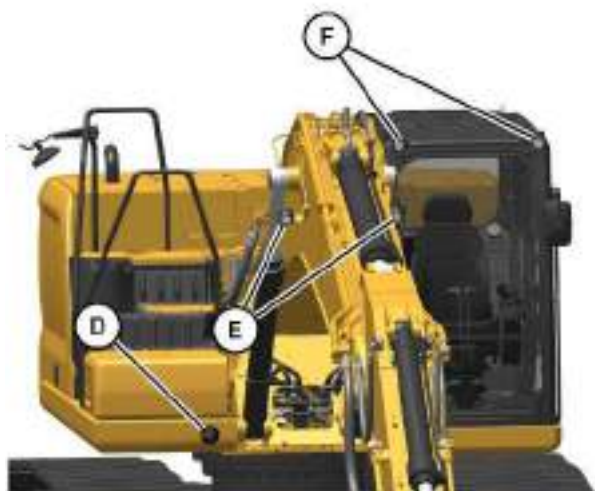


Illustration 392

g06178337



Light Switch – Push the switch to turn on the work lights.

Whenever you push the switch, you change the pattern of the work lights that are turned on. The indicator lights in the cab indicate the pattern of the work lights.

Pattern 1 – When you press the light switch once, the first indicator light turns on. When the first indicator light is on, the following work lights are turned on: work light (D), which is mounted on the chassis, and work lights (F), which are mounted on the cab.

Pattern 2 – When you press the light switch twice, the first indicator light and the second indicator light turn on. When the first indicator light and the second indicator lights are on, the following work lights are turned on: work light (D), which is mounted on the chassis, work lights (F), which are mounted on the cab, and work lights (E), which are mounted on the boom.

OFF – When both of the indicator lights are off, all the work lights are off.

Note: Your machine may be equipped with a premium surrounding lighting package with left side, right side and rear lights. Refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”, Function List Screen, Work Light Control for more information.

Note: Your machine may be equipped with a lighting system that has a time delay. When this system is installed, the exterior lights will not turn off for a predetermined amount of time after the engine start switch has been turned to the OFF position. Refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”, Application Menu, Lighting Shutdown Timer for more information.

Window Washer (12)



Window Washer (12) – Push the switch to activate the window washer. While the switch is depressed, the indicator light will come on and washer fluid will spray from the nozzle. The window wiper will also operate while the switch is depressed. After the switch is released for approximately 3 seconds, the window wiper will stop.

NOTICE

If the wiper does not operate with the switch in the ON position, turn the switch off immediately. Check the cause. If the switch remains on, motor failure can result.

NOTICE

If the washer is used continuously for more than 20 seconds or used when no washer solution comes out, motor failure can result.

Operator Information (13)



Operator Information Button (13) – Press and hold this button to view the operator information screen. The indicator light will illuminate when the button is pressed.

Jog Dial (14)

The jog dial can be used to select items displayed on the monitor screen. The dial can be rotated 360 degrees. The dial can also be moved left, right, up, and down. The dial can be pushed in to make a selection.

Air Conditioning and Heating (15)



Air Conditioning and Heating (15) – Press this button to bring up the air conditioning and heating menu. The indicator light will illuminate when the heating and cooling system is active. The jog dial (13) can be used to make selections. If equipped with a touch screen, the selections can be made by touching the monitor.

Refer to Air Conditioning and Heating Control for more information.

Next Menu (16)



Next Menu (16) – Press the next menu button to access the next higher menu. If there is not a menu above the current screen being viewed, the button will not do anything.

Refer to Monitoring System for more information.

Travel Alarm Mute Switch (17) (If Equipped)



Travel Alarm Mute Switch (17) – Press travel alarm mute switch (17) to mute the travel alarm.

Note: The travel alarm will sound when the travel levers or the travel pedals are activated.

Home (18)



Home Key (18) – Press the home key to return to the default display at any time.

Refer to Monitoring System for more information.

Radio Control (19)



Radio Control (19) – Press this button to display the radio controls on the monitor. The indicator light on the button will illuminate when the radio is turned on. Use jog dial (14) to make selections. If equipped with a touch screen display, touch the icons on the screens to make a selection.

Refer to Radio for more information.

Radio Mute (20)



Radio Mute (20) – Press radio mute switch (20) to mute the radio. The indicator light will illuminate when mute is activated. Press the button again to unmute the radio.

Window Wiper (21)



Window Wiper (21) – Push the switch to activate the window wiper. Whenever the switch is depressed, the mode of the window wiper will change according to the indicator light that is illuminated.

6 Second Delay – When the window wiper switch is depressed one time, the first indicator light will turn on. The window wiper will operate intermittently at six second intervals.

3 Second Delay – When the window wiper switch is depressed two times, the second indicator light will turn on. The window wiper will operate intermittently at three second intervals.

Continuous Operation – When the window wiper switch is depressed three times, the first indicator light and the second indicator light will turn on. The window wiper will operate continuously.

OFF – When the window wiper switch is depressed four times, the indicator lights will turn off. The window wiper stops.

Travel Speed Control (22)

WARNING

Do not change the setting of the travel speed control switch while you travel. Machine stability may be adversely affected.

Personal injury can result from sudden changes in machine stability.



Travel Speed Control Switch (22) – Press the travel speed control switch to select automatic travel speed or low travel speed. When the engine start switch is on, the travel speed control switch is always set at the LOW SPEED position. Whenever the travel speed control switch is pressed, the travel speed changes. The indicator lights illuminate to show which speed selection is active.



LOW SPEED – Select the LOW SPEED position if you travel on rough surfaces or on soft surfaces or if you require a great drawbar pull. Also, select the LOW SPEED position if you are loading a machine onto a trailer or you are unloading a machine from a trailer.



AUTOMATIC – If you travel on a hard, level surface at a fast speed, select the AUTO position.

Continuous driving at high speed should be limited to 2 hours. If you need to continue driving at high speed for more than 2 hours, stop the machine for 10 minutes. This process will cool down the travel drives before you resume driving.

USB/Aux Ports (23)



Illustration 393

g06204950

(23) USB/AUX/MIC port
(24) 12 V socket



USB – The USB port is used to play music from a portable device. The USB symbol on the radio screen on the monitor must be selected.

AUX – The AUX port is used to play music from a portable device. AUX must be selected on the radio screen on the monitor.

12V Power Receptacle (24) & (25)

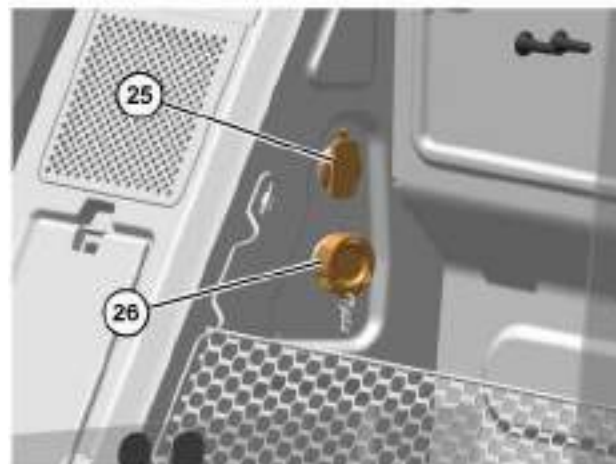


Illustration 394

g06178354

Electronic Technician service port



12V Power Receptacle – The power receptacles can be used to power automotive electrical equipment or accessories. The power receptacle only operates when the engine start switch is in the ON position.

Service Port (26)

An Electronic Technician (ET) service port is located inside the cab behind the seat. This service port allows service personnel to connect a laptop computer that is equipped with Electronic Technician. Service personnel can use electronic technician to diagnose machine and engine systems.

Contact your Cat dealer for additional information.

Dome Light (27)



Illustration 395

g06208645

The dome light has three different positions. When the dome light is in the center position (horizontal), the light will come on when the door is open and shut off when the door is closed.

When the left side of the light is pressed, the lamp will be inoperable.

When the right side of the light is pressed, the lamp will be illuminated until the lamp is switched to another position.

i07945377

Battery Disconnect Switch

SMCS Code: 1411-B11



Illustration 396

g06179792

The battery disconnect switch is on the left side of the machine behind the rear access door.

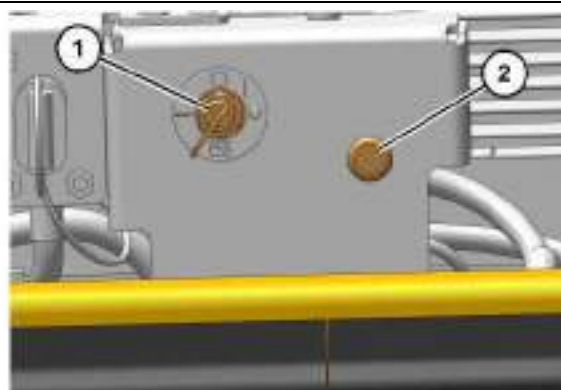


Illustration 397

g06179801

- (1) Battery disconnect switch
- (2) DEF purge light

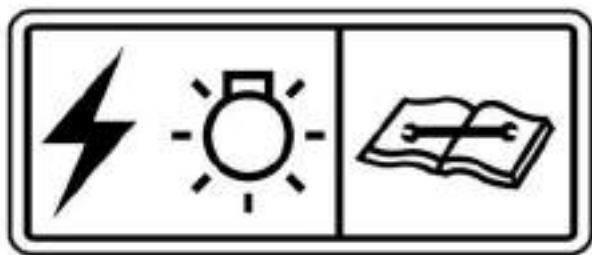


Illustration 398

g03796564

For machines equipped with a circuit that stays activated for DEF purge with the disconnect switch off.

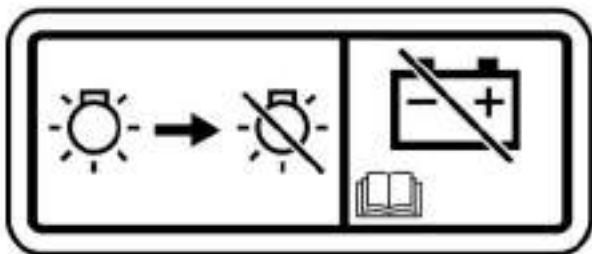


Illustration 399

g03408962

For machines not equipped with a circuit that stays activated for DEF purge with the disconnect switch off.

NOTICE

Do not conduct any service procedures on the DEF system until the DEF purge indicator lamp is not illuminated. The indicator lamp may remain illuminated for several minutes, even though the battery disconnect switch is OFF and the engine start switch is OFF. When the indicator lamp is on, the DEF system is still powered.



Battery Disconnect Switch – The battery disconnect switch can be used to disconnect the battery from the machines electrical system. The key must be inserted into the battery disconnect switch before the battery disconnect switch can be turned.



ON – To activate the electrical system, insert the disconnect switch key and turn the battery disconnect switch clockwise. The battery disconnect switch must be turned to the ON to enable battery power to start the engine.



OFF – To deactivate the electrical system, turn the battery disconnect switch counterclockwise to the OFF position.

The battery disconnect switch and the engine start switch perform different functions. The entire electrical system is disabled when you turn the battery disconnect switch to the OFF position. The battery remains connected to the electrical system when you turn the engine start switch to the OFF position.

Turn the battery disconnect switch to the OFF position and remove the key when you service the electrical system or any other machine components. If installed with a cover lock, close the cover and install a padlock.

Turn the battery disconnect switch to the OFF position and remove the key if you do not operate the machine for extended periods of a month or more. Turning off the disconnect switch will prevent the battery from being discharged.

A good practice is to use the disconnect switch after you operate the machine. Turning off the disconnect switch will prevent the battery from being discharged. The following problems can cause battery discharge:

- short circuits
- current draw via some components
- vandalism

Machines equipped with circuit for DEF purge: A special circuit stays energized with the disconnect switch in the OFF position. The Diesel Exhaust Fluid (DEF) purge circuit is supplied power from this circuit with the disconnect switch off. The special circuit ensures that the system will purge and no damage will be done to engine components.

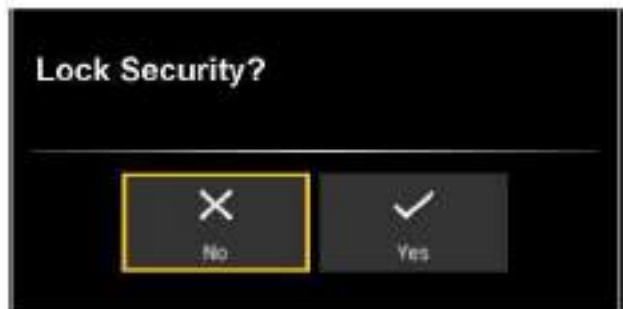


Illustration 400

g06217247

Note: Do not turn off the battery disconnect switch until 5 seconds have elapsed after turning the engine start switch to the OFF position. Do not turn off the battery disconnect switch while the “Lock Security?” screen is displayed on the monitor. Both conditions would prevent the Diesel Exhaust Fluid (DEF) system from being purged and could cause DEF fluid to freeze in the lines.

NOTICE

Do not conduct any service procedures on the DEF system until the DEF purge indicator lamp is not illuminated. The indicator lamp may remain illuminated for several minutes, even though the battery disconnect switch is OFF and the engine start switch is OFF. When the indicator lamp is on, the DEF system is still powered.

Machines not equipped with circuit for DEF

purge: If DEF purge light (2) is illuminated, wait for the light to turn off before turning the disconnect switch to the OFF position. Waiting for the light to turn off allows the DEF system to fully purge the exhaust fluid and prevents damage to engine components.

i08001446

Product Link

SMCS Code: 7490; 7606

Note: Your machine may be equipped with the Cat® Product Link™ system.

The Cat Product Link communication device utilizes cellular and/or satellite technology to communicate equipment information. This information is communicated to Caterpillar, Cat dealers, and Caterpillar customers. The Cat Product Link communication device uses Global Positioning System (GPS) satellite receivers.

The capability of two-way communication between the equipment and a remote user is available with the Cat Product Link communication device. The remote user can be a dealer or a customer.

Data Broadcasts

Data concerning this machine, the condition of the machine, and the operation of the machine is being transmitted by Cat Product Link to Caterpillar and/or Cat dealers. The data is used to serve the customer better and to improve upon Cat products and services. The information transmitted may include: machine serial number, machine location, and operational data, including but not limited to: fault codes, emissions data, fuel usage, service meter hours, software, and hardware version numbers and installed attachments.

Caterpillar and/or Cat dealers may use this information for various purposes. Refer to the following list for possible uses:

- Providing services to the customer and/or the machine
- Checking or maintaining Cat Product Link equipment
- Monitoring the health of the machine or performance
- Helping maintain the machine and/or improve the efficiency of the machine
- Evaluating or improving Cat products and services
- Complying with legal requirements and valid court orders
- Performing market research
- Offering the customer new products and services

Caterpillar may share some or all the collected information with Caterpillar affiliated companies, dealers, and authorized representatives. Caterpillar will not sell or rent collected information to any other third party and will exercise reasonable efforts to keep the information secure. Caterpillar recognizes and respects customer privacy. For more information, please contact your local Cat dealer.

Operation in a Blast Site for Product Link Radios

WARNING

This equipment is equipped with a Cat® Product Link communication device. When electric detonators are being used for blasting operations, radio frequency devices can cause interference with electric detonators for blasting operations which can result in serious injury or death. The Product Link communication device should be deactivated within the distance mandated under all applicable national or local regulatory requirements. In the absence of any regulatory requirements Caterpillar recommends the end user perform their own risk assessment to determine safe operating distance.

Refer to your products Operation and Maintenance Manual Supplement, "Regulatory Compliance Information" for more information.

For information regarding the methods to disable the Cat Product Link communication device, please refer to your specific Cat Product Link manual listed below:

- Operation and Maintenance Manual, SEBU8142, "Product Link - PL121, PL321, PL522, and PL523"
- Operation and Maintenance Manual, SEBU8832, "Product Link PLE702, PLE602, PLE601, PL641, PL631, PL542, PL240, PL241, PL243, PL141, PL131, PL161, PL083 and PL042 Systems"

Note: If no radio disable switch is installed and the equipment will be operating near a blast zone, a Product Link radio disable switch may be installed on the equipment. The switch will allow the Cat Product Link communication device to be shut off by the operator from the equipment control panel. For more details and installation procedures, refer to the following:

- Special Instruction, REHS7339, "Installation Procedure for Product Link PLE640 Systems"
- Special Instruction, REHS8850, "Installation Procedure for the Elite Product Link PLE601, PLE641, and PLE631 Systems"
- Special Instruction, SEHS0377, "Installation Procedure for the Product Link PL131, PL141, and PL161 Systems"

- Special Instruction, REHS9111, "Installation Procedure for the Pro Product Link PL641 and PL631 Systems"
- Special Instruction, M0098124, "Installation Procedure for Pro Product Link PL243 Systems"
- Special Instruction, M0109130, "Installation Procedure for Product Link PL683 and PL783 Systems"

i07785958

Machine Security System

SMCS Code: 7631

General Information

NOTICE

This machine may be equipped with a Cat® Machine Security System (MSS) and may not start under certain conditions.



Illustration 401

g06223917

Machines that are equipped with Cat MSS can be identified by a decal in the operator station. Read the following information and know your machines settings. Your Cat dealer can identify your machine settings.

The Cat Machine Security System (MSS) discourages unwanted operation of a machine. When armed, the MSS requires operator login to start the engine. The following methods of operator login to disarm the security system are available:

- Cat Bluetooth® key fob
- Cat App: Fleet Management
- Passcode

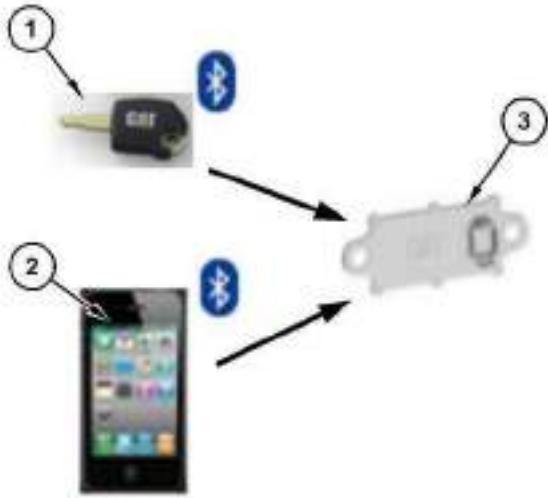


Illustration 402

g06212167

Bluetooth Connections

- (1) Cat Bluetooth key fob (CATBTFOB)
- (2) Cat App: Fleet Management mobile application
- (3) Cat Bluetooth transceiver (CATBTNT)

The Cat Bluetooth key fob (1) contains an electronic chip. The electronic chip has a unique identification number (ID). A Bluetooth transceiver is mounted in the cab to read the ID of the key. The Bluetooth transceiver module translates the information received from the key fob into a J1939 message. This message is sent to the Electronic Control Module (ECM) that is connected to the MSS. The ECM is typically the Machine ECM. The ECM is set up with the ID of the keys of the intended users.

When the MSS is armed, the ECM validates the ID of the key fob. If the key ID is on the list of authorized keys in the ECM and the key is valid, the machine will operate normally. If the key ID is not on the list of authorized keys in the ECM or is not valid, the MSS will keep the critical machine functions disabled.

Note: A Bluetooth enabled phone can disarm MSS if the phone is on the list. Operator Management System (OMS) is necessary to be able to add phones to the vehicle ID list. After the phone is added, OMS is no longer needed for the Cat App: Fleet management mobile application to function as a valid key.

If the MSS is not installed, the operator can skip the login and the machine will operate normally.

Components

The Machine Security System (MSS) consists of the following components:

- Electronic Control Module (ECM)

- Cat Bluetooth key fob (CATBTFOB)
- Machine display
- Bluetooth transceiver module (CATBTNT)
- Engine start switch

System Overview

The Machine Security System (MSS) is designed to restrict operation of a machine. A list of the authorized electronic keys and passcodes for a machine is contained in the ECM for the MSS. A valid Bluetooth key fob, mobile application, or passcode can disarm the MSS. If the MSS is disabled or not installed, any operator may access critical machine functions.

The Cat® Electronic Technician (Cat ET) Service Tool can be used to program the ECM with the authorized keys and passcodes. Bluetooth devices and passcodes can be registered using the in-cab display if the operator is logged in to the system using a master access account.

When the engine start switch is turned to the ON position, the display boots up. If Bluetooth detection is enabled, the transceiver will receive a signal from any Bluetooth key that is present or from the mobile application. The ECM will then compare this ID to the list of authorized keys.

Note: If multiple devices are present, the first valid device detected by the transceiver will be read.

If the ID of the key matches an authorized key, the status indicator on the engine start switch will turn a green color and the MSS will disarm. This disarming will allow the operator access to critical functions of the machine.

If the ID of the key that is read does not match the list in the ECM, the status indicator will remain a red color. The MSS remains in the “armed” state and the machine will remain disabled.

If the MSS is disabled and the ID of the key matches an authorized key, the operator will be identified and allowed access to the critical machine functions. The operator will be able to save configurations and start the machine.

If the MSS is disabled and the ID of the key that is read does not match an authorized key, the operator must log in as a guest. The operator will not be able to save custom configurations but will have access to starting the engine.

Activating Bluetooth Functionality

For shipping purposes, Bluetooth functionality is deactivated. Ensure that Bluetooth functionality is active on your machine using the following procedure:

1. Ensure that the function is active from the radio screen:

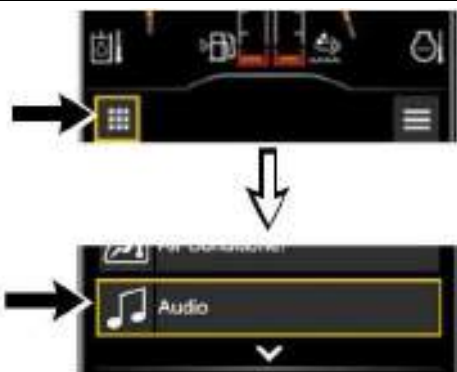


Illustration 403

g06319669

- a. From the home screen, press the navigation button in the lower left corner, then select "Audio" .

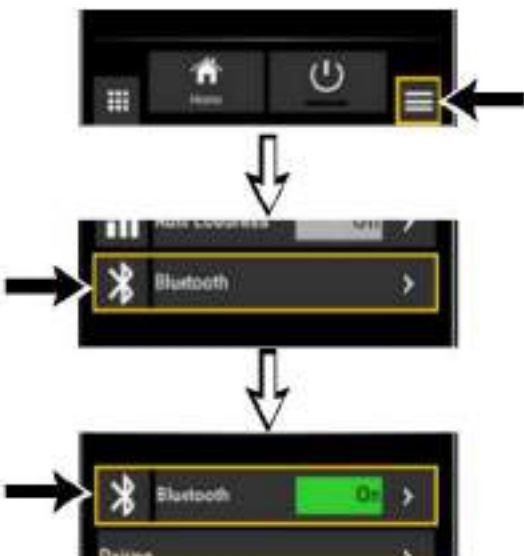


Illustration 404

g06319667

- b. Press the radio function list menu button in the lower right corner, then select "Bluetooth" . Ensure that "Bluetooth" is set to "ON" .

Pairing Your Device to the Machine

Use to following procedure to pair your device to the machine:

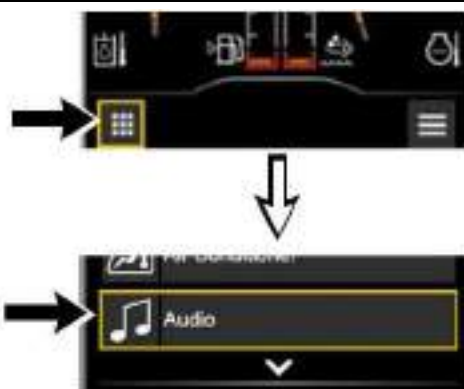


Illustration 405

g06319669

1. From the home screen, press the navigation button in the lower left corner, then select "Audio" .



Illustration 406

g06319672

2. Press the radio function list menu button in the lower right corner, then select "Bluetooth" .

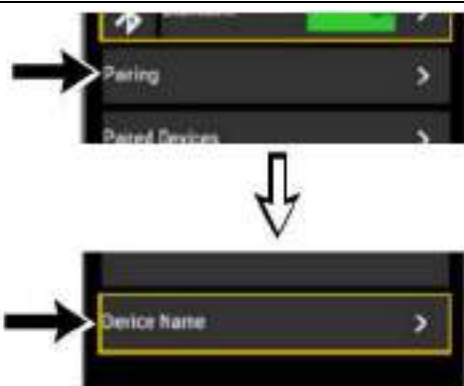


Illustration 407

g06319676

3. Select "Pairing" , then "Device Name" .



Illustration 408

g06319681

Machine name on monitoring system and operator device

4. Find your device on the list and pair the devices. Ensure that the devices are paired on your phone as well.

Note: The device name on your phone should be “## CAT RADIO”, with the number being from “00” to “99”.

Reading the ID of a Key

The Machine Security System (MSS) must identify a valid passcode, Bluetooth key fob ID, or Cat App: Fleet management mobile application ID.

When the engine start switch ring is turned to the ON position, the MSS will check the ID of any key fob or mobile application. If the ID matches a key ID stored in the machine ECM, the critical ECM functions are enabled. An enable message is also sent via the Cat[®] data link or J1939 data link to the other ECMs on the machine. The machine will operate normally.

Note: If the machine ECM has failed or has been removed, the critical machine operations controlled by the other electronic control modules will not operate.

Armed

When the MSS is armed, critical machine functions are disabled. The MSS disables the power that is supplied to each component that is powered by the output drivers. The machine will not be able to operate normally.

There are two states of operation within the “armed” mode:

Engine Start Switch Ring Position OFF – When no power is applied to the MSS, the MSS will default to “armed” state. When power is applied to the MSS and the grace period has expired, the MSS will return to the “MSS Armed”.

Engine Start Ring Switch Position ON – When the engine start switch ring is first moved to the ON position, the display boots up and the system attempts to detect a Bluetooth key ID or mobile application ID. The ECM will continue reading until a valid key ID is read or a passcode is entered. If a valid key ID or passcode is not read, the MSS status indicator will remain red and the MSS remains armed.

Disarmed

When the MSS is disarmed, normal machine operation is allowed. A message is sent to the other machine ECMs over the Cat data link, or J1939 data link. The machine will be able to start. The green LED on the status indicator will illuminate.

There are multiple ways to disarm the machine:

- Use a valid passcode
- Use a valid Bluetooth key fob
- Use the Cat App: Fleet management mobile application
- Use the Cat[®] Electronic Technician (Cat ET) Service Tool to configure the MSS bypass schedule to allow machine operations during scheduled periods of time during the week.

Grace Period

After a machine has been started successfully, the operator will have a grace period after the machine is turned off before the MSS is automatically armed. The operator is not required to arm the system manually.

During the grace period an operator can start the machine without a key ID or passcode. When the grace period expires, the MSS will rearm automatically.

If the MSS is unable to read a key ID, the system will remain armed. When the MSS identifies a key with an invalid key ID, the system will remain armed.

The grace period for a machine can be configured with Cat ET if a factory password or master level account is available.

Navigating the User Interface
Touchscreen Display

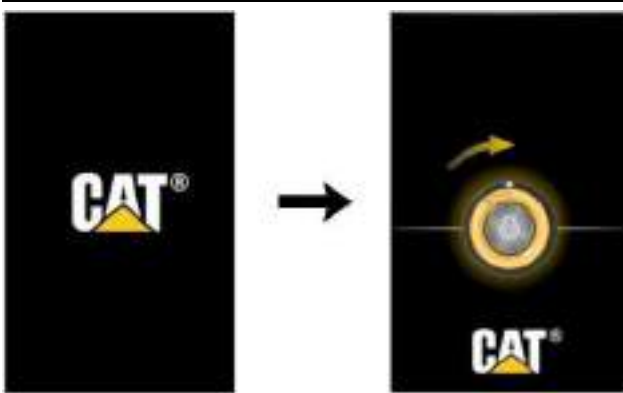


Illustration 409 g06210561

Startup sequence screens

The display will start up automatically after turning the battery disconnect switch to the ON position. The screen will prompt the operator to turn the engine start ring to the ON position. When the engine start ring is turned to the ON position, the display will navigate to the passcode entry screen.

If the operator has not turned the engine start ring to the ON position, the display will time out after 1 minute.

If the battery disconnect switch is already on and the display is off, the display will startup automatically after the engine start ring is powered on. The Cat screen will appear for a short time and then login keypad will appear.

Note: Avoid touching the screen with sharp objects.

The access level assigned to the operator can limit or expand the amount of freedom the user has to manage the system. The following paragraphs explain access levels.

There are three levels of operator access recognized by the touchscreen display. The following levels are available:

- Guest
- Standard
- Master

Guest – If an operator does not have an authenticated key or passcode, the user is able to bypass log in as a guest. Some menu features will not be available such as the options for saving

configurations and operator management. If the Machine Security System (MSS) is enabled, guest operators cannot start the machine.

Standard – A standard operator is a registered user of the machine. Operators with this access level can start the engine whether or not the MSS is installed. This user may save a control configuration for future application.

Master – Master accounts can perform operator management in addition to all standard level functions.

Any “Standard” or “Master” account may be created or removed by a “Master” level operator.

Table 26

Selections and Access for the Touchscreen Display	
Access Level	Operator Setting
Guest	“Operator Input Configuration” “Response” “Change Operator”
Standard	“Operator Input Configuration” “Response” “Controls Setup” “Change Operator”
Master	“Operator Input Configuration” “Response” “Controls Setup” “Change Operator” “Manage Operator”

Operation of Status Indicator



Illustration 410 g06215426

Engine start switch with integrated MSS indicator

The Machine Security System (MSS) uses a status indicator that is integrated into the engine start switch in the cab. This indicator provides a visible alert for the security system.

The operator can use the status indicator to determine the status of the system or for troubleshooting.



Illustration 411

g06226442

Status indicator when the MSS is armed

When the MSS is armed, the status indicator will be red. The red light warns the operator that the machine is armed with the security system and that an operator login is required. The red LED will remain ON until a valid key is read while the key switch or engine start switch ring is in the ON position.



Illustration 412

g06226444

Status indicator when the MSS is disarmed or uninstalled

When the MSS is disarmed, the status indicator will be green. The green light notifies the operator that an operator is logged on the machine and the security system has been disarmed. The status indicator will be green if the MSS is not installed on the machine. Also, the green LED will remain ON after power down for the duration of the grace period. After the grace period, the MSS automatically returns to the “armed” mode.

For machines with a standard key switch, a separate status indicator will be available.

Operator Login

Any user may start the engine if the Machine Security System (MSS) is disabled.

If the MSS is active, only a “Standard” or “Master” account can start the machine engine. Before starting the machine engine, the security system must identify a registered operator. An operator can access the machine display using one of the following methods:

- Passcode
- Cat Bluetooth key
- Cat App: Fleet management mobile application

Passcode Entry

To log in using a passcode, refer to the following steps:

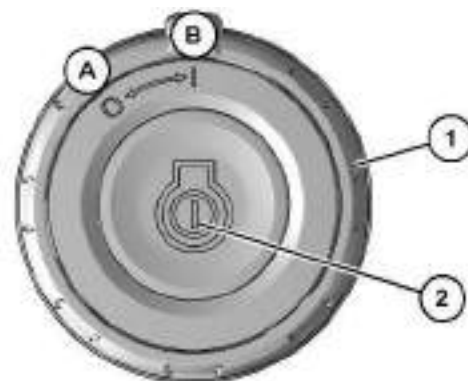


Illustration 413

g06180554

- (A) Off
(B) On
(1) Engine start switch ring
(2) Engine start button

1. Turn engine start switch (1) to the ON position (B).



Illustration 414

g06209470

2. Enter a registered passcode using the monitor keypad and then press “Enter” .

Note: The jog dial or the numbered buttons on the right-side switch panel can also be used to enter the code.



Illustration 415

g06211194

3. Select “Enter” to confirm the passcode. If a registered passcode is recognized, the operator information screen will appear on the display. If the MSS is not installed, the passcode screen will be bypassed automatically after 10 seconds. The operator will be logged in to the system as a guest. Refer to Illustration 415 .
4. Select “OK” to continue to the display homescreen.



Illustration 416 g06209482

5. After an operator logs in to the system successfully, an “Engine Start Allowed” message will appear across the top of the monitor. Refer to Operation and Maintenance Manual, Engine Starting for instructions on starting the engine.

Invalid Passcode



Illustration 417 g06209472

Invalid passcode screen

If a passcode is not recognized, the display will notify the user with an “Invalid code” message. Refer to Illustration 417 .

The operator has five tries to enter a valid passcode successfully. After a fifth unsuccessful attempt, a lockout screen will appear and remain on the display for a duration of 5 minutes.

Note: If the Machine Security System (MSS) is not active, the user can select the “Skip Login” button to avoid the lockout period. Refer to the “Bypass login” section for further information.

Bypass login

Operator login can be bypassed if the user selects the “Skip Login” button on the display. The operator will be logged in to the machine with “Guest” level access.

If the MSS is inactive, the operator will be able to start the engine as normal and view all display screens.

If the MSS is active on the machine, the operator is able to view all display screens but will not have access to starting the engine.

Bluetooth Entry

Alternatively, a Bluetooth Operator ID can be used to log in to the machine. For a Bluetooth key to be detected by the system, ensure that the following qualifications are met:

- The key must be registered with the machine
- The key must be within the cab
- Bluetooth setting must be enabled on the display

Refer to the following steps when logging in to a machine using the Bluetooth key:

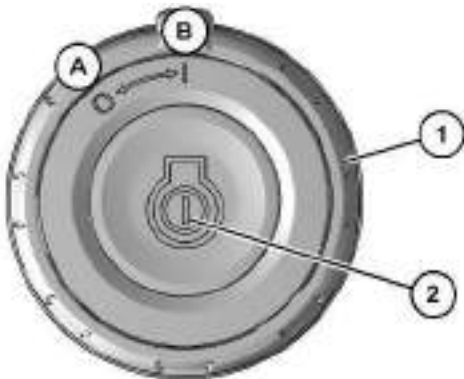


Illustration 418 g06180554

- (A) Off
(B) On
(1) Engine start switch ring
(2) Engine start button

1. Turn engine start switch (1) to the ON position (B).

- Wait several seconds for the system to detect the key when the passcode dialog appears. Once the key is detected, the “Operator Information” screen will display.

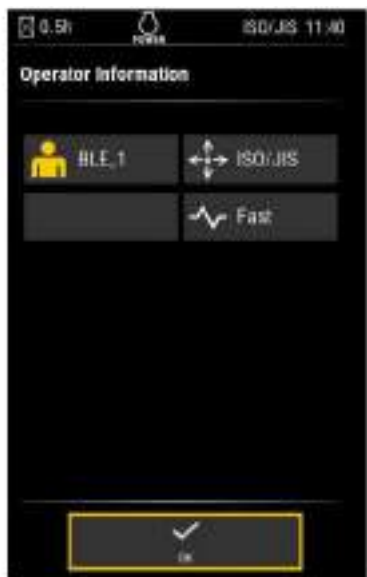


Illustration 419

g06209615

Bluetooth operator information screen

- Select the “OK” button if the proper operator has been displayed.



Illustration 420

g06209482

- After an operator logs in to the system successfully, an “Engine Start Allowed” message will appear across the top of the monitor. Refer to Operation and Maintenance Manual, Engine Starting for instructions on starting the engine.

NOTICE

The access level will change to a “Guest” account automatically if the key is removed from the cab at any time. If the MSS is enabled and the engine is on when the key is removed from the cab, the operator will not be able to start the engine if turned off. To turn on the engine again without the Bluetooth key, the operator will need to log in a registered account using either the smart phone application or passcode.

Note: If multiple Bluetooth devices are within the cab, the system will select the first device detected by the Bluetooth transceiver as the active operator.

Cat® Fleet Management Mobile Application

Operators can also log in to a machine using the Cat App: Fleet management mobile application. For the application to be detected by the system, ensure that the following qualifications are met:

- The Mobile Device ID (MDID) of the mobile application must be assigned to the machine in the Operator Management System (OMS)

Note: It is not possible to assign mobile devices through the MSS interface.

- The mobile device must be within the cab

- “Operator Management Bluetooth Device Enabled Status” is enabled (Cat ET Configuration)

Note: For adding an operator, adding MDID to the machine key list, and pushing the machine key list from the OMS to the machine refer to the OMS documentation at:

<https://myoperators.cat.com/>

Mobile Device / Operating Software Compatibility

Table 27

Mobile Device / Operating Software Compatibility		
Make	Model	Operating Software
Android	Varies	Android 8.x Oreo and up (Preferably Android 9.x Pie)
Apple	iPhone 6/ iPhone 6 Plus and up	iOS 11.0 and up (Preferably 12)

Note: Android mobile hardware support for Bluetooth can vary, so it is possible that a mobile device running Android 8.x software or higher could have hardware that does not support Bluetooth 4.1.

Mobile Application Entry (Android Devices)

To log in using the Cat App: Fleet management mobile application, refer to the following steps:

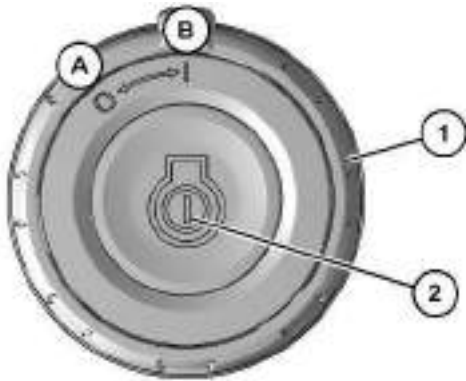


Illustration 421 g06180554

- (A) Off
- (B) On
- (1) Engine start switch ring
- (2) Engine start button

1. Turn engine start switch (1) to the ON position (B).



Illustration 422 g06400799

2. Open the Cat App: Fleet management mobile application on the mobile device.



Illustration 423 g06433500

3. Click “I Agree” to agree with the End-User License Agreement and sign in with Cat eCustomer account credentials.

Note: If you do not have a Cat eCustomer account, click “Get Started” to create one.

4. Select “Login” .



Illustration 424
CWS login screen

5. Enter Cat eCustomer account credentials.



Illustration 426
(1) Menu Icon
(2) Map Icon
(3) Add Equipment Icon

7. Upon login, the Cat App: Fleet management mobile application will open onto the equipment tab.

Note: The “Equipment” tab will be empty on the first login.

The Menu Icon (1) includes “Preferences” , “Notifications and Alerts” , along with various documents covering the end-user license agreement, and privacy notice.

By tapping the Map Icon (2), the operator will be able to see the location of each vehicle on their “Equipment” tab on a map.

If a vehicle is not equipped with a telematics device, or the vehicle is a non-caterpillar machine it may be necessary to add it manually using the Add Equipment Icon (3).



Illustration 425

6. Click “Allow” to enable the Cat App: Fleet management mobile application to work as designed.

Note: Cat App: Fleet management requires access to the mobile device location to use Bluetooth radio to connect to Cat machines.



Illustration 427

g06433513

8. Go to the keys tab.

Keys are necessary to connect to vehicles. If no keys are given, the screen in Illustration 427 will be displayed. The MDID is necessary to assign keys to the user account. The MDID of Android devices is linked to the app.

Note: Uninstalling the Cat App: Fleet management mobile application will result in the MDID and keys being deleted. The Sim card of the phone contains the MDID information, damage to the Sim card may result in keys being lost, if the phone is replaced transfer the Sim card to avoid key loss.

9. Pull the “Equipment Keys” down to refresh the list once the Fleet/Key configuration process has finished. Wait up to 30 seconds until the keys populate. If the keys do not populate, check that the MDID is correct and that the list was pushed properly through OMS.

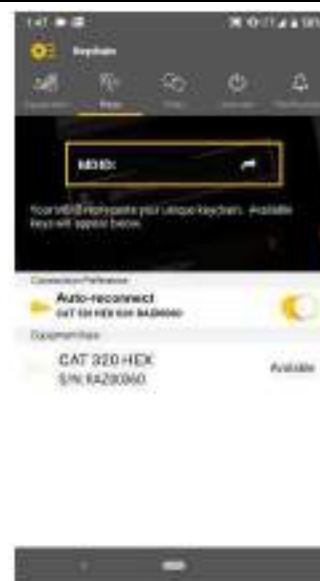


Illustration 428

g06433520

10. Ensure that the machine is ON to connect to a machine.

The machine display should prompt the operator for a password. In the “Keys” tab in the Cat App: Fleet management mobile application and tap the key that matches the machine. If a vehicle is OFF, currently occupied, or too far away the key will be grayed out and say “Out of range” .

Note: If the machine has a user signed in that is not in the cab and it has become necessary for a different user to operate the machine, change the operator in operator setting.



Illustration 429 g06433521

11. Access the “Equipment” tab by tapping it. The machines associated with the keys should have populated. To learn more about a machine, tap it.



Illustration 430 g06433525

12. Press the “Disconnect” button if the user wishes to disconnect. If the user wants to switch machines, tap the arrow in the upper left then tap the machine the operator wants to be switched to.

Note: Disconnect from vehicles if the user is not going to be using a vehicle again within a short time.



Illustration 431 g06209482

13. After an operator logs in to the system successfully, an “Engine Start Allowed” message will appear across the top of the monitor. Refer to Operation and Maintenance Manual, Engine Starting for instructions on starting the engine.

Mobile Application Entry (iOS Devices)

To log in using the Cat App: Fleet management mobile application, refer to the following steps:

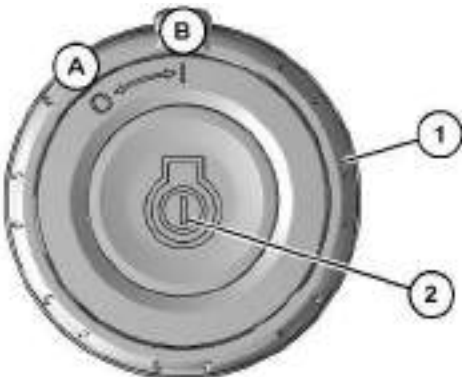


Illustration 432 g06180554

- (A) Off
- (B) On
- (1) Engine start switch ring
- (2) Engine start button

1. Turn engine start switch (1) to the ON position (B).
2. Ensure that Bluetooth detection is enabled on the in-cab display.



Illustration 433 g06400799
Cat App: Fleet management mobile application icon

3. Open the Cat App: Fleet management mobile application on the mobile device.



Illustration 434 g06433528

4. Click “I Agree” to agree with the End-User License Agreement and sign in with Cat eCustomer account credentials.

Note: If you do not have a Cat eCustomer account, click “Get Started” to create one.

5. Select “Login” .



Illustration 435 g06214875

6. Enter eCustomer account credentials to log in.



Illustration 436 g06433533

- (1) Menu Icon
- (2) Map Icon
- (3) Add Equipment Icon

7. Upon login the Cat App: Fleet management mobile application will open onto the equipment tab.

Note: The equipment tab will be empty on the first login.

The Menu Icon (1) includes “Preferences”, “Notifications and Alerts”, along with various documents covering the end-user license agreement, and privacy notice.

By tapping the Map (2), the operator will be able to see the location of each vehicle on the equipment tab on a map.

When using this for the first time, the Cat App: Fleet management mobile application will ask if it can use the location feature. Allow the location feature to use the map.

If a vehicle is not equipped with a telematics device, or the vehicle is a non-caterpillar machine it may be necessary to add it manually using the add equipment button (3).



Illustration 437

g06433536

8. Access the “Keys” tab. Keys are necessary to connect to vehicles. If no keys are given, the screen in Illustration 437 will be displayed. The MDID is necessary to assign keys to the user account. The MDID of IOS devices is linked to the user account.
9. Once the Fleet/Key configuration process has finished, tap the refresh button in the Cat App: Fleet management mobile application top right. Wait up to 30 seconds until the keys populate. If the keys do not populate, check that the MDID is correct and that the list was pushed properly in OMS.



Illustration 438

g06433538

10. To connect to a machine, ensure that the machine is ON. The machine display should prompt the operator for a password. At this point on the Cat App: Fleet management mobile application go to the “Keys” tab and tap the key that matches the machine. If a vehicle is OFF, currently occupied, or too far away the key will be grayed out and say “Out of range”.



Illustration 439

g06433540

11. Press the disconnect button if the user wishes to disconnect. If the user wants to switch machines tap the arrow in the upper left, then tap the key of the machine to switch to.

Note: Disconnect from vehicle if the user will not be using the vehicle again within a short time.



Illustration 440 g06209482

12. After an operator logs in to the system successfully, an “Engine Start Allowed” message will appear across the top of the monitor. Refer to Operation and Maintenance Manual, Engine Starting for instructions on starting the engine.

Engine Start Switch Troubleshooting

Table 28

Switch Status	Possible Cause	Resolution
Engine startswitch is not illuminated	Engine start accessory power not on	Turn engine start switch ring to ON position
	Power management triggered	Cycle engine start switch ring and try to restart
Engine start switch is green	Starting component failure	Contact your Cat dealer
Engine start switch is red	Machine interlock conditions not met	Hydraulic lock in LOCKED position
	Engine shut down without cycling start switch ring	Cycle engine start switch ring and try restart
	Operator not authenticated (Passcode login)	Add operator to machine authorized user list
		Switch operator from guest mode using display
	Operator not authenticated (Bluetooth key)	Add operator to machine authorized user list
		Replace key fob battery
		Ensure more than 4.5 m (15 ft) from other Bluetooth equipped machine

(continued)

(Table 28, contd)

Switch Status	Possible Cause	Resolution
	Operator not authenticated (Cat Fleet management app)	Alternately login with display passcode or contact local Cat dealer
		Add operator to machine authorized user list
		Ensure more than 4.5 m (15 ft) from other Bluetooth equipped machine
		Enable phone Bluetooth and connect Cat Fleet management app
		Change Bluetooth system enable status to enabled (Cat dealer)
		Contact local Cat dealer if unable to see machine Bluetooth device

i08257805

Monitoring System

SMCS Code: 7451; 7490

WARNING

Do not operate the machine if the monitor is not functioning (for example, monitor has a black screen or is not responding) when the key switch is in the ON position.

The monitor provides images from the camera system and other information for safe machine operation. Operating the machine without a properly functioning monitor may result in injury or death. If the monitor is not functioning, place the machine in a safe state by following the procedures for stopping and parking the machine. Determine the cause of the monitor malfunction and correct before returning the machine to service.

NOTICE

When the monitor provides a warning, immediately check the monitor and perform the required action or maintenance as indicated by the monitor.

The monitor indicator does not guarantee that the machine is in a good condition. Do not use the monitor panel as the only method of inspection. Maintenance and inspection of the machine must be performed on a regular basis. See the Maintenance Section of this Operation and Maintenance Manual.

The monitoring system is an input and an output of the Machine Control System. The monitor has a multi-touch 8 inch or 10 inch display. The Machine Control System communicates back and forth on the data link. The monitoring system consists of the following components:

- Display (with numerous screens and menus)
- Indicators
- Gauges
- Soft Switch Panel
- Jog Dial

General Information

Reference: For complete monitoring system information, refer to M0109053, "Next Generation Hydraulic Excavator Monitoring System Supplement".



Illustration 441

g06397192

- (1) Action Lamps
- (2) Status Information Area
- (3) Event Indicator Area
- (4) Camera View Area
- (5) Gauge Area
- (6) Navigation Area
- (7) Function List
- (8) Shortcuts

The monitoring system displays various warnings and information about the condition of the machine, and the machines surrounding with various camera views. There are gauges and several alert indicators included on monitoring system display. Each gauge is dedicated to a parameter within a machine system. The monitoring system will allow the user to do the following:

- View Surroundings
- Interpret status information
- Interpret parameters
- View OMM
- View service intervals
- Perform calibrations
- Troubleshoot machine systems

Action Lamps (1)

The action lamps illuminate to show that a problem has occurred with the machine.

Status Information Area (2)

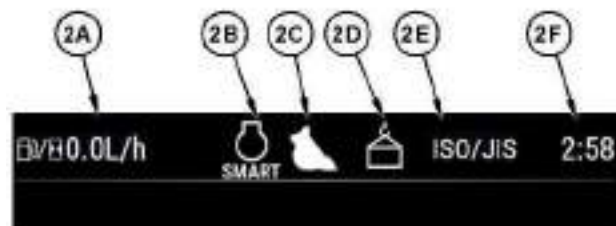


Illustration 442

g06464039

- (2A) Multi Status Information
- (2B) Seatbelt Switch Status (if equipped) / Power Mode
- (2C) Work tool
- (2D) Heavy Lift / Cat® Dig Boost (if equipped)
- (2E) Lever Pattern / Fuel Level
- (2F) Clock

Reference: For complete status information, refer to M0109053, "Next Generation Hydraulic Excavator Monitoring System Supplement".

Event Indicator Area (3)

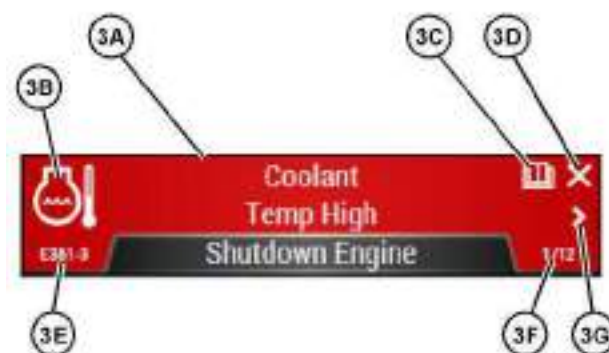


Illustration 443

g06223056

- (3A) Event Description
- (3B) Event Symbol
- (3C) e-OMM
- (3D) Close
- (3E) Event ID
- (3F) Order Number / Total Number
- (3G) Arrow (Next)

Event Description (3A) – This area will display the description of pop-up message of the impending problem.

- Line 1 : System
- Line 2 : Condition
- Line 3 : Action to be taken

Event Symbol (3B) – This area will display the symbol of the problem

e-OMM (3C) – This area will indicate that the e-OMM is available for this event. If e-OMM is not available for this event, the area will be blank.

Close (3D) – Select this to hide the pop-up message and display the event icon list.

Event ID (3E) – The identification number for the event will be shown here.

Priority Number / Total Number (3F) – The priority number of the pop-up message is displayed next to the total number of messages. The messages are ordered from the highest to the lowest priority.

Arrow (3G) – The arrow is shown when there is a next or previous message. Select the arrow to display the next or previous message.

Camera View (4)

This area on the monitor displays the view of the cameras. A rear view camera mounted on top of the counterweight and an optional side view camera mounted in the side panel next to the hydraulic tank.

If both rear view camera and side view camera are equipped, the monitor screen can be toggled to show:

- Rear only
- Side only
- Split vertically
- Split horizontally

The camera view can be toggled when the cursor is on the camera view area and the area is touched or the jog dial is turned.

Gauge Area (5)



Fuel Level – This gauge indicates the amount of fuel that is remaining in the fuel tank. When the fuel gauge is in the red range, add fuel immediately.



Hydraulic Oil Temperature – This gauge indicates the temperature of the hydraulic oil. The normal operating range is the green range. If the gauge is in the white range, the engine and machine warm-up is required. Refer to Operation and Maintenance Manual, “Engine and Machine Warm-up”. If the gauge reaches the red range, reduce the load on the system. If the gauge stays in the red range, stop the machine and investigate the cause of the problem.



Engine Coolant Temperature – This gauge indicates the temperature of the engine coolant. The normal operating range is the green range. If the gauge is in the white range, the engine and machine warm-up is required. Refer to Operation and Maintenance Manual, “Engine and Machine Warm-up”. If the gauge reaches the red range, stop the machine and investigate the cause of the problem.



Diesel Exhaust Fluid (DEF) Gauge – This gauge indicates the level of DEF fluid in the DEF tank. When the DEF gauge is in the red range, add DEF immediately.

Navigation Bar (6)



Apps Key – Allows you to display different information in the gauge area related to operation. Also contains air conditioner and audio controls. This key includes the settings screen allowing change of a multitude of parameters, some password protected.



Function List Key – Allows you to turn on and off various functions related to the active screen. This icon only appears in certain screens where additional settings are necessary.

Shortcuts – Allows you to set certain shortcuts on the navigation bar.

Machine Warnings



Illustration 444

g06223766

Event Indicator Area

- (1) Event Icon List
(2) Pop-up Event Message

The Monitor will display warnings, and log events for machine conditions that are not within normal operating parameters.

The event warnings are classified into three warning levels. Warning Level 1 represents the least severe problem and Warning Level 3 represents the most severe problem. The warning levels, monitor response, and the required operator actions are given below.

Warning Level 1 (Gray) – Requires operator awareness. The icon and pop-up message will both appear gray.

Warning Level 2 (Amber) – Requires a change in the operation of the machine or a change in the maintenance of the machine to correct the condition. The icon and pop-up message will both appear amber and the action lamp will blink.

Warning Level 3 (Red) – Requires immediate shutdown of the machine to prevent damage to the machine or personnel. The icon and pop-up message will both appear red, the action lamp will blink, and the buzzer will sound.

When multiple warnings are present in the system, the highest warning is shown first. Press the right or left key to view all the logged warnings. If no keys are pressed within a few seconds, the display will return to the highest warning.

Reference: For complete machine warning information, refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”.

Logging In



Illustration 445

g06242074

There are different ways to access the monitor which include:

- Guest access
- Passcode access
- Bluetooth access
- Cat® Fleet Management app

For more information on logging in, refer to Operation and Maintenance Manual, Machine Security System - Operator Login.

Navigation

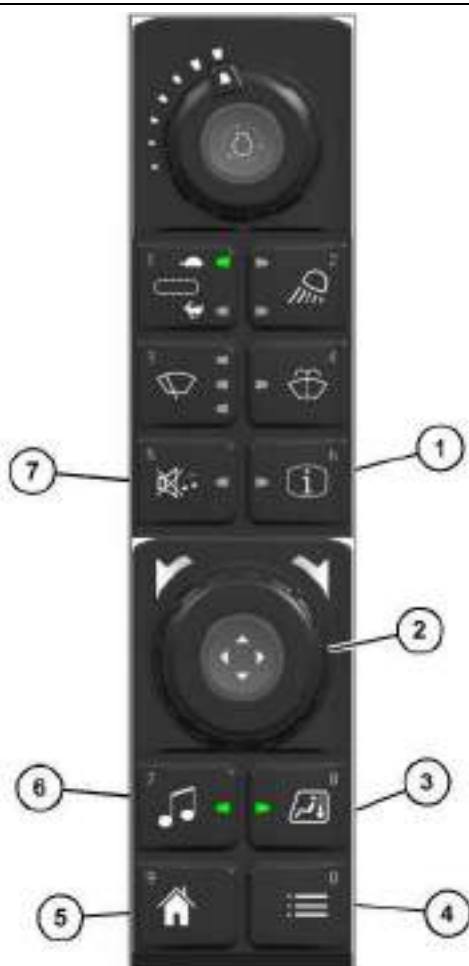


Illustration 446

g06464384

Right side switch panel

- (1) Operator information button
- (2) Jog dial
- (3) Air conditioner button
- (4) Next menu button
- (5) Home button
- (6) Audio button
- (7) Mute button

The monitor can be navigated by touch screen or the switch panel. Switch panel components can be used to interface with the monitor in the following ways:

Operator information button (1) – Press and hold this button to access the operator information screen. This screen shows information such as operator settings.

Jog dial (2) – Rotate the jog dial to highlight menu items in the monitor. Push the jog dial down to select the highlighted item.

Air conditioner button (3) – Press the button to access the air conditioner controls.

Next menu button (4) – This button is equivalent to the function list key on the monitor. This button can only be used on screens where the function list key is shown.

Home button (5) – Press this button to return to the main screen.

Audio button (6) – Press this button to access the audio controls.

Mute button (7) – Press this button to mute the audio. Press the button again to unmute the audio.

Each of the buttons is also assigned a number which is imprinted in the top corner of the button. These buttons can be used to enter the numerical passcodes used to log in to the monitor.

Application Menu

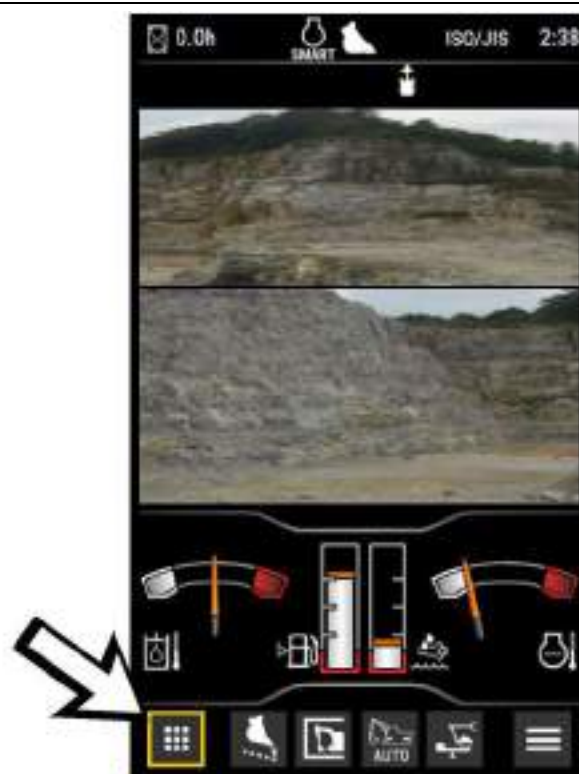


Illustration 447

g06391692

Press the application menu button to access the Application menu.



Illustration 448g06247445

The Application menu “(Apps)” contains the list of applications. The available applications may differ depending on the machine configuration or software version.



Illustration 449g06263058

Once an “App” has been selected, an operator can sort through the available “Apps” by swiping the selected area left or right.

Reference: For complete application menu information, refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”.

Setting Menu



Illustration 450g06213909

From the main screen, press the application menu button. In the Application menu, select “Setting” .



Illustration 451g06217518

A warning will appear notifying the operator that the camera is not visible in the Setting menu. After you have read the warning and understand the content, press the “OK” button.



Illustration 452 g06213929

The Setting menu contains the following menu items:

- Display Setting
- Information
- Machine Setting
- Operator Setting
- Service

Note: A dealer password is necessary to access the Service menu.

Machine Setting



Illustration 454 g06261306

The Machine Setting menu may differ depending on the machine configuration or software version. Basic settings are described below.

Reference: For complete machine settings information, refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”.

Audio

The Audio screen allows the user to choose the radio region, enable Bluetooth, pair devices, and scan for Digital Audio Broadcast (DAB) channels.



Illustration 453 g06217583

From the main screen, press the application menu button. In the Application menu, select “Setting” . Next, select “Machine Setting” .

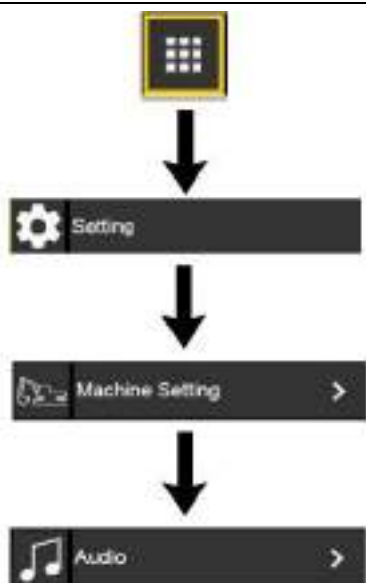


Illustration 455

g06261319

From the main screen, press the application menu button. In the Application menu, select “Setting”. Next, select “Machine Setting”, and then “Audio”.



Illustration 456

g06241515

The Audio screen contains the following menu items:

- Radio Region Setting - Choose the radio region from the list of locations around the world.
- Bluetooth - Allows the operator to enable Bluetooth and pair a phone. This menu is also available through the main Audio screen. Refer to Operation and Maintenance Manual, Monitoring System - Bluetooth for information on the Bluetooth screen.

- DAB Channel Scan - Start a scan to find DAB channels in the area with good reception.
- Volume Gain - Allows the user to individually adjust the gain on various outputs such as the AM radio, FM radio, and the phone.

Auto Warm Up

The Auto Warm Up screen allows the user to enable and set the auto warm-up feature. This feature automatically starts a warm-up period when the engine is started and the hydraulic oil is below the set temperature.

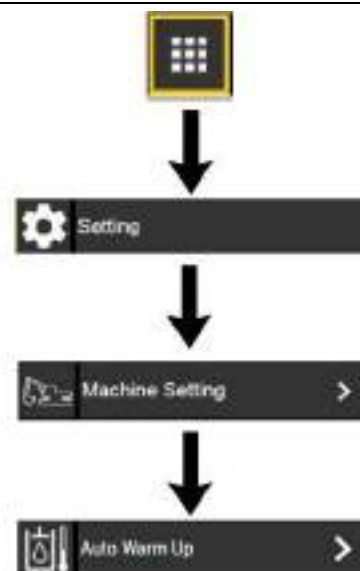


Illustration 457

g06261321

From the main screen, press the application menu button. In the Application menu, select “Setting”. Next, select “Machine Setting”, and then “Auto Warm Up”.



Illustration 458 g06219830

To set the auto warm-up temperature, press the “Auto Warm Up Target Temp” window, then enter the temperature. If the hydraulic oil is below the set temperature, the auto warm-up feature will activate after the engine is started.

Sleep Time



Illustration 459 g06360101

The Sleep time Setting feature allows the user to set a sleep timer for the engine start switch. If the start switch is left in the ON position, power will automatically shut off once the chosen timer interval has elapsed.

Security

Note: Master level access is required to adjust security settings.

The Security screen allows the user to set the operator lockout time. The lockout time is the amount of time after engine shutdown that an operator can start the engine without logging back in to the monitor.

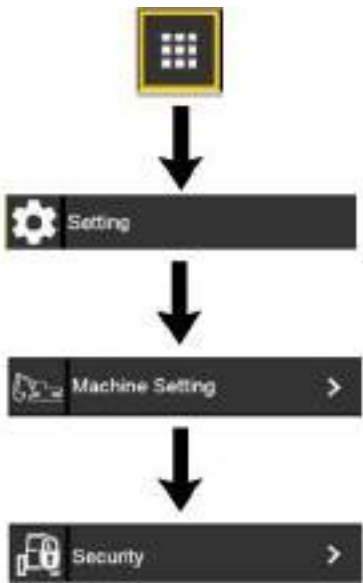


Illustration 460 g06261324

From the main screen, press the application menu button. In the Application menu, select “Setting” . Next, select “Machine Setting” , and then “Security” .



Illustration 461 g06242069

Select “Operator ID Timeout” to select the amount of time before the operators passcode times out after engine shutdown. To block out periods of time throughout the week to bypass security, select “Security Bypass Time” .



Illustration 462 g06242071

Enter the times and the days to bypass the security system.

Operator Setting



Illustration 463 g06217328

From the main screen, press the application menu button. In the Application menu, select “Setting” . Next, select “Operator Setting” .



Illustration 464 g06261330

The Operator Setting menu contains the following menu items:

- Operator Inputs Configuration
- Response

Reference: For complete operator settings information, refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”.

Operator Inputs Configuration

The operator inputs configuration screen allows the operator to configure the joystick buttons for personal preference. The settings will be stored in the preferences for that login ID.



Illustration 465

g06261334

From the main screen, press the application menu button. In the Application menu, select “Setting”. Next, select “Operator Setting”, then “Operator Inputs Configuration” .

Lever Pattern

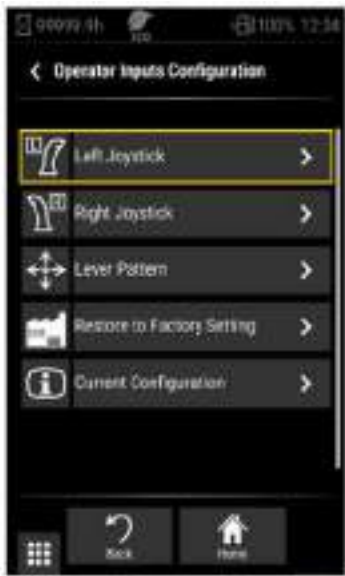


Illustration 467

g06222026

From the Operator Inputs Configuration screen, select “Lever Pattern” .

Select the desired lever pattern from the menu items. Press the “Home” button to return to the main screen.

Restore to Factory Setting

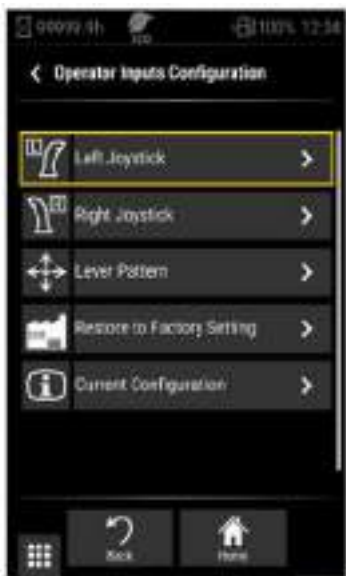


Illustration 466

g06222026

Choose the operator input to configure.

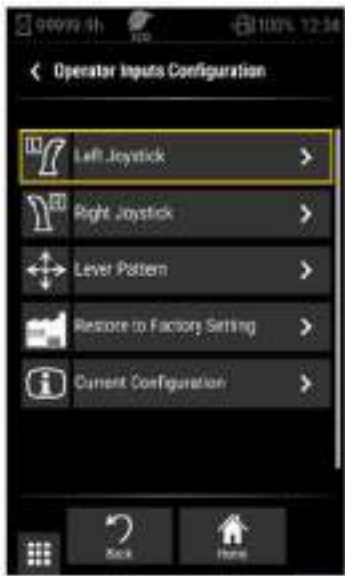


Illustration 468

g06222026

From the Operator Inputs Configuration screen, select “Restore to Factory Setting” .



Illustration 469

g06222053

Select the item from the list to be restored.



Illustration 470

g06222061

A warning will appear asking if you want to proceed. Press “Restore” to restore the settings or press “Cancel” to abort. After pressing “Restore”, the operator information screen will appear to show the new button assignments. Press the “OK” button to return to the main screen.

Air Conditioner

Refer to Operation and Maintenance Manual, Air Conditioning and Heating Control for coverage of the air conditioner screen in the monitor.

Audio

Refer to Operation and Maintenance Manual, Radio for coverage of the audio screen in the monitor.

Phone

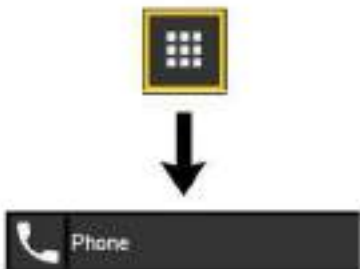


Illustration 471

g06213901

From the main screen, press the application menu button. In the Application menu, select “Phone”.



Illustration 472

g06340470

To make a call, select “Call”, then use the keypad to dial the number.

Note: There must be a paired phone in the cab and Bluetooth enabled to make a call.

Reference: For phone and Bluetooth settings, refer to M0109053, “Next Generation Hydraulic Excavator Monitoring System Supplement”.

Bucket/Work Tool Setting

The Bucket/Work Tool Setting screen allows the operator to choose the tool being used for work.



Illustration 473

g06220041

From the main screen, press the application menu button. In the Application menu, select "Setting". Next, select "Information" and then "Electronic OMM".



Illustration 474

g06220034

Select the desired tool.

Reference: For complete Bucket/Work Tool settings, refer to M0109053, "Next Generation Hydraulic Excavator Monitoring System Supplement".

Electronic OMM

The Electronic OMM screen allows the operator to view the Operation and Maintenance Manual (OMM) for the machine.

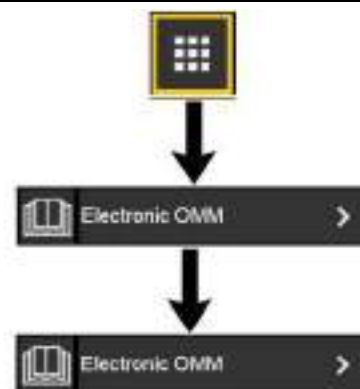


Illustration 475

g06261340

From the main screen, press the application menu button. In the Application menu, select "Setting". Next, select "Information" and then "Electronic OMM".

Note: Contact your Cat Dealer for information to update the electronic OMM.

Function List Screen



Illustration 476

g06213511

Press the Function List icon to access the Function List screen.

The Function List screen allows the user to turn on or off the available functions on the machine. Tap or select with the jog dial the item to turn on or off.

The Function list menu may differ depending on the machine configuration or software version.

Reference: For complete function list information, refer to M0109053, "Next Generation Hydraulic Excavator Monitoring System Supplement".

i06951931

Fuel Transfer Pump (Refueling) (If Equipped)

SMCS Code: 1256

Use the following procedure to pump fuel and store the hose.

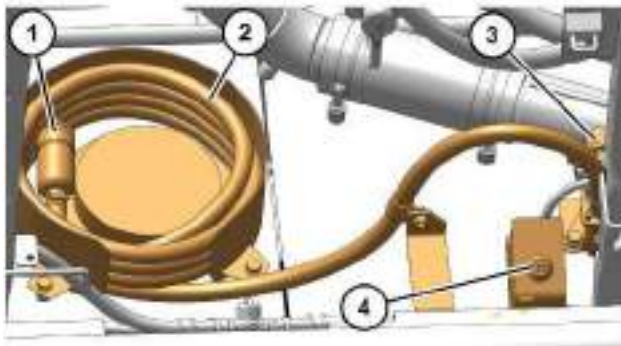


Illustration 477

g06180565

- (1) Suction valve
- (2) Suction hose
- (3) Electric refueling pump
- (4) ON/OFF switch



ON/OFF Switch – Push the ON/OFF switch to activate or deactivate the fuel transfer pump. A red indicator on the switch will illuminate when the fuel transfer pump is activated.

If one of the following conditions occur, the fuel transfer pump will not activate and/or stop operating:

- Battery disconnect switch is in the OFF position
- Engine is operating
- Engine start switch is moved to the START position.
- Engine start switch is in the OFF position
- Hydraulic lockout control is not in the LOCKED position
- Fuel tank level is full
- Fuel is not detected at the suction valve.
- 30 seconds following a detection of no fuel at the suction valve.

Use the following procedure to pump fuel and store the hose.

1. Park the machine on a level surface. Move the hydraulic lockout control to the LOCKED position. Stop the engine
2. Turn the engine start switch to the ON position without starting the engine.
3. Remove the fuel tank cap from the fuel tank.
4. Open the access that is on the right side of the machine.



Illustration 478

g06180748

Suction valve (1) is at the end of hose (2).

Turn the end of the suction valve clockwise to open the valve

5. Uncoil the hose and turn the end of the suction valve clockwise to open the suction valve.
6. Properly insert the end of the suction valve into a container of fuel.
7. Push switch (4) to activate the fuel transfer pump and supply the fuel to the tank. A red indicator on the switch will illuminate when the fuel transfer pump is activated.

When the fuel tank is full, the fuel transfer pump will automatically stop.

When the fuel container is empty, push the switch again to stop refueling. If additional fuel is needed, wait 30 seconds and return to step 6.

Note: The red indicator on the switch will no longer illuminate when the fuel transfer pump has stopped refueling.

Note: The fuel transfer pump will not activate for 30 seconds following a detection of no fuel at the suction valve.

8. Push the switch at any time to deactivate the fuel transfer pump.

Note: The red indicator on the switch will no longer illuminate when the fuel transfer pump has stopped refueling.

9. Drain excess fuel from the hose and turn the end of the suction valve counter-clockwise to close the suction valve.
10. Wind the hose and store in the hose container.

NOTICE

To prevent hose damage, do not coil the hose in a tight radius.

11. Close the access door.
12. Install the fuel tank cap onto the fuel tank.
13. Turn the engine start switch to the OFF position.

i07092848

Radio

SMCS Code: 7338

The radio is integrated into the monitoring system. All the radio controls are adjusted using the monitor. The actual radio is mounted in the right rear console behind the operator seat.



Illustration 479

g06213193

- (1) Jog dial
- (2) Home button
- (3) Radio button

The audio menu can be directly accessed by pressing radio button (3) on the right side switch panel. Input selections can be made using jog dial (1) or using the monitor touch screen. Home button (2) can be used to return to the main screen.



Illustration 480

g06213076

Press radio button (3) to go directly to the radio screen. To navigate to the radio screen from the main screen, press application menu button (4).



Illustration 481

g06213198

Use jog dial (1) to highlight "Audio" and then press the jog dial downward to select the entry. You may also access the screen by simply touching the "Audio" box on the touch screen.

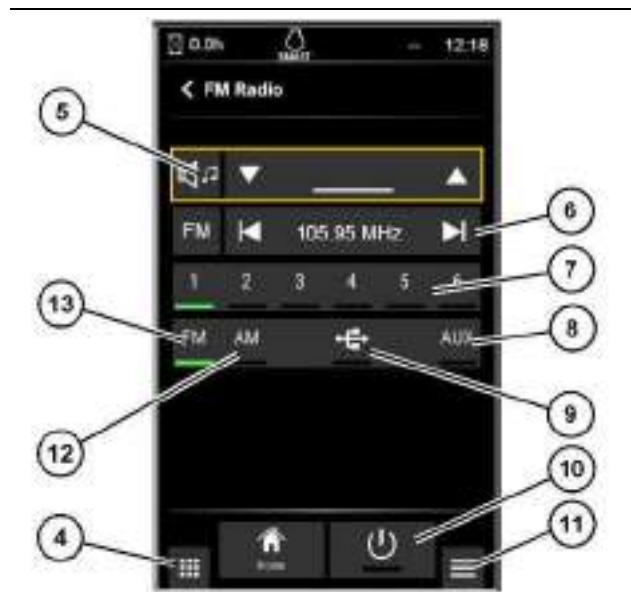


Illustration 482

g06213200

Radio screen

- (4) Application menu button
- (5) Volume control
- (6) Tuner
- (7) Preset stations
- (8) Auxiliary button
- (9) USB button
- (10) Power button
- (11) Radio Function List menu
- (12) AM button
- (13) FM button

Application menu button (4) – Use this button to return to the application menu.

Volume control (5) – The volume control is used to raise or lower the audio volume.

Tuner (6) – The tuner is used to tune the radio to the desired station.

Preset stations (7) – The preset stations store favorite radio stations for the operator. To set a station, tune to the desired station. Press and hold the preset number you want to assign to that station. Once a beep is heard, release the button. The indicator light for the active preset station will illuminate.

Auxiliary button (8) – When a device is plugged into the auxiliary port, press the auxiliary button to connect the device to the radio. The indicator light will illuminate when this mode is active.

USB button (9) – When a device is plugged into the USB port, press the USB button to connect the device to the radio. The indicator light will illuminate when this mode is active.

Power button (10) – Pressing this button turns the radio on and off. The indicator light on the monitor and on button (3) will illuminate when the power is on.

Radio function list menu button (11) – Pressing this button leads to the radio function list menu.

AM button (12) – Press this button to access AM radio. The indicator light will illuminate when this mode is active.

FM button (13) – Press this button to access FM radio. The indicator light will illuminate when this mode is active.

Radio Function List

To access the radio function list, press the function list button (11) in the lower right corner of the radio screen.



Illustration 483

g06223378

The function list menu consists of the following items:

Treble – Allows the user to adjust the treble.

Bass – Allows the user to adjust the bass.

Balance – Allows the user to adjust the balance between speakers.

Auto Loudness – When on, this feature automatically adjust treble and bass levels when reducing the volume setting. This effect allows the user to hear more clearly at a lower volume.

Bluetooth – Allows the user to pair a phone, view paired devices, and edit device names.

Refer to Operation and Maintenance Manual, Monitoring System - Bluetooth for information on the Bluetooth screen.

Selection Method

All settings can be made using the touch screen or by using the jog dial. The method depends on the preference of the operator. When using the touch screen, simply touch the icon you want to select. When using the jog dial, rotate the dial to switch to different selections within the screen. Press the jog dial downward to choose a selection.



Illustration 484

g06213233

When using the jog dial to set the volume or tuner, rotate the dial clockwise to increase and counter-clockwise to decrease. Press downward on the dial to enter the desired setting.

Radio Operation

1. To operate the system, press power button (10).
2. Select between the AM button for AM stations or select the FM button for FM stations.
3. Use tuner (6) to adjust to the desired station. If presets stations (7) are set, press the desired preset station.
4. Use volume control (5) to adjust the volume.

5. When the machine is in operation turn down the volume of the radio.

USB/AUX Operation



Illustration 485

g06213245

1. To play music from a device such as an MP3 player or a phone, connect the device using an auxiliary cable or a USB cable. Depending on the cable being used, plug the cable into the appropriate port on the console.
2. Select either USB or AUX depending on which type of cable was used. Play the music from the device. The music should be playing over the radio speakers if properly connected. Adjust the volume as necessary.

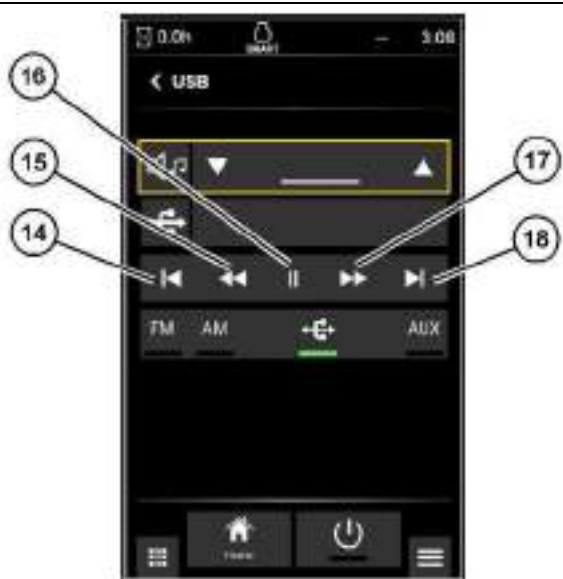


Illustration 486

g06213254

Note: If USB is selected, extra controls appear on the screen for playing music. They include the following:

- 14** – Skip to the beginning of the track
- 15** – Rewind the track
- 16** – Pause/Play the track
- 17** – Fast forward the track
- 18** – Skip to the end of the track

i08031142

Air Conditioning and Heating Control

SMCS Code: 7304; 7320; 7337

Consult with your Cat dealer for periodic maintenance of the heating and air conditioning system.



Illustration 487

g06178710

- (1) Jog dial
- (2) Heating and air conditioning button
- (3) Home button

Air conditioning and heating functions are controlled through the monitor. The heating and cooling menu can be directly accessed by pressing button (2) on the right side switch panel. Input selections can be made using jog dial (1) or using the monitor touch screen. Home button (3) can be used to return to the main screen.



Illustration 488

g06213076

Press the Air Conditioning and Heating button (2) to go directly to the air conditioner screen. To navigate to the air conditioner screen from the main screen, press application menu button (4).



Illustration 489

g06213088

Use jog dial (1) to highlight “Air Conditioner” and then press the jog dial downward to select the entry. You may also access the screen by simply touching the “Air Conditioner” box on the touch screen.

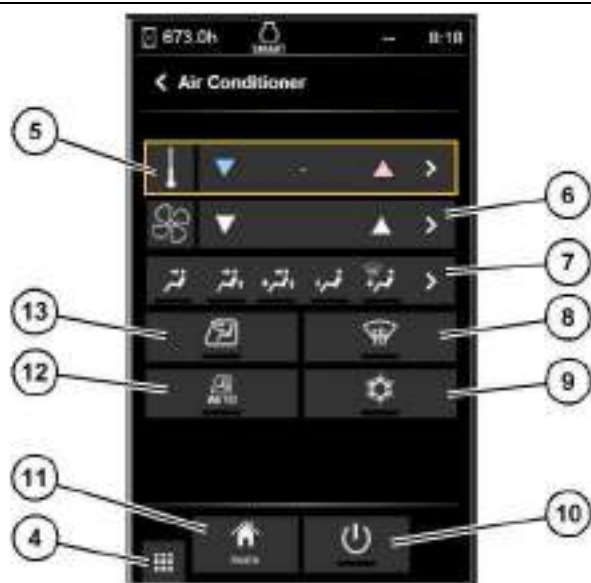


Illustration 490

g06213092

Air conditioner screen

- (4) Application menu button
- (5) Temperature control
- (6) Fan blower speed control
- (7) Air outlet settings
- (8) Defroster
- (9) Compressor ON/OFF
- (10) Power
- (11) Home button
- (12) Auto
- (13) Recirculation

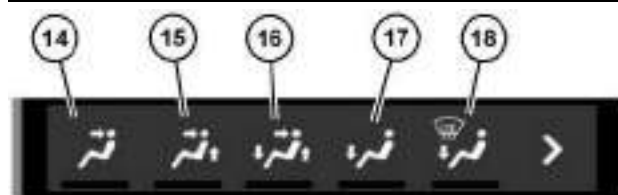


Illustration 491

g06213104

- (14) Front vents
- (15) Front and rear vents
- (16) Front, foot, and rear vents
- (17) Foot vents
- (18) Defrost and foot mode

Application menu button (4) – Use this button to return to the application menu.

Temperature control (5) – The temperature control is used to raise or lower the desired temperature.

Fan blower speed control (6) – The blower control is used to increase or decrease the desired blower speed.

Air outlet settings (7) – The desired air outlet setting can be chosen from this panel. The indicator light will illuminate to show the active setting.

Defroster (8) – Pressing this button activates and deactivates the defroster. Use the defroster to remove steam and frost from the windows. The indicator light will illuminate when this mode is active.

Compressor ON/OFF (9) – Pressing this button activates and deactivates the air conditioner. The indicator light will illuminate when this mode is active.

Power (10) – Pressing this button turns the heating and cooling system on and off. Push and hold the power button for 3 seconds to turn OFF the HVAC system. The indicator light on the monitor and on button (2) will turn green when power is on.

Home button (11) – Use this button to return to the main screen.

Auto (12) – Select this option to hold the system at the desired temperature. The system will modulate to keep the cab at the temperature that the system is set to. The indicator light will illuminate when this mode is active.

Recirculation (13) – This option recirculates air from the cab instead of pulling air from the outside. This mode is more efficient because the system is recycling conditioned air from the cab. However, no fresh air is coming into the cab when in this mode. The indicator light will illuminate when this mode is active.

Front vents (14) – In this mode, air will only circulate from the front vents.

Front and rear vents (15) – In this mode, air will circulate from the front vents and rear vents.

Front, foot, and rear vents (16) – In this mode, air will circulate from the front vents, rear vents, and foot vents.

Foot vents (17) – In this mode, air will only circulate from the foot vents.

Defrost and foot mode (18) – In defrost mode, air will circulate from the defrost vents and foot vents.

Selection Method

All settings can be made using the touch screen or by using the jog dial. The method depends on the preference of the operator. When using the touch screen, simply touch the icon you want to select. When using the jog dial, rotate the dial to switch to different selections within the screen. Press the jog dial downward to choose a selection.



Illustration 492

g06213141

When using the jog dial to set the temperature or blower speed, rotate the dial clockwise to increase and counter-clockwise to decrease. Press downward on the dial to enter the desired setting.

Operation

1. To operate the system, press power button (10).
2. Use temperature control (5) to adjust to the desired temperature.
3. Select the desired mode and outlet vents.
4. Use fan blower speed control (6) to adjust the blower. If the system is in "Auto" mode, the blower fan speed and air outlet setting will automatically be adjusted. But recirculation air intake is not changed automatically.

Note: In cold ambient temperature condition, fan speed is stopped or restricted depend on coolant temperature.

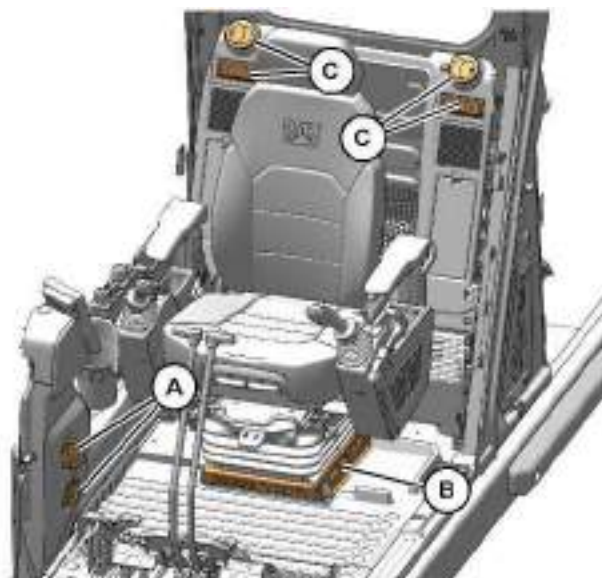


Illustration 493

g06178705

- (A) Defrost vent (front window)
- (B) Foot air vents
- (C) Rear air vents

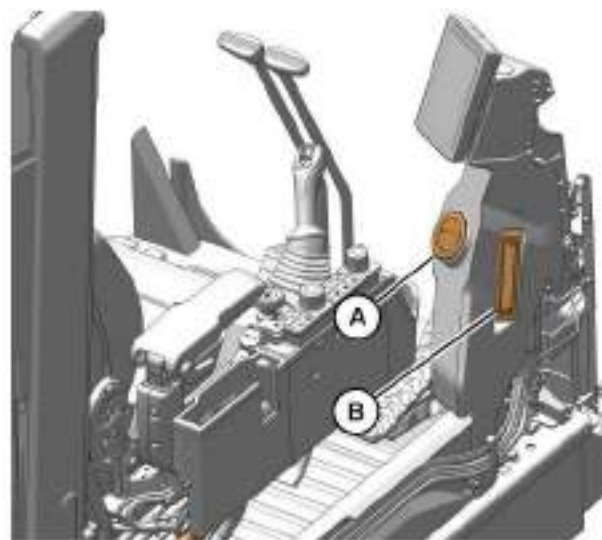


Illustration 494

g06287632

- (A) Front vent
- (B) Defrost vent (RH window)

5. Redirect the louvers for air outlets (A) and (C) by hand to the desired direction. The louvers for air outlet (B) cannot be redirected.

i07088628

Mirror

SMCS Code: 7319

WARNING

Adjust all mirrors as specified in the Operation and Maintenance Manual. Failure to heed this warning can lead to personal injury or death.

Note: Your machine may not be equipped with all the mirrors that are described in this topic.



Illustration 495

g06220616

- (1) Right Side View Mirror on the Cab
(2) Left Side View Mirror on the Cab



Illustration 496

g06220627

- (3) Tank mirror

Mirrors provide additional visibility around your machine. Make sure that the mirrors are in proper working condition and that the mirrors are clean. Adjust all mirrors at the beginning of each work period and adjust the mirrors when you change operators.

Modified machines or machines that have additional equipment or attachments may influence your visibility.

Mirror Adjustment

- Park the machine on a level surface.
- Lower the work tool to the ground.
- Move the hydraulic lockout control to the LOCKED position. For further details on this procedure, refer to Operation and Maintenance Manual, "Operator Controls".
- Stop the engine.
- Adjust rear view mirrors to provide visibility behind the machine at a maximum distance of 30 m (98 ft) from the rear corners of the machine.



Illustration 497

g06220634

Tightening sequence

After adjustment of the mirror angle, make sure that the CAT logo is at the top.

It may be necessary to periodically tighten the mirror mounting bolts. If the bolts are loose, tighten the bolts in the sequence shown in Illustration 497. Tighten bolts (1) and (2) to $11 \pm 2 \text{ N}\cdot\text{m}$ ($8.1 \pm 1.5 \text{ lb ft}$).

Tighten the bolts (3) through (6) to $2 \pm 0.4 \text{ N}\cdot\text{m}$ ($1.5 \pm 0.3 \text{ lb ft}$).

Right Side View Mirror on the Cab (3)



Illustration 498

g06223277

If equipped, adjust the right side view mirror on the cab (1) so the front of the right track can be seen from the operator seat. A view of at least 1 m (3.3 ft) from the right front of the machine should be seen from the operator seat.

Left Side View Mirror on the Cab (4)



Illustration 499

g06223279

If equipped, adjust the left side view mirror on the cab (4) so the left side of the cab, access door, and rear of left track can be seen from the operator seat. A view of at least 1 m (3.3 ft) from the side of the machine should be seen from the operator seat. Additionally, provide as much visibility to the rear as possible.

Tank Mirror (3)



Illustration 500

g06223284

If equipped, adjust the tank mirror so the fuel tank and the hydraulic tank can be seen from the operator seat. A view of at least 1 m (3.3 ft) from the side of the machine should be seen from the operator seat.

i07491391

Camera

SMCS Code: 7347; 7348

Rear View Camera



Illustration 501

g06184579

The rear view camera system consists of a camera that is located in the middle of the top of the counterweight.

Note: The rear view camera system has been set up by the factory or by a Cat dealer to provide views which comply with specified guidelines. Consult your Cat dealer before any adjustments are made to the system.

For more information refer to Operation and Maintenance Manual, "Monitoring System".

Side View Camera (If Equipped)



Illustration 502

g06214504

The side view camera system consists of a camera mounted on the panel next to the pump compartment.

Note: The side view camera system has been set up by the factory or by a Cat dealer to provide views which comply with specified machine side views. Consult your Cat dealer before any adjustments are made to the system.

For more information refer to Operation and Maintenance Manual, "Monitoring System".

360 Visibility (If Equipped)



Illustration 503

g06263449

Front camera



Illustration 504

g06263435

Left side camera

If equipped with the optional side and front cameras, the 360 visibility feature can be toggled on and off to enable or disable the 360 degree visibility view in the monitor.



Illustration 505

g06263096

From the main screen, press the Function List icon. In the Function List menu, select "360 Visibility".

Note: The 360 visibility system has been set up by the factory or by a Cat dealer to provide views which comply with specified machine side views. Consult your Cat dealer before any adjustments are made to the system.

For more information refer to Operation and Maintenance Manual, "Monitoring System".

i07427990

Window (Front)

SMCS Code: 7310-FR

To provide full ventilation inside the cab, the upper window and the lower window can be fully opened.

WARNING

When opening or closing the windows, be extra careful to prevent any personal injury. The hydraulic lockout control must be in the LOCKED position in order to prevent any possibility of sudden movement of the machine due to inadvertent contact with the hydraulic control(s).

Do not change the position of the window until the following items have been done:

- Park the machine on a level surface.
- Lower the work tool to the ground.
- Move the hydraulic lockout control to the LOCKED position.
- Stop the engine.

Perform Step 1 through Step 3 to open the upper window.

Note: If equipped, the Cat Grade Control monitor may interfere with the window when opening. Ensure that the monitor is adjusted out of the way before opening the window.

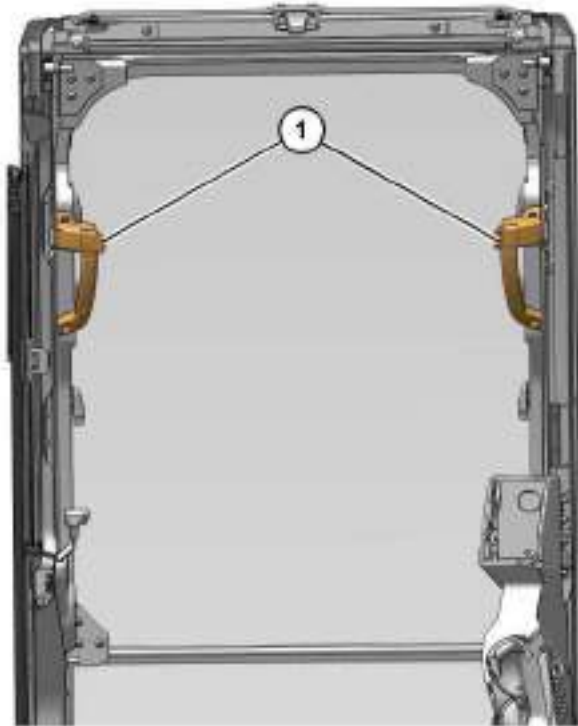


Illustration 506

g06185052

(1) Release lever

1. Release the auto-lock latches by pressing release levers (1) on the window handles.
2. Holding both handles on the window frame, pull the window upward.
3. Hold both grips that are provided on the window frame and move the window into the storage position until the auto-lock latches near the ceiling are engaged.

Perform Steps 4 through 5 to close the upper window.

Note: If equipped, the Cat Grade Control monitor may interfere with the window when closing. Ensure that the monitor is adjusted out of the way before closing the window.

4. Release the auto-lock latches by pressing release levers (1) on the window handles.

5. Reverse Steps 1 through 3 to close the upper window.

Perform Steps 6 through 8 to open the lower window and close the lower window.

6. Raise the lower window out of the window frame.

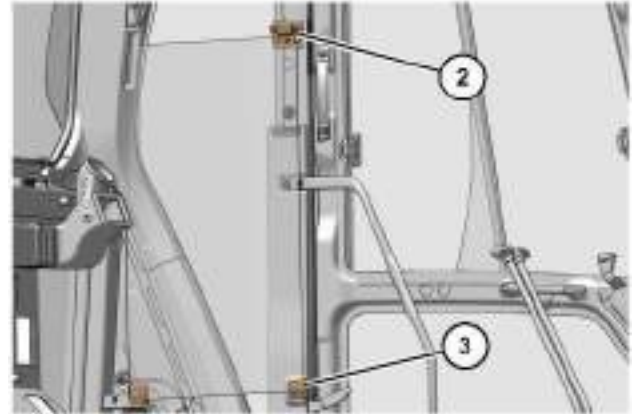


Illustration 507

g06185076

(2) Catch
(3) Brackets

7. Store the lower window in the holder that is located in the rear of the left side cab frame. To store the lower window, locate one end of the lower window into brackets (3). Secure the opposite end of the lower window with catch (2).

8. To close the lower window, reverse the procedure that is used for opening the lower window.

Note: The lower window is curved. The lower window can only be positioned one way in the holders.

i07538754

Sun Screen

SMCS Code: 7165-ZZ

WARNING

When opening or closing the windows, be extra careful to prevent any personal injury. The hydraulic lockout control must be in the LOCKED position in order to prevent any possibility of sudden movement of the machine due to inadvertent contact with the hydraulic control(s).

NOTICE

Do not change the position of the sun screen without performing the following actions:

- Park the machine on a level surface.
- Lower the work tool to the ground.
- Move the hydraulic lockout control to the LOCKED position.
- Stop the engine.

i07538807

Roof Hatch**SMCS Code:** 7303**WARNING**

When opening or closing the windows, be extra careful to prevent any personal injury. The hydraulic lockout control must be in the LOCKED position in order to prevent any possibility of sudden movement of the machine due to inadvertent contact with the hydraulic control(s).

NOTICE

Do not change the position of the roof hatch without performing the following actions:

- Park the machine on a level surface.
- Lower the work tool to the ground.
- Move the hydraulic lockout control to the LOCKED position.
- Stop the engine.



Illustration 508

g06179846

Pull sun screen (1) down from the ceiling. Hook the sun screen to the brackets (2) at both sides of the front window. The sun screen may be positioned at two different heights.

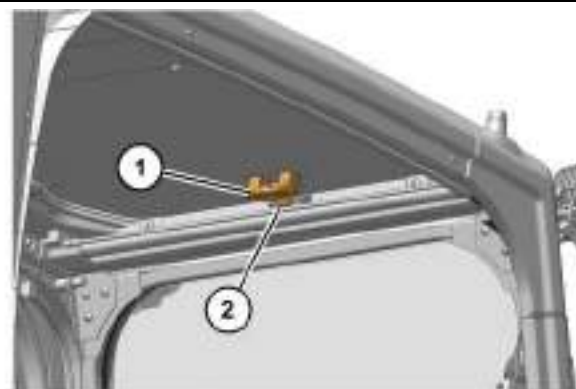


Illustration 509

g06179871

- (1) Grip
(2) Lock

To open the roof hatch, release lock (2). Hold grip (1) and push the roof hatch upward.

To close the roof hatch, hold grip (1) and pull the roof hatch downward. Engage lock (2) securely.

NOTICE

Do not stand or walk on the hatch or the roof of the cab. Serious damage may occur.

i06949447

Cab Door

SMCS Code: 7308



Illustration 510

g06180275

To open the cab door from the outside of the cab, pull outward on the door handle.



Illustration 511

g06179959

To open the cab door while inside the cab, push forward on the lever for the cab door latch.

For additional ventilation, open the cab door all the way to engage the catch on the exterior wall of the cab.

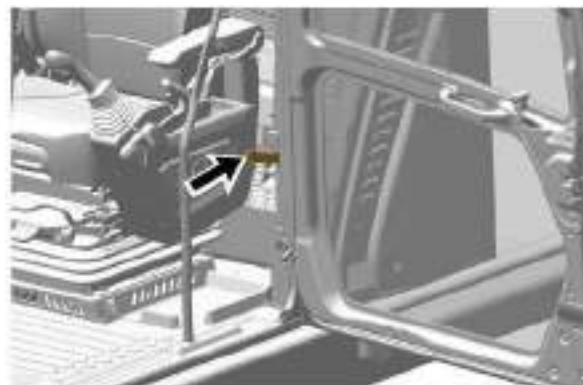


Illustration 512

g06180267

To release the cab door from the catch, pull downward on the cab door release lever.

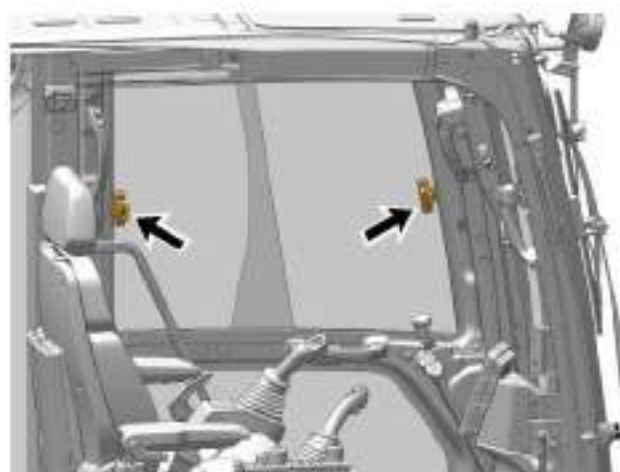


Illustration 513

g06179957

To open a window, release the window latch, and then slide the window to the desired position.

i06946782

Travel Control (Straight Travel Pedal (If Equipped))

SMCS Code: 5462

WARNING

With certain attachment combinations, the third pedal can have different functions. Always check for third pedal function before using the third pedal. Improper operation of the third pedal could result in serious injury or death.



Illustration 514

g06178249

Position for normal travel

- (A) Rear of machine
- (B) Final drive
- (C) Idler

When you travel, make sure that final drive sprockets (B) are under the rear of the machine.

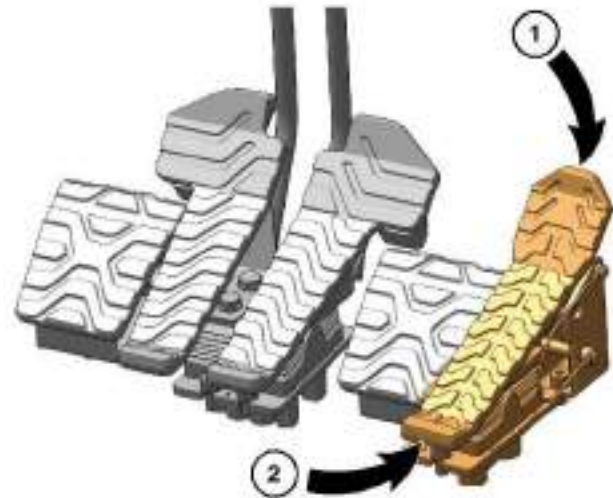


Illustration 515

g06178758

- (1) Forward Travel
- (2) Reverse Travel

The third pedal is to the right of the right travel pedal. The third pedal controls the forward and backward movement of the machine.

Note: If the third pedal is depressed and a travel pedal or a travel lever is operated, the machine will turn accordingly.

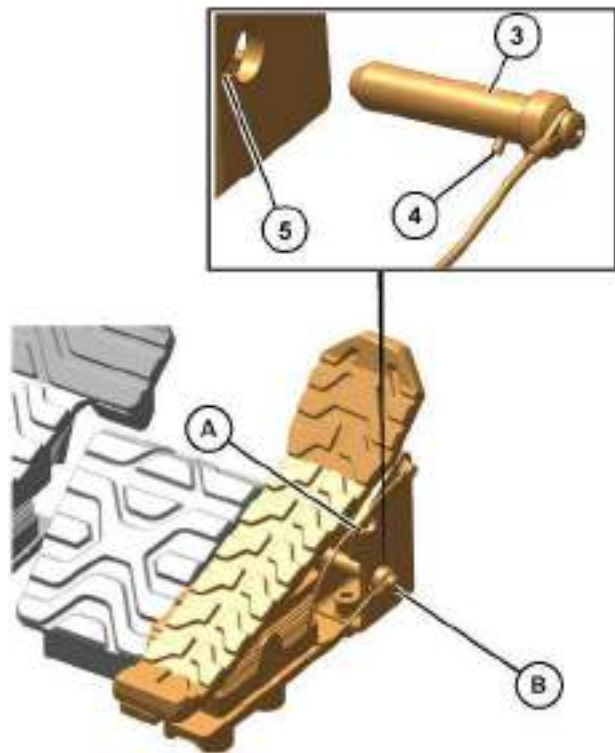


Illustration 516

g06178798

- (3) Lock pin
- (4) Pin
- (5) Notch
- (A) LOCKED position
- (B) UNLOCKED position

When the machine is not operated with the third pedal, install lock pin (3) at the LOCKED position to prevent accidental operation.

Note: To prevent lock pin (3) from being pulled out, insert pin (4) through notch (5) and turn lock pin (3) counterclockwise by 1/4 turn.

i07233941

Shovel Crane Control (If Equipped)

SMCS Code: 7451

S/N: HEX1-Up

WARNING

Do not perform a lifting operation with the shovel crane on a slope greater than 5 degrees. Lifting on a slope greater than 5 degrees may cause the machine to become unstable or roll over. Lifting with the shovel crane on a slope greater than 5 degrees may result in property damage, personal injury, or death.

WARNING

Do not operate the shovel crane with a hook that has cracks or deformities. Failure to follow these instructions may cause the load to fall and result in injury or death. Replace the shovel crane hook if there are any signs of cracks or deformities.

WARNING

Do not operate the bucket without securing the shovel crane hook. Operating the bucket without properly securing the hook may result in personal injury or machine damage.

NOTICE

Do not perform a lifting operation if the monitors external display light does not come on. Do not perform a lifting operation if the monitor does not display the suspended load information.

NOTICE

Lifting capacities are different than rated load capacities. Lifting capacities should not be used to determine the rated load of the shovel crane function. Only use the rated load capacities found in Operation and Maintenance Manual, "Specifications" to determine the rated load capacity for the shovel crane function.

NOTICE

Do not operate the shovel crane if the beacon light does not function properly. Before operation, check that the beacon light is flashing/rotating properly.

In some regions, regulations require a shovel crane configuration to lift certain objects. Always obey the local regulations in your region.

Refer to this Operation and Maintenance Manual, "Shovel Crane Operation" for additional information regarding the operation of the shovel crane.

Operate the machine according to the rated load table for your machine. Refer to Operation and Maintenance Manual, "Specifications (Shovel Crane)" for more information.

Activating the Shovel Crane



Illustration 517

g06224668

Use pitch indicator (1) and roll indicator (2) on the monitor to position the machine on a level surface.

Once the machine is in position on firm level ground, retract the bucket and lower the bucket to the ground.

Move the hydraulic lockout control to the LOCKED position. Stop the engine.

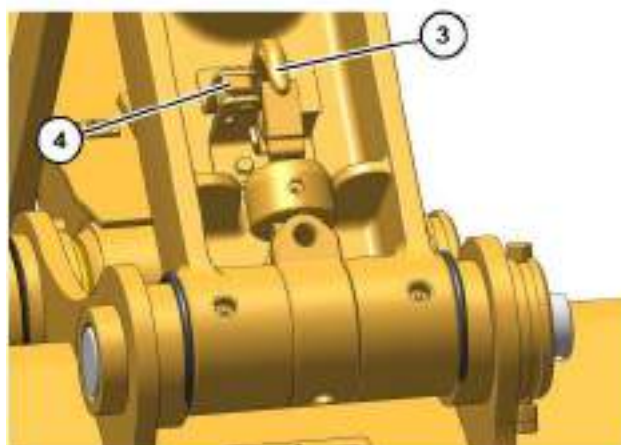


Illustration 518

g06205428

(3) Shovel crane hook
(4) Shovel crane latch

Release latch (4). Lower hook (3) into operating position.

Inspect the shovel crane hook for damage. See Operation and Maintenance Manual, "Shovel Crane - Inspect"



Illustration 519

g06224672

Press the function list button on the home screen.



Illustration 520

g06210063

Tap the crane option or if using the jog dial highlight the crane option and press down on the jog dial. The crane option should now say "ON"

Note: Tool Control should be turned OFF before the shovel crane can be activated. Select "Bucket" instead of any other tools. If any other work tool is selected, the shovel crane mode cannot be activated.

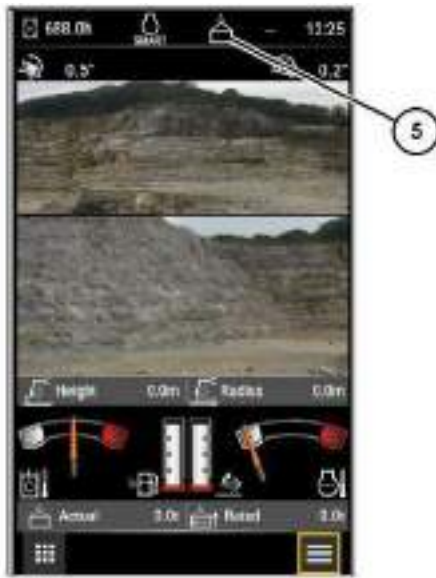


Illustration 521

g06224673

(5) Shovel crane indicator

Once the crane option is enabled, press the home button.

The monitor will display the following on the home page:

- Height
- Radius
- Actual Load
- Rated Load (not traveling)
- Rated Load (traveling)

Note: The rated load icon will turn yellow and a track will appear when the machine is travelling.

Shovel crane indicator (5) will display at the top of the monitor to indicate that the shovel crane feature is active.

Fully retract the bucket to activate the shovel crane mode. The bucket will lock into place and will not move, and the engine speed will then be reduced.

Note: If the hydraulic lockout control is moved to the LOCKED position, the shovel crane will become inoperable.

Refer to Operation and Maintenance Manual, "Shovel Crane Operation" for additional information regarding the operation of the shovel crane.

Deactivating the Shovel Crane



Illustration 522

g06191883

Return to the function list screen. Tap the crane option or if using the jog dial highlight the crane option and press down on the jog dial. The crane option should now say "OFF".

Position the stick so that it is perpendicular to the ground and slowly lower the boom until the bucket contacts the ground.

Return the hook to the stored position and latch the hook.

Indicators and Warnings

Below is a list of indicators and warnings that may appear on the monitor during the shovel crane operation.

Travel With Lift Out of Work Area – If the radius is over 70% of maximum radius when traveling, the monitor will display this warning. If this warning occurs, stop traveling until the machine center of gravity becomes stable or reduce the suspended load.

Crane 90% Load – If the suspended load is 90% of the rated load, the monitor will display this warning and an alarm will sound continuously. If this warning occurs, stop operation until the machine center of gravity becomes stable or reduce the suspended load.

Crane 100% Load – If the suspended load is 100% of the rated load, the monitor will display this warning.

and an alarm will sound continuously. If this warning occurs, stop operation until the machine center of gravity becomes stable or reduce the suspended load.

Crane Hook Interference – When the lifting height is too high, the monitor will display this warning. If this warning is present, stop lifting and lower the boom or retract the stick.

Boom Out of Work Area – When the boom cylinder is extended to the point near stroke end, the monitor will display this warning. If this warning is present, lower the boom.

Stick Out of Work Area – When the stick is retracted from the vertical position, the monitor will display this warning. If this warning occurs, return the stick to vertical.

Crane Malfunction – When an abnormality occurs with a shovel crane-related component, the monitor will display this warning. If this warning occurs, stop lifting and contact your Cat dealer.

Crane Mode Off – When the crane system has been stopped or disabled, the monitor will display this indicator.

i08015260

Joystick Controls

SMCS Code: 5705

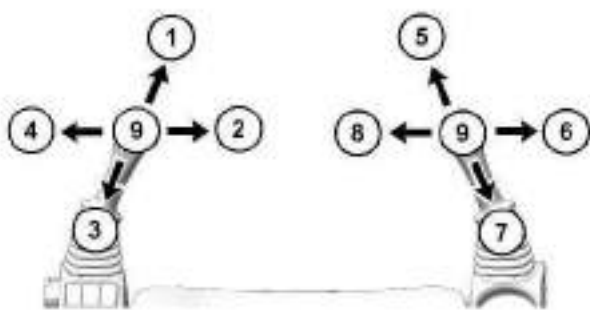


Illustration 523

g06180324

- (1) STICK OUT
- (2) SWING RIGHT
- (3) STICK IN
- (4) SWING LEFT
- (5) BOOM LOWER
- (6) BUCKET DUMP
- (7) BOOM RAISE
- (8) BUCKET CLOSE
- (9) HOLD

WARNING

The joystick and controls on the joystick can be configured with different functions. Always make sure to check the joystick configuration on the monitor before using the machine to avoid unexpected machine movement. These unexpected machine movements could cause a hazard resulting in serious injury or death.

WARNING

The Fine Swing Control delays the engagement of the swing parking brake.

If the machine is operating on a slope with the Fine Swing Control in the ON position, the swing motion may become uncontrollable which could result in property damage, personal injury or death.

Turn the Fine Swing Control to the OFF position when the machine is operating on a slope.

When you release the joysticks from any position, the joysticks will return to HOLD position (9). Movement of the upper structure will stop unless the fine swing control (if equipped) is ON. When the fine swing control is ON, the swing parking brake will not activate until 6.5 seconds after the joystick control for the swing function returns to the HOLD position.

Two functions may be performed at the same time by moving a joystick diagonally.

The machine control pattern is initially set at the factory to the SAE system, as shown. The pattern on the left pertains to the left joystick and the pattern on the right pertains to the right joystick.

The machine control pattern can be varied. Refer to Operation and Maintenance Manual, "Joystick Controls Alternate Patterns" for more information.

Manual Low Idle – Activate the manual low idle to reduce the engine speed to approximately 1000 rpm. Pressing the switch again will allow the engine speed to return to the original setting of the engine speed dial.

The manual low idle allows the operator to reduce the rpm without touching the engine speed dial. Manual low idle is useful when the operator wants to reduce the engine speed to talk to someone or while the operator is waiting for a truck.

Vertical Slider Joystick Controls



Illustration 524

g06224200

Vertical Slider Joystick Controls

- (1) Left joystick switch 3
(2) Left joystick switch 4
(3) Left joystick switch 1
(4) Left joystick switch 2
- (5) Left joystick thumbwheel
(6) Right joystick thumbwheel
(7) Right joystick switch 2
(8) Right joystick switch 1
- (9) Right joystick switch 4
(10) Right joystick switch 3

Table 29

Joystick Configurations				
Switch Location	Machine Configuration			
	Grade	Grade Assist	Grade Payload	Grade Assist Payload
1 ⁽¹⁾	Hammer	Hammer	Hammer	Hammer
2	Standby / Weigh	Standby / Weigh	Standby / Weigh	Standby / Weigh
3	Horn	Horn	Horn	Horn
4	Store	Store	Store	Store
5 ⁽¹⁾	Work Tool Rotation	Work Tool Rotation	Work Tool Rotation	Work Tool Rotation
6 ⁽¹⁾	Work Tool Open / Close	Work Tool Open / Close	Work Tool Open / Close	Work Tool Open / Close
7	Bucket Recall and Store	Bucket Recall and Store	Bucket Recall and Store	Bucket Recall and Store
8	Bench Mark	Bench Mark	Bench Mark	Bench Mark
9	Grade and Bucket Assist Activation	Grade and Bucket Assist Activation	Grade and Bucket Assist Activation	Grade and Bucket Assist Activation
10	Laser and Touch Point	Laser and Touch Point	Laser and Touch Point	Laser and Touch Point

⁽¹⁾ Button is configurable on machines without tool control.

Note: The following functions can be assigned to the configurable buttons: radio mute, one-touch low idle, work tool select, and HVAC.



Illustration 525

g06225116

Vertical Slider Joystick Controls

- (7) Left joystick switch 3
- (8) Left joystick switch 4
- (9) Left joystick switch 1
- (10) Left joystick switch 2
- (11) Left joystick thumbwheel
- (12) Right joystick thumbwheel
- (13) Right joystick switch 2
- (14) Right joystick switch 1
- (15) Right joystick switch 4
- (16) Right joystick switch 3

Table 30

Joystick Configurations	
Switch Location	Joystick With Tool Control Sliders
7 ⁽¹⁾	Hammer
8	Configurable
9	Horn
10	Configurable
11 ⁽¹⁾	Work Tool Rotation
12 ⁽¹⁾	Work Tool Open / Close
13	Configurable
14	Configurable
15	Configurable
16	Configurable

(Table 30, contd)

(1) Button is configurable on machines without tool control.

Medium Pressure (If Equipped)

Rotating Tool Control

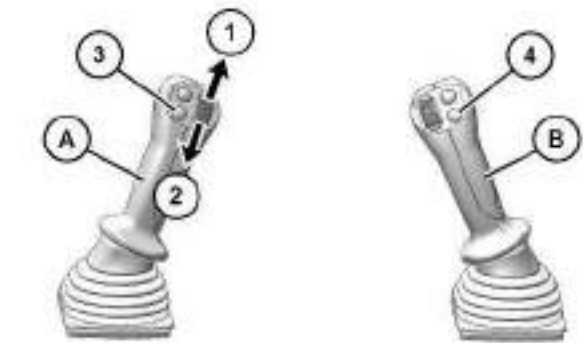


Illustration 526

g06260903

- (A) Left joystick
- (B) Right joystick
- (1) Thumb wheel (Clockwise)
- (2) Thumb wheel (Counterclockwise)
- (3) Horn switch
- (4) AEC switch

(continued)



(1) **ROTATE CLOCKWISE** – Move the thumb wheel upward to rotate the work tool clockwise.



(2) **ROTATE COUNTERCLOCKWISE** – Move the thumb wheel downward to rotate the work tool counterclockwise.



(3) **HORN** – Press the horn switch on the left joystick to activate the horn.



(4) **AEC SWITCH** – Press the AEC switch on the right joystick to activate low engine speed. Press the switch again to activate high engine speed.

i08233391

Joystick Controls (Tilt-Rotator System (TRS) (If Equipped))

SMCS Code: 5705

WARNING

Verify the joystick control pattern and functionality parameters before operating machine.

Ensure that the joystick control film on the cab window matches the control functions of your machine. Consult your Cat dealer for additional information regarding the joystick control film.

Become familiar with the joystick controls before operating the machine.

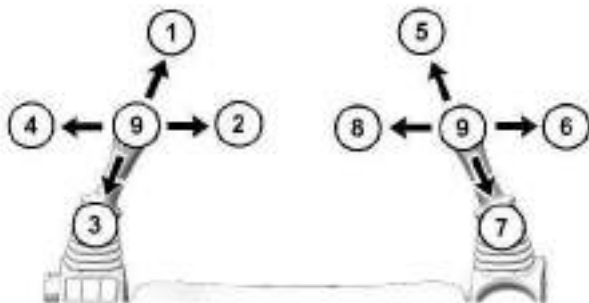


Illustration 527

g06180324

- (1) STICK OUT
- (2) SWING RIGHT
- (3) STICK IN
- (4) SWING LEFT
- (5) BOOM LOWER
- (6) BUCKET DUMP
- (7) BOOM RAISE
- (8) BUCKET CLOSE
- (9) HOLD

WARNING

The Fine Swing Control delays the engagement of the swing parking brake.

If the machine is operating on a slope with the Fine Swing Control in the ON position, the swing motion may become uncontrollable which could result in property damage, personal injury or death.

Turn the Fine Swing Control to the OFF position when the machine is operating on a slope.

When you release the joysticks from any position, the joysticks will return to HOLD position (9). Movement of the upper structure will stop unless the fine swing control (if equipped) is ON. When the fine swing control is ON, the swing parking brake will not activate until 6.5 seconds after the joystick control for the swing function returns to the HOLD position.

Two functions may be performed at the same time by moving a joystick diagonally.

The machine control pattern is initially set at the factory to the SAE system, as shown. The pattern on the left pertains to the left joystick and the pattern on the right pertains to the right joystick.

The machine control pattern can be varied. Refer to Operation and Maintenance Manual, "Joystick Controls Alternate Patterns" for more information.

Manual Low Idle – Activate the manual low idle to reduce the engine speed to approximately 1000 rpm. Pressing the switch again will allow the engine speed to return to the original setting of the engine speed dial.

The manual low idle allows the operator to reduce the rpm without touching the engine speed dial. Manual low idle is useful when the operator wants to reduce the engine speed to talk to someone or while the operator is waiting for a truck.

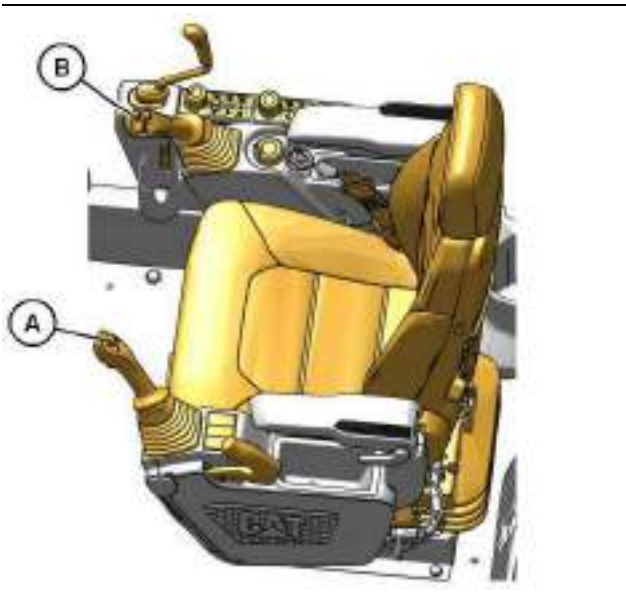


Illustration 528 g06595876

- (A) Left Joystick
- (B) Right Joystick

Rear View of Joystick Controls

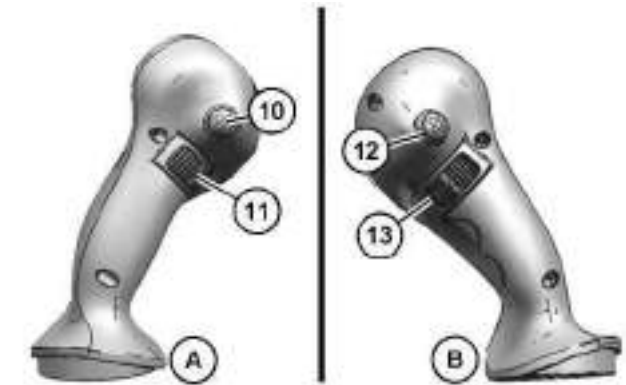


Illustration 529 g06596328

- (A) Rear View Left Joystick
- (B) Rear View Right Joystick
- (10) Left Joystick Switch
- (11) Left Joystick Thumb Wheel
- (12) Right Joystick Switch
- (13) Right Joystick Thumb Wheel

Left Joystick Control Patterns

The left joystick has seven buttons.

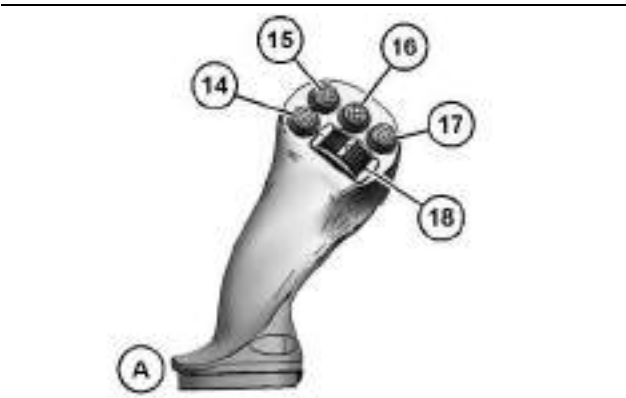


Illustration 530 g06595878

Typical Example
(A) Left Joystick

Table 31

Joystick Control	Left Joystick Function
(14)	Tilt-Rotator Bucket
(15)	Horn
(16)	Weigh Mode
(17)	One Touch
(18)	-

Right Joystick Control Patterns

The right joystick has seven buttons.



Illustration 531 g06595879

Typical Example
(B) Right Joystick

Table 32

Joystick Control	Right Joystick Patterns
19	Laser and Touch
20	Grade and Bucket
21	Bench Mark
22	Tilt-Rotator Rec
23	-

i06951419

SmartBoom Control (If Equipped)

SMCS Code: 5461-ZS; 7332

WARNING

Personal injury or death can result from not following the proper procedures.

To avoid the possibility of injury or death, follow the established procedure.

WARNING

Activating the SmartBoom function and using the work tool joystick control while the front of the machine is elevated could result in unexpected machine motion. Unexpected machine motion could result in serious injury or death. Do not activate the SmartBoom function if the front of the machine is elevated by the front linkage.

WARNING

Do not elevate or lower the track when in the SmartBoom mode. Follow the operation procedures for the SmartBoom in the Operation and Maintenance Manual. Failure to follow these instructions can result in serious injury or death.

WARNING

Always make sure that the boom control joystick is in the NEUTRAL position before activating the SmartBoom control. Activating the SmartBoom control with the joystick out of the neutral position could result in unexpected machine motion which could result in serious injury or death.

WARNING

Do not select any SmartBoom mode, using the SmartBoom selector switch located on the console, while the tracks are elevated. Selecting the SmartBoom mode with the tracks elevated could result in a sudden drop of the machine which could result in serious injury or death.

WARNING

If any SmartBoom mode is active and the boom control joystick is in the BOOM DOWN position (forward) with a bucket or a work tool on the ground, pressing the disable button that is located on the front of the right hand joystick could cause a sudden boom down motion. This control function could lift the machine upward, with unexpected machine movement that could result in serious injury or death. Do not press the disable button while the SmartBoom mode is active and the boom control joystick is in the BOOM DOWN position (forward) with a bucket or a work tool on the ground.

WARNING

Do not attempt to lift the tracks of the machine by using the disable button and applying downward force with the boom lowering control while the machine is in any SmartBoom mode. Releasing the disable button will immediately return the machine to the active SmartBoom mode. This action could cause the machine to drop down abruptly which could result in serious injury or death.



Illustration 532

g06210059

From the home screen, press the function list button.



Illustration 533

g06210090

Tap the smart boom option, or if using the jog dial, select the smart boom option and press down on the dial.



Illustration 534

g06210095

Select either “Up & Down” or “Down Only” from the list. The two options work as follows:

“Up & Down” mode: When the joystick is moved to the BOOM DOWN position, the boom will lower by the weight of the boom. The boom can move upward freely.

“Down Only” mode: The boom will lower by the weight of the boom when the control lever is moved to the BOOM LOWER position. This mode prevents the boom from moving upward. To move the boom upward, the operator must use the boom control joystick to activate the BOOM UP mode.

Press the home button when finished.

During operation of the SmartBoom function, the operator may wish to apply downward force to the boom. The operator can disable the SmartBoom function temporarily with the SmartBoom disable switch. While the trigger switch is pressed, BOOM RAISE and BOOM LOWER will operate in the normal modes. The SmartBoom disable switch is a configurable switch.

i08222281

Work Tool Control (One-Way Flow) (If Equipped)

SMCS Code: 6700

The following information pertains to work tools that require hydraulic oil flow in one direction. Hydraulic hammers are an example of work tools that require hydraulic oil flow in one direction.

Note: For information that pertains to work tools that require hydraulic oil flow in two directions, refer to Operation and Maintenance Manual, "Work Tool Control (Two-Way Flow)".

Joystick

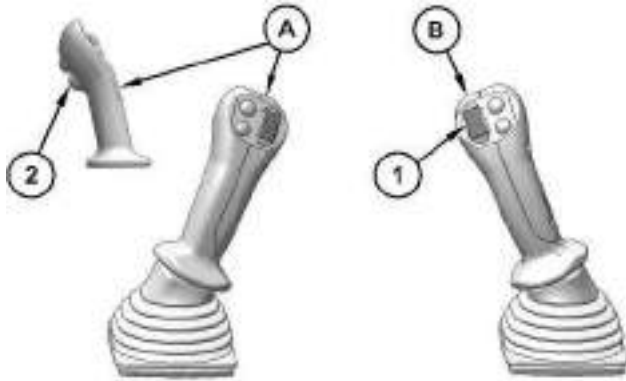


Illustration 535

g06588792



Illustration 536

g06588793

(A) Left joystick
(B) Right joystick



(1) Variable Speed – Move the thumb wheel downward to activate the work tool. Move the thumb wheel further to increase the speed of the work tool.



(2-1) On/Off (Momentary) – While pressing this switch, the work tool will remain activate at a constant rate. Release the switch to turn off the work tool.



(2-2) On/Off (Toggled) – Press the switch once to activate the work tool. Press the switch again to turn off the work tool.

Work Tool Pedal

WARNING

With certain attachment combinations, the work tool pedal can have different functions. Always check for work tool pedal function before using the work tool pedal. Improper operation of the work tool pedal could result in serious injury or death.

The work tool pedal can be installed on either side of the travel pedals. The work tool pedal allows the operator to modulate the speed of the work tool.

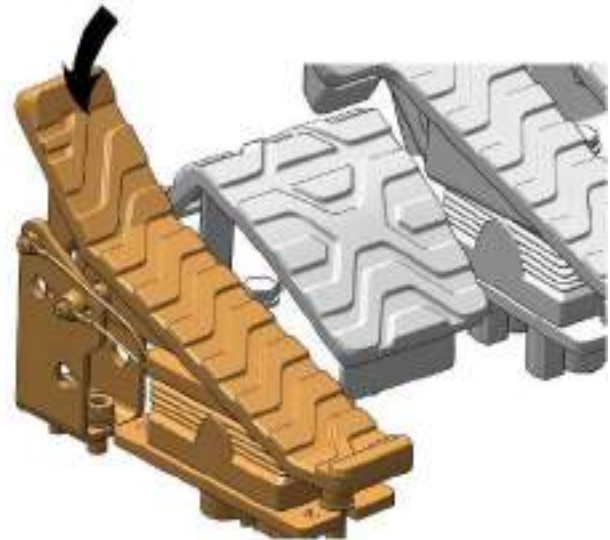


Illustration 537

g06180447



Variable Speed – Push down on the front of the pedal to activate the work tool. Move the pedal further to increase the speed of the work tool. Release the pedal to turn off the work tool.

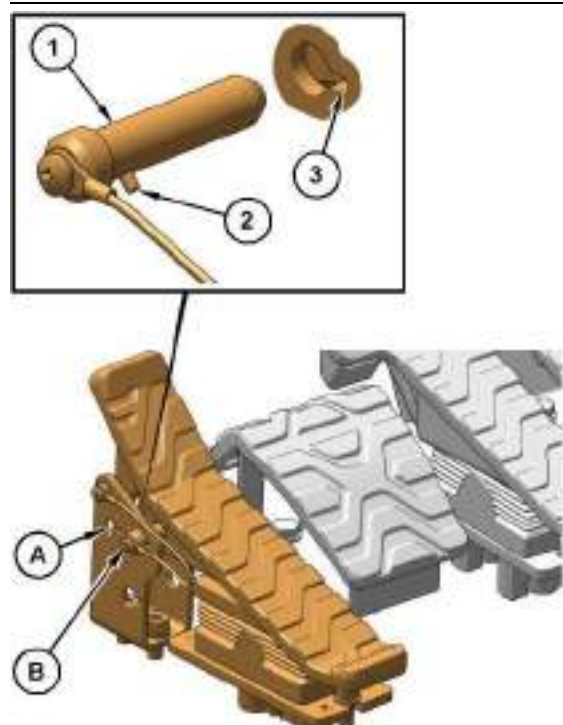


Illustration 538

g06591307

- (1) Lock pin
- (2) Pin
- (3) Notch
- (A) UNLOCKED position
- (B) LOCKED position

When you are not using the work tool, put the lock pin (1) in LOCKED position (B). This will lock the work tool pedal to prevent any unexpected operation of the work tool.

Note: To prevent lock pin (1) from being accidentally pulled out, insert pin (2) through notch (3) and turn lock pin (1) counterclockwise by 1/4 turn.

i07243676

Work Tool Control (Two-Way Flow) (If Equipped)

SMCS Code: 6700

WARNING

The joystick and controls on the joystick can be configured with different functions. Always make sure to check the joystick configuration on the monitor before using the machine to avoid unexpected machine movement. These unexpected machine movements could cause a hazard resulting in serious injury or death.

The following information pertains to work tools that require hydraulic oil flow in two directions. These work tools can also be equipped with a rotate circuit. Hydraulic shears, pulverizers, crushers, and grapples are examples of work tools that require hydraulic oil flow in two directions.

Note: For information that pertains to hydraulic hammers, refer to Operation and Maintenance Manual, "Work Tool Control (One-Way)".

Joystick

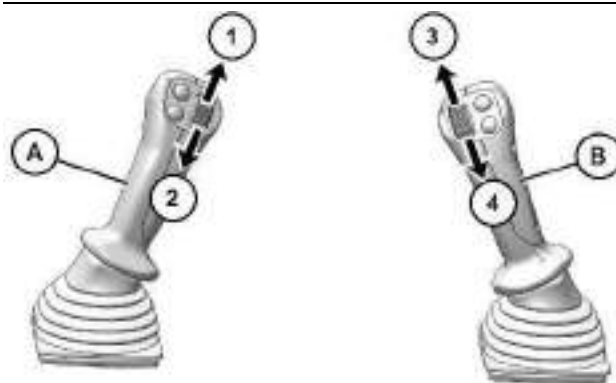


Illustration 539

g06180488

- (A) Left joystick
- (B) Right joystick



(1) ROTATE CLOCKWISE – Move the thumb wheel upward to rotate the work tool clockwise.



(2) ROTATE COUNTERCLOCKWISE – Move the thumb wheel downward to rotate the work tool counterclockwise.



(3) CLOSE – Move the thumb wheel upward to close the work tool.



(4) OPEN – Move the thumb wheel downward to open the work tool.

Work Tool Pedal

WARNING

With certain attachment combinations, the work tool pedal can have different functions. Always check for work tool pedal function before using the work tool pedal. Improper operation of the work tool pedal could result in serious injury or death.

The work tool pedal can be installed on either side of the travel pedals. The work tool pedal allows the operator to vary the speed of the work tool.

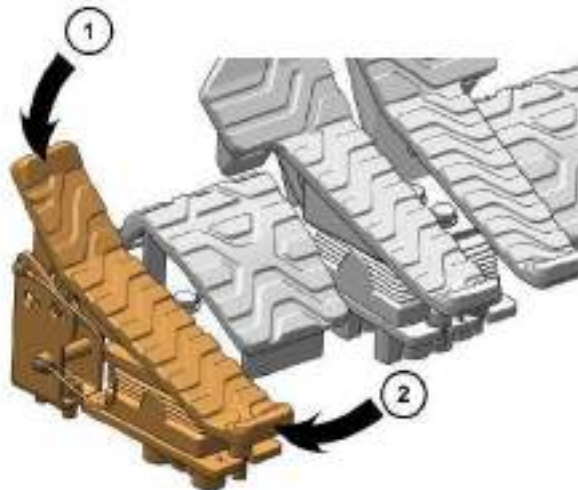


Illustration 540

g06180510



(1) CLOSE – Push down on the front of the pedal to close the work tool.



(2) OPEN – Push down on the rear of the pedal to open the work tool.

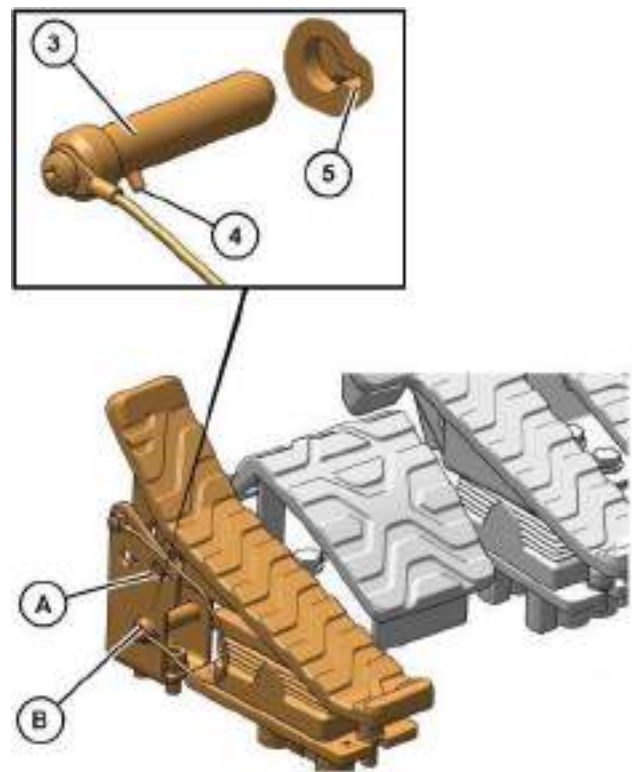


Illustration 541

g06180514

- (3) Lock pin
- (4) Pin
- (5) Notch
- (A) LOCKED position
- (B) UNLOCKED position

When you are not using the work tool, put the lock pin (3) in LOCKED position (A). This will lock the work tool pedal to prevent any unexpected operation of the work tool.

Note: To prevent lock pin (3) from being accidentally pulled out, insert pin (4) through notch (5) and turn lock pin (3) by 1/4 turn.

i08209549

Joystick Controls Alternate Patterns

SMCS Code: 5059; 5137

Changing Machine Control Pattern (If Equipped)

WARNING

Whenever a change is made to the machine control pattern, also exchange the pattern card in the cab to match the new pattern.

Check the machine control pattern for conformance to the pattern on the card in the cab. If the pattern does not match, change the card to match the machine control pattern before you operate the machine. Failure to do so could result in personal injury.

The machine control pattern can be changed to the ISO/JIS pattern, BHL pattern, MHI pattern, KOBE pattern, or the former SCM pattern. To change the joystick controls between the patterns, refer to Operation and Maintenance Manual, Monitoring System for more information.

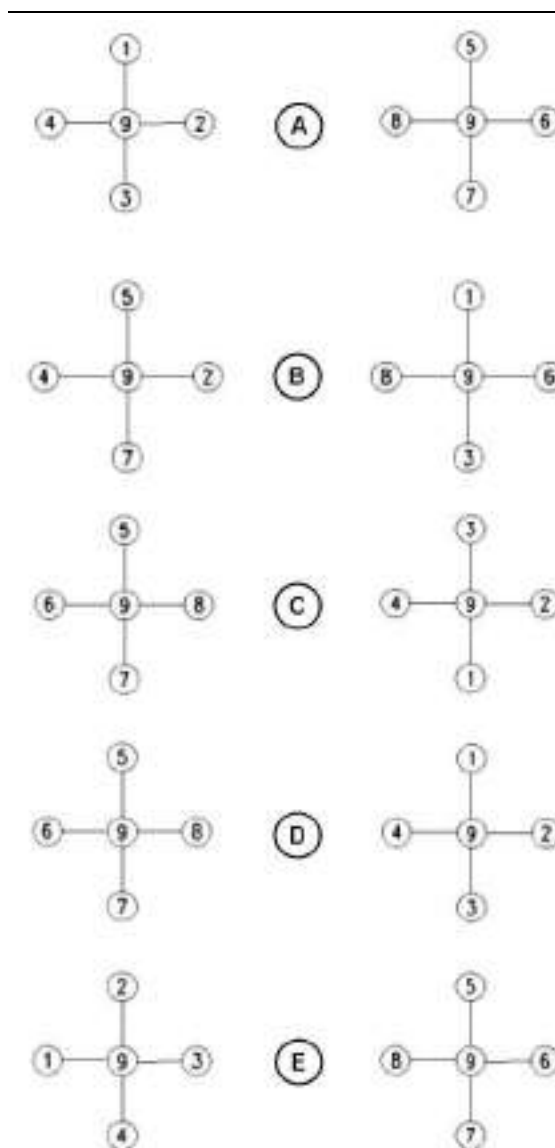


Illustration 542

g06136699

- (A) ISO/JIS machine control pattern
- (B) BHL machine control pattern
- (C) MHI machine control pattern
- (D) KOBE machine control pattern
- (E) Former SCM machine control pattern

The patterns on the left side of the Illustration show the possible configurations for the left control lever. The patterns on the right side of the Illustration show the possible configurations for the right control lever.



STICK OUT (1) – Move the control lever to this position to move the stick outward.



SWING RIGHT (2) – Move the control lever to this position to swing the upper structure to the right.



STICK IN (3) – Move the control lever to this position to move the stick inward.



SWING LEFT (4) – Move the control lever to this position to swing the upper structure to the left.



BOOM LOWER (5) – Move the control lever to this position to lower the boom.



BUCKET DUMP (6) – Move the control lever to this position to dump the bucket.



BOOM RAISE (7) – Move the control lever to this position to raise the boom.



BUCKET CLOSE (8) – Move the control lever to this position to close the bucket.

HOLD (9) – When the control lever is released from any position, the control lever will return to the HOLD position. Movement of the upper structure will stop.

Two functions may be performed at the same time by moving a control lever diagonally.

If the machine is equipped with a hydraulic hammer, the function of position (6) and of position (8) is different.

HYDRAULIC HAMMER RAISE (6) – Move the control lever to this position to raise the hydraulic hammer.

HYDRAULIC HAMMER LOWER (8) – Move the control lever to this position to lower the hydraulic hammer.

If the machine is equipped with a grapple, the function of position (6) and of position (8) is different.

GRAPPLE OPEN (6) – Move the control lever to this position to open the grapple arms.

GRAPPLE CLOSE (8) – Move the control lever to this position to close the grapple arms.

If the machine is equipped with a clamshell, the function of position (6) and of position (8) is different when in “Clamshell” mode.

Note: When in “Clamshell” mode, the work tool can only be operated if Work Tool Select also has clamshell selected.

CLAMSHELL OPEN (6) – Move the control lever to this position to open the clamshell.

CLAMSHELL CLOSE (8) – Move the control lever to this position to close the clamshell.

i06951805

Fuel Tank Shutoff and Drain Control

SMCS Code: 1273

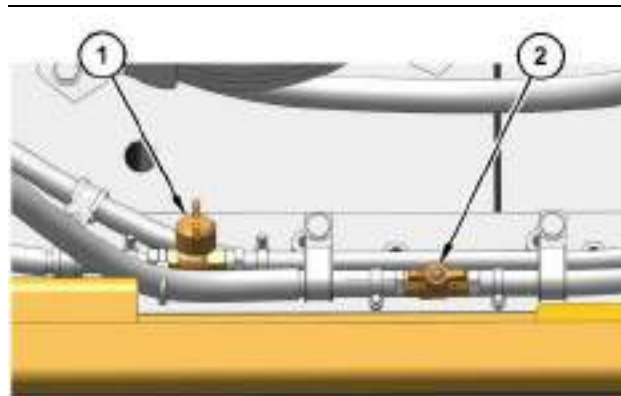


Illustration 543

g06214473

Fuel tank drain valve and shutoff

Fuel Tank Drain Valve (1) – The drain valve for the fuel tank is located behind the right side access door. To drain the water and sediment from the fuel tank, turn the fuel drain valve counterclockwise. To close the fuel tank drain valve, turn the drain valve clockwise.

Fuel Shutoff Valve (2) – The fuel shutoff valve is located behind right side access door. To shut off the fuel supply, turn the fuel shutoff valve clockwise. To turn on the fuel supply, turn the fuel shutoff valve counterclockwise.

Note: For more detailed information that pertains to draining the water and sediment from the fuel tank, refer to Operation and Maintenance Manual, “Fuel Tank Water and Sediment - Drain”.

Engine Starting

i08185791

Engine Starting

SMCS Code: 1000; 1090; 1456; 7000

NOTICE

This machine is equipped with a Cat[®] Machine Security System (MSS) and may not start under certain conditions.

NOTICE

The engine start switch must be in the ON position and the engine must be running in order to maintain electrical functions and hydraulic functions. This procedure must be followed in order to prevent serious machine damage.

Note: The engine can start in areas that have temperatures as low as -18°C (0°F). For areas that are colder, a starting kit for cold weather is available.

1. Move the hydraulic lockout control to the LOCKED position.

This machine is equipped with an engine neutral start system. The system only allows the engine to start when the lever for the hydraulic lockout control is in the LOCKED position.

2. Ensure that the joysticks and travel controls are in the HOLD position.



Illustration 544

g06209482

3. The operator passcode, Bluetooth key or Cat App: Fleet management app must be authenticated before starting the engine. Once authenticated, an "Engine Start Allowed" message will appear across the top of the monitor and the start switch LED will turn green.



Illustration 545 g06579150

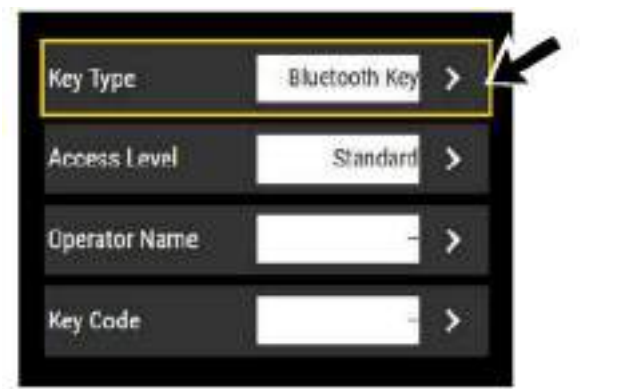


Illustration 546 g06579155

4. Bluetooth devices and passcodes can be registered using the in-cab display if the operator is logged in to the system using a master access account. Contact your Cat dealer for additional information.

Reference: Refer to Operation and Maintenance Manual, Machine Security System, Operator Login for instructions.

5. Before you start the engine, check for the presence of bystanders or maintenance personnel. Ensure that all personnel are clear of the machine. Briefly sound the horn before you start the engine.

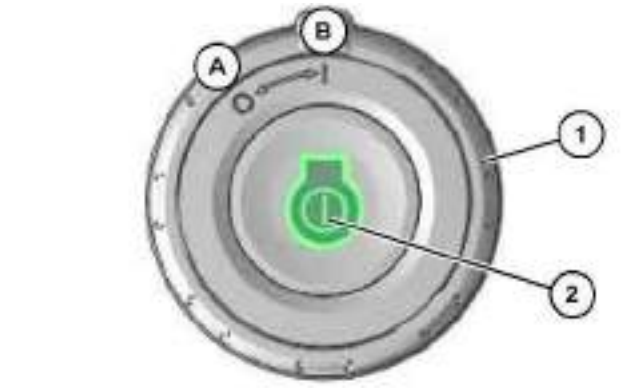


Illustration 547 g06226447

- (A) Off
- (B) On
- (1) Engine start ring
- (2) Engine start button

6. Turn the engine start ring(1) to the ON position, then press and hold start button (2) to start the engine. Release the button after the engine has started.

If the engine is having trouble starting, do not crank the engine for more than 30 seconds. Cranking the engine for more than 30 seconds can damage starting system components.

i07552385

Engine and Machine Warm-Up

SMCS Code: 1000; 7000

NOTICE

Keep engine speed low and do not operate until the message "Warm-Up Mode Power Derate" on the monitor goes out. If it does not go out within thirty seconds, stop the engine and investigate the cause before starting again. Failure to do so, can cause engine damage.

NOTICE

Always run the engine at low idle for at least ten minutes before performing any other operations in cold conditions or each time the engine oil and oil filter are changed in order to protect your engine and hydraulic components.

NOTICE

Depending on the ambient temperature, in order to prevent the machine operation with high speed without sufficient lubrication at the turbo bearing, the engine speed may be set to low speed and the hydraulic power minimized for a pre-determined time after the engine starts. Refer to turbo protection feature.

The engine may automatically change speeds when the machine is stationary and idling in cold ambient temperature for an extended time. This is to:

- Maintain desired coolant temperature.
- Maintain desired operation of engine systems.

During extended idling in cold ambient conditions, engine speed may operate between 900 rpm and 1000 rpm. Operation at 1000 rpm is minimal and will only last for up to 20 minutes.

Hydraulic System

Automatic Warm-Up

This machine comes with an automatic warm-up feature that can be enabled or disabled. If the feature is enabled, and the hydraulic oil temperature is below the threshold that has been set, a prompt will appear on the monitor after starting the machine. Follow the prompts on the monitor. If this feature is disabled or you would like to change the temperature setting, refer to Operation and Maintenance Manual, Monitoring System.



Illustration 548

g06219830

Manual Warm-Up

⚠ WARNING

When you cycle the machine controls, the machine can move suddenly. Contact between the machine and external objects or ground personnel can result in serious injury or death. Before you cycle the machine controls, the machine should be located in an unobstructed, hazard-free work area that is away from external objects and ground personnel.

1. Make sure that the area is clear of personnel and equipment.

Note: The hydraulic lockout control must be in the UNLOCKED position before the hydraulic controls will function.

2. Allow the engine to warm up at low idle for at least 5 minutes. Engage the work tool controls and disengage the work tool controls. This will speed up the warm-up of the hydraulic components.

When you idle the machine for warm-up, observe the following recommendations:

- If the temperature is greater than 0°C (32°F), warm up the engine for approximately 15 minutes.
- If the temperature is less than 0°C (32°F), warm up the engine for approximately 30 minutes.

- If the temperature is less than -18°C (0°F) or if hydraulic functions are sluggish, additional time may be required.

NOTICE

The hydraulic oil temperature should be higher than 25°C (77°F) before performing work with the machine. Make sure that the warm-up procedure is performed.

If the hydraulic oil temperature is less than 25°C (77°F) and the machine is operated abruptly, serious damage to the hydraulic components may occur.

Note: The recommended operating temperature of the hydraulic fluid for this machine is 55°C (131°F).

3. To warm up the hydraulic oil, turn the engine speed dial to the medium engine speed. Run the engine for approximately 5 minutes and move the joystick intermittently from the BUCKET DUMP position to the HOLD position. Do not hold the joystick in the BUCKET DUMP position with the bucket cylinder fully extended for more than 10 seconds.

This allows the oil to attain relief pressure, which causes the oil to warm up more rapidly.

4. Turn the engine speed dial to the maximum engine speed and repeat Step 3.
5. Cycle all controls to circulate warm oil through all hydraulic cylinders and all hydraulic lines, and through the swing motor and travel motors.
6. Observe the gauges and the indicators frequently during the operation.



Turbo Protection Power Derate – After an engine start, the engine speed will be set to low speed and the hydraulic power limited for a time period. During this period, the monitor displays the message "Warm-Up Mode Power Derate". (Maximum is around 30 seconds). After the turbo bearing lubrication is sufficient, the engine speed goes to the setting dial speed and the monitor stops to display the message.

Improve Cold-Weather Performance

Covers installed over the vents in the radiator compartment door will help to control overcooling in ambient temperatures below -15°C (5°F).

The materials used for the covers and the method used to install the covers is at the installers discretion.

Install the covers if overcooling is observed while the machine is idling in ambient temperatures below -15°C (5°F).

Stop the machine, and remove the covers under the following conditions:

- The ambient temperature is above -15°C (5°F).
- The engine temperature gauge indicates overheating.
- The hydraulic oil temperature gauge indicates overheating.

Recommendation for Crankcase Breather Protection (Machines with C4.4 and C7.1)

Crankcase ventilation gases contain a large quantity of water vapor. This water vapor can freeze in cold ambient conditions and can plug or damage the crankcase ventilation system. If the engine is operated in temperatures below -25°C (-13°F), measures must be taken to prevent freezing and plugging of the breather system. Insulated hoses and a heated canister assembly should be installed.

Consult with your Cat dealer for the recommended breather components for operation from -25° to -40°C (-13° to -40°F).

Installation



Illustration 549

g06181368

Vent locations on the radiator compartment door.

1. Clean the surface of the radiator compartment door.

2. Install the covers in the locations shown in Illustration 549 . The covers should fully cover the door vents.

Operation

i07348768

Operation Information

SMCS Code: 7000

Note: Operating Temperature Range for the Machine The machine must function satisfactorily in the anticipated ambient temperature limits that are encountered during operation. The standard machine configuration is intended for use within an ambient temperature range of -18°C (0°F) to 43°C (109°F). Special configurations for different ambient temperatures may be available. Consult your Cat dealer for additional information on special configurations of your machine.

Make sure that no personnel are on the machine or near the machine to prevent any personal injury. Keep the machine under control at all times to prevent injury.

Sound the horn and allow adequate time for bystanders to clear the area before moving the machine into a restricted visibility area. Follow local practices for your machine application. For more information refer to Operation and Maintenance Manual, "Restricted Visibility".

Reduce the engine speed when you maneuver the machine in tight quarters and when you drive over an incline.

Select the necessary travel speed range before you drive downgrade. Do not change the travel speed range while you drive downhill.

Use the same travel speed on a downgrade and on an upgrade.

When you travel for any distance, keep the stick inward and carry the boom in a low position.

When you drive up a steep grade, keep the boom as close to the ground as possible.

When you travel uphill or you travel downhill, keep the boom on the uphill side of the machine.

1. Adjust the operator seat.
2. Fasten the seat belt.



Illustration 550

g06181515

3. Turn the engine speed dial to the desired operating range.
4. Move the hydraulic lockout control to the UNLOCKED position.



Illustration 551

g06181525

5. Raise the boom enough to provide sufficient ground clearance.



Illustration 552

g06181517

6. Select the desired travel speed by operating the travel speed control switch. The indicator will light to display the active mode.
7. Make sure that the position of the upper structure and of the undercarriage is known before you move the machine. The drive sprockets should be at the rear of the machine.

Note: The directional steering controls will operate normally if the drive sprockets are at the rear of the machine and the idlers are at the front of the machine and under the cab. When the sprockets are under the cab, the travel controls will operate backward.

8. Turn the engine speed dial to increase the engine speed (rpm) to the desired speed.
9. Push both travel levers forward at the same time to travel forward. If both travel levers are pushed farther, the travel speed at the selected engine speed (rpm) will be faster.

Note: If the machine does not operate or if the machine does not travel in a straight line, consult your Cat dealer.

10. See Operation and Maintenance Manual, "Operator Controls" for information about spot turning and about pivot turns.

11. When you make turns in soft material, travel in a forward direction occasionally to clear the tracks.
12. Slowly move both of the travel levers or both of the travel pedals to the CENTER position to stop the machine.

Lifting Objects

If the machine is equipped with the CE plate per requirements for the European Union, used to lift objects, then the machine must be equipped with a boom lowering control valve, a stick lowering control valve, and an overload warning device.

A fit for purpose test was completed to confirm that a properly equipped machine meets the requirements of the European Union Machinery Directive "2006/42/EC" for lifting objects.

The overload warning device (if equipped) must be adjusted for the bucket linkage and bucket size that is installed on the machine. Adjust the overload warning device for proper operation.

The setting for the overload warning device (if equipped) should be checked by an authorized dealer.

i06981624

Frozen Ground Conditions

SMCS Code: 7000



Illustration 553

g06185895

To free the tracks from frozen ground, swing the boom to the front of the machine. Use boom down pressure to free the idler end of the machine.

Swing the boom to the rear of the machine. Use boom down pressure to free the sprocket end of the machine.

i07474576

Equipment Lowering with Engine Stopped

SMCS Code: 7000

To lower the boom, place the hydraulic lockout control in the UNLOCKED position. Move the joystick to the BOOM LOWER position. If the accumulator is still charged, the boom will lower.

If the boom does not lower, the accumulator is empty. Use one of the following procedures to lower the boom.

Machines Equipped with Boom Lowering Control Valves

WARNING

Boom load may cause cylinder oil pressure to reach relief pressure of the boom lowering control device when the boom is supported by one cylinder. Boom can lower suddenly, causing possible injury or death.

To avoid possible injury or death, be sure no one is under or near the work tool before manually lowering the boom.

Keep all personnel away from the boom drop area when lowering the boom with the engine stopped.

WARNING

Be sure no one is under or near the work tools before manually lowering the boom. Keep all personnel away from the boom drop area when lowering the boom with the engine stopped in order to avoid possible personal injury.

WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the work tools have been lowered to the ground, and the oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

If the engine or the hydraulic system is disabled and the boom is up, the boom can be lowered manually. Boom lowering control valves allow the boom to be manually lowered. The boom lowering control valves are located at the head end port on the boom cylinders.

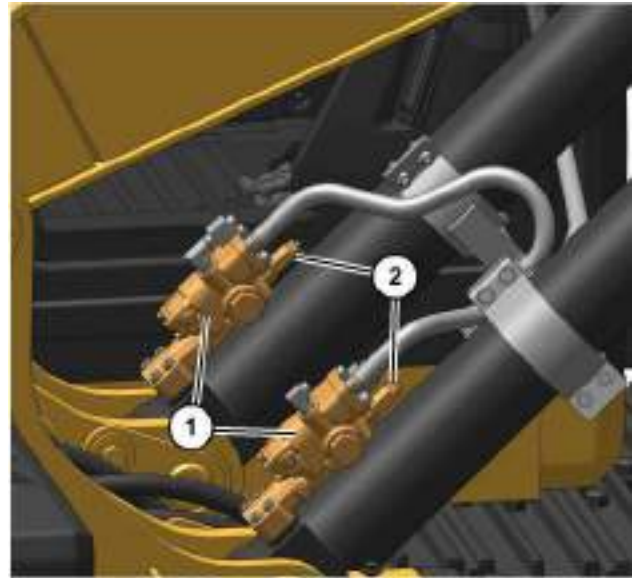


Illustration 554

g06263683

- (1) Boom lowering control valve
- (2) Line relief

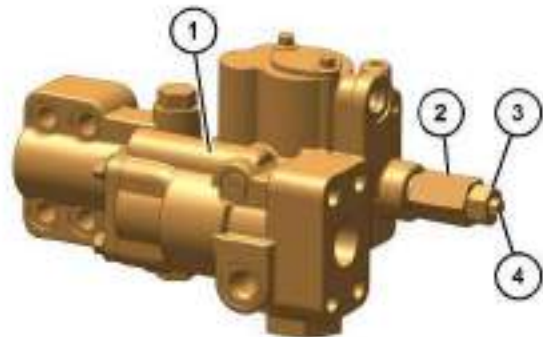


Illustration 555

g06263723

- (1) Boom lowering control valve
- (2) Line relief
- (3) Locknut
- (4) Set screw

1. Loosen the locknut (3) at each of the boom lowering control valves.

2. Slowly, turn set screw (4) counterclockwise until the boom begins to lower onto the ground.

Note: Once the boom begins to lower, stop turning set screw (4).

3. After the boom has lowered completely onto the ground, turn set screw (4) back to the original position.
4. Tighten locknut (3) to 40 ± 4 N·m (29.5 ± 3 b ft).
5. Before operating the machine, make any necessary repairs.

For additional information, consult your Cat dealer.

Machines without a Boom Lowering Control Valve

WARNING

Be sure no one is under or near the work tools before manually lowering the boom. Keep all personnel away from the boom drop area when lowering the boom with the engine stopped in order to avoid possible personal injury.

Use the following procedure to manually lower the boom due to an engine malfunction.

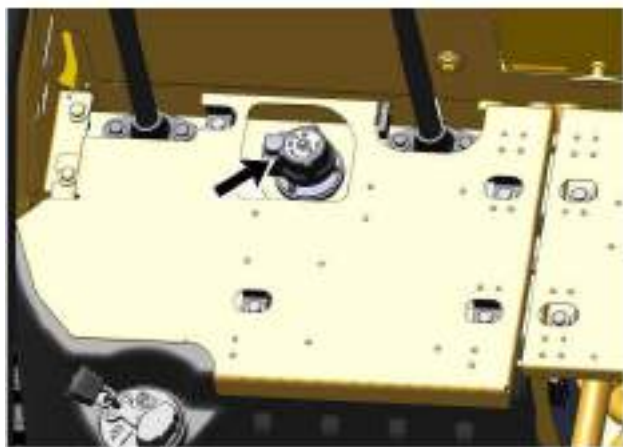


Illustration 556

g06184080

Hydraulic tank filler cap location

WARNING

Pressurized system!

The hydraulic tank contains hot oil under pressure. To prevent burns from the sudden release of hot oil, relieve the tank pressure with the engine off. Relieve pressure by slowly turning the cap until the cap reaches the secondary stop.

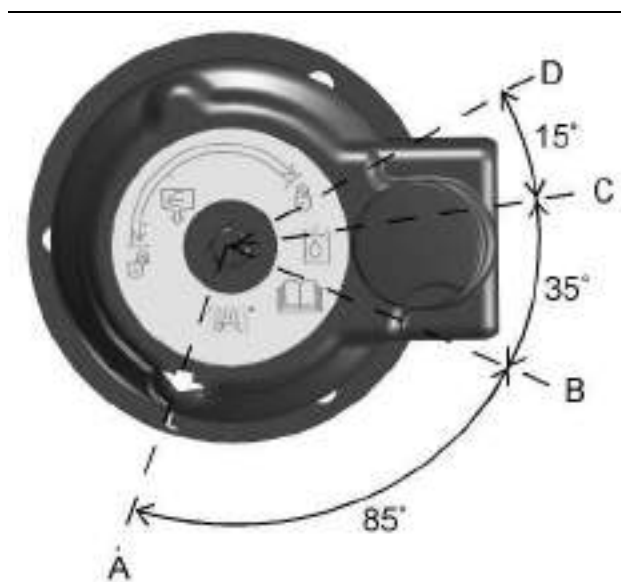


Illustration 557

g06184990

Filler cap

- (A) LOCK position
- (B) PRESSURE RELEASE - START position
- (C) PRESSURE RELEASE - END position
- (D) OPEN position

1. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 557 for filler cap positions.
 - a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).
 - b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
 - c. Move the arrow from position (C) to position (D).
 - d. After the tank pressure is relieved, remove the filler cap.

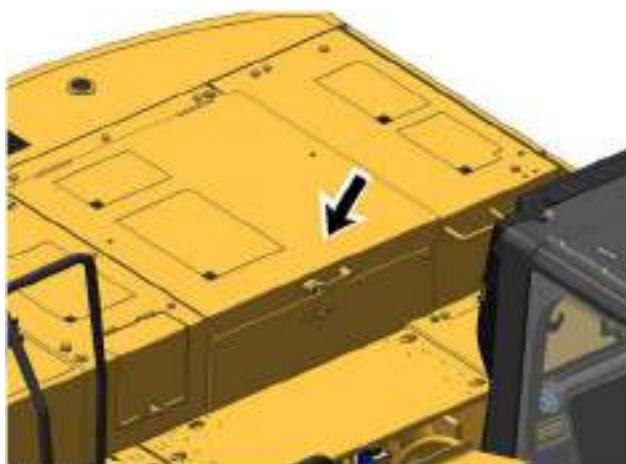


Illustration 558

g06225770

2. Open the engine hood.

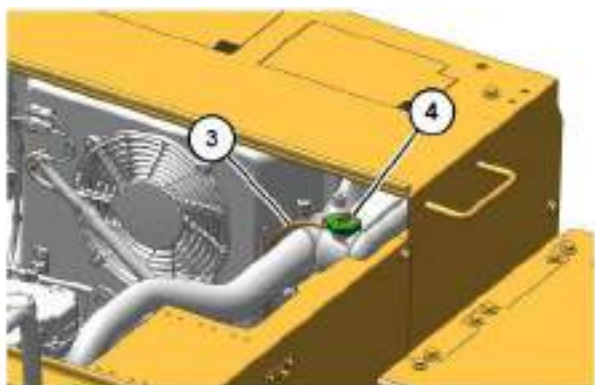


Illustration 559

g06184089

(3) Hose
(4) Clamp

3. Loosen clamp (4) and disconnect hose (3) from the reservoir. Remove the clamps and the cable straps that secure the hose to the machine.

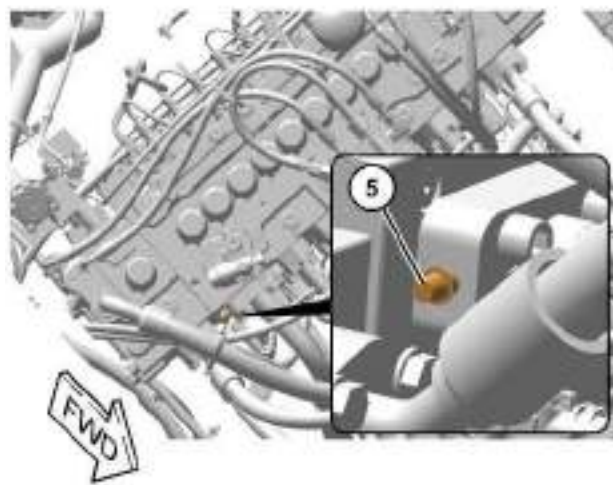


Illustration 560

g06205184

Main control valve

(5) Screw

4. Connect one end of the radiator hose to screw (5). Put the other end of the hose into the hydraulic tank opening. The screw is located at the front, right side of the main control valve.
5. Slowly loosen screw (5) by a maximum of 1/2 turn. This allows the hydraulic oil in the boom circuit to drain into the hydraulic tank. The boom will now start to lower.
6. Make sure that the work tool has lowered all the way to the ground. Tighten screw (5) to $13 \pm 2 \text{ N}\cdot\text{m}$ ($9 \pm 1 \text{ lb ft}$).
7. Disconnect the hose from the screw. Do not allow the oil that is contained in the hose to spill. Drain the oil into a suitable container.
8. Connect the hose to the original position on the radiator and install the hydraulic tank filler cap.
9. Close the engine hood.

After completion of the manual boom lowering, make necessary repairs before you operate the machine again.

Pressure Release of Auxiliary Lines

WARNING

Personal injury can result from hot oil spray and raised work tools.

Make sure all the work tools have been lowered, the oil is cool and the pressure has been released from the hydraulic system before removing any components or lines.

Do not allow hot oil or components to contact skin.

Note: Refer to Operation and Maintenance, "General Hazard Information" for information on containing fluid spillage.

Refer to the procedure below before any of the following conditions.

- The work tool is changed.
 - The position of the ball valve is changed.
1. Turn the engine start switch to the OFF position.
 2. Place the hydraulic lockout lever in the UNLOCKED position.
 3. Release the pressure in the auxiliary lines by pressing the auxiliary control buttons or the auxiliary control pedal three times.
 4. Place the hydraulic lockout lever in the LOCKED position.
 5. Change the work tool.

Note: There should be movement in the auxiliary hydraulic lines as the pressure is released. If there is no movement in the auxiliary hydraulic lines, start the engine and run the engine for 20 seconds. Repeat steps 1 to 5.

For additional information, consult your Cat dealer.

Operating Techniques

i08069554

Operating Technique Information

SMCS Code: 7000

WARNING

Know the maximum height and the maximum reach of your machine. Serious injury or death by electrocution can occur if the machine or the work tools are not kept a safe distance from electrical power lines. Keep a distance of at least 3000 mm (118 inch) plus an additional 10 mm (0.4 inch) for each 1000 volts over 50000 volts.

For safety, one of the following may require a greater distance:

- Local codes
- State codes
- Requirements of the job site

NOTICE

When swinging into a ditch, do not use the ditch to stop the swinging motion. Inspect the machine for damage if the boom is swung into a bank or an object.

Repeated stopping by an object can cause structural damage if the boom is swung into a bank or an object.

With certain boom-stick-bucket combinations, the bucket or worktool can hit the cab and/or the front structure of the machine. Always check for interference when first operating a new bucket or a new work tool. Keep the bucket or work tool away from the cab and away from the front structure during operation.

Whenever the tracks of the machine raise off the ground while digging, lower the machine back to the ground smoothly. **DO NOT DROP OR CATCH IT WITH THE HYDRAULICS.** Damage to the machine can result.

With certain combinations of work tools, the third pedal can have different functions. Always check the function of the third pedal before you use the third pedal.

Know the location of any buried cables. Mark the locations clearly before you dig.

Consult your Cat dealer for special work tool tips that are available for use in severe applications.

Move the machine whenever the position for operating the machine is not efficient. The machine can be moved forward or backward during the operating cycle.

When you operate the machine in close places, utilize the bucket or the other work tool to perform the following functions:

- Pushing the machine
- Pulling the machine
- Lifting the tracks

Use a comfortable travel speed while you operate the machine.

Operating efficiency can be increased by using more than one machine control to perform a task.

Never swing a load over a truck cab or workers.

Position the truck so that material can be loaded from the rear of the truck or from the side of the truck. Load the truck evenly so that the rear axles are not overloaded.

An oversize bucket or a bucket that is equipped with side cutters should not be used in rocky material. These types of buckets slow down the cycle. Damage to the bucket and to other machine components could result.

Coaching Tips

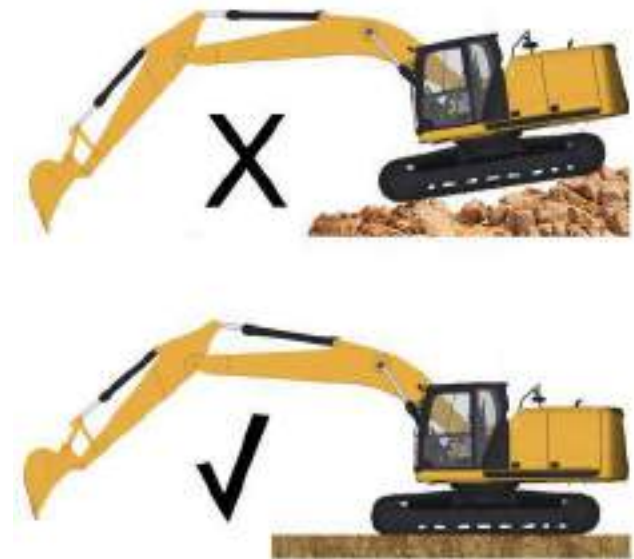


Illustration 561

g06223763

Digging with a stable machine increases productivity. Create a stable work platform.

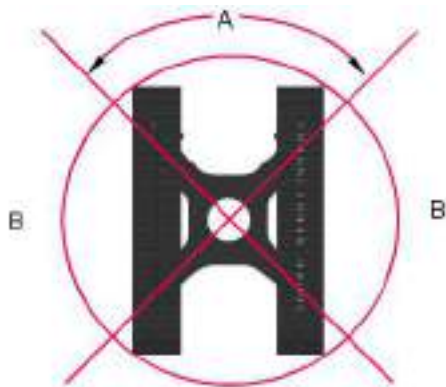


Illustration 562

g06210141

- (A) Most stable dig
(B) Dump

For improved stability: Do not dig over the drives or perpendicular to the tracks.

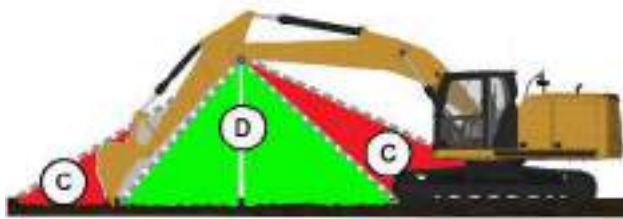


Illustration 563

g06212328

- (C) Weak crowd force
(D) Ideal crowd force

Dig from the top down in layers. Try to have a full bucket by the time the stick is vertical, but do not reach too far with the stick. The most crowd force is generated with the stick +/- 30 degrees from vertical.

Minimize unneeded movement. Only curl/dump the bucket as much as required to hold/dump material.



Illustration 564

g06210334

Minimize unneeded movement. During backfilling, start with the material closest to the trench.



Illustration 565

g06210343

Watch your surroundings. Swing left to dump material for better visibility.

Watch the bucket. The bucket can contact the tracks or the cab.

Concentrate on being smooth, speed will come with practice.

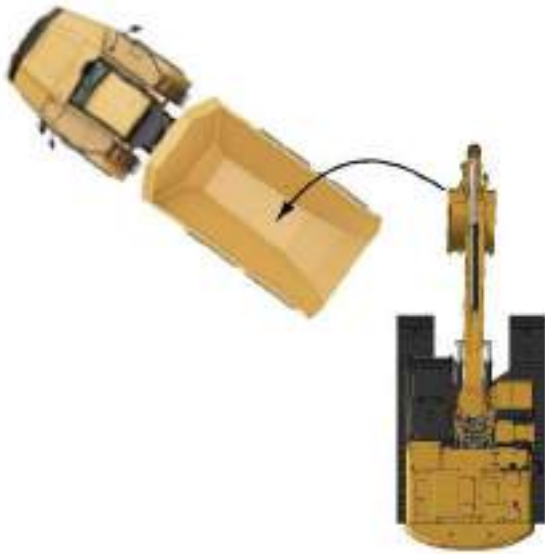


Illustration 566

g06212604

Truck placement will affect efficiency: 45 degree truck loading is more efficient than 90 degree. Spotting the truck too far from the excavator causes excessive motion.

Load from a bench when possible. Bench loading is more efficient.

Restricted Operation



Illustration 567

g06222487

Do not use the swing force to perform the following operations:

- Soil compaction
- Ground breaking
- Demolition

Do not swing the machine while the bucket tips are in the soil.

These operations will damage the boom, the stick, and the work tool and the operations will reduce the life of the equipment.



Illustration 568

g06212594

Do not use the dropping force of the bucket or work tool as a hammer. This will bring excessive force on the rear of the machine. Possible damage to the machine could result.



Illustration 569

g06222492

If the cylinder is operated at the end of the stroke during operations, excessive force will occur on the stopper on the inside of the cylinder. This will reduce the life of the cylinder and structures. To avoid this problem, always leave a small margin of play when the cylinder is operated.

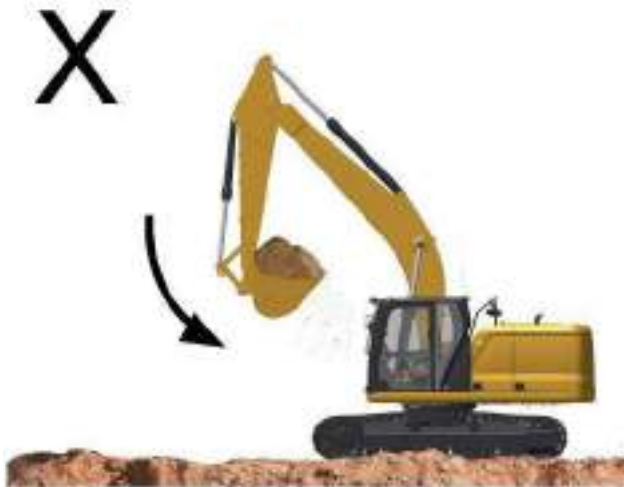


Illustration 570

g06222498

If the stick IN function is operated at full speed with a fully loaded bucket or heavy work tool attachment to the end of the cylinder stroke, excessive force will occur inside the stick cylinder. This action will reduce the life of the stick cylinder. To avoid this problem, always operate a stick IN function with moderate speed towards the end of cylinder stroke.



Illustration 572

g06222505

Do not use the dropping force of the rear of the machine for excavation. This operation will damage the machine.

Operating Precaution



Illustration 571

g06222500

While the bucket is in the ground, do not use the travel force for any excavation. This operation will cause excessive force on the rear of the machine.



Illustration 573

g06222507

NOTICE

Do not allow the machine to swing from the force of traveling when you use the bucket, the stick, or the boom to assist in travel. If the force from traveling causes the machine to swing, damage may occur to the swing motor and to the swing drive.

Do not use the force of the bucket, the stick, or the boom to assist in turning the machine while the machine is traveling. This technique is referred to as "jump steering". This technique will damage the swing motor and the swing brake.



Illustration 574

g06222509

When deep holes are dug, do not lower the boom so that the bottom side of the boom touches the ground.

When deep holes are dug, do not allow the boom to interfere with the tracks.

Grade and Assist Operating Tips

The following image displays proper stick speeds for accurate grading. One of the factors to accuracy is the grade assist speed setting. The setting can be set to Quick, Normal, or Fine. The settings are found in the Grade Assist screen on the monitor.

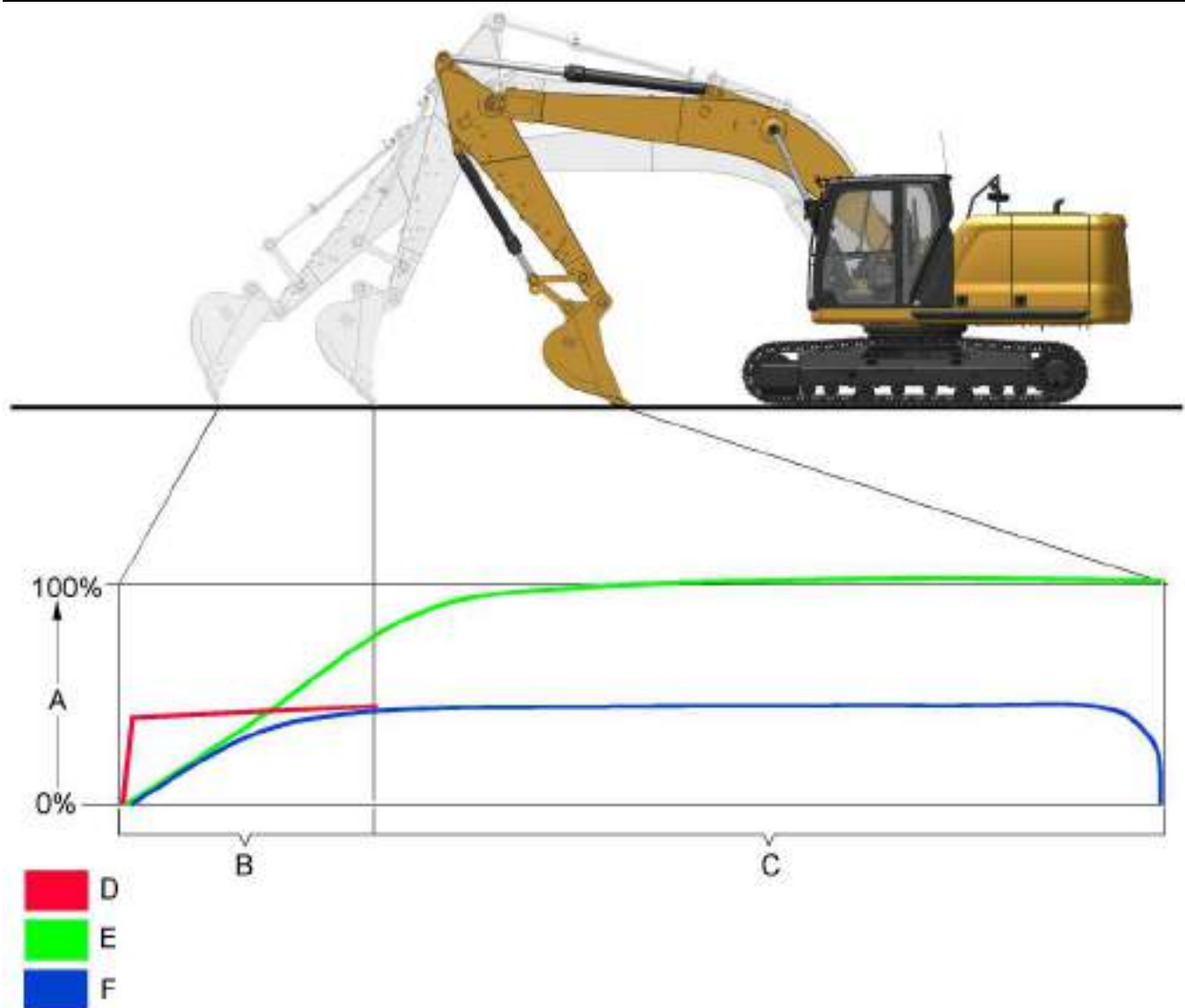


Illustration 575

g06250472

Stick speed vs. accuracy

(A) Stick joystick movement

(B) Joystick movement must be gradual

(C) In fine or normal mode, joystick input can be at 100 percent.

(D) Joystick movement too fast

(E) Good for fine or normal mode

(F) Good in any mode

For best results using Cat Grade, it is important to be cognizant of stick speed in order to obtain accurate results. When the bucket touches the ground before digging, the initial dig movement must be gradual. If operating in Fine or Normal mode, joystick speed can be increased to 100 percent after the slow initial engagement. Approximately 50 percent stick speed is necessary to maintain accuracy when in Quick mode after the slow initial engagement.

Cut/fill error can be minimized if the work tool is properly calibrated and the operator maintains a deliberate speed of operation.

Smart Boom

When using the smart boom function, it is necessary to slow down stick speed. Smart boom cannot keep up when the stick is operating at a rapid pace. This is particularly noticeable when the boom is at the raise/lower transition point and the stick is near vertical.

i07058371

Travel in Water and Mud

SMCS Code: 7000-V6

NOTICE

When working in or around any body of water, around a stream or river, or in conditions of heavy mud, be careful that the swing bearing, the swing drive gear, and the swivel joint do not dip into water, mud, sand, or gravel. If the swing bearing dips into water, mud, sand, or gravel, immediately grease the swing bearing until the used grease leaks from the outer circle of the swing bearing. Failure to carry out this procedure may cause premature wear in the swing bearing.



Illustration 576

g06223764

Depth of water to the center of the track carrier roller.

The following guidelines pertain to travel across water and travel through mud, sand, or gravel.

The machine can travel across a river only under the following conditions:

- The bed of the river is flat.
- The flow of the river is slow.
- The machine dips into the water only to the center of the track carrier roller (dimension A).

NOTICE

Do not allow the fan on the engine to contact the water while the machine travels through the water. Do not allow the fan on the engine to contact the water during a swing while the machine is in the water. Damage to the fan may occur if the fan contacts the water.

While you cross the river, carefully confirm the depth of the water with the bucket. Do not move the machine into an area that has a water depth that is greater than Dimension A.

The machine may sink gradually on soft ground. Therefore, you should frequently check the height of the undercarriage from ground level and the depth of water on the ground.

Check the swing gear by looking through the port for inspection that is on the upper frame. If there is water in the swing gear, contact your Cat dealer for the required maintenance on the swing gear.

After you travel through water, carefully clean the machine to remove any salt, sand, or other foreign matter.

Procedure for Removing the Machine from Water or Mud

NOTICE

Do not allow the machine to swing from the force of traveling when you use the bucket, the stick, or the boom to assist in travel. If the force from traveling causes the machine to swing, damage may occur to the swing motor and to the swing drive.



Illustration 577

g06222519

1. You may not be able to move the machine by using the travel controls only. In this case use both the travel control levers/pedals and the stick to pull the machine out of the water or ground.



Illustration 578

g06222525

2. The machine may slip because of a steep slope. The procedure in Step 1 may not work. In this case, first rotate the upper structure by 180°. Then use both the travel control levers/pedals and the stick to move the machine up the slope.



Illustration 579

g06212337

3. It may be impossible to travel because the bottom of the frame comes into contact with the ground or the undercarriage is clogged with mud or gravel. In this case, operate the boom and the stick together. Raise the track and rotate the track forward and backward to remove the mud and the gravel.

i07058860

Boom, Stick and Bucket Operation

SMCS Code: 7000

Digging



Illustration 580

g06212506

1. Position the stick at a 70 degree angle to the ground.



Illustration 581

g06212513

2. Position the bucket cutting edge at a 120 degree angle to the ground. Maximum breakout force can now be exerted with the bucket.



Illustration 582

g06222533

3. Move the stick toward the cab and keep the bucket parallel to the ground.



Illustration 583

g06222535

4. If the stick stops due to the load, raise the boom and/or perform a curl to adjust the depth of the cut.

5. To apply the greatest force at the cutting edge, decrease the down pressure as you move the stick toward the cab.

6. Maintain a bucket attitude that ensures a continuous flow of material into the bucket.

7. Continue the pass in a horizontal direction so that material peels into the bucket.



Illustration 584

g06222538

8. Close the bucket and raise the boom when the pass has been completed.



Illustration 585

g06223077

9. Engage the swing control when the bucket is clear of the excavation.



Illustration 586

g06223078

10. To dump a load, move the stick outward and open the bucket in a smooth motion.

Lifting Objects

WARNING

To prevent injury, do not exceed the rated load capacity of the machine. If the machine is not on level ground, load capacities will vary.

NOTICE

Damage to bucket cylinder, bucket or linkage could result if slings are placed incorrectly.

There may be local regulations and/or government regulations that govern the use of machines which lift heavy objects. Obey all local and government regulations.

If this machine is used to lift objects within an area that is controlled by the European Directive "2006/42/EC", the machine must be equipped with a boom lowering control valve, a stick lowering control valve, and an overload warning device.

Japan regulations require a shovel crane configuration to lift certain objects.

Contact your Cat dealer for additional information.

Short slings will prevent excessive load swing.



Illustration 587

g06212526

Use the lifting eye that is provided on the linkage to lift objects.

If the lifting eye is used, the connection must be made with a sling or with a shackle.



Illustration 588

g06212532

An unstable condition can exist if a load exceeds the machine load rating or if a heavy load is swung over an end or over a side.



Illustration 589

g06212530

The most stable lifting position is over a corner of the machine.

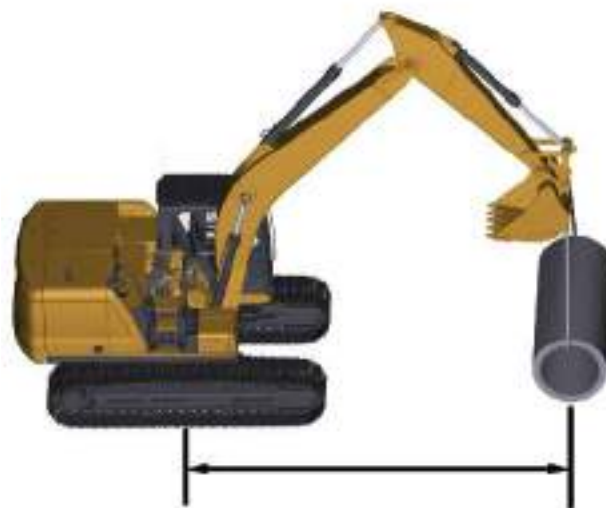


Illustration 591

g06212539

Lift capacity decreases as the distance from the swing centerline is increased.

Machines that are Equipped with a Long Reach Configuration

Machines with a long reach configuration require larger swing drift than standard machines when stopping, because inertial force in time of swing is large. Taking this into account, adjustments are made in timing for applying the swing brakes and speed of swinging.

Machines with a long reach configuration could be damaged and stability of the machine would be adversely affected if a control was suddenly operated, because inertial force of work tool is large.

i08036890



Illustration 590

g06212535

For the best stability, carry a load close to the machine and to the ground.

Shovel Crane Operation

SMCS Code: 6500

S/N: HEX1-Up

WARNING

Operating the machine using the correct method when performing a lifting operation is important. Incorrectly operating the machine may result in serious injury or death. Be sure to observe the following precautions.

Refer to this Operation and Maintenance Manual, "Shovel Crane Control" for additional information regarding machine controls for the shovel crane.

Operate the machine according to the rated load table of your machine. Refer to Operation and Maintenance Manual, "Specifications (Shovel Crane Specifications)" for more information.

Traveling with a Suspended Load



Illustration 592

g06222543

The rated load when traveling while suspending a load is limited to 50 percent of a stationary suspension. Make sure that the suspending load weight is within this limit before starting to move.

When traveling while suspending a load:

1. Make sure that the load is in the front of the machine, and operate within the maximum operating radius of 70° or less.
2. The height below the suspended load shall be 30 cm (12 inch) or less above ground.
3. The traveling speed shall be 3 km/h or less.
4. The traveling ground shall be level and of firm ground.

Traveling while suspending a swinging load may cause the machine to roll over. Wait until the load stops swinging before starting to move.

Dragging of load is prohibited.

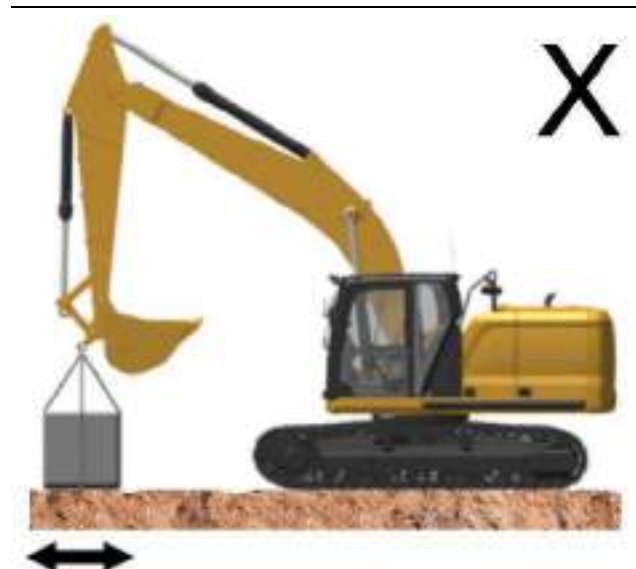


Illustration 593

g06222544

Horizontally, vertically or diagonally dragging a load may cause the machine to roll over, the wire rope to break, or the load to collapse, resulting in personal injuries.

Be sure to lift the load directly above.

Lifting operation on a slope is prohibited.



Illustration 594

g06222549

Performing an operation on a slope of 5° or more and on a soft ground may cause the machine to roll over or the load to collapse, resulting in personal injuries. Make sure that the machine is on level and firm ground before performing the operation.

Place an iron plate or other appropriate material on a soft ground.

Use in non-standard specification conditions is prohibited.



Illustration 595

g06222553

Use of the machine in non-standard specification conditions is prohibited as the lifting load and the operation range cannot be accurately displayed.

Sudden lifting operation is prohibited.



Illustration 596

g06222555

WARNING

Suddenly lifting a suspended load will apply an abnormal force on the hook and cause the hook to break, resulting in serious injuries or death. Never perform such an operation.

Sudden swinging operation while suspending a load is prohibited.



Illustration 597

g06222561

WARNING

Never perform a swinging operation suddenly while suspending a load. Doing so may cause the suspended load to be pulled with a centrifugal force, causing the machine to roll over.

Never suddenly swing, and stop or lower the hook suddenly, as doing so may cause the wire rope to come off the hook latch.

Factors that cause the wire rope to come off

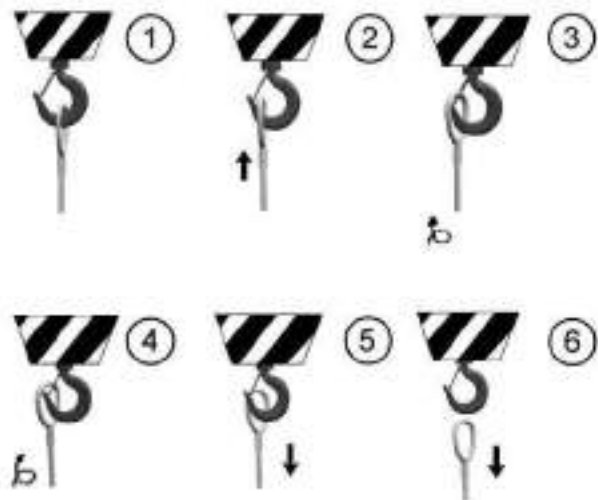


Illustration 598

g06222768

(1) Wire rope in normal condition.

- (2) The wire rope gets raised when the hook is lowered too suddenly.
- (3) If the wire rope gets twisted and gets untwisted in the direction of the arrow, the wire rope will go beyond the tip of the hook.
- (4) The wire rope goes around the back of the hook.
- (5) The hook goes up or the wire rope goes down.
- (6) The wire rope simply falls off.

Diagonal dragging operation is prohibited.



Illustration 599

g06222769

WARNING

Diagonal dragging operation applies an abnormal force on the hook and may cause the hook to break, resulting in serious injuries or death. Never perform such an operation.

Leaving the seat while suspending a load is prohibited.



Illustration 600

g06222773

Do not leave the operator seat while a load is being suspended. The load may fall, resulting in personal injuries. Do not allow a slinging operator or another worker under the load.

Temporarily stopping the lifting operation.



Illustration 601

g06191918

When there is a large area to stop the machine, fully retract the stick cylinder and slowly lower the boom until the bucket contacts the ground.

Note: Make sure that the lifting tool does not get caught under the bucket.



Illustration 602

g06191883

When there is small area to stop the machine, press the shovel crane button on the switch panel in order to disable the operation. Position the stick so that it is perpendicular to the ground and slowly lower the boom until the bucket contacts the ground.

Note: Make sure that the lifting tool does not get caught under the bucket.



Illustration 603

g06191925

Hook in latched position

Be sure to store the hook during a bucket use. Otherwise, the bucket, the hook, and other parts may be damaged.

Wire rope suspension angle

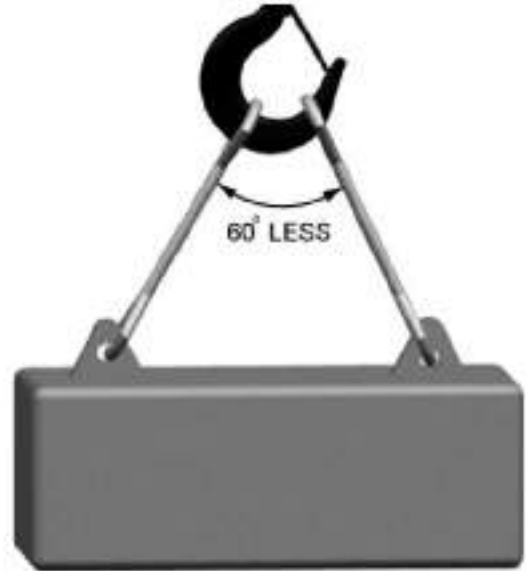


Illustration 604

g06222775

As a rule, try to set the wire rope suspension angle at 60° or below.

Keep out of the area under a suspended load.



Illustration 605

g06222777

Never allow anyone to enter an area under a suspended load.

Lifting operation with a quick coupler device is prohibited.

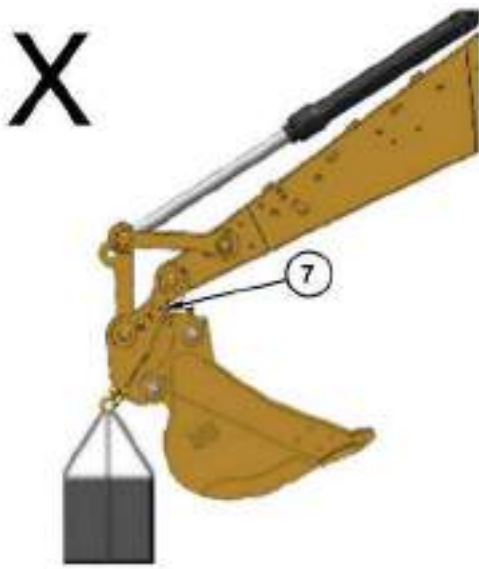


Illustration 606

g06222779

(7) Quick Coupler

Accurate lifting load and operation range cannot be displayed in a lifting operation equipped with a quick coupler. Furthermore, never perform a lifting operation with a quick coupler attached, as doing so causes the hook to contact the quick coupler, applies an abnormal force on the hook, and causes the hook to become damaged and/or the machine to roll over.

i03875131

SmartBoom Operation (If Equipped)

SMCS Code: 5461-ZS; 7332

WARNING

Personal injury or death can result from not following the proper procedures.

To avoid the possibility of injury or death, follow the established procedure.

WARNING

Activating the SmartBoom function and using the work tool joystick control while the front of the machine is elevated could result in unexpected machine motion. Unexpected machine motion could result in serious injury or death. Do not activate the SmartBoom function if the front of the machine is elevated by the front linkage.

WARNING

Do not elevate or lower the track when in the SmartBoom mode. Follow the operation procedures for the SmartBoom in the Operation and Maintenance Manual. Failure to follow these instructions can result in serious injury or death.

WARNING

Always make sure that the boom control joystick is in the NEUTRAL position before activating the SmartBoom control. Activating the SmartBoom control with the joystick out of the neutral position could result in unexpected machine motion which could result in serious injury or death.

WARNING

Do not select any SmartBoom mode, using the SmartBoom selector switch located on the console, while the tracks are elevated. Selecting the SmartBoom mode with the tracks elevated could result in a sudden drop of the machine which could result in serious injury or death.

WARNING

If any SmartBoom mode is active and the boom control joystick is in the BOOM DOWN position (forward) with a bucket or a work tool on the ground, pressing the disable button that is located on the front of the right hand joystick could cause a sudden boom down motion. This control function could lift the machine upward, with unexpected machine movement that could result in serious injury or death. Do not press the disable button while the SmartBoom mode is active and the boom control joystick is in the BOOM DOWN position (forward) with a bucket or a work tool on the ground.

WARNING

Do not attempt to lift the tracks of the machine by using the disable button and applying downward force with the boom lowering control while the machine is in any SmartBoom mode. Releasing the disable button will immediately return the machine to the active SmartBoom mode. This action could cause the machine to drop down abruptly which could result in serious injury or death.

The Caterpillar SmartBoom provides significant advantages in the following operations:

Excavation and Loading

The operator should select the SmartBoom UP AND DOWN mode for excavating operations and for loading operations. This mode is effective during the return cycle. The BOOM DOWN movement is assisted by gravity, and the pump flow that is normally required for the boom circuit is available for faster STICK OUT and SWING functions. More work is performed for the amount of hydraulic oil flow that is provided by the pumps. This results in faster cycle times and improved fuel efficiency.

When the SmartBoom UP AND DOWN mode is active, no downward hydraulic force is applied to the boom. The operator can intermittently apply downward hydraulic force when the force is required for bucket penetration. A button on the right joystick enables the operator to override the SmartBoom mode.

Hammering

The operator should select the SmartBoom DOWN mode for hammering operations. In the SmartBoom DOWN mode, the weight of the hammer plus the boom and the stick provides sufficient downward force for effective hammering. This mode prevents the hammer from rebounding. The boom follows the hammer downward freely as the tool penetrates the rock. This mode also reduces strain on the machine structures.

Advantages

In hammering, the SmartBoom provides the following advantages:

- The mode reduces shock in the cab.
- The mode reduces strain on machine structures.
- The mode prevents blank shots.
- The mode maintains optimum frequency.

In rock cleaning, the SmartBoom provides the following advantages:

- The mode reduces strain on machine structures.
- The mode reduces wear on the bucket and the teeth.
- The mode maintains optimum frequency.

In a leveling operation, the SmartBoom eases the operation. Only STICK IN actuation and bucket actuation are required to level the surface.

In material handling, the SmartBoom reduces the chance of damage below the material.

i06978374

Bucket - Remove and Install

SMCS Code: 6001-011; 6001; 6001-012; 6101; 6102; 6523

Removal Procedure

WARNING

Failure to follow the instruction below for the installation of a work tool may result in personal injury or death. Special care must be taken if more than one person is installing the work tool.

- **Confirm the verbal communication and the hand signals that will be used during the installation.**
- **Be alert for sudden movement of the front linkage and the work tool.**
- **Do not insert fingers into the bores of the support pins when the support pins and the bores are being aligned.**

NOTICE

To facilitate removal of the bucket pins without causing damage to the pins, the bearings, and/or the O-ring seals put the bucket on the floor and the stick in a vertical position, as shown.



Illustration 607

g06181120

1. Start the engine. Park the machine on a hard, level surface. Position the bucket, the stick, and the bucket control linkage, as shown. Shut off the engine.

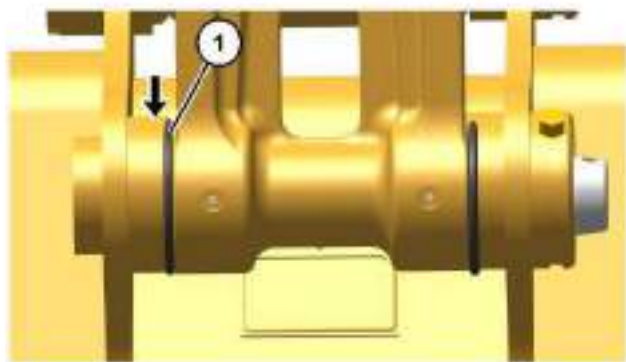


Illustration 608

g06192508

2. Slide O-ring seals (1) off the pin joints and onto the flanges of the bucket.

WARNING

When the pin assembly is removed, the linkage assembly may swing out of the bucket. To prevent possible personal injury, do not stand in front of the linkage assembly when the pin assembly is being removed.

Note: Removing the support pin may be difficult due to excessive pressure on the support pin. Remove the pressure on the support pin by adjusting the front linkage.

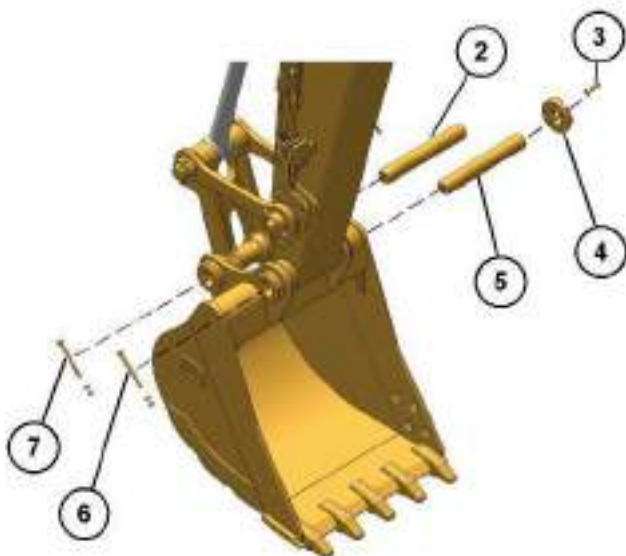


Illustration 609

g06186090

3. Remove nuts and retaining bolt (7) from support pin (2). Remove the support pin.

4. Remove bolts (3) and adapter plate (4). Remove the shims.
5. Remove nuts and retaining bolt (6) from support pin (5). Remove the support pin.
6. Start the engine and raise the stick out of the bucket.
7. Remove the O-ring seals (1) from the flanges on the bucket.

Note: After the support pins have been removed, make sure that the support pins do not become contaminated with sand or dirt. Make sure that the seals on the end of the stick and the seals on the end of the link do not become damaged.

Installation Procedure

1. Clean each pin and each pin bore. Lubricate each pin bore with molybdenum grease.

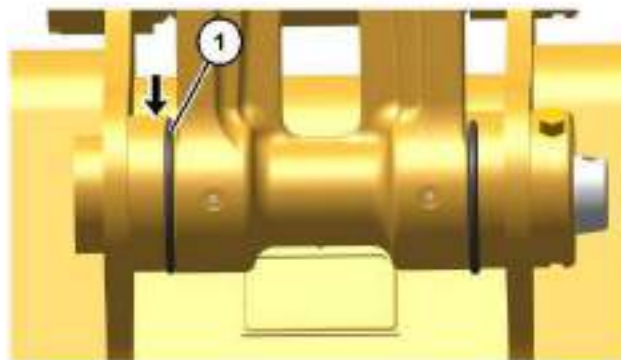


Illustration 610

g06192508

2. Position the O-ring seals (1) onto the flanges of the bucket.
3. Start the engine and lower the stick into the bucket until the pin bores are in alignment with each other. Stop the engine.

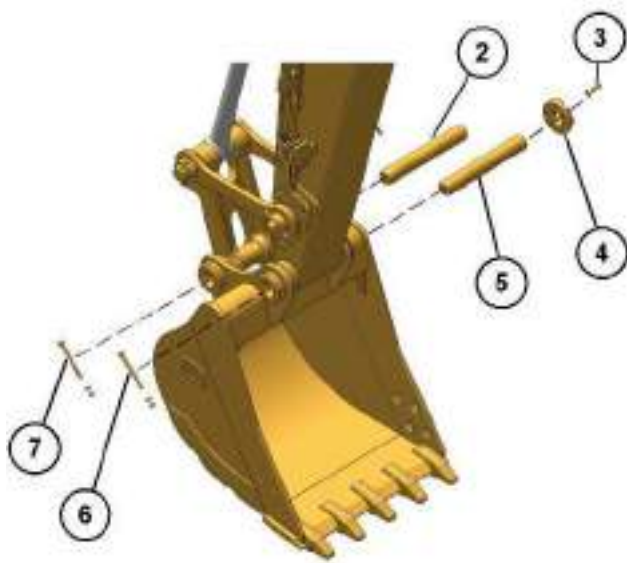


Illustration 611

g06186090

4. Install support pin (5). Align the retaining bolt hole in the support pin with the retaining bolt hole in the bucket.
5. Install the retaining bolt and nuts (6). Install adapter plate (4) without the shims, and without bolts (3) that hold the adapter plate.
6. Refer to Operation and Maintenance Manual, "Bucket Linkage - Inspect/Adjust" to adjust the bucket clearance.
7. Slide O-ring seals (1) in position over the pin joints between the bucket and the stick.
8. Start the engine and position the bucket linkage into the bucket until the pin bores are in alignment with each other. Stop the engine.
9. Install support pin (2). Align the retaining bolt hole in the bucket pin with the retaining bolt hole in the bucket.
10. Install retaining bolt and nuts (7).
11. Slide the O-ring seals (1) over the pin joints between the bucket and the link assembly.

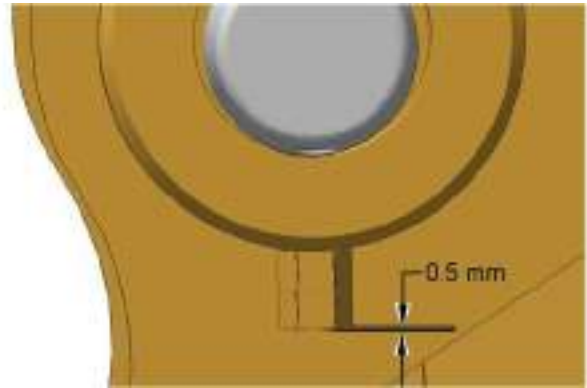


Illustration 612

g06192530

12. Tighten retaining nuts (6) and (7). Position the outside nut even with the end of the retaining bolt or 0.5 mm (0.02 inch) beyond the end of the retaining bolt. Tighten the inside nut against the outside nut.

13. Lubricate the bucket pins. Refer to Operation and Maintenance Manual, "Bucket Linkage - Lubricate".

i07757775

Quick Coupler Operation (Circuit for CW Coupler (If Equipped))

SMCS Code: 6129; 6522; 7000

NOTICE

The Cat Quick Coupler (CW Coupler) is not designed to be used in applications where there is long exposure to excessive vibration. The vibration caused by extensive use of a hydraulic hammer as well as the added weight of certain demolition tools such as shears, crushers, and pulverizers may cause premature wear and decreased service life of the coupler.

Be sure to carefully inspect the coupler daily for cracks, bent components, wear, distressed welds, etc. when operating with any of the above work tools.

General Operation

The quick coupler is used to change work tools while the operator remains in the cab. The quick coupler can be used with a broad range of buckets and work tools. Each work tool must have a set of pins in order for the quick coupler to work properly.

The work tools are held onto the quick coupler by hydraulic pressure. If pressure is lost, a locking bar keeps the work tool locked with the force of built-in springs. Ensure that the hydraulic system and the blocking bar are working properly before using the quick coupler.

A lifting eye is included on the quick coupler. Release the work tool from the quick coupler to use the lifting eye to pick up loads. To lift a load with the lifting eye, extend the bucket cylinder until the quick coupler is in a VERTICAL position. Do not exceed the rated load for the machine.

NOTICE

Once the work tool has been properly attached to the coupler, no loosening of the work tool should occur. Refer to the "Quick Coupler Installation and Removal" section of the quick coupler Operation and Maintenance Manual for additional information. If at any point after the proper attachment and back drag testing of the work tool, should the work tool then become loose or if the rear pin of the work tool detaches from the movable hook, stop work immediately and safely ground and detach the work tool. Consult your Cat dealer to inspect the coupler prior to putting the coupler back into service. This situation could indicate potential coupler damage that may not be readily visible to the customer or operator of the machine and coupler.

Electric Switch Operation

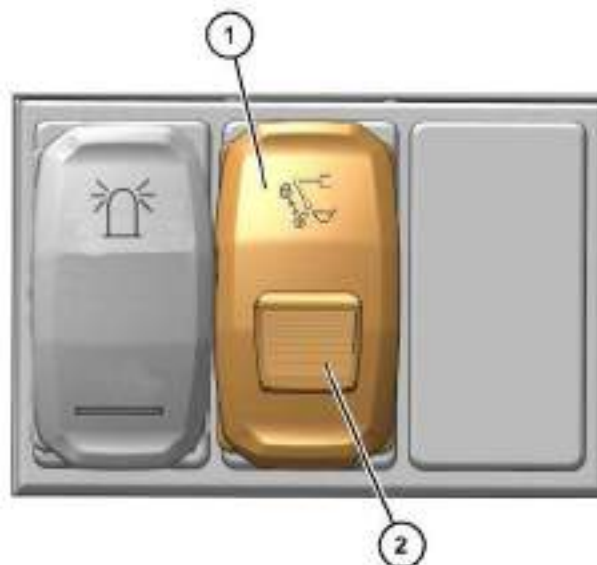


Illustration 613

g06184557

Quick coupler switch (1) is located inside the cab on the switch panel to the left of the operator's seat. The electric switch has only one position for coupling the work tool and uncoupling the work tool. The switch is equipped with a safety lock (2). The locking tab must be pushed forward before the switch can be pressed.

Coupling the Work Tool

WARNING

Inspect the coupler wedge engagement before you operate the excavator.

Serious injury or death may result from an improperly engaged coupler.

Inspect coupler wedge engagement from the cab by rotating the bucket or the work tool inward. Extend the bucket cylinder to bring the coupler actuator into view and bring the stick in until the wedges are visible.



Illustration 614

g06220881

1. Position the work tool on a level surface.
2. Retract the bucket cylinder. Position the quick coupler in alignment between the hinges of the work tool.

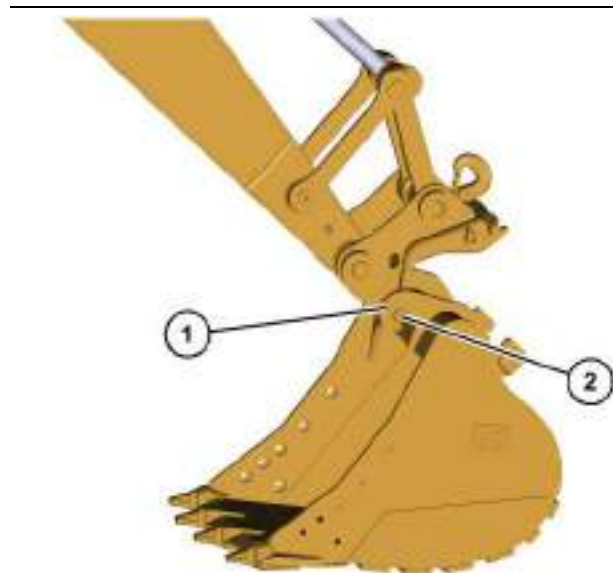


Illustration 615

g06220883

- (1) Hinges
(2) Lower bosses

3. Move the stick forward and raise the stick until the lower bosses (2) engage the hinges (1) of the work tool.
4. There are two possible settings for locking the quick coupler in Cat ET, "Alarm" and "Hold to Run". For the "Hold to Run" setting, push the locking tab on the switch forward and then depress and hold the rear of the switch until the quick coupler engages the work tool. The message "Quick Coupler Locking" will display on the monitor. For the "Alarm" setting, push the locking tab on the switch forward and then depress, but do not hold, the rear of the switch until the quick coupler engages the work tool. The message "Quick Coupler Locking" will display on the monitor.

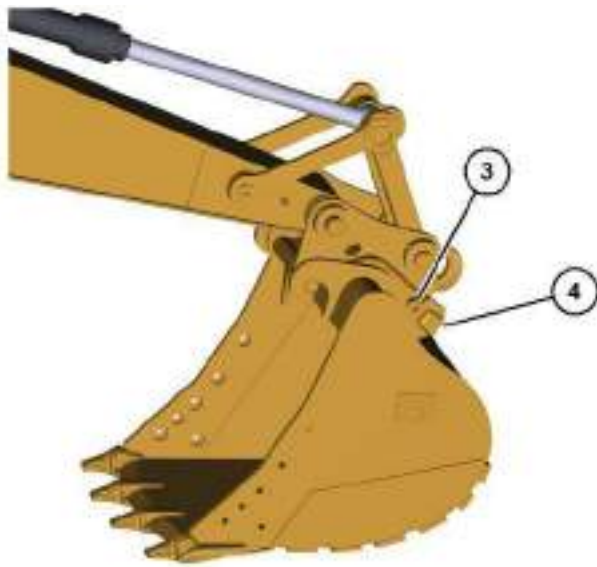


Illustration 616

g06220887

- (3) Center bosses
(4) Locking area

5. Extend the bucket cylinder to rotate the quick coupler toward the work tool.

Center bosses (3) must engage with the cutout of the hinge.

6. Release the quick coupler switch to lock the work tool.

The springs in the quick coupler will move the wedge into the locking area (4).

The monitor will display the message "Verify Tool Locking" .

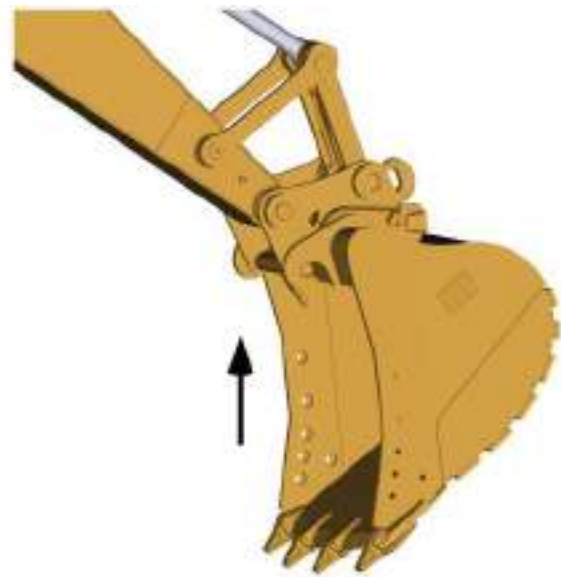


Illustration 617

g06220888

7. Raise the boom or raise the stick. Retract the bucket cylinder to confirm that the wedge is fully engaged. If the wedge is fully engaged, the work tool is locked in place. The work tool is ready to use.

Uncoupling the Work Tool

WARNING

Place the work tool or bucket in a safe position before disengaging the coupler. Disengaging the coupler will release the work tool or bucket from control of the operator.

Serious injury or death may result from disengaging the work tool or bucket when it is in an unstable position or carrying a load.



Illustration 618

g06220889

1. Level the bucket or level the work tool on the ground.
2. Push the locking tab on the switch forward and then depress the rear of the switch until the quick coupler releases the work tool. The message "Quick Coupler Unlock" will display on the monitor. For the "Alarm" setting, push the locking tab on the switch forward and then depress, but do not hold, the rear of the switch until the quick coupler engages the work tool. The message "Quick Coupler Locking" will display on the monitor.



Illustration 619

g06220891

3. Retract the bucket cylinder to move the quick coupler toward the machine.
4. Release the quick coupler switch to retract the wedge.

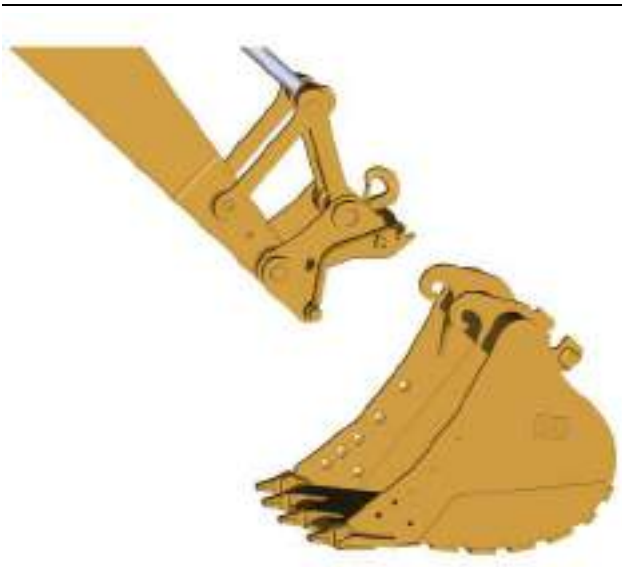


Illustration 620

g06220892

5. Lower the stick and move the stick toward the machine to disengage the quick coupler.

i07852525

Quick Coupler Operation (Hydraulic Pin Grabber Quick Coupler (If Equipped))

SMCS Code: 6129; 6522; 7000

NOTICE

The Cat Quick Coupler (Hydraulic Pin Grabber) is not designed to be used in applications where there is long exposure to excessive vibration. The vibration caused by extensive use of a hydraulic hammer as well as the added weight of certain demolition tools such as shears, crushers, and pulverizers may cause premature wear and decreased service life of the coupler.

Be sure to carefully inspect the coupler daily for cracks, bent components, wear, distressed welds, etc. when operating with any of the above work tools.

General Operation

The quick coupler is used to change work tools while the operator remains in the cab. The quick coupler can be used with a broad range of buckets and work tools. Each work tool must have a set of pins in order for the quick coupler to work properly.

The work tools are held onto the quick coupler by hydraulic pressure. If pressure is lost, a check valve in the hydraulic cylinder traps oil in the cylinder. Ensure that the hydraulic system is working properly before using the quick coupler.

A lifting eye is included on the quick coupler. Release the work tool from the quick coupler to use the lifting eye to pick up loads. To lift a load with the lifting eye, extend the bucket cylinder until the quick coupler is in a VERTICAL position. Do not exceed the rated load for the machine.

NOTICE

Once the work tool has been properly attached to the coupler, no loosening of the work tool should occur. Refer to the "Quick Coupler Installation and Removal" section of the quick coupler Operation and Maintenance Manual for additional information. If at any point after the proper attachment and back drag testing of the work tool, should the work tool then become loose or if the rear pin of the work tool detaches from the movable hook, stop work immediately and safely ground and detach the work tool. Consult your Cat dealer to inspect the coupler prior to putting the coupler back into service. This situation could indicate potential coupler damage that may not be readily visible to the customer or operator of the machine and coupler.

Quick Coupler Operation

Electric Switch Operation

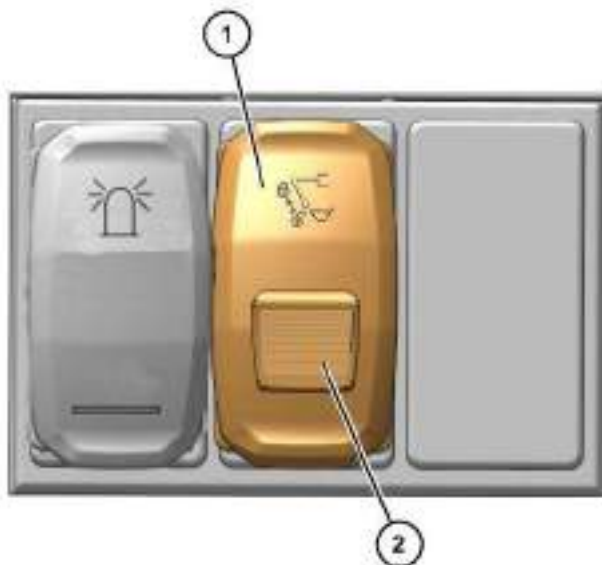


Illustration 621

g06184557

Quick coupler switch (1) is located inside the cab on the switch panel to the left of the operator's seat. The electric switch has only one position for coupling the work tool and uncoupling the work tool. The switch is equipped with safety lock (2). The locking tab must be pushed forward before the switch can be pressed.

Coupling the Work Tool

WARNING

Place the work tool or bucket in a safe position before engaging the quick coupler. Ensure that the work tool or bucket is not carrying a load.

Serious injury or death may result from engaging the work tool or bucket when it is in an unstable position or carrying a load.

WARNING

Inspect the quick coupler engagement before operating the machine.

Serious injury or death may result from improperly engaged coupler.

WARNING

Crush injury. Could cause serious injury or death. Always confirm that the quick coupler is engaged onto the pins. Read the Operator's Manual.

WARNING

The alarm will go off when the coupler is locked and the monitor will display a message. These assurances do not confirm that the coupler pins are engaged. A physical test is required by dragging the attachment on the ground to confirm the coupler pins are engaged.

NOTICE

With certain work tool combinations, including quick couplers, the work tool can hit the cab or the front of the machine. Always check for interference when first operating a new work tool.

1. Position the bucket or the work tool on a level surface.
2. Make sure that the pins are in the bucket or the work tool. Make sure that the pin keepers are installed correctly.



Illustration 622

g06187057

3. Extend the stick cylinder and fully extend the bucket cylinder until the quick coupler is curled past a vertical position. This action must be performed before pressing the switch.



Illustration 623

g06300078

4. Push the lock mechanism on the switch forward and then depress the rear of the switch. The alarm will sound continuously and the message "Quick Coupler Unlock" will display on the monitor.
5. Hold the control lever for the bucket cylinder in the EXTEND position for 5 seconds to unlock the hook.



Illustration 624

g06187063

6. Align the quick coupler with the work tool.



Illustration 625

g06187068

7. Rotate the quick coupler to grab the top pin.



Illustration 626

g06187086

8. Rotate the quick coupler downward to grab the bottom pin.



Illustration 627

g06187108

9. Extend the stick cylinder and extend the bucket cylinder until the work tool is curled past a vertical position.

Operation Section
Hydraulic Pin Grabber Quick Coupler (If Equipped)



Illustration 628

g06300088

10. Push the lock mechanism on the switch forward and then depress the rear of the switch. The alarm will sound continuously and the message “Quick Coupler Locking” will display on the monitor.
11. Hold the control lever for the bucket cylinder in the EXTEND position for 5 seconds to lock the hook. The monitor will display the message “Verify Tool Locking” and the alarm will stop.



Illustration 629

g06187115

WARNING

Inspect the quick coupler engagement before operating the machine.

Verify that the quick coupler is engaged per the procedure in the Operation and Maintenance Manual. Verify prior to operating the machine, after every engine start, and after an extended time of inactivity.

Serious injury or death may result from improperly engaged coupler.

12. Verify that the quick coupler and the work tool are locked together.
 - a. Retract the bucket cylinder and place the work tool on the ground.
 - b. Apply pressure to the work tool against the ground.

NOTICE

Back drag the work tool on the ground to ensure the quick coupler is properly locked.

Do Not strike the work tool on the ground to ensure the quick coupler is properly locked. Striking the work tool on the ground will result in damage to the coupler cylinder.

- c. Drag the work tool backward.

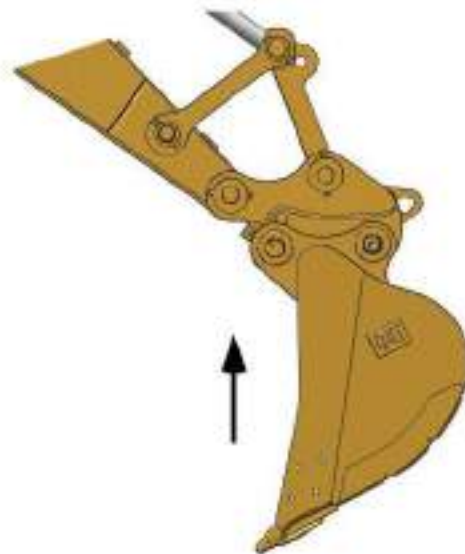


Illustration 630

g06458083

13. Raise the boom or raise the stick. Retract the bucket cylinder to confirm that the coupler is fully engaged. If the coupler is fully engaged, the work tool is locked in place. The work tool is ready to use.

Uncoupling the Work Tool

WARNING

Place the work tool or bucket in a safe position before disengaging the coupler. Disengaging the coupler will release the work tool or bucket from control of the operator.

Serious injury or death may result from disengaging the work tool or bucket when it is in an unstable position or carrying a load.

NOTICE

Auxiliary hoses for work tools must be disconnected before the Hydraulic Quick Coupler is disengaged.

Pulling the work tool with the auxiliary hoses could result in damage to the host machine or the work tool.



Illustration 631

g06187108

1. Extend the stick cylinder and fully extend the bucket cylinder until the work tool is curled past a vertical position. This action must be performed before pressing the switch.
2. Push the lock mechanism on the switch forward and then depress the rear of the switch to unlock the work tool. The alarm will sound continuously and the message "Quick Coupler Unlock" will display on the monitor.



Illustration 632

g01231447

3. Hold the control lever for the bucket cylinder in the EXTEND position for 5 seconds to unlock the hook.



Illustration 633

g06187142

4. Move the boom and the stick until the tool or the bucket is in the storage position. Keep the tool close to the ground.



Illustration 634

g06187151

5. Rotate the quick coupler upward to release the bottom pin.

Operation Section
Hydraulic Pin Grabber Quick Coupler (If Equipped)

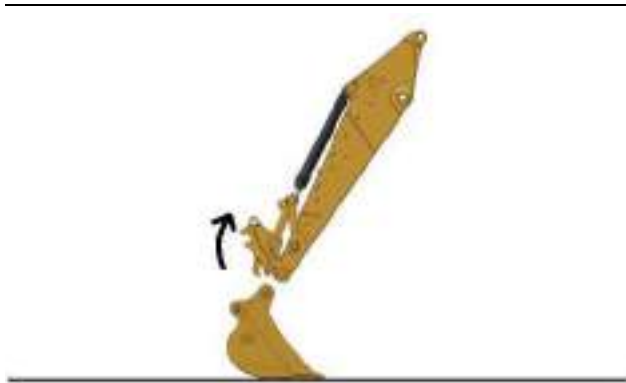


Illustration 635

g06187156

6. Continue to rotate the quick coupler upward to release the top pin and completely release the work tool from the quick coupler.
7. Move the stick to a position that is clear of the work tool.

Note: To lift objects with the lifting eye of the quick coupler, refer to “Coupler Lifting Eye Operation without Bucket” later in this chapter.

Coupling a Bucket that is Reversed



Illustration 636

g06187159

1. When you use a hydraulic pin grabber quick coupler, you can connect to a bucket that is in a reversed position. Refer to Illustration 636 for an example of connecting to a bucket that is in a reversed position.



Illustration 637

g06187057

2. Extend the stick cylinder and fully extend the bucket cylinder until the quick coupler is curled past a vertical position. This action must be performed before pressing the switch.
3. Follow the same steps for coupling the work tool to couple the host machine to a bucket that is reversed. Refer to “Coupling the Work Tool” for the proper procedure.

NOTICE

When some Caterpillar buckets are used in the reverse position, it can be more difficult to couple the bucket and uncouple the bucket than in the normal position.

Care must be taken to ensure that the position of the boom, stick, and bucket are aligned to ensure smooth coupling. The coupler must be in position between the bucket bosses.

If the bucket is not fully engaged in the jaw of the coupler, the quick coupler can become snagged on the bucket bosses. The full weight of the bucket is then carried by the quick coupler sideplates, which can cause damage to the quick coupler.



Illustration 638

g06187418

- (1) Quick coupler
- (2) Bucket
- (3) Boss
- (4) Hook

Coupler Lifting Eye Operation without Bucket

1. Remove the work tool. Refer to "Uncoupling the Work Tool" for the proper procedure.



Illustration 639

g06187057

2. The quick coupler must be curled past a vertical position before depressing the switch. Extend the stick cylinder and fully extend the bucket cylinder until the quick coupler is curled past a vertical position.



Illustration 640

g06300088

3. Push the lock mechanism on the switch forward and then depress the rear of the switch to lock the quick coupler. The alarm will sound continuously and the message "Quick Coupler Locking" will display on the monitor.

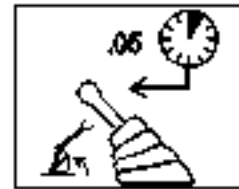


Illustration 641

g01231447

4. Hold the control lever for the bucket cylinder in the EXTEND position for 5 seconds to lock the hook. The monitor will display the message "Verify Tool Locking" and the alarm will stop.



Illustration 642

g06187164

5. Rotate the quick coupler downward and move the stick to a position that is clear of the work tool.



Illustration 643

g06223888

6. Use the lifting eye of the quick coupler, as needed.

7. To reinstall the bucket or the work tool, refer to “Coupling the Work Tool” for the proper procedure.

i07349163

Work Tool Operation (If Equipped)

SMCS Code: 6700; 7000

Hammer Operation (If Equipped)



Illustration 644

g06222793

NOTICE

Use only a hydraulic hammer that is recommended by Caterpillar.

The use of a hydraulic hammer that is not recommended by Caterpillar could result in structural damage to the host machine.

Consult your Cat dealer for information on recommended hydraulic hammers.

Only use the hydraulic hammer to break rocks, concrete, and other hard objects. Before you start hydraulic hammer operation, place the machine on a level, stable surface.

Before you start hydraulic hammer operation, close the front window. Caterpillar recommends the installation of a window guard on the front window for protection from flying debris.

NOTICE

In order to avoid structural damage to the host machine or the hydraulic hammer, comply with the following:

Do not attempt to break rocks or concrete by burying the hammer tool completely into the rocks or concrete.

Do not apply a prying force to the hammer tool in order to remove the hammer tool from the material.

Do not allow the hydraulic hammer to operate at one location and for more than 15 seconds. Change the location of the hydraulic hammer and repeat the procedure. Failure to change the location of the hydraulic hammer could cause the hydraulic oil to overheat. Overheated hydraulic oil could damage the accumulator.

Stop the hydraulic hammer immediately if the jumper lines are pulsating violently. This indicates that the accumulator nitrogen charge is lost. Consult your Cat dealer for the necessary repair.

NOTICE

Do not use the dropping force of the hydraulic hammer to break rocks or other hard objects. This could cause structural damage to the machine.

Do not use the sides or back of the hydraulic hammer to move rocks or other hard objects. Doing this could cause damage not only to the hammer but to stick or boom cylinder.

Do not operate the hydraulic hammer with any of the cylinders fully retracted or extended. Doing this could cause structural damage to the machine, resulting in reduced machine life.

Do not use the hydraulic hammer to lift an object.

Do not operate the hydraulic hammer while the stick is vertical to the ground. This could allow the stick cylinder to vibrate excessively.

Operate the attachment control levers carefully to keep the hydraulic hammer tool from hitting the boom.

Do not operate the hydraulic hammer under water unless the hydraulic hammer is properly equipped. Operating the hydraulic hammer under water could damage the machine hydraulic system. Consult your Cat dealer for information on underwater operation.

Do not operate the hydraulic hammer with the upper structure sideways to the undercarriage. Before you start hydraulic hammer operation, place the upper structure in the recommended positions that are shown in illustration 645 . Any other operating positions could make the machine unstable. Any other operating positions could place excessive loads on the undercarriage.

Refer to the following for any additional questions about the operation and care of your Cat hydraulic hammer:

- Operation and Maintenance Manual, SEBU7346, "Hydraulic Hammers"
- The Operation and Maintenance Manual specific to your machine

An operation and maintenance decal, SMEU7397, is available for all hydraulic hammers. The decal provides procedures for operation and maintenance of the hydraulic hammers. The decal can be placed on the machine or the hammer. The decal can be obtained through the normal literature ordering channels.

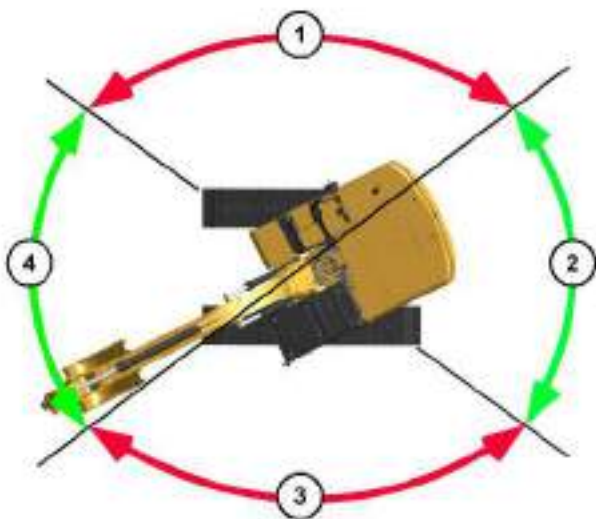


Illustration 645

g06192837

- (1) Incorrect working position
- (2) Correct working position
- (3) Incorrect working position
- (4) Correct working position

Shear Operation (If Equipped)



Illustration 646

g06222798

WARNING

Do not operate or work on this work tool unless you have read and understand the instructions and warnings in the Operation And Maintenance Manual for both the work tool and the host machine.

Failure to follow the instructions or heed the warnings could result in injury or death.

Contact your Caterpillar dealer for replacement manuals. Proper care is your responsibility.

NOTICE

Selection of a hydraulic shear must be done with extra care.

Use of a hydraulic shear not recommended by Caterpillar could result in structural damage to the host machine.

Consult your Cat dealer for hydraulic shear information.

 WARNING

Serious injury or death could occur from the demolition of pipes, vessels, tanks or other containers that may contain gas, flammable materials or hazardous chemicals.

Do not perform any demolition work on these items until all of their contents have been removed.

Follow all regulations for the removal and disposal of these materials.

NOTICE

Using the demolition tool to level the work site or push over standing structures may damage the machine or the demolition tool. Use appropriate equipment to do site preparation or maintenance operations.

NOTICE

To avoid structural damage to the machine, do not break road surfaces by placing the cutting edge of the hydraulic shear on the ground and moving the machine.

Be sure that no one is near the work tool to prevent injury. Keep the work tool under control at all times to prevent injury. When a demolition tool is used, all personnel must maintain a minimum distance of 10 m (33 ft).

Close all windows. Make sure that all required guards are in place. Wear all required protective equipment. Follow the instructions in the Operation and Maintenance Manual for the work tool.

Crusher Operation (If Equipped)

 WARNING

Improper operation and maintenance of the crusher could cause personal injury or death. Observe the following procedures for safe operation of the crusher.

Consult your Cat dealer for more information on the operation and maintenance of the crusher.

Do not operate the host machine with the work tool unless you have read and understood the instructions and warnings in the Operation and Maintenance Manual. Failure to follow the instructions or heed the warnings could result in machine or work tool damage, and/or serious injury or death. Contact your Cat dealer for a replacement manual, if needed.

When the crusher is installed on the host machine, always make sure that the protective guarding is in place.

Using the crusher in an incorrect manner can damage the machine and/or cause personal injury or death.

Always ensure that the work area is clear of ground personnel, due to the potential crush hazards with falling debris and machine movement.

Resting or placing your foot on the work tool pedal could result in unexpected movement of the machine / work tool which could result in personal injury or death. Always lock the crusher when not in use.

NOTICE

Selection of a hydraulic crusher must be done with extra care.

Use of a hydraulic crusher not recommended by Caterpillar could result in structural damage to the host machine.

Consult your Cat dealer for hydraulic crusher information.

Close all windows. Make sure that all required guards are in place. Wear all required protective equipment. Follow the instructions in the Operation and Maintenance Manual for the work tool.



Illustration 647

g06222800

Demolition work on the roof of a building could lead to serious personal injury if the building were to collapse and the excavator turned over or fell off the roof. The demolition work must be started **ONLY AFTER** surveying the building for its structural integrity.

X



Illustration 649

g06222806

Do not perform demolition work at the base of the machine, because the ground could be unstable and cause the machine to fall.

X



Illustration 648

g06222803

Crushing work above your head must be avoided because objects can fall and damage the machine.

X



Illustration 650

g06222809

Do not suddenly lower or stop the work tool, otherwise the excavator could turn over.

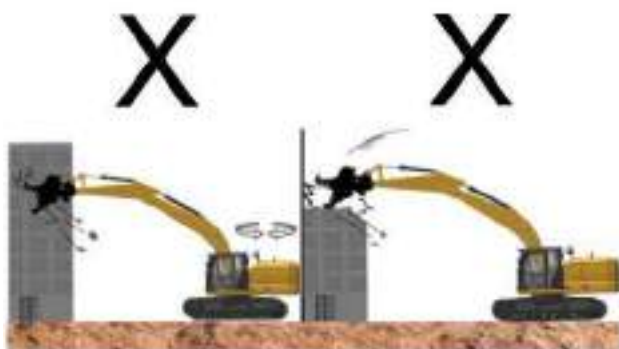


Illustration 651

g06222813

Crushing work using impact, swing, or dropping forces of the excavator could cause damages to the machine and also could lead to personal injury. As such, NEVER perform such an operation.



Illustration 653

g06222821

When performing work at elevated positions, always use care for the surroundings as well as for falling objects to avoid personal injury. Use guide personnel and signs as required.



Illustration 652

g06222817

Crushing work with hydraulic cylinder at stroke end position could damage the excavator, resulting in shortening of lifespan. Also, as it could lead to unexpected personal injury due to breakage of the machine, do not perform any work at stroke end.



Illustration 654

g06222829

When working sideways, the track can lift. Avoid abrupt operation and operate slowly.



Illustration 655

g06222831

Crushers could interfere with the boom and the cab depending on the type and method of usage. Know the working range of the crusher being used.

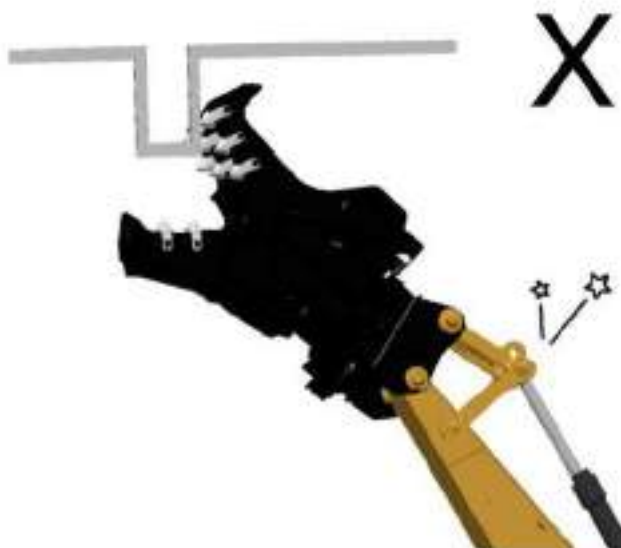


Illustration 656

g06222833

If the tooth of the crusher engages an object at a slant, excessive forces could be applied to the front regions. As such move the crusher to the front.



Illustration 657

g06222836

Never extend the boom cylinder suddenly. Sudden extension of the boom could cause tip backwards.



Illustration 658

g06222831

Sudden extension of the bucket cylinder, or sudden extension of the stick cylinder could cause damages at the stroke end position, resulting in personal injury. Operations that cause sudden extension of the cylinders is PROHIBITED!

Parking

i07868085

Stopping the Machine

SMCS Code: 7000

WARNING

Leaving the machine unattended when the engine is running may result in personal injury or death. Before leaving the machine operator station, neutralize the travel controls, lower the work tools to the ground and deactivate all work tools, and place the lever for the hydraulic lockout control in the LOCKED position.

Note: There may be regulations that define the requirements for the operator and/or support personnel to be present when the engine is running.

Park on a level surface. If the machine must be parked on a grade, chock the tracks securely.

Note: The swing parking brake is automatically applied when the machine is stopped. The swing parking brake is released when the engine is running and the joystick is activated.

1. Turn the engine speed dial counterclockwise to reduce engine speed.



Illustration 659

g06181402

2. Release the travel levers/pedals to stop the machine.

3. Lower the work tool to the ground. Apply a slight downward pressure.
4. Move the hydraulic lockout control to the LOCKED position.

i07088536

Freezing Conditions

SMCS Code: 7000

If freezing temperatures are expected, remove the mud and the dirt from each track roller frame. Park the machine on wood planks. Use the following procedure to clean each track roller frame.



Illustration 660

g06188791

1. Position the boom over one side of the machine.
2. Use boom down pressure to lift the track on one side off the ground. Operate the track in the forward direction. Then operate the track in reverse. Continue this procedure until the maximum amount of material is thrown off the track.
3. Lower the track onto the wood planks.
4. Repeat the procedure for the other track.
5. Clean the area around the carrier rollers and around the track rollers.
6. Lower the work tool onto a wood plank to prevent the work tool from touching the ground.

i07867959

Stopping the Engine

SMCS Code: 1000; 7000**NOTICE**

Stopping the engine immediately after it has been working under load can result in overheating and accelerated wear of engine components.

Refer to the following procedure to allow the engine to cool and to prevent excessive temperatures in the turbocharger housing, which could cause oil coking problems.

NOTICE

Never turn the battery disconnect switch to the OFF position while the engine is running. Serious damage to the electrical system may result.

1. Park the machine on level ground. Refer to Operation and Maintenance Manual, "Stopping the Machine" for the recommended procedure.
2. While the machine is stopped, run the engine for 5 minutes at low idle. Idling the engine allows hot areas of the engine to cool gradually.
3. Turn the engine start switch to the OFF position.

Note: If the "Regen Active" indicator is illuminated, do not shut off the engine. Refer to Operation and Maintenance Manual, "Monitoring System" for more information on indicators.

Engine Shutdown Switch

NOTICE

Perform a walk around inspection after actuation of a shutdown device.

Take necessary corrective action to resolve the cause of the shutdown.

Ensure that no additional damage has been done or could occur before returning to operation.

Turn the engine start switch to the OFF position. If the engine does not stop, perform the following procedure.

Note: Always use the engine start switch to stop the engine. Use the engine shutdown switch as an alternate method to stop the engine if the start switch fails.

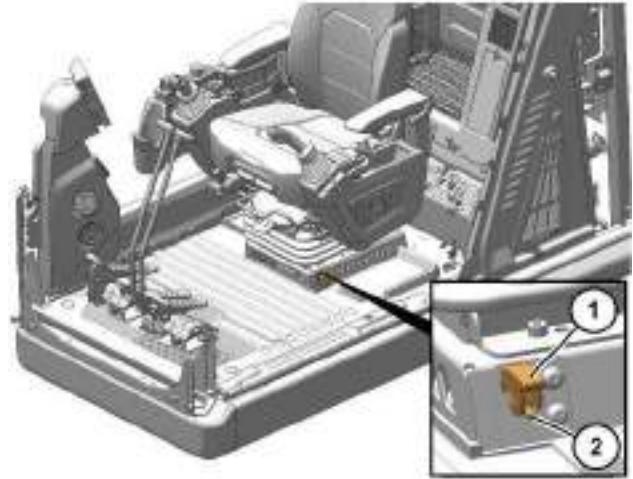


Illustration 661

g06181487

1. The engine shutdown switch is located below the left side of the operator seat.
2. Lift cover (1).
3. Push switch (2) upward. Pushing the switch upward should stop the engine and prevent the engine from being started again.
4. Return the switch to the original position. The engine will be enabled to start.

Note: Do not operate the machine again until the malfunction has been corrected.

5. Use the method that follows if the previous steps do not stop the engine.

Stop the Engine if an Electrical Malfunction Occurs

Turn the engine start switch to the OFF position. If the engine does not stop, perform the following procedure.

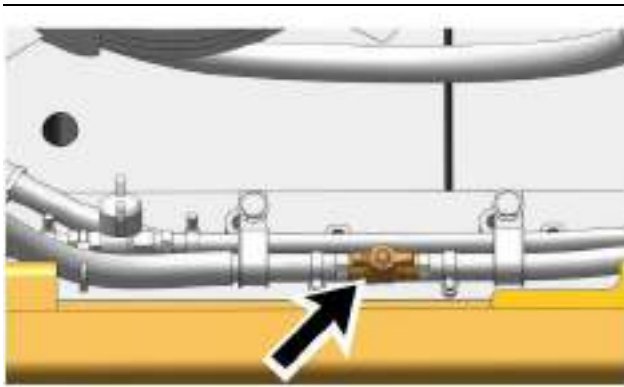


Illustration 662

g06214480

The fuel shutoff valve is located behind the right access door.

Shut off the fuel supply by turning the fuel shutoff valve clockwise. The engine will stop after consuming the fuel in the fuel line. The engine may continue to run for a few minutes.

Repair the engine before you restart the engine. The fuel system may need to be primed. See Operation and Maintenance Manual, "Fuel System - Prime" for instructions.

i07103299

Leaving the Machine

SMCS Code: 7000



Illustration 663

g06224270

1. Use the steps and the hand holds when you dismount. When you dismount, face the machine and use both hands.

2. Inspect the engine compartment for debris. Clean out any debris to avoid a fire hazard.
3. Remove all flammable debris from the front bottom guard through the access doors to reduce a fire hazard. Discard the debris properly.
4. Always turn the battery disconnect switch to the OFF position before leaving the machine.
5. If the machine will not be operated for a month or more, remove the battery disconnect switch key.
6. Lock all compartments and all vandalism covers (if equipped).

i07735116

Machine Storage and Specified Storage Period

SMCS Code: 7000

Machine Storage

The Safety Section of this Operation and Maintenance Manual contains storage information for fuels, lubricants, and ether (if equipped).

The Operation Section of this Operation and Maintenance Manual contains information for short-term storage of this machine, including engine shutdown, parking, and instructions for leaving the machine.

For detailed steps on long-term storage refer to Special Instruction, SEHS9031, "Storage Procedure for Caterpillar Products".

Specified Storage Period

The specified storage period of this machine is 1 year.

After the specified storage period has expired, consult your Cat dealer for inspect, repair, rebuild, install remanufactured, or install new components, and disposal options, and to establish a new specified storage period.

If a decision is made to remove the machine from service, refer to Decommissioning and Disposal for further information.

Transportation Information

i08163556

Shipping the Machine

SMCS Code: 7000; 7500

WARNING

Automatic Engine Speed Control (AEC) will increase engine speed automatically when you operate the control levers and/or travel pedals with AEC switch on.

When loading and unloading the machine from the truck or working in close quarters always turn AEC switch off to prevent any possibility of sudden movement of machine, which could result in serious injury or death.

Set the travel speed control switch to LOW before loading the machine. Never operate this switch when loading the machine on a trailer.

Investigate the travel route for overpass clearances. Make sure that there will be adequate clearance for the machine.

Remove ice, snow, or other slippery material from the loading dock and from the truck bed before you load the machine onto the transport machine. Removing ice, snow, or other slippery material will help to prevent the machine from slipping in transit.

Note: Obey all laws that govern the characteristics of a load (height, weight, width, and length). Observe all regulations that govern wide loads. Certain regions may require the removal of door hooks and cab bumpers, if equipped. Consult all local and regional regulations

Choose the flattest ground when you load the machine or when you unload the machine.

1. Before you load the machine, chock the trailer wheels or the rail car wheels.
2. When you use loading ramps, make sure that the loading ramps have adequate length, adequate width, adequate strength, and an adequate slope.
3. Maintain the slope of the loading ramps within 15 degrees of the ground.
4. Position the machine so that the machine can drive straight up the loading ramps. The final drives should be toward the rear of the machine. Do not operate the control levers while the machine is on the loading ramps.
5. When you drive over the loading ramp joint areas, maintain the balance point of the machine.

6. Lower the work tool to the bed or to the floor of the transport machine.
7. To prevent rolling of the machine or sudden movement of the machine, perform the following items:
 - Chock both tracks.
 - Install sufficient tie-downs at several locations.
 - Fasten wire cables.
8. If equipped, remove door hooks, cab bumpers, and fuel tank step as necessary. Refer to local regulations.

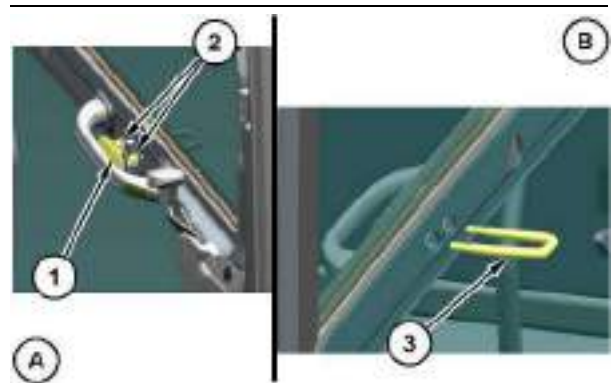


Illustration 664

g06516462

Typical example of door hook

- (A) Inside
(B) Outside
(1) Cover
(2) Nuts
(3) Door Hook

- a. Remove cover (1) and nuts (2) to remove door hook (3).



Illustration 665

g06516469

Typical example of cab bumper

- b. Remove any bumpers on your cab.

NOTICE

Do not allow the chrome surface of the bucket cylinder rod to touch any part of the trailer. Damage to the rod can occur from impact with the trailer during transport.

Note: Refer to Operation and Maintenance Manual, "Specifications".

Shipping a Machine that is not Completely Assembled

If the machine must be shipped when the boom, stick, or counterweight is not assembled on the machine, follow the instructions in Operation and Maintenance Manual, "Operation".

WARNING

The ROPS structural certification depends on the support of the boom, stick, and counterweight in the event of a machine tip over or a machine rollover incident.

When the machine needs to be moved without the boom, stick, or counterweight being installed, avoid any machine operations which could affect machine stability as a machine tip over or a machine rollover incident could result in serious injury or death.

The machine should be operated slowly on flat, stable ground or pavement by qualified operators.

i07539618

Securing the Machine

SMCS Code: 7000

WARNING

Do not transport the machine if there is a malfunction of the swing parking brake system.

The machine may swing during transportation if the swing parking brake system is not functioning properly which could result in injury or death.

Contact your Cat dealer for service.

Comply with any laws that govern the characteristics of a load (length, width, height, and weight).

1. Move the hydraulic lockout control to the LOCKED position.
2. Turn the engine start ring to the OFF position or press button to stop engine.

3. Turn the battery disconnect switch to OFF and remove the disconnect switch key.
4. Remove the ether starting aid cylinder. See Operation and Maintenance Manual, "Ether Starting Aid Cylinder - Replace" for the removal procedure.
5. Lock the door and the access covers. Attach any vandalism protection.

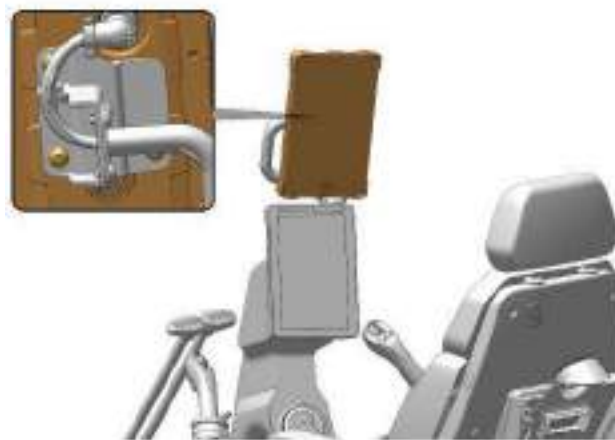


Illustration 666

g06181075

Note: Caterpillar strongly recommends removing the Cat Grade Control monitor (if equipped) before transporting the machine to protect the monitor from damage or theft.

6. Disconnect the wiring harness from the Cat Grade Control monitor. Remove the three screws that secure the monitor to the mounting bracket and remove the monitor. Remove the monitor from the cab and store in the monitor carrying case.
7. Cover the exhaust opening.

NOTICE

Do not allow the turbocharger to rotate while the engine is not operating. Damage to the turbocharger can result.

Note: Before you unload the excavator from the transport machine, remove the protective covering from the exhaust opening.

i07785977

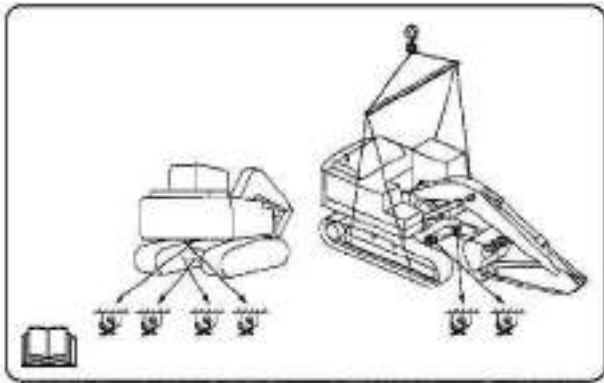


Illustration 667

g06289667

8. Chock the tracks and secure the machine with tie-downs. Make sure that you use the proper rated wire cable.

Use the front towing eyes on the lower frame, the rear towing eyes on the lower frame, and the rear towing eye that is on the upper frame.

Securely fasten all loose parts and all removed parts to the trailer or to the rail car.

When the engine is stopped, the swing parking brake is automatically applied. The swing brake prevents the upper structure from rotating.

NOTICE

In freezing weather, protect the cooling system with antifreeze, to the lowest outside expected temperature on the travel route. Or, drain the cooling system completely.

Lifting and Tying Down the Machine

SMCS Code: 7000; 7500

WARNING

Improper lifting and tie-down techniques can allow the load to shift or fail resulting in personal injury or property damage. Use only properly rated cables and slings with lift and tie down points provided on the machine. Keep the deck of the transport vehicle clean and use anti-slip mats on steel decks.

Follow the instructions in Operation and Maintenance Manual, "Lifting and Tying Down the Machine" for the proper technique for securing the machine. Refer to Operation and Maintenance Manual, "Specifications" for specific weight information.

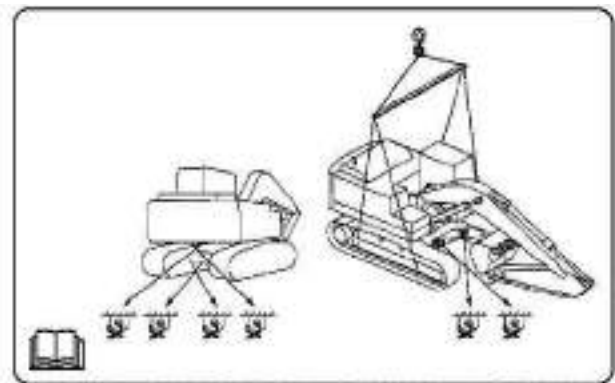


Illustration 668

g06289667

The lift and tie-down film is located near the base of the boom.

Lifting the machine



Illustration 669

g06184026

The machine center of gravity is located at the center of the swing gear.



Lifting Point – To lift the machine, attach the lifting devices to the lifting points.

The weight and the instructions that are given herein describe the machine as the machine is manufactured by Caterpillar.

Refer to the Operation and Maintenance, “Specifications” for specific weight information.

Note: Only lift objects from approved lifting points and with approved lifting devices

1. Use proper rated cables and slings for lifting. The crane should be positioned so that the machine is lifted parallel to the ground.
2. To prevent contact with the machine, lifting cables should have sufficient length.
3. Move the hydraulic lockout control to the LOCKED position.
4. Thread the cable between the first and second rollers at each end of the track.
5. Do not use the foot step as a lifting point.

6. If the full length roller guard is equipped, remove the guard.

7. Apply the proper protector to prevent machine/wire damage and slippage. Make sure that the rollers are not affected by the load.

Tying Down the Machine

There are two methods that can be used to tie down a machine. Local and/or regional regulations will determine which method to use.

Note: Obey all local and regional governmental regulations.

Frictional and Direct Lashing

When allowed, a combination of frictional lashing and direct lashing is the preferred method to tie down a machine.



Illustration 670

g06184145

(A) Front of the machine

(B) Rear of the machine

Diagonal Lashing

In areas where frictional lashing is not allowed, diagonal lashing can be used as shown below.

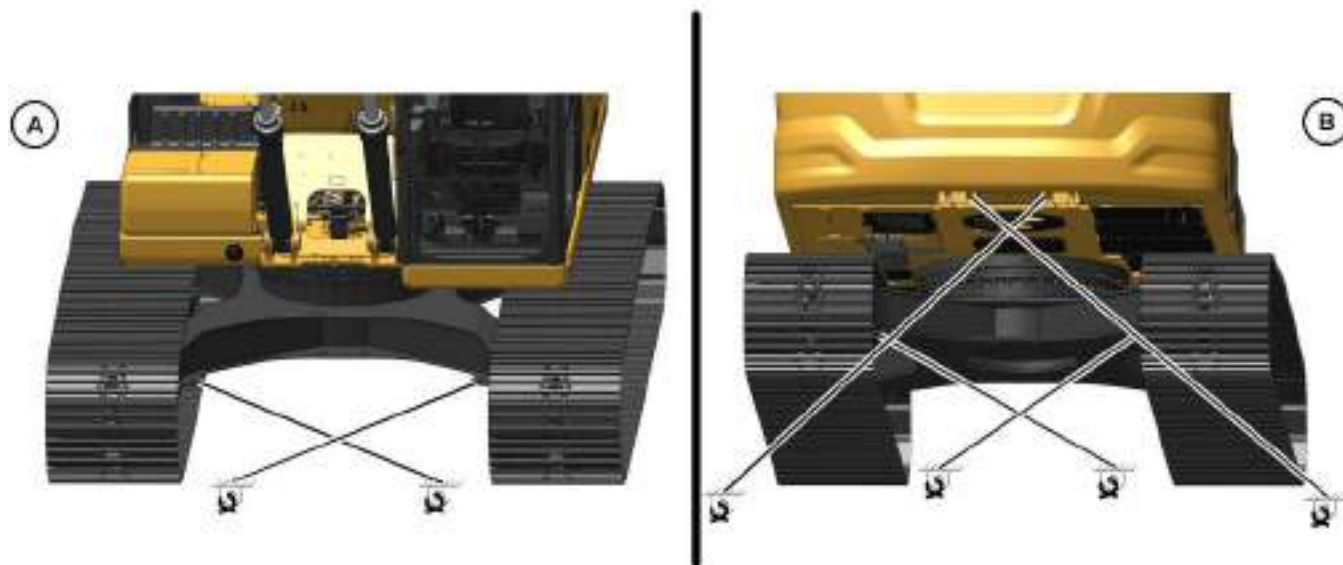


Illustration 671

g06435607

(A) Front of the machine

(B) Rear of the machine

Tying Down the Machine



Tie Down Point – To tie down the machine, attach the tie-downs to the tie-down points.

The weight and the instructions that are given herein describe the machine as the machine is manufactured by Caterpillar.

Refer to the Operation and Maintenance, "Specifications" for specific weight information.

1. Use proper rated cables and shackles for tying down the machine.
2. Use the rear eyes and the front eyes that are provided on the lower frame to fasten tie-downs. Use corner protectors for sharp corners.
3. Move the hydraulic lockout control to the LOCKED position.
4. If there is a requirement of diagonal lashing for tying down, use the proper tie-down point on the lower frame. Set the lashing angle which is on the longitudinal axis of the machine and the cable, at 30 to 50 degrees.
5. Keep the transport vehicle surface clean (for example, trailer deck).
6. For steel deck transport vehicles use skid-inhibiting or anti-slip mats (for example, rubber mats) with a friction coefficient of at least 0.3.

Lifting the Machine Segments

Bucket

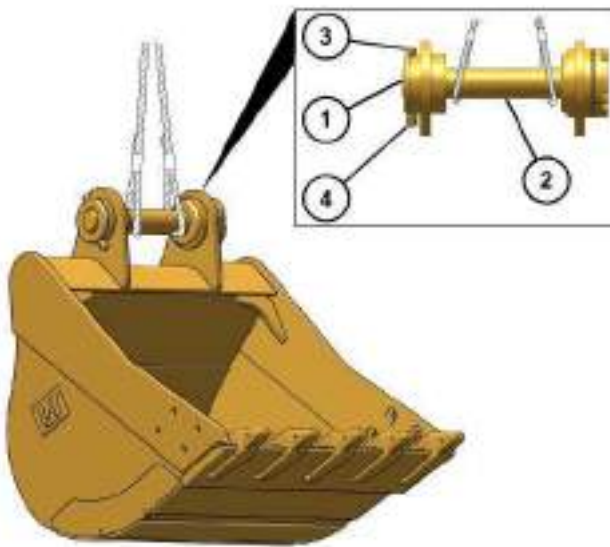


Illustration 672

g06184591

(1) Pin.(2) Sleeve.(3) Bolts.(4) Nuts.

Install pin (1) and install sleeve (2) in the brackets of the bucket. The previous illustration indicates the method to secure pin (1) with bolts (3) and nuts (4). Fasten two proper rated wire cables to pin (1).

Towing Information

i07348733

Towing the Machine

SMCS Code: 7000

 WARNING

Personal injury or death could result when towing a disabled machine incorrectly.

Block the machine to prevent movement before final drives are disengaged. The machine can roll free if it is not blocked. With final drives disengaged, the machine cannot be stopped or steered.

Follow the recommendations below, to properly perform the towing procedure.

Relieve the hydraulic tank and line pressure before any disassembly.

Even after the machine has been turned off, the hydraulic oil can still be hot enough to burn. Allow the hydraulic oil to cool before draining.

NOTICE

To tow the machine, both final drives must be disengaged.

Do not operate the travel motors with the final drives disengaged. Damage could result.

These towing instructions are for moving a disabled machine for a short distance at low speed. Move the machine at a speed of 2 km/h (1.2 mph) or less to a convenient location for repair. Always haul the machine if long distance moving is required.

Shields must be provided on both machines. This will protect the operator if the tow line or the tow bar breaks.

Do not allow an operator to be on the machine that is being towed.

Before you tow the machine, make sure that the tow line or the tow bar is in good condition. Do not use a wire rope that is kinked, twisted, or damaged. Make sure that the tow line or the tow bar has enough strength for the towing procedure that is involved. The strength of the tow line or of the tow bar should be at least 150 percent of the gross weight of the towed machine. This requirement is for a disabled machine that is stuck in the mud and for being towed on a grade.

Do not use a chain for pulling a disabled machine. A chain link can break. This may cause personal injury. Use a wire rope with ends that have loops or rings. Put an observer in a safe position to watch the pulling procedure. The observer can stop the procedure if the wire rope starts to break. Stop pulling whenever the towing machine moves without moving the towed machine.

During towing, do not allow anyone to step between the towing and the towed machines.

Do not allow the wire rope to be straddled while the machine is being towed.

Keep the tow line angle to a minimum. Do not exceed a 30 degree angle from the straight ahead position.

Avoid towing the machine on a slope.

Quick machine movement could overload the tow line or the tow bar. This could cause the tow line or the tow bar to break. Gradual, steady machine movement will be more effective.

Prior to releasing the brake of the final drive, firmly lock both tracks to prevent the machine from moving suddenly. When the machine is ready to be towed, release the brake of the final drive. Refer to Operation and Maintenance Manual, "Final Drive Sun Gear Removal".

Normally, the towing machine should be as large as the disabled machine. Make sure that the towing machine has enough brake capacity, enough weight, and enough power. The towing machine must be able to control both machines for the grade that is involved and for the distance that is involved.

You must provide sufficient control and sufficient braking when you are moving a disabled machine downhill. This may require a larger towing machine or additional machines that are connected to the rear of the disabled machine. This will prevent the machine from rolling away out of control.

All situation requirements cannot be listed. Minimal towing machine capacity is required on smooth, level surfaces. Maximum towing machine capacity is required on an incline or on a surface that is in poor condition.

Do not tow a loaded machine.

Consult your Cat dealer for the equipment that is necessary for towing a disabled machine.

Retrieval and Towing of Machine

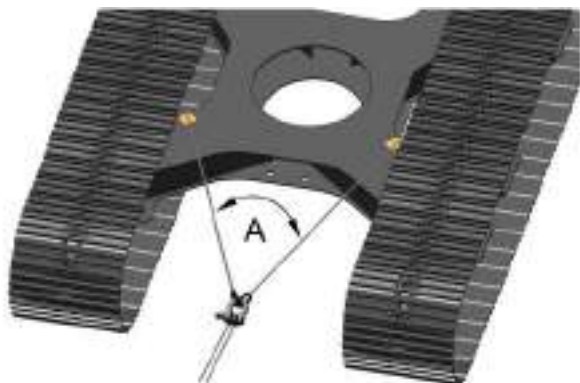


Illustration 673

g06289671

Note: Shackles must be used for towing the machine. The wire rope should be horizontal and straight to the track frame.

Install a properly rated wire rope to the lower frame of the towing machine and the lower frame of the towed machine. The permissible force for the lower frame is 100 percent of the gross weight of the towed machine.

Note: To prevent damage to the wire rope or the lower frame of the machines, use protective sleeves on the corners of the lower frame.

Retrieve the disabled machine carefully. The applied load for each wire rope should be equal. The angle (A) between each wire rope should be 60 degree maximum. Operate the machine at a low speed.

Lightweight Towing

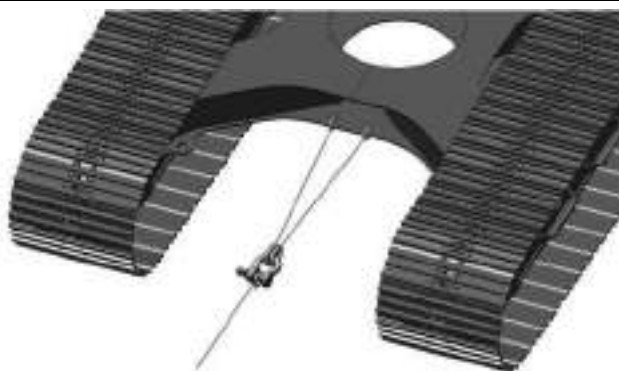


Illustration 674

g06186106

The maximum load for lightweight towing is 102900 N (75895 lb).

Shackles must be used for towing the machine. The wire rope should be horizontal and straight to the track frame.

Install a properly rated wire rope to the lower frame of the towing machine and the lower frame of the towed machine. Operate the machine at a low speed.

i06954175

Final Drive Sun Gear Removal

SMCS Code: 4050

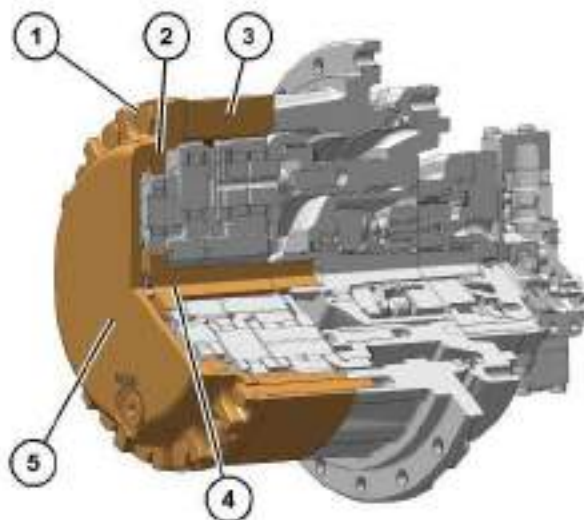


Illustration 675

g06188195

- (1) Bolt
- (2) Ring gear
- (3) Ring gear
- (4) Sun gear
- (5) Final drive cover

WARNING

Without the sun gear in place, the brakes are ineffective. Personal injury or death could result. Provide other means to hold or stop the machine.

1. Thoroughly clean the area around the final drive. Make sure that you also clean the track shoes that are positioned above the final drive.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

2. Drain the final drive oil into a suitable container. See Operation and Maintenance Manual, "Final Drive Oil - Change" for the procedure.

Operation Section
Final Drive Sun Gear Removal

3. Remove 14 of 16 cover bolts (1) from final drive cover (5). Do not leave a bolt in the top hole of the cover.
4. Insert an alignment dowel through the top hole of the cover and into the threads in the final drive housing. This is necessary to support ring gear (2) and ring gear (3) while you are removing the final drive cover.
5. Remove one track shoe to allow access to the face between final drive cover (5) and ring gear (2).
6. Loosen remaining two cover bolts (1).
7. Use a hammer and a wedge to separate final drive cover (5) and ring gear (2). Make sure that ring gear (2) and ring gear (3) stay in place.
8. Remove remaining two cover bolts (1) and final drive cover (5).
9. Remove sun gear (4) from final drive.
10. Install final drive cover (5) and 16 cover bolts (1).
11. Fill the final drive with new oil. See Operation and Maintenance Manual, "Final Drive Oil - Change" for the procedure.
12. Repeat Steps 1 through 11 for the other final drive.
13. Refer to the Service Manual for information on the installation of the final drive sun gear.

Engine Starting (Alternate Methods)

i06953771

Engine Starting with Jump Start Cables (If Equipped)

SMCS Code: 1000; 7000

 WARNING

Failure to properly service the batteries may cause personal injury.

Prevent sparks near the batteries. They could cause vapors to explode. Do not allow the jump start cable ends to contact each other or the machine.

Do not smoke when checking battery electrolyte levels.

Electrolyte is an acid and can cause personal injury if it contacts skin or eyes.

Always wear eye protection when starting a machine with jump start cables.

Improper jump start procedures can cause an explosion resulting in personal injury.

Always connect the battery positive (+) to battery positive (+) and the battery negative (-) to battery negative (-).

Jump start only with an energy source with the same voltage as the stalled machine.

Turn off all lights and accessories on the stalled machine. Otherwise, they will operate when the energy source is connected.

NOTICE

To prevent damage to engine bearings and to electrical circuits when you jump-start a machine, do not allow the stalled machine to touch the machine that is used as the electrical source.

Turn on (close) the battery disconnect switch prior to the boost connection to prevent damage to electrical components on the stalled machine.

Use only equal voltage for starting. Check the battery and starter voltage rating of your machine. Use only the same voltage for jump starting. Use of a welder or higher voltage will damage the electrical system.

Severely discharged maintenance free batteries do not fully recharge from the alternator after jump starting. The batteries must be charged to proper voltage with a battery charger. Many batteries thought to be unusable are still rechargeable.

Refer to Special Instruction, SEHS7633, "Battery Test Procedure" for complete testing and charging information. This publication is available from your Cat dealer.

When the auxiliary start receptacles are not available, use the following procedure.

1. Lower the equipment to the ground. Move all controls to the HOLD position. Move the hydraulic lockout control to the LOCKED position.
2. Turn the start switch on the stalled machine to the OFF position. Turn off all accessories.
3. Turn the battery disconnect switch on the stalled machine to the ON position.
4. Move the machine that is being used as an electrical source near the stalled machine so that the jump-start cables reach the stalled machine.
Do not allow the machines to contact each other.
5. Stop the engine of the machine that is being used as an electrical source. If you are using an auxiliary power source, turn off the charging system.



Illustration 676

g06181546

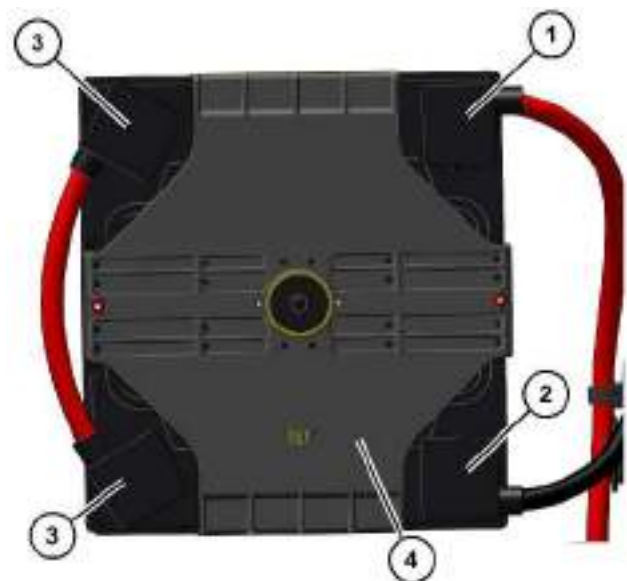


Illustration 677

g06181551

- (1) Red positive post to starter
- (2) The black negative post connects to the battery disconnect switch.
- (3) Do not use these two connections for jump starting. The red positive post is connected in series to the black negative post.
- (4) Cover

6. Ensure that battery caps on both machines are tight and correctly placed. Ensure that batteries in the stalled machine are not frozen. Make sure that the batteries have enough electrolyte.

Note: The positive terminal of the 24 V system of the source and the negative terminal of the 24 V system of the source must be identified correctly before the jumper cables are connected. The positive terminal of the 24 V system of the discharged battery must be identified correctly before the jumper cables are connected.

7. The positive ends of the jump-start cable are red. Connect one positive end of the jump-start cable to the positive cable terminal of the discharged battery. Some machines have battery sets.

Note: Batteries that are in series may be in separate compartments. Use the terminal that is connected to the starter solenoid. This battery or battery set is normally on the same side of the machine as the starter.

Do not allow the positive cable clamps to contact any metal except for the battery terminals.

8. Connect the other positive end of the jump-start cable to the positive cable terminal of the electrical source.
9. Connect one negative end of the jump-start cable to the negative cable terminal of the electrical source.
10. Finally, connect the other negative end of the jump-start cable to the frame of the stalled machine. Do not connect the jump-start cable to the battery post. Do not allow the jump-start cables to contact the battery cables, the fuel lines, the hydraulic lines, or any moving parts.
11. Start the engine of the machine that is being used as an electrical source or energize the charging system on the auxiliary power source.
12. Wait at least two minutes before you attempt to start the stalled machine. This will allow the batteries in the stalled machine to partially charge.
13. Attempt to start the stalled engine. See Operation and Maintenance Manual, "Engine Starting" for the correct starting procedure.
14. Immediately after you start the stalled engine, disconnect the jump-start cables in reverse order.

i07349165

Engine Starting with Auxiliary Start Receptacle (If Equipped)

SMCS Code: 1000; 7000



Illustration 678

g06179792

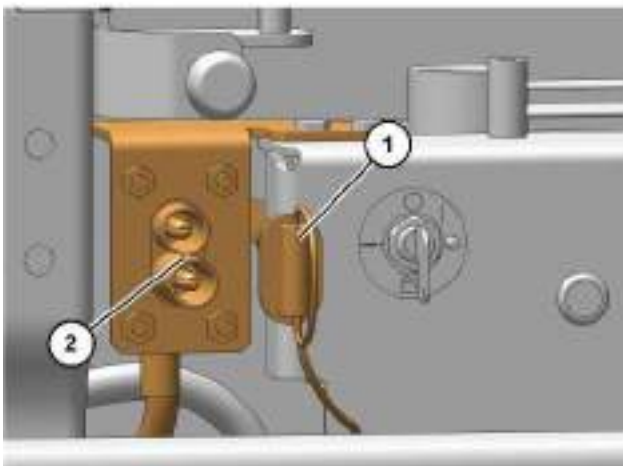


Illustration 679

g06181572

- (1) Cover
(2) Receptacle

Some Cat products may be equipped with an auxiliary start receptacle as a standard part. If your machine is not equipped with an auxiliary start receptacle, the machine can be equipped with an auxiliary start receptacle from parts service. This will ensure that a permanent receptacle is always available to jump-start the machine.

There are two cable assemblies that can be used to jump-start the stalled machine. You can jump-start the stalled machine from another machine that is equipped with an auxiliary start receptacle or with an auxiliary power pack. Your Cat dealer can provide the correct cable lengths for your application.

1. Determine the reason that the engine will not start.

Reference: Refer to Special Instruction, SEHS7633, "Battery Test Procedure" for more information.

2. Ensure that the travel control levers on the stalled machine are in the CENTER position. Engage the hydraulic lockout control. Engage the parking brake. Lower all work tools to the ground. Move all controls to HOLD.
3. Turn the engine start switch key on the stalled machine to the OFF position. Turn off all accessories.
4. Turn the battery disconnect switch on the stalled machine to ON.
5. Move the machine that is being used as a power source close to the stalled machine. The jump-start cables should reach the batteries on both machines. **DO NOT ALLOW THE MACHINES TO CONTACT EACH OTHER.**
6. Stop the engine on the machine that is being used as a power source. If you use an auxiliary power source, turn off the charging system.
7. Connect the appropriate jump-start cable to the auxiliary start receptacle on the stalled machine.
8. Connect the other end of the jump-start cable to the auxiliary start receptacle of the machine that is being used as a power source.
9. Start the engine on the machine that is being used as a power source or energize the charging system on the auxiliary power source.
10. Wait for a minimum of 2 minutes while the batteries in the stalled machine partially charge.
11. Attempt to start the stalled engine.
12. Immediately after the stalled engine starts, disconnect the jump-start cable from the power source.
13. Disconnect the other end of the jump-start cable from the stalled machine.
14. Conclude the failure analysis on the starting charging system of the stalled machine, as required. Check the machine while the engine is running and the charging system is in operation.

Maintenance Section

Maintenance Access

i07945068

Access Door and Cover Locations

SMCS Code: 726A-CH

Engine Hood

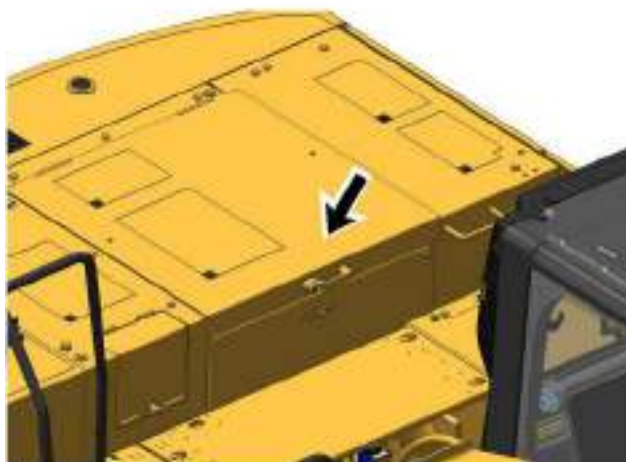


Illustration 680

g06225770

Allows access to engine and coolant tank.

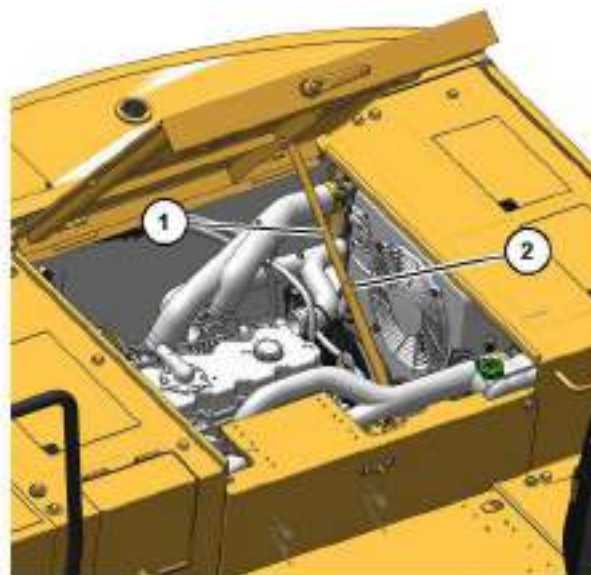


Illustration 681

g06225771

1. Open the engine hood.
2. Gas spring (1) will lock in place to hold the engine hood open.

**WARNING****Operation of the Push Button Release for the Engine Hood**

When closing the engine hood, only operate the push-button release by hand.

Failure to remove hands from the push-button release before closing the engine hood could result in personal injury.

Be sure to remove hands from the push-button release before completely closing the engine hood.

NOTICE

Do not add pressure to the engine hood when open.

**WARNING**

When closing the engine hood, Do Not operate the push-button release by foot.

Operation of the push-button release by foot could result in damage to the gas spring of the closing mechanism and/or personal injury.

Only operate the push-button release for closing the engine hood by hand.

3. To close the engine hood, support the engine hood with the door handle. Press the push-button release (2) to unlock the gas spring. Release the push button and slowly close the engine hood.

Engine Service Door

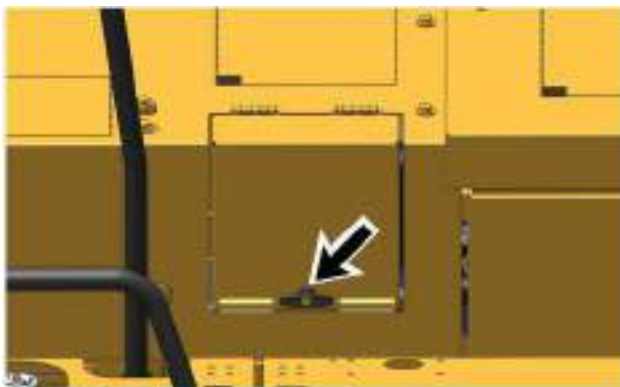


Illustration 682

g06183460

Allows access to the engine oil filler cap and the upper dipstick.

Left Rear Access Door



Illustration 683

g06179792

Allows access to the coolant sample port, coolant drain, cooling cores, power fuses, battery disconnect switch, and coolant reservoir.

Left Front Access Door



Illustration 684

g06181546

Allows access to engine air filter, batteries, and window washer reservoir.

Right Side Access Door



Illustration 685

g06182545

Allows access to the engine oil filter, engine oil sampling port, and ground level dipstick. Additionally, the compartment houses the hydraulic pump, fuel filters, refueling pump, fuel tank drain valve, and hydraulic tank sight gauge.

Right Front Access Door



Illustration 686

g06183098

Allows access to the Diesel Exhaust Fluid (DEF) tank.

Lubricant Viscosities and Refill Capacities

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Lubricant Viscosities (Fluids Recommendations)

SMCS Code: 7581

General Information for Lubricants

When you are operating the machine in temperatures below -20°C (-4°F), refer to Special Publication, SEBU5898, "Cold Weather Recommendations". This publication is available from your Cat dealer.

Refer to the "Lubricant Information" section in the latest revision of the Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for a list of Cat engine oils and for detailed information. This manual may be found on the web at Safety.Cat.com.

The footnotes are a key part of the tables. Read all footnotes that pertain to the machine compartment in question.

Selecting the Viscosity

To select the proper oil for each machine compartment, refer to the "Lubricant Viscosity for Ambient Temperature" table. Use the oil type AND oil viscosity for the specific compartment at the proper ambient temperature.

The proper oil viscosity grade is determined by the minimum ambient temperature (the air in the immediate vicinity of the machine). Measure the temperature when the machine is started and while the machine is operated. To determine the proper oil viscosity grade, refer to the "Min" column in the table. This information reflects the coldest ambient temperature condition for starting a cold machine and for operating a cold machine. Refer to the "Max" column in the table for operating the machine at the highest temperature that is anticipated. Unless specified otherwise in the "Lubricant Viscosities for Ambient Temperatures" tables, use the highest oil viscosity that is allowed for the ambient temperature.

Machines that are operated continuously should use oils that have the higher oil viscosity in the final drives and in the differentials. The oils that have the higher oil viscosity will maintain the highest possible oil film thickness. Refer to "General Information for Lubricants" article, "Lubricant Viscosities" tables, and any associated footnotes. Consult your Cat dealer if additional information is needed.

NOTICE

Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Engine Oil

Cat oils have been developed and tested to provide the full performance and life that has been designed and built into Cat engines.

Cat DEO-ULS or oils that meet the Cat ECF-3 specification, API CJ-4, and the newer API CK-4, are required for use in the applications listed below. Cat DEO-ULS and oils meeting Cat ECF-3 specification, API CJ-4, the newer API CK-4, and ACEA E9 oil categories have been developed with limited sulfated ash, phosphorus, and sulfur. These chemical limits are designed to maintain the expected aftertreatment devices life, performance, and service interval. If oils meeting the Cat ECF-3 specification, API CJ-4, the newer API CK-4 specifications are not available, oils meeting ACEA E9 may be used. ACEA E9 oils meet the chemical limits designed to maintain aftertreatment device life. ACEA E9 oils are validated using some but not all ECF-3 and API CJ-4 standard engine performance tests. Consult your oil supplier when considering use of an oil that is not Cat ECF-3, API CJ-4, or the newer API CK-4 qualified.

Failure to meet the listed requirements will damage aftertreatment-equipped engines and can negatively impact the performance of the aftertreatment devices. The Diesel Particulate Filter (DPF) if equipped, will plug sooner and require more frequent DPF ash service intervals.

Typical aftertreatment systems may include the following:

- Diesel Particulate Filters (DPF)
- Diesel Oxidation Catalysts (DOC)
- Selective Catalytic Reduction (SCR)
- Lean NOx Traps (LNT)

Other systems may apply.

Oils that are per API CI-4 or prior categories are not allowed. These oils have high ash and are not appropriate for use in engines with aftertreatment devices.

Table 33

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Engine Crankcase for all Machines	Cat DEO-ULS Cold Weather	SAE 0W-40	-40	40	-40	104
	Cat DEO-ULS SYN	SAE 5W-40	-30	50	-22	122
	Cat DEO-ULS	SAE 10W-30	-18	40	0	104
		SAE 15W-40	-10	50	14	122
Pump Coupling (If Equipped)	Cat DEO-ULS	SAE 10W-30	-18	40	0	104

Hydraulic Systems

Refer to the “Lubricant Information” section in the latest revision of the Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for detailed information. This manual may be found on the web at Safety.Cat.com.

The following are the preferred oils for use in most Cat machine hydraulic systems:

- Cat HYDO Advanced 10 SAE 10W
- Cat HYDO Advanced 20 SAE 20W
- Cat HYDO Advanced 30 SAE 30
- Cat BIO HYDO Advanced, Multigrade

Cat HYDO Advanced oils allow 6000 hours or higher oil drain intervals for most applications. S·O·S Services oil analysis is recommended when the oil drain interval is increased to 6000 hours or higher. In comparison, non-Cat commercial hydraulic oils (second choice oils) allow 2000 hours oil drain interval. It is recommended to follow the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your particular machine. Consult your Cat dealer for details. When switching to Cat HYDO Advanced fluids, cross contamination with the previous oil should be kept to less than 10%.

Second choice oils are listed below.

- Cat MTO
- Cat DEO
- Cat DEO-ULS
- Cat TDTO
- Cat TDTO Cold Weather
- Cat TDTO-TMS
- Cat DEO-ULS Cold Weather

Note: Oil drain intervals of the oils listed above are less than those of Cat HYDO Advanced oils. The oil drain interval of these oils is typically 2000 hours and up to a maximum of 4000 hours. An exception is Cat TDTO Cold Weather oil which allows 6000 hours or higher oil drain interval. S·O·S Services oil analysis is required when the oils listed above are used in Cat hydraulic system components and hydrostatic transmissions.

Table 34

Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Hydraulic System	Cat HYDO Advanced 10 Cat TDTO	SAE 10W	-20	40	-4	104
	Cat HYDO Advanced 20 Cat TDTO	SAE 20W	-5	45	23	113
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	10	50	50	122
	Cat BIO HYDO Advanced	"ISO 46" Multi-Grade	-30	50	-22	122
	Cat MTO Cat DEO-ULS Cat DEO	SAE10W-30	-20	40	-4	104
	Cat DEO-ULS Cat DEO	SAE15W-40	-15	50	5	122
	Cat TDTO-TMS	Multi-Grade	-15	50	5	122
	Cat DEO-ULS SYN	SAE 5W-40	-25	40	-13	104
	Cat DEO-ULS Cold Weather	SAE0W-40	-40	40	-40	104
	Cat TDTO Cold Weather	SAE 0W-20	-40	40	-40	104

For applications requiring the use of fire-resistant hydraulic fluids, Caterpillar recommends the use of EcoSafe FR-46. This product is a fully synthetic, non-aqueous hydraulic fluid. Water based and glycol-based hydraulic products are not recommended for use in Cat machine hydraulic systems. EcoSafe FR-46 is an ISO 46 multi-grade product. For more information refer to the product information provided by the manufacturer of EcoSafe hydraulic oil. Refer to your Cat dealer for availability.

Other Fluid Applications

Table 35

Excavators, Front Shovels, Mass Excavators, Demolition Excavators, and Track Material Handlers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosity Grade	°C		°F	
			Min	Max	Min	Max
Final Drives and Swing Drives	Cat TDTO Cold Weather	SAE 0W-20	-40	0	-40	32
	Cat TDTO	SAE 10W	-30	0	-22	32
		SAE 30 ⁽¹⁾	-30	35	-22	95
		SAE 50	-15	50	5	122
	Cat TDTO-TMS	Multi-Grade	-25	25	-13	77
Track Roller Frame Recoil Spring and Pivot Shaft Bearings	Cat TDTO Cold Weather	SAE 0W-20	-40	0	-40	32
	Cat TDTO	SAE 10W	-30	0	-22	32
		SAE 30 ⁽¹⁾	-20	25	-4	77

(continued)

(Table 35, contd)

Excavators, Front Shovels, Mass Excavators, Demolition Excavators, and Track Material Handlers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosity Grade	°C		°F	
			Min	Max	Min	Max
		SAE 50	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	-25	25	-13	77
Track Idlers and Track Rollers	Cat DEO-ULS Cat DEO	SAE 15W-40 ⁽¹⁾	-10	50	14	122
	Cat DEO Cold Weather	SAE 0W-40	-40	40	-40	104
	Cat DEO-ULS SYN	SAE 5W-40	-35	40	-31	104
	Cat FDAO SYN	Multi-Grade	-30	50	-22	122

⁽¹⁾ Factory fill oil for standard configuration machines.

Special Lubricants

Grease

To use a non-Cat grease, the supplier must certify that the lubricant is compatible with Cat grease.

Each pin joint should be flushed with the new grease. Ensure that all old grease is removed. Failure to meet this requirement may lead to failure of a pin joint.

Table 36

Recommended Grease						
Compartment or System	Grease Type	NLGI Grade	°C		°F	
			Min	Max	Min	Max
External Lubrication Points	Cat Prime Application Grease	NLGI Grade 2	-20	140	-4	284
	Cat Extreme Application Grease	NLGI Grade 1	-20	140	-4	284
		NLGI Grade 2	-15	140	+5	284
	Cat Extreme Application Grease-Artic	NLGI Grade 0.5	-50	130	-58	266
	Cat Extreme Application Grease-Desert	NLGI Grade 2	-10	140	+14	284
	Cat Utility Grease	NLGI Grade 2	-20	140	-4	284
	Cat Ball Bearing Grease	NLGI Grade 2	-20	160	-4	320

Diesel Fuel Recommendations

Diesel fuel must meet Caterpillar Specification for Distillate Fuel and the latest revisions of "ASTM D975" and "EN 590" to ensure optimum engine performance. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for the latest fuel information and for Cat fuel specification. This manual may be found on the Web at Safety.Cat.com.

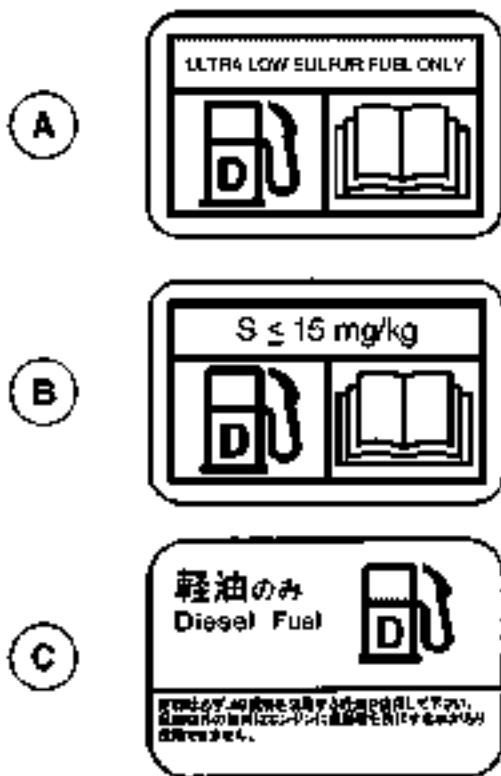


Illustration 687

g03218956

(A) North America film

(B) Europe, Africa, Middle East film

(C) Japan film

NOTICE

Ultra Low Sulfur Diesel (ULSD) fuel 0.0015 percent (≤ 15 ppm (mg/kg)) sulfur is required by regulation for use in engines certified to nonroad Tier 4 standards (U.S. EPA Tier 4 certified) and that are equipped with exhaust aftertreatment systems.

European ULSD 0.0010 percent (≤ 10 ppm (mg/kg)) sulfur fuel is required by regulation for use in engines certified to European Stage V type-approved nonroad engines. Additionally, the Cetane number should not be less than 45, and the bio-diesel content should not be greater than 20% volume/volume.

Misfueling with fuels of higher sulfur level can have the following negative effects:

- Shorten the time interval between aftertreatment device service intervals (cause the need for more frequent service intervals)
- Adversely impact the performance and life of aftertreatment devices (cause loss of performance)
- Reduce regeneration intervals of aftertreatment devices
- Reduce engine efficiency and durability.
- Increase the wear.
- Increase the corrosion.
- Increase the deposits.
- Lower fuel economy
- Shorten the time period between oil drain intervals (more frequent oil drain intervals).
- Increase overall operating costs.

Failures that result from the use of improper fuels are not Caterpillar factory defects. Therefore, the cost of repairs would not be covered by a Caterpillar warranty.

Caterpillar does not require the use of ULSD in non-road and machine applications that are not Tier 4/Stage V certified engines and are not equipped with aftertreatment devices. For Tier 4/Stage V certified engines, always follow operating instructions to ensure that the correct fuels are used.

Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more details about fuels, lubricants, and Tier 4/Stage V requirements. This manual may be found on the web at Safety.Cat.com.

Diesel Exhaust Fluid

General Information

Diesel Exhaust Fluid (DEF) is a liquid that is injected into the exhaust system of engines equipped with Selective Catalytic Reduction (SCR) systems. SCR reduces emissions of nitrogen oxides (NOx) in diesel engine exhaust.

Diesel Exhaust Fluid (DEF) is also known under other names including Aqueous Urea Solution (AUS) 32, AdBlue, or generically referred to as urea.

In engines equipped with SCR emissions reduction system, DEF is injected in controlled amounts into the engine exhaust stream. At the elevated exhaust temperature, urea in DEF is converted to ammonia. The ammonia chemically reacts with NOx in diesel exhaust in the presence of the SCR catalyst. The reaction converts NOx into harmless nitrogen (N₂) and water (H₂O).

DEF Recommendations

For use in Cat engines, DEF must meet all the requirements defined by "ISO 22241-1" Requirements.

Caterpillar recommends the use of DEF available through the Cat parts ordering system for use in Cat engines equipped with SCR systems.

In North America, commercial DEF that is API approved and meets all the requirements defined in "ISO 22241-1" may be used in Cat engines that are equipped with SCR systems.

Outside of North America, commercial DEF that meets all requirements defined in "ISO 22241-1" may be used in Cat engines that are equipped with SCR systems.

The supplier should provide documentation to prove that the DEF is compliant with the requirements of "ISO 22241-1".

NOTICE

Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE

Do not use agriculture grade urea solutions. Do not use any fluids that do not meet "ISO 22241-1" Requirements in SCR emissions reduction systems. Use of these fluids can result in numerous problems including damage to SCR equipment and a reduction in NOx conversion efficiency.

DEF is a solution of solid urea that is dissolved in demineralized water to produce a final concentration of 32.5% urea. DEF concentration of 32.5% is optimal for use in SCR systems. DEF solution of 32.5% urea has the lowest attainable freeze point of -11.5° C (11.3° F). DEF concentrations that are higher or lower than 32.5% have higher freeze points. DEF dosing systems and "ISO 22241-1" specifications are designed for a solution that is approximately 32.5%.

Caterpillar offers a refractometer, Cat part number 360-0774, that can be used to measure DEF concentration. Follow the instructions provided with the instrument. Appropriate commercial portable refractometers can be used to determine urea concentration. Follow the instructions from the manufacturer.

DEF Guidelines

DEF solution is typically colorless and clear. Changes to color or clarity are indicators of quality issues. Quality of DEF can degrade when stored and handled inappropriately or if DEF is not protected from contamination. Details are provided below.

If quality issues are suspected, testing of DEF should focus on urea percentage, alkalinity as NH₃ and biuret content. DEF that does not pass all these tests or that is no longer clear should not be used.

Materials compatibility

DEF is corrosive. Due to the corrosion caused, DEF must be stored in tanks constructed of approved materials. Recommended storage materials:

Stainless Steels:

- 304 (S30400)
- 304L (S30403)
- 316 (S31600)
- 316L (S31603)

Alloys and metals:

- Chromium Nickel (CrNi)
- Chromium Nickel Molybdenum (CrNiMo)
- Titanium

Non-metallic materials:

- Polyethylene
- Polypropylene
- Polyisobutylene
- Teflon (PFA)
- Polyfluoroethylene (PFE)
- Polyvinylidene fluoride (PVDF)
- Polytetrafluoroethylene (PTFE)

Materials NOT compatible with DEF solutions include Aluminum, Magnesium, Zinc, Nickel coatings, Silver and Carbon steel, and Solders containing any of the above. Unexpected reactions may occur if DEF solutions come in contact with any non-compatible material or unknown materials.

Bulk storage

Follow all local regulations covering bulk storage tanks. Follow proper tank construction guidelines. Tank volume typically should be 110% of planned capacity. Appropriately vent indoor tanks. Plan for control of overflow of the tank. Heat tanks that dispense DEF in cold climates.

Bulk tank breathers should be fitted with filtration to keep airborne debris from entering the tank. Desiccant breathers should not be used because water will be absorbed, which potentially can alter DEF concentration.

Handling

Follow all local regulations covering transport and handling. DEF transport temperature is recommended to be -5° C (23° F) to 25° C (77° F). All transfer equipment and intermediate containers should be used exclusively for DEF. Containers should not be reused for any other fluids. Ensure that transfer equipment is made from DEF-compatible materials. Recommended material for hoses and other non-metallic transfer equipment includes:

- Nitrile Rubber (NBR)
- Fluoroelastomer (FKM)
- Ethylene Propylene Diene Monomer (EPDM)

The condition of hoses and other nonmetallics that are used with DEF should be monitored for signs of degradation. DEF leaks are easily recognizable by white urea crystals that accumulate at the site of the leak. Solid urea can be corrosive to galvanized or unalloyed steel, aluminum, copper, and brass. Leaks should be repaired immediately to avoid damage to surrounding hardware.

Cleanliness

Contaminants can degrade the quality and life of DEF. Filtering DEF is recommended when dispensed into the DEF tank. Filters should be compatible with DEF and should be used exclusively with DEF. Check with the filter supplier to confirm compatibility with DEF before using. Mesh-type filters using compatible metals, such as stainless steel, are recommended. Paper (cellulose) media and some synthetic filter media are not recommended because of degradation during use.

Care should be taken when dispensing DEF. Spills should be cleaned immediately. Machine or engine surfaces should be wiped clean and rinsed with water. Caution should be used when dispensing DEF near an engine that has recently been running. Spilling DEF onto hot components will cause harmful vapors.

Stability

NOTICE

Storing Diesel Exhaust Fluid in high heat areas is not recommended. Do not store DEF in high heat generating areas on the machine, such as the pump compartment or the engine compartment. The quality of the DEF can degrade when exposed to high temperatures.

DEF fluid is stable when stored and handled properly. The quality of DEF rapidly degrades when stored at high temperatures. The ideal storage temperature for DEF is between -9°C (15.8°F) and 25°C (77°F). DEF that is stored above 35°C (95°F) for longer than 1 month must be tested before use. Testing should evaluate Urea Percentage, Alkalinity as NH_3 and Biuret content.

The length of storage of DEF is listed in the following table:

Table 37

Storage Temperature	Expected DEF Life
Below 25°C (77°F)	18 months
25°C (77°F) to 30°C (86°F)	12 months
30°C (86°F) to 35°C (95°F)	6 months
Above 35°C (95°F)	test quality before use

Refer to "ISO 22241" document series for more information about DEF quality control.

Note: Dispose of all fluids according to applicable regulations and mandates.

Fuel Additives

Cat Diesel Fuel Conditioner and Cat Fuel System Cleaner are available for use when needed. These products are applicable to diesel and biodiesel fuels. Consult your Cat dealer for availability.

Biodiesel

Biodiesel is a fuel that can be made from various renewable resources that include vegetable oils, animal fat, and waste cooking oil. Soybean oil and rapeseed oil are the primary vegetable oil sources. To use any of these oils or fats as fuel, the oils, or fats are chemically processed (esterified). The water and contaminants are removed.

U.S. distillate diesel fuel specification "ASTM D7467" includes up to B20 (20 percent) biodiesel. Any diesel fuel in the U.S. may contain up to B20 biodiesel fuel.

European distillate diesel fuel specification "EN16709:2015" includes up to B20 (20 percent) biodiesel. Any diesel fuel in Europe may contain up to B20.

Note: The diesel portion used in the biodiesel blend must be Ultra Low Sulfur Diesel (15 ppm sulfur or less, per "ASTM D975"). In Europe the diesel fuel portion used in the biodiesel blend must be sulfur free diesel (10 ppm sulfur or less, per "EN 590"). The final blend must have 15 ppm sulfur or less.

When biodiesel fuel is used, certain guidelines must be followed. Biodiesel fuel can influence the engine oil, aftertreatment devices, non-metallic, fuel system components, and others. Biodiesel fuel has limited storage life and has limited oxidation stability. Follow the guidelines and requirements for engines that are seasonally operated.

To reduce the risks associated with the use of biodiesel, the final biodiesel blend and the biodiesel fuel used must meet specific blending requirements.

All the guidelines and requirements are provided in the latest revision of Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations". This manual may be found on the web at Safety.Cat.com.

Coolant Information

The information provided in this "Coolant Recommendation" section should be used with the "Lubricants Information" provided in the latest revision of Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations". This manual may be found on the web at Safety.Cat.com.

The following two types of coolants may be used in Cat diesel engines:

Preferred – Cat ELC (Extended Life Coolant)

Acceptable – Cat DEAC (Diesel Engine Antifreeze/Coolant)

NOTICE

Never use water alone as a coolant. Water alone is corrosive at engine operating temperatures. In addition, water alone does not provide adequate protection against boiling or freezing.

Table 38

Recommendations for Finished Coolants for use in Cat engines				
Coolant Type	Recommendations	Product	Service hours ⁽¹⁾⁽²⁾	Required Maintenance ⁽³⁾
Cat ELC, Cat ELI, or commercial coolant that meets Cat EC-1	Preferred	Cat ELC	12000 hours or 6 years	Add Cat ELC Extender at 6000 service hours or one half of service life
		Cat ELI (water based) ⁽⁴⁾	12000 hours or 6 years	Add Cat ELC Extender at 6000 service hours or one half of service life
	Min requirements	Cat EC-1 specification and "ASTM D6210" and Organic Additive Technology (OAT) based on a combination of a monocarboxylic acid and a dicarboxylic acid. Free of phosphate, borate, and silicate. Tolyltriazole: minimum typical concentration of 900 ppm Nitrite: minimum typical concentration of 500 ppm in new coolants	6000 hours or 6 years	Add Extender at 3000 service hours or one half of service life
Cat DEAC, Cat SCA, conventional coolants, and commercial extended life coolants that do not meet EC-1	Acceptable	Cat DEAC	3000 hours or 3 years	SCA at maintenance intervals
		Cat SCA (water based) ⁽⁴⁾	3000 hours or 2 years	SCA at maintenance intervals

(continued)

(Table 38, contd)

Recommendations for Finished Coolants for use in Cat engines				
	Min requirements for fully formulated Heavy-Duty Commercial coolants	"ASTM D6210" and Nitrite (as NO ₂) concentration: Minimum of 1200 ppm (70 grains/US gal) and maximum of 2400 ppm (140 grains/US gal) Silicon concentration: minimum of 100 ppm and maximum of 275 ppm	3000 hours or 2 years	SCA at maintenance intervals
	Min requirements for SCA and water ⁽⁴⁾ ⁽⁵⁾	Commercial supplemental coolant additive and water having Nitrite (as NO ₂) concentration: Minimum of 1200 ppm (70 grains/US gal) and maximum of 2400 ppm (140 grains/US gal) Silicon concentration: minimum of 100 ppm and maximum of 275 ppm	3000 hours or 1 year	Per manufacturer recommendations

- (1) New Coolants at 50 volume percent diluted. Coolants that are prediluted at the coolant manufacturer must be diluted with water that meets Reagent 4 "ASTM D1193" requirements.
- (2) Maintain the in-service coolant at the given limits.
- (3) For appropriate maintenance procedures, refer to the details given in this Chapter. For applications that allow the use of Cat ELI in water, a minimum of 7.5 percent of Cat ELI is recommended. For applications that allow the use of SCA and water, a minimum of 6 percent to a maximum of 8 percent concentration of Cat SCA are recommended.
- (4) Water-based coolants are not allowed for use in machines that has NO_x reduction aftertreatment devices, in engines that has AATAC and in Marine engines that have SWAC
- (5) There are currently no industry standards to define the quality of water-based conventional coolants. To control the quality of SCA and water coolants, the commercial SCA additive package should pass ASTM D6210 when this package is used in a glycol-based coolant. Do not use a commercial SCA additive package that only meets the ASTM D3306 or equivalent specification when used in a glycol-based coolant.

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Capacities (Refill)

SMCS Code: 1000; 7000

S/N: NDL1–Up

S/N: LTN1–Up

S/N: HDT1–Up

S/N: RAZ1–Up

Table 39

323 Approximate Capacities (Refill)			
Component or System	Liters	US gal	Recommended Type
Cooling System	25	6.6	Refer to Operation and Maintenance Manual, "Lubricant Viscosities".
Fuel Tank	345	91	
Engine Crankcase with Filter - C7.1 Engine	25	6.6	
Hydraulic System ⁽¹⁾	115	30	
Swing Drive	5	1.3	

(continued)

Maintenance Section
Capacities (Refill)

(Table 39, contd)

323 Approximate Capacities (Refill)			
Component or System	Liters	US gal	Recommended Type
Each Final Drive	5	1.3	
DEF Tank	41	10.8	
	kg	lbs	
Swing Gear	13.5	29.8	Refer to Operation and Maintenance Manual, "Lubricant Viscosities - Special Lubricants".
Refrigerant ⁽²⁾	0.9	1.98	R-134a
	mL	oz	
Refrigerant Oil ⁽²⁾	240	8	Polyalkylene Glycol (PAG) Oil

⁽¹⁾ The amount of hydraulic fluid that is needed to refill the hydraulic system after performing Operation and Maintenance Manual, "Hydraulic System Oil - Change"

⁽²⁾ Refer to Service Manual, "Air Conditioning and Heating R-134a for All Caterpillar Machines" for additional information

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Capacities (Refill)

SMCS Code: 1000; 7000

S/N: KFE1-Up

S/N: MYK1-Up

S/N: YCP1-Up

S/N: HEX1-Up

Table 40

320 Approximate Capacities (Refill)			
Component or System	Liters	US gal	Recommended Type
Cooling System	25	6.6	Refer to Operation and Maintenance Manual, "Lubricant Viscosities".
Fuel Tank	345	91	
Engine Crankcase with Filter - C4.4 Engine	15	4.0	
Hydraulic System ⁽¹⁾	115	30	
Swing Drive	5	1.3	
Each Final Drive	5	1.3	
DEF Tank	39	10.3	
	kg	lbs	
Swing Gear	13.5	29.8	Refer to Operation and Maintenance Manual, "Lubricant Viscosities - Special Lubricants".
Refrigerant ⁽²⁾	0.9	1.98	R-134a
	mL	oz	
Refrigerant Oil ⁽²⁾	240	8	Polyalkylene Glycol (PAG) Oil

⁽¹⁾ The amount of hydraulic fluid that is needed to refill the hydraulic system after performing Operation and Maintenance Manual, "Hydraulic System Oil - Change"

⁽²⁾ Refer to Service Manual, "Air Conditioning and Heating R-134a for All Caterpillar Machines" for additional information

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S·O·S Information

SMCS Code: 1000; 1348; 3080; 4050; 5050; 7000;
7542-008

S·O·S Services is a highly recommended process for Cat customers to use in order to minimize owning and operating cost. Customers provide oil samples, coolant samples, and other machine information. The dealer uses the data in order to provide the customer with recommendations for management of the equipment. In addition, S·O·S Services can help determine the cause of an existing product problem.

Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluid Recommendations" for detailed information concerning S·O·S Services.

The effectiveness of S·O·S Services is dependent on timely submission of the sample to the laboratory at recommended intervals.

Refer to the Operation and Maintenance Manual, "Maintenance Interval Schedule" for a specific sampling location and a service hour maintenance interval.

Consult your Cat dealer for complete information and assistance in establishing an S·O·S program for your equipment.

Maintenance Support

i07477696

Prepare the Machine for Maintenance

SMCS Code: 1000; 7000

Refer to the following procedure before you perform any maintenance to the machine.

 WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the attachments have been lowered, oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.

Note: Permit only one operator on the machine. Keep all other personnel away from the machine or in view of the operator.

1. Park the machine on a dry, level, solid surface that is free of any debris.

Note: The surface must be solid enough to support the weight of the machine and any tooling that is used to support the machine.

2. Engage the parking brake. Place wheel blocks in front and behind the wheels or tracks.
3. Lower all work tools to the ground.
4. Stop the engine.

5. Release the pressure in the hydraulic system. Refer to Operation and Maintenance Manual, "System Pressure Release" for more information.

Perform a visual inspection first. If the visual checks are completed but the problem has not been identified, perform operational checks. If the problem has not been identified, perform instrument tests. This procedure will help to identify system problems.

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Service Interval Chart

SMCS Code: 7000

The service interval chart is inside the battery compartment door on the left side of the machine.

Refer to the Operation and Maintenance Manual, "Maintenance Interval Schedule" for the correct maintenance intervals and procedures that are specific to your machine.

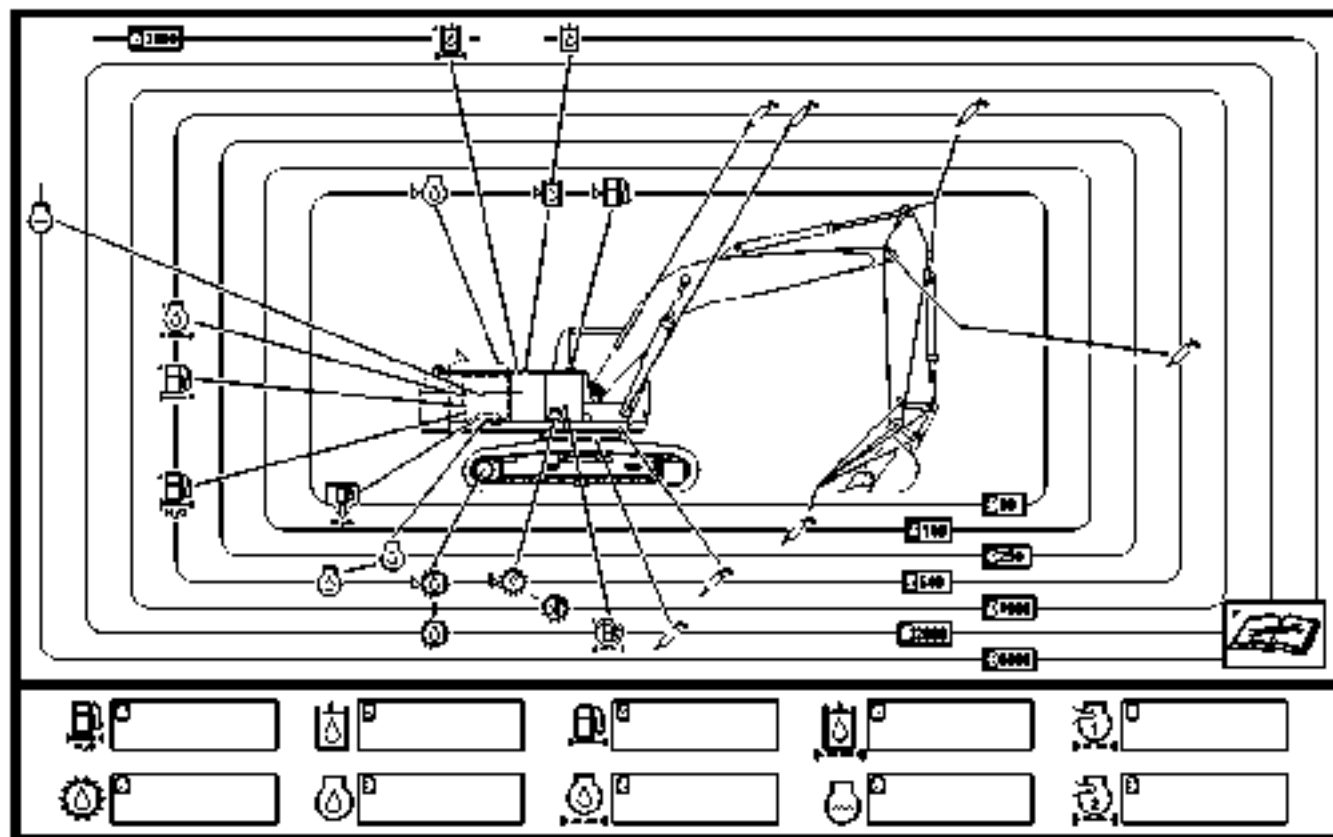


Illustration 688

g06255255



Fuel system water separator element – Replace the fuel system water separator element.



Fumes Disposal Filter Element – Replace the fumes disposal filter element.



Grease zerk – Lubricate the designated locations.



Hydraulic oil level – Check the hydraulic oil level.



Hydraulic oil – Change the hydraulic oil.



Hydraulic oil filter – Change the hydraulic oil filter.



Swing drive oil level – Check the swing drive oil level.



Swing drive oil – Change the swing drive oil.

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System Pressure Release

SMCS Code: 1250-553-PX; 1300-553-PX; 1350-553-PX; 5050-553-PX; 6700-553-PX; 7540-553-PX

WARNING

Personal injury or death can result from sudden machine movement.

Sudden movement of the machine can cause injury to persons on or near the machine.

To prevent injury or death, make sure that the area around the machine is clear of personnel and obstructions before operating the machine.

Coolant System

WARNING

Pressurized system: Hot coolant can cause serious burn. To open cap, stop engine, wait until radiator is cool. Then loosen cap slowly to relieve the pressure.

To relieve the pressure from the coolant system, turn off the machine. Allow the cooling system pressure cap to cool. Remove the cooling system pressure cap slowly to relieve pressure.

Hydraulic System

The release of hydraulic pressure in a hydraulic circuit is required before service is performed to that hydraulic circuit. Release the pressure in the following hydraulic circuits before any hydraulic lines are disconnected or removed from that hydraulic circuit.

- Boom hydraulic circuit
- Stick hydraulic circuit
- Bucket hydraulic circuit
- Swing hydraulic circuit
- Travel hydraulic circuit
- Attachment hydraulic circuits (if equipped)
- Pilot hydraulic circuit
- Return hydraulic circuit

Note: Refer to the Disassembly and Assembly Manual for additional information concerning service of the components of specific hydraulic circuits.

Release of Hydraulic Pressure from the Main Hydraulic System

WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the work tools have been lowered to the ground, and the oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.

Perform the following steps to release the hydraulic system pressure from the main hydraulic system.

Note: For additional safety, wrap hydraulic joint with material that could absorb/reduce any residual pressure of oil when released. Loosen the joint slowly, pause, and carefully check hydraulic joint for tensions indicating presence of pressure or spring force in lines or components.

1. Position the machine on level ground.



Illustration 689

g06185115

2. Fully retract the stick cylinder rod. Adjust the position of the bucket so that the bucket is parallel to the ground. Lower the boom until the work tool is flat on the ground. Refer to Illustration 689 .

3. Release the system pressure from the implement and swing hydraulic circuits.

Note: Perform Step 3b through Step 3d immediately after the engine is shut off to insure adequate pilot system pressure is available to release the pressure in the hydraulic circuits.

- a. Shut off the engine.
- b. Turn the engine start switch to the ON position without starting the engine.

c. Place the hydraulic activation control lever in the UNLOCKED position.

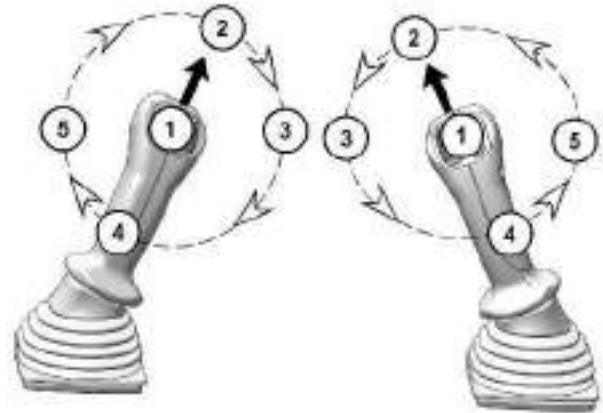


Illustration 690

g06184822

d. Move both joysticks in a circular motion to the FULL STROKE positions multiple times until the pilot accumulator has been exhausted.

Note: Pilot pressure is required to relieve hydraulic system pressure.

e. Place the hydraulic activation control lever in the LOCKED position.

f. Start the engine to recharge the pilot accumulator.

Note: Do not activate any controls when recharging the pilot accumulator.

g. Shut off the engine.

h. Repeat Step 3b through Step 3g until the high-pressure lines have been released.

Note: Each time the accumulator is recharged, start the joysticks at different positions or rotate in the reverse direction. Doing so will ensure that the same circuit is not being relieved each time.

Note: You can also move only the joysticks or pedals of the hydraulic circuit that requires service to the full stroke positions after moving joysticks in a circular motion multiple times. This action will release the high pressure only in that single hydraulic circuit. This action will also release any pressure that might be present in the pilot hydraulic circuit.

Maintenance Section
System Pressure Release

4. Release hydraulic system pressure in the attachment circuits.

- a. Start the engine to charge pilot accumulator.
- b. Shut off the engine.

Note: Perform Step 4c through Step 4e immediately after the engine is shut off to insure adequate pilot system pressure is available to release the pressure in the hydraulic circuits.

- c. Turn the engine start switch to the ON position without starting the engine.
- d. Place the hydraulic activation control lever in the UNLOCKED position.
- e. Activate the switch or pedal for the attachment circuit.
- f. Place the hydraulic activation control lever in the LOCKED position.
- g. Start the engine to recharge pilot accumulator.

Note: Do not activate any controls when recharging pilot accumulator.

- h. Shut off the engine.
- i. Repeat Step 4c through Step 4f for each attachment circuit.

5. After releasing the hydraulic pressure in each of the desired hydraulic circuits, place the hydraulic activation control lever in the LOCKED position.

6. Turn the engine start switch to the OFF position.

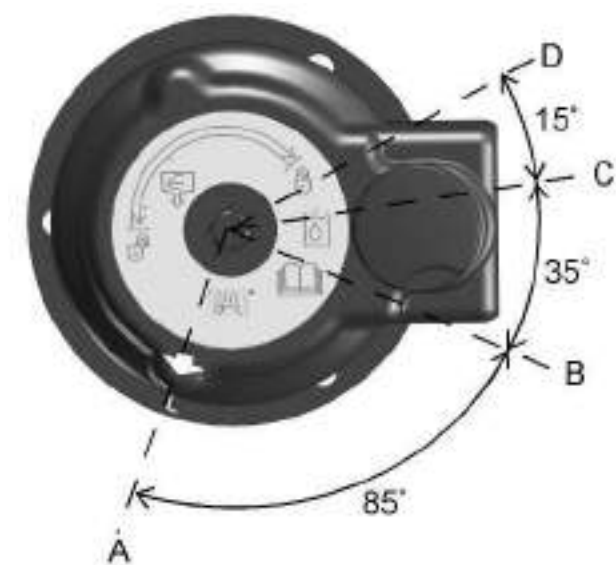


Illustration 691

g06184990

Hydraulic oil tank filler cap

- (A) LOCK position
- (B) PRESSURE RELEASE - START position
- (C) PRESSURE RELEASE - END position
- (D) OPEN position

7. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 691 for filler cap positions.

- a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).
- b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
- c. Push down the filler cap and move the arrow from position (C) to position (D).

Note: The travel hydraulic circuit is open to the hydraulic tank. Pressure from the travel circuit is released by releasing pressure from the return circuit.

8. Release the pressure that may be present in the boom circuit to remove the risk of residual pressure in the line. Make sure that the engine start switch is in the OFF position and the pressure in the hydraulic tank has been released.

- a. Remove the hydraulic oil tank filler cap.
- b. Remove any covers to access the main control valve.

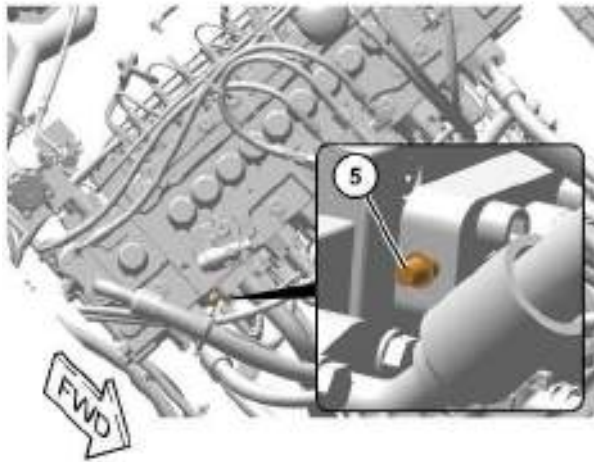


Illustration 692

g06205184

Main control valve

(5) Screw

- c. Use a clean hose of adequate length and size and connect to screw (5). Put the other end of the hose in the filler cap opening.
- d. Slowly loosen screw (5) by a maximum of 1/2 turn. Loosening the screw allows the hydraulic oil in the boom circuit to drain into the hydraulic tank.
- e. Tighten screw (5) to 13 ± 2 N·m (9 ± 1 lb ft).

Note: Refer to the Operation and Maintenance Manual, Equipment Lowering with Engine Stopped for information on lowering the work tool with the engine off.

- f. Disconnect the hose from screw (5). Do not allow the oil that is contained in the hose to spill. Drain the oil into a suitable container.

9. The pressure in the multiple hydraulic circuits that require service is now released and lines and components can be disconnected or removed from those hydraulic circuits.

i07746333

Welding on Machines and Engines with Electronic Controls

SMCS Code: 1000; 7000

Do not weld on any protective structure. If it is necessary to repair a protective structure, contact your Cat dealer.

Proper welding procedures are necessary to avoid damage to the electronic controls and to the bearings. When possible, remove the component that must be welded from the machine or the engine and then weld the component. If you must weld near an electronic control on the machine or the engine, temporarily remove the electronic control to prevent heat related damage. The following steps should be followed to weld on a machine or an engine with electronic controls.

1. Turn off the engine. Place the engine start switch in the OFF position.
2. If equipped, turn the battery disconnect switch to the OFF position. If there is no battery disconnect switch, remove the negative battery cable at the battery.

NOTICE

Do NOT use electrical components (ECM or sensors) or electronic component grounding points for grounding the welder.

3. Clamp the ground cable from the welder to the component that will be welded. Place the clamp as close as possible to the weld. Make sure that the electrical path from the ground cable to the component does not go through any bearing. Use this procedure to reduce the possibility of damage to the following components:

- Bearings of the drive train
- Hydraulic components
- Electrical components
- Other components of the machine

4. Protect any wiring harnesses and components from the debris and the spatter which is created from welding.

5. Use standard welding procedures to weld the materials together.

i04807435

Severe Service Application

SMCS Code: 1000

An engine which operates outside of normal conditions is operating in a severe service application.

An engine that operates in a severe service application may need more frequent maintenance intervals in order to maximize the following conditions:

- Reliability
- Service life

The number of individual applications cause the impossibility of identifying all of the factors which may contribute to severe service operation. Consult your Caterpillar dealer for the unique maintenance that may be necessary for your engine.

An application is a severe service application if any of the following conditions apply:

Severe Environmental Factors

- Frequent operation in dirty air
- Frequent operation at an altitude which is above 1525 m (5000 ft)
- Frequent operation in ambient temperatures which are above 32° C (90° F)
- Frequent operation in ambient temperatures which are below 0° C (32° F)

Severe Operating Conditions

- Frequent operation with inlet air which has a corrosive content
- Operation with inlet air which has a combustible content
- Operation which is outside of the intended application
- Operation with a plugged fuel filter
- Extended operation at low idle (more than 20% of hours)
- Frequent cold starts at temperatures below 0° C (32° F)

- Frequent dry starts (starting after more than 72 hours of shutdown)
- Frequent hot shutdowns (shutting down the engine without the minimum of 2 minutes to 5 minutes of cool down time)
- Operation above the engine rated speed
- Operation below the peak torque speed
- Operating with fuel which does not meet the standards for distillate diesel fuel as stated in Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" "Distillate Diesel Fuel"
- Operating with a blend of distillate fuel which contains more than 20 percent biodiesel

Improper Maintenance Procedures (Maintenance Procedures Which May Contribute to a Severe Service Application)

- Inadequate maintenance of fuel storage tanks from causes such as excessive water, sediment, and microorganism growth.
- Extending maintenance intervals beyond the recommended intervals
- Using fluids which are not recommended in Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations"
- Extending maintenance intervals for changing the engine oil and engine coolant without S·O·S validation
- Extending maintenance intervals for changing air filters, oil filters, and fuel filters
- Failure to use a water separator
- Using filters which are not recommended by Special Publication, PEWJ0074, "2008 Cat Filter and Fluid Application Guide"
- Storing the engine for more than 3 months but less than 1 yr (For information about engine storage, refer to Special Publication, SEHS9031, "Storage Procedure for Caterpillar Products")

i08271903

Maintenance Interval Schedule

SMCS Code: 7000

S/N: RAZ1-10000

Ensure that all safety information, warnings, and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance. All adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging are included. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Use mileage, fuel consumption, service hours, or calendar time, WHICH EVER OCCURS FIRST, to determine the maintenance intervals. Products that operate in severe operating conditions may require more frequent maintenance. Refer to the maintenance procedure for any other exceptions that may change the maintenance intervals.

Note: The aftertreatment system can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation. All prescribed maintenance requirements must be followed.

Note: Before each consecutive interval is performed, all maintenance from the previous interval must be performed.

The following guidelines should be followed if the service hours are not met:

Items listed between 10 and 100 service hours should be performed at least every 3 months.

Items listed between 250 and 500 service hours should be performed at least every 6 months.

Items listed between 1000 service hours and 2500 service hours should be performed at least every year.

When Required

" Air Conditioner/Cab Heater Filter (Recirculation) - Inspect/Replace"	433
" Battery Electrolyte Level - Check"	433
" Battery or Battery Cable - Inspect/Replace"	434
" Bucket Linkage - Inspect/Adjust"	438
" Bucket Tips - Inspect/Replace"	439
" Cab Air Filter (Fresh Air) - Clean/Replace"	445

" Camera - Clean"	446
" Condenser (Refrigerant) - Clean"	447
" DEF Filler Screen - Clean"	453
" Diesel Exhaust Fluid - Drain"	455
" Diesel Exhaust Fluid - Fill"	455
" Engine Air Filter Primary and/or Secondary Element - Replace"	460
" Ether Starting Aid Cylinder - Replace"	470
" Film (Product Identification) - Clean"	472
" Fuel System - Prime"	475
" Fuel Tank Strainer - Clean"	481
" Fuses - Replace"	483
" High Intensity Discharge Lamp (HID) - Replace"	489
" Oil Filter - Inspect"	502
" Radiator, Aftercooler and Oil Cooler Cores - Clean"	502
" Rollover Protective Structure (ROPS) - Inspect"	504
" Shovel Crane - Inspect"	505
" Shovel Crane - Lubricate"	508
" Track Adjustment - Adjust"	513
" Undercarriage - Check"	517
" Window Washer Reservoir - Fill"	517
" Window Wiper - Inspect/Replace"	517
" Windows - Clean"	518

Every 10 Service Hours or Daily for First 50 Hours

" Boom and Stick Linkage - Lubricate"	435
" Boom and Stick Linkage - Lubricate"	436
" Bucket Linkage - Lubricate"	439

Every 10 Service Hours or Daily

" Cooling System Coolant Level - Check"	451
" Engine Oil Level - Check"	464
" Fuel System Water Separator - Drain"	480
" Fuel Tank Water and Sediment - Drain"	481
" Hydraulic System Oil Level - Check"	497

“ Indicators and Gauges - Test”	500
“ Seat Belt - Inspect”	504
“ Track Adjustment - Inspect”	515
“ Travel Alarm - Test”	516

Every 100 Service Hours

“ Bucket Linkage - Lubricate”	439
“ Oil Filter (Hydraulic Hammer) - Replace”	500

Every 250 Service Hours

“ Cooling System Coolant Sample - Obtain”	452
“ Engine Oil Sample - Obtain”	465

Initial 500 Service Hours

“ Engine Oil and Filter - Change”	466
“ Final Drive Oil - Change”	472
“ Swing Drive Oil - Change”	509

Every 500 Service Hours

“ Boom and Stick Linkage - Lubricate”	436
“ Boom and Stick Linkage - Lubricate”	435
“ Final Drive Oil Level - Check”	473
“ Final Drive Oil Sample - Obtain”	474
“ Fuel Lift Pump Strainer - Replace”	474
“ Fuel System Primary Filter (Water Separator) Element - Replace”	476
“ Fuel System Secondary Filter - Replace”	478
“ Hydraulic System Oil Sample - Obtain”	499
“ Swing Bearing - Lubricate”	509
“ Swing Drive Oil Level - Check”	511
“ Swing Drive Oil Sample - Obtain”	512

Every 1000 Service Hours

“ Battery - Clean”	433
“ Battery Hold-Down - Tighten”	434
“ Belt - Inspect/Adjust/Replace”	434
“ Engine Oil and Filter - Change”	466

“ Swing Drive Oil - Change”	509
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Every 2000 Service Hours

“ Final Drive Oil - Change”	472
“ Fuel Cap Filter - Replace”	474
“ Fumes Disposal Filter Element - Replace”	482
“ Swing Gear - Lubricate”	512

Every 3000 Service Hours

“ Cooling System Coolant Sample - Obtain”	452
“ Hydraulic System Oil Filter (Return) - Replace”	495

Every 3 Years

“ Seat Belt - Replace”	505
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Every 5000 Service Hours

“ Diesel Exhaust Fluid Filter - Replace”	457
“ Diesel Exhaust Fluid Injector - Replace”	458
“ Receiver Dryer (Refrigerant) - Replace”	503

Every 6000 Service Hours or 3 Years

“ Cooling System Coolant Extender (ELC) - Add”	450
“ Hydraulic System Oil - Change”	489

Every 10 000 Service Hours

“ DEF Manifold Filters - Replace”	455
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Every 12 000 Service Hours or 6 Years

“ Cooling System Coolant (ELC) - Change”	447
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i08271904

Maintenance Interval Schedule

SMCS Code: 7000

S/N: HEX1-10000

Ensure that all safety information, warnings, and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance. All adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging are included. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Use mileage, fuel consumption, service hours, or calendar time, **WHICH EVER OCCURS FIRST**, to determine the maintenance intervals. Products that operate in severe operating conditions may require more frequent maintenance. Refer to the maintenance procedure for any other exceptions that may change the maintenance intervals.

Note: The aftertreatment system can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation. All prescribed maintenance requirements must be followed.

Note: Before each consecutive interval is performed, all maintenance from the previous interval must be performed.

The following guidelines should be followed if the service hours are not met:

Items listed between 10 and 100 service hours should be performed at least every 3 months.

Items listed between 250 and 500 service hours should be performed at least every 6 months.

Items listed between 1000 service hours and 2500 service hours should be performed at least every year.

When Required

" Air Conditioner/Cab Heater Filter (Recirculation) - Inspect/Replace"	433
" Battery Electrolyte Level - Check"	433
" Battery or Battery Cable - Inspect/Replace"	434
" Bucket Linkage - Inspect/Adjust"	438
" Bucket Tips - Inspect/Replace"	439
" Cab Air Filter (Fresh Air) - Clean/Replace"	445

" Camera - Clean"	446
" Condenser (Refrigerant) - Clean"	447
" DEF Filler Screen - Clean"	453
" Diesel Exhaust Fluid - Drain"	455
" Diesel Exhaust Fluid - Fill"	455
" Engine Air Filter Primary and/or Secondary Element - Replace"	460
" Ether Starting Aid Cylinder - Replace"	470
" Film (Product Identification) - Clean"	472
" Fuel System - Prime"	475
" Fuel Tank Strainer - Clean"	481
" Fuses - Replace"	483
" High Intensity Discharge Lamp (HID) - Replace"	489
" Oil Filter - Inspect"	502
" Radiator, Aftercooler and Oil Cooler Cores - Clean"	502
" Rollover Protective Structure (ROPS) - Inspect"	504
" Shovel Crane - Inspect"	505
" Shovel Crane - Lubricate"	508
" Track Adjustment - Adjust"	513
" Undercarriage - Check"	517
" Window Washer Reservoir - Fill"	517
" Window Wiper - Inspect/Replace"	517
" Windows - Clean"	518

Every 10 Service Hours or Daily for First 50 Hours

" Boom and Stick Linkage - Lubricate"	436
" Boom and Stick Linkage - Lubricate"	435
" Bucket Linkage - Lubricate"	439

Every 10 Service Hours or Daily

" Cooling System Coolant Level - Check"	451
" Engine Oil Level - Check"	464
" Fuel System Water Separator - Drain"	480
" Fuel Tank Water and Sediment - Drain"	481
" Hydraulic System Oil Level - Check"	497

“ Indicators and Gauges - Test“	500
“ Seat Belt - Inspect“	504
“ Track Adjustment - Inspect“	515
“ Travel Alarm - Test“	516

Every 100 Service Hours

“ Bucket Linkage - Lubricate“	439
“ Oil Filter (Hydraulic Hammer) - Replace“	500

Every 250 Service Hours

“ Cooling System Coolant Sample - Obtain“	452
“ Engine Oil Sample - Obtain“	465

Initial 500 Service Hours

“ Engine Oil and Filter - Change“	466
“ Final Drive Oil - Change“	472
“ Swing Drive Oil - Change“	509

Every 500 Service Hours

“ Boom and Stick Linkage - Lubricate“	436
“ Boom and Stick Linkage - Lubricate“	435
“ Final Drive Oil Level - Check“	473
“ Final Drive Oil Sample - Obtain“	474
“ Fuel Lift Pump Strainer - Replace“	474
“ Fuel System Primary Filter (Water Separator) Element - Replace“	476
“ Fuel System Secondary Filter - Replace“	478
“ Hydraulic System Oil Sample - Obtain“	499
“ Swing Bearing - Lubricate“	509
“ Swing Drive Oil Level - Check“	511
“ Swing Drive Oil Sample - Obtain“	512

Every 1000 Service Hours

“ Battery - Clean“	433
“ Battery Hold-Down - Tighten“	434
“ Belt - Inspect/Adjust/Replace“	434
“ Engine Oil and Filter - Change“	466

“ Swing Drive Oil - Change“	509
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Every 1500 Service Hours

“ Diesel Exhaust Fluid Filter - Replace“	456
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Every 2000 Service Hours

“ Final Drive Oil - Change“	472
“ Fuel Cap Filter - Replace“	474
“ Fumes Disposal Filter Element - Replace“	482
“ Swing Gear - Lubricate“	512

Every 3000 Service Hours

“ Cooling System Coolant Sample - Obtain“	452
“ Hydraulic System Oil Filter (Return) - Replace“	495

Every 3 Years

“ Seat Belt - Replace“	505
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Every 5000 Service Hours

“ Receiver Dryer (Refrigerant) - Replace“	503
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Every 6000 Service Hours or 3 Years

“ Cooling System Coolant Extender (ELC) - Add“	450
“ Hydraulic System Oil - Change“	489

Every 10 000 Service Hours

“ DEF Manifold Filters - Replace“	455
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Every 12 000 Service Hours or 6 Years

“ Cooling System Coolant (ELC) - Change“	447
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i08271905

Maintenance Interval Schedule

SMCS Code: 7000

S/N: YCP1–Up

S/N: HEX10001–Up

Ensure that all safety information, warnings, and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance. All adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging are included. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Use mileage, fuel consumption, service hours, or calendar time, **WHICH EVER OCCURS FIRST**, to determine the maintenance intervals. Products that operate in severe operating conditions may require more frequent maintenance. Refer to the maintenance procedure for any other exceptions that may change the maintenance intervals.

Note: The aftertreatment system can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation. All prescribed maintenance requirements must be followed.

Note: Before each consecutive interval is performed, all maintenance from the previous interval must be performed.

The following guidelines should be followed if the service hours are not met:

Items listed between 10 and 100 service hours should be performed at least every 3 months.

Items listed between 250 and 500 service hours should be performed at least every 6 months.

Items listed between 1000 service hours and 2500 service hours should be performed at least every year.

When Required

" Air Conditioner/Cab Heater Filter (Recirculation) - Inspect/Replace"	433
" Battery Electrolyte Level - Check"	433
" Battery or Battery Cable - Inspect/Replace"	434
" Bucket Linkage - Inspect/Adjust"	438
" Bucket Tips - Inspect/Replace"	439

" Cab Air Filter (Fresh Air) - Clean/Replace"	445
" Camera - Clean"	446
" Condenser (Refrigerant) - Clean"	447
" DEF Filler Screen - Clean"	453
" Diesel Exhaust Fluid - Drain"	455
" Diesel Exhaust Fluid - Fill"	455
" Engine Air Filter Primary and/or Secondary Element - Replace"	460
" Ether Starting Aid Cylinder - Replace"	470
" Film (Product Identification) - Clean"	472
" Fuel System - Prime"	475
" Fuel Tank Strainer - Clean"	481
" Fuses - Replace"	483
" High Intensity Discharge Lamp (HID) - Replace"	489
" Oil Filter - Inspect"	502
" Radiator, Aftercooler and Oil Cooler Cores - Clean"	502
" Rollover Protective Structure (ROPS) - Inspect"	504
" Shovel Crane - Inspect"	505
" Shovel Crane - Lubricate"	508
" Track Adjustment - Adjust"	513
" Undercarriage - Check"	517
" Window Washer Reservoir - Fill"	517
" Window Wiper - Inspect/Replace"	517
" Windows - Clean"	518

Every 10 Service Hours or Daily for First 50 Hours

" Boom and Stick Linkage - Lubricate"	436
" Boom and Stick Linkage - Lubricate"	435
" Bucket Linkage - Lubricate"	439

Every 10 Service Hours or Daily

" Cooling System Coolant Level - Check"	451
" Engine Oil Level - Check"	464
" Fuel System Water Separator - Drain"	480
" Fuel Tank Water and Sediment - Drain"	481

“Hydraulic System Oil Level - Check”	497
“Indicators and Gauges - Test”	500
“Seat Belt - Inspect”	504
“Track Adjustment - Inspect”	515
“Travel Alarm - Test”	516

Every 100 Service Hours

“Bucket Linkage - Lubricate”	439
“Oil Filter (Hydraulic Hammer) - Replace”	500

Every 250 Service Hours

“Cooling System Coolant Sample - Obtain”	452
“Engine Oil Sample - Obtain”	465

Initial 500 Service Hours

“Engine Oil and Filter - Change”	466
“Final Drive Oil - Change”	472
“Swing Drive Oil - Change”	509

Every 500 Service Hours

“Boom and Stick Linkage - Lubricate”	436
“Boom and Stick Linkage - Lubricate”	435
“Final Drive Oil Level - Check”	473
“Final Drive Oil Sample - Obtain”	474
“Hydraulic System Oil Sample - Obtain”	499
“Swing Bearing - Lubricate”	509
“Swing Drive Oil Level - Check”	511
“Swing Drive Oil Sample - Obtain”	512

Every 1000 Service Hours

“Battery - Clean”	433
“Battery Hold-Down - Tighten”	434
“Belt - Inspect/Adjust/Replace”	434
“Engine Oil and Filter - Change”	466
“Fuel System Primary Filter (Water Separator) Element - Replace”	477
“Fuel System Secondary Filter - Replace”	479

“Swing Drive Oil - Change”	509
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Every 1500 Service Hours

“Diesel Exhaust Fluid Filter - Replace”	456
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Every 2000 Service Hours

“Final Drive Oil - Change”	472
“Fuel Cap Filter - Replace”	474
“Fumes Disposal Filter Element - Replace”	482
“Swing Gear - Lubricate”	512

Every 3000 Service Hours

“Cooling System Coolant Sample - Obtain”	452
“Hydraulic System Oil Filter (Return) - Replace”	495

Every 3 Years

“Seat Belt - Replace”	505
---------------------------------	-----

Every 5000 Service Hours

“Receiver Dryer (Refrigerant) - Replace”	503
--	-----

Every 6000 Service Hours or 3 Years

“Cooling System Coolant Extender (ELC) - Add”	450
“Hydraulic System Oil - Change”	489

Every 10 000 Service Hours

“DEF Manifold Filters - Replace”	455
--	-----

Every 12 000 Service Hours or 6 Years

“Cooling System Coolant (ELC) - Change”	447
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i08271906

Maintenance Interval Schedule

SMCS Code: 7000

S/N: KFE1–Up

S/N: MYK1–Up

S/N: NDL1–Up

S/N: LTN1–Up

S/N: HDT1–Up

S/N: RAZ10001–Up

Ensure that all safety information, warnings, and instructions are read and understood before any operation or any maintenance procedures are performed.

The user is responsible for the performance of maintenance. All adjustments, the use of proper lubricants, fluids, filters, and the replacement of components due to normal wear and aging are included. Failure to adhere to proper maintenance intervals and procedures may result in diminished performance of the product and/or accelerated wear of components.

Use mileage, fuel consumption, service hours, or calendar time, **WHICH EVER OCCURS FIRST**, to determine the maintenance intervals. Products that operate in severe operating conditions may require more frequent maintenance. Refer to the maintenance procedure for any other exceptions that may change the maintenance intervals.

Note: The aftertreatment system can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation. All prescribed maintenance requirements must be followed.

Note: Before each consecutive interval is performed, all maintenance from the previous interval must be performed.

The following guidelines should be followed if the service hours are not met:

Items listed between 10 and 100 service hours should be performed at least every 3 months.

Items listed between 250 and 500 service hours should be performed at least every 6 months.

Items listed between 1000 service hours and 2500 service hours should be performed at least every year.

When Required

“ Air Conditioner/Cab Heater Filter (Recirculation) - Inspect/Replace” 433

“ Battery Electrolyte Level - Check”	433
“ Battery or Battery Cable - Inspect/Replace”	434
“ Bucket Linkage - Inspect/Adjust”	438
“ Bucket Tips - Inspect/Replace”	439
“ Cab Air Filter (Fresh Air) - Clean/Replace”	445
“ Camera - Clean”	446
“ Condenser (Refrigerant) - Clean”	447
“ DEF Filler Screen - Clean”	453
“ Diesel Exhaust Fluid - Drain”	455
“ Diesel Exhaust Fluid - Fill”	455
“ Engine Air Filter Primary and/or Secondary Element - Replace”	460
“ Ether Starting Aid Cylinder - Replace”	470
“ Film (Product Identification) - Clean”	472
“ Fuel System - Prime”	475
“ Fuel Tank Strainer - Clean”	481
“ Fuses - Replace”	483
“ High Intensity Discharge Lamp (HID) - Replace”	489
“ Oil Filter - Inspect”	502
“ Radiator, Aftercooler and Oil Cooler Cores - Clean”	502
“ Rollover Protective Structure (ROPS) - Inspect”	504
“ Shovel Crane - Inspect”	505
“ Shovel Crane - Lubricate”	508
“ Track Adjustment - Adjust”	513
“ Undercarriage - Check”	517
“ Window Washer Reservoir - Fill”	517
“ Window Wiper - Inspect/Replace”	517
“ Windows - Clean”	518

Every 10 Service Hours or Daily for First 50 Hours

“ Boom and Stick Linkage - Lubricate”	436
“ Boom and Stick Linkage - Lubricate”	435

“ Bucket Linkage - Lubricate” 439

Every 10 Service Hours or Daily

“ Cooling System Coolant Level - Check” 451

“ Engine Oil Level - Check” 464

“ Fuel System Water Separator - Drain” 480

“ Fuel Tank Water and Sediment - Drain” 481

“ Hydraulic System Oil Level - Check” 497

“ Indicators and Gauges - Test” 500

“ Seat Belt - Inspect” 504

“ Track Adjustment - Inspect” 515

“ Travel Alarm - Test” 516

Every 100 Service Hours

“ Bucket Linkage - Lubricate” 439

“ Oil Filter (Hydraulic Hammer) - Replace” 500

Every 250 Service Hours

“ Cooling System Coolant Sample - Obtain” 452

“ Engine Oil Sample - Obtain” 465

Initial 500 Service Hours

“ Engine Oil and Filter - Change” 466

“ Final Drive Oil - Change” 472

“ Swing Drive Oil - Change” 509

Every 500 Service Hours

“ Boom and Stick Linkage - Lubricate” 435

“ Boom and Stick Linkage - Lubricate” 436

“ Final Drive Oil Level - Check” 473

“ Final Drive Oil Sample - Obtain” 474

“ Hydraulic System Oil Sample - Obtain” 499

“ Swing Bearing - Lubricate” 509

“ Swing Drive Oil Level - Check” 511

“ Swing Drive Oil Sample - Obtain” 512

Every 1000 Service Hours

“ Battery - Clean” 433

“ Battery Hold-Down - Tighten” 434

“ Belt - Inspect/Adjust/Replace” 434

“ Engine Oil and Filter - Change” 466

“ Fuel System Primary Filter (Water Separator)
Element - Replace” 477

“ Fuel System Secondary Filter - Replace” 479

“ Swing Drive Oil - Change” 509

Every 2000 Service Hours

“ Final Drive Oil - Change” 472

“ Fuel Cap Filter - Replace” 474

“ Fumes Disposal Filter Element - Replace” 482

“ Swing Gear - Lubricate” 512

Every 3000 Service Hours

“ Cooling System Coolant Sample - Obtain” 452

“ Hydraulic System Oil Filter (Return) -
Replace” 495

Every 3 Years

“ Seat Belt - Replace” 505

Every 5000 Service Hours

“ Diesel Exhaust Fluid Filter - Replace” 457

“ Diesel Exhaust Fluid Injector - Replace” 458

“ Receiver Dryer (Refrigerant) - Replace” 503

Every 6000 Service Hours or 3 Years

“ Cooling System Coolant Extender (ELC) -
Add” 450

“ Hydraulic System Oil - Change” 489

Every 10 000 Service Hours

“ DEF Manifold Filters - Replace” 455

Every 12 000 Service Hours or 6 Years

“ Cooling System Coolant (ELC) - Change” 447

i06954215

Air Conditioner/Cab Heater Filter (Recirculation) - Inspect/Replace

SMCS Code: 1054-040-A/C; 1054-510-A/C

NOTICE

An air recirculation filter element plugged with dust will result in decreased performance and service life to the air conditioner or cab heater.

To prevent decreased performance, clean the filter element, as required.

The air conditioner filter is on the lower left side of the cab behind the seat.

1. Slide the operator seat forward.

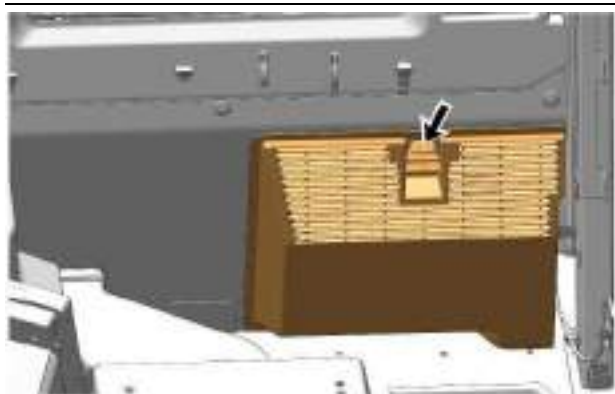


Illustration 693

g06181599

2. Release the cover latch.

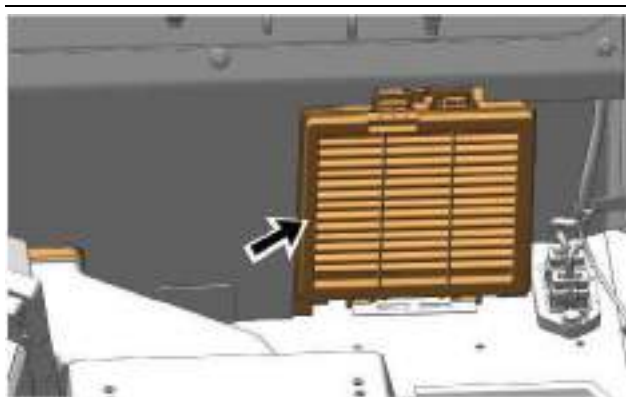


Illustration 694

g06181603

3. Slide the filter element upward.
4. Tap the air filter to remove the dirt. Do not use compressed air to clean the filter.

5. After you clean the filter element, inspect the filter element. If the filter element is damaged or badly contaminated, use a new filter element. Make sure that the filter element is dry.

6. Install the filter element.

7. Install the cover.

NOTICE

Failure to reinstall the filter element for the air conditioning system will contaminate and damage the system components.

i00934864

Battery - Clean

SMCS Code: 1401-070

Clean the battery surface with a clean cloth. Keep the terminals clean and keep the terminals coated with petroleum jelly. Install the post cover after you coat the terminal post with petroleum jelly.

i06019968

Battery Electrolyte Level - Check

SMCS Code: 1401-535-FLV; 1401; 1401-535

WARNING

All lead-acid batteries contain sulfuric acid which can burn the skin and clothing. Always wear a face shield and protective clothing when working on or near batteries.

Note: If the machine is operated in extreme temperatures, check the electrolyte level Every 500 Service Hours or 3 months.

When the engine is not run for long periods of time or when the engine is run for short periods, the batteries may not fully recharge. Ensure a full charge in order to help prevent the battery from freezing.

1. Clean the battery surface with a clean cloth. Clean the terminals and the cable clamps. Coat the clamps and the terminals with silicone lubricant or petroleum jelly. Install the post cover.
2. Inspect the electrolyte level in each battery cell. Maintain the electrolyte level to the bottom of the filler openings. Use distilled water. If distilled water is not available, use clean drinking water.

i00934872

Battery Hold-Down - Tighten

SMCS Code: 7257

Tighten the hold-downs for the battery in order to prevent the batteries from moving during machine operation.

i07592714

Battery or Battery Cable - Inspect/Replace

SMCS Code: 1401; 1401-561; 1401-040; 1401-510; 1402-040; 1402-510

WARNING

Personal injury can result from battery fumes or explosion.

Batteries give off flammable fumes that can explode. Electrolyte is an acid and can cause personal injury if it contacts the skin or eyes.

Prevent sparks near the batteries. Sparks could cause vapors to explode. Do not allow jumper cable ends to contact each other or the engine. Improper jumper cable connections can cause an explosion.

Always wear protective glasses when working with batteries.

1. Turn all the switches to the OFF position. Turn the engine start switch key to the OFF position.
2. Turn the battery disconnect switch to the OFF position. Remove the key.
3. Remove the battery hold-down.

Note: The machine may contain more than one set of batteries.

4. Disconnect the negative battery cable at the battery.
5. Disconnect the positive battery cable at the battery.
6. Disconnect the battery cable at the battery disconnect switch.
7. Inspect the battery terminals for corrosion. Clean the battery terminals and the surfaces of the batteries with a clean cloth.
8. Inspect the battery cables for wear or damage.

9. Make any necessary repairs. If necessary, replace the battery cables and/or the battery.

10. Connect the positive battery cable at the battery.

11. Connect the negative battery cable at the battery.

12. Coat the battery terminals with petroleum jelly to prevent corrosion and install the terminal covers.

13. Reinstall the battery hold-down. Tighten the hold-downs for the battery to prevent the batteries from moving during machine operation.

14. Connect the battery cable at the battery disconnect switch.

15. Install the key and turn the battery disconnect switch ON.

Recycle the Battery

Always recycle a battery. Never discard a battery.

Always return used batteries to one of the following locations:

- A battery supplier
- An authorized battery collection facility
- Recycling facility

i07041934

Belt - Inspect/Adjust/Replace

SMCS Code: 1357-040; 1357-510; 1357-025; 1397-025; 1397-040; 1397-510

Note: This engine is equipped with a belt tightener that automatically adjusts the belt to the correct tension.

1. Unlatch the engine hood and raise the engine hood.

i07531952

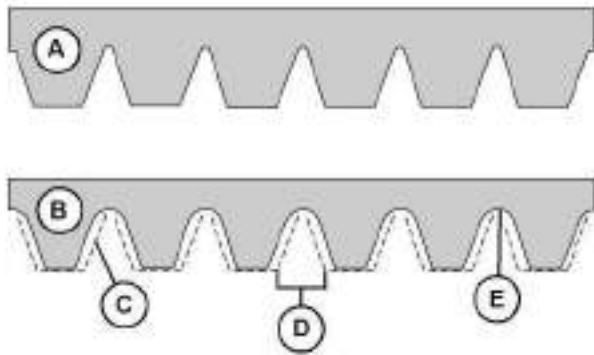


Illustration 695

g06206193

(A) New belt
(B) Worn belt

2. Inspect the condition of the serpentine belt. Over time the belt ribs will lose material (C). The space between the ribs will increase (D). The loss of material will cause the pulley sheave to contact the belt valley. This will lead to belt slippage and accelerated wear (E). Replace the belt if the belt is worn or frayed.
3. If the belt requires replacement, perform Step 3a through Step 3f.
 - a. Remove the upper fan guard.



Illustration 696

g06206669

- b. Rotate the belt tensioner clockwise to remove the belt.
- c. Remove the belt.
- d. Install a new belt.
- e. Rotate the belt tensioner clockwise to install the belt.
- f. Install the upper fan guard.
- g. Lower the engine hood and latch the engine hood.

Boom and Stick Linkage - Lubricate (VA Boom (If Equipped))

SMCS Code: 6501-086; 6502-086

Note: Caterpillar recommends the use of 5% molybdenum grease for lubricating the boom, and stick linkage. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on molybdenum grease.

Apply lubricant through all fittings after operation under water.

Wipe all fittings before you apply lubricant.

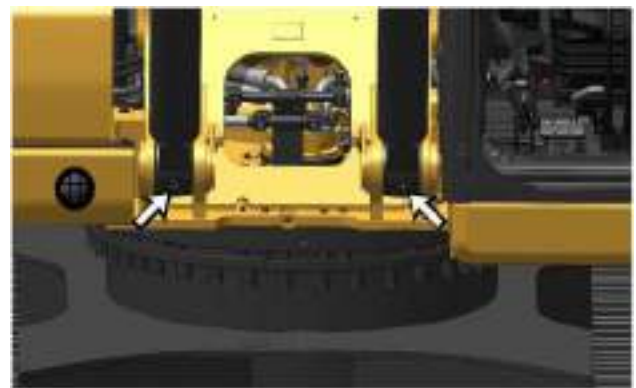


Illustration 697

g06183509

1. Apply lubricant through the fitting at the base of each boom cylinder.

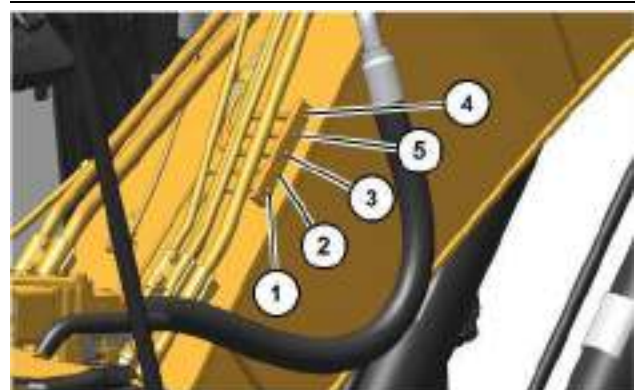


Illustration 698

g06183534

Note: Your machine may have the fittings mounted in a vertical position as shown in the illustration above, or your machine may have the fittings mounted in a horizontal position.

2. The fittings are at the base of the boom. The fittings can be serviced from the platform on the fuel tank. To lubricate the lower boom bearings, apply lubricant through fittings (1) and (2).
3. Apply lubricant through fitting (3) for the head end of the VA boom cylinder.
4. Apply lubricant through fittings (4) and (5) for the boom cylinder rod.

Note: To ensure proper lubrication of the lower boom bearings and of the boom cylinder rod end bearings, lubricant should be applied through fittings (1),(2),(4), and (5). Apply lubricant first when the boom is raised and any attachment is suspended. Then apply lubricant when the boom is lowered and the attachment is rested on the ground with a slight downward pressure.

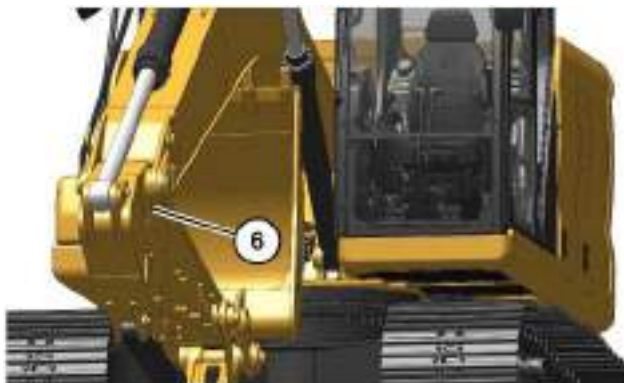


Illustration 699

g06183854

Typical example

5. Apply lubricant through fitting (6). Fitting (6) is at the connection point of the boom and of the stick.

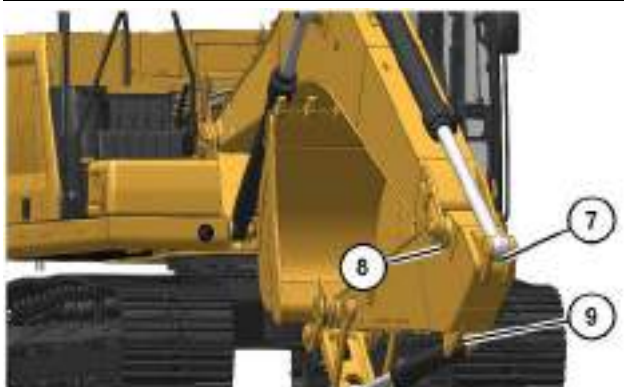


Illustration 700

g06183864

Typical example

6. Apply lubricant through fitting (7) on the stick cylinder rod. Apply lubricant through fitting (8) at the connection point of the boom and of the stick. Apply lubricant through fitting (9) at the bucket cylinder head end.

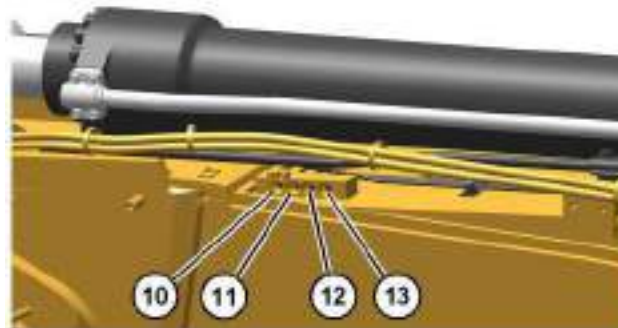


Illustration 701

g06192051

End of fore boom

7. Apply lubricant through fitting (10) and fitting (13) to lubricate the connection point between the stub boom and the fore boom.
8. Apply lubricant through fitting (11) to lubricate the stick cylinder head end.
9. Apply lubricant through fitting (12) to lubricate the rod end of the VA boom cylinder.

i07531958

Boom and Stick Linkage - Lubricate

SMCS Code: 6501-086; 6502-086

Note: Caterpillar recommends the use of 5% molybdenum grease for lubricating the boom, and stick linkage. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on molybdenum grease.

Apply lubricant through all fittings after operation under water.

Wipe all fittings before you apply lubricant.

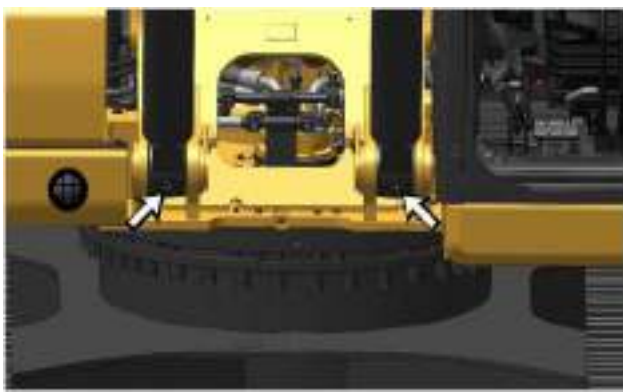


Illustration 702

g06183509

1. Apply lubricant through the fitting at the base of each boom cylinder.

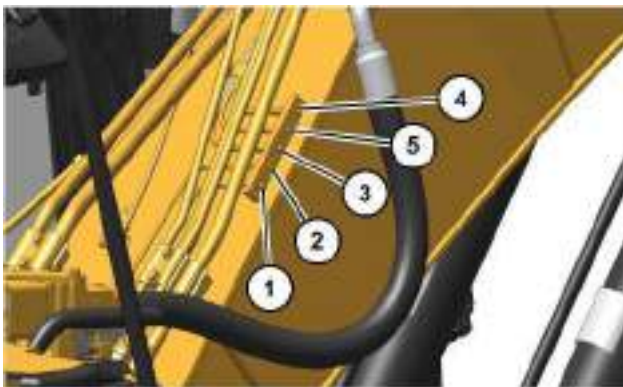


Illustration 703

g06183534

Note: Your machine may have the fittings mounted in a vertical position as shown in the illustration above, or your machine may have the fittings mounted in a horizontal position.

2. The fittings are at the base of the boom. The fittings can be serviced from the platform on the fuel tank. To lubricate the lower boom bearings, apply lubricant through fittings (1) and (2).
3. Apply lubricant through fittings (3) and (4) for the boom cylinder rod.
4. Apply lubricant through fitting (5) for the stick cylinder head.

Note: To ensure proper lubrication of the lower boom bearings and of the boom cylinder rod end bearings, lubricant should be applied through fittings (1), (2), (3), and (4). Apply lubricant first when the boom is raised and any attachment is suspended. Then apply lubricant when the boom is lowered and the attachment is rested on the ground with a slight downward pressure.

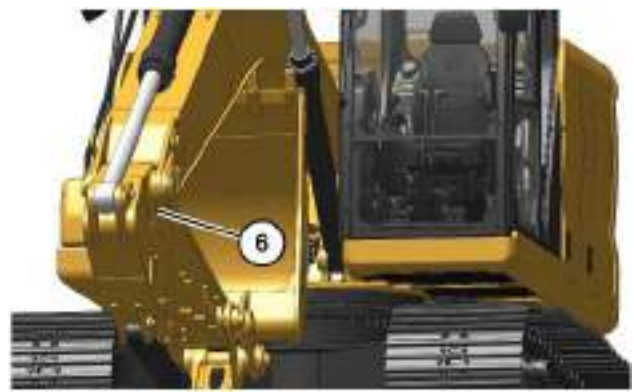


Illustration 704

g06183854

5. Apply lubricant through fitting (6). Fitting (6) is at the connection point of the boom and of the stick.

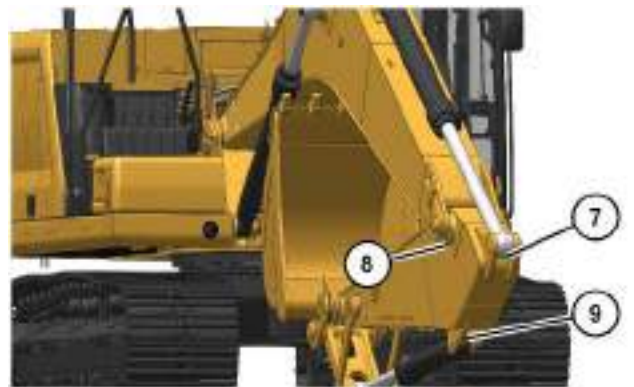


Illustration 705

g06183864

6. Apply lubricant through fitting (7) on the stick cylinder rod. Apply lubricant through fitting (8) at the connection point of the boom and of the stick. Apply lubricant through fitting (9) at the bucket cylinder head end.

i06980273

Bucket Linkage - Inspect/Adjust

SMCS Code: 6513-025; 6513-040

WARNING

Unexpected machine movement can cause injury or death.

To avoid possible machine movement, move the hydraulic lockout control to the LOCKED position and attach a Special Instruction, SEHS7332, "Do Not Operate" or similar warning tag to the hydraulic lockout control.

NOTICE

Improperly adjusted bucket clearance could cause galling on the contact surfaces of the bucket and stick, resulting in excessive noise and/or damaged O-ring seals.



Illustration 706

g06185692

Area for linkage adjustment

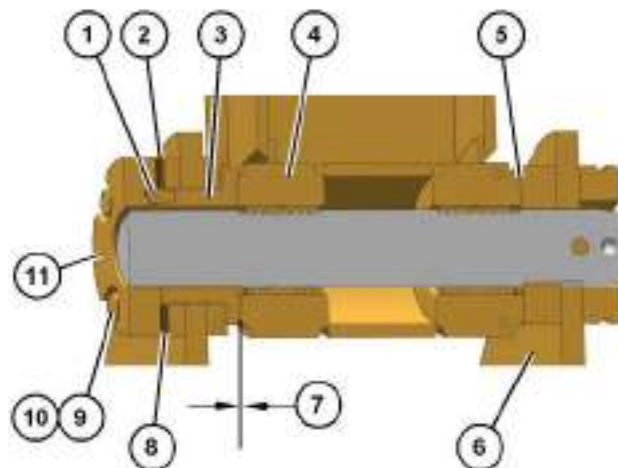


Illustration 707

g06185866

- (1) Pin
- (2) Shims
- (3) Flange
- (4) Stick boss
- (5) No gap
- (6) Bucket boss
- (7) Bucket clearance
- (8) Location
- (9) Bolts
- (10) Washers
- (11) Plate

The clearance of the bucket control linkage on this machine can be adjusted by shimming. If the gap between the bucket and the stick becomes excessive, adjust bucket clearance (7) to 0.5 to 1 mm (0.02 to 0.04 inch).

Two shims of different thickness are used at location (8). The thicknesses of the shims are 0.5 mm (0.02 inch) and 1.0 mm (0.04 inch).

1. Position the machine on a level surface and lower the bucket to the ground.
2. Slowly operate the swing control lever until stick boss (4) and the bucket boss (6) are in full face contact at no gap (5). This will help to determine the total clearance of the connection point of the stick and of the bucket.
3. Move the hydraulic lockout control to the LOCKED position. Stop the engine.
4. Measure bucket clearance (7), which is the existing total clearance.
5. Determine the number of shims that need to be removed from shims (2) by using the following calculation:

Subtract 0.5 mm (0.02 inch) or 1.0 mm (0.04 inch) from bucket clearance (7).

6. Remove the appropriate number of shims at location (8) to meet the above thickness. Make sure that you use a minimum of three 0.5 mm (0.02 inch) shims. To remove the shims, remove bolts (9), washers (10), and plate (11).
7. After the correct number of shims has been removed and pin (1) is aligned with the pin hole, install plate (11), bolts (9), and washers (10). Tighten the bolts to 240 ± 40 N·m (175 ± 30 lb ft).
8. After installation, make sure that bucket clearance (7) is still correct.

i06970647

Bucket Linkage - Lubricate

SMCS Code: 6513-086

Note: Caterpillar recommends the use of 5% molybdenum grease for lubricating the bucket linkage. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on grease.

Wipe all fittings before you apply lubricant.

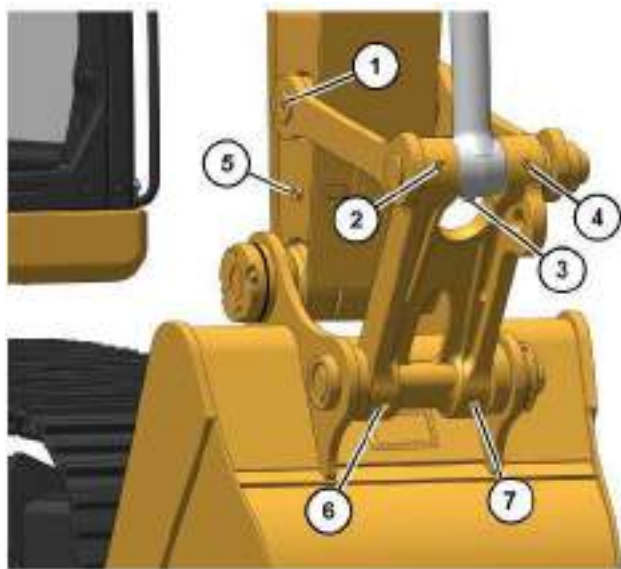


Illustration 708

g06183336

Note: Completely fill all cavities of the bucket control linkage with grease when you initially install a bucket.

1. Apply lubricant through fittings for the linkages (1), (2), (3), and (4).
2. Apply lubricant through fittings for the bucket (5), (6), and (7).

Note: Service the above fittings after you operate the bucket under water.

i08059817

Bucket Tips - Inspect/Replace

SMCS Code: 6805-510; 6805-040



Personal injury or death can result from bucket falling.

Block the bucket before changing bucket tips or side cutters.

K Series GET Drive-through System Bucket Tips

Note: To maximize the life of the bucket tip and the penetration of the bucket tip, the bucket tip can be rotated.



Illustration 709

g01055179

Acceptable wear



Illustration 710

g01055196

Replace this bucket tip.

Check the bucket tips for wear. If the bucket tip has a hole, replace the bucket tip.

Removal Procedure

WARNING

Retainer pin, when struck with force, can fly out and cause injury to nearby people.

Make sure the area is clear of people when driving retainer pins.

To avoid injury to your eyes, wear protective glasses when striking a retainer pin.



Illustration 711

g06528662

- (1) Bucket tip
- (2) Retainer
- (3) Adapter

Note: Retainers are often damaged during the removal process. Caterpillar recommends the installation of a new retainer when bucket tips are rotated or replaced.



Illustration 712

g01054386

Internal view

1. Use a hammer and a punch to drive out the retainer. The retainer can be removed from the top of the bucket tip or from the bottom of the bucket tip.
2. Remove the bucket tip from the adapter with a slight counterclockwise rotation.

Installation Procedure

1. Clean the adapter, if necessary.
2. Install the new bucket tip or the rotated bucket tip onto the adapter with a slight clockwise rotation.

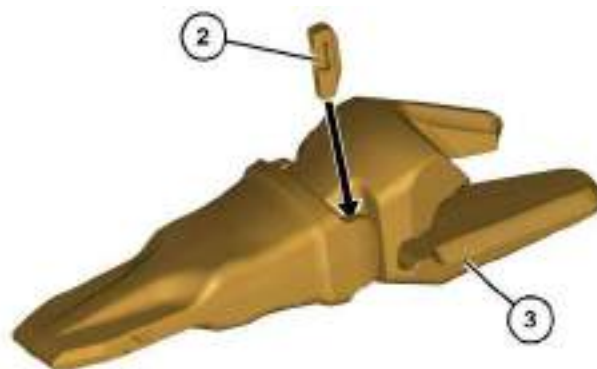


Illustration 713

g06528668

Proper location for installing the retainer

3. The retainer can be installed from the top of the bucket tip or from the bottom of the bucket tip. Use a hammer and a 1 inch X 1 inch X 8 inch steel bar stock to drive retainer (2) into adapter (3).

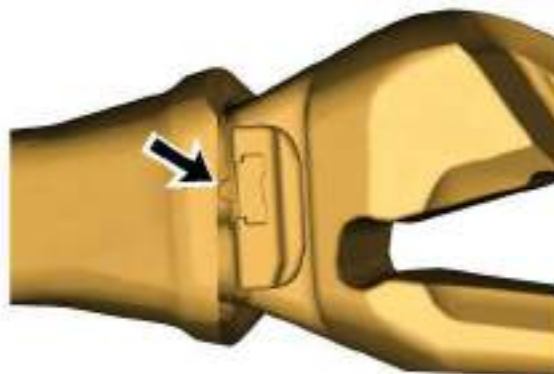


Illustration 714

g06528672

Internal View

The latch of the retainer is properly seated in the recess of the bucket tip.



Illustration 715

g06528674

A properly installed retainer does not extend beyond the ear of the bucket tip.

4. The retainer is properly seated if the retainer can be moved slightly by the technicians hand. If the retainer cannot be moved, adjust the retainer, as needed. The ends of the retainer should not extend beyond the ear of the bucket tip.

J Series GET Bucket Tips

WARNING

Block the bucket before changing the bucket teeth.

To prevent possible injury to the eyes, wear a protective face shield when striking the pin.

The pin, when struck, can fly out and cause injury to nearby personnel.

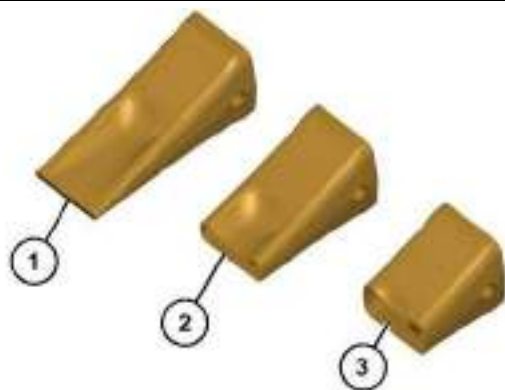


Illustration 716

g06528680

- (1) Usable tip
- (2) Replaceable bucket tip
- (3) Overworn tip

Check the bucket tips for wear. If the bucket tip has a hole, replace the bucket tip.

1. Remove the pin from the bucket tip. The pin can be removed by one of the following methods.

- Use a hammer and a punch from the retainer side of the bucket to drive out the pin.
- Use a Pin-Master. Follow Step 1a through Step 1c for the procedure.

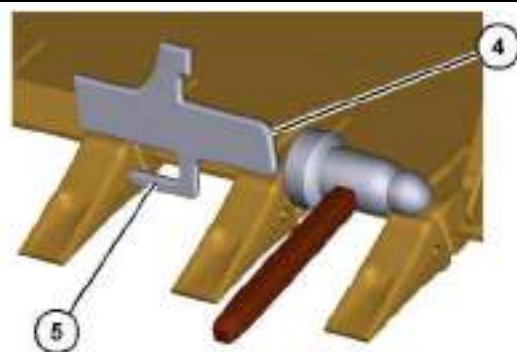


Illustration 717

g06214793

- (4) Back of Pin-Master
- (5) Extractor

- a. Place the Pin-Master on the bucket tip.
- b. Align extractor (5) with the pin.
- c. Strike the Pin-Master at the back of the tool (4) and remove the pin.

Note: Discard the old pin and the retainer assembly. When you change tips, use a new pin and a new retainer assembly. Refer to the appropriate parts manual for your machine.

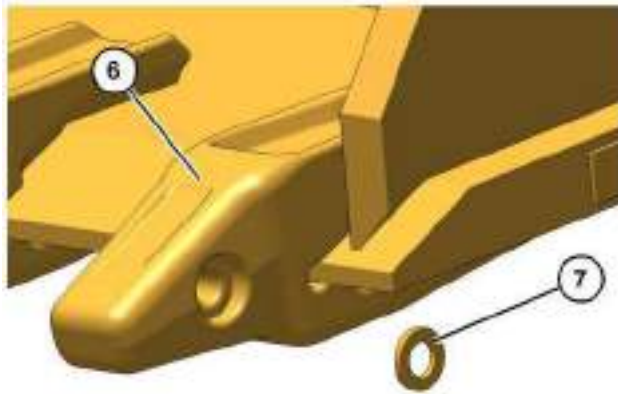


Illustration 718

g06214921

(6) Retainer assembly
(7) Adapter

2. Clean the adapter and the pin.

3. Fit retainer assembly (6) into the counterbore that is in the side of adapter (7). Make sure that the face of the retainer assembly with the marking "OUTSIDE" is visible.



Illustration 719

g06214795

4. Install the new bucket tip onto the adapter.

Note: The bucket tips can be rotated by 180 degrees to allow the tip to wear evenly. You may also move the tips from the outside teeth to the inside teeth. Check the tips often. If wear is present on the tips, rotate the tips. The outside teeth generate the most wear.

5. Drive the pin through the bucket tip. The pin can be installed by using one of the following methods:

- From the same side of the retainer, drive the pin through the bucket tip, the retainer assembly, and the adapter.
- Use a Pin-Master. Follow Step 5a through Step 5e for the procedure.

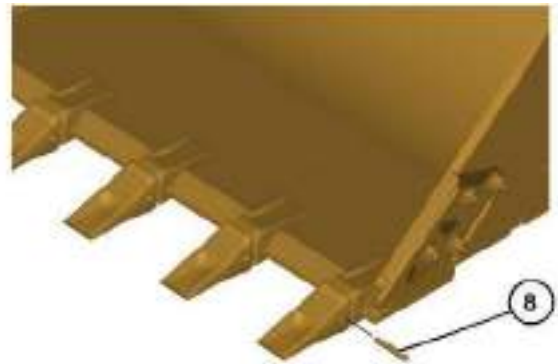


Illustration 720

g06214803

(8) Pin

a. Insert pin (8) through the bucket tip.

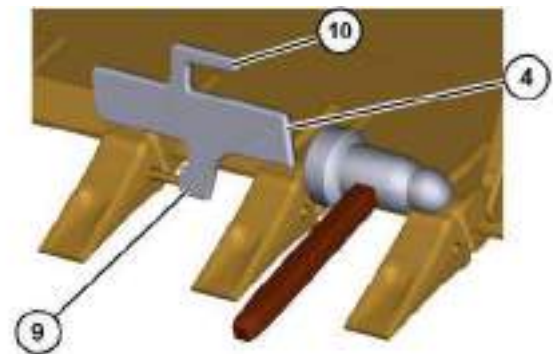


Illustration 721

g06214807

- Place the Pin-Master over the bucket tips so that the pin will fit into the counterbore of the pin holder (9).
- Strike the Pin-Master with a hammer at the back of the tool (4) to insert the pin.
- Slide pin holder (9) away from the pin and rotate the tool slightly to align pin setter (10) with the pin.



Illustration 722

g06214812

Final assembly of pin into bucket tip

- e. Strike the end of the tool until the pin is fully inserted.

Bucket Tips (Cat® Advansys) - If Equipped

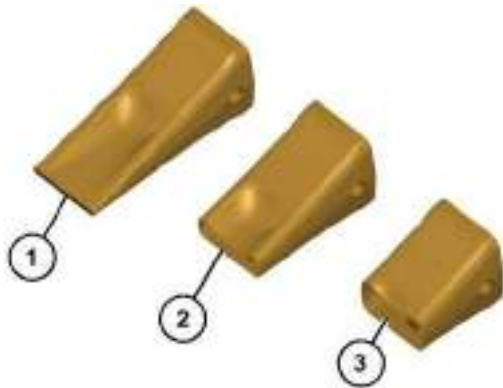


Illustration 723

g06528680

- (1) Usable tip
(2) Replaceable bucket tip
(3) Overworn tip

Check the bucket tips for wear. If the bucket tip has a hole, replace the bucket tip.

Removal

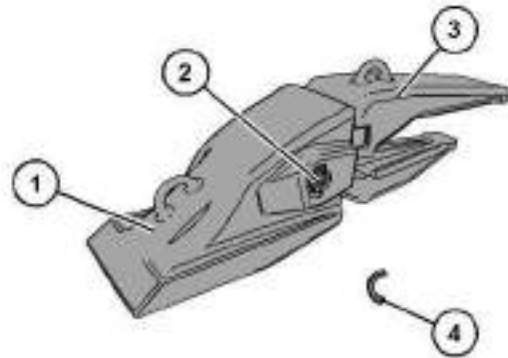


Illustration 724

g06528701

- (1) Bucket Tip
(2) Retainer
(3) Adapter
(4) Compression Sleeve

1. Use a 1/2" ratchet and rotate the retainer (2) 180 degrees to the unlocked position.
2. Remove the bucket tip (1) from adapter (3).
3. Clean adapter (3).

Installation

1. Clean the adapter and the area around the latch, if necessary.
2. Install the new bucket tip (1) onto the adapter (3).

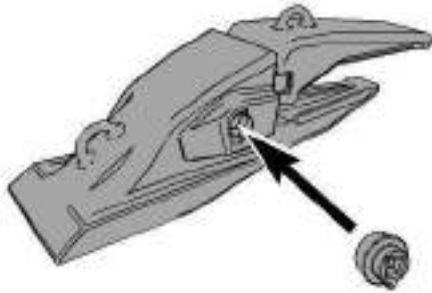


Illustration 725

g06528728

3. Use a 1/2" ratchet to rotate the retainer (2) 180 degrees to the locked position.

Side Cutters

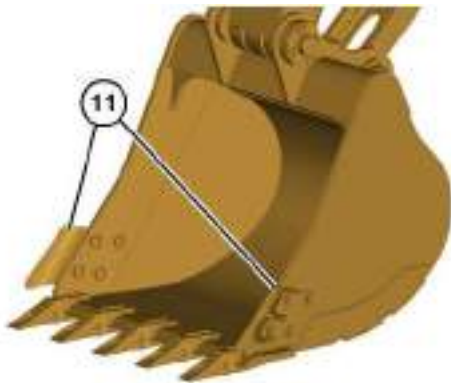


Illustration 726

g06214814

Bucket With Side Cutters

1. Remove the mounting bolts and the side cutters (11).
2. Clean the mounting surface of the side plate on the bucket and of the side cutter. Remove any burrs or protrusions on the mating surfaces.

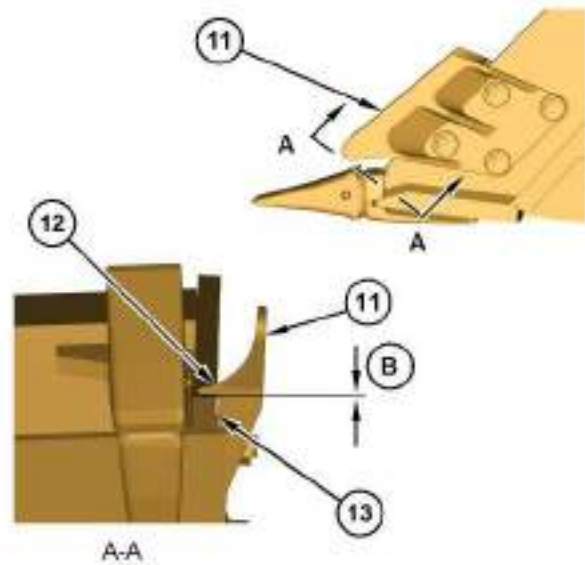


Illustration 727

g06214887

- (12) Shear ledge on a side cutter
(13) Side plate on a bucket
(B) 0.0 mm (0.0 inch)

Note: Some side cutters may be rotated for additional wear.

3. Install the side cutter.

Note: Certain bolts may require thread compound.

4. Hand tighten the bolts.
5. Make sure that there is not a gap between the side plate on the bucket and the shear ledge on the side cutter.
6. Torque the mounting bolts to the correct specification.

Side Protectors (If Equipped)

Inspect the wear of the side protector. When too much wear is present, replace the protector.

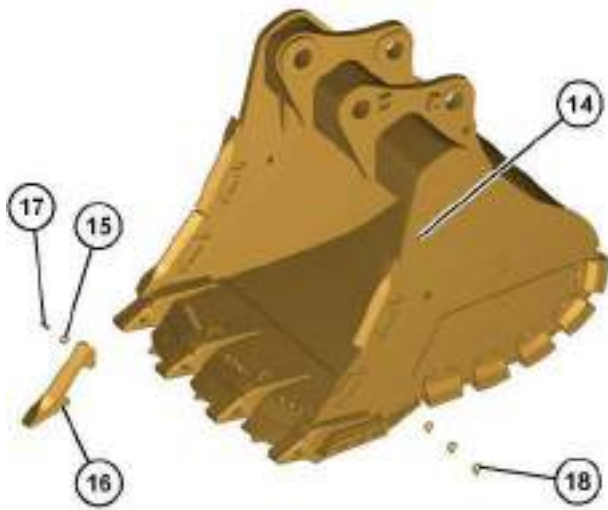


Illustration 728

g06219766

1. Hit pin (17) from the side of the bucket without the retainer to remove side protector (16) from side plate (14).
2. Clean side protector (16), pin (17), retainer (15), and side plate (14) before installation.

Note: Lateral clearance between the side plate and the side protector should not exceed 1 mm (0.04 inch). Shims (18) may be required to decrease the lateral clearance which will decrease movement. Install the shims between the side plate and the side protector on the opposite side of the retainer.

3. Put retainer (15) in side plate (14).
4. Align two pin holes of the new protector and the side plate. Hit the pin from the retainer side of the bucket.

Note: If the pin and/or the retainer are worn, replace the pin and/or the retainer.

i07092323

Cab Air Filter (Fresh Air) - Clean/Replace

SMCS Code: 7342-070; 7342-510

The cab air filter is on the left side of the cab.

1. Use the ignition key to open the access panel.



Illustration 730

g06182115

2. Remove air filter (1).
3. Tap the air filter to remove the dirt. Do not use compressed air to clean the filter.
4. After you clean the air filter, inspect the air filter. If the air filter is damaged or badly contaminated, use a new air filter.
5. Install the air filter.
6. Close and lock the access panel.

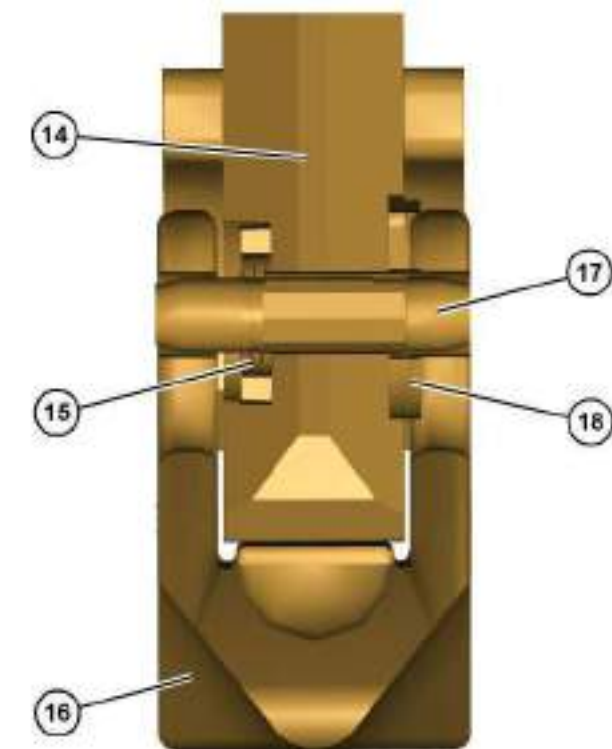


Illustration 729

g06219767

- (14) Side plate
- (15) Retainer
- (16) Side protector
- (17) Pin
- (18) Shim

i07243995

Camera - Clean

SMCS Code: 7348-070

WARNING

Failure to use an appropriate external ladder or an appropriate platform for direct access to the rear view camera could result in slipping and falling which could result in personal injury or death. Be sure to use an appropriate external ladder or an appropriate platform for direct access to the rear view camera.

The machine's counterweight and the engine hood are not approved as a maintenance platforms.

WARNING

Unexpected machine movement can cause injury or death.

In order to avoid possible machine movement, move the hydraulic lockout control to the LOCKED position and attach a Special Instruction, SEHS7332, "Do Not Operate" or similar warning tag to the hydraulic lockout control.

Note: When you access the camera for cleaning, be sure to observe safe procedures for access. Maintain a three-point contact and/or use a body harness.



Illustration 731

g06184579

The rear view camera is on top of the counterweight.

If necessary, use a damp cloth to clean the glass of the camera. The camera is sealed. The camera is not affected by a wash with high-pressure spray.

Note: Alternatively, cameras may be cleaned from ground level by using a wash with a high-pressure spray or a damp rag on a wand.



Illustration 732

g06214504

If equipped, clean the right side view camera.



Illustration 733

g06263435

If equipped, clean the left side view camera.



Illustration 734

g06263449

If equipped, clean the front view camera.

i06969907

Condenser (Refrigerant) - Clean

SMCS Code: 1805-070

NOTICE

If excessively dirty, clean condenser with a brush. To prevent damage or bending of the fins, do not use a stiff brush.

Repair the fins if found defective.



Illustration 735

g06179792

1. Open the access door on the left side of the machine. The condenser is located in front of the radiator.

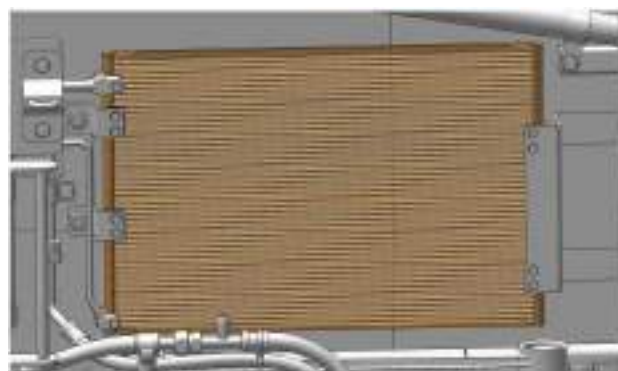


Illustration 736

g06183025

2. Inspect the condenser for debris. Clean the condenser, if necessary.
3. Use clean water to wash off all dust and dirt from the condenser.
4. Close the access door.

i06971000

Cooling System Coolant (ELC) - Change

SMCS Code: 1350-044

WARNING

Engine hood and engine hood parts can be hot while engine is running or immediately after engine shutdown. Hot parts or hot components can cause burns or personal injury. Do not allow these parts to contact your skin, when engine is running or immediately after engine shutdown. Use protective clothing or protective equipment to protect your skin.

WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

NOTICE

Do not change the coolant until you read and understand the cooling system information in Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations".

Failure to do so could result in damage to the cooling system components.

NOTICE

Mixing ELC with other products will reduce the effectiveness of the coolant.

This could result in damage to cooling system components.

If Caterpillar products are not available and commercial products must be used, make sure they have passed the Caterpillar EC-1 specification for pre-mixed or concentrate coolants and Caterpillar Extender.

Note: This machine was filled at the factory with Cat Extended Life Coolant.

If the coolant in the machine is changed to Extended Life Coolant from another type of coolant, see Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations".

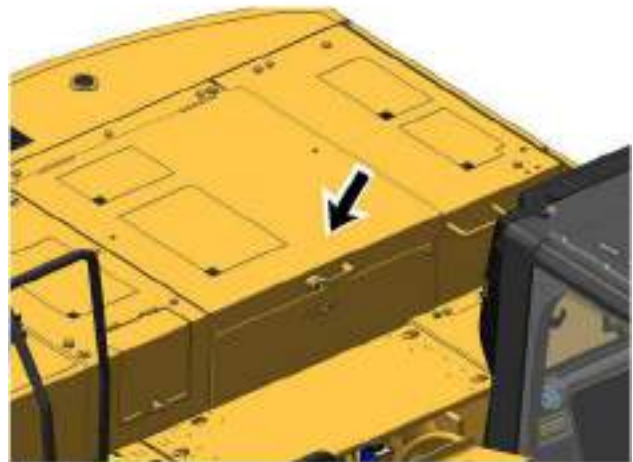


Illustration 737

g06225770

1. Unlatch the engine hood and raise the engine hood.
-



Illustration 738

g06183808

2. Slowly loosen the pressure cap that is on the coolant reservoir to release pressure from the cooling system.
3. Remove the pressure cap.
4. Inspect the gasket on the pressure cap. If the gasket is damaged, replace the pressure cap.



Illustration 739

g06179792

5. Open the rear access door on the left side of the machine.

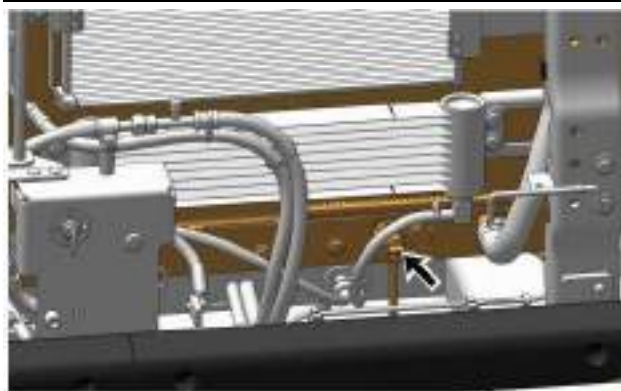


Illustration 740

g06183531

6. Open the drain valve and allow the coolant to drain into a suitable container. The drain valve is on the bottom of the radiator.

Note: Refer to Operation and Maintenance Manual, “General Hazard Information” for information that pertains to containing fluid spillage.

7. Flush the cooling system. Follow Step 7a through Step 7h to flush the cooling system.
- Close the drain valve.
 - Fill the cooling system with clean water.
 - Install the pressure cap.
 - Start the engine and run the engine until the engine reaches operating temperature.
 - Stop the engine and allow the engine to cool.
 - Loosen the pressure cap slowly to relieve any pressure in the cooling system.

- Open the drain valve that is on the bottom of the radiator and allow the coolant to drain into a suitable container.

- Flush the radiator with clean water until the draining water is transparent.

8. Close the drain valve.

9. Add the Extended Life Coolant. Refer to the following topics:

- Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations”
- Operation and Maintenance Manual, “Capacities (Refill)”

10. After the cooling system has been filled, perform the following procedures during initial start-up:

- Start the engine without the filler cap.
- Run the engine at low idle for 10 minutes.
- Then, increase the engine speed to a high idle until the water temperature regulator is open and the coolant level is stabilized.
- Maintain the coolant at the proper level as the water temperature regulator opens and air is purged from the system. Refer to Operation and Maintenance Manual, “Cooling System Coolant Level - Check”.

11. Install the cooling system pressure cap.

12. Stop the engine.

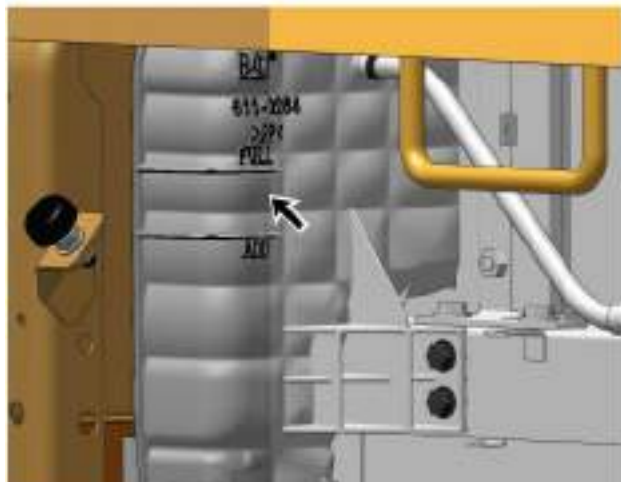


Illustration 741

g06205196

13. Check the coolant reservoir. Maintain the coolant level so that coolant is between the "ADD" and "FULL" lines. If more coolant is needed, see Operation and Maintenance Manual, "Cooling system Coolant Extender (ELC) - Add".
14. If more coolant is necessary, remove the pressure cap and add the appropriate coolant solution.
15. Install the pressure cap.
16. Close and latch the engine hood. Close the left access door.

i06972485

Cooling System Coolant Extender (ELC) - Add

SMCS Code: 1352; 1353; 1395

WARNING

Engine hood and engine hood parts can be hot while engine is running or immediately after engine shutdown. Hot parts or hot components can cause burns or personal injury. Do not allow these parts to contact your skin, when engine is running or immediately after engine shutdown. Use protective clothing or protective equipment to protect your skin.

WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

Use Cat Extended Life Coolant (ELC) when you add coolant to the cooling system. See Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for all cooling system requirements.

Use a Coolant Conditioner Test Kit to check the concentration of the coolant.

NOTICE

Mixing ELC with other products will reduce the effectiveness of the coolant.

This could result in damage to cooling system components.

If Caterpillar products are not available and commercial products must be used, make sure they have passed the Caterpillar EC-1 specification for pre-mixed or concentrate coolants and Caterpillar Extender.

Note: This machine was filled at the factory with Cat Extended Life Coolant.

1. Park the machine on level ground.
2. Stop the engine.

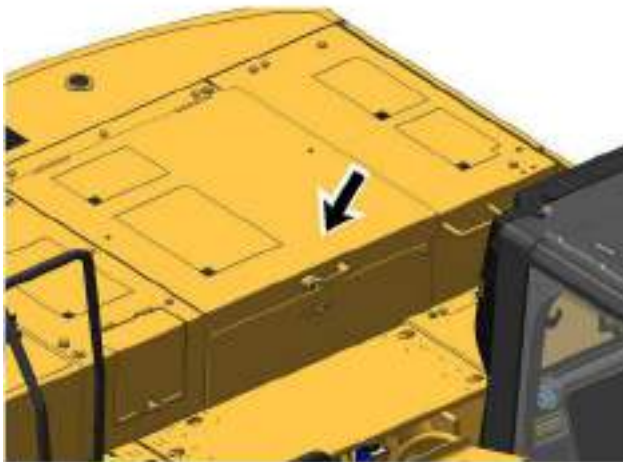


Illustration 742

g06225770

3. Unlatch the engine hood and raise the hood.



Illustration 743

g06183808

4. Make sure that the cooling system has cooled down. Loosen the cooling system pressure cap slowly to relieve system pressure. Remove the pressure cap.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

5. You may need to drain some coolant from the radiator so that Cat Extender can be added to the cooling system.

Note: Always discard drained fluids according to local regulations.

6. Add Cat Extended Life Coolant (ELC) to the cooling system. Refer to the following topics for the proper amount of Cat Extender:
 - Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations"
 - Operation and Maintenance Manual, "Capacities (Refill)"
7. Inspect the o-ring of the cooling system pressure cap. If the o-ring is damaged, replace the pressure cap.
8. Install the cooling system pressure cap.
9. Close and latch the engine hood.

i07038988

Cooling System Coolant Level - Check

SMCS Code: 1350-040; 1350-535-FLV; 1395-535-FLV

WARNING

Engine hood and engine hood parts can be hot while engine is running or immediately after engine shutdown. Hot parts or hot components can cause burns or personal injury. Do not allow these parts to contact your skin, when engine is running or immediately after engine shutdown. Use protective clothing or protective equipment to protect your skin.

⚠ WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

1. Park the machine on level ground.
2. Stop the engine.
3. Open the rear access door on the left side of the machine.

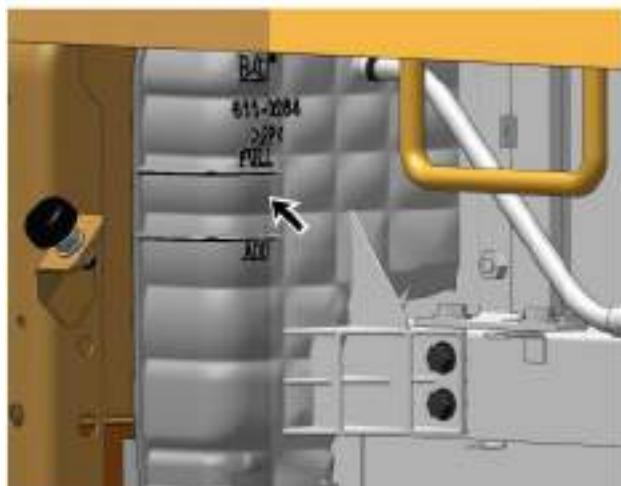


Illustration 744

g06205196

4. Maintain the coolant level so that coolant is between the “ADD” and “FULL” marks. If more coolant is needed, see Operation and Maintenance Manual, “Cooling system Coolant Extender (ELC) - Add”.
5. Close the access door.

i07836536

Cooling System Coolant Sample - Obtain

SMCS Code: 1395-008; 1395-554

The cooling system coolant should be sampled and monitored with regular frequency. The samples should be analyzed per the following guidelines:

- Level 1 analysis: Every 250 hours
- Level 2 analysis: Every Year

Note: It is not necessary to obtain a Coolant Sample (Level 1) if the cooling system is filled with Cat ELC (Extended Life Coolant). Cooling systems that are filled with Cat ELC only require Level 2 analysis.

Note: Obtain a Coolant Sample (Level 1) if the cooling system is filled with any other coolant instead of Cat ELC. This includes the following types of coolants.

- Commercial long life coolants that meet the Caterpillar Engine Coolant Specification -1 (Caterpillar EC-1)
- Cat Diesel Engine Antifreeze/Coolant (DEAC)
- Commercial heavy-duty coolant/antifreeze

Note: A level 1 analysis may indicate the need for a Level 2 analysis.

Note: A Level 2 analysis is required after 500 hours of operation for the following reasons:

- The cooling system is new
- The cooling system has been refilled
- The cooling system has been converted to a new coolant

NOTICE

Always use a designated pump for oil sampling, and use a separate designated pump for coolant sampling. Using the same pump for both types of samples may contaminate the samples that are being drawn. This contaminate may cause a false analysis and an incorrect interpretation that could lead to concerns by both dealers and customers.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.



Illustration 745

g06179792

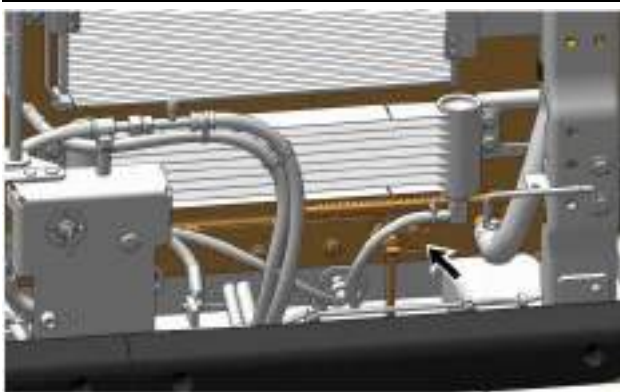


Illustration 746

g06183525

The coolant sampling port is on the radiator.

Obtain the sample of the coolant as close as possible to the recommended sampling interval. To receive the full effect of S·O·S analysis, a consistent trend of data must be established. To establish a pertinent history of data, perform consistent samplings that are evenly spaced. Supplies for collecting samples can be obtained from your Cat dealer.

Use the following guidelines for proper sampling of the coolant:

- Complete the information on the label for the sampling bottle before you begin to take the samples.
- Keep the unused sampling bottles stored in plastic bags.
- Obtain coolant samples directly from the coolant sample port. You should not obtain the samples from any other location.
- Keep the lids on empty sampling bottles until you are ready to collect the sample.
- Place the sample in the mailing tube immediately after obtaining the sample to avoid contamination.
- Never collect samples from expansion bottles.
- Never collect samples from the drain for a system.

Submit the sample for the appropriate analysis.

For additional information about coolant analysis, see Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" or consult your Cat dealer.

i06970259

DEF Filler Screen - Clean

SMCS Code: 108K-070-Z3

NOTICE

Ensure that the engine is stopped before any servicing or repair is performed.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Cat Dealer Service Tool Catalog" or refer to Special Publication, PECJ0003, "Cat Shop Supplies and Tools Catalog" for tools and supplies suitable to collect and contain fluids on Cat products.

Dispose of all fluids according to local regulations and mandates.

The filler neck adapter filter screen in the diesel exhaust fluid tank will need to be cleaned or replaced if contaminated.

Maintenance Section
DEF Filler Screen - Clean



Illustration 747

g06183098

1. Open the Diesel Exhaust Fluid (DEF) compartment.

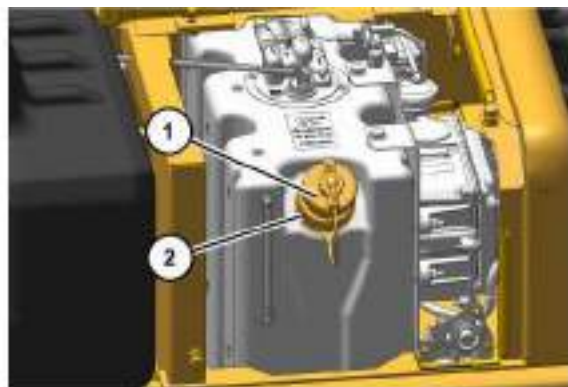


Illustration 748

g06183119

2. Remove fill cap (1).
3. Use a screwdriver or pick to press the tabs on the strainer. Pull the strainer upward from both sides of fill neck adapter (2).

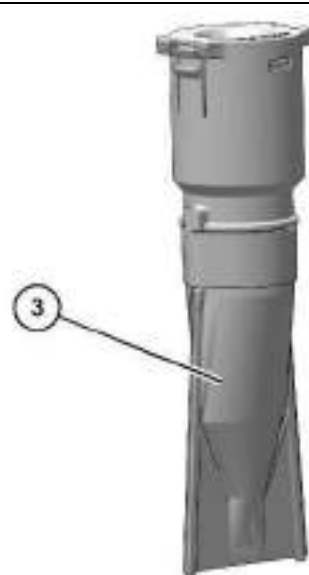


Illustration 749

g06183125

4. Use water or compressed air to clean out the filter screen (3). If there is any debris inside, dry and remove the debris by turning the screen upside down and dumping debris out. If the debris cannot be removed or if the filter screen is damaged, replace the filter screen.

i06970250

DEF Manifold Filters - Replace

SMCS Code: 108K-510-FI

Illustration 750

g06183110

Replace the manifold filter (1). Refer to Disassembly and Assembly, "Manifold (DEF Heater) - Remove and Install" for more information.

i08258668

Diesel Exhaust Fluid - Drain

SMCS Code: 108K-543

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Dispose of all fluids according to local regulations and mandates.

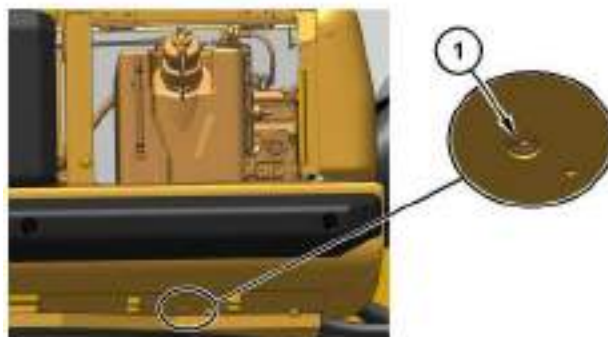


Illustration 751

g06604316

(1) Drain plug

1. Remove drain plug (1).
2. Drain the diesel exhaust fluid into a suitable container.
3. Install and tighten the drain plug to a torque of $20 \pm 3 \text{ N}\cdot\text{m}$ ($177 \pm 26 \text{ lb in}$).

i07391329

Diesel Exhaust Fluid - Fill

SMCS Code: 108K-544

Note: Stop the engine and turn the engine start switch to the OFF position before filling the DEF tank. Failure to stop the engine may cause fault codes.

Refer to Operation and Maintenance Manual, "Selective Catalytic Reduction Warning System" for more information.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Dispose of all fluids according to local regulations and mandates.

Reference: See Operation and Maintenance Manual, "Capacities (Refill)" for the capacity of the DEF tank for your machine.

Maintenance Section
Diesel Exhaust Fluid Filter - Replace



Illustration 752

g06183098

The DEF tank is located inside the storage box on the right side of the machine.

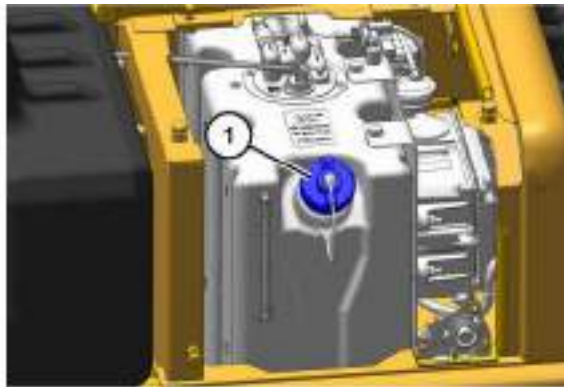


Illustration 753

g06306032



Illustration 754

g06306029

Blue DEF Tank Filler Cap

1. Clean the blue DEF tank filler cap and the surrounding area.
2. Remove the blue DEF tank filler cap.

3. Fill the tank with diesel exhaust fluid (DEF). DEF level gauge is located on the front side of DEF tank.

Note: Do not fill the DEF tank from a contaminated container or funnel.

Note: Do not over fill the tank. DEF can freeze and needs room for expansion.

4. Install the blue DEF tank filler cap.

Refer to Operation and Maintenance Manual, "Lubricant Viscosities" for more information on diesel exhaust fluid (DEF) guidelines.

i07942245

Diesel Exhaust Fluid Filter - Replace

SMCS Code: 108K-510-FI

S/N: YCP1-Up

S/N: HEX1-Up



Illustration 755

g06183098

1. Open the Diesel Exhaust Fluid (DEF) compartment.
2. Ensure that the area around the DEF filter is clean and free from dirt. The DEF filter threaded cap and the filter element are a combined assembly.

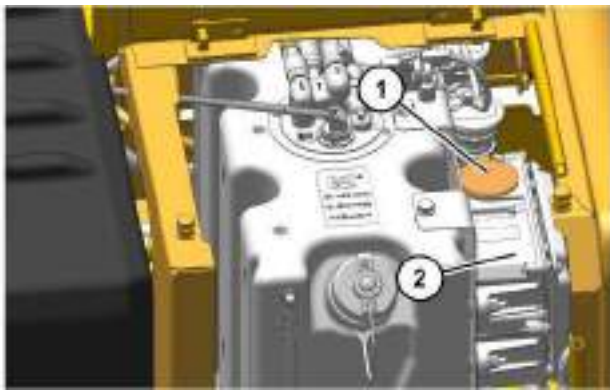


Illustration 756

g06183089

3. Remove the protective cover (1) from DEF pump housing (2).

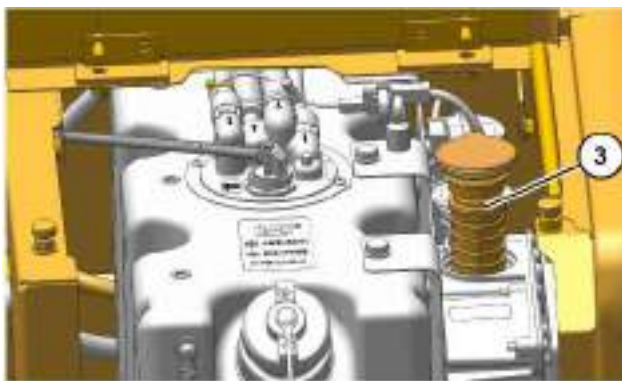


Illustration 757

g06183092

4. Remove the DEF filter assembly (3) and discard.
5. Install a new DEF filter assembly into the DEF pump housing.
6. Tighten the filter assembly to 10 N·m (88.5 lb in). Install the protective cover.
7. Turning on the power will automatically prime the DEF system.

i07942258

Diesel Exhaust Fluid Filter - Replace

SMCS Code: 108K-510-FI

S/N: KFE1-Up

S/N: MYK1-Up

S/N: NDL1-Up

S/N: LTN1-Up

S/N: HDT1-Up

S/N: RAZ1-Up



Illustration 758

g06183098

1. Open the Diesel Exhaust Fluid (DEF) compartment.



Illustration 759

g06210724

For clarity, the DEF assembly has been removed from the compartment.

2. Ensure that the area around the DEF filter is clean and free from dirt.

i07942263

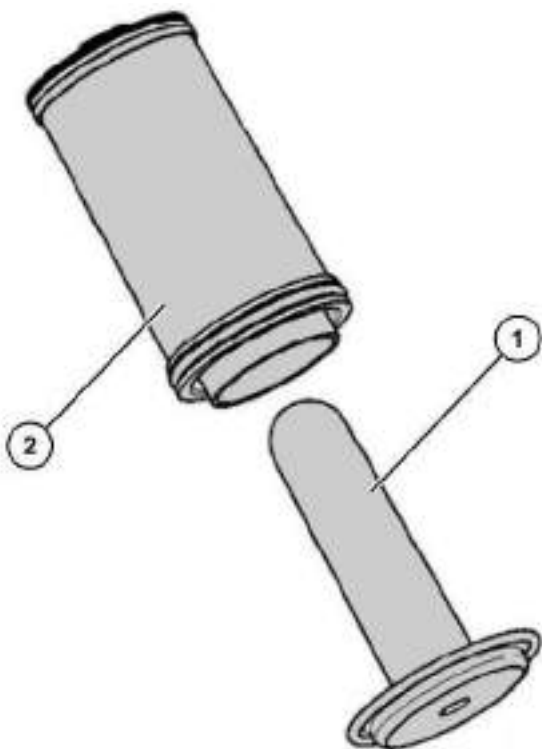


Illustration 760

g06216533

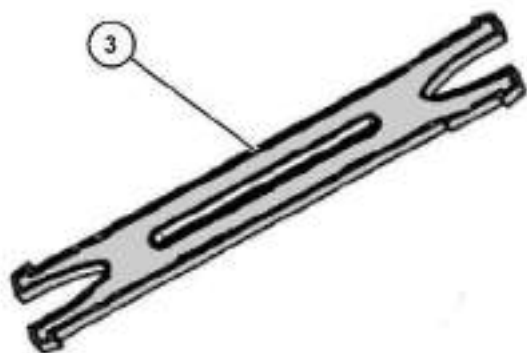


Illustration 761

g06216531

Diesel Exhaust Fluid Injector - Replace

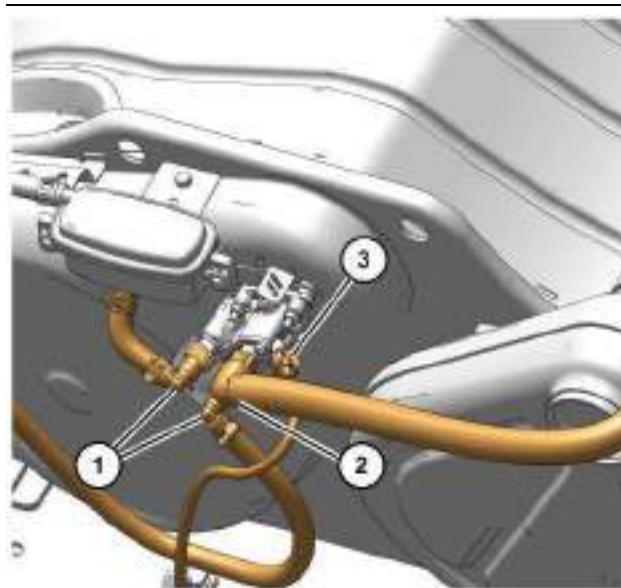
SMCS Code: 108I-510**S/N:** KFE1-Up**S/N:** MYK1-Up**S/N:** NDL1-Up**S/N:** LTN1-Up**S/N:** HDT1-Up**S/N:** RAZ1-Up

Illustration 762

g06226112

Typical example

- (1) Coolant lines
- (2) DEF line
- (3) Electrical connector

3. Unscrew the cap, remove expansion device (1) from DEF filter (2). Then, remove the DEF filter using special tool (3) that comes with the new filter. Discard the old filter and expansion device.
4. Install a new DEF filter assembly into the DEF pump housing.
5. Tighten the filter cap to $20 \pm 5 \text{ N}\cdot\text{m}$ ($15 \pm 4 \text{ lb ft}$).
6. Turning on the power will automatically prime the DEF system.

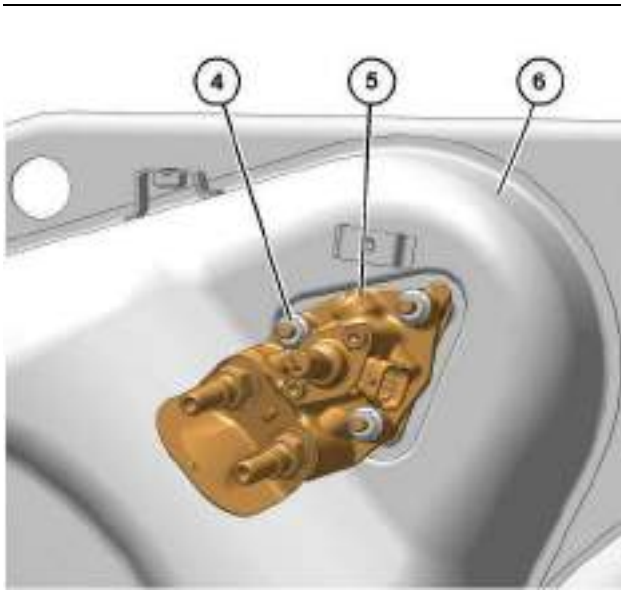


Illustration 763

g06226151

Typical example

- (4) Nuts
- (5) DEF injector
- (6) Clean Emission Module

1. Drain the coolant to remove coolant lines (1). Remove cooling lines (1) and remove Diesel Exhaust Fluid (DEF) line (2).
2. Remove electrical connector (3) from DEF injector (5) and install protection caps to DEF injector (5). Remove nuts (4) and washers (9) from DEF injector (5) and remove DEF injector from Clean Emission Module (CEM) (6).

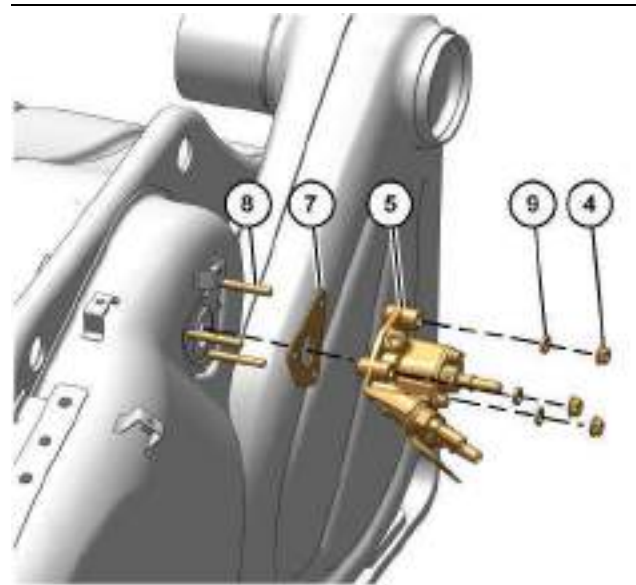


Illustration 764

g06226155

Typical example

- (4) Nuts
- (5) DEF injector
- (7) Gasket
- (8) Studs
- (9) Washers

3. Remove gasket (7) and remove studs (8). Discard the gasket and the studs.
4. Ensure that the sealing face of the CEM is clean and free from dirt.
5. Apply bostik pure nickel ant-seize compound to ends of the new studs (8). Install the coated ends of the new studs into the CEM, and tighten to a torque of 5 N·m (44 lb in).
6. Install new gasket (7), ensure that the metal side of the gasket is to the CEM. Install new DEF injector (5) to CEM (6).
7. Install washers (9) and apply bostik pure nickel ant-seize compound to the threads of studs (8).
8. Install nuts (4) and tighten the nuts to a torque of 5 N·m (44 lb in). Retighten the nuts to 5 N·m (44 lb in). Then, turn the nuts an additional 90 degrees.
9. Install electrical connector (3), DEF line (2), and coolant lines (1).
10. Fill cooling system to the correct level. Ensure that the correct specification of coolant is used.

If available, using the electronic service tool (ET) perform DEF Dosing System Verification test.

i08165097

Engine Air Filter Primary and/or Secondary Element - Replace

SMCS Code: 1054-510-PY; 1054-510-SE

Primary Air Filter Element - Replace

NOTICE

Service the air cleaner only with the engine stopped. Engine damage could result.

NOTICE

Service the engine air filter elements only when a message or a warning is displayed on the monitor display. Do not open the filter compartment unless service is indicated. Opening the filter compartment when not necessary to do so increases the chance of dirt contamination in engine air intake system components.

NOTICE

Short air filter life can result if the pre-cleaner system malfunctions. If air filter life is drastically reduced from typical for the operating conditions, consult your Cat dealer.

NOTICE

Do not use the air filter elements longer than 1 year.

The engine air cleaner assembly is located behind the front access door on the left side of the machine.

1. Park the machine on a level surface. Stop the engine.



Illustration 765

g06181546

2. Open the front access door on the left side of the machine.

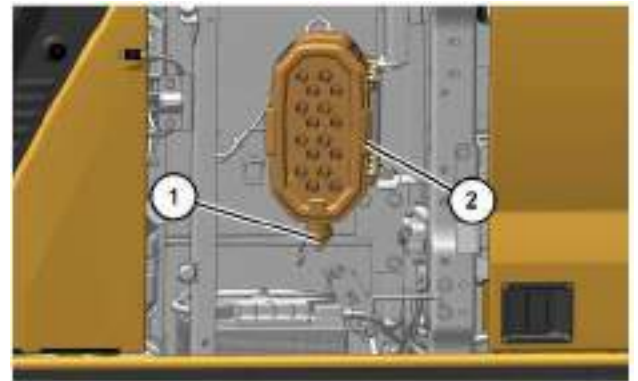


Illustration 766

g06183302

3. Squeeze outlet tube (1) to purge the dirt from the outlet tube.

Note: Purge the dirt from the outlet tube every 10 service hours or daily in a dirty environment.

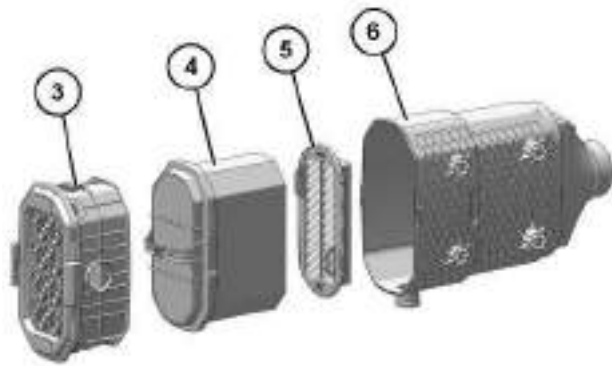


Illustration 767

g06222467

4. Release latches (2) that secure pre-cleaner (3) to engine air filter housing (6).
5. Remove pre-cleaner (3).
6. Clean inside the air filter housing where the pre-cleaner was removed.

NOTICE

Caterpillar does not recommend cleaning the primary air filter element. Caterpillar only recommends to replace the primary air filter element. Caterpillar does not cover costs for damage to engine components caused by cleaning the primary air filter element.

Observe the following guidelines if you attempt to clean the primary filter element:

Do not tap or strike the filter element in order to remove dust.

Do not wash the filter element.

Use low pressure compressed air in order to remove the dust from the filter element. Air pressure must not exceed 500 kPa (73 psi). Direct the air flow up the pleats and down the pleats from the inside of the filter element. Take extreme care in order to avoid damage to the pleats.

Do not clean the air filter element more than three times. The air filter element must be replaced if the filter has been in use for one year, regardless of the number of times the filter has been cleaned.

Do not use air filters with damaged pleats, gaskets, or seals. Dirt entering the engine will cause damage to engine components.

NOTICE

Do not clean the air filter elements by bumping or tapping. This could damage the seals. Do not use elements with damaged pleats, gaskets, or seals. Damaged elements will allow dirt to pass through. Engine damage could result.

7. Remove primary air filter element (4). Replace the filter element as necessary.

Note: Replace the primary filter if the filter has been in use for 1 year.

8. Clean inside the air filter assembly housing.

Note: Do not allow any dirt or debris to contact the secondary air filter element (5).

9. Inspect the seal area ensure that no foreign debris has fallen into the seal area. Clean the air cleaner interior to remove remaining dust or debris
10. Without removing secondary air filter element (5), inspect the filter element for damage. Replace if necessary or dirty. Refer to Secondary Air Filter Element - Replace.

11. Install the secondary air filter element.

12. Install the primary filter.

Note: Filters must be fully installed before the pre-cleaner can be attached. If the pre-cleaner cannot be fully latched, verify that the filter elements are properly seated.

13. Install the pre-cleaner and secure the latches that hold the pre-cleaner to the air filter housing.

14. Close the access door.

Secondary Air Filter Element - Replace

NOTICE

Always replace the secondary element. Do not attempt to reuse it by cleaning. Engine damage could result.

NOTICE

Do not use the air filter elements longer than 1 year.

NOTICE

Replace the secondary filter element when you service the primary element for the third time. If a clean primary element has been installed and the filter element indicator is still flashing, replace the secondary filter element. Also if the exhaust smoke remains black and a clean primary filter element has been installed, replace the secondary filter element.

1. Open the front access door on the left side of the machine.

Maintenance Section

Engine Air Filter Primary and/or Secondary Element - Replace

2. Refer to the section "Primary Air Filter Element - Replace". Remove the pre-cleaner from the engine air filter housing. Remove the primary air filter element from the air filter housing.



Illustration 768

g06183315

3. Secondary air filter element (5) is pressed into the rear portion of engine air filter housing (6). Pull forward on the secondary air filter element to remove the element from the engine air filter housing.
4. Cover the air inlet opening. Clean inside the air cleaner housing.
5. Clean all surfaces of the pre-cleaner cover and body.
6. Uncover the air inlet opening.
7. Carefully press the secondary air filter element into the rear portion of the engine air filter housing.

Note: Be certain that the new secondary air filter element is properly seated in the filter housing. Also, check to see that no damage to the filter element has occurred during installation.

8. Install the primary air filter element and the pre-cleaner.
9. Close the access door.

Engine Air Pre-Cleaner - Clean

Note: Do not attempt to clean the pre-cleaner by hitting the filter against another object. Damage to the filter is likely to occur.



Illustration 769

g06183310

After removing the pre-cleaner from the air filter housing, inspect the interior of the pre-cleaner through the ejection ports around the perimeter.

In general, dust and debris will clear automatically through normal pre-cleaner operation and no further service will be required.

If the ejection ports are blocked, or dust is packed between the pre-cleaner tubes, first try to clear any accumulation by vigorously shaking the pre-cleaner.

If the pre-cleaner is still blocked with dirt, the pre-cleaner may be separated by releasing the snap features holding the front cover to the pre-cleaner body.



Illustration 770

g06069263

1. Place the pre-cleaner assembly on a suitable work surface. The interior of the pre-cleaner may have accumulated debris. Protect your work surface to collect excess debris and to avoid scattering debris.



Illustration 771

g06069266

2. Lift the locking tabs on one end of the cover away from the retainers. Move the locking tabs only far enough to clear the retention posts.

Note: Do not bend the locking tabs farther than necessary to release the tabs. Damage to the air cleaner can result. Do not use tools to force the snap features off the retainers.

3. Pull one end of the top cover away from the bottom half.
4. Hold one end of the cover sections apart and separate one of the locking tabs on the other end of the pre-cleaner.



Illustration 772

g06069264

5. Pull the top cover up and away from the bottom.
6. Clear any blockage by shaking the pre-cleaner components and/or brushing away accumulated debris.

Note: Do not use picks or other stiff implements to clear debris, you may damage the pre-cleaner components

Note: Do not attempt to remove the pre-cleaner tubes from the top cover or you will damage the air cleaner.

7. If the blockage is still not cleared, you may attempt to clear the debris by use of an air nozzle limited to 500 kPa (73 psi).

Alternately, each part of the pre-cleaner may be submerged in water to loosen mud or other debris that may have dried on the interior surface of the air cleaner.

Note: Never attempt to use a pressure washer or other high-pressure water sources to clean the pre-cleaner. Use of high-pressure water may damage the pre-cleaner tubes and reduce the pre-cleaner effectiveness.

8. After cleaning, reassemble the pre-cleaner by lining up the tubes on the pre-cleaner top with the tubes in the pre-cleaner bottom.
9. Allow the pre-cleaner top to rest on the pre-cleaner bottom and ensure that the four snap features are aligned.



Illustration 773

g06069247

10. If all the tabs are aligned, gently push the pre-cleaner top down into place. Ensure that all the snap features have engaged.

11. Before installing the pre-cleaner to the air filter housing, inspect the pre-cleaner gasket for damage. Replace if damaged.

i06970869

Engine Oil Level - Check

SMCS Code: 1000-535

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

NOTICE

Do not overfill the crankcase. Engine damage can result.

This machine is equipped with both an automated function for checking fluid levels and dipsticks. Refer to Operation and Maintenance Manual, "Monitoring System" regarding the automated system. If the machine is on an incline or the engine has been stopped only for a short time, all engine oil may not be in the crankcase. The fluid level cannot be properly checked by either method during these instances. Park the machine on level ground. The engine oil level can be checked after the engine has been stopped for at least 30 minutes. Do not check the oil level while the engine is running.

The machine is equipped with a ground level dipstick and a dipstick on top of the engine.



Illustration 774

g06183460

1. Open the access door on top of the machine.

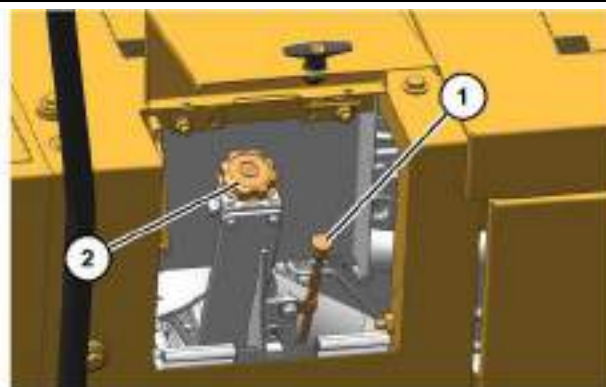


Illustration 775

g06183463

2. Remove dipstick (1). Wipe the oil off the dipstick and reinsert the dipstick.

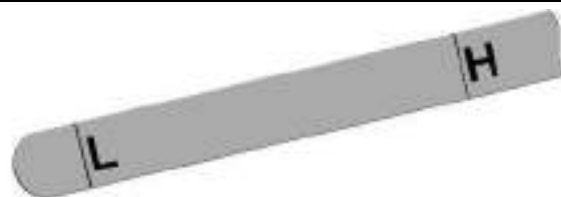


Illustration 776

g06183475

3. Remove the dipstick and check the dipstick. The oil level should be between the "L" mark and the "H" mark.

NOTICE

Operating your engine when the oil level is above the "H" mark could cause the crankshaft to dip into the oil. This could lead to excessively high oil temperatures which can reduce the lubricating characteristics of the oil, lead to bearing damage, and could result in loss of engine power.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on Containing Fluid Spillage.

4. Remove oil filler plug (2) to add oil, if necessary. See Operation and Maintenance Manual, "Lubricant Viscosities".

Note: If the oil is deteriorated or badly contaminated, change the oil regardless of the maintenance interval.

5. Clean the oil filler plug. Install the oil filler plug.
6. Close the access door.

Ground Level Dipstick



Illustration 777

g06182545

To access the ground level dipstick, open the right access door.

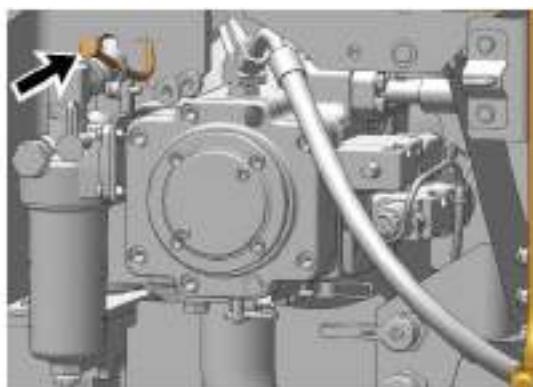


Illustration 778

g06211545

The ground level dipstick is located near the engine oil filter and main hydraulic pump.

i07088417

Engine Oil Sample - Obtain

SMCS Code: 1000-008; 1000; 1348-554-SM; 1348-008; 7542-008; 7542-554-OC; 7542-554-SM

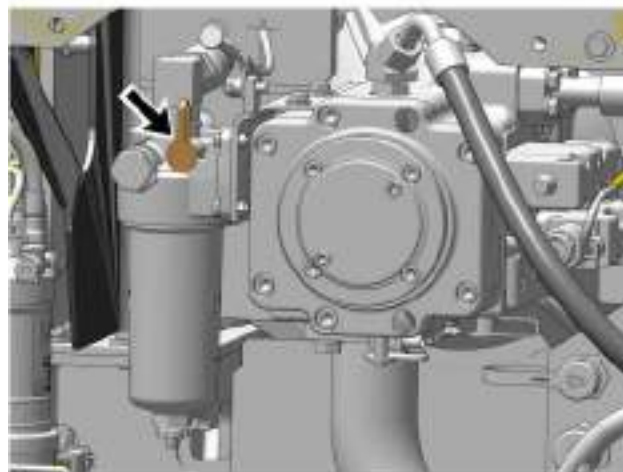


Illustration 779

g06220379

Obtain a sample of the engine oil from the engine oil sampling valve that is located on the engine oil filter housing. Refer to Special Publication, SEBU6250, "S·O·S Oil Analysis" for information that pertains to obtaining a sample of the engine oil. Refer to Special Publication, PEGJ0047, "How To Take A Good Oil Sample" for more information about obtaining a sample of the engine oil.

i08271902

Engine Oil and Filter - Change

SMCS Code: 1318-510

NOTICE

The engine oil and filter change interval for standard service application is every 1000 hours when the following requirements are met:

- Utilize Cat Recommended Fluids
- Utilize Cat Filters
- Utilize S·O·S Services at recommended interval

When these requirements are not met, the oil and filter change interval should be every 500 hours, or use S·O·S Services oil sampling and analysis program to determine an acceptable oil change interval.

If you select an interval for oil and filter change that is too long, you may damage the engine.

NOTICE

When operating in any of the conditions or environments outlined in this Operation and Maintenance Manual, Severe Service Application, use S·O·S Services oil analysis to determine the best oil and filter change interval.

When S·O·S Services are not used in severe service applications, the oil and filter change interval should be every 250 hours..

If you select an interval for oil and filter change that is too long, you may damage the engine.

Reference: Operation and Maintenance Manual, "Lubricant Viscosities"

Reference: Operation and Maintenance Manual, "Maintenance Interval Schedule"

Reference: Operation and Maintenance Manual, "Severe Service Application"

Reference: Operation and Maintenance Manual, "S·O·S Information"

Use the table below to determine the appropriate oil and filter change interval.

Table 41

Selection of Oil and Filter Change Interval				
	Conditions			Interval
	Cat [®] Recommended Fluids	Cat [®] Filters	S·O·S Services	
Standard Service Application	YES	YES	YES	1000 hours
	YES	YES	NO	500 hours
	YES	NO	YES	500 hours
	NO	YES	YES	500 hours
	NO	NO	NO	250 hours
Severe Service Application	NO	NO	NO	250 hours
	YES	YES	NO	250 hours
	YES	YES	YES	Use S·O·S ⁽¹⁾
	YES	NO	YES	Use S·O·S ⁽¹⁾
	NO	YES	YES	Use S·O·S ⁽¹⁾

⁽¹⁾ If operating in any of the conditions or environments outlined in the Severe Service Application, use S·O·S Services oil analysis to determine the best oil change interval.

Note: Initial oil and filter change is required at initial 500 hours.

Procedure for Changing Engine Oil and Filter

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

Note: If the sulfur content in the fuel is greater than 1.5 percent by weight, use an oil that has a Total Base Number (TBN) of 30 and reduce the oil change interval by one-half.

Note: Drain the crankcase while the oil is warm. This allows waste particles that are suspended in the oil to drain. As the oil cools, the waste particles will settle to the bottom of the crankcase. The particles will not be removed by draining the oil and the particles will recirculate in the engine lubrication system with the new oil.

1. Prepare the machine for maintenance. Refer to Operation and Maintenance Manual, "Prepare the Machine for Maintenance".

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on Containing Fluid Spillage.



Illustration 780

g06183508

Crankcase drain valve

2. Open the crankcase drain valve. Allow the oil to drain into a suitable container.

Note: Discard any drained fluids according to local regulations.

3. Close the drain valve.

Maintenance Section
Engine Oil and Filter - Change



Illustration 781

g06182545

Access door

4. Open the access door at the right side of the machine.

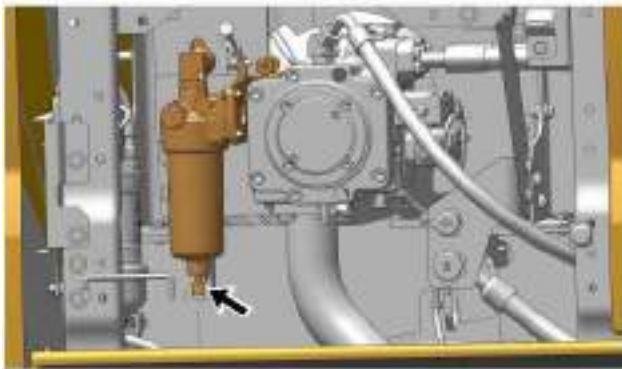


Illustration 782

g06183511

Drain valve

5. Loosen drain valve and allow the oil to drain out of the housing. The drain valve is on the bottom of the engine oil filter housing.

Note: If equipped with a fast fill system, the oil can be drained and filled using the fast fill coupler. Refer to "Fast Fill (If Equipped)" in this chapter.



Illustration 783

g06183515

Oil filter

6. Remove the oil filter housing. Refer to Operation and Maintenance Manual, "Oil Filter - Inspect". Dispose the used filter according to local regulations.
7. Remove the oil filter from the housing.
8. Clean the oil filter housing and the base thread.
9. Install the new oil filter element into the housing.
10. After installing the new element, hand tighten the drain valve at the bottom of the housing.



Illustration 784

g06604084

Oil filter gasket

11. Apply a thin coat of engine oil to the gasket of the filter. Refer to Illustration 784 for lubrication points on the gasket.

12. Install the filter housing with element to the filter base utilizing a socket wrench.

Note: Apply a tightening torque of 90 N·m (66 lb ft) and tighten until metal to metal contact is achieved.

13. Close the access door.



Illustration 785

g06183460

Access door

14. Open the access door on top of the machine. Refer to “Access Door and Cover Locations”.



Illustration 786

g06606526

(1) Oil level gauge
(2) Oil filler cap

15. Remove oil filler cap (2). Fill the crankcase with new oil. See Operation and Maintenance Manual, “Capacities (Refill)”. Clean the oil filler plug and install the oil filler plug.

NOTICE

Do not under fill or overfill engine crankcase with oil. Either condition can cause engine damage.

16. START the engine and allow the oil to warm. Refer to “Engine Starting”. Check the engine for leaks. STOP the engine.

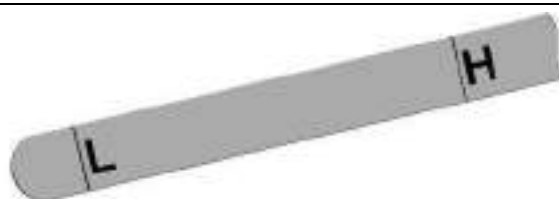


Illustration 787

g06183475

Oil level gauge

17. Wait for 30 minutes to allow the oil to drain back into the crankcase. Check the oil level with oil level gauge (1). Maintain the oil between the “L” and “H” marks on the oil level gauge(1). If necessary, add oil.

18. Close the access door.

Fast Fill (If Equipped)

If your machine is equipped with a deluxe service center, you may drain and add the engine oil through the fast fill port.



Illustration 788

g06182545

Access door

1. Open the access door on the right side of the machine.

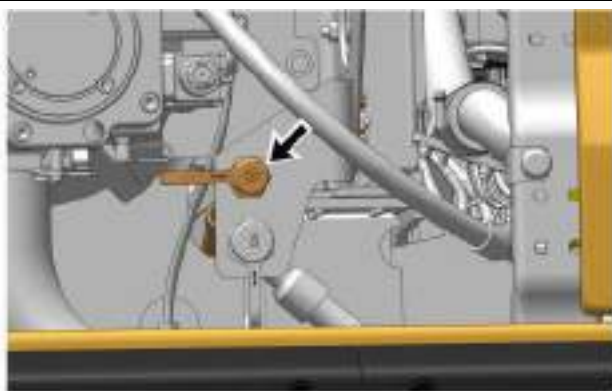


Illustration 789

g06183504

Dust cover

2. Remove the dust cover.
3. Attach a hose that is equipped with a nozzle.

Note: Make sure that the nozzle and the receiver are free from debris before attaching.

4. Drain the oil or add the oil, as needed.
5. Install the dust cover.

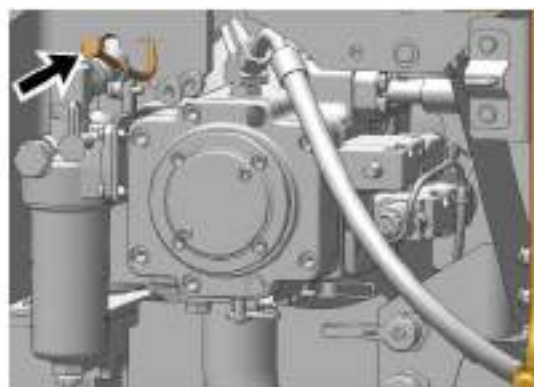


Illustration 790

g06211545

Ground level oil level gauge.

6. Verify that the correct amount of oil was added and add or remove oil as necessary. A ground level oil level gauge is in the same compartment as the fast fill port.

i07051358

Ether Starting Aid Cylinder - Replace (If Equipped)

SMCS Code: 1456-510-CD

WARNING

Breathing ether vapors or repeated contact of ether with skin can cause personal injury. Personal injury may occur from failure to adhere to the following procedures.

Use ether only in well ventilated areas.

Do not smoke while changing ether cylinders.

Use ether with care to avoid fires.

Do not store replacement ether cylinders in living areas or in the operator's compartment.

Do not store ether cylinders in direct sunlight or at temperatures above 49 °C (120 °F).

Discard cylinders in a safe place. Do not puncture or burn cylinders.

Keep ether cylinders out of the reach of unauthorized personnel.

The ether cylinder is located inside the rear access door on the left side of the machine.

Refer to Operation and Maintenance, "Fire Prevention and Explosion Prevention" before you replace the ether cylinder.



Illustration 791

g06179792

1. Open the rear access door at the left side of the machine.

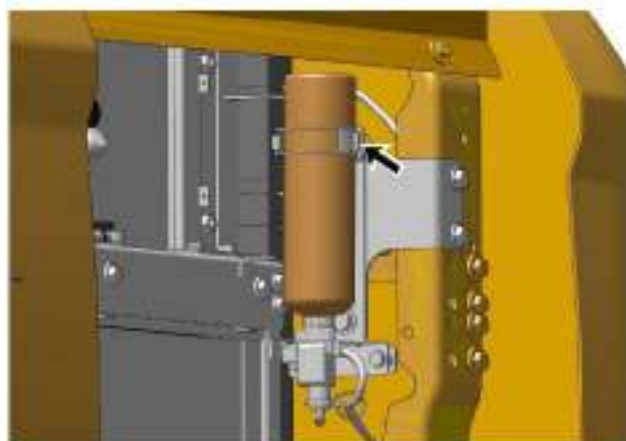


Illustration 792

g06211094

2. Loosen the cylinder retaining clamp.
3. Unscrew the empty ether starting aid cylinder and remove the empty ether starting aid cylinder.

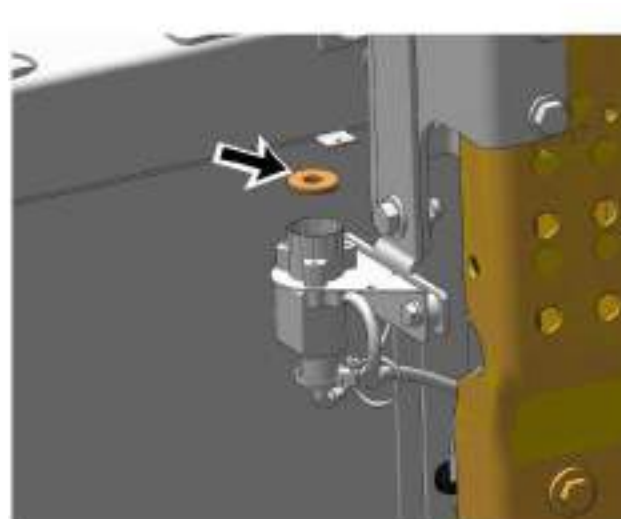


Illustration 793

g06211096

4. Remove the used gasket.
 5. Install a new gasket.
- Note:** A new gasket and O-ring is provided with each new ether starting aid cylinder.
6. Install the new ether starting aid cylinder.
 7. Tighten the ether starting aid cylinder hand tight.
 8. Tighten the cylinder retaining clamp securely.
 9. Close the access door.

i07786001

Film (Product Identification) - Clean

SMCS Code: 7405-070; 7557-070



Illustration 794

g06435629

Cleaning of the Films

Make sure that all the product identification films are legible. Make sure that the recommended procedures are used to clean the product identification films. Ensure that all the product identification films are not damaged or missing. Clean the product identification films or replace the films.

Hand Washing

Use a wet solution with no abrasive material that contains no solvents and no alcohol. Use a wet solution with a "pH" value between 3 and 11. Use a soft brush, a rag, or a sponge to clean the product identification films. Avoid wearing down the surface of the product identification films with unnecessary scrubbing. Ensure that the surface of the product identification films is flushed with clean water and allow the product identification films to air dry.

Power Washing

Power washing or washing with pressure may be used to clean product identification films. However, aggressive washing can damage the product identification films.

Excessive pressure during power washing can damage the product identification films by forcing water underneath the product identification films. Water lessens the adhesion of the product identification film to the product, allowing the product identification film to lift or curl. These problems are magnified by wind. These problems are critical for the perforated film on windows.

To avoid lifting of the edge or other damage to the product identification films, follow these important steps:

- Use a spray nozzle with a wide spray pattern.
- A maximum pressure of 83 bar (1200 psi)
- A maximum water temperature of 50° C (120° F)
- Hold the nozzle perpendicular to the product identification film at a minimum distance of 305 mm (12 inch).
- Do not direct a stream of water at a sharp angle to the edge of the product identification film.

i06969803

Final Drive Oil - Change

SMCS Code: 4050-044-FLV

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

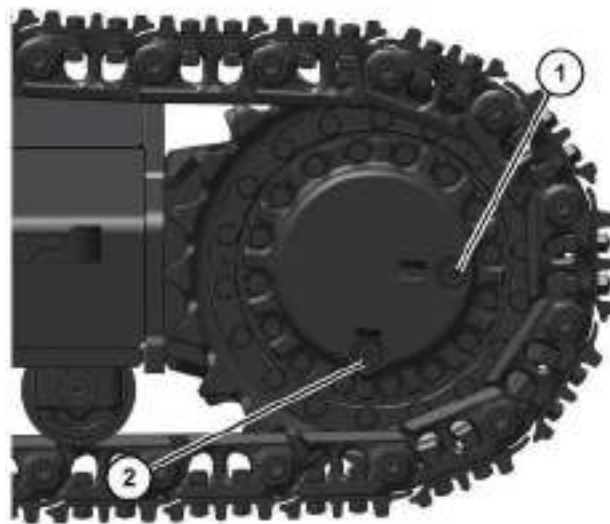


Illustration 795

g06182944

- (1) Oil level plug
(2) Oil drain plug

1. Position one final drive so that oil drain plug (2) is at the bottom.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on Containing Fluid Spillage.

2. Remove drain plug (2) and level plug (1). Allow the oil to drain into a suitable container.
3. Clean the plugs and inspect the O-ring seals. If wear or damage is evident, replace the drain plug, the level plug, and/or the O-ring seals.
4. Install drain plug (2).
5. Fill the final drive to the bottom of the opening on level plug (1). See Operation and Maintenance Manual, "Lubricant Viscosities" and Operation and Maintenance Manual, "Capacities (Refill)".

Note: If the oil fills slowly, the fill hole may be blocked by the planetary gear. Rotate the final drive to move the planetary gear away from the fill hole.

Note: Overfilling the final drive will cause the seals on the travel motor to allow hydraulic oil or water to enter the final drive. The final drive may become contaminated.

6. Install level plug (1).
7. Perform Step 1 to Step 6 on the other final drive. Use a different container for the oil so that the oil samples from the final drives will be separate.
8. Completely remove the oil that has spilled onto surfaces.
9. Start the machine and allow the final drives to run through several cycles.
10. Stop the machine. Check the oil level.
11. Check the drained oil for metal chips or for particles. If there are any chips or particles, consult your Cat dealer.
12. Properly dispose of the drained material. Obey local regulations for the disposal of the material.

i06969810

Final Drive Oil Level - Check

SMCS Code: 4050-535-FLV



WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

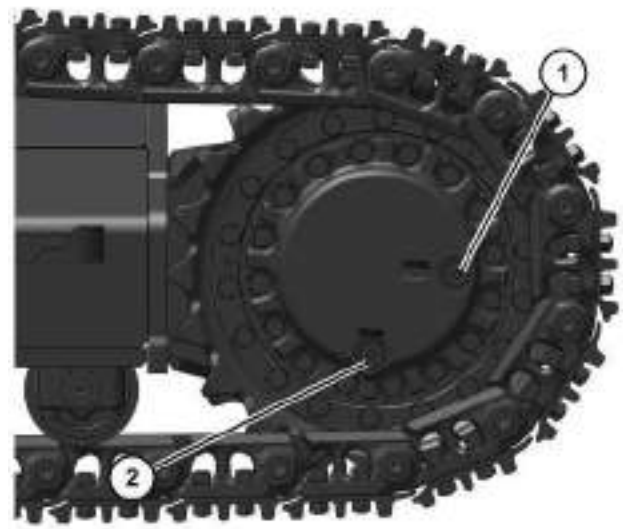


Illustration 796

g06182944

- (1) Oil level plug
(2) Oil drain plug

1. Position one final drive so that oil drain plug (2) is at the bottom.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on Containing Fluid Spillage.

2. Remove oil level plug (1).
3. Check the oil level. The oil should be near the bottom of the level plug opening.
4. Add oil through the level plug opening, if necessary. See Operation and Maintenance, "Lubricant Viscosities".

Note: If the oil fills slowly, the fill hole may be blocked by the planetary gear. Rotate the final drive to move the planetary gear away from the fill hole.

Note: Overfilling the final drive will cause the seals on the travel motor to allow hydraulic oil or water to enter the final drive. The final drive may become contaminated.

5. Clean oil level plug (1). Inspect the O-ring seal. Replace the O-ring seal if the O-ring seal is worn or damaged.
6. Install oil level plug (1).
7. Repeat the procedure for the other final drive.

i06969816

i07088575

Final Drive Oil Sample - Obtain

SMCS Code: 4011-008; 4050-SM; 4050-008; 7542-008

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

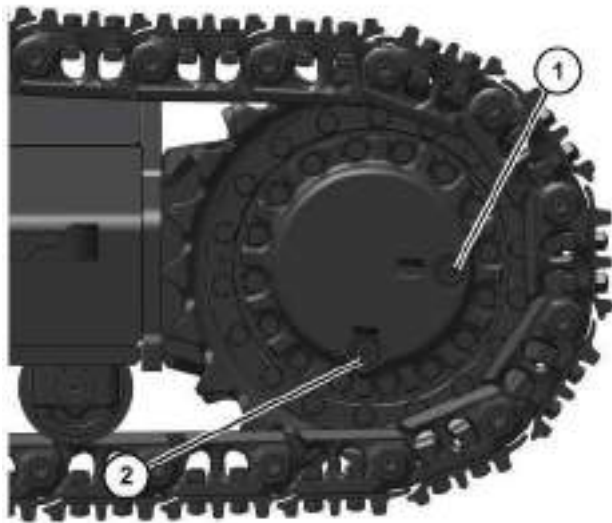


Illustration 797

g06182944

- (1) Oil level plug
(2) Oil drain plug

1. Position the final drive so that oil drain plug (2) is at the bottom.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on Containing Fluid Spillage.

2. Remove oil level plug (1).
3. Obtain a sample of the final drive oil through the hole for the oil level plug.
4. Install oil level plug (1).

Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" "S·O·S Oil Analysis" for more information on obtaining a sample of the final drive oil. For additional information about taking an oil sample, refer to Special Publication, PEGJ0047, "How To Take A Good Oil Sample".

Fuel Cap Filter - Replace

SMCS Code: 1261-510

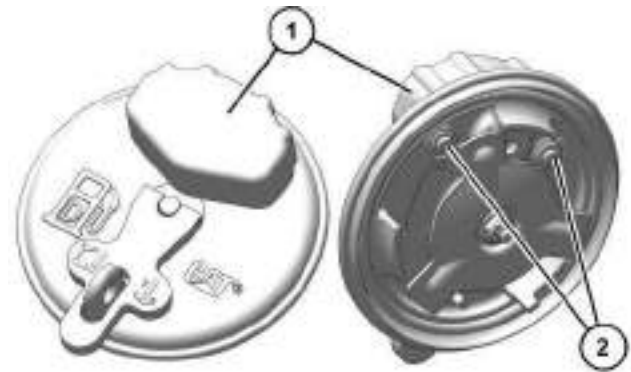


Illustration 798

g06220524

1. Remove the fuel cap.
2. Remove filter element screws (2) from the underside of the fuel cap and remove filter element (1).
3. Install a clean fuel cap filter element.
4. Install the screws to secure the filter element to the fuel cap.
5. Install the fuel tank cap.

i07088554

Fuel Lift Pump Strainer - Replace (In-line Filter)

SMCS Code: 1256; 1256-510-STR

S/N: HEX1-10000

S/N: RAZ1-10000

WARNING

Personal injury or death may result from failure to adhere to the following procedures.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch OFF or disconnect the battery when changing fuel filters.

NOTICE

Do not fill the fuel filters with fuel before installing the fuel filters. The fuel will not be filtered and could be contaminated. Contaminated fuel will cause accelerated wear to fuel system parts.

i06674757

Fuel System - Prime

SMCS Code: 1250-548

WARNING

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire. To help prevent possible injury, turn the start switch off when changing fuel filters or water separator elements. Clean up fuel spills immediately.



Illustration 799

g06182545

The fuel lift pump strainer is located behind the access door on the right side of the machine.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE

Do not loosen the fuel lines at the fuel manifold. The fittings may be damaged and/or a loss of priming pressure may occur when the fuel lines are loosened.

NOTICE

Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over any disconnected fuel system components.

Prime the fuel system in order to fill the fuel filter, and prime the fuel system in order to purge trapped air. The fuel system should be primed under the following conditions:

- The fuel tank is running low.
- The machine has been stored.
- The fuel filter is being replaced.

1. Turn the engine start switch to the ON position. Leave the engine start switch in the ON position for 2 minutes.

2. Verify that the water separator is full of fuel.

3. If the water separator is not full of fuel, turn the engine start switch OFF and then turn the engine start switch ON. Turning the engine start switch off and on will cycle the fuel priming pump again.

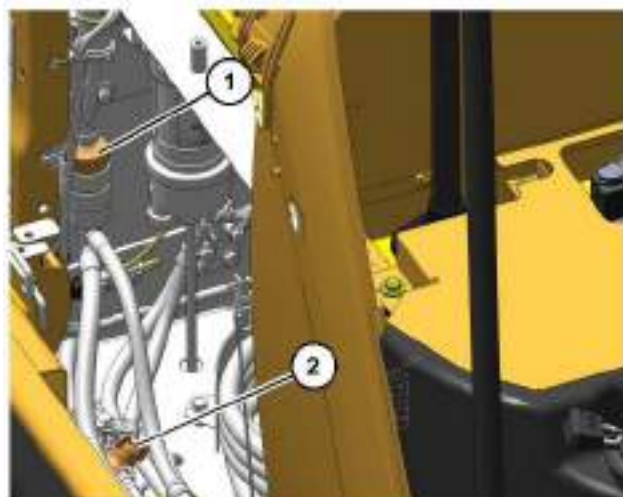


Illustration 800

g06220476

- 1.** Open the access door and close the fuel shut off valve (2) by turning the valve clockwise.
- 2.** Disconnect the hoses from the strainer (1) and remove the strainer.
- 3.** Install a new strainer, and reconnect the hoses.
- 4.** Open the fuel shut off valve by turning the valve counterclockwise.

4. When the water separator is full of fuel, attempt to start the engine. If the engine starts and the engine runs rough or the engine misfires, operate at low idle until the engine is running smoothly. If the engine cannot be started, or if the engine continues to misfire or smoke, repeat Step 1.

i06969829

Fuel System Primary Filter (Water Separator) Element - Replace

SMCS Code: 1263-510-FQ

S/N: HEX1-10000

S/N: RAZ1-10000

WARNING

Personal injury or death may result from failure to adhere to the following procedures.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch OFF or disconnect the battery when changing fuel filters.

NOTICE

Do not fill the fuel filters with fuel before installing the fuel filters. The fuel will not be filtered and could be contaminated. Contaminated fuel will cause accelerated wear to fuel system parts.

The primary filter/water separator is located behind the access door on the right side of the machine.



Illustration 801

g06182545

1. Open the rear access door on the right side of the machine.

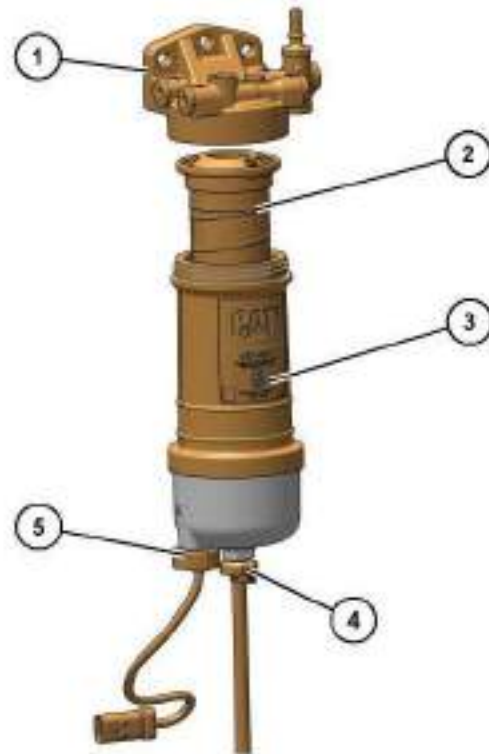


Illustration 802

g06182966

- (1) Filter base
- (2) Filter
- (3) Filter housing
- (4) Drain valve
- (5) Sensor

2. Turn drain valve (4) counterclockwise to open. The drain valve is on the bottom of the water separator.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information that pertains to containing fluid spillage.

3. Drain the water and the sediment into a suitable container.

Note: Dispose of used fluids according to local regulations.

4. Close the drain valve (4).

5. Disconnect the water sensor (5) from the harness.

Note: Do not attempt to remove the sediment bowl from the housing. The sediment bowl is permanently attached to the housing. Attempting to remove the sediment bowl may damage the bowl.

6. Unscrew filter housing (3) and remove primary filter (2). A filter wrench may be used to loosen the filter housing. Properly discard the used filter.

7. Clean the mounting base (1).

8. Lubricate the seal of the new filter (2) with clean diesel fuel.
9. Install the new filter (2) into the housing.
10. Tighten the filter housing approximately 1/6 of a turn. Do not use tools to tighten the filter housing to the filter base.
11. Ensure that sensor (5) is in the correct position and connect to the wiring harness. If the sensor was removed from the bowl, install the sensor and tighten to $2.5 \pm 0.5 \text{ N}\cdot\text{m}$ ($22 \pm 4 \text{ lb in}$).
12. Open the fuel shutoff valve.
13. Close the access door.

i07511456

Fuel System Primary Filter (Water Separator) Element - Replace

SMCS Code: 1263-510-FQ**S/N:** KFE1-Up**S/N:** MYK1-Up**S/N:** NDL1-Up**S/N:** LTN1-Up**S/N:** YCP1-Up**S/N:** HDT1-Up**S/N:** HEX10001-Up**S/N:** RAZ10001-Up

WARNING

Personal injury or death may result from failure to adhere to the following procedures.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch **OFF** or disconnect the battery when changing fuel filters.

NOTICE

Do not fill the fuel filters with fuel before installing the fuel filters. The fuel will not be filtered and could be contaminated. Contaminated fuel will cause accelerated wear to fuel system parts.

The primary filter/water separator is located behind the access door on the right side of the machine.



Illustration 803

g06182545

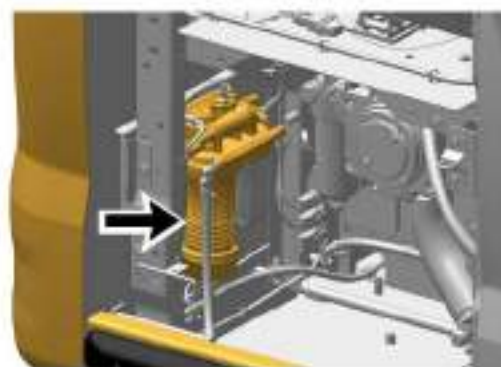


Illustration 804

g06342852

Typical Example

1. Open the rear access door on the right side of the machine.

Maintenance Section
Fuel System Secondary Filter - Replace

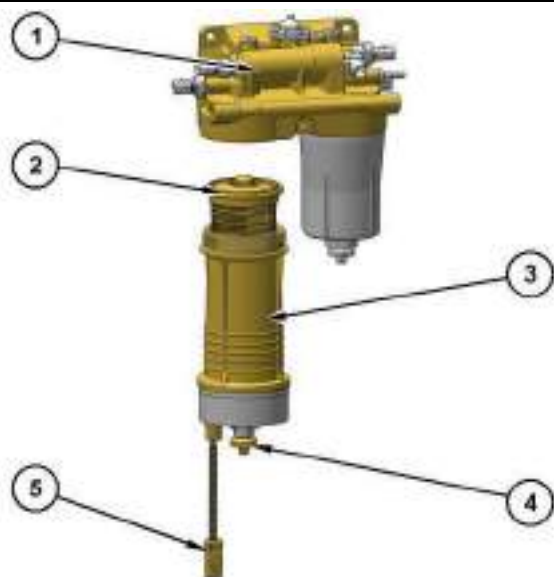


Illustration 805

g06344343

Typical example of primary filter water separator

- (1) Filter base
- (2) Filter
- (3) Filter housing
- (4) Drain valve
- (5) Sensor

2. Turn drain valve (4) counterclockwise to open. The drain valve is on the bottom of the water separator.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information that pertains to containing fluid spillage.

3. Drain the water and the sediment into a suitable container.

Note: Dispose of used fluids according to local regulations.

4. Close the drain valve (4).
5. Disconnect the water sensor (5) from the harness.

Note: Do not attempt to remove the sediment bowl from the housing. The sediment bowl is permanently attached to the housing. Attempting to remove the sediment bowl may damage the bowl.

6. Unscrew filter housing (3) and remove primary filter (2). A filter wrench may be used to loosen the filter housing. Properly discard the used filter.
7. Clean the mounting base (1).
8. Lubricate the seal of the new filter (2) with clean diesel fuel.
9. Install the new filter (2) into the housing.

10. Tighten the filter housing approximately 1/6 of a turn. Do not use tools to tighten the filter housing to the filter base.
11. Ensure that sensor (5) is in the correct position and connect to the wiring harness. If the sensor was removed from the bowl, install the sensor and tighten to $2.5 \pm 0.5 \text{ N}\cdot\text{m}$ ($22 \pm 4 \text{ lb in}$).
12. Open the fuel shutoff valve.
13. Close the access door.

i06969842

Fuel System Secondary Filter - Replace

SMCS Code: 1261-510

S/N: HEX1-10000

S/N: RAZ1-10000

WARNING

Personal injury or death can result from a fire.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch OFF or disconnect the battery when changing fuel filters.

NOTICE

Do not fill fuel filters with fuel before installing them. Contaminated fuel will cause accelerated wear to fuel system parts.

NOTICE

Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over any disconnected fuel system components.



Illustration 806

g06182545

1. Open access door on the right side of the machine.
2. Shut off the fuel supply. Refer to Fuel Tank Shutoff and Drain Control for additional information.

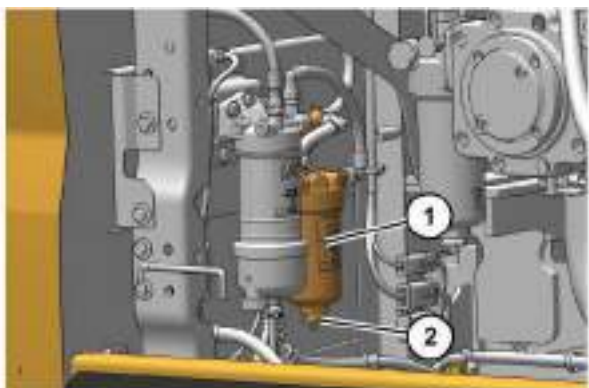


Illustration 807

g06182979

3. Loosen drain valve (2) and allow the fuel to drain out of the housing into a suitable container.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

4. After all the fuel has been removed, tighten the drain.
5. Remove the secondary fuel filter housing (1) from the base.
6. Remove the filter element from the housing.
7. Clean the filter housing and the base.
8. Install the new filter element into the housing.
9. Apply a thin coat of clean diesel fuel to the sealing surface of the new fuel filter.
10. Install the new filter and tighten by hand.

11. Turn on the fuel supply.

12. Close the access door.

i07506478

Fuel System Secondary Filter - Replace

SMCS Code: 1261-510

S/N: KFE1-Up

S/N: MYK1-Up

S/N: NDL1-Up

S/N: LTN1-Up

S/N: YCP1-Up

S/N: HDT1-Up

S/N: HEX10001-Up

S/N: RAZ10001-Up

WARNING

Personal injury or death can result from a fire.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch **OFF** or disconnect the battery when changing fuel filters.

NOTICE

Do not fill fuel filters with fuel before installing them. Contaminated fuel will cause accelerated wear to fuel system parts.

NOTICE

Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over any disconnected fuel system components.



Illustration 808

g06182545

1. Open access door on the right side of the machine.
2. Shut off the fuel supply. Refer to Fuel Tank Shutoff and Drain Control for additional information.

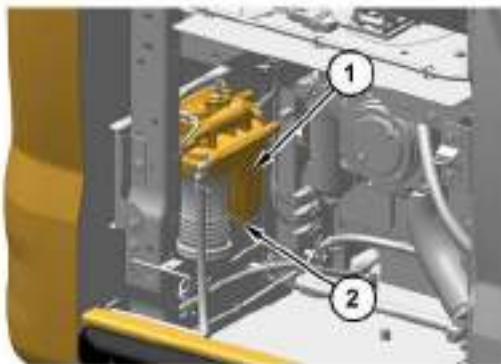


Illustration 809

g06342848

Typical Example

3. Loosen drain valve (2) and allow the fuel to drain out of the housing into a suitable container.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

4. After all the fuel has been removed, tighten the drain.
5. Remove the secondary fuel filter housing (1) from the base.
6. Remove the filter element from the housing.
7. Clean the filter housing and the base.
8. Install the new filter element into the housing.
9. Apply a thin coat of clean diesel fuel to the sealing surface of the new fuel filter.

10. Install the new filter and tighten to 50 +/- 5 N·m (36.9 +/- 4 ft lb).

11. Turn on the fuel supply.

12. Close the access door.

i06969852

Fuel System Water Separator - Drain

SMCS Code: 1263

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.



Illustration 810

g06182545

1. Open the access door on the right side of the machine.

i06969894

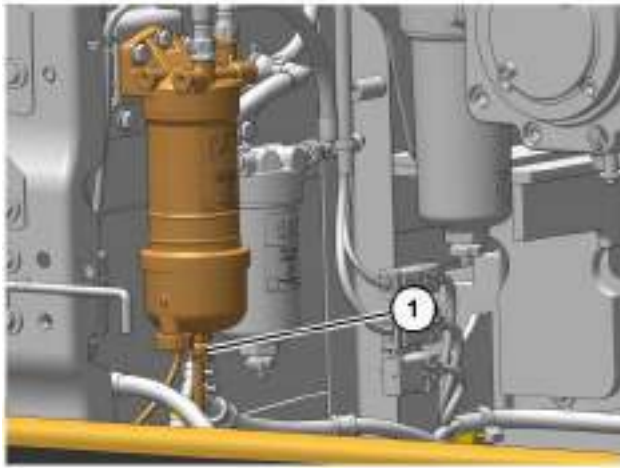


Illustration 811

g06182984

2. Open drain valve (1) on the bottom of the fuel/water separator element. Drain the water into a suitable container.
3. Close the drain valve when all the water has been drained.

Note: When water is not drained from the primary filter sufficiently, water will collect in the secondary fuel filter. Trapped water will eventually overflow. Draining water from the secondary fuel filter will prevent water damaging the fuel system. The procedure for the secondary filter is the same as the primary filter.

4. Close the access door.

Fuel Tank Strainer - Clean

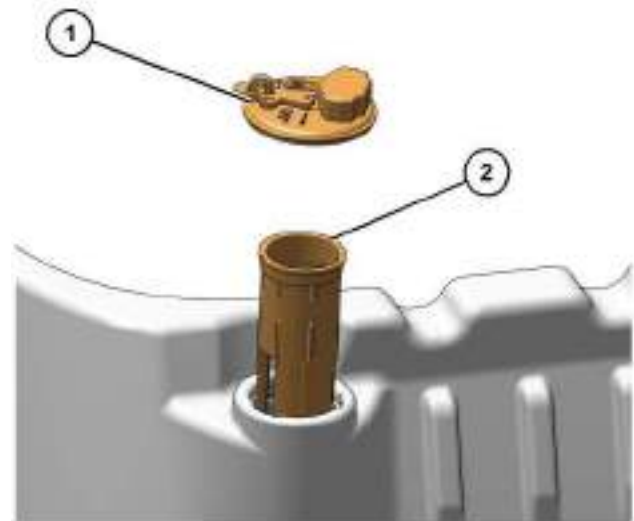
SMCS Code: 1273-070-STR

Illustration 812

g06183008

1. Remove fuel tank cap (1).
2. Remove strainer (2) from the filler opening.
3. Wash the strainer in a clean, nonflammable solvent.
4. Install the strainer into the filler opening.
5. Install the fuel tank cap.

i06954978

Fuel Tank Water and Sediment - Drain

SMCS Code: 1273-543

The drain valve for the fuel tank is located in the right compartment.

i07047559



Illustration 813

g06182545

1. Open the right compartment door.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

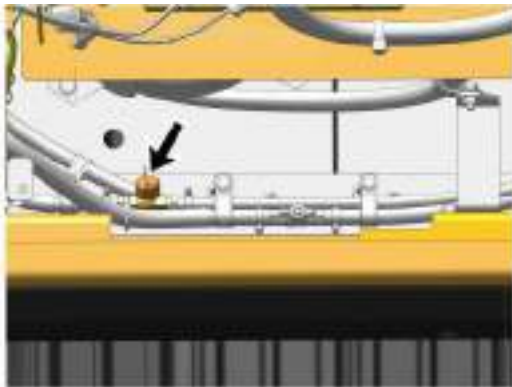


Illustration 814

g06182139

2. Open the drain valve by turning the valve counterclockwise. Allow the water and the sediment to drain into a suitable container.

Note: Dispose of drained fluids according to local regulations.

3. Close the drain valve by turning the valve clockwise.
4. Close the compartment door.

Fumes Disposal Filter Element - Replace

SMCS Code: 1074



Illustration 815

g06209196

Typical example

Note: Perform the maintenance with the engine shut off.

1. Open the engine hood.
2. Remove the lid that holds the canister to the filter base assembly.
3. Remove the filter element. Dispose of the used element properly.
4. Install the new filter.
5. Install the lid.
6. Close the engine hood.

i07946786

Fuses - Replace

SMCS Code: 1417-510

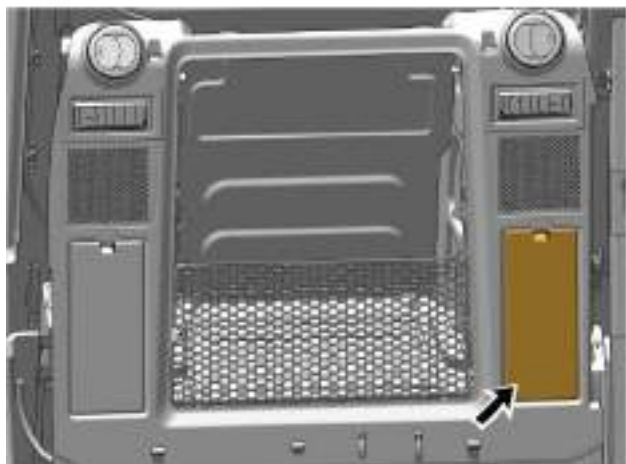


Illustration 816

g06181624

The fuse panel is on the left side of the interior storage box. Remove the cover to access the fuses.



Fuses – Fuses protect the electrical system from damage that is caused by overloaded circuits. Change a fuse if the element separates. If the element of a new fuse separates, check the circuit and/or repair the circuit.

NOTICE

Always replace fuses with the same type and capacity fuse that was removed. Otherwise, electrical damage could result.

NOTICE

If it is necessary to replace fuses frequently, an electrical problem may exist.

Contact your Cat dealer.

Machines Equipped with C4.4 Engines

To replace a fuse, use the puller that is stored in the fuse panel.

The following list identifies the circuits that are protected by each fuse. The amperage for each fuse is included with each circuit.

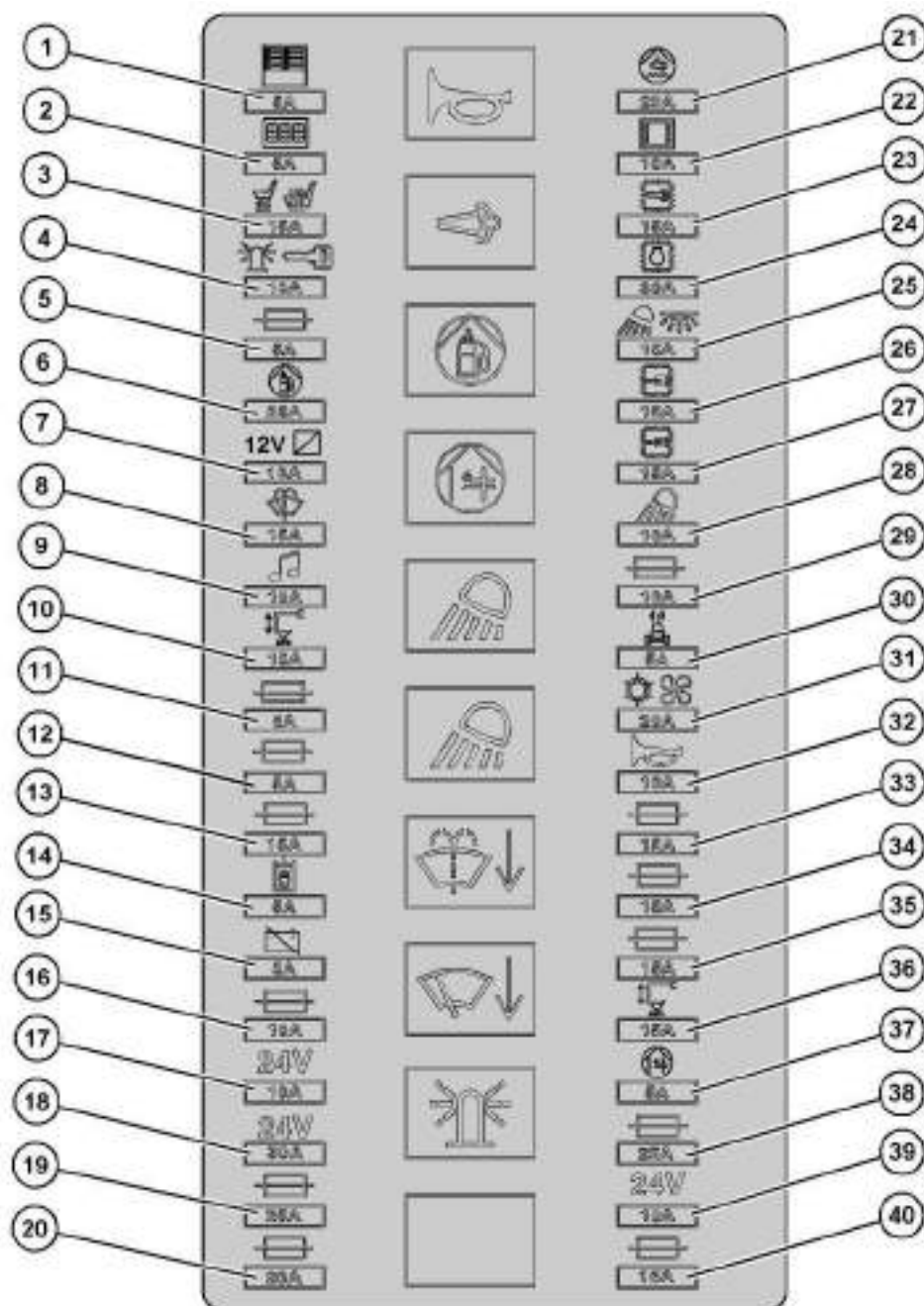


Illustration 817

g06223509

(1) Heater and Air Conditioner Control and Monitor – 5 Amp

(2) Electronic Switch Control Panel – 5 Amp

(3) Grip, Seat Heater, and Air Suspension Seat – 15 Amp

(4) Beacon – 10 Amp

(5) Spare – 5 Amp

(6) Electric Refueling Pump – 25 Amp

(7) 12V Converter – 10 Amp

(8) Window Wiper and Window Washer – 15 Amp

(9) Radio – 10 Amp

(10) Cat Grade Control – 15 Amp

(11) Spare – 5 Amp

(12) Spare – 5 Amp

- (13) **Spare** – 15 Amp
- (14) **Hydraulic Lock** – 5 Amp
- (15) **DEF Keep Alive** – 5 Amp
- (16) **Spare** – 10 Amp
- (17) **24V Auxiliary Circuit** – 10 Amp
- (18) **24V Auxiliary Circuit** – 30 Amp
- (19) **Spare** – 25 Amp
- (20) **Spare** – 20 Amp
- (21) **DEF Pump** – 25 Amp
- (22) **Display and Electronic Technician Connector** – 10 Amp
- (23) **Body Control Module** – 15 Amp
- (24) **Engine Electronic Control Module** – 30 Amp
- (25) **Dome Light** – 15 Amp
- (26) **Primary Electronic Control Module** – 15 Amp
- (27) **Secondary Electronic Control Module** – 15 Amp
- (28) **Boom Lamp** – 10 Amp
- (29) **Spare** – 10 Amp
- (30) **Product Link Module** – 5 Amp
- (31) **Air Conditioner and Heater Blower** – 20 Amp
- (32) **Horn** – 10 Amp
- (33) **Spare** – 15 Amp
- (34) **Spare** – 15 Amp
- (35) **12V Converter** – 10 Amp
- (36) **Cat Grade Control** – 15 Amp
- (37) **Fuel Lifting Pump** – 5 Amp
- (38) **Spare** – 25 Amp
- (39) **Auxiliary Circuit** – 10 Amp
- (40) **Spare** – 15 Amp

Relays

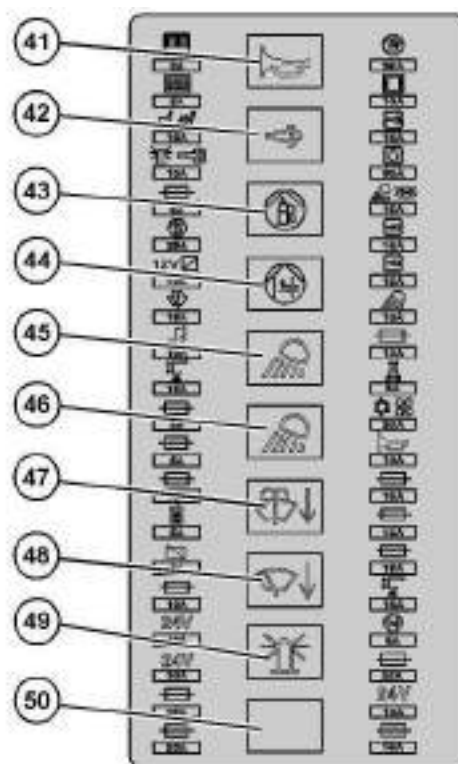


Illustration 818

g06223510

- (41) **Horn** – Relay
- (42) **DEF Pump** – Relay
- (43) **Priming Pump** – Relay
- (44) **Lifting Pump** – Relay
- (45) **Boom Light** – Relay
- (46) **Chassis Light and Cab Light** – Relay
- (47) **Lower Washer** – Relay
- (48) **Lower Wiper** – Relay
- (49) **Caution** – Relay
- (50) **Spare** – Relay

Machines Equipped with C7.1 Engines

To replace a fuse, use the puller that is stored in the fuse panel.

The following list identifies the circuits that are protected by each fuse. The amperage for each fuse is included with each circuit.

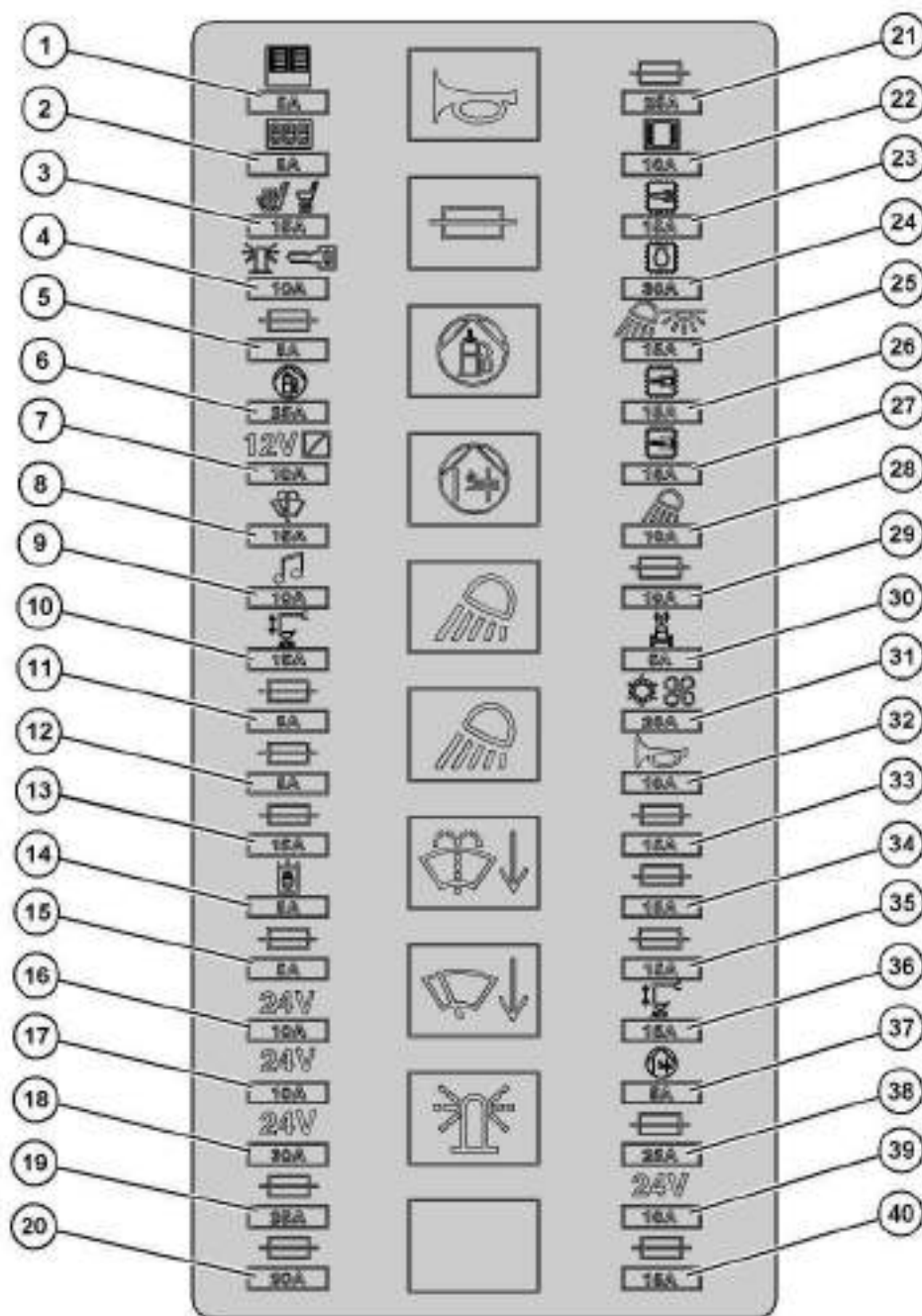


Illustration 819

g06223505

(1) Heater and Air Conditioner Control and Monitor – 5 Amp

(2) Electronic Switch Control Panel – 5 Amp

(3) Grip, Seat Heater, and Air Suspension Seat – 15 Amp

(4) Beacon – 10 Amp

(5) Spare – 5 Amp

(6) Electric Refueling Pump – 25 Amp

(7) 12V Converter – 10 Amp

(8) Window Wiper and Window Washer – 15 Amp

(9) Radio – 10 Amp

(10) Cat Grade Control – 15 Amp

(11) Spare – 5 Amp

(12) Spare – 5 Amp

- (13) **Spare** – 15 Amp
- (14) **Hydraulic Lock** – 5 Amp
- (15) **Spare** – 5 Amp
- (16) **24V Auxiliary Circuit** – 10 Amp
- (17) **24V Auxiliary Circuit** – 10 Amp
- (18) **24V Auxiliary Circuit** – 30 Amp
- (19) **Spare** – 20 Amp
- (20) **Spare** – 10 Amp
- (21) **Spare** – 25 Amp
- (22) **Display and Electronic Technician Connector** – 10 Amp
- (23) **Body Control Module** – 15 Amp
- (24) **Engine Electronic Control Module** – 30 Amp
- (25) **Dome Light** – 15 Amp
- (26) **Primary Electronic Control Module** – 15 Amp
- (27) **Secondary Electronic Control Module** – 15 Amp
- (28) **Boom Lamp** – 10 Amp
- (29) **Spare** – 10 Amp
- (30) **Product Link Module** – 5 Amp
- (31) **Air Conditioner and Heater Blower** – 20 Amp
- (32) **Horn** – 10 Amp
- (33) **Spare** – 15 Amp
- (34) **Spare** – 15 Amp
- (35) **12V Converter** – 10 Amp
- (36) **Cat Grade Control** – 15 Amp
- (37) **Fuel Lifting Pump** – 5 Amp
- (38) **Spare** – 25 Amp
- (39) **Auxiliary Circuit** – 10 Amp
- (40) **Spare** – 15 Amp

Relays

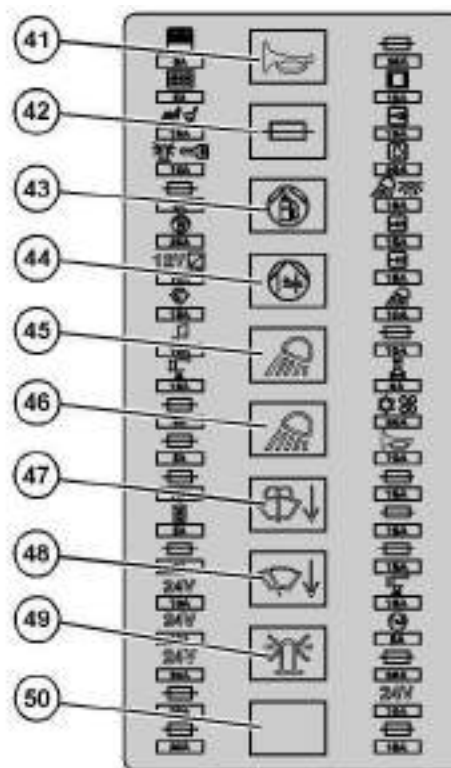


Illustration 820

g06223517

- (41) **Horn** – Relay
- (42) **DEF Pump** – Relay
- (43) **Priming Pump** – Relay
- (44) **Lifting Pump** – Relay
- (45) **Boom Light** – Relay
- (46) **Chassis Light and Cab Light** – Relay
- (47) **Lower Washer** – Relay
- (48) **Lower Wiper** – Relay
- (49) **Caution** – Relay
- (50) **Spare** – Relay

Power Fuse Module



Illustration 821

g06181546

The power fuse module is located behind the front access door on the left side of the machine. Remove the cover to access the fuses.

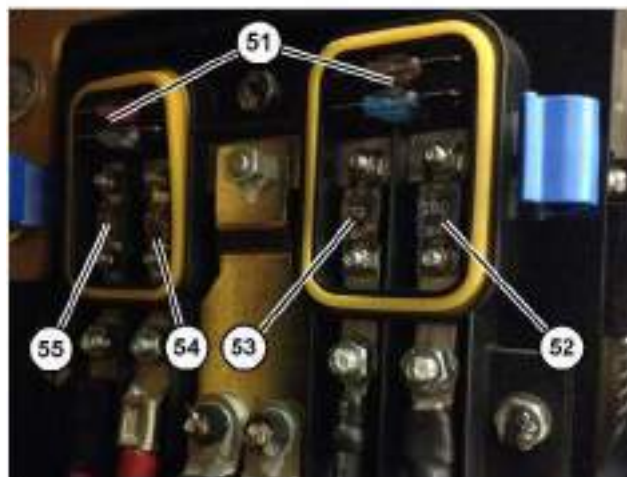


Illustration 822

g06225641



Spare (51) – The fuse module includes spare fuses which can be used if one of the installed fuses opens. One spare fuse is provided for each fuse in use.



Main Circuit 100 Amp (52) – This fuse is designed to protect the wires between the batteries and the fuses. If the wires are shorted to the machine body, this fuse would minimize the damage to the wires.



Glow Plug Circuit 70 Amp (53) – This fuse is designed to protect the glow plugs.



Alternator Circuit 150 Amp (54) – This fuse is designed to protect the alternator. If the batteries are installed with reversed polarity, the fuse would prevent the alternator from damaging the rectifier.



Electric Cooling Fans 125 Amp (55) – This fuse is designed to protect the cooling fans.

Secondary Power Fuse Module (If Equipped)



Illustration 823

g06179792

The secondary power fuse module is located behind the rear access door on the left side of the machine. Remove the cover to access the fuses.

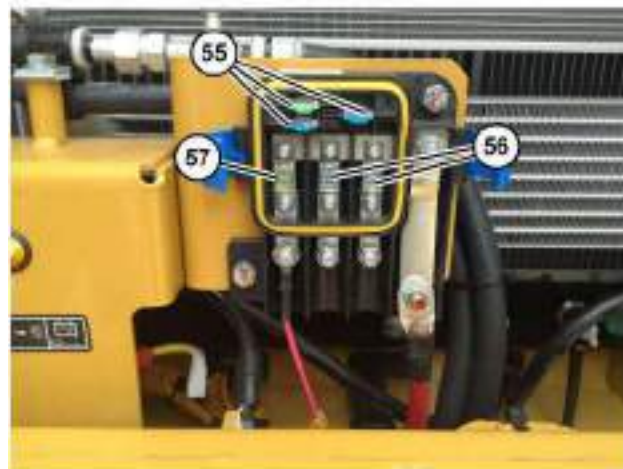


Illustration 824

g06225651



Spare (55) – The fuse module includes spare fuses which can be used if one of the installed fuses opens. One spare fuse is provided for each fuse in use.

Not Used (56) – These fuses are currently not used.



Diesel Exhaust Fluid (DEF) Pump Circuit 40 Amp (57) – This fuse is designed to protect the DEF pump circuit.

i08067436

High Intensity Discharge Lamp (HID) - Replace (If Equipped)

SMCS Code: 1434-510

WARNING

HID lamps operate at very high voltages. To avoid electrical shock and personal injury, disconnect power before servicing HID lamps.

WARNING

HID bulbs become very hot during operation. Before servicing, remove power from lamp for at least five minutes to ensure lamp is cool.

NOTICE

Although HID bulb materials may change over time, HID bulbs produced at the time of the printing of this manual contain mercury. When disposing of this component, or any waste that contains mercury, please use caution and comply with any applicable laws.

1. Prepare the machine for maintenance. Refer to Operation and Maintenance Manual, "Prepare the Machine for Maintenance".
2. Remove the electrical power from the high intensity discharge lamp (HID). The electrical power must be removed from the HID lamp for at least 5 minutes, to ensure that the bulb is cool.
3. Disassemble the housing for the HID lamp to have access to the bulb.

Note: On some HID lamps, the bulb is a part of the lens assembly. The bulb is not removed separately from the lens assembly. Replace the entire lens assembly on these HID lamps.

4. Remove the bulb from the HID lamp.
5. Install the replacement bulb in the HID lamp.

If the bulb is a part of the lens assembly, install the replacement lens assembly in the HID lamp.

Note: To avoid failure to the bulb that is premature, avoid touching the bulb's surface with your bare hands. Clean any fingerprints from the bulb with alcohol prior to operation.

6. Reassemble the housing for the HID lamp. Ensure that any printing on the lens is oriented correctly for the HID lamp's mounting position on the machine.
7. Reattach the electrical power to the HID lamp.
8. Check the HID lamp for proper operation.

Note: Consult your Cat dealer for additional information on HID lamps.

i07833138

Hydraulic System Oil - Change

SMCS Code: 5056-044

Cat HYDO Advanced 10 Oil Change Interval

The standard Cat HYDO Advanced 10 oil change interval is every 6000 service hours or 3 years. But a 6000 service hour or 3-year maintenance interval for hydraulic oil (change) is strongly recommended with S·O·S monitoring of the hydraulic oil after 3000 service hours. The interval for S·O·S monitoring is every 500 hours. The hydraulic oil change is strongly recommended when the oil deterioration or contamination is detected. The maintenance interval for the hydraulic oil filter is not changed.

Hydraulic Hammer Use

The use of hydraulic hammers shortens the life of hydraulic oil. If a hydraulic hammer is used, the maintenance interval is shortened, refer to Table 42 for the intervals.

Table 42

Percentage of Hammer Use	Hydraulic System Oil - Change
50%	Every 1000 service hours
100%	Every 600 service hours

Procedure to Change the Hydraulic Oil

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

Maintenance Section
Hydraulic System Oil - Change

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, "Dealer Service Tool Catalog" for tools and supplies suitable to collect and contain fluids on Cat® products.

Dispose of all fluids according to local regulations and mandates.

1. Park the machine on level ground. Lower the bucket to the ground so that the stick is vertical.

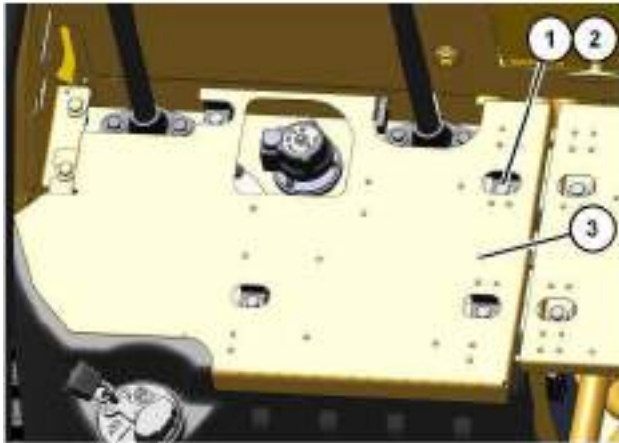


Illustration 825

g06182169

2. Remove five bolts (1) and washers (2). Remove cover (3) from the top of the hydraulic tank.

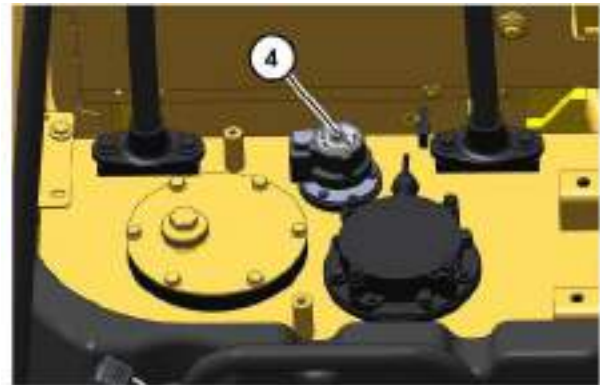


Illustration 826

g06182174

3. Clean the area thoroughly to keep dirt out of the screen cover and filler cap (4).

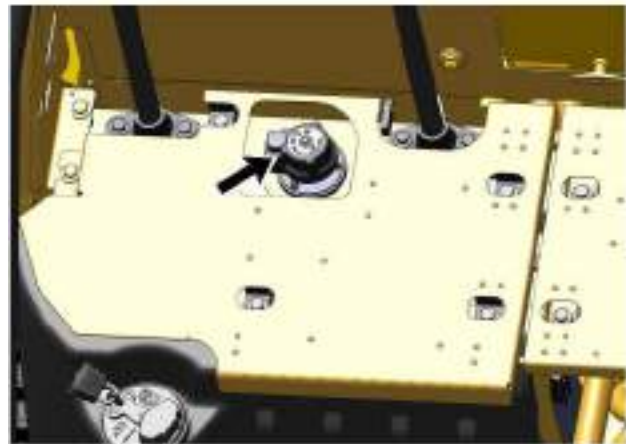


Illustration 827

g06184080

Hydraulic tank filler cap location



Pressurized system!

The hydraulic tank contains hot oil under pressure. To prevent burns from the sudden release of hot oil, relieve the tank pressure with the engine off. Relieve pressure by slowly turning the cap until the cap reaches the secondary stop.

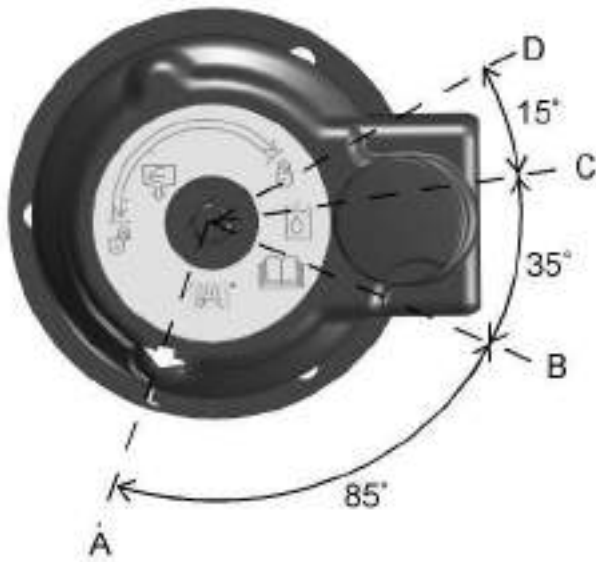


Illustration 828

g06184990

Filler cap

- (A) LOCK position
 (B) PRESSURE RELEASE - START position
 (C) PRESSURE RELEASE - END position
 (D) OPEN position

4. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 828 for filler cap positions.
 - a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).
 - b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
 - c. Move the arrow from position (C) to position (D).
 - d. After the tank pressure is relieved, tighten the filler cap.



Illustration 829

g06182179

5. Remove the hydraulic tank access cover that is located under the upper structure. Removing the cover will allow access to the drain valve.

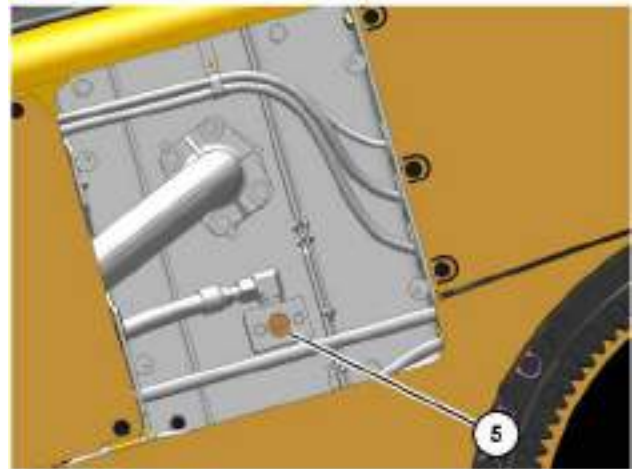


Illustration 830

g06182182

(5) Plug

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information that pertains to containing fluid spillage.

6. Remove plug (5).
7. Inspect the O-ring. Replace the O-ring if wear or damage is evident.

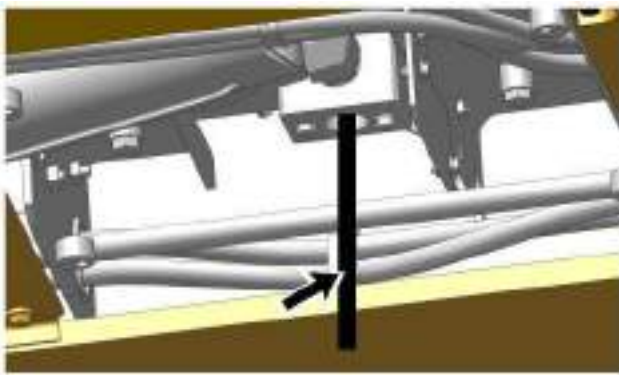


Illustration 831

g06182192

8. Use a bar to push the plunger up to allow the oil to drain.

9. Drain the oil into a suitable container.

Note: Dispose of used fluids according to local regulations.

10. After the oil has been drained, clean drain plug (5) and install. Tighten the plug to $68 \pm 7 \text{ N}\cdot\text{m}$ ($50 \pm 5 \text{ lb ft}$).

11. Open the access door on the right side of the machine.

12. Clean the pump, the hydraulic lines, and the hydraulic tank.

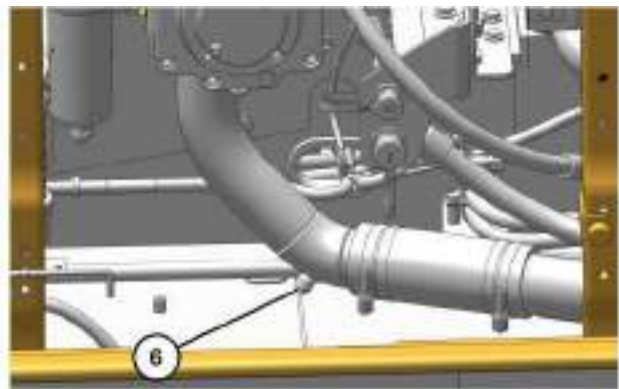


Illustration 832

g06182196

13. Remove plug (6) from the tube. Allow the oil to drain into a container.

14. Inspect the O-ring. Replace the O-ring if wear or damage is evident.

15. Clean the plug. Install the plug and the O-ring into the drain port.

Hydraulic Tank Screen - Clean



Illustration 833

g06182201

(7) Bolts
(8) Washers
(9) Cover

1. Remove bolts (7), washers (8), and cover (9).



Illustration 834

g06182213

(10) Spring
(11) Screen

2. Remove spring (10) and screen (11).

Note: Do not allow spring (10) to fall back into the tank.



Illustration 835

g06182515

(12) O-ring seal

3. Remove O-ring seal (12) from the screen.

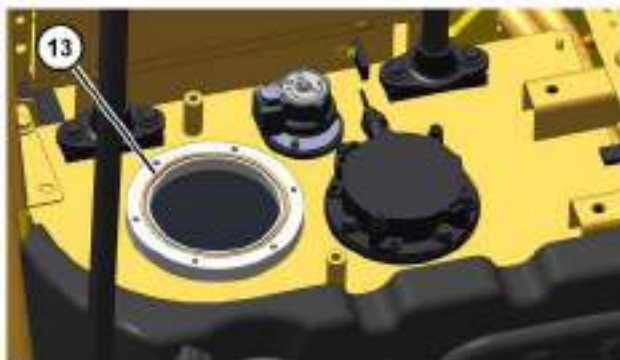


Illustration 836

g06182229

(13) O-ring seal

4. Remove O-ring seal (13) from the tank.
5. Inspect O-ring seals (12) and (13). Replace the O-ring seals if wear or damage is evident.
6. Wash the screen in a clean nonflammable solvent. Allow the screen to dry. Inspect the screen. Replace the screen, if the screen is damaged.
7. Install O-ring seal (12) on screen (11).
8. Install screen (11) and spring (10). Then install cover (9), washers (8), and bolts (7).

Note: Make sure that the O-ring seals and the spring are properly positioned during installation.

Case Drain Filter - Clean

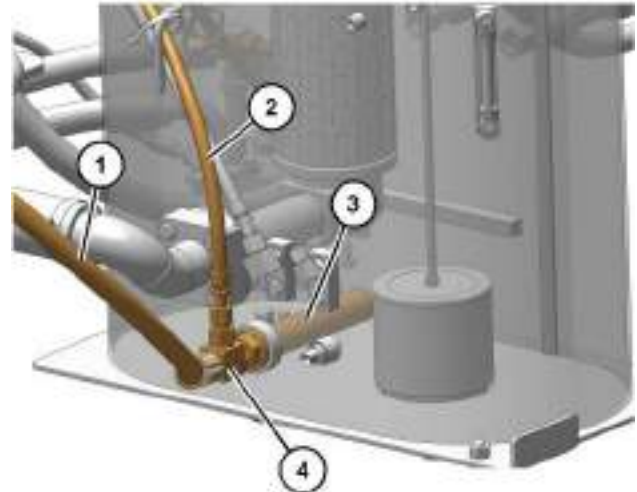


Illustration 837

g06220559

- (1) Hose
- (2) Hose
- (3) Case drain filter
- (4) Tee

1. Remove hose (1) and hose (2) from tee (4). Remove tee (4).
2. Remove case drain filter (3) from the hydraulic tank.
3. Wash the screen of the case drain filter in a clean nonflammable solvent. Allow the filter to dry. Inspect the filter. Replace the filter if the filter is damaged.
4. Inspect the O-ring seal on the filter. Replace the O-ring seal if wear or damage is evident.
5. Install the filter in the hydraulic tank. Tighten the filter to $175 \pm 26 \text{ N}\cdot\text{m}$ ($129 \pm 19 \text{ lb ft}$).
6. Install the tee onto the filter. Tighten the tee to $65 \pm 10 \text{ N}\cdot\text{m}$ ($48 \pm 7 \text{ lb ft}$).
7. Install the two hoses onto the tee.

Hydraulic System Oil - Fill

1. Fill the hydraulic system oil tank. Refer to Operation and Maintenance Manual, "Capacities (Refill)".
2. Inspect the O-ring seal on the filler cap for damage. Replace the O-ring, if necessary. Clean the filler cap. Install the filler cap.

Note: Make no attempt to start the engine until the pump has been filled with hydraulic oil. Serious damage to the hydraulic components can result.

Main Pump and Hydraulic System Air Purge

1. Access the hydraulic pump. The hydraulic pump is located behind the right access door.

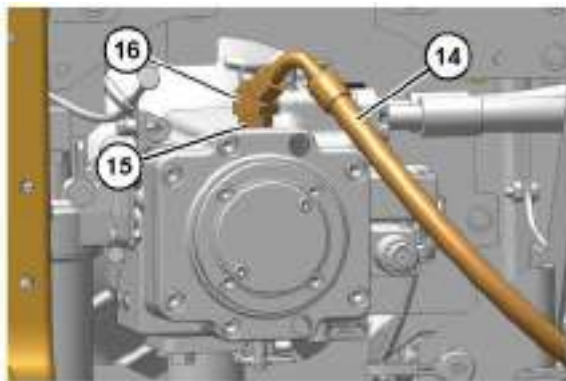


Illustration 838

g06205207

- (14) Hose
(15) Connector
(16) Elbow

2. While the engine is stopped, remove hose (14), elbow (15), and connector (16) from the top of the pump. Add hydraulic oil through the opening.
3. Check the condition of the seals. If a seal is damaged, replace the seal.
4. After the pump has been filled with oil, install drain hose (14), connector (15), and seal (16) to the original locations.
5. Start the engine. When the engine is at low idle, raise the boom. Hold the boom in this position.
6. Stop the engine. Slowly lower the boom until the work tool is on the ground. The hydraulic tank will pressurize.
7. Slowly loosen hose (14) until hydraulic oil flows from the connection. Oil flowing from the connection indicates that the air has been released from the pump.
8. Tighten hose (14).
9. Start the engine. Operate the engine at idling speed for 5 minutes.



Illustration 839

g06181120

10. Operate the joysticks to circulate the hydraulic oil. Lower the bucket to the ground so that the stick is vertical to the ground. Stop the engine.

11. Check the hydraulic oil level.

Reference: For the correct procedure, refer to Operation and Maintenance Manual, "Hydraulic System Oil Level - Check".

12. Close the access door.

13. Close the engine hood and latch the engine hood.

Fast Fill

If your machine is equipped with a deluxe service center, you may drain the hydraulic oil through the fast fill port. You may also add the hydraulic oil through a fast fill port.



Illustration 840 g06182545

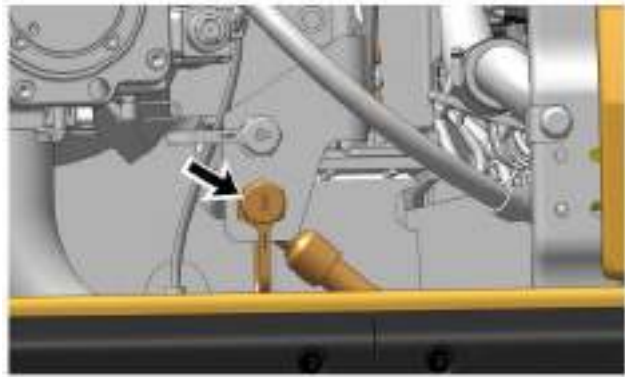


Illustration 841 g06182544

1. Remove the dust cover.
2. Attach the hose to the male coupling.
3. Drain the oil or add the oil, as needed.

i07833122

Hydraulic System Oil Filter (Return) - Replace

SMCS Code: 5068-510-RJ

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

The return filter is a cartridge type filter. The amount of foreign material that enters the hydraulic system is reduced when the filter element is replaced.

Two different filters are available for the return filter. One filter is used for standard applications such as digging and normal use of a hammer. The second filter is used for an application such as demolishing a ceiling in a tunnel with a hammer.

Note: If the message display shows that the hydraulic return filter is plugged, turn off the machine. After you make sure that the warning has disappeared, start the machine and run the machine on level ground for approximately 10 minutes. If the warning still appears in the message display, inspect the filter and replace the filter, if necessary.

Hydraulic Hammer Use

The use of hydraulic hammers shortens the life of hydraulic oil. If a hydraulic hammer is used, the maintenance interval is shortened, refer to Table 43 for the intervals.

Table 43

Percentage of Hammer Use	Hydraulic System Oil Filter (Return) - Replace
50%	Every 500 service hours
100%	Every 250 service hours

Return Filter Replacement Procedure

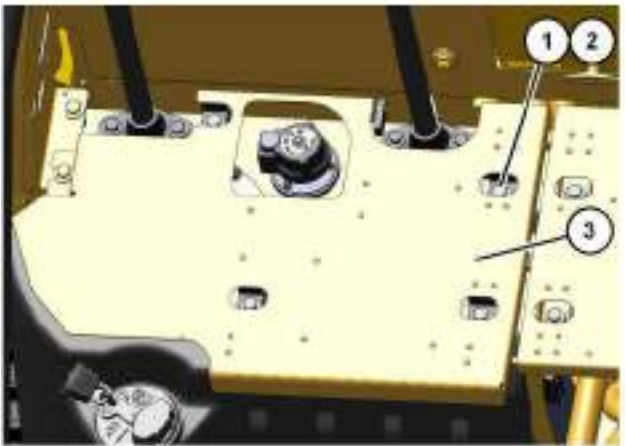


Illustration 842 g06182169

1. Remove five bolts (1) and washers (2). Remove cover (3) from the top of the hydraulic tank.
2. Clean the area thoroughly to keep dirt out of the return filter and filler cap.

WARNING

Pressurized system!

The hydraulic tank contains hot oil under pressure. To prevent burns from the sudden release of hot oil, relieve the tank pressure with the engine off. Relieve pressure by slowly turning the cap until the cap reaches the secondary stop.

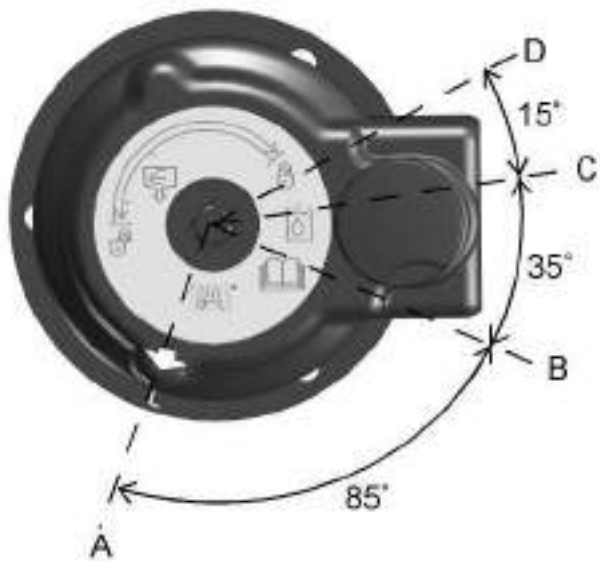


Illustration 843

g06184990

Filler cap

- (A) LOCK position
- (B) PRESSURE RELEASE - START position
- (C) PRESSURE RELEASE - END position
- (D) OPEN position

3. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 843 for filler cap positions.
 - a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).
 - b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
 - c. Move the arrow from position (C) to position (D).
 - d. After the tank pressure is relieved, tighten the filler cap on the hydraulic tank to position (A).
4. Check the hydraulic system oil level.

Reference: For the correct procedure, refer to Operation and Maintenance Manual, "Hydraulic System Oil Level - Check".



Illustration 844

g06254537

5. Remove four bolts (4), disconnect the harness connector from the filter bypass switch, and remove cover assembly (5) from the tank. Inspect the O-ring on the cover for damage and replace as necessary.



Illustration 845

g06254829

6. Remove filter element (6) and discard. Install a new element into the filter case.



Illustration 846

g06254537

7. Place cover assembly (5) into position in the tank. Install four bolts (4) and tighten to $30 \pm 7 \text{ N}\cdot\text{m}$ ($22 \pm 5 \text{ lb ft}$). Install the harness connector on the filter bypass switch.

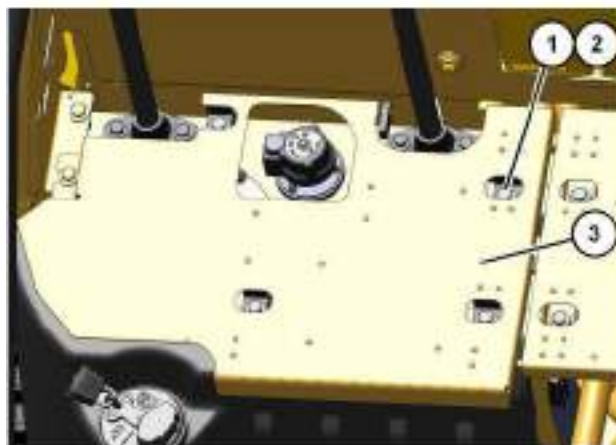


Illustration 847

g06182169

8. Position cover (3) in place on the top of the hydraulic tank. Install five bolts (1) and washers (2).

i07174987

Hydraulic System Oil Level - Check

SMCS Code: 5050-535

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

NOTICE

Never remove the fill/vent plug from the hydraulic tank if the oil is hot.

Air can enter the system and cause pump damage.

Maintenance Section
Hydraulic System Oil Level - Check



Illustration 848

g06181120

1. Park the machine on level ground. Lower the bucket to the ground with the stick in a vertical position, as shown.



Illustration 849

g06219991

2. Open the access door on the right side of the machine.

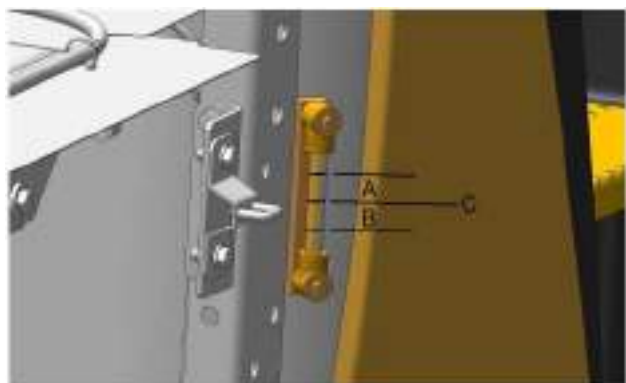


Illustration 850

g06182648

- (A) High temperature range
(B) Low temperature range
(C) Mid temperature range

3. If the hydraulic oil temperature is between 10° to 30° C (50° to 86° F), maintain the oil level in low temperature range (B). If the hydraulic oil temperature is between 50° to 80° C (122° to 187° F), maintain the oil level in high temperature range (A). If the hydraulic oil temperature is between 31° to 49° C (87° to 121° F), maintain the oil level in mid temperature range (C).

4. Close the access door.

Note: Perform Step 5 through Step 8 if the oil level is low.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information that pertains to Containing Fluid Spillage.



WARNING

Pressurized system!

The hydraulic tank contains hot oil under pressure. To prevent burns from the sudden release of hot oil, relieve the tank pressure with the engine off. Relieve pressure by slowly turning the cap until the cap reaches the secondary stop.

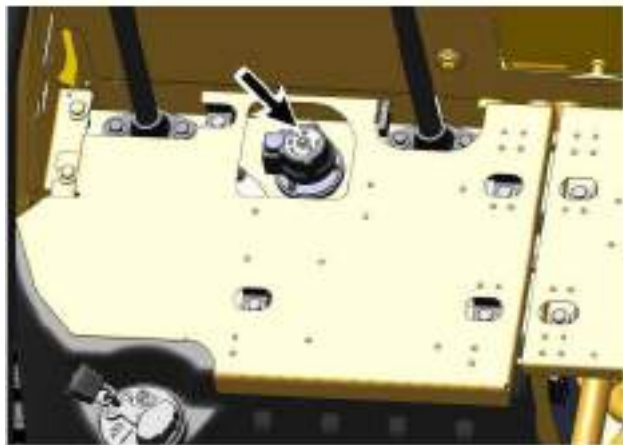


Illustration 851

g06182653

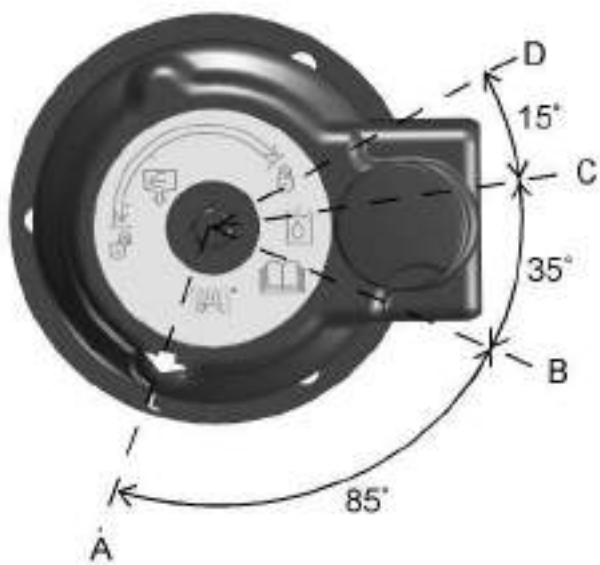


Illustration 852

g06184990

Filler cap

- (A) LOCK position
(B) PRESSURE RELEASE - START position
(C) PRESSURE RELEASE - END position
(D) OPEN position

5. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 852 for filler cap positions.

- a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).

- b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
- c. Move the arrow from position (C) to position (D).
- d. After the tank pressure is relieved, remove the filler cap.

6. Add oil if necessary. See Operation and Maintenance, "Lubricant Viscosities"
7. Check the O-ring seal on the filler cap. Replace the O-ring seal if the seal is damaged.
8. Clean the filler cap and install on the tank. Tighten the filler cap on the hydraulic tank to position (A).

i06972505

Hydraulic System Oil Sample - Obtain

SMCS Code: 5050-008-OC; 5095-008; 5095-SM; 7542-008; 7542

Note: If Cat HYDO Advanced hydraulic oils are used, the hydraulic oil change interval is extended to 6000 hours. S-O-S services after 3,000 hours is strongly recommended. Consult your Cat dealer for details.

The hydraulic oil sampling valve is near the accumulator on the main valve.

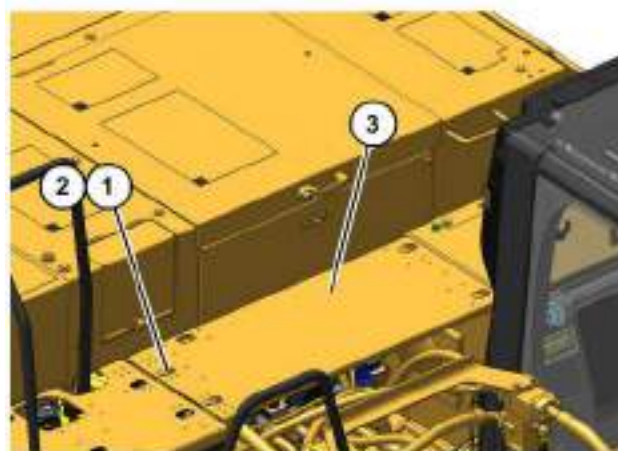


Illustration 853

g06225772

1. Remove bolts (1), washers (2), and cover (3).

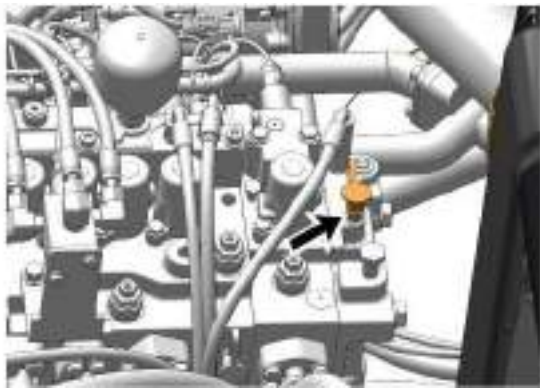


Illustration 854 g06183830

2. Obtain a sample of the hydraulic oil from the hydraulic oil sampling valve that is on the main valve next to the accumulator.
3. Install cover (3), washers (2), and bolts (1).

Refer to Special Publication, SEBU6250, "S·O·S Oil Analysis" for information that pertains to obtaining a sample of the hydraulic oil. Refer to Special Publication, PEGJ0047, "How To Take A Good Oil Sample" for more information about obtaining a sample of the hydraulic oil.

i03753191

Indicators and Gauges - Test

SMCS Code: 7450-081; 7490-081

1. Look for broken lenses on the gauges, broken indicator lights, broken switches, and other broken components in the cab.
2. Start the engine.
3. Look for inoperative gauges.
4. Turn on all machine lights. Check for proper operation.
5. Move the machine forward. Release the travel levers and the travel pedals. The machine should stop.
6. Stop the engine.
7. Make any repairs that are required before operating the machine.

i07833447

Oil Filter (Hydraulic Hammer) - Replace (If Equipped)

SMCS Code: 5068-510

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

The use of hydraulic hammers shortens the life of hydraulic oil. If a hydraulic hammer is used, the maintenance interval is shortened, refer to Table 44 for the intervals.

Table 44

Percentage of Hammer Use	Oil Filter (Hydraulic Hammer) - Replace
50%	Every 250 service hours
100%	Every 100 service hours



Illustration 855 g06181120

1. Park the machine on level ground in the service position as shown.
2. Move the hydraulic lockout control to the UNLOCKED position.
3. Turn the engine start switch to the ON position.
4. Move the joysticks and the travel levers/pedals to the full stroke positions to relieve the pressure in the hydraulic lines.
5. Turn the engine start switch to the OFF position and return the lever for the hydraulic lockout control to the LOCKED position.

⚠ WARNING**Pressurized system!**

The hydraulic tank contains hot oil under pressure. To prevent burns from the sudden release of hot oil, relieve the tank pressure with the engine off. Relieve pressure by slowly turning the cap until the cap reaches the secondary stop.

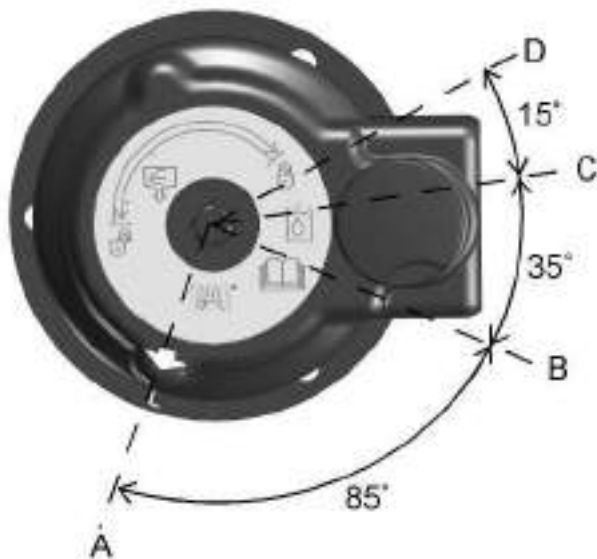


Illustration 856

g06184990

Filler cap

- (A) LOCK position
- (B) PRESSURE RELEASE - START position
- (C) PRESSURE RELEASE - END position
- (D) OPEN position

6. Release the pressure that may be present in the return hydraulic circuit with the following procedure. Refer to Illustration 856 for filler cap positions.
 - a. Turn the filler cap counterclockwise and move the arrow from position (A) to position (B).
 - b. Release the pressure for a minimum of 45 seconds by moving the arrow from position (B) to position (C).
 - c. Move the arrow from position (C) to position (D).
 - d. After the tank pressure is relieved, remove the filler cap.
7. The oil filter for the hammer is located near the base of the boom.

Note: Some configurations may invert the hammer filter installation.

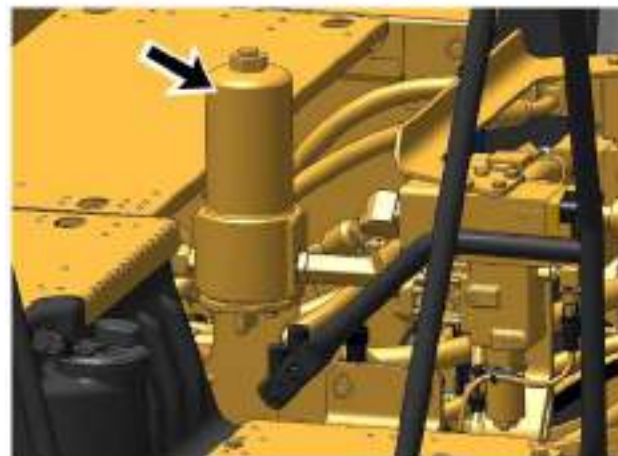


Illustration 857

g06211151

8. Position a suitable container to contain the oil.

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information on containing fluid spillage.

9. Loosen the filter housing and remove. Inspect the plug on the top of the housing. If there are signs of leakage, replace the O-ring on the plug. Tighten the plug to $59 \pm 5 \text{ N}\cdot\text{m}$ ($44 \pm 4 \text{ lb ft}$).

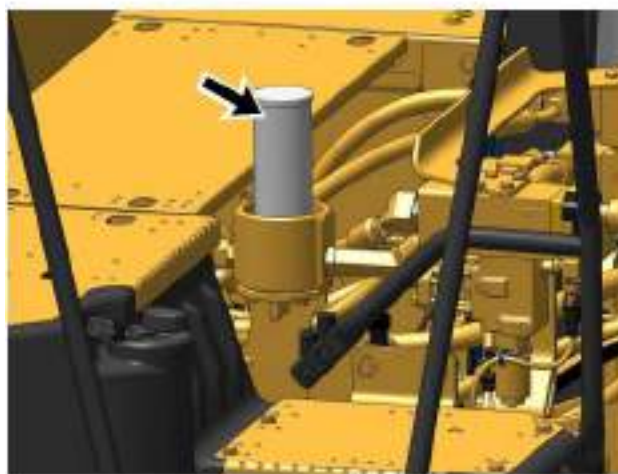


Illustration 858

g06211156

10. Remove the filter element and discard. The element cannot be reused.

Note: Used filter elements should always be disposed of according to local regulations.

11. Install the new filter element.
12. Clean the filter housing and install on the filter base. Tighten the nut on the filter case to $98 \pm 10 \text{ N}\cdot\text{m}$ ($72 \pm 7 \text{ lb ft}$).
13. Start the engine and operate the machine slowly for 10 to 15 minutes. Move each cylinder evenly through several cycles and operate the hammer.
14. Return the machine to the service position. Check the machine for oil leaks.
15. Stop the engine.
16. Check the hydraulic oil level.

Reference: For the correct procedure, refer to Operation and Maintenance Manual, "Hydraulic System Oil Level - Check".

i07349186

Oil Filter - Inspect

SMCS Code: 1308-507; 5068-507

Inspect a Used Filter for Debris



Illustration 859

g06224663

The element is shown with debris.

Use a filter cutter to cut the filter element open. Spread apart the pleats and inspect the element for metal and for other debris. An excessive amount of debris in the filter element can indicate a possible failure.

If metals are found in the filter element, a magnet can be used to differentiate between ferrous metals and nonferrous metals.

Ferrous metals can indicate wear from steel parts and on cast iron parts.

Nonferrous metals can indicate wear from the aluminum parts of the engine such as main bearings, rod bearings, or turbocharger bearings.

Small amounts of debris may be found in the filter element. This debris could be caused by friction and by normal wear. Consult your Cat dealer to arrange for further analysis if an excessive amount of debris is found.

Using an oil filter element that is not recommended by Caterpillar can result in severe engine damage to engine bearings, to the crankshaft, and to other parts. This can result in larger particles in unfiltered oil. The particles could enter the lubricating system and the particles could cause damage.

i06972489

Radiator, Aftercooler and Oil Cooler Cores - Clean

SMCS Code: 1063-070-KO; 1353-070-KO; 1374-070-KO

WARNING

Personal injury can result from air pressure.

Personal injury can result without following proper procedure. When using pressure air, wear a protective face shield and protective clothing.

Maximum air pressure at the nozzle must be less than 205 kPa (30 psi) for cleaning purposes.



Illustration 860

g06179792

1. Open the access door on the left side of the machine.

i08192239

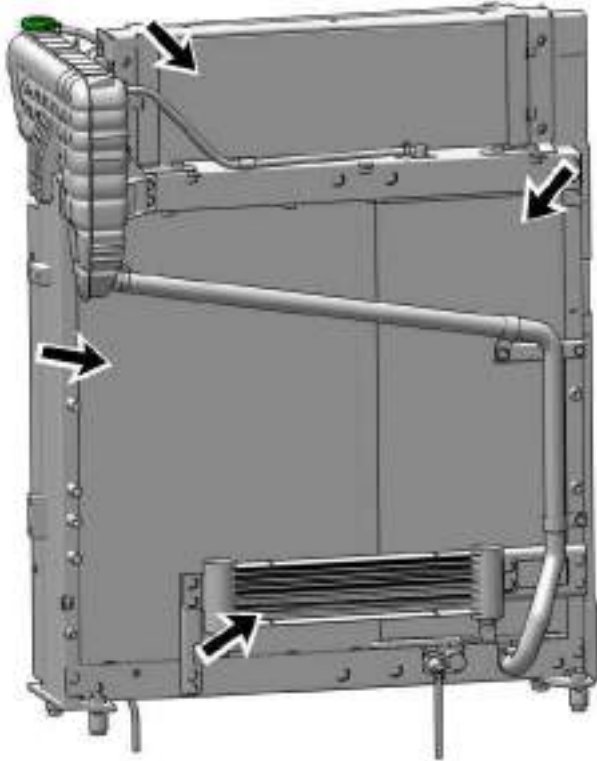


Illustration 861

g06183814

2. Remove dust and debris from all the core fins.

Compressed air is preferred, but high-pressure water or steam can be used to remove dust and general debris from a core.

See Special Publication, SEBD0518, "Know Your Cooling System" for more detailed information about cleaning core fins.

3. Close the access door on the left side of the machine.

Receiver Dryer (Refrigerant) - Replace

SMCS Code: 7322-510; 7322-710

WARNING

Personal injury can result from contact with refrigerant.

Contact with refrigerant can cause frost bite. Keep face and hands away to help prevent injury.

Protective goggles must always be worn when refrigerant lines are opened, even if the gauges indicate the system is empty of refrigerant.

Always use precaution when a fitting is removed. Slowly loosen the fitting. If the system is still under pressure, release it slowly in a well ventilated area.

Personal injury or death can result from inhaling refrigerant through a lit cigarette.

Inhaling air conditioner refrigerant gas through a lit cigarette or other smoking method or inhaling fumes released from a flame contacting air conditioner refrigerant gas, can cause bodily harm or death.

Do not smoke when servicing air conditioners or wherever refrigerant gas may be present.

Use a certified recovery and recycling cart to properly remove the refrigerant from the air conditioning system.

NOTICE

If the refrigerant system has been open to the outside air (without being plugged) for more than 30 minutes, the receiver-dryer must be replaced. Moisture will enter an open refrigerant system and cause corrosion which will lead to component failure.

Prepare the machine for maintenance. Refer to Operation and Maintenance Manual, "Prepare the Machine for Maintenance".

Refer to Service Manual, "Air Conditioning and Heating R-134a for All Caterpillar Machines" for the proper procedure to change the receiver-dryer assembly and for the procedure to reclaim the refrigerant gas.

i07349192

i07103309

Rollover Protective Structure (ROPS) - Inspect

SMCS Code: 7323-040; 7325-040



Illustration 862

g06184357

Consult your Cat dealer for repair of any cracks in the ROPS.

Inspect the ROPS for loose bolts or for damaged bolts. Replace any damaged bolts or missing bolts with original equipment parts only. Tighten the M24 bolt (1) to $425 \pm 50 \text{ N}\cdot\text{m}$ ($315 \pm 40 \text{ lb ft}$).

Note: Apply oil to all ROPS bolt threads before you install the bolts. Failure to apply oil to the bolt threads can result in improper bolt torque.

Do not straighten the ROPS. Do not repair the ROPS by welding reinforcement plates to the ROPS.

Consult your Cat dealer for inspection of any potential damage or repair of any damage to any operator protective structure. (Including ROPS, FOPS, TOPS, OPS, and OPG) Refer to Special Instruction, SEHS6929, "Inspection, Maintenance, and Repair of Operator Protective Structures (OPS) and Attachment Installation Guidelines for All Earthmoving Machinery"

Seat Belt - Inspect

SMCS Code: 7327-040

Always inspect the condition of the seat belt and the condition of the seat belt mounting hardware before you operate the machine. Replace any parts that are damaged or worn before you operate the machine.



Illustration 863

g06224278

Typical example

Inspect buckle (2) for wear or for damage. If the buckle is worn or damaged, replace the seat belt.

Inspect seat belt (1) for webbing that is worn or frayed. Replace the seat belt if the webbing is worn or frayed.

Inspect all seat belt mounting hardware for wear or for damage. Replace any mounting hardware that is worn or damaged. Make sure that the mounting bolts are tight.

If your machine is equipped with a seat belt extension, also perform this inspection procedure for the seat belt extension.

Contact your Cat dealer for the replacement of the seat belt and the mounting hardware.

Note: The seat belt should be replaced within 3 years of the date of installation. A date of installation label is attached to the seat belt retractor and buckle. If the date of installation label is missing, replace belt within 3 years from the year of manufacture as indicated on belt webbing label, buckle housing, or installation tags (non-retractable belts).

i06970675

Seat Belt - Replace

SMCS Code: 7327-510

The seat belt should be replaced within 3 years of the date of installation. A date of installation label is attached to the seat belt retractor and buckle. If the date of installation label is missing, replace the belt within 3 years from the year of manufacture as indicated on the belt webbing label, buckle housing, or installation tags (non-retractable belts).

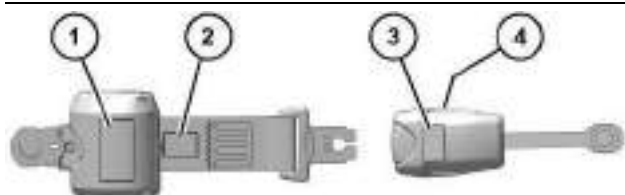


Illustration 864

g06183390

- (1) Date of installation (retractor)
- (2) Year of manufacture (tag) (fully extended web)
- (3) Date of installation (buckle)
- (4) Year of manufacture (underside) (buckle)

Consult your Cat dealer for the replacement of the seat belt and the mounting hardware.

Determine the age of a new seat belt before installing on seat. A manufacture label is on the belt webbing and imprinted on the belt buckle. Do not exceed the install by date on the label.

A complete seat belt system should be installed with new mounting hardware.

Date of installation labels should be marked and affixed to the seat belt retractor and buckle.

Note: Date of installation labels should be permanently marked by punch (retractable belt) or stamp (non-retractable belt).

If your machine is equipped with a seat belt extension, also perform this replacement procedure for the seat belt extension.

i06991878

Shovel Crane - Inspect (If Equipped)

SMCS Code: 6500-040

S/N: HEX1-Up

WARNING

Do not operate the shovel crane with a hook that has cracks or deformities. Failure to follow these instructions may cause the load to fall and result in injury or death. Replace the shovel crane hook if there are any signs of cracks or deformities.

1. Position the machine on a level surface and retract the bucket. Lower the bucket to the ground.
2. Move the hydraulic lockout control to the LOCKED position. Stop the engine.

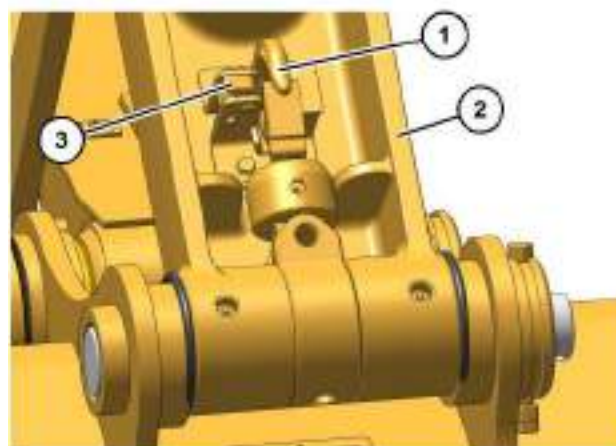


Illustration 865

g06191360

Type 2

- (1) Shovel crane hook
- (2) Bucket linkage
- (3) Latch assembly

3. Unlock shovel crane hook (1) from latch assembly (3) for inspection.
4. Inspect the shovel crane hook and the hook latch. Make any repairs before operation of the shovel crane.

Inspect Shovel Crane Hook

Ensure that the shovel crane is properly lubricated. Refer to this Operation and Maintenance Manual, "Shovel Crane - Lubricate" for the correct procedure.



Illustration 866

g06191406

Inspect for notable scratches, tears, or welded parts.



Illustration 869

g06191411

Inspect the tip of shovel crane hook and ensure that the tip is not bent to either side.



Illustration 867

g06191407

Visually inspect for corrosion and rust. If corrosion or rust is found, polish the area with a grinder and lubricate the hook with oil.

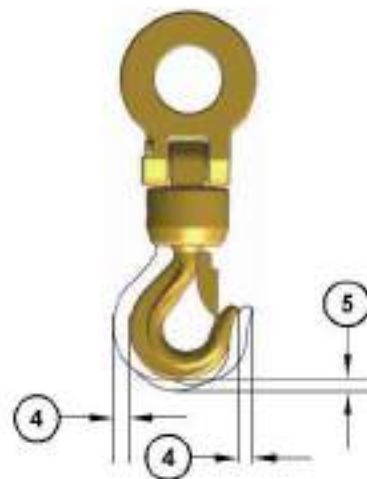


Illustration 870

g06191428

(4) Horizontal movement
(5) Vertical movement

Inspect the shovel crane hook for excessive movement. The horizontal movement (4) of the hook should not exceed 5 mm (0.2 inch). The vertical movement (5) of the hook should not exceed 4 mm (0.2 inch.) vertically .



Illustration 868

g06191409

Inspect the shovel crane swivel. The swivel should rotate smoothly and not rattle.

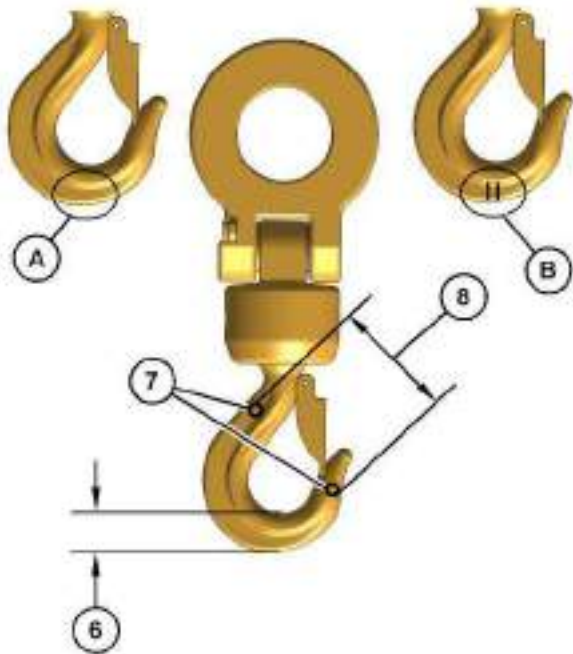


Illustration 871

g06191434

(6) Width
(7) Punch marks
(8) Length

There are two types of hooks:

Hook (A) – This type of hook will not have a mark to indicate the area intended to bear the weight of a suspended load.

Hook (B) – This type of hook will have a mark to indicate the area intended to bear the weight of a suspended load.

Measure the shovel crane hook to ensure that the specifications do not exceed the thresholds found in the table below.

Table 45

Disposal Reference Chart				
Hook Type	New Shovel Hook Specifications for Reference		Shovel Hook Specification Threshold	
	Length (8)	Width (6)	Length (8)	Width (6)
A	66 mm (2.60 inch)	31.8 mm (1.25 inch)	69.3 mm (2.73 inch)	30.2 mm (1.19 inch)
B	65 mm (2.55 inch)	31.8 mm (1.25 inch)	68.2 mm (2.69 inch)	30.2 mm (1.19 inch)

Note: Length (8) is the distance between punch marks (7).

Replace shovel crane hook for any of the following reasons:

- Deep scratches, tears, or welded parts
- Corrosion or rust that cannot be removed with a grinder
- Hook swivel does not rotate smoothly
- Tip of the hook is bent to either side
- Hook has excessive movement in the swivel
- Hook exceeds the specification thresholds found in the disposal reference chart.

Inspect the Shovel Crane Hook Latch



Illustration 872

g06191466

(C) Open
(D) Closed
(E) Operating range

Inspect the latch to ensure that the latch moves smoothly in operating range (E).

X



Illustration 873

g06191481

Inspect the latch for excessive play (F). Ensure that the spring is not broken and the latch does not rattle.

X



Illustration 874

g06191492

Inspect the latch to ensure that the latch closes completely and there is no gap between the latch and the hook. The latch should rest firmly on the tip of the hook.

X



Illustration 875

g06191696

Inspect the latch to ensure that the latch is not bent or deformed. The latch should be centered on the tip of the hook.

Replace shovel crane latch for any of the following reasons:

- Latch does not move smoothly within the operating range
- Latch has excessive movement or rattles
- Latch does not fully close
- Latch is bent or deformed

i06995214

Shovel Crane - Lubricate (If Equipped)

SMCS Code: 6500

S/N: HEX1–Up

Note: Caterpillar recommends the use of 5% molybdenum grease for lubricating the shovel crane linkage. Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on molybdenum grease.

Wipe all fittings before you apply lubricant.

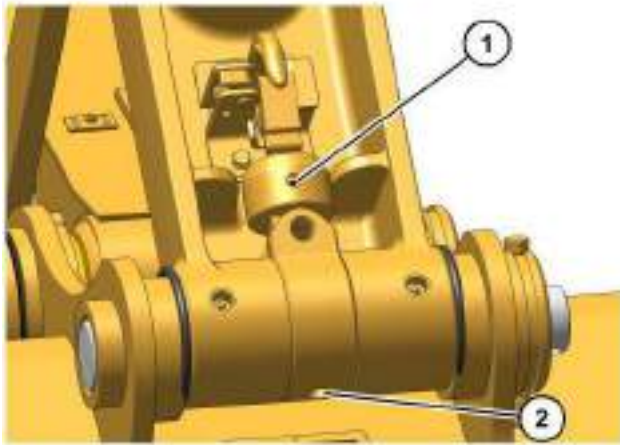


Illustration 876

g06191976

1. Apply lubricant through fittings (1) and (2). Continue to fill with grease until the grease is visible.

Note: Service the above fittings after you operate the shovel crane under water.

2. Check the hook for cracks and deformities. Replace the hook if necessary.

Note: Refer to this Operation and Maintenance Manual, "Shovel Crane - Inspect"

i06987212

Swing Bearing - Lubricate

SMCS Code: 7063-086

Note: Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on grease.

Note: Do not overgrease the swing bearings. Do not grease more than the recommended maintenance interval. Refer to Operation and Maintenance Manual, "Maintenance Interval Schedule" for more information.

Wipe the fittings before you lubricate the swing bearing.

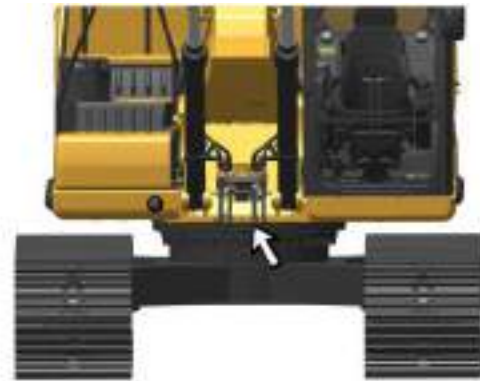


Illustration 877

g06210366

The swing bearing grease zerks are located at the front of the swing drive housing near the boom cylinders.

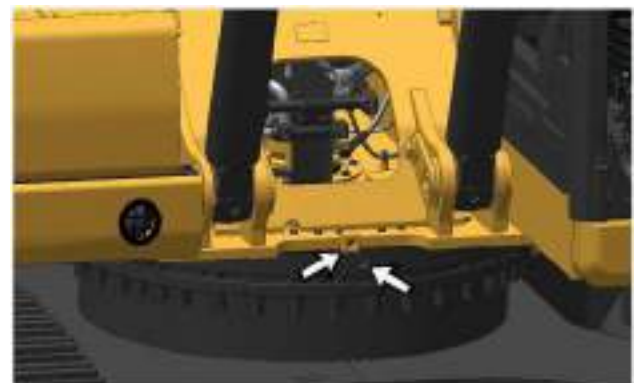


Illustration 878

g06188212

Apply lubricant through the fittings until the lubricant overflows the bearing seals.

i07349198

Swing Drive Oil - Change

SMCS Code: 5459-044

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.



Illustration 879

g06188541

1. Remove the access cover that is located below the swing drives.

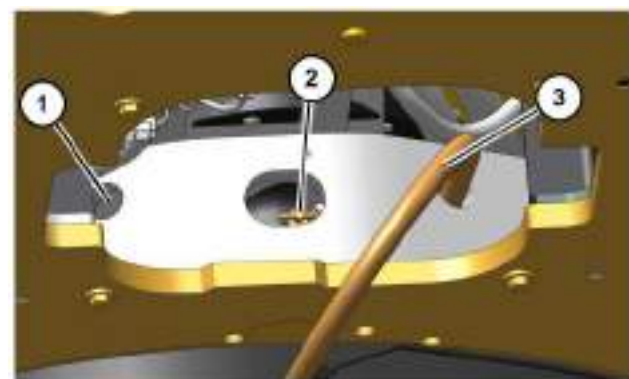


Illustration 880

g06192018

Note: Refer to Operation and Maintenance Manual, "General Hazard Information" for information that pertains to Containing Fluid Spillage.

2. Remove drain hose (3) from holder (1) on the upper frame. Face the end of the hose toward the container.
3. Loosen drain valve (2). Drain the oil into a suitable container.

Note: Drained fluids should always be disposed of according to local regulations.

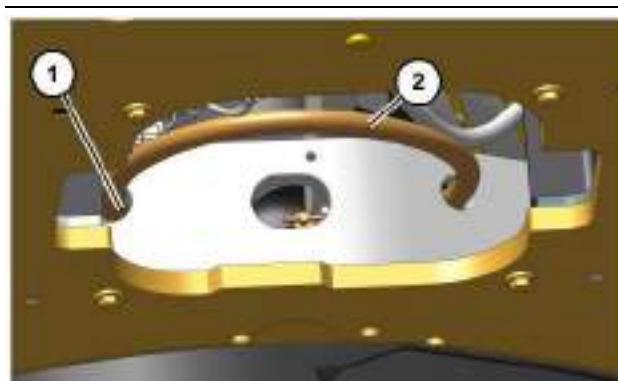


Illustration 881

g06192024

4. Tighten the drain valve. Return the drain hose to holder (1). Make sure that the end of the hose is facing upward.

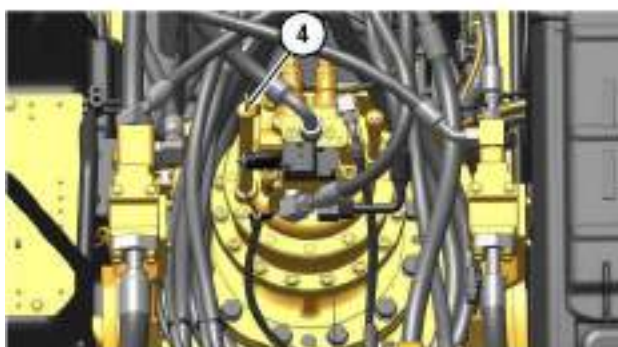


Illustration 882

g06188581

5. Remove dipstick (4).
6. Add the specified quantity of oil through the dipstick tube. See Operation and Maintenance, "Capacities (Refill)".



Illustration 883

g06188660

7. Maintain the oil level between the “ADD” and “FULL” marks on the dipstick.
8. Check the oil that has been drained for metal chips or metal particles. Consult your Cat dealer if any metal chips or metal particles are found.
9. Drained materials should always be disposed of according to local regulations.

i06988089

Swing Drive Oil Level - Check

SMCS Code: 5459-535-FLV

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.



Illustration 884

g06188672

The dipstick for the swing drive oil is on the swing drive at the rear base of the boom.

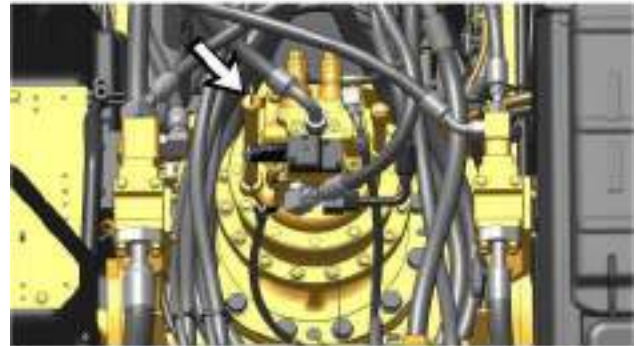


Illustration 885

g06188677

1. Remove the dipstick.



Illustration 886

g06188660

Note: Refer to Operation and Maintenance Manual, “General Hazard Information” for information that pertains to Containing Fluid Spillage.

2. Check the dipstick. Maintain the oil level between the “ADD” and “FULL” marks on the dipstick. Add oil through the dipstick tube, if necessary. Refer to Operation and Maintenance Manual, “Lubricant Viscosities” when you select an oil. If the oil level is above the “FULL” line, then remove oil from the system. Restore the oil to the correct level position.
3. Insert the dipstick.

i06988143

Swing Drive Oil Sample - Obtain

SMCS Code: 5459-554-OC; 5459-008; 5459-008-OC; 5459-OC; 7542-008

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

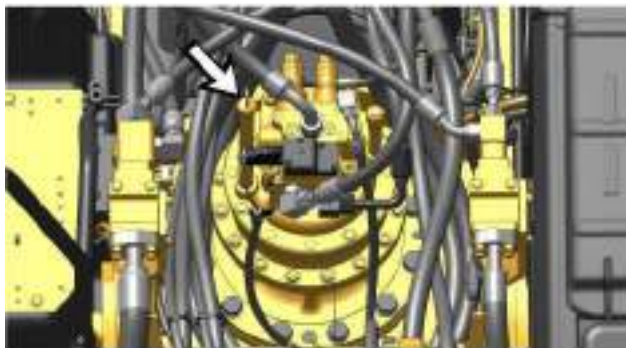


Illustration 887

g06188677

Obtain an oil sample of the swing drive oil through the opening for the dipstick. Refer to Special Publication, SEBU6250, "S·O·S Oil Analysis" for information that pertains to obtaining an oil sample from the swing drive housing. Refer to Special Publication, PEGJ0047, "How To Take A Good Oil Sample" for more information about obtaining an oil sample from the swing drive housing.

i06988169

Swing Gear - Lubricate

SMCS Code: 7063-086

Note: Refer to Special Publication, SEBU6250, "Caterpillar Machine Fluids Recommendations" for more information on grease.

NOTICE

Improper lubrication can cause damage to machine components.

To avoid damage, make sure that the proper amount of grease is applied to the swing drive.

When the amount of grease in the compartment becomes too large, the agitation loss becomes large, thereby accelerating grease deterioration.

Grease deterioration can cause damage to the pinion gear of the swing drive and swing internal gear.

Not enough grease will result in poor gear lubrication.

Remove the inspection cover that is located near the boom base. Inspect the grease.

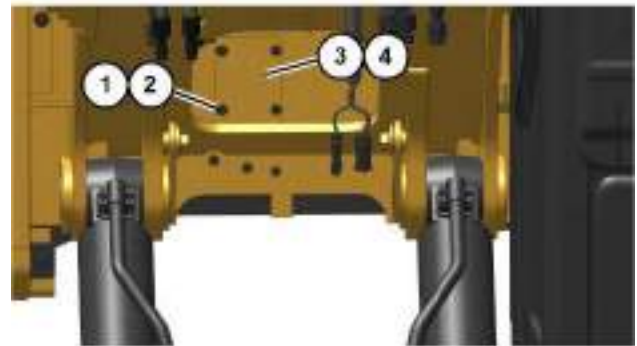


Illustration 888

g06188728

- (1) Bolts
- (2) Washers
- (3) Cover
- (4) Gasket

1. Remove bolts (1) and washers (2). Remove cover (3) and gasket (4).

2. Inspect gasket (4). Replace the gasket if damage is evident.

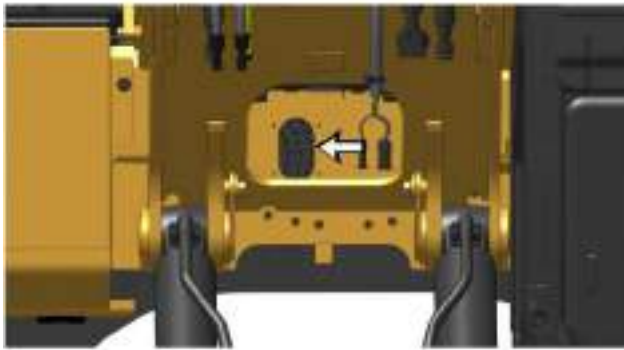


Illustration 889

g06188736

3. Check the level of grease. The level of grease is correct when:

- Waves of grease are present from the rotating swing drive pinion.
- The grease is evenly distributed on the floor of the pan.

Note: Smeared or waveless areas are evidence for a lack of grease.

Note: Add grease, as needed. Remove grease, as needed. Too much grease will result in the deterioration of the grease because of excessive movement of the grease. Too little grease will result in poor lubrication of the swing gear.

Refer to Operation and Maintenance Manual, "Capacities (Refill)" for the size of the pan.

4. Check for contamination and for discolored grease.

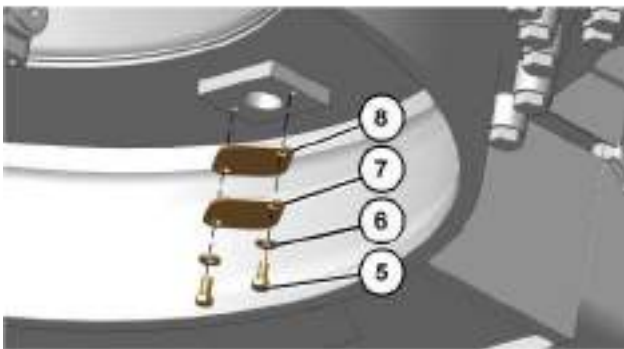


Illustration 890

g06188788

- (5) Bolts
(6) Washers
(7) Cover
(8) Gasket

5. If the grease is contaminated or discolored with water, change the grease. Remove the covers from below the swing drive underneath the undercarriage frame.
6. Remove bolts (5), washers (6), cover (7), and gasket (8) to allow the water to drain. When you reinstall cover (7), inspect gasket (8). Replace the gasket if damage is evident.



Illustration 891

g06188791

7. Raise the boom and turn the upper structure by 1/4 turn. Lower the bucket to the ground.
8. Repeat Step 7 at every 1/4 turn in four places. Add grease, as needed.
9. Install gasket (4), cover (3), washers (2), and bolts (1).

i06988628

Track Adjustment - Adjust

SMCS Code: 4170-025

WARNING

Personal injury or death can result from grease under pressure.

Grease coming out of the relief valve under pressure can penetrate the body causing injury or death.

Do not watch the relief valve to see if grease is escaping. Watch the track or track adjustment cylinder to see if the track is being loosened.

Loosen the relief valve one turn only.

If track does not loosen, close the relief valve and contact your Caterpillar dealer.

NOTICE

Keeping the track properly adjusted will increase the service life of the track and drive components.

Note: The track tension must be adjusted according to the current operating conditions. Keep the track as slack as possible if the soil is heavy.

Measuring Track Tension

1. Operate the machine in the direction of the idlers.
-

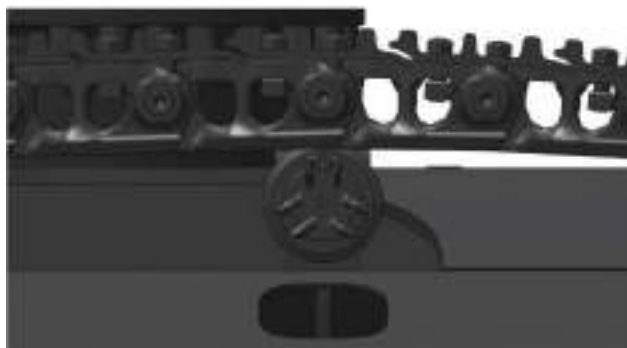


Illustration 892

g06188816

2. Stop with one track pin directly over the front carrier roller. Park the machine and turn off the engine.



Illustration 893

g06208711

3. Place a straight edge on top of the track grousers between the front carrier roller and the idler. The straight edge should be long enough to reach from the front carrier roller to the idler.

Note: If your machine is equipped with three carrier rollers, place a straight edge on the tracks between the carrier rollers. The straight edge should be long enough to reach from one carrier roller to another carrier roller.

4. Measure the maximum amount of sag in the track. The sag is measured from the highest point of the track grouser to the bottom of the straight edge. A track that is properly adjusted will have a sag of 40.0 to 55.0 mm (1.57 to 2.17 inch).

5. If the track is too tight, or if the track is too loose, adjust the track tension according to the appropriate procedure below.

Adjusting Track Tension



Illustration 894

g06188820

Typical example

The track adjuster is located on the track frame.

Tightening the Track

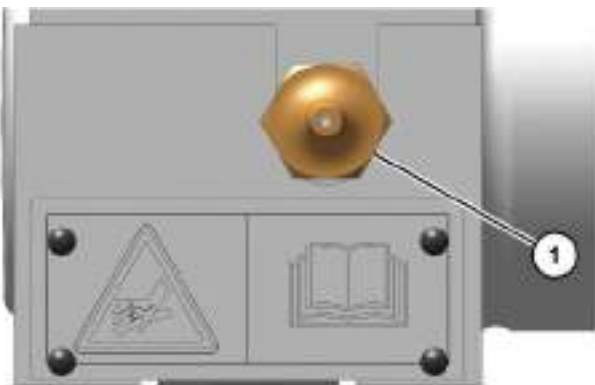


Illustration 895

g06188830

(1) Grease valve

Wipe the fitting before you add grease.

1. Add grease through grease valve (1) until the correct track tension is reached.
2. Operate the machine back and forth in order to equalize the pressure.
3. Check the amount of sag. Adjust the track, as needed.

Loosening the Track

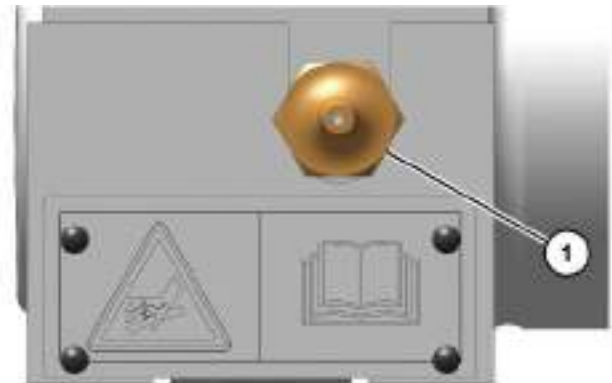


Illustration 896

g06188830

(1) Grease valve

1. Loosen grease valve (1) carefully until the track begins to loosen. One turn should be the maximum.
2. Tighten grease valve (1) to 34 ± 5 N·m (25 ± 4 lb ft) when the desired track tension is reached.
3. Operate the machine back and forth in order to equalize the pressure.
4. Check the amount of sag. Adjust the track, as needed.

i06969791

Track Adjustment - Inspect

SMCS Code: 4170-040



Illustration 897

g06182929

Check the track adjustment. Check the track for wear and for excessive dirt buildup.

If the track appears to be too tight or too loose, refer to Operation and Maintenance Manual, "Track Adjustment - Adjust".

i06954313

Travel Alarm - Test (If Equipped)

SMCS Code: 7429-081

Move the machine to test the travel alarm.

1. Start the engine. Move the hydraulic lockout control to the UNLOCKED position.
2. Raise the work tool to avoid any obstacles. Make sure that there is adequate overhead clearance.



Illustration 898

g06181402

3. Use the travel levers or the travel pedals to move the machine forward. The travel alarm should sound.
4. Release the travel levers and the travel pedals to stop the machine.
5. Use the travel levers and the travel pedals to move the machine backward. The travel alarm should sound.



Illustration 899

g06181631

6. Press the alarm mute button. The travel alarm should shut off.
7. Stop the machine. Lower the work tool to the ground. Move the hydraulic lockout control to the LOCKED position. Stop the engine.

i08233399

Undercarriage - Check

SMCS Code: 4150-535

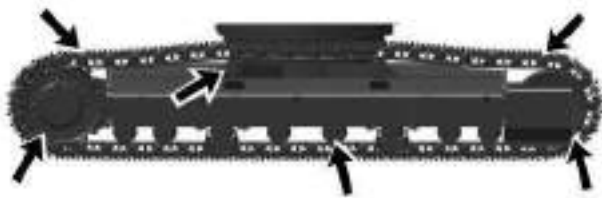


Illustration 900

g06182923

1. Check the carrier rollers, the track rollers, and the idler wheels for possible leakage.
2. Check the surface of the track, the carrier rollers, the track rollers, the idler wheels, the track shoes, and the drive sprockets. Look for signs of wear and loose mounting bolts.
3. Listen for any abnormal noises while you are moving slowly in an open area.
4. If required, clean the undercarriage to keep excess material from building up and solidifying.
5. If abnormal wear exists or abnormal noises or leaks are found, consult your Cat® dealer.

i06954326

Window Washer Reservoir - Fill

SMCS Code: 7306-544-KE

NOTICE

When operating in freezing temperatures, use Caterpillar or any commercially available nonfreezing window washer solvent.



Illustration 901

g06181546

1. Open the access door on the left side of the machine.

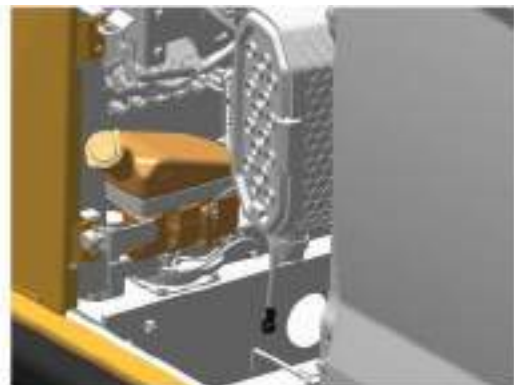


Illustration 902

g06181644

2. Remove the filler cap.
3. Fill the window washer reservoir with washer fluid through the filler opening.
4. Install the filler cap.
5. Close the access door.

i01258249

Window Wiper - Inspect/Replace

SMCS Code: 7305-040; 7305-510

Inspect the condition of the wiper blades. Replace the wiper blades if the wiper blades are worn or damaged or if streaking occurs.

i07103291

Windows - Clean

SMCS Code: 7310-070; 7340-070

Clean the outside of the windows from the ground, unless handholds are available.



Illustration 903

g06224268

Typical example

Cleaning Methods

Aircraft Window Cleaner

Apply the cleaner with a soft cloth. Rub the window with moderate pressure until all the dirt is removed. Allow the cleaner to dry. Wipe off the cleaner with a clean soft cloth.

Soap and Water

Use a clean sponge or a soft cloth. Wash the windows with a mild soap or with a mild detergent. Also use plenty of lukewarm water. Rinse the windows thoroughly. Dry the windows with a moist chamois or with a moist cellulose sponge.

Stubborn Dirt and Grease

Wash the windows with a good grade of naphtha, of isopropyl alcohol, or of Butyl Cellosolve. Then, wash the windows with soap and with water.

Polycarbonate Windows (If equipped)

Special care is needed to clean polycarbonate windows.

Wash polycarbonate windows with mild soap and warm water that does not exceed 50° C (122° F). Use a soft sponge, or damp cloth. Never use a dry cloth or paper towels on polycarbonate windows. Rinse the windows with a sufficient amount of clean cold water.

Note: Naphtha or kerosene can be used to remove labels, films, paint, or marking pen from polycarbonate windows.

Note: Do not use abrasive, or highly alkaline cleaners. Do not use sharp instruments, such as squeegees or razor blades on polycarbonate windows. Do not clean polycarbonate windows in the hot sun or at elevated temperatures.

Warranty Section

Warranty Information

i06044323

Emissions Warranty Information

SMCS Code: 1000

The certifying engine manufacturer warrants to the ultimate purchaser and each subsequent purchaser that:

1. New non-road diesel engines and stationary diesel engines less than 10 liters per cylinder (including Tier 1 and Tier 2 marine engines < 37 kW, but excluding locomotive and other marine engines) operated and serviced in the United States and Canada, including all parts of their emission control systems ("emission related components"), are:

- a. Designed, built, and equipped so as to conform, at the time of sale, with applicable emission standards prescribed by the United States Environmental Protection Agency (EPA) by way of regulation.
- b. Free from defects in materials and workmanship in emission-related components that can cause the engine to fail to conform to applicable emission standards for the warranty period.

2. New non-road diesel engines (including Tier 1 and Tier 2 marine propulsion engines < 37 kW and Tier 1 through Tier 4 marine auxiliary engines < 37 kW, but excluding locomotive and other marine engines) operated and serviced in the state of California, including all parts of their emission control systems ("emission related components"), are:

- a. Designed, built, and equipped so as to conform, at the time of sale, to all applicable regulations adopted by the California Air Resources Board (ARB).
- b. Free from defects in materials and workmanship which cause the failure of an emission-related component to be identical in all material respects to the component as described in the engine manufacturer's application for certification for the warranty period.

3. New non-road diesel engines installed in construction machines conforming to the South Korean regulations for construction machines manufactured after January 1, 2015, and operated and serviced in South Korea, including all parts of their emission control systems ("emission related components"), are:

- a. Designed, built, and equipped so as to conform, at the time of sale, with applicable emission standards prescribed in the Enforcement Rule of the Clean Air Conservation Act promulgated by South Korea MOE.
- b. Free from defects in materials and workmanship in emission-related components that can cause the engine to fail to conform to applicable emission standards for the warranty period.

The aftertreatment system can be expected to function properly for the lifetime of the engine (emissions durability period) subject to prescribed maintenance requirements being followed.

A detailed explanation of the Emission Control Warranty that is applicable to new non-road and stationary diesel engines, including the components covered and the warranty period, is found in a supplemental Special Publication. Consult your authorized Cat dealer to determine if your engine is subject to an Emission Control Warranty and to obtain a copy of the applicable Special Publication.

Reference Information Section

Reference Materials

i07422648

Reference Material

SMCS Code: 1000; 7000

Additional literature regarding your product may be purchased from your local Cat dealer or by visiting publications.cat.com. Use the product name, sales model, and serial number to obtain the correct information for your product.

publications.cat.com

i07743978

Decommissioning and Disposal

SMCS Code: 1000; 7000

When the product is removed from service, local regulations for the product decommissioning will vary. Disposal of the product will vary with local regulations.

Improperly disposing of waste can threaten the environment. Obey all local regulations for the decommissioning and disposal of materials.

Utilize appropriate personal protective equipment when decommissioning and disposing product.

Consult the nearest Cat dealer for additional information. Including information for component remanufacturing and recycling options.

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Product and Dealer Information

Note: For product identification plate locations, see the section “Product Identification Information” in the Operation and Maintenance Manual.

Delivery Date: _____

Product Information

Model: _____

Product Identification Number: _____

Engine Serial Number: _____

Transmission Serial Number: _____

Generator Serial Number: _____

Attachment Serial Numbers: _____

Attachment Information: _____

Customer Equipment Number: _____

Dealer Equipment Number: _____

Dealer Information

Name: _____ Branch: _____

Address: _____

	<u>Dealer Contact</u>	<u>Phone Number</u>	<u>Hours</u>
Sales:	_____	_____	_____
Parts:	_____	_____	_____
Service:	_____	_____	_____

